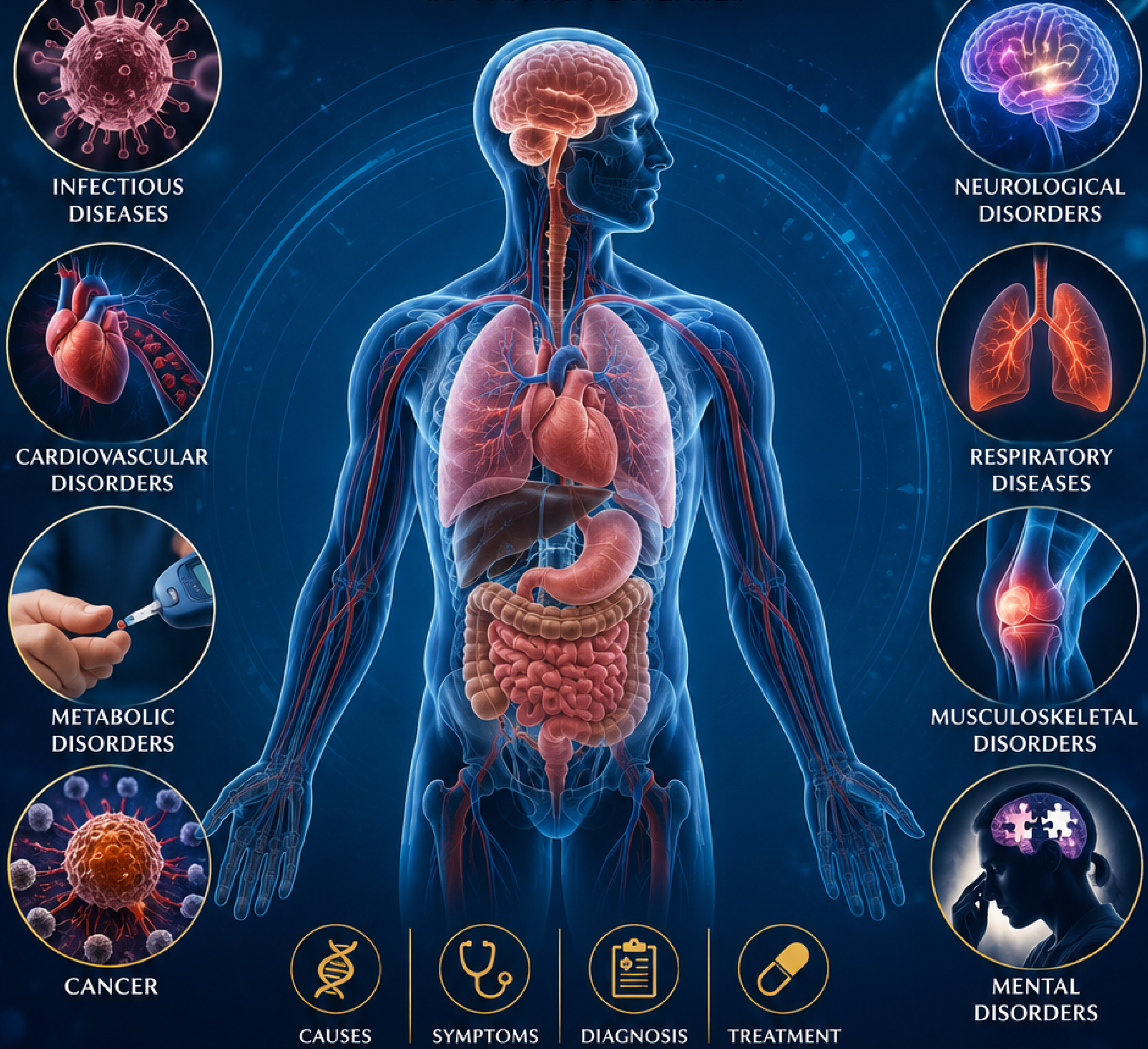
ALL ABOUT HUMAN DISEASES

A COMPLETE GUIDE TO UNDERSTANDING THE CAUSES, SYMPTOMS, DIAGNOSIS AND TREATMENT OF HUMAN DISEASES



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All About Human Diseases

An Organ-Categorized Medical Reference

A Complete Catalog of Diseases and Disorders

Affecting Each Organ and System of the Human Body

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This comprehensive report catalogs human diseases organized by the primary organ or organ system affected. Each chapter provides an overview of the organ's anatomy and function, followed by detailed entries covering the most significant diseases—their etiology, pathophysiology, clinical presentation, diagnosis, and treatment. The report spans all major organ systems of the human body, from the brain and nervous system to the reproductive organs, providing a thorough medical reference for students, educators, and practitioners.

Contents

Chapter 1

Abnormal Uterine Bleeding

1.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Abnormal uterine bleeding (AUB) is any deviation from normal menstrual volume, duration, or regularity. The PALM-COEIN classification system provides standardized terminology: structural causes include Polyp, Adenomyosis, Leiomyoma, Malignancy and abnormal increase in cells
- Non-structural causes include Coagulopathy, Ovulatory dysfunction, Endometrial, Iatrogenic, and Not yet classified.

1.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

1.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

1.4 Signs and Symptoms

- You may experience abnormal menstrual bleeding pattern. Diagnosis includes complete blood count, serum hCG to exclude pregnancy, TSH, prolactin, and pelvic ultrasound
- Hysteroscopy and endometrial biopsy are indicated for persistent AUB. Treatment depends on the cause: hormonal therapy, tranexamic acid, NSAIDs, or surgical management.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

1.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

1.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

1.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

1.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

1.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

1.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

1.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 2

Abscess

2.1 Overview

A cutaneous abscess is a localized collection of pus in the dermis or subcutaneous tissue, often caused by *S. aureus* (including MRSA). This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

2.2 Causes

This condition happens because of infection of a hair follicle (folliculitis progressing to furuncle to carbuncle), blocked sweat gland, or direct inoculation. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

2.3 Risk Factors

- Most people recover well with adequate removal of fluid
- Recurrent abscesses should prompt evaluation for risk factors (diabetes, HIV, nasal *S. aureus* carriage, community MRSA).
- Close contact with infected individuals
- Compromised immune system
- Travel to endemic areas
- Poor sanitation and hygiene practices

- Lack of vaccination
- Exposure to contaminated food or water
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

2.4 Signs and Symptoms

You may experience a painful, fluctuant, erythematous nodule with surrounding induration. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

2.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

2.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)

- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

2.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

2.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures

- Travel precautions and pre-travel consultations

2.9 Treatment Options

- Treatment typically involves incision and removal of fluid (I&D), which is the primary intervention
- Antibiotics are not routinely required for simple abscesses after adequate I&D but are indicated for surrounding cellulitis, immunocompromised patients, multiple lesions, or facial location.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

2.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

2.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 3

Acetaminophen (Paracetamol) Overdose

3.1 Overview

Acetaminophen overdose is the most common pharmaceutical overdose in the United States and the leading cause of acute liver failure. A toxic dose in adults is typically >7.5–10 g (or >150 mg/kg). This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

3.2 Causes

This condition happens because of depletion of liver glutathione and accumulation of the toxic metabolite N-acetyl-p-benzoquinone imine (NAPQI), which covalently binds to hepatocyte proteins, causing centrilobular liver tissue death. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

3.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

3.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

3.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Clinical features progress through four phases: Phase I (0–24 hours): nausea, vomiting, malaise, pallor
- Phase II (24–72 hours): right upper quadrant pain, elevated transaminases, PT/INR prolongation
- Phase III (72–96 hours): peak hepatotoxicity, liver encephalopathy, coagulopathy, jaundice, acute liver failure
- Phase IV (4 days–2 weeks): recovery or progression to death.

3.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

3.7 Diagnosis

Doctors diagnose this using serum acetaminophen level at >4 hours post-ingestion plotted on the Rumack-Matthew nomogram. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

3.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

3.9 Treatment Options

Treatment focuses on N-acetylcysteine (NAC), which replenishes glutathione and provides substrate for non-toxic sulfate conjugation, most effective if started within 8–10 hours. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

3.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

3.11 Outlook

- Your outlook depends on timely treatment
- The King's College Criteria guide liver transplantation decisions.

Chapter 4

Achalasia

4.1 Overview

Achalasia is a primary esophageal motility disorder characterized by impaired lower esophageal sphincter (LES) relaxation and absent esophageal peristalsis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

4.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

4.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

4.4 Signs and Symptoms

You may experience progressive difficulty swallowing to both solids and liquids, regurgitation of undigested food, chest pain, and weight loss. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

4.5 Progression and Stages

This condition happens because of autoimmune deterioration of inhibitory nitrergic ganglion cells in the myenteric (Auerbach) plexus of the esophagus. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

4.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

4.7 Diagnosis

- Your doctor confirms the diagnosis with esophageal manometry showing incomplete LES relaxation and aperistalsis
- Barium swallow demonstrates a characteristic bird's beak appearance of the distal esophagus. Management options include using small incisions and a camera Heller myotomy with fundoplication, peroral using a thin camera tube myotomy (POEM), pneumatic balloon dilation, and botulinum toxin injection into the LES.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

4.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

4.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

4.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

4.11 Outlook

The outlook is good for symptom control, though patients have a slightly increased risk of esophageal squamous cell carcinoma. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 5

Achondroplasia

5.1 Overview

Achondroplasia is the most common form of disproportionate short stature, caused by a gain-of-function mutation in FGFR3 (most commonly G380R), with an incidence of 1 in 25,000 live births. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

5.2 Causes

This condition happens because of constitutive activation of FGFR3, which negatively regulates endochondral ossification in the growth plate, leading to impaired long bone growth. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

5.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

5.4 Signs and Symptoms

You may experience rhizomelic limb shortening (proximal segments more affected), macrocephaly with frontal bossing, midface underdevelopment, trident hand, genu varum, thoracolumbar kyphosis, and foramen magnum stenosis. Intelligence is normal. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

5.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

5.6 Complications

Complications include cervicomedullary compression, hydrocephalus, spinal stenosis, and obesity.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

5.7 Diagnosis

Doctors diagnose this based on clinical and radiographic findings with FGFR3 genetic confirmation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

5.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

5.9 Treatment Options

Treatment focuses on monitoring for hydrocephalus, decompression surgery for spinal stenosis, limb lengthening procedures (controversial), and vosoritide (CNP analog, FDA-approved in 2021) to improve growth velocity. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

5.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

5.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 6

Acne Vulgaris

6.1 Overview

Acne vulgaris is a chronic inflammatory disorder of the pilosebaceous unit, affecting up to 85% of adolescents and young adults. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

6.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of four pathogenic factors: follicular hyperkeratinization, increased sebum production (androgen-mediated), *Cutibacterium acnes* (formerly *Propionibacterium acnes*) colonization, and inflammation. Clinical lesions include open comedones (blackheads), closed comedones (whiteheads), papules, pustules, nodules, and cysts
- Acne predominantly affects the face, chest, and back.

6.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

6.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

6.5 Progression and Stages

Severity is classified as mild (comedonal), moderate (papulopustular), or severe (nodulocystic, conglobata). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

6.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

6.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

6.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

6.9 Treatment Options

Isotretinoin requires iPLEDGE program enrollment due to teratogenicity. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

6.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

6.11 Outlook

- In most cases, the outlook is favorable
- Many patients improve with age.

Chapter 7

Acoustic Neuroma (Vestibular Schwannoma)

7.1 Overview

Vestibular schwannoma is a non-cancerous tumor arising from the Schwann cells of the vestibular portion of cranial nerve VIII and is the most common tumor of the cerebellopontine angle. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

7.2 Causes

This condition happens because of neoplastic growth of Schwann cells. Bilateral tumors are pathognomonic of neurofibromatosis type 2. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

7.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

7.4 Signs and Symptoms

- You may experience unilateral sensorineural hearing loss, ringing in the ears, and imbalance
- Larger tumors may compress adjacent structures causing facial numbness (CN V), facial weakness (CN VII), and brainstem compression.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

7.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

7.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

7.7 Diagnosis

Doctors diagnose this based on MRI with gadolinium, which is the gold standard. Management options include observation with serial imaging (for small, slow-growing tumors), stereotactic radiosurgery, and microsurgical removal.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

7.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

7.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

7.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

7.11 Outlook

The outlook is generally positive for small tumors, with hearing preservation as a key goal. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 8

Acromegaly

8.1 Overview

Acromegaly is a chronic disorder caused by excessive growth hormone (GH) secretion, almost always from a pituitary somatotroph adenoma (95%), with GH stimulating liver production of insulin-like growth factor 1 (IGF-1) which mediates most of the somatic effects, and onset being insidious with a mean delay of 7–10 years between symptom onset and diagnosis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

8.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

8.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

8.4 Signs and Symptoms

You may experience coarsening of facial features, prognathism (jaw protrusion), frontal bossing, enlarged hands and feet, soft tissue swelling, macroglossia, excessive sweating, deepening of the voice, arthropathy, carpal tunnel syndrome, high blood pressure, diabetes mellitus, sleep apnea, cardiomyopathy, and increased risk of colorectal polyps and cancer. Diagnosis shows elevated age- and sex-matched IGF-1 level and failure to suppress GH below 1 ng/mL after 75-g oral glucose tolerance test, with MRI showing a pituitary macroadenoma in most cases. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

8.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

8.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

8.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

8.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

8.9 Treatment Options

Treatment options include transsphenoidal surgery (first-line, curative in 80–90% of microadenomas), somatostatin analogs (octreotide LAR, lanreotide, pasireotide) for persistent disease, pegvisomant (GH receptor antagonist), and dopamine agonists (cabergoline) for mild cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

8.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

8.11 Outlook

outlook is improved with early treatment, which reduces cardiovascular morbidity and mortality. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 9

Actinic Keratosis

9.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

9.2 Causes

This condition happens because of UV-induced keratinocyte dysplasia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

9.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

9.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight

changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

9.5 Progression and Stages

Actinic keratosis is a precancerous lesion caused by chronic UV radiation exposure, representing the earliest stage of the squamous cell carcinoma spectrum. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

9.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

9.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Biopsy may be required to exclude carcinoma. Management options depend on the number and location of lesions: freezing treatment (liquid nitrogen), topical 5-fluorouracil, imiquimod, ingenol mebutate, photodynamic therapy, and field therapy for extensive actinic damage.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

9.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

9.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

9.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

9.11 Outlook

- Most people recover well with treatment
- Regular skin surveillance is recommended.

Chapter 10

Acute Bacterial Sinusitis

10.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- Acute bacterial sinusitis is a bacterial infection of the paranasal sinuses lasting less than 4 weeks, most commonly following a viral upper respiratory infection
- The most frequent pathogens are *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*.

10.2 Causes

This condition happens because of bacterial superinfection after viral mucosal inflammation impairs sinus removal of fluid. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

10.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

10.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

10.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- You may experience facial pain or pressure, nasal obstruction, purulent rhinorrhea, and loss of smell
- To make the diagnosis, doctors look for persistent symptoms for more than 10 days without improvement, worsening after initial improvement, or severe symptoms with high fever and purulent discharge for 3–4 consecutive days.

10.6 Complications

- Most people recover well with appropriate antibiotic therapy
- Complications (orbital, intracranial) are rare but serious.
- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

10.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

10.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer

- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

10.9 Treatment Options

Treatment options include amoxicillin-clavulanate as first-line therapy, with nasal saline irrigation, intranasal corticosteroids, and decongestants as adjunctive measures. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

10.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

10.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 11

Acute Bronchitis

11.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Acute bronchitis is an inflammation of the bronchial mucosa, most commonly caused by viral infections (rhinoviruses, coronaviruses, influenza, parainfluenza, adenovirus), accounting for approximately 90% of cases
- Bacterial causes include *Mycoplasma pneumoniae*, *Chlamydophila pneumoniae*, and *Bordetella pertussis*.

11.2 Causes

This condition happens because of infection of the bronchial mucosa leading to inflammation and increased mucus production. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

11.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

11.4 Signs and Symptoms

- You may experience acute onset of cough (productive or nonproductive), often following an upper respiratory infection, with or without sputum production, chest discomfort, and low-grade fever
- The cough may persist for 2–3 weeks.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

11.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

11.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

11.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with chest X-ray typically normal (helping to differentiate from pneumonia). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

11.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

11.9 Treatment Options

- Treatment typically involves symptomatic: rest, hydration, analgesics, cough suppressants for sleep-disturbing cough, and bronchodilators for wheezing
- Antibiotics are not indicated for most cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

11.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

11.11 Outlook

Most people recover well, with resolution in 1–3 weeks. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 12

Acute Decompensated Heart Failure

12.1 Overview

This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

12.2 Causes

This condition happens because of precipitating factors including medication non-adherence, dietary indiscretion, myocardial reduced blood flow, arrhythmias, infections, and anemia. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

12.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol

- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

12.4 Signs and Symptoms

You may experience shortness of breath at rest, orthopnea, peripheral swelling, and signs of lung congestion. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

12.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

- Acute decompensated heart failure is a sudden worsening of heart failure signs and symptoms, most commonly presenting as lung congestion with shortness of breath, orthopnea, and crackles on auscultation
- It is the leading cause of hospitalization in patients over 65.

12.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia

- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

12.7 Diagnosis

Doctors diagnose this based on clinical assessment, chest imaging, and natriuretic peptide levels. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

12.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

12.9 Treatment Options

- Treatment focuses on intravenous loop diuretics for decongestion, vasodilators for hypertensive crisis, and inotropic agents for cardiogenic shock
- Non-invasive positive pressure ventilation reduces the need for intubation.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk

- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

12.10 Living with It

- Your outlook depends on the underlying cause and response to therapy
- Early post-discharge follow-up and transition-of-care programs reduce readmission rates.
- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

12.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 13

Acute Gastritis

13.1 Overview

Acute gastritis is a self-limited inflammation of the gastric mucosa. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

13.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Common causes include NSAIDs, alcohol, stress (in critically ill patients), and infections
- Stress-related mucosal disease (SRMD) occurs in critically ill patients, particularly those requiring mechanical ventilation or with coagulopathy.

13.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

13.4 Signs and Symptoms

You may experience epigastric pain, nausea, vomiting, and hematemesis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

13.5 Progression and Stages

This condition happens because of disruption of the gastric mucosal barrier by offending agents, typically affecting the fundus and body with potential progression to erosion or ulceration. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

13.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

13.7 Diagnosis

- Doctors diagnose this based on clinical presentation
- Endoscopy may be performed when indicated.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

13.8 Prevention

- Treatment focuses on removal of the offending agent and acid suppression with PPIs or H2 receptor antagonists
- Prophylaxis is recommended for high-risk ICU patients.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

13.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

13.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

13.11 Outlook

Most people recover well with resolution upon removal of the causative factor. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 14

Acute Interstitial Nephritis

14.1 Overview

Acute interstitial nephritis (AIN) is an immune-mediated inflammatory condition of the kidney tubules and interstitium, a common cause of AKI in hospitalized patients. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

14.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common cause is drug reaction (antibiotics [penicillins, cephalosporins, fluoroquinolones], NSAIDs, PPIs, diuretics)
- Other causes include infections, autoimmune diseases, and idiopathic. This condition happens because of a hypersensitivity reaction to the offending agent.

14.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

14.4 Signs and Symptoms

- Clinical features may include fever, rash, eosinophilia, and eosinophiluria (the classic triad of fever, rash, and arthralgias is uncommon)
- AKI with sterile pyuria and mild proteinuria should raise suspicion.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

14.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

14.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

14.7 Diagnosis

To make the diagnosis, doctors look for kidney biopsy showing lymphocytic infiltration, swelling, and tubulitis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

14.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

14.9 Treatment Options

- Treatment focuses on discontinuation of the offending drug
- Corticosteroids are used for severe or delayed recovery.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

14.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

14.11 Outlook

In most cases, the outlook is favorable with early identification and drug withdrawal. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 15

Acute Kidney Injury

15.1 Overview

Acute kidney injury (AKI) is a rapid decline in kidney function over hours to days, resulting in accumulation of nitrogenous waste products (urea, creatinine) and inability to maintain fluid, electrolyte, and acid-base homeostasis. The KDIGO criteria define AKI as an increase in serum creatinine $\geq$0.3 mg/dL within 48 hours, increase in creatinine to $\geq$1.5 times baseline within 7 days, or urine volume <0.5 mL/kg/h for 6 hours. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

15.2 Causes

This condition happens because of the specific underlying cause leading to reduced GFR. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

15.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

15.4 Signs and Symptoms

You may experience oliguria or anuria, fluid overload, electrolyte disturbances, and metabolic acidosis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

15.5 Progression and Stages

Causes are classified as prerenal (decreased kidney perfusion: dehydration, heart failure, sepsis, NSAIDs), intrinsic (acute tubular tissue death from reduced blood flow or nephrotoxins, glomerulonephritis, interstitial nephritis, thrombotic microangiopathy), and postrenal (obstruction: kidney stones, enlarged prostate, pelvic malignancy). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

15.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

15.7 Diagnosis

Doctors diagnose this based on serum creatinine elevation, urine output criteria, urinalysis, and imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

15.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

15.9 Treatment Options

Treatment focuses on addressing the underlying cause, fluid and electrolyte management, and kidney replacement therapy (dialysis) for severe cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

15.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

15.11 Outlook

Mortality is significant, particularly in hospitalized patients. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 16

Acute Lymphoblastic Leukemia

16.1 Overview

Acute lymphoblastic leukemia is the most common childhood cancer (peak age 2–5 years), characterized by growth of immature lymphoid precursors (blasts) in the bone marrow, with B-cell ALL accounting for approximately 85% of childhood cases. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

16.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

16.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

16.4 Signs and Symptoms

You may experience bone marrow failure: anemia (fatigue, pallor), neutropenia (fever, infections), and thrombocytopenia (bleeding, bruising), along with bone pain, lymphadenopathy, hepatosplenomegaly, and testicular enlargement, with CNS involvement possibly presenting with headache, vomiting, or cranial nerve palsies. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

16.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer

prognosis.

16.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

16.7 Diagnosis

Doctors diagnose this using bone marrow biopsy showing >20% blasts, immunophenotyping (flow cytometry), cytogenetics [hyperdiploidy, t(12

- 21) ETV6-RUNX1—favorable
- T(9
- 22) BCR-ABL—unfavorable
- MLL rearrangements, Philadelphia chromosome-like ALL], and molecular studies.
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

16.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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- Participate in recommended cancer screening programs
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- Avoid exposure to known carcinogens in the workplace and environment

16.9 Treatment Options

Treatment typically involves risk-stratified and prolonged (2–3 years): induction (vincristine, corticosteroids, L-asparaginase, ± anthracyclines), consolidation, interim maintenance, delayed intensification, and maintenance therapy, with CNS-directed therapy (intrathecal chemotherapy, ± cranial radiation) essential. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
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- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

16.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Plan for work and financial considerations during treatment
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16.11 Outlook

Most people recover well in children, with cure rates exceeding 90% in developed countries, though adult ALL has a worse outlook (40–50% long-term survival), and CAR-T cell therapy (tisagenlecleucel) and blinatumomab are used for relapsed/refractory B-cell ALL. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 17

Acute Lymphoblastic Leukemia (ALL) in Children

17.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- ALL is the most common childhood cancer (25% of all pediatric malignancies) and the most common leukemia in children (see Chapter 10, Diseases of the Blood for comprehensive coverage). Peak incidence is 2–5 years. The majority of childhood ALL is B-cell precursor (B-ALL), and 25% have the t(12
- 21) translocation (ETV6-RUNX1 fusion) which confers a favorable outlook.

17.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

17.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)

- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

17.4 Signs and Symptoms

You may experience pallor, fatigue, fever, bone pain, tiny red or purple spots, bruising, and hepatosplenomegaly. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

17.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

17.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

17.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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17.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin

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- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

17.9 Treatment Options

Treatment usually involves risk-stratified chemotherapy with 5-year survival rates >90% in children. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

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17.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

17.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 18

Acute Myeloid Leukemia

18.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

18.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

18.3 Risk Factors

Acute myeloid leukemia is the most common acute leukemia in adults (median age 68), characterized by clonal growth of immature myeloid precursors (blasts) in the bone marrow, with risk factors including prior chemotherapy (alkylating agents, topoisomerase II inhibitors), radiation, myelodysplastic syndrome, and myeloproliferative tumors. The outlook varies from person to person depending on cytogenetic and molecular risk factors.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)

- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

18.4 Signs and Symptoms

You may experience bone marrow failure (anemia, neutropenia, thrombocytopenia) and symptoms of organ infiltration (hepatosplenomegaly, gum abnormal increase in cells, skin infiltration). Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
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- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

18.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer

prognosis.

18.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

18.7 Diagnosis

To make the diagnosis, doctors look for bone marrow biopsy showing ≥20% blasts, immunophenotyping, cytogenetics (t(15

- 17) APL—favorable
- Inv(16), t(8
- 21)—favorable
- Complex karyotype, monosomy 5/7—unfavorable), and molecular studies (FLT3, NPM1, CEBPA, IDH1/2, TP53 mutations).
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

18.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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18.9 Treatment Options

- Treatment focuses on induction chemotherapy (cytarabine + anthracycline ["7+3"] or CPX-351 for secondary AML) followed by consolidation (cytarabine, allogeneic stem cell transplantation for intermediate/high-risk), with acute promyelocytic leukemia (APL, t(15
- 17), PML-RARA) treated with all-trans retinoic acid (ATRA) + arsenic trioxide with excellent cure rates (>90%).
- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

18.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies

(acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

18.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 19

Acute Otitis Media

19.1 Overview

Acute otitis media is a bacterial or viral infection of the middle ear that affects 80% of children by age 3. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

19.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of Eustachian tube dysfunction secondary to upper respiratory infection, allowing bacterial or viral entry
- The most common pathogens are *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*.

19.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

19.4 Signs and Symptoms

- You may experience ear pain, fever, irritability, and hearing loss
- Otoscopic examination reveals a bulging, erythematous tympanic membrane with loss of landmarks and possible purulent effusion.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

19.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

19.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

19.7 Diagnosis

Doctors diagnose this based on clinical and otoscopic examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

19.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

19.9 Treatment Options

- Treatment options include analgesics and observation in mild cases, with amoxicillin as first-line antibiotic therapy for moderate-to-severe disease or treatment failure
- Recurrent acute otitis media may warrant myringotomy with tympanostomy tube placement.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

19.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

19.11 Outlook

Most people recover well with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 20

Acute Otitis Media

20.1 Overview

Acute otitis media (AOM) is the most common bacterial infection in children (see Chapter 24, Diseases of the Ear for comprehensive coverage). It affects 80% of children by age 3, with peak incidence at 6–24 months. Pathogens include Streptococcus pneumoniae, Haemophilus influenzae (nontypeable), and Moraxella catarrhalis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

20.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

20.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

20.4 Signs and Symptoms

You may experience otalgia, fever, irritability, and bulging tympanic membrane with decreased mobility. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

20.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

20.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

20.7 Diagnosis

To make the diagnosis, doctors look for acute onset of symptoms and middle ear effusion on otoscopy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

20.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

20.9 Treatment Options

Treatment options include observation (watchful waiting for children >6 months with mild symptoms) or antibiotics (amoxicillin first-line). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

20.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

20.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 21

Acute Pancreatitis

21.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Acute pancreatitis is acute inflammation of the pancreas characterized by sudden onset of severe epigastric pain radiating to the back, often with nausea and vomiting. The two most common causes worldwide are gallstones (40%) and alcohol excess (30%)
- Other causes include hypertriglyceridemia (triglycerides >1000 mg/dL), medications (azathioprine, valproic acid, thiazides), hypercalcemia, trauma, ERCP, and genetic factors (PRSS1, SPINK1 mutations).

21.2 Causes

This condition happens because of premature activation of digestive enzymes within the pancreas, leading to autodigestion and inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

21.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

21.4 Signs and Symptoms

You may experience characteristic abdominal pain, nausea, and vomiting. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

21.5 Progression and Stages

Severity is assessed using APACHE II, Ranson's criteria, and modified Marshall score. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

21.6 Complications

- Treatment options include aggressive IV fluid resuscitation (lactated Ringer's), pain control, early enteral nutrition, and monitoring for complications
- Infected pancreatic tissue death may require using a thin camera tube, through the skin, or surgical necrosectomy. Severe acute pancreatitis is marked by organ failure (persistent for >48 hours) and/or local complications (tissue death, pseudocyst, abscess).
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

21.7 Diagnosis

To make the diagnosis, doctors look for two of three criteria: characteristic abdominal pain, serum amylase or lipase >3 times the upper limit of normal, and characteristic findings on imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

21.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

21.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

21.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

21.11 Outlook

- In most cases, the outlook is favorable
- Most cases (80%) are mild and self-limiting.

Chapter 22

Acute Stress Disorder

22.1 Overview

Acute stress disorder has a variable prevalence (5–20% depending on trauma type). The condition describes the development of anxiety, dissociative, and other symptoms within 1 month of a traumatic event, with required symptom clusters similar to PTSD (intrusion, negative mood, dissociation, avoidance, and arousal) but with an emphasis on dissociative symptoms. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

22.2 Causes

Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

22.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

22.4 Signs and Symptoms

- Clinical features last for 3–30 days after trauma
- If symptoms persist beyond 1 month, the diagnosis transitions to PTSD.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

22.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

22.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

22.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

22.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

22.9 Treatment Options

Treatment options include trauma-focused CBT initiated within the first month. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

22.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

22.11 Outlook

- The outlook varies from person to person
- Approximately 50–80% of patients with ASD develop PTSD within 6 months.

Chapter 23

Acute/Short-Term Insomnia

23.1 Overview

Acute insomnia is a sleep disturbance lasting less than three months, typically triggered by a specific stressor (acute medical illness, hospitalization, surgery, bereavement, work stress, jet lag). It is extremely common, affecting 30–50% of adults annually. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

23.2 Causes

This condition happens because of transient hyperarousal and maladaptive behavioral responses (extended time in bed, napping, irregular sleep schedules). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

23.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

23.4 Signs and Symptoms

You may experience difficulty initiating or maintaining sleep that is temporally linked to an identifiable stressor.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

23.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

23.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

23.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

23.8 Prevention

Treatment focuses on sleep hygiene, stimulus control, and short-term pharmacotherapy limited to 1–2 weeks to prevent progression to chronic insomnia.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

23.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

23.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

23.11 Outlook

- The outlook is generally positive
- Most cases resolve spontaneously when the stressor resolves.

Chapter 24

Addison Disease

24.1 Overview

Addison disease (primary adrenal insufficiency) is chronic adrenocortical destruction leading to deficiency of cortisol, aldosterone, and adrenal androgen production. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

24.2 Causes

This condition happens because of autoimmune adrenalitis in Western nations (80% of cases, often associated with APS-1 or APS-2) and tuberculosis worldwide. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

24.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

24.4 Signs and Symptoms

You may experience fatigue, weight loss, hyperpigmentation of skin folds and palmar creases (due to elevated ACTH and MSH cleavage from POMC), postural hypotension, hyponatremia, hyperkalemia, and cravings for salt. Adrenal crisis is a life-threatening acute presentation with severe hypotension, vomiting, and confusion. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

24.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

24.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

24.7 Diagnosis

Your doctor confirms the diagnosis with low morning serum cortisol with elevated ACTH level, and failure to stimulate cortisol in response to the cosyntropin (synthetic ACTH) test. Management requires lifelong glucocorticoid replacement (hydrocortisone) and mineralocorticoid replacement (fludrocortisone). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

24.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

24.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

24.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

24.11 Outlook

Most people recover well with adequate replacement therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 25

Addison's Disease

25.1 Overview

Addison's disease (primary adrenal insufficiency) is destruction or dysfunction of the adrenal cortex, leading to deficiency of cortisol, aldosterone, and adrenal androgens. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

25.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common cause in developed countries is autoimmune adrenalitis (often associated with other autoimmune diseases in APS type 2), while tuberculosis remains the most common cause worldwide
- Other causes include infections (CMV in HIV/AIDS, fungal), adrenal bleeding (Waterhouse-Friderichsen syndrome—meningococcal sepsis), metastatic disease, and drugs (ketoconazole, etomidate).

25.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

25.4 Signs and Symptoms

You may experience fatigue, weight loss, anorexia, nausea, vomiting, abdominal pain, hyperpigmentation (from elevated ACTH and MSH stimulating melanocytes, a hallmark of primary adrenal insufficiency), salt craving, hypotension, and orthostatic dizziness. Diagnosis shows low morning cortisol, elevated ACTH (primary), and failure to suppress cortisol with high-dose dexamethasone. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

25.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

25.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

25.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

25.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

25.9 Treatment Options

Treatment typically involves glucocorticoid replacement (hydrocortisone 15–25 mg/day in divided doses) and mineralocorticoid replacement (fludrocortisone 0.05–0.2 mg/day). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

25.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

25.11 Outlook

The outlook is good with appropriate replacement therapy and education about stress dosing. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 26

Adhesive Capsulitis

26.1 Overview

Adhesive capsulitis (frozen shoulder) is a chronic condition characterized by progressive pain and global restriction of passive and active range of motion of the glenohumeral joint, affecting 2–5% of the population. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

26.2 Causes

This condition happens because of synovial inflammation followed by capsular thickening, pericapsular contracture, and dense fibrotic adhesions leading to severe loss of joint volume. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

26.3 Risk Factors

Risk factors include diabetes mellitus (most significant), thyroid dysfunction, cerebrovascular accidents, prolonged shoulder immobilization, and autoimmune disease. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

26.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

26.5 Progression and Stages

Clinical progression follows three overlapping phases: freezing (painful, 2–9 months), frozen (stiffness spredominates, 4–12 months), and thawing (gradual motion recovery, 5–24 months). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

26.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

26.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam based on equal restriction of active and passive external rotation
- Radiographs rule out glenohumeral osteoarthritis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

26.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

26.9 Treatment Options

Treatment focuses on physical therapy, intra-articular corticosteroid injections, oral NSAIDs, and manipulation under anesthesia or capsular release for recalcitrant cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

26.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

26.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 27

Adjustment Disorder

27.1 Overview

Adjustment disorder is one of the most commonly diagnosed psychiatric conditions in medical settings, with prevalence of 5–20% in outpatient mental health clinics and up to 12% in hospitalized medical patients. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

27.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

27.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

27.4 Signs and Symptoms

Once the stressor is removed, symptoms typically resolve within 6 months. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

27.5 Progression and Stages

You may experience the development of emotional or behavioral symptoms in response to an identifiable stressor within 3 months of onset, with marked distress that is out of proportion to the severity of the stressor, or significant functional impairment, not meeting criteria for another mental disorder. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

27.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

27.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

27.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

27.9 Treatment Options

Treatment typically involves primarily supportive psychotherapy, crisis intervention, and stress management. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

27.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

27.11 Outlook

Most people recover well, with most cases resolving with removal of the stressor. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 28

Adjuvant Analgesics

28.1 Overview

Adjuvant analgesics are medications whose primary indication is not pain but have analgesic properties in specific pain conditions. Gabapentinoids (gabapentin, pregabalin) bind to the alpha-2-delta subunit of voltage-gated calcium channels, reducing excitatory neurotransmitter release. They are first-line for neuropathic pain. Gabapentin is titrated from 300 mg to 900–3600 mg/day in three divided doses. Pregabalin is dosed at 150–600 mg/day in two to three divided doses. Side effects include sedation, dizziness, loss of coordination, peripheral swelling, and weight gain. Dose adjustment is required in kidney impairment. Tricyclic antidepressants (TCAs)—amitriptyline, nortriptyline, desipramine—inhibit reuptake of serotonin and norepinephrine, and block sodium channels and NMDA receptors. Doses for pain (10–150 mg at bedtime) are lower than those for depression. Side effects include anticholinergic effects (dry mouth, constipation, urinary retention, blurred vision), sedation, weight gain, and QT prolongation (caution in cardiac disease). Serotonin-norepinephrine reuptake inhibitors (SNRIs)—duloxetine, venlafaxine, milnacipran—are effective for diabetic peripheral neuropathy, fibromyalgia, and chronic musculoskeletal pain. Duloxetine is FDA-approved for diabetic peripheral neuropathy (60–120 mg/day), fibromyalgia (60–120 mg/day), and chronic MSK pain. Venlafaxine (150–225 mg/day) has a similar profile. Lidocaine patichtes (5%) are topical agents effective for postherpetic neuralgia and localized neuropathic pain. The patch is applied to the painful area for 12 hours on, 12 hours off. Systemic absorption is minimal. Capsaicin is a TRPV1 agonist that depletes substance P from peripheral nociceptors. Low-dose capsaicin creams and high-dose capsaicin 8% patches (applied in clinic under medical supervision for 30–60 minutes) are effective for postherpetic neuralgia and peripheral neuropathic pain. Ketamine is an NMDA receptor antagonist with potent analgesic and antihyperalgesic properties. Sub-anesthetic doses (IV infusion 0.1–0.5 mg/kg/hour) are used in acute pain (postoperative, opioid-tolerant patients, complex regional pain syndrome) and refractory chronic pain. The mechanism involves blockade of central sensitization. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

28.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

28.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

28.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

28.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

28.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries

- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

28.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

28.8 Prevention

They are effective for neuropathic pain, fibromyalgia, chronic tension-type headache, and migraine prevention.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

28.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

28.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups

- Stay informed about your condition

28.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 29

Adrenal Incidentaloma

29.1 Overview

Adrenal incidentalomas are adrenal masses discovered incidentally on imaging performed for unrelated reasons, found in approximately 4% of CT scans. Most are non-cancerous, non-functioning cortical adenomas, but they may represent functioning tumors (cortisol-producing adenoma causing subclinical Cushing's syndrome, aldosterone-producing adenoma, pheochromocytoma) or rarely, adrenocortical carcinoma or metastases. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

29.2 Causes

This condition happens because of incidental detection of adrenal masses. Evaluation requires hormonal workup (cortisol after 1 mg dexamethasone suppression test, plasma metanephrines, aldosterone/renin ratio) and imaging characteristics (size, density, enhancement pattern). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

29.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

29.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

29.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

29.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

29.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

29.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

29.9 Treatment Options

Treatment options include surgery for suspicious masses (>4 cm, HU >10, irregular borders, rapid growth), functioning tumors, and suspected malignancy, with surveillance for indeterminate masses. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

29.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

29.11 Outlook

Most people recover well for non-cancerous non-functioning adenomas.

Chapter 30

Adult-Onset Still's Disease

30.1 Overview

Adult-onset Still's disease is a rare systemic inflammatory disorder of unknown cause, considered the adult counterpart of systemic juvenile idiopathic arthritis (sJIA), characterized by a classic triad of high-spiking fevers, evanescent salmon-pink rash, and joint pain/arthritis, with an incidence of 0.16–0.4 per 100,000, a bimodal age distribution (15–25 and 36–46 years), and a slight female predominance. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

30.2 Causes

This condition happens because of dysregulated innate immunity with activation of the inflammasome, leading to overproduction of IL-1, IL-6, IL-18, and other pro-inflammatory cytokines, with macrophage activation being a key feature. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

30.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

30.4 Signs and Symptoms

You may experience quotidian or double-quotidian fevers, an evanescent, salmon-pink, non-pruritic maculopapular rash (Koebner phenomenon), joint pain and arthritis (initially pauciarticular, may progress to severe destructive polyarthritis, particularly involving wrists and carpal bones—carpal ankylosis is characteristic), sore throat (intense, exudative, a highly distinctive feature), lymphadenopathy and splenomegaly, serositis, and muscle pain. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

30.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

30.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

30.7 Diagnosis

Doctors diagnose this using Yamaguchi criteria, with serum ferritin characteristically markedly elevated (often >5–10 times normal, >1000 ng/mL), with decreased glycosylated ferritin (<20%). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

30.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

30.9 Treatment Options

Treatment typically involves stratified: NSAIDs for mild disease, corticosteroids for moderate disease, and IL-1 inhibitors (anakinra, canakinumab) or IL-6 inhibitors (tocilizumab) for refractory or severe disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

30.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

30.11 Outlook

- The outlook varies from person to person
- Some patients achieve remission, while others have a chronic relapsing course.

Chapter 31

Advance Care Planning

31.1 Overview

ACP reduces unwanted hospitalizations, improves goal-concordant care, and reduces distress for families and surrogates. In older adults with dementia, ACP should be initiated early in the disease course while the patient retains decisional capacity. Serious illness conversations should address outlook, expected functional decline, goals of care, and specific treatment preferences. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

31.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

31.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

31.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

31.5 Progression and Stages

Advance care planning (ACP) is the ongoing process of discussing values, goals, and preferences for future medical care with patients and their surrogates before the patient loses decision-making capacity. Components include designation of a healthcare proxy, advance directives, do-not-resuscitate (DNR) orders, do-not-intubate (DNI) orders, and Physician Orders for Life-Sustaining Treatment (POLST forms). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

31.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

31.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

31.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

31.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

31.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

31.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 32

Advanced Sleep-Wake Phase Disorder

32.1 Overview

It is less common than DSWPD, affecting approximately 1% of the population, predominantly older adults (though familial cases with PER2 or PER3 gene mutations exist). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

32.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

32.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

32.4 Signs and Symptoms

You may experience early evening sleepiness and early morning awakening. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

32.5 Progression and Stages

Advanced sleep-wake phase disorder (ASWPD) is a circadian rhythm sleep-wake disorder characterized by sleep-wake timing that is several hours earlier than desired, with sleep onset between 6:00–9:00 PM and spontaneous awakening between 2:00–5:00 AM. This condition happens because of a short intrinsic circadian period (<24 hours) or an exaggerated phase-advance response to morning light. Treatment focuses on bright light therapy in the evening to phase-delay the circadian clock and low-dose melatonin upon early morning awakening. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

32.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

32.7 Diagnosis

Doctors diagnose this using sleep diaries and actigraphy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

32.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

32.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

32.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

32.11 Outlook

The outlook is generally positive with treatment compliance. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 33

African Trypanosomiasis (Sleeping Sickness)

33.1 Overview

African trypanosomiasis (sleeping sickness) is a protozoan infection caused by *Trypanosoma brucei gambiense* (West African, 98% of cases, chronic) and *T. b. rhodesiense* (East African, acute), transmitted by the tsetse fly, with fewer than 2000 cases annually, endemic in sub-Saharan Africa. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

33.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

33.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

33.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

33.5 Progression and Stages

You may experience hemolymphatic stage (Stage 1): chancre at the tsetse fly bite site, intermittent fever, headache, joint pain, generalized lymphadenopathy (Winterbottom's sign), and itching The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Meningoencephalitic stage (Stage 2): progressive confusion, personality changes, daytime somnolence, nighttime insomnia, headache, tremor, loss of coordination, and finally coma and death. The right treatment depends on staging: Stage 1 doctors typically treat it with pentamidine or suramin
- Stage 2 doctors typically treat it with nifurtimox-eflornithine combination therapy (NECT) or melarsoprol. The outlook is good with early treatment
- Stage 2 disease has higher mortality.

33.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

33.7 Diagnosis

Doctors diagnose this using microscopic identification of motile trypomastigotes in blood, lymph node aspirate, or CSF. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

33.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

33.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

33.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

33.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 34

Age-Related Bone Loss

34.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- It affects 10–20% of women over 50 and 4–5% of men
- Prevalence increases steeply with age.

34.2 Causes

This condition happens because of an imbalance between bone resorption and bone formation, driven by estrogen deficiency, secondary hyperparathyroidism, decreased mechanical loading, and reduced anabolic hormone levels. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

34.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

34.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

34.5 Progression and Stages

Osteoporosis is a systemic skeletal disorder characterized by low bone mass and microarchitectural deterioration of bone tissue, leading to increased fracture risk. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

34.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

34.7 Diagnosis

- Doctors diagnose this using dual-energy X-ray absorptiometry (DXA): a T-score of $\leq$-2.5 defines osteoporosis
- -1.0 to -2.5 defines osteopenia. The FRAX tool estimates 10-year fracture probability.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

34.8 Prevention

Treatment options include calcium and vitamin D supplementation, weight-bearing exercise, fall prevention, bisphosphonates (first-line), denosumab, teriparatide, and romosozumab.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

34.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

34.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

34.11 Outlook

The outlook is generally positive with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 35

Age-Related Cataract

35.1 Overview

Age-related cataract is an opacification of the crystalline lens and the leading cause of preventable blindness worldwide, resulting from cumulative oxidative damage to lens proteins. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

35.2 Causes

This condition happens because of aggregation of damaged lens proteins that scatter light. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

35.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

35.4 Signs and Symptoms

You may experience painless progressive blurring of vision, glare sensitivity, decreased color perception, and dimming of vision. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

35.5 Progression and Stages

Classification by location includes nuclear (yellowing and hardening of the lens nucleus), cortical (peripheral wedge-shaped opacities), and posterior subcapsular (opacity near the posterior capsule, often associated with steroid use). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

35.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

35.7 Diagnosis

Doctors diagnose this using slit-lamp examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

35.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

35.9 Treatment Options

Treatment typically involves surgical: phacoemulsification with implantation of an intraocular lens. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

35.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

35.11 Outlook

Most people recover well, with cataract surgery being one of the most commonly performed and cost-effective surgical procedures globally. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 36

Age-Related Macular Degeneration

36.1 Overview

This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

36.2 Causes

This condition happens because of age-related degenerative changes in the retinal pigment epithelium and photoreceptors. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

36.3 Risk Factors

Risk factors include age, smoking, family history, and genetic variants (complement factor H).

- Advancing age
- Genetic predisposition
- Repetitive use or overuse (joints)
- Previous injuries
- Obesity
- Smoking and alcohol use
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

36.4 Signs and Symptoms

You may experience progressive central vision loss and distortion. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

36.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Age-related macular deterioration is the leading cause of irreversible central vision loss in individuals over 50 in developed countries, affecting the macula. Early AMD is marked by drusen and pigmentary changes
- Advanced AMD manifests as either dry (atrophic) AMD with geographic shrinkage or wasting or wet (neovascular) AMD with choroidal neovascularization.

36.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

36.7 Diagnosis

- Doctors diagnose this based on fundus examination and optical coherence tomography. Treatment for wet AMD involves anti-VEGF intravitreal injections (ranibizumab, aflibercept, bevacizumab)
- Complement inhibitors are under investigation for dry AMD.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

36.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

36.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

36.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

36.11 Outlook

The outlook has improved significantly with anti-VEGF therapy for wet AMD. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 37

Age-Related Macular Degeneration

37.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Wet (exudative/neovascular) AMD is less common but more aggressive, driven by choroidal neovascularization. Treatment for dry AMD includes the AREDS2 formulation (vitamins C and E, lutein/zeaxanthin, zinc) for intermediate disease
- Wet AMD doctors typically treat it with intravitreal anti-VEGF injections.

37.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

37.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

37.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

37.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Age-related macular deterioration (AMD) is the leading cause of irreversible central vision loss in adults over 65. Dry (nonexudative) AMD accounts for 85–90% of cases, characterized by drusen, retinal pigment epithelium shrinkage or wasting, and photoreceptor deterioration. outlook is guarded
- Low-vision rehabilitation is essential for all stages.

37.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

37.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

37.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

37.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

37.10 Living with It

In older adults, AMD severely impacts quality of life—loss of reading, face recognition, and driving independence—and is linked to depression and falls.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

37.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 38

Agoraphobia

38.1 Overview

Agoraphobia has a lifetime prevalence of 1–3%, with a mean age of onset in the late 20s. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

38.2 Causes

This condition happens because of conditioned avoidance behaviors and interoceptive hypersensitivity, with Disease mechanism overlapping with panic disorder. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

38.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

38.4 Signs and Symptoms

You may experience marked fear or anxiety about being in situations from which escape might be difficult or help might not be available in the event of panic-like symptoms, with typical feared situations including using public transportation, being in open spaces, being in enclosed spaces, standing in line or being in a crowd, or being outside of the home alone.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

38.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

38.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

38.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

38.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

38.9 Treatment Options

Treatment options include CBT with in vivo exposure and SSRIs/SNRIs. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

38.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

38.11 Outlook

The outlook is good with treatment. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 39

Alcohol Intoxication and Withdrawal

39.1 Overview

Alcohol (ethanol) is the most widely used psychoactive substance. Intoxication produces dose-dependent CNS depression: disinhibition, loss of coordination, slurred speech, involuntary eye movements, impaired judgment, and at higher levels, stupor, coma, and respiratory depression. Hypoglycemia (from inhibition of gluconeogenesis) is a critical consideration. Alcohol withdrawal syndrome occurs after reduction or cessation of chronic heavy alcohol use, typically starting 6–12 hours after last drink. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

39.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

39.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

39.4 Signs and Symptoms

- Symptoms can range from mild withdrawal (tremor, anxiety, tachycardia, high blood pressure) to delirium tremens (profound confusion, agitation, hallucinations, autonomic hyperactivity), with withdrawal seizures (generalized tonic-clonic) occurring 12–48 hours after cessation. Treatment: benzodiazepines are first-line, administered using a symptom-triggered protocol (CIWA-Ar scale)
- Severe DTs may require ICU admission.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

39.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

39.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

39.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

39.8 Prevention

Treatment for intoxication is supportive: monitoring, IV fluids, dextrose (if hypoglycemic), thiamine (100 mg IV) to prevent Wernicke encephalopathy, and consideration of co-ingestants.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

39.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

39.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

39.11 Outlook

The outlook is generally positive with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 40

Alcohol Use Disorder

40.1 Overview

Alcohol use disorder is one of the most prevalent psychiatric disorders, with a 12-month prevalence of 5–8% and lifetime prevalence of 20–30% in the United States, more common in men but increasing in women. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

40.2 Causes

This condition happens because of dysregulation of the brain reward system (mesolimbic dopamine pathway), with chronic alcohol use producing tolerance, withdrawal (GABA receptor downregulation and NMDA receptor upregulation), and sensitization of stress systems, with genetic factors contributing 50–60% of risk. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

40.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

40.4 Signs and Symptoms

You may experience at least 2 of 11 DSM-5-TR criteria occurring within a 12-month period, including alcohol taken in larger amounts or over longer periods than intended, persistent desire to cut down, craving, recurrent use resulting in failure to fulfill obligations, continued use despite problems, tolerance, and withdrawal. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

40.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

40.6 Complications

Medical complications include alcoholic hepatitis, cirrhosis, pancreatitis, Wernicke encephalopathy (confusion, loss of coordination, ophthalmoplegia—treated with IV thiamine), Korsakoff syndrome, alcoholic cardiomyopathy, peripheral neuropathy, and increased cancer risk.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

40.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

40.8 Prevention

- Treatment options include detoxification (benzodiazepines for withdrawal prevention), pharmacotherapy for relapse prevention (naltrexone, acamprosate, disulfiram), and psychosocial interventions (CBT, MET, 12-step facilitation). outlook: chronic relapsing condition
- 30–50% achieve sustained sobriety with treatment.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

40.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

40.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

40.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 41

Alcoholic Liver Disease

41.1 Overview

Alcoholic liver disease is a spectrum of liver injury due to excessive alcohol consumption that includes simple steatosis, alcoholic hepatitis, and cirrhosis. Simple steatosis is present in over 90% of heavy drinkers. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

41.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of direct hepatotoxicity of alcohol and its metabolites. Alcoholic hepatitis presents with jaundice, fever, hepatomegaly, and leukocytosis, carrying significant mortality (up to 40% in severe cases)
- Alcoholic cirrhosis develops in 10–20% of heavy drinkers.

41.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

41.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

41.5 Progression and Stages

Your symptoms depend on the stage of disease. Management centers on complete alcohol abstinence, nutritional support, corticosteroids for severe alcoholic hepatitis, and liver transplantation for end-stage disease. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

41.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

41.7 Diagnosis

Doctors diagnose this based on history of significant alcohol intake (typically more than 50 g/day in men, more than 40 g/day in women for more than 5 years), clinical and laboratory findings, and exclusion of other causes. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

41.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

41.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

41.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

41.11 Outlook

The outlook is generally positive with abstinence but poor with continued drinking. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 42

Alkaptonuria — Ochronosis

42.1 Overview

Alkaptonuria is a rare autosomal recessive disorder caused by deficiency of homogentisate 1,2-dioxygenase (HGD), the enzyme responsible for the breakdown of homogentisic acid (HGA) in the tyrosine degradation pathway. The incidence is approximately 1 in 250,000–1,000,000. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

42.2 Causes

This condition happens because of accumulation of HGA which polymerizes to form a dark brown-black pigment that deposits in connective tissues throughout the body. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

42.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

42.4 Signs and Symptoms

You may experience homogentisic aciduria (urine turns dark brown/black upon exposure to air), ochronosis (bluish-black pigmentation of ears, nose, sclera, and skin), ochronotic arthropathy (progressive degenerative joint disease affecting the spine and large joints), and cardiovascular involvement including aortic valve stenosis and coronary artery calcium buildup. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

42.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment typically involves symptomatic: nitisinone reduces HGA production and slows disease progression but does not reverse existing ochronosis
- Pain management, physical therapy, and joint replacement surgery for advanced arthropathy.

42.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

42.7 Diagnosis

Doctors diagnose this using elevated urinary homogentisic acid and HGD gene sequencing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

42.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

42.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

42.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

42.11 Outlook

- The outlook varies from person to person
- The disease is progressive.

Chapter 43

Allergic Fungal Rhinosinusitis (AFRS)

43.1 Overview

Allergic fungal rhinosinusitis (AFRS) is a non-invasive, IgE-mediated hypersensitivity reaction to dematiaceous fungal elements (Aspergillus, Bipolaris, Curvularia) inhabiting the paranasal sinuses, accounting for 5–10% of chronic rhinosinusitis cases in immunocompetent young adults. Fungal spores elicit an intense type I and type IV inflammatory response characterized by eosinophilic recruitment, thick 'peanut-butter-like' allergic mucin production, and nasal polyposis. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

43.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

43.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

43.4 Signs and Symptoms

You may experience chronic nasal obstruction, loss of smell, facial pressure, and expanded sinus osteitis causing orbital or anterior cranial fossa erosion. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

43.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

43.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

43.7 Diagnosis

Diagnosis relies on the Bent-Kuhn criteria: nasal polyposis, allergic mucin with Charcot-Leyden crystals and fungal hyphae on microscopy, characteristic hyperdense sinus opacification with central attenuation on CT, and positive anti-fungal IgE. Management requires functional using a thin camera tube sinus surgery (FESS) to evacuate allergic mucin followed by prolonged topical and systemic corticosteroid therapy. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

43.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification

- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

43.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

43.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

43.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 44

Allergic Rhinitis

44.1 Overview

Allergic rhinitis is an IgE-mediated inflammatory condition of the nasal mucosa that affects 10–30% of the global population. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

44.2 Causes

This condition happens because of exposure to airborne allergens (pollens, dust mites, animal dander, molds) triggering IgE-mediated mast cell degranulation. Seasonal allergic rhinitis (hay fever) correlates with pollen seasons, while perennial allergic rhinitis is year-round. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

44.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

44.4 Signs and Symptoms

- You may experience sneezing, nasal congestion, rhinorrhea (clear, watery discharge), nasal itching, and nasal obstruction
- Ocular symptoms (itching, tearing, conjunctival injection) are common.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

44.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

44.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

44.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by skin prick testing or specific serum IgE levels. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

44.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

44.9 Treatment Options

Treatment options include allergen avoidance, intranasal corticosteroids (first-line for moderate-to-severe disease), second-generation oral antihistamines, leukotriene receptor antagonists, and allergen immunotherapy for refractory or severe cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

44.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

44.11 Outlook

The outlook is generally positive with appropriate treatment, though the condition is chronic. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 45

Alopecia Areata

45.1 Overview

Alopecia areata is an autoimmune dermatological disorder characterized by non-scarring patchy hair loss affecting hair-bearing areas of the body, most commonly the scalp and beard area. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

45.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

45.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

45.4 Signs and Symptoms

You may experience sudden onset of smooth, round or oval, circumscribed patches of hair loss without redness of the skin or scaling, featuring pathognomonic 'exclamation mark' hairs (short broken hairs tapering near the scalp surface) at active patch margins. Severe variants include alopecia totalis (complete loss of scalp hair) and alopecia universalis (complete loss of all scalp and body hair). Nail changes (pitting, trachyonychia) occur in 10–20% of patients. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

45.5 Progression and Stages

This condition happens because of CD8+ T cell-mediated autoimmune attack directed against hair follicles in the anagen (growth) phase, with collapse of hair follicle immune privilege. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

45.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

45.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam based on characteristic morphology and dermoscopy (trichoscopy). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

45.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

45.9 Treatment Options

Treatment options include topical or intralesional corticosteroid injections (triamcinolone acetonide) for localized disease, topical minoxidil, immunotherapy, and oral Janus kinase (JAK) inhibitors (baricitinib, ritlecitinib) for severe diffuse disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

45.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

45.11 Outlook

- The outlook varies from person to person
- Spontaneous regrowth occurs in localized disease, but relapse is common.

Chapter 46

Alpha-1 Antitrypsin Deficiency

46.1 Overview

Alpha-1 antitrypsin (AAT) deficiency is an autosomal codominant disorder caused by mutations in the SERPINA1 gene, most commonly the Pi*Z allele, leading to reduced serum levels and accumulation of misfolded AAT in the liver. The incidence is approximately 1 in 2,500–5,000 live births in Northern European populations. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

46.2 Causes

This condition happens because of accumulation of polymerized AAT in hepatocytes causing liver disease and lack of functional AAT in the lungs permitting unchecked proteolytic destruction of alveolar walls. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

46.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

46.4 Signs and Symptoms

You may experience neonatal cholestasis, hepatitis, cirrhosis, and hepatocellular carcinoma (liver disease)

- Early-onset panacinar emphysema particularly in smokers (lung disease)
- And widespread panniculitis.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

46.5 Progression and Stages

Treatment options include intravenous augmentation therapy (weekly infusions of pooled human plasma-derived AAT) to slow lung function decline The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Smoking cessation is critical
- Liver transplantation is curative for end-stage liver disease.

46.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

46.7 Diagnosis

Doctors diagnose this using low serum AAT level, phenotyping (Pi*ZZ, Pi*SZ), and genotyping.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

46.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

46.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

46.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

46.11 Outlook

- The outlook varies from person to person
- Avoidance of hepatotoxins is essential. Disorders of lipid metabolism encompass both primary (genetic) and secondary dyslipidemias that contribute substantially to the global burden of atherosclerotic cardiovascular disease. Genetic dyslipidemias result from mutations affecting lipoprotein production, processing, or clearance, while secondary causes include diabetes, hypothyroidism, nephrotic syndrome, cholestasis, and medications. Early identification and aggressive lipid-lowering therapy can substantially modify cardiovascular risk.

Chapter 47

Altitude Sickness

47.1 Overview

Altitude illness occurs at elevations above 2,500 m (8,000 ft) due to decreased barometric pressure and hypoxia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

47.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of compensatory hyperventilation, increased sympathetic tone, and lung and cerebral vasoconstriction and vasodilation
- Inadequate acclimatization leads to three syndromes. Acute mountain sickness (AMS) is the most common form: headache, nausea, dizziness, anorexia, fatigue, and sleep disturbance.

47.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

47.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

47.5 Progression and Stages

High-altitude cerebral swelling (HACE) is a life-threatening progression with loss of coordination, altered mental status, hallucinations, seizures, and coma. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

47.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

47.7 Diagnosis

Doctors use the Lake Louise Score. Treatment: immediate descent, supplemental oxygen, dexamethasone. High-altitude lung swelling (HAPE) is non-cardiogenic lung swelling with cough, shortness of breath at rest, cyanosis, and crackles. Treatment: immediate descent, supplemental oxygen, nifedipine or phosphodiesterase-5 inhibitors. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

47.8 Prevention

- Treatment: acetazolamide for prophylaxis and mild treatment
- Descent for persistent symptoms.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

47.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

47.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

47.11 Outlook

The outlook is generally positive with prompt descent. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 48

Alzheimer Disease

48.1 Overview

Alzheimer disease (AD) is a progressive neurodegenerative disorder and the most common cause of dementia (60–80% of cases), with prevalence doubling every 5 years after age 65. In the geriatric population, AD presents with a higher burden of comorbidity, polypharmacy, and functional dependence. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

48.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

48.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

48.4 Signs and Symptoms

You may experience progressive memory loss, impaired executive function, language difficulties, and behavioral and psychological symptoms of dementia (BPSD: agitation, aggression, psychosis, depression, apathy). Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

48.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

48.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

48.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam based on history, examination, and thinking and memory testing. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

48.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

48.9 Treatment Options

- Treatment options include non-pharmacological approaches first (person-centered care, environmental modification, structured routines)
- Antipsychotics carry a black box warning for increased mortality in elderly dementia patients and should be used only for severe, refractory symptoms.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

48.10 Living with It

- Late-onset AD (>65 years) is distinguished from early-onset AD by its sporadic nature, strong association with the APOE epsilon-4 allele, and slower progression. outlook is progressive and ultimately fatal
- End-of-life considerations include advance care planning and hospice referral.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

48.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 49

Alzheimer's Disease

49.1 Overview

Alzheimer's disease is a progressive neurodegenerative disorder and the most common cause of dementia, accounting for 60–70% of cases. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

49.2 Causes

This condition happens because of the accumulation of amyloid-beta (Aβ) plaques between neurons and neurofibrillary tangles composed of hyperphosphorylated tau protein within neurons. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

49.3 Risk Factors

Risk factors include advancing age, family history, the apolipoprotein E4 allele, cardiovascular risk factors, and low educational attainment.

- Advanced age
- Family history and genetic predisposition
- Head trauma or brain injury
- Vascular risk factors (hypertension, diabetes)
- Exposure to environmental toxins
- Chronic stress
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

49.4 Signs and Symptoms

You may experience insidious memory impairment, particularly of recent events, progressing to global thinking and memory decline, disorientation, language difficulties, and eventual loss of independence. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

49.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

49.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries

- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

49.7 Diagnosis

Doctors diagnose this based on clinical assessment, neuropsychological testing, neuroimaging showing hippocampal shrinkage or wasting and reduced glucose metabolism on PET, and cerebrospinal fluid biomarkers (Aβ42, tau, phospho-tau). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

49.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

49.9 Treatment Options

- Treatment focuses on symptomatic treatment with cholinesterase inhibitors (donepezil, rivastigmine, galantamine) and memantine
- Disease-modifying therapies targeting amyloid remain under investigation.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

49.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

49.11 Outlook

outlook is progressive decline with modest benefit from available therapies. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 50

Ambiguous Genitalia / Disorders of Sex Development

50.1 Overview

Disorders of sex development (DSD) are congenital conditions in which the development of chromosomal, gonadal, or anatomic sex is atypical. Incidence is approximately 1 in 4500 live births. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

50.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

50.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

50.4 Signs and Symptoms

- You may experience atypical genitalia at birth. Evaluation includes karyotype, hormone levels, imaging, and genitourinary anatomy assessment. Management requires a multidisciplinary team
- Salt-wasting congenital adrenal abnormal increase in cells is a life-threatening emergency requiring glucocorticoid and mineralocorticoid replacement.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

50.5 Progression and Stages

DSD is classified by karyotype: 46,XX DSD (most commonly congenital adrenal abnormal increase in cells due to 21-hydroxylase deficiency), 46,XY DSD (androgen insensitivity syndrome, 5-alpha-reductase deficiency, gonadal dysgenesis), and sex chromosome DSD (Turner syndrome, Klinefelter syndrome, mixed gonadal dysgenesis). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

50.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

50.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

50.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

50.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

50.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

50.11 Outlook

Your outlook depends on the specific diagnosis. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 51

Amblyopia

51.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Amblyopia is reduced best-corrected visual acuity in one or both eyes resulting from abnormal visual development during the critical period of visual maturation (birth to approximately 8 years)
- It is the most common cause of monocular vision loss in children.

51.2 Causes

This condition happens because of strabismus, anisometropia, form deprivation (congenital cataract, ptosis), or bilateral high refractive error, leading to failure of normal neural pathway development in the amblyopic eye. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

51.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

51.4 Signs and Symptoms

You may experience reduced visual acuity that cannot be corrected with lenses. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

51.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

51.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

51.7 Diagnosis

Doctors diagnose this based on visual acuity testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

51.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

51.9 Treatment Options

Treatment options include corrective lenses, patching or penalization (atropine drops) of the dominant eye, and treatment of the underlying cause. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

51.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

51.11 Outlook

Most people recover well with early intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 52

Amebiasis (Entamoeba histolytica)

52.1 Overview

Amebiasis is an intestinal and extraintestinal infection caused by *Entamoeba histolytica*, a protozoan parasite, with 50 million symptomatic cases and 100,000 deaths annually worldwide, particularly in the Indian subcontinent, sub-Saharan Africa, and Central/South America. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

52.2 Causes

This condition happens because of cysts excysting in the terminal ileum, releasing trophozoites that colonize the large intestine, with *E. histolytica* trophozoites adhering to colonic mucin, secreting cysteine proteinases that degrade extracellular matrix, inducing host cell apoptosis, and invading the colonic epithelium. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

52.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

52.4 Signs and Symptoms

- You may experience intestinal amebiasis: amoebic dysentery—bloody mucoid diarrhea (strawberry jelly stools), abdominal pain, tenesmus, and weight loss
- Extraintestinal amebiasis: amebic liver abscess—most common extraintestinal form with fever, right upper quadrant pain, and "anchovy paste" pus.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

52.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

52.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

52.7 Diagnosis

Doctors diagnose this using fresh stool microscopy (trophozoites with ingested RBCs), antigen detection, PCR, and serology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

52.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

52.9 Treatment Options

Treatment options include tinidazole or metronidazole followed by a luminal agent (paromomycin) to eradicate cysts. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

52.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

52.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 53

Amenorrhea (Primary, Secondary)

53.1 Overview

Amenorrhea is the absence of menstruation. Primary amenorrhea is the absence of menarche by age 15 with normal secondary sexual development, or by age 13 with no secondary sexual characteristics. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

53.2 Causes

Causes include hypothalamic/pituitary disorders, gonadal disorders (Turner syndrome, Swyer syndrome, premature ovarian insufficiency), and outflow tract obstruction (imperforate hymen, Müllerian agenesis). Secondary amenorrhea is the absence of menstruation for three months or more in a woman with previously normal cycles The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common cause is pregnancy, with other causes including functional hypothalamic amenorrhea, PCOS, hyperprolactinemia, premature ovarian insufficiency, and Asherman syndrome. Diagnosis includes β-hCG, prolactin, TSH, FSH, and estrogen levels
- Progestin challenge test and MRI for pituitary lesions or outflow tract anomalies.

53.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

53.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

53.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

53.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

53.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

53.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

53.9 Treatment Options

The right treatment depends on the underlying cause. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

53.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

53.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 54

Amyloidosis

54.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Amyloidosis is a group of diseases characterized by the extracellular deposition of abnormal, insoluble fibrillar proteins (amyloid) in various tissues and organs, disrupting normal structure and function, with three main types clinically important: AL amyloidosis (immunoglobulin light chain—the most common systemic form, associated with plasma cell dyscrasia/multiple myeloma), AA amyloidosis (secondary/reactive
- Serum amyloid A protein derived from chronic inflammation), and ATTR amyloidosis (transthyretin
- Hereditary or wild-type/senile systemic amyloidosis). The incidence of AL amyloidosis is 3–5 per million, typically presenting at age 60–70.

54.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

54.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

54.4 Signs and Symptoms

Your symptoms depend on the organs involved: AL amyloidosis commonly presents with nephrotic syndrome (kidney amyloidosis—most common presentation, 70%), restrictive cardiomyopathy (heart failure with preserved ejection fraction, low-voltage QRS on ECG, thickened ventricular walls on echocardiography with a "speckled" appearance), hepatomegaly, peripheral and autonomic neuropathy, macroglossia (tongue enlargement—pathognomonic for AL), periorbital purple spots on the skin, and carpal tunnel syndrome. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

54.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment for AL amyloidosis is anti-plasma cell therapy (similar to multiple myeloma: bortezomib-based regimens, autologous stem cell transplantation in selected patients)
- AA amyloidosis is treated by controlling the underlying inflammatory condition
- ATTR amyloidosis doctors typically treat it with tafamidis (stabilizes transthyretin, slows neuropathy and cardiomyopathy progression).

54.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

54.7 Diagnosis

To make the diagnosis, doctors look for tissue biopsy with Congo red staining and apple-green birefringence under polarized light—the diagnostic gold standard. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

54.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

54.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

54.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

54.11 Outlook

Unfortunately, the outlook is often poor for AL amyloidosis without treatment but has improved significantly with modern therapies. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 55

Amyotrophic Lateral Sclerosis

55.1 Overview

About 10% of cases are familial, with mutations in SOD1, C9orf72, TARDBP, and FUS genes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

55.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

55.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

55.4 Signs and Symptoms

- You may experience focal muscle weakness, shrinkage or wasting, fasciculations, spasticity, and hyperreflexia
- Bulbar involvement causes slurred speech and difficulty swallowing.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

55.5 Progression and Stages

Amyotrophic lateral sclerosis is a fatal motor neuron disease characterized by progressive deterioration of both upper and lower motor neurons in the motor cortex, brainstem, and spinal cord. This condition happens because of progressive motor neuron deterioration. Treatment focuses on riluzole and edaravone to modestly slow progression, non-invasive ventilation, and multidisciplinary care. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

55.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

55.7 Diagnosis

Diagnosis relies on the revised El Escorial criteria, supported by electromyography and exclusion of mimics. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

55.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

55.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

55.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

55.11 Outlook

Unfortunately, the outlook is often poor, with respiratory failure typically causing death within 3–5 years of onset. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 56

Anal Fissure

56.1 Overview

Anal fissure is a painful linear tear or ulceration in the distal anal canal lining extending from the dentate line to the anal verge. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

56.2 Causes

This condition happens because of trauma caused by passage of hard, dry stool or severe diarrhea, precipitating internal anal sphincter spasm and reduced blood flow of the posterior anal midline (where 90% of fissures occur due to poor microvascular perfusion). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

56.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

56.4 Signs and Symptoms

- You may experience severe, sharp, tearing pain during and after defecation, accompanied by bright red rectal bleeding on toilet paper or stool surface. Physical examination reveals a visible longitudinal fissure
- Chronic fissures feature a sentinel skin tag and hypertrophied anal papilla.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

56.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

56.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

56.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

56.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

56.9 Treatment Options

- Treatment options include stool softeners, high-fiber diet, warm sitz baths, and topical nitrates (0.2% nitroglycerin) or calcium channel blockers (2% diltiazem) to relax the sphincter
- Lateral internal sphincterotomy is reserved for chronic medically refractory fissures.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

56.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

56.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 57

Anaphylaxis

57.1 Overview

Anaphylaxis is a severe, life-threatening systemic hypersensitivity reaction (usually IgE-mediated) that occurs rapidly and can involve multiple organ systems. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

57.2 Causes

This condition happens because of exposure to triggers including foods (nuts, shellfish, milk, eggs), insect stings (Hymenoptera), medications (penicillin, NSAIDs), latex, and contrast media. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

57.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

57.4 Signs and Symptoms

You may experience skin involvement (urticaria, angioedema, flushing in 80–90%), respiratory compromise (laryngeal swelling, bronchospasm), cardiovascular collapse (hypotension, distributive shock), digestive tract symptoms (nausea, vomiting, diarrhea), and neurological symptoms (anxiety, altered consciousness). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

57.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

57.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

57.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam: acute onset (minutes to hours) of skin/mucosal involvement plus respiratory compromise or hypotension after exposure to a likely allergen, with serum tryptase possibly elevated but not required for diagnosis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

57.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

57.9 Treatment Options

Treatment typically involves with intramuscular epinephrine (0.3–0.5 mg of 1:1000 in the mid-ant thigh, repeat every 5–15 minutes as needed), IV fluid resuscitation, H1 and H2 antihistamines, corticosteroids, and inhaled bronchodilators for bronchospasm. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

57.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

57.11 Outlook

Most people recover well with prompt epinephrine administration. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 58

Angelman Syndrome

58.1 Overview

Angelman syndrome is a neurogenetic disorder caused by loss of function of the maternal copy of the UBE3A gene on chromosome 15q11-q13. Mechanisms include maternal deletion (70%), paternal uniparental disomy (2–3%), imprinting defects (3–5%), and UBE3A mutations (10%). Incidence is 1 in 12,000–20,000. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

58.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

58.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

58.4 Signs and Symptoms

- You may experience severe intellectual disability, loss of coordination, seizures, frequent laughter, happy demeanor, hand-flapping, and absent speech
- Microcephaly develops after age 2 years.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

58.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

58.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

58.7 Diagnosis

Doctors diagnose this using methylation analysis of the 15q11-q13 region.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

58.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

58.9 Treatment Options

Treatment typically involves supportive, with antiepileptic drugs for seizures and physical/occupational therapy. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

58.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 59

Angina Pectoris

59.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Angina pectoris is chest discomfort caused by transient myocardial reduced blood flow without myocardial tissue death. Stable angina occurs predictably with exertion or emotional stress and is relieved by rest or nitroglycerin, reflecting fixed coronary atherosclerotic stenosis
- Unstable angina occurs unpredictably or at rest and represents acute coronary syndrome without troponin elevation
- Microvascular angina results from coronary microvascular dysfunction.

59.2 Causes

This condition happens because of an imbalance between myocardial oxygen supply and demand. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

59.3 Risk Factors

- Treatment for stable angina includes antianginals (beta-blockers, calcium channel blockers, nitrates), risk factor modification, and revascularization when medical therapy is inadequate
- Unstable angina requires urgent antiplatelet therapy, anticoagulation, and early invasive strategy. Your outlook depends on the extent of coronary disease and risk

factor control.
- Advanced age
- Cigarette smoking
- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease
- Obesity and physical inactivity
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

59.4 Signs and Symptoms

You may experience chest pressure or discomfort that may radiate to the left arm or jaw. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

59.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

59.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

59.7 Diagnosis

Diagnosis typically involves ECG (may show ST depression during episodes), stress testing, and coronary angiography for definitive anatomic assessment. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

59.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

59.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

59.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

59.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 60

Ankylosing Spondylitis

60.1 Overview

Ankylosing spondylitis (AS) is a chronic inflammatory disease primarily affecting the axial skeleton (sacroiliac joints and spine), representing the prototypical spondyloarthropathy. It predominantly affects young men (20–40 years) and is strongly associated with HLA-B27 (present in 90% of patients). This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

60.2 Causes

This condition happens because of immune-mediated inflammation of the entheses and axial joints. The hallmark is sacroiliitis presenting with inflammatory back pain (insidious onset, morning stiffness >30 minutes, improvement with activity, onset before age 40). In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

60.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease

- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

60.4 Signs and Symptoms

- You may experience progressive spinal involvement leading to syndesmophyte formation, vertebral fusion ("bamboo spine"), and spinal rigidity
- Peripheral arthritis (hips, shoulders, knees), enthesitis (Achilles tendinitis, plantar fasciitis), anterior uveitis, and aortic regurgitation are associated features.
- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

60.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

60.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

60.7 Diagnosis

Diagnosis typically involves clinical assessment, HLA-B27 testing, and imaging (MRI for early sacroiliitis, X-rays showing sacroiliac joint erosions and fusion, syndesmophytes). Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

60.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

60.9 Treatment Options

- Treatment options include NSAIDs (first-line, used continuously), TNF inhibitors (adalimumab, etanercept, infliximab, certolizumab), and IL-17 inhibitors (secukinumab, ixekizumab) for inadequate response
- Physical therapy is essential.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function

- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

60.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

60.11 Outlook

- The outlook varies from person to person
- Early treatment improves outcomes.

Chapter 61

Anorexia Nervosa

61.1 Overview

Anorexia nervosa has a lifetime prevalence of 0.5–1% in women and 0.1–0.3% in men, with a 10:1 female-to-male ratio, and age of onset typically in adolescence (peak 15–19 years). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

61.2 Causes

This condition happens because of serotonergic and dopaminergic dysregulation, aberrant reward processing, and sociocultural factors (thin-ideal internalization). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

61.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

61.4 Signs and Symptoms

You may experience restriction of energy intake relative to requirements leading to significantly low body weight, intense fear of gaining weight or becoming fat (even when underweight), and disturbance in self-perceived weight or shape, with subtypes including restricting type and binge-eating/purging type.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

61.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

61.6 Complications

Medical complications include bradycardia, hypotension, hypothermia, osteopenia/osteoporosis, lanugo, irregular heartbeats (prolonged QT interval), electrolyte disturbances, refeeding syndrome, and structural brain changes.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

61.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

61.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

61.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

61.10 Living with It

Treatment options include medical stabilization, nutritional rehabilitation, psychotherapy (family-based treatment for adolescents, CBT-E for adults), and olanzapine for weight gain.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

61.11 Outlook

The outlook varies from person to person: 50% recover, 30% improve, 20% remain chronic, with mortality of 5–10%. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 62

Anthrax

62.1 Overview

Anthrax is a zoonotic infection caused by *Bacillus anthracis*, a large, gram-positive, spore-forming, encapsulated rod, endemic in agricultural regions worldwide, acquired through contact with infected animals or animal products, and is a category A bioterrorism agent. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

62.2 Causes

This condition happens because of spores entering through skin (cutaneous), inhalation (lung), or ingestion (digestive tract), germinating and producing three virulence factors: protective antigen (PA), lethal factor (LF), and swelling factor (EF), with LF being a metalloproteinase that cleaves MAPKKs, disrupting signaling and causing macrophage lysis and cytokine storm. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

62.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

62.4 Signs and Symptoms

You may experience cutaneous anthrax (95% of natural cases): pruritic papule developing into a painless, ulcerated, black eschar surrounded by significant swelling (cancerous pustule)

- Inhalational anthrax: initial flu-like symptoms followed abruptly by respiratory failure, involving bleeding mediastinitis, pleural effusions, and shock
- Digestive tract anthrax: abdominal pain, nausea, vomiting, bloody diarrhea, and bowel perforation.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

62.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

62.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

62.7 Diagnosis

Doctors diagnose this using culture and Gram stain. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

62.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

62.9 Treatment Options

Treatment options include anthrax antitoxin plus antibiotics (ciprofloxacin or doxycycline). outlook: cutaneous $< 1\%$ mortality

- Inhalational 40–80%
- Digestive tract 25–60%.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

62.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

62.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 63

Anti-GBM Disease

63.1 Overview

Anti-glomerular basement membrane disease, also known as Goodpasture syndrome, is a rare, life-threatening small-vessel vasculitis characterized by autoantibodies directed against the non-collagenous domain of the $\alpha 3$ chain of type IV collagen ($\alpha 3$(IV)NC1), found in the basement membranes of glomeruli and lung alveoli, with an incidence of 0.5–1 per million and a bimodal age distribution (young adults 20–30 years with male predominance, and older adults 60–70 years with female predominance). The condition involves anti-GBM antibodies binding to the target antigen in the GBM and alveolar basement membrane, activating complement and recruiting neutrophils, leading to crescentic glomerulonephritis and alveolar bleeding, with genetic susceptibility strongly associated with HLA-DRB1*1501. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

63.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

63.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

63.4 Signs and Symptoms

You may experience rapidly progressive glomerulonephritis (hematuria, proteinuria, oliguria, rising creatinine, crescentic GN on biopsy), with or without lung involvement (hemoptysis, shortness of breath, alveolar bleeding—seen in 60–80%, more common in smokers and young patients). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

63.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

63.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

63.7 Diagnosis

Doctors diagnose this based on detection of circulating anti-GBM antibodies by ELISA and kidney biopsy showing linear IgG deposition along the GBM on immunofluorescence (the pathognomonic finding). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

63.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

63.9 Treatment Options

Treatment typically involves urgent: high-dose corticosteroids combined with cyclophosphamide and plasmapheresis (daily for 7–14 days to remove pathogenic antibodies). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

63.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

63.11 Outlook

- Your outlook depends on the degree of kidney impairment at presentation
- If dialysis-dependent at diagnosis, kidney recovery is rare.

Chapter 64

Antidotes Overview

Key antidotes include:

- **Naloxone:** Opioid receptor antagonist; 0.04–2 mg IV/IM/IN for respiratory depression from opioids; may repeat or infuse.
- **N-Acetylcysteine (NAC):** Replenishes liver glutathione for acetaminophen toxicity; IV (20-hour protocol) or oral (72-hour protocol) within 8–10 hours.
- **Flumazenil:** $GABA_A$ receptor antagonist for benzodiazepine overdose; used with extreme caution due to risk of seizures in mixed overdoses or chronic benzodiazepine users.
- **Fomepizole:** Alcohol dehydrogenase inhibitor for methanol and ethylene glycol poisoning; preferred over ethanol.
- **Physostigmine:** Acetylcholinesterase inhibitor for anticholinergic toxidrome (pure anticholinergic delirium, supraventricular tachycardia); contraindicated in TCA overdose.
- **Digoxin Immune Fab (Digibind):** Antibody fragments that bind digoxin for life-threatening digoxin toxicity.
- **Glucagon:** Positive chronotrope/inotrope via cAMP activation for beta-blocker and calcium channel blocker overdose.
- **Hydroxocobalamin:** Vitamin B_{12a} that binds cyanide; first-line antidote for cyanide poisoning.
- **Atropine:** Antimuscarinic for organophosphate poisoning and symptomatic bradycardia.
- **Pralidoxime (2-PAM):** Reactivates acetylcholinesterase in organophosphate poisoning.
- **Octreotide:** Somatostatin analog used for hypoglycemia from sulfonylurea overdose.
- **Sodium bicarbonate:** Alkalinizes serum to treat TCA-induced QRS widening and salicylate toxicity.

64.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

64.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

64.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

64.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

64.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

64.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries

- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

64.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

64.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

64.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

64.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

64.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 65

Antiphospholipid Syndrome

65.1 Overview

Antiphospholipid syndrome is an autoimmune thrombophilic disorder characterized by arterial or venous blood clot formation and/or pregnancy morbidity in the presence of persistent antiphospholipid antibodies: anticardiolipin antibodies, anti-β2-glycoprotein I antibodies, and/or lupus anticoagulant, and may be primary (no associated disease) or secondary (associated with SLE). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

65.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

65.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

65.4 Signs and Symptoms

You may experience deep vein blood clot formation, lung blockage by a blood clot, stroke, heart attack, recurrent pregnancy loss, livedo reticularis, thrombocytopenia, and cardiac valve disease, with catastrophic APS (CAPS) being a rare, life-threatening variant with widespread blood clot formation affecting multiple organs simultaneously.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

65.5 Progression and Stages

Your outlook depends on the site and severity of blood clot formation. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

65.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

65.7 Diagnosis

To make the diagnosis, doctors look for at least one clinical criterion (blood clot formation or pregnancy morbidity) and one laboratory criterion (persistent antiphospholipid antibodies on two occasions $\geq$12 weeks apart).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

65.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

65.9 Treatment Options

Treatment typically involves long-term anticoagulation with warfarin (INR target 2–3 for venous blood clot formation, 2–3 or 3–4 for arterial blood clot formation), with DOACs not recommended for APS. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

65.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 66

Antisocial Personality Disorder (ASPD)

66.1 Overview

Antisocial personality disorder has a prevalence of 0.5–3% in the general population, with a male-to-female ratio of 3:1, and is higher in prison populations (50–80%). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

66.2 Causes

This condition happens because of reduced amygdala volume and reactivity, ventromedial prefrontal cortex dysfunction, reduced autonomic responsivity to distress cues, and reduced gray matter volume in the orbitofrontal cortex and anterior insula, with genetic factors being strong (heritability 50–60%). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

66.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

66.4 Signs and Symptoms

You may experience a pervasive pattern of disregard for and violation of the rights of others, occurring since age 15, with at least three of the following: failure to conform to social norms, deceitfulness, impulsivity, irritability and aggressiveness, reckless disregard for safety, consistent irresponsibility, and lack of remorse, with the individual being at least 18 years old and having evidence of conduct disorder before age 15.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

66.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

66.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

66.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

66.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

66.9 Treatment Options

Treatment typically involves challenging—no FDA-approved medications for ASPD, with pharmacotherapy targeting aggression and impulsivity (lithium, valproate, atypical antipsychotics). Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

66.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

66.11 Outlook

- Unfortunately, the outlook is often poor
- Symptoms may attenuate with age but functional impairment persists.

Chapter 67

Aortic Dissection

67.1 Overview

Aortic dissection is a life-threatening medical emergency caused by a tear in the aortic intima that allows blood to surge into the aortic media, creating a false lumen. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

67.2 Causes

This condition happens because of chronic high blood pressure, connective tissue disorders (Marfan syndrome, Ehlers-Danlos syndrome), or bicuspid aortic valve. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

67.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

67.4 Signs and Symptoms

You may experience sudden, severe, tearing or ripping chest pain radiating to the back (interscapular region), asymmetric blood pressure between arms, and neurological deficits. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

67.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

67.6 Complications

- Stanford Type A (ascending aorta) requires immediate emergency surgical repair
- Stanford Type B (descending aorta) is managed medically with IV beta-blockers (esmolol, labetalol) to control heart rate and blood pressure unless complications occur.
- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events

- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

67.7 Diagnosis

Diagnosis is confirmed urgently by CT angiography of the chest and abdomen or transesophageal echocardiography (TEE). Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

67.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

67.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy

(pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

67.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

67.11 Outlook

outlook is guarded, with high early mortality. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 68

Aortic Regurgitation

68.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Aortic regurgitation is incompetence of the aortic valve allowing diastolic backflow of blood from the aorta into the left ventricle
- Causes include bicuspid aortic valve, endocarditis, aortic root dilation (Marfan syndrome, aortic dissection, high blood pressure), and rheumatic disease.

68.2 Causes

This condition happens because of aortic valve leaflet or aortic root abnormalities. The chronic volume overload leads to left ventricular dilation and eccentric enlargement, eventually causing heart failure. Patients may be asymptomatic for years despite severe regurgitation. Clinical signs include a wide pulse pressure, water-hammer pulse, decrescendo diastolic murmur, and de Musset's sign. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

68.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

68.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

68.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

68.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection

- Chronic kidney disease from reduced perfusion

68.7 Diagnosis

Doctors diagnose this using echocardiography. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

68.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

68.9 Treatment Options

Treatment focuses on surgical aortic valve repair or replacement, indicated for symptomatic patients or when left ventricular ejection fraction falls below 50% or left ventricular end-systolic diameter exceeds 50 mm. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting)

Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

68.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

68.11 Outlook

Most people recover well after surgical correction. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 69

Aortic Stenosis

69.1 Overview

Calcific (degenerative) aortic stenosis (AS) is the most common valvular heart disease in older adults, affecting 3–5% of those over 75. It is marked by progressive thickening, scarring, and calcium buildup of the aortic valve cusps, driven by lipid deposition, inflammation, and osteogenic differentiation of valvular interstitial cells. Once symptomatic (classic triad: shortness of breath, angina, syncope), outlook without intervention is poor (50% mortality at 2 years). In older adults, transcatheter aortic valve replacement (TAVR) has largely replaced surgical aortic valve replacement and is the standard of care for most elderly patients, with excellent outcomes and shorter recovery times. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

69.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

69.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

69.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

69.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

69.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

69.7 Diagnosis

History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

69.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

69.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)

- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

69.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

69.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 70

Aortic Stenosis

70.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Aortic stenosis is a narrowing of the aortic valve opening and the most common valvular disease in developed countries
- Degenerative calcific aortic stenosis is the most common cause in older adults, while bicuspid aortic valve (present in 1–2% of the population) is the most common cause in younger patients.

70.2 Causes

This condition happens because of progressive calcium buildup and restriction of the valve leaflets, causing pressure overload of the left ventricle, compensatory enlargement, and eventually heart failure. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

70.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

70.4 Signs and Symptoms

You may experience the classic triad of angina, syncope, and shortness of breath. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

70.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

70.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

70.7 Diagnosis

Doctors diagnose this using echocardiography showing severe aortic stenosis (aortic valve area less than 1.0 cm^2, mean gradient greater than 40 mmHg). Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

70.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

70.9 Treatment Options

- Treatment typically involves aortic valve replacement with surgical (SAVR) or transcatheter (TAVR) approaches
- TAVR is increasingly preferred in intermediate- and high-risk patients.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)

- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

70.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

70.11 Outlook

Unfortunately, the outlook is often poor without intervention (mortality 50% at 2 years after symptom onset) but excellent after successful valve replacement. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 71

Aphthous Ulcers (Canker Sores)

71.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Aphthous ulcers are recurrent, painful, shallow ulcers of the oral mucosa that affect up to 25% of the population
- They are not contagious and are distinct from cold sores (herpes labialis).

71.2 Causes

This condition happens because of a multifactorial cause including local trauma, stress, nutritional deficiencies (iron, folate, B12), hormonal changes, and immune dysregulation. Clinical forms include minor aphthous (most common, small ulcers less than 1 cm, heal without scarring), major aphthous (larger, deeper ulcers, may scar), and herpetiform aphthous (multiple pinpoint ulcers that may coalesce). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

71.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

71.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

71.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

71.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

71.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

71.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

71.9 Treatment Options

Treatment typically involves symptomatic, including topical anesthetics (lidocaine), antimicrobial rinses (chlorhexidine), topical corticosteroids (triamcinolone paste), and for severe cases, systemic agents (colchicine, thalidomide, apremilast). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

71.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

71.11 Outlook

The outlook is good, with lesions healing spontaneously but frequently recurring. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 72

Aplastic Anemia

72.1 Overview

Aplastic anemia is a life-threatening bone marrow failure syndrome characterized by pancytopenia (anemia, leukopenia, thrombocytopenia) with a hypocellular bone marrow. The condition may be acquired (most common), inherited (Fanconi anemia, dyskeratosis congenita), or idiopathic, with acquired causes including autoimmune (T-cell mediated attack on hematopoietic stem cells), infections (hepatitis-associated aplastic anemia, parvovirus B19), drugs and toxins (chloramphenicol, chemotherapy agents, benzene), radiation, and pregnancy. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

72.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

72.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

72.4 Signs and Symptoms

You may experience fatigue, pallor (anemia), recurrent infections (neutropenia), and bleeding (thrombocytopenia). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

72.5 Progression and Stages

To make the diagnosis, doctors look for bone marrow biopsy showing markedly hypocellular marrow with increased fat cells, with severity classified as severe (marrow cellularity <25%, or 25–50% with <30% residual hematopoiesis and two of three cytopenias: ANC <500, platelets <20,000, reticulocytes <1%) or very severe (ANC <200). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

72.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

72.7 Diagnosis

Management options include allogeneic hematopoietic stem cell transplantation (curative, preferred for young patients with matched donors) and immunosuppressive therapy (antithymocyte globulin + cyclosporine, effective for older patients), with eltrombopag (thrombopoietin receptor agonist) showing efficacy in combination with immunosuppressive therapy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

72.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

72.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

72.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

72.11 Outlook

outlook is guarded without treatment but improved with stem cell transplantation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 73

Appendicitis

73.1 Overview

Appendicitis is inflammation of the vermiform appendix, the most common surgical emergency of the abdomen, with a lifetime risk of approximately 7–8%. It typically occurs between ages 10 and 30, with a slight male predominance. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

73.2 Causes

The condition usually results from luminal obstruction of the appendix by a fecalith, lymphoid abnormal increase in cells, or rarely tumors, leading to increased intraluminal pressure, bacterial overgrowth, reduced blood flow, and perforation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

73.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

73.4 Signs and Symptoms

- You may experience the classic presentation of periumbilical pain that migrates to the right lower quadrant (McBurney's point), accompanied by anorexia, nausea, vomiting, and low-grade fever
- However, the classic The condition usually appears as present in only 50–60% of cases. Physical findings include tenderness at McBurney's point, guarding, and rebound tenderness. The Alvarado score (MANTRELS) aids clinical assessment.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

73.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

73.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

73.7 Diagnosis

- Diagnosis typically involves laboratory findings of leukocytosis with left shift and CT abdomen/pelvis as the preferred imaging modality in adults (sensitivity and specificity >95%)
- Ultrasound may be used as first-line imaging in children and pregnant women.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

73.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

73.9 Treatment Options

- Treatment typically involves appendectomy (using small incisions and a camera preferred over open), with preoperative IV antibiotics
- Antibiotics alone may be considered for uncomplicated appendicitis in selected patients, though recurrence rates of 15–40% exist. Complicated appendicitis (gangrenous, perforated, with abscess or peritonitis) requires IV antibiotics and possibly interval appendectomy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

73.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

73.11 Outlook

Most people recover well with timely surgical intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 74

Arboviral Encephalitis

74.1 Overview

Arboviral encephalitis encompasses viral infections transmitted by arthropod vectors that cause brain inflammation, with major causes in North America including La Crosse encephalitis, St. Louis encephalitis, and Western equine encephalitis.

La Crosse Encephalitis: Caused by La Crosse virus (bunyavirus), transmitted by *Aedes triseriatus*. It is the most common cause of pediatric arboviral encephalitis in the United States (50–150 cases annually). This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

74.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

74.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)

- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

74.4 Signs and Symptoms

- Clinical features: incubation 5–15 days, with most infections being asymptomatic or mild febrile illness
- Neuroinvasive disease presents with fever, headache, seizures, altered mental status, and focal deficits. Case fatality rate is less than 1%.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

74.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

74.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts

- Multi-organ failure in severe cases

74.7 Diagnosis

To diagnose arboviral encephalitis is by serology (IgM in CSF or serum by ELISA). Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

74.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

74.9 Treatment Options

No specific antiviral therapy is available. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

74.10 Living with It

Treatment typically involves supportive care.

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

74.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 75

Arsenic Poisoning

75.1 Overview

Arsenic is a naturally occurring metalloid found in groundwater, rice, and some industrial products. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

75.2 Causes

This condition happens because of inorganic arsenic (trivalent arsenite) inhibiting pyruvate dehydrogenase, disrupting the Kreb's cycle and oxidative phosphorylation, and impairing DNA repair. Acute arsenic poisoning presents with severe digestive tract symptoms (vomiting, profuse watery diarrhea), hypotension, cardiac dysfunction, acute kidney injury, and death within 24 hours. Subacute/chronic toxicity presents with peripheral neuropathy, Mee's lines (transverse white bands on fingernails), diffuse hyperpigmentation, palmar-plantar hyperkeratosis, and squamous cell carcinoma. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

75.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

75.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

75.5 Progression and Stages

Your outlook depends on severity and timing of treatment. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

75.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

75.7 Diagnosis

Doctors diagnose this using 24-hour urine arsenic level. Treatment: dimercaprol (BAL) or succimer (DMSA) for acute symptomatic poisoning. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

75.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

75.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

75.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

75.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 76

Asbestosis

76.1 Overview

Asbestosis is a chronic interstitial lung disease caused by the inhalation of asbestos fibers, typically occurring after occupational exposure (shipbuilding, construction, insulation manufacturing) with a latency period of 20–30 years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

76.2 Causes

This condition happens because of macrophage phagocytosis of inhaled asbestos fibers, leading to chronic inflammation, fibrogenic growth factor release, and diffuse lung scarring. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

76.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

76.4 Signs and Symptoms

You may experience progressive exertional shortness of breath, dry cough, end-inspiratory bibasilar crackles, and digital clubbing. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

76.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

76.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

76.7 Diagnosis

Doctors diagnose this based on occupational history, high-resolution CT showing pleural plaques, lower lobe predominant reticular opacities, and asbestos bodies (ferruginous bodies) on bronchoalveolar lavage or lung biopsy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

76.8 Prevention

- Treatment typically involves supportive, including smoking cessation, oxygen therapy, and vaccination
- Patients are at high risk for bronchogenic carcinoma and cancerous mesothelioma.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

76.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

76.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

76.11 Outlook

outlook is progressive and irreversible. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 77

Ascites

77.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

77.2 Causes

This condition happens because of portal high blood pressure (sinusoidal pressure greater than 12 mmHg) and splanchnic arterial vasodilation, leading to neurohormonal activation (RAAS, sympathetic nervous system, ADH) and sodium and water retention. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

77.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

77.4 Signs and Symptoms

You may experience abdominal distension and discomfort. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

77.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

77.6 Complications

- Ascites is the accumulation of fluid in the peritoneal cavity and is the most common complication of decompensated cirrhosis. Your outlook depends on the response to treatment and underlying liver disease severity
- Spontaneous bacterial peritonitis is a serious complication.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

77.7 Diagnosis

- Doctors diagnose this using clinical examination (shifting dullness, fluid thrill) and abdominal ultrasound
- Diagnostic paracentesis shows a serum-ascites albumin gradient (SAAG) of 1.1 g/dL or greater, confirming portal high blood pressure.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

77.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

77.9 Treatment Options

- Treatment options include sodium restriction (less than 2 g/day), diuretics (spironolactone and furosemide), and large-volume paracentesis for tense ascites
- Refractory ascites may require transjugular intrahepatic portosystemic shunt (TIPS) or liver transplantation.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

77.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

77.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 78

Aspergillosis (Invasive)

78.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

78.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of conidia (spores) being inhaled and reaching the alveoli
- In immunocompromised hosts, conidia germinate into hyphae that invade lung blood vessels (angioinvasion), causing blood clot formation, tissue tissue death due to lack of blood supply, and involving bleeding tissue death.

78.3 Risk Factors

Invasive aspergillosis is a life-threatening infection caused by *Aspergillus* species, most commonly *A. fumigatus*, and is the most common mold infection in immunocompromised patients, with mortality 30–80%, and risk factors including prolonged neutropenia, HSCT, SOT, high-dose corticosteroids, and advanced HIV/AIDS. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

78.4 Signs and Symptoms

You may experience invasive lung aspergillosis (most common): fever refractory to antibiotics, cough, pleuritic chest pain, and shortness of breath, with chest CT showing nodules with halo sign and air crescent sign. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

78.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

78.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

78.7 Diagnosis

Doctors diagnose this using culture, galactomannan (GM) antigen, and CT chest. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

78.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

78.9 Treatment Options

Treatment options include voriconazole (IV or oral) as first-line, with isavuconazole as an alternative. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

78.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

78.11 Outlook

- Unfortunately, the outlook is often poor without treatment
- Mortality 30–80%.

Chapter 79

Asplenia

79.1 Overview

Asplenia is the absence of splenic function, which may be congenital (asplenia syndrome, often associated with complex congenital heart disease and situs abnormalities) or acquired (splenectomy, sickle cell disease, celiac disease, inflammatory bowel disease). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

79.2 Causes

This condition happens because of anatomical absence or functional impairment of the spleen, which normally clears encapsulated bacteria through opsonin-dependent phagocytosis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

79.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

79.4 Signs and Symptoms

You may experience increased risk for overwhelming post-splenectomy infection (OPSI), a fulminant sepsis with 50–70% mortality. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

79.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

79.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

79.7 Diagnosis

Doctors diagnose this based on history of splenectomy or conditions causing functional asplenia, confirmed by the presence of Howell-Jolly bodies on blood smear or absence of splenic uptake on nuclear medicine scan. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

79.8 Prevention

Treatment focuses on pneumococcal, meningococcal, and Hib vaccination, prophylactic antibiotics (penicillin V or amoxicillin), and patient education about infection risks.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

79.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

79.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

79.11 Outlook

- The outlook is good with preventive measures
- OPSI carries high mortality. Functional asplenia (impaired splenic function without anatomical absence) occurs in sickle cell disease (autosplenectomy from repeated splenic tissue death due to lack of blood supply), celiac disease, and inflammatory bowel disease.

Chapter 80

Asplenia and Hyposplenism

80.1 Overview

Asplenia (absence of splenic tissue) and hyposplenism (impaired splenic reticuloendothelial function) result from surgical splenectomy, congenital anomalies (Ivemark syndrome), or autosplenectomy secondary to repeated microvascular tissue death due to lack of blood supply in sickle cell anemia. The spleen filters encapsulated bacteria, senescent erythrocytes, and intraerythrocytic inclusions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

80.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

80.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

80.4 Signs and Symptoms

You may experience vulnerability to fulminant encapsulated bacterial sepsis (Streptococcus pneumoniae, Neisseria meningitidis, Haemophilus influenzae type b) and capnocytophaga infections following dog bites. Peripheral blood smears demonstrate Howell-Jolly bodies (nuclear remnants), Pappenheimer bodies, Heinz bodies, and target cells. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

80.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

80.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

80.7 Diagnosis

Your doctor confirms the diagnosis with blood smear morphology and 99mTc-labeled heat-damaged RBC radionuclide scintigraphy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

80.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

80.9 Treatment Options

Treatment options include mandatory immunizations (pneumococcal, meningococcal, Hib, annual influenza), prophylactic oral penicillin in children, and immediate empiric parenteral antibiotics for febrile episodes. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

80.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

80.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 81

Asthma

81.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Asthma is a heterogeneous chronic inflammatory disease of the airways characterized by wheezing, breathlessness, chest tightness, and cough, particularly at night or in the early morning
- Triggers include allergens, respiratory infections, exercise, cold air, and irritants.

81.2 Causes

This condition happens because of IgE-mediated mast cell degranulation, eosinophilic infiltration, goblet cell abnormal increase in cells, smooth muscle enlargement, and airway remodeling, leading to variable airflow obstruction and airway hyperresponsiveness. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

81.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

81.4 Signs and Symptoms

You may experience episodic wheezing, breathlessness, chest tightness, and cough. Diagnosis is supported by variable airflow limitation on spirometry (reversibility of more than 12% and 200 mL in FEV_1 after bronchodilator), peak flow variability, and bronchoprovocation testing. Management follows stepwise therapy: inhaled corticosteroids (cornerstone of controller therapy), long-acting beta-agonists, leukotriene receptor antagonists, long-acting muscarinic antagonists, and biologic therapies (anti-IgE, anti-IL5, anti-IL4Rα) for severe eosinophilic asthma. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

81.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

81.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

81.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

81.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

81.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

81.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

81.11 Outlook

Most people recover well with appropriate controller therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 82

Asthma-COPD Overlap

82.1 Overview

Asthma-COPD overlap is a condition in which patients share features of both asthma and COPD, including persistent airflow limitation, airway inflammation that is both eosinophilic and neutrophilic, and significant exacerbation burden. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

82.2 Causes

This condition happens because of overlapping pathophysiological mechanisms of both asthma and COPD, with greater morbidity than either disease alone. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

82.3 Risk Factors

To make the diagnosis, doctors look for a history consistent with asthma (early onset, atopy, variability) combined with risk factors for COPD (smoking, occupational exposures). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

82.4 Signs and Symptoms

You may experience persistent airflow limitation with variability and significant exacerbation burden. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

82.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

82.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

82.7 Diagnosis

- Management combines inhaled corticosteroids with long-acting bronchodilators
- Biologic therapies are considered for severe disease.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

82.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

82.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

82.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

82.11 Outlook

outlook is worse than for asthma or COPD alone. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 83

Astigmatism

83.1 Overview

Astigmatism is a refractive error caused by irregular curvature of the cornea or lens, resulting in different refractive powers in different meridians. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

83.2 Causes

This condition happens because of irregular corneal or lenticular curvature. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

83.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

83.4 Signs and Symptoms

You may experience blurred or distorted vision at both near and distance. Most astigmatism is corneal and regular. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

83.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

83.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

83.7 Diagnosis

Doctors diagnose this based on refraction testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

83.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

83.9 Treatment Options

- Treatment focuses on correction with cylindrical lenses (glasses or contact lenses) or refractive surgery
- Irregular astigmatism may require rigid gas-permeable contact lenses or scleral lenses.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

83.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

83.11 Outlook

Most people recover well with appropriate correction. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 84

Atelectasis

84.1 Overview

Atelectasis is the collapse or incomplete expansion of lung parenchyma, leading to impaired gas exchange. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

84.2 Causes

This condition happens because of airway obstruction by mucus plugs, foreign bodies, or tumors (resorptive atelectasis) The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Pleural effusion or pneumothorax (compression atelectasis)
- Or loss of lung surfactant (microatelectasis). It is the most common cause of early postoperative fever (within 24–48 hours).

84.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

84.4 Signs and Symptoms

You may experience tachypnea, cough, hypoxemia, and decreased breath sounds over the affected region. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

84.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

84.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

84.7 Diagnosis

Doctors diagnose this using chest radiograph showing opacification with volume loss and tracheal deviation toward the affected side. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

84.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

84.9 Treatment Options

Treatment focuses on incentive spirometry, chest physiotherapy, early postoperative ambulation, and therapeutic bronchoscopy for persistent mucus plugging. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

84.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

84.11 Outlook

Most people recover well when treated promptly. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 85

Atopic Dermatitis

85.1 Overview

Atopic dermatitis (AD, eczema) is a chronic, relapsing inflammatory skin disease characterized by intense itching, xerosis, and eczematous lesions with a characteristic distribution. It affects 10–20% of children and 3–7% of adults. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

85.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of epidermal barrier dysfunction (filaggrin mutations in 30–50% of cases), immune dysregulation (Th2-predominant inflammation), and microbial dysbiosis (colonization with *Staphylococcus aureus*)
- The "atopic march" links AD to food allergy, allergic rhinitis, and asthma.

85.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

85.4 Signs and Symptoms

Symptoms vary depending on age: infantile AD involves the face and extensor surfaces

- Childhood AD affects flexural areas (antecubital and popliteal fossae)
- Adult AD presents with lichenified plaques on flexural surfaces.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

85.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

85.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

85.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

85.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

85.9 Treatment Options

Treatment options include regular emollient use, topical corticosteroids (first-line for flares), topical calcineurin inhibitors (tacrolimus, pimecrolimus for sensitive areas), and for moderate-to-severe disease, biologic therapies (dupilumab [anti-IL-4Rα], tralokinumab [anti-IL-13]) and JAK inhibitors (abrocitinib, baricitinib). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

85.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

85.11 Outlook

- The outlook varies from person to person
- The condition tends to improve with age.

Chapter 86

Atrial Fibrillation

86.1 Overview

Atrial fibrillation is the most common sustained irregular heartbeat, affecting 2–4% of the adult population and increasing in prevalence with age. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

86.2 Causes

This condition happens because of disorganized electrical activity in the atria, leading to irregular and often rapid ventricular response. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

86.3 Risk Factors

Risk factors include high blood pressure, heart failure, valvular disease, obesity, obstructive sleep apnea, hyperthyroidism, and alcohol excess. Your outlook depends on risk factor management and stroke prevention.

- Advanced age
- Cigarette smoking
- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease

- Obesity and physical inactivity
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

86.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

86.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

86.6 Complications

- You may experience palpitations, fatigue, shortness of breath, and decreased exercise tolerance
- Complications include thromboembolism (stroke), tachycardia-mediated cardiomyopathy, and impaired quality of life.
- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events

- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

86.7 Diagnosis

Doctors diagnose this using ECG showing absent P waves, irregularly irregular R-R intervals, and variable QRS morphology. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

86.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

86.9 Treatment Options

- Treatment focuses on rate control (beta-blockers, calcium channel blockers, digoxin) and rhythm control (antiarrhythmic drugs, cardioversion, catheter ablation)
- The CHA_2DS_2-VASc score guides anticoagulation decisions, with direct oral anticoagulants preferred over warfarin for non-valvular atrial fibrillation.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)

- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

86.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

86.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 87

Atrial Fibrillation in the Elderly

87.1 Overview

Atrial fibrillation (AF) prevalence increases from 2% in adults under 65 to 15–20% in those over 80. Geriatric-specific management challenges include higher thromboembolic risk, increased bleeding risk, kidney impairment affecting DOAC dosing, high rates of polypharmacy, and increased risk of falls complicating anticoagulation decisions. Rate control is the default strategy in older adults, reserving rhythm control for symptomatic patients despite adequate rate control. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

87.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

87.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus

- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

87.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

87.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

87.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

87.7 Diagnosis

History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

87.8 Prevention

Anticoagulation for stroke prevention is indicated for most older adults with AF, using DOACs in preference to warfarin due to lower intracranial bleeding risk.

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

87.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk

- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

87.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

87.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 88

Atrophic Vaginitis (Genitourinary Syndrome of Menopause)

88.1 Overview

Atrophic vaginitis, now termed genitourinary syndrome of menopause (GSM), is a chronic progressive condition resulting from estrogen deficiency after menopause, affecting up to 50% of postmenopausal women. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

88.2 Causes

This condition happens because of estrogen withdrawal leading to thinning of the vaginal epithelium, decreased vaginal secretions, loss of rugae, shortening and narrowing of the vaginal canal, and changes in the vulvar and lower urinary tract tissues. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

88.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

88.4 Signs and Symptoms

You may experience vaginal dryness, burning, itching, dyspareunia, and urinary symptoms (frequency, urgency, recurrent UTIs).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

88.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

88.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

88.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam: pale, thin, smooth vaginal mucosa with loss of rugae, elevated vaginal pH (>5.0), and maturation index showing predominance of parabasal cells on cytology.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

88.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

88.9 Treatment Options

Treatment options include vaginal moisturizers and lubricants for mild symptoms, and low-dose vaginal estrogen therapy for moderate to severe symptoms. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

88.10 Living with It

Most people recover well with treatment, significantly improving quality of life and sexual function.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 89

Attention-Deficit/Hyperactivity Disorder (ADHD)

89.1 Overview

Attention-deficit/hyperactivity disorder has a worldwide prevalence of 5–7% in children and 2.5–3% in adults, more common in boys in childhood (3:1 ratio) but approaching parity in adulthood. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

89.2 Causes

This condition happens because of striatal and prefrontal dopaminergic and noradrenergic dysregulation, with reduced brain volume in prefrontal cortex, caudate, and cerebellum, delayed cortical maturation, and high genetic heritability (75–80%). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

89.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

89.4 Signs and Symptoms

You may experience a persistent pattern of inattention, hyperactivity, and impulsivity that interferes with functioning, with three presentations: predominantly inattentive, predominantly hyperactive-impulsive, and combined presentation, with onset before age 12.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

89.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

89.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

89.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, based on detailed developmental history, collateral reports, and behavior rating scales.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

89.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

89.9 Treatment Options

- Treatment options include pharmacotherapy (stimulants as first-line: methylphenidate and amphetamine formulations
- Non-stimulants as second-line: atomoxetine, guanfacine, clonidine) and psychosocial interventions (behavioral parent training, CBT, executive function coaching). outlook: 50–65% of children continue to have impairing symptoms into adulthood.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

89.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

89.11 Outlook

With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 90

Atypical Facial Pain

90.1 Overview

Persistent idiopathic facial pain (PIFP, formerly atypical facial pain) is a persistent facial pain that does not fit the diagnostic criteria for any other cranial neuralgia or headache disorder. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

90.2 Causes

This condition happens because of poorly understood Disease mechanism, possibly involving central sensitization and dysfunction of descending pain modulatory systems. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

90.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

90.4 Signs and Symptoms

You may experience daily or nearly continuous deep, poorly localized, aching or burning pain that may cross the midline. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

90.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

90.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

90.7 Diagnosis

To make the diagnosis, doctors look for exclusion of dental and neurological causes after thorough investigation including imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

90.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

90.9 Treatment Options

Treatment typically involves multimodal: TCAs, SNRIs, gabapentin, pregabalin, and thinking and memory behavioral therapy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

90.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

90.11 Outlook

- The outlook varies from person to person
- Invasive procedures are generally not recommended.

Chapter 91

Autism Spectrum Disorder (ASD)

91.1 Overview

Autism spectrum disorder has a prevalence of 1–2% worldwide, with a male-to-female ratio of 3–4:1. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

91.2 Causes

This condition happens because of dysregulation of synaptic pruning and neural connectivity, an imbalance of excitation/inhibition (particularly GABAergic and glutamatergic systems), aberrant brain growth, and alterations in frontotemporal, frontoparietal, and interhemispheric connectivity, with strong genetic factors (heritability 80–90%). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

91.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

91.4 Signs and Symptoms

Social communication deficits include reduced social-emotional reciprocity, deficits in nonverbal communicative behaviors, and deficits in developing relationships. Restricted, repetitive patterns manifest as stereotyped speech/gait, insistence on sameness, highly restricted fixated interests, and hyper- or hyporeactivity to sensory input. Intellectual disability is present in 30–50%, and language impairment in 25–30%.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

91.5 Progression and Stages

You may experience persistent deficits in social communication and social interaction across multiple contexts, and restricted, repetitive patterns of behavior, interests, or activities, with three levels of severity requiring support. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

91.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

91.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

91.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

91.9 Treatment Options

Treatment options include early intensive behavioral intervention (EIBI/ABA), social skills training, speech and language therapy, and pharmacotherapy for comorbid symptoms (atypical antipsychotics for irritability/aggression, SSRIs for anxiety/depression). Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

91.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

91.11 Outlook

- Outlook is highly variable
- Early intervention significantly improves functional outcomes.

Chapter 92

Autoimmune Hepatitis

92.1 Overview

Autoimmune hepatitis is a chronic inflammatory liver disease characterized by autoimmune destruction of hepatocytes, predominantly affecting women. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

92.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of autoimmune-mediated hepatocyte destruction. Type 1 (positive ANA or anti-smooth muscle antibodies) is more common and occurs at any age
- Type 2 (positive anti-LKM-1 antibodies) is rarer and typically affects children and young women.

92.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

92.4 Signs and Symptoms

Symptoms can range from asymptomatic elevation of aminotransferases to acute hepatitis or fulminant failure. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

92.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

92.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

92.7 Diagnosis

To make the diagnosis, doctors look for elevated transaminases, hypergammaglobulinemia (IgG), characteristic autoantibodies, and liver biopsy showing interface hepatitis with plasma cell infiltration. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

92.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification

- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

92.9 Treatment Options

Treatment focuses on corticosteroids (prednisone) combined with azathioprine to induce and maintain remission. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

92.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

92.11 Outlook

The outlook is good with treatment, although lifelong immunosuppression is often necessary. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 93

Autoimmune Pancreatitis

93.1 Overview

Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions.

- Autoimmune pancreatitis (AIP) is a form of chronic pancreatitis caused by autoimmune inflammation of the pancreatic parenchyma. Type 1 AIP is linked to IgG4-related disease and involves other organs (biliary strictures, retroperitoneal scarring, salivary gland enlargement)
- Type 2 AIP is limited to the pancreas and is linked to inflammatory bowel disease.

93.2 Causes

This condition happens because of autoimmune-mediated inflammation. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

93.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection

- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

93.4 Signs and Symptoms

Clinical features often mimic pancreatic cancer with painless jaundice, weight loss, and a pancreatic mass. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

93.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

93.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

93.7 Diagnosis

Doctors diagnose this based on the international consensus diagnostic criteria (ICDC): imaging (sausage-shaped pancreas on CT, delayed enhancement), serology (elevated IgG4 in type 1), histology (lymphoplasmacytic infiltrate with storiform scarring), and response to corticosteroids. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

93.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

93.9 Treatment Options

Treatment typically involves with oral prednisone, with rituximab for relapsing type 1 AIP. Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

93.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

93.11 Outlook

- The outlook is generally positive
- Unlike pancreatic cancer, AIP responds dramatically to corticosteroids.

Chapter 94

Autoimmune Pancreatitis (AIP)

94.1 Overview

Type 1 AIP affects predominantly older males and presents with painless obstructive jaundice, weight loss, and a diffuse 'sausage-shaped' enlargement of the pancreas with a capsule-like rim on CT/MRI, frequently mimicking pancreatic ductal adenocarcinoma. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

94.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

94.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)

- Vitamin D deficiency

94.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

94.5 Progression and Stages

Autoimmune pancreatitis (AIP) is a distinct form of chronic pancreatitis driven by autoimmune inflammation, classified into Type 1 (IgG4-related disease, systemic involvement with dense IgG4+ plasma cell infiltration and storiform scarring) and Type 2 (pancreas-specific, neutrophilic phlebotomy lesions without IgG4 systemic disease). Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

94.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

94.7 Diagnosis

Diagnosis relies on the International Consensus Diagnostic Criteria (ICDC): elevated serum IgG4 levels ($> 2\times$ normal), characteristic parenchymal enlargement, main pancreatic duct narrowing, and IgG4+ plasma cells on core biopsy. AIP demonstrates dramatic responsiveness to systemic corticosteroid therapy (prednisone), avoiding unnecessary pancreaticoduodenectomy (Whipple procedure). Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

94.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

94.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief

- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

94.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

94.11 Outlook

Autoimmune diseases are generally chronic conditions requiring long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 95

Autosomal Dominant Polycystic Kidney Disease

95.1 Overview

Autosomal dominant polycystic kidney disease (ADPKD) is the most common hereditary kidney disease, affecting 1 in 400–1000 individuals. It is caused by mutations in PKD1 (chromosome 16, 85% of cases) or PKD2 (chromosome 4, 15% of cases). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

95.2 Causes

This condition happens because of defective polycystin proteins leading to progressive bilateral kidney cyst formation, massive kidney enlargement, declining GFR, and ESRD typically by the fifth to seventh decade. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

95.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

95.4 Signs and Symptoms

- You may experience flank pain, hematuria, high blood pressure, and abdominal distension
- Extrarenal manifestations include liver cysts (most common), intracranial berry aneurysms (5–10%), cardiac valvular abnormalities, and colonic diverticulosis.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

95.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

95.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

95.7 Diagnosis

Doctors diagnose this based on imaging (CT or MRI showing ≥3 cysts per kidney) and genetic testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

95.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

95.9 Treatment Options

Treatment options include BP control (ACE inhibitors/ARBs), salt restriction, adequate hydration, and tolvaptan (vasopressin V2 receptor antagonist) to slow cyst growth and kidney function decline. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

95.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

95.11 Outlook

- The outlook varies from person to person
- Screening of first-degree relatives is recommended.

Chapter 96

Bacterial Meningitis

96.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- Bacterial meningitis is an acute, life-threatening infection of the leptomeninges and cerebrospinal fluid
- The most common causative organisms vary by age: *Streptococcus pneumoniae* and *Neisseria meningitidis* in adults, *Group B Streptococcus* and *Escherichia coli* in neonates, and *Haemophilus influenzae* type b (in unvaccinated populations).

96.2 Causes

This condition happens because of bacterial invasion of the subarachnoid space. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

96.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

96.4 Signs and Symptoms

- You may experience severe headache, high fever, neck stiffness (nuchal rigidity), photophobia, altered mental status, and petechial rash (meningococcal disease)
- Kernig's and Brudzinski's signs may be positive.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

96.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

96.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment

- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

96.7 Diagnosis

To make the diagnosis, doctors look for lumbar puncture showing elevated opening pressure, turbid CSF with neutrophilic pleocytosis, elevated protein, low glucose, and positive Gram stain and culture. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

96.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures

- Travel precautions and pre-travel consultations

96.9 Treatment Options

- Treatment focuses on empiric treatment with ceftriaxone plus vancomycin, which should not be delayed
- Dexamethasone improves outcomes in pneumococcal meningitis.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

96.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

96.11 Outlook

Your outlook depends on the causative organism and timeliness of treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 97

Bacterial Vaginosis

97.1 Overview

Bacterial vaginosis (BV) is a gynecologic infection and the most common cause of vaginal discharge in reproductive-age women, affecting 20–50% depending on population. It is a polymicrobial dysbiosis characterized by a reduction in hydrogen peroxide-producing *Lactobacillus* species and overgrowth of anaerobic bacteria. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

97.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

97.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices

- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

97.4 Signs and Symptoms

You may experience a thin, homogenous, gray-white vaginal discharge with a fishy odor (especially after intercourse or menses), but 50% of women are asymptomatic. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

97.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

97.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment

- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

97.7 Diagnosis

Doctors diagnose this using Amsel criteria (≥3 of: thin white discharge, pH >4.5, positive whiff test, clue cells on saline wet mount) or Nugent score. BV in pregnancy is linked to preterm labor, PROM, and postpartum endometritis. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

97.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures

- Travel precautions and pre-travel consultations

97.9 Treatment Options

Treatment typically involves metronidazole (500 mg BID for 7 days or 2 g single dose) or clindamycin cream. Recurrence is common. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

97.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

97.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 98

Balanitis and Balanoposthitis

98.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Balanitis is inflammation of the glans penis
- Balanoposthitis additionally involves the foreskin, and is common in uncircumcised men.

98.2 Causes

This condition happens because of infectious (Candida albicans—most common, bacterial, STIs), irritant (soap, detergent), or skin (lichen planus, psoriasis, contact dermatitis) factors, with diabetes mellitus predisposing to candida balanitis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

98.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

98.4 Signs and Symptoms

You may experience redness of the skin, swelling, itching, discharge, dysuria, and tender inguinal lymphadenopathy. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

98.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

98.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

98.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Swab for KOH preparation (candida), bacterial culture, and STI screening.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

98.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

98.9 Treatment Options

Treatment options include improved hygiene, topical antifungal (clotrimazole, miconazole) for candida, topical antibiotic/steroid combinations, and addressing underlying causes. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

98.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

98.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 99

Barotrauma of the Ear

99.1 Overview

Ear barotrauma is tissue injury to the tympanic membrane and middle ear cavity caused by an uncompensated pressure differential between ambient atmospheric pressure and the air-filled middle ear space, occurring during rapid altitude changes (scuba diving descent, commercial aviation, blast injuries). This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

99.2 Causes

This condition happens because of Eustachian tube dysfunction (occlusion due to upper respiratory infection, mucosal swelling, or allergic rhinitis) preventing equalizer air flow into the middle ear. During ambient pressure increases, negative middle ear pressure draws the tympanic membrane inward, precipitating mucosal hyperemia, subepithelial bleeding, middle ear effusion, or tympanic membrane rupture. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

99.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

99.4 Signs and Symptoms

Symptoms often appear as sudden severe ear pain (otalgia), a sensation of ear fullness/pressure, conductive hearing loss, ringing in the ears, and occasionally feeling of spinning or otorrhagia (bloody ear discharge). Otoscopy demonstrates tympanic membrane retraction, hemotympanum, or perforation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

99.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

99.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

99.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

99.8 Prevention

- Treatment focuses on prevention (Valsalva maneuver, chewing, oral/nasal decongestants prior to flight/dive)
- Acute traumatic perforations usually heal spontaneously, with surgical tympanoplasty reserved for persistent non-healing tears.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

99.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

99.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

99.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 100

Barrett Esophagus

100.1 Overview

Barrett esophagus is a metaplastic condition of the distal esophagus resulting from chronic gastroesophageal reflux disease (GERD). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

100.2 Causes

This condition happens because of chronic acid and bile exposure inducing intestinal metaplasia, where normal stratified squamous epithelium is replaced by non-ciliated columnar epithelium with goblet cells. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

100.3 Risk Factors

Barrett esophagus is a major risk factor for esophageal adenocarcinoma (40-fold increased risk). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

100.4 Signs and Symptoms

Clinical features mirror those of GERD (heartburn, regurgitation), though some patients are asymptomatic. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

100.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment options include high-dose proton pump inhibitors (PPIs) and using a thin camera tube surveillance to screen for dysplasia
- Using a thin camera tube radiofrequency ablation or mucosal removal is indicated for high-grade dysplasia.

100.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

100.7 Diagnosis

To make the diagnosis, doctors look for upper endoscopy showing salmon-pink mucosa extending proximal to the gastroesophageal junction, with histopathological confirmation of intestinal goblet cells. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

100.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

100.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

100.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

100.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 101

Bartholin Cyst

101.1 Overview

Bartholin cyst (and Bartholin abscess) results from fluid accumulation secondary to obstruction of the main duct of the greater vestibular (Bartholin) gland located at the 4 and 8 o'clock positions of the posterior vaginal introitus. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

101.2 Causes

This condition happens because of ductal occlusion caused by local inflammation, trauma, thickened mucus, or prior infection (Neisseria gonorrhoeae, Chlamydia trachomatis, Escherichia coli). Uninfected cysts present as painless, unilateral labial swellings. Secondary bacterial infection causes a Bartholin abscess, characterized by rapid onset of severe localized vulvar pain, dyspareunia, redness of the skin, and a fluctuant mass. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

101.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

101.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

101.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

101.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

101.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam based on characteristic anatomic location
- Biopsy is indicated in women > 40 years to rule out rare Bartholin gland carcinoma. Treatment for asymptomatic small cysts is conservative
- Symptomatic cysts or abscesses require incision and removal of fluid with Word catheter placement or marsupialization to maintain ductal patency.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

101.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

101.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

101.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

101.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 102

Basal Cell Carcinoma

102.1 Overview

Basal cell carcinoma (BCC) is the most common skin cancer worldwide, arising from basal cells of the epidermis. It is strongly associated with cumulative ultraviolet radiation exposure and predominantly affects fair-skinned individuals. BCC is locally invasive but rarely metastasizes. Clinical subtypes include nodular (most common, pearly papule with telangiectasia and central ulceration—"rodent ulcer"), superficial (erythematous, scaly plaque, often on the trunk), and morpheaform (sclerosing, whitish, scar-like). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

102.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

102.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

102.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

102.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- The right treatment depends on subtype, size, and location: surgical removal, Mohs micrographic surgery (for high-risk locations such as the face, with highest cure rate of 99%), electrodesiccation and curettage (for superficial, low-risk lesions), and topical therapy (5-fluorouracil, imiquimod) for superficial BCC
- Vismodegib and sonidegib (Hedgehog pathway inhibitors) are used for advanced or metastatic BCC.

102.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

102.7 Diagnosis

Doctors diagnose this based on biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

102.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

102.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

102.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

102.11 Outlook

- Most people recover well
- Metastasis is extremely rare.

Chapter 103

Becker Muscular Dystrophy

103.1 Overview

Becker muscular dystrophy (BMD) is an X-linked recessive disorder caused by in-frame mutations in the dystrophin gene, producing a truncated but partially functional dystrophin protein. It affects approximately 1 in 18,000–30,000 males. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

103.2 Causes

This condition happens because of dystrophin deficiency that is less severe than DMD. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

103.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

103.4 Signs and Symptoms

Cardiomyopathy is a significant cause of morbidity and mortality, often presenting before significant skeletal muscle weakness.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

103.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- You may experience proximal muscle weakness with later onset (late childhood to early adulthood) and slower progression compared to DMD
- Patients may remain ambulatory into their 30s or beyond.

103.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

103.7 Diagnosis

Doctors diagnose this based on genetic testing and muscle biopsy.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

103.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

103.9 Treatment Options

Treatment typically involves similar to DMD, with emphasis on cardiac surveillance and management. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

103.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

103.11 Outlook

outlook is better than DMD, with longer ambulation and survival.

Chapter 104

Beers Criteria / STOPP/START Criteria

104.1 Overview

Two major tools guide prescribing in older adults. The **Beers Criteria** (American Geriatrics Society, updated 2023) lists potentially inappropriate medications (PIMs) to avoid or use with caution in older adults, including anticholinergics, benzodiazepines, Z-drugs, NSAIDs, sulfonylureas, and PPIs. The **STOPP/START criteria** are a European alternative that identifies PIMs (STOPP) and potential prescribing omissions (START—eg., missing statin in diabetes with cardiovascular disease, missing bisphosphonate after osteoporotic fracture). Both criteria reduce inappropriate prescribing and ADRs when integrated into clinical practice. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

104.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

104.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

104.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

104.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

104.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

104.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

104.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

104.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

104.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

104.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 105

Behçet's Disease

105.1 Overview

Behçet's disease is a chronic, relapsing systemic vasculitis of unknown cause capable of affecting vessels of all sizes, with a predilection for the oral and genital mucosa, eyes, skin, and central nervous system. It is discussed in detail in Chapter **??** (Section 106). Briefly, it is marked by recurrent oral aphthous ulcers, genital ulcers, uveitis, skin lesions (redness of the skin nodosum, pseudofolliculitis, pathergy reaction), vascular involvement (arterial aneurysms, venous blood clot formation), and neurological involvement. It is most prevalent along the Silk Road. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

105.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

105.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

105.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

105.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

105.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

105.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (ICBD criteria). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

105.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

105.9 Treatment Options

Treatment typically involves tailored to organ involvement: colchicine for mucocutaneous disease, azathioprine, cyclosporine, or TNF inhibitors for ocular disease, and cyclophosphamide for severe vascular/neurological disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

105.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

105.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 106

Behçet's Disease

106.1 Overview

Behçet's disease is a systemic vasculitis of unknown cause, characterized by recurrent oral aphthous ulcers (the hallmark feature, present in >98% of patients), genital ulcers, uveitis (which can be sight-threatening), and skin lesions (redness of the skin nodosum, pseudofolliculitis, pathergy), with a predilection for the Silk Road populations (Turkey, Iran, Japan, China), with a prevalence of up to 1 in 250 in Turkey. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

106.2 Causes

This condition happens because of systemic vasculitis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

106.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

106.4 Signs and Symptoms

Clinical features also include vascular involvement (arterial aneurysms, venous blood clot formation), neurological involvement (neuro-Behçet's with brainstem/diencephalic lesions), digestive tract involvement (ileocecal ulcers), and arthritic involvement. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

106.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

106.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

106.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, based on the International Criteria for Behçet's Disease (ICBD), with no pathognomonic laboratory test. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

106.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

106.9 Treatment Options

Treatment typically involves tailored to organ involvement: topical agents for oral/genital ulcers, colchicine for mucocutaneous disease, azathioprine or TNF inhibitors for ocular or systemic involvement, and cyclophosphamide for severe vascular or neurological disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

106.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

106.11 Outlook

The outlook varies from person to person depending on organ involvement. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 107

Bell's Palsy

107.1 Overview

Bell's palsy is an acute, idiopathic, unilateral facial nerve (cranial nerve VII) paralysis of lower motor neuron type and is the most common cause of acute facial palsy. The condition is thought to result from reactivation of herpes simplex virus type 1. Onset is typically abrupt, with complete facial weakness developing over hours to days. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

107.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

107.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

107.4 Signs and Symptoms

You may experience inability to close the affected eye, drooping mouth, hyperacusis, and loss of taste on the anterior two-thirds of the tongue. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

107.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

107.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

107.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, after excluding causes such as Ramsay Hunt syndrome, Lyme disease, and stroke. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

107.8 Prevention

- Management with oral corticosteroids (prednisone) started within 72 hours of onset significantly improves outcomes
- Eye protection is essential to prevent corneal damage.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

107.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

107.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

107.11 Outlook

The outlook is generally positive, with most patients recovering fully within weeks to months. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 108

Benign Paroxysmal Positional Vertigo

108.1 Overview

non-cancerous paroxysmal positional feeling of spinning is the most common cause of peripheral feeling of spinning, resulting from displaced otoliths from the utricle into the semicircular canals, most commonly the posterior canal. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

108.2 Causes

This condition happens because of movement of calcium carbonate crystals within the semicircular canals. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

108.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

108.4 Signs and Symptoms

You may experience brief episodes (less than 1 minute) of severe feeling of spinning triggered by head movement, with characteristic involuntary eye movements. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

108.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

108.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

108.7 Diagnosis

Doctors diagnose this based on the Dix-Hallpike test, which reproduces feeling of spinning and reveals torsional-upbeating involuntary eye movements. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

108.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

108.9 Treatment Options

Treatment focuses on canalith repositioning maneuvers (Epley maneuver for posterior canal BPPV), which are highly effective. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

108.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

108.11 Outlook

Most people recover well, although recurrence occurs in 20–50% of cases. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 109

Benign Prostatic Hyperplasia

109.1 Overview

non-cancerous prostatic abnormal increase in cells is the most common non-cancerous tumor in men, characterized by growth of stromal and epithelial cells in the transition zone of the prostate, causing urethral obstruction, affecting over 50% of men by age 60 and up to 90% by age 85. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

109.2 Causes

This condition happens because of age-related hormonal changes driving prostatic growth. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

109.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

109.4 Signs and Symptoms

You may experience lower urinary tract symptoms (LUTS): urinary frequency, urgency, nocturia, weak stream, hesitancy, straining, incomplete emptying, and dribbling, with acute urinary retention possibly occurring. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

109.5 Progression and Stages

The right treatment depends on severity: watchful waiting for mild symptoms, alpha-blockers (tamsulosin, alfuzosin) for smooth muscle relaxation, 5-alpha-reductase inhibitors (finasteride, dutasteride) for gland shrinkage in larger prostates, and surgical intervention (TURP, holmium laser enucleation) for refractory cases. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

109.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

109.7 Diagnosis

Diagnosis typically involves digital rectal examination (DRE) showing smooth, symmetrically enlarged prostate, American Urological Association Symptom Score (AUASS), PSA, urinalysis, post-void residual volume, and uroflowmetry. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

109.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

109.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

109.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

109.11 Outlook

Most people recover well with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 110

Benzodiazepine Overdose

110.1 Overview

Benzodiazepine overdose is a pharmaceutical overdose involving positive allosteric modulators of $GABA_A$ receptors. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

110.2 Causes

This condition happens because of excessive GABAergic inhibition producing CNS depression. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

110.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

110.4 Signs and Symptoms

- You may experience drowsiness, slurred speech, loss of coordination, and coma
- Respiratory depression is mild in pure benzodiazepine overdose but is synergistic with ethanol, opioids, and other CNS depressants.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

110.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

110.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

110.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Urine drug screens have limited utility.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

110.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

110.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

110.10 Living with It

- Treatment typically involves supportive care (airway protection, respiratory support)
- Flumazenil, a $GABA_A$ receptor antagonist, is reserved for pure benzodiazepine overdose with severe respiratory depression and is contraindicated in chronic benzodiazepine users or mixed ingestions due to risk of seizures. The outlook is generally positive with supportive care.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

110.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 111

Beta-Blocker Overdose

111.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Beta-blocker overdose (propranolol most lethal, also atenolol, metoprolol, sotalol) is a pharmaceutical overdose causing bradycardia, hypotension, and decreased myocardial contractility. Propranolol also has sodium channel blockade contributing to QRS widening and seizures
- Sotalol causes QT prolongation and torsades de pointes.

111.2 Causes

This condition happens because of beta-adrenergic receptor blockade. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

111.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

111.4 Signs and Symptoms

You may experience bradyarrhythmias, low cardiac output, lung swelling, hypoglycemia, seizures, and altered mental status. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

111.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

111.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

111.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

111.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

111.9 Treatment Options

Treatment options include IV fluids, atropine, glucagon (5–10 mg IV bolus, then 1–10 mg/hour infusion) which increases cyclic AMP via non-adrenergic activation of adenylate cyclase, high-dose insulin euglycemia therapy, vasopressors, and calcium salts. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

111.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

111.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 112

Binge-Eating Disorder

112.1 Overview

Binge-eating disorder has a lifetime prevalence of 2–3% in the general population and is the most common eating disorder, more common in women but with a narrower sex ratio than anorexia and bulimia (3:2 female-to-male). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

112.2 Causes

This condition happens because of altered reward sensitivity (dopamine dysfunction in striatum), impulsivity, and emotional dysregulation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

112.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

112.4 Signs and Symptoms

You may experience recurrent episodes of binge eating (loss-of-control eating of objectively large amounts of food) without regular compensatory behaviors, associated with eating much more rapidly than normal, eating until uncomfortably full, eating large amounts when not physically hungry, eating alone due to embarrassment, and feeling disgusted/depressed/guilty afterward, occurring at least once a week for 3 months. BED is strongly associated with obesity (60–80% are obese) and metabolic syndrome. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

112.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

112.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

112.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

112.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

112.9 Treatment Options

Treatment options include CBT-E, IPT, and lisdexamfetamine (FDA-approved for moderate-to-severe BED). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

112.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

112.11 Outlook

- Outlook is better than for anorexia or bulimia
- 50–70% achieve remission with structured treatment.

Chapter 113

Bipolar I Disorder

113.1 Overview

Bipolar I disorder has a lifetime prevalence of approximately 1% worldwide, with equal distribution across sexes. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

113.2 Causes

This condition happens because of dysregulation of intracellular signaling pathways (GSK-3, PKC, protein kinase A) and circadian rhythm disruption, with strong genetic factors (80–90% heritability). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

113.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

113.4 Signs and Symptoms

You may experience at least one lifetime manic episode lasting at least 1 week (or any duration if hospitalization is required), characterized by elevated, expansive, or irritable mood plus increased goal-directed activity or energy, with at least three of the following (four if mood is only irritable): inflated self-esteem or grandiosity, decreased need for sleep, pressured speech, flight of ideas or racing thoughts, distractibility, psychomotor agitation, and excessive involvement in high-risk activities.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

113.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

113.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

113.7 Diagnosis

- Doctors diagnose this using clinical interview using DSM-5-TR criteria. Treatment for acute mania includes a mood stabilizer plus an atypical antipsychotic or benzodiazepine
- Lithium remains the gold-standard mood stabilizer with demonstrated antisuicidal effects, with other first-line treatments including valproate and atypical antipsychotics (olanzapine, quetiapine, risperidone, aripiprazole).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

113.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

113.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

113.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

113.11 Outlook

The outlook varies from person to person, with most patients requiring long-term maintenance therapy. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 114

Bipolar II Disorder

114.1 Overview

This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

114.2 Causes

This condition happens because of similar pathophysiological mechanisms as Bipolar I. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

114.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

114.4 Signs and Symptoms

You may experience patients spending significantly more time in depressive episodes than hypomanic episodes, often leading to underdiagnosis.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

114.5 Progression and Stages

Bipolar II disorder has a lifetime prevalence of approximately 0.5%, characterized by one or more hypomanic episodes (distinct period of elevated, expansive, or irritable mood lasting at least 4 consecutive days, plus similar symptoms as mania but of lesser severity, without marked functional impairment or psychosis) and one or more major depressive episodes. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

114.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

114.7 Diagnosis

To make the diagnosis, doctors look for the absence of full manic episodes, distinguishing it from Bipolar I.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

114.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

114.9 Treatment Options

Treatment options include mood stabilizers (lithium, lamotrigine, valproate) and atypical antipsychotics (quetiapine, which is FDA-approved for bipolar II depression), with careful use of antidepressants with mood stabilizer coverage. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

114.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

114.11 Outlook

The outlook varies from person to person but generally better than Bipolar I. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 115

Bladder Cancer

115.1 Overview

Bladder cancer is the fourth most common cancer in men, with 550,000 new cases/year worldwide. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

115.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

- Urothelial (transitional cell) carcinoma accounts for >90%
- Squamous cell carcinoma (3–5%) and adenocarcinoma (1–2%) are less common. This condition happens because of multifocal field cancerization with mutations in FGFR3 (low-grade) and p53/Rb (high-grade, CIS).

115.3 Risk Factors

Risk factors include smoking (50% of cases), occupational exposure (aromatic amines, aniline dyes), chronic cystitis (squamous cell carcinoma), *Schistosoma haematobium* infection (squamous cell in endemic areas), cyclophosphamide, pelvic radiation, and family history.

- Advancing age
- Family history of cancer

- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

115.4 Signs and Symptoms

You may experience gross, painless hematuria (most common), microscopic hematuria, and irritative voiding symptoms with CIS. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

115.5 Progression and Stages

Recovery varies by stage: 5-year survival is 96% for Ta, 70–90% for T1, 50–60% for T2, 30–50% for T3, and <15% for metastatic. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

115.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

115.7 Diagnosis

Doctors diagnose this based on cystoscopy (gold standard) with biopsy/removal, urine cytology, FISH (UroVysion), and CT urography. Staging distinguishes non-muscle-invasive (Ta, T1, CIS) from muscle-invasive (T2+) disease. Treatment for NMIBC includes TURBT with risk-adapted intravesical therapy (mitomycin C, BCG)

- BCG-unresponsive NMIBC requires radical cystectomy or pembrolizumab. MIBC doctors typically treat it with neoadjuvant cisplatin-based chemotherapy followed by radical cystectomy with urinary diversion
- Trimodal therapy is an option for patients unfit for cystectomy. Metastatic disease doctors typically treat it with platinum-based chemotherapy, checkpoint inhibitors, FGFR inhibitors (erdafitinib for FGFR3-mutant), and antibody-drug conjugates (enfortumab vedotin, sacituzumab govitecan).
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status

- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

115.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

115.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

115.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

115.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 116

Blastomycosis

116.1 Overview

Blastomycosis is a systemic fungal infection caused by *Blastomyces dermatitidis*, a dimorphic fungus endemic to the Ohio and Mississippi River Valleys and the Great Lakes region, with 1–2 cases per 100,000 population in endemic areas. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

116.2 Causes

This condition happens because of conidia being inhaled, converting to broad-based budding yeasts in the alveoli, and disseminating hematogenously. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

116.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

116.4 Signs and Symptoms

- You may experience lung blastomycosis: acute pneumonia or chronic pneumonia (resembles TB or lung cancer)
- Extrapulmonary blastomycosis: skin is the most common site (verrucous or ulcerative lesions), bone (lytic lesions, osteomyelitis), and genitourinary (prostatitis, epididymitis).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

116.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

116.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

116.7 Diagnosis

Doctors diagnose this using culture, direct examination of KOH wet mount (large, broad-based budding yeasts), and antigen testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

116.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

116.9 Treatment Options

Treatment options include itraconazole for mild-moderate lung disease and liposomal amphotericin B for severe, disseminated, or CNS disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

116.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

116.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 117

Body Dysmorphic Disorder

117.1 Overview

Body dysmorphic disorder has a prevalence of 1.7–2.4% in the general population, with higher rates in dermatology and cosmetic surgery settings (8–15%). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

117.2 Causes

This condition happens because of distorted body image perception. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

117.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

117.4 Signs and Symptoms

You may experience a preoccupation with one or more perceived defects or flaws in physical appearance that are not observable or appear only slightly noticeable to others, with repetitive behaviors (mirror checking, excessive grooming, skin picking, reassurance seeking) or mental acts (comparing appearance with others) in response to the appearance concerns, with the preoccupation being time-consuming (at least 1 hour per day). Common concerns involve the skin, hair, nose, eyes, teeth, and weight or muscle size (in muscle dysmorphia). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

117.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

117.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

117.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

117.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

117.9 Treatment Options

Treatment options include SSRIs at high doses (similar to OCD) and CBT tailored for BDD. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

117.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

117.11 Outlook

- Outlook is chronic without treatment
- Cosmetic procedures generally do not help.

Chapter 118

Borderline Personality Disorder (BPD)

118.1 Overview

Borderline personality disorder has a prevalence of 1–2.5% in the general population and 10–20% in outpatient psychiatric settings. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

118.2 Causes

This condition happens because of a hyperreactive amygdala (heightened emotional sensitivity), impaired prefrontal regulation of emotion, reduced hippocampal and amygdala volumes, and altered oxytocin and endogenous opioid systems, with adverse childhood experiences being strongly associated. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

118.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

118.4 Signs and Symptoms

You may experience a pervasive pattern of instability in interpersonal relationships, self-image, affect, and marked impulsivity, with DSM-5-TR criteria requiring at least five of the following: frantic efforts to avoid real or imagined abandonment, unstable and intense interpersonal relationships (alternating between idealization and devaluation—splitting), marked identity disturbance, impulsivity, recurrent suicidal behavior or non-suicidal self-injury (NSSI), affective instability, chronic feelings of emptiness, inappropriate anger, and transient stress-related paranoid ideation or dissociative symptoms.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

118.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

118.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

118.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

118.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

118.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

118.10 Living with It

- Treatment options include dialectical behavior therapy (DBT) as first-line psychotherapy, mentalization-based treatment (MBT), and symptom-targeted pharmacotherapy (SSRIs, mood stabilizers, low-dose atypical antipsychotics). outlook: 80% achieve symptomatic remission over 10-year follow-up
- Suicide risk is 5–10%.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

118.11 Outlook

With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 119

Botulism (Clostridium botulinum)

119.1 Overview

Botulism is a rare, potentially fatal neuroparalytic disease caused by botulinum neurotoxin (BoNT) produced by *Clostridium botulinum*, with approximately 200 cases per year in the US, including foodborne, infant botulism (most common form in US), wound botulism (injection drug users), and iatrogenic forms. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

119.2 Causes

This condition happens because of BoNT (a zinc-dependent metalloproteinase) cleaving SNARE proteins (SNAP-25, synaptobrevin, syntaxin) at the presynaptic neuromuscular junction, preventing acetylcholine release and causing flaccid paralysis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

119.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

119.4 Signs and Symptoms

You may experience symmetric, descending flaccid paralysis beginning with cranial nerve palsies (diplopia, slurred speech, dysphonia, difficulty swallowing, ptosis) and progressing to descending symmetric weakness of neck, trunk, and extremities, with no sensory deficits and no fever. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

119.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

119.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

119.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by detection of BoNT in serum, stool, or food (mouse bioassay or PCR). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

119.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

119.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

119.10 Living with It

Treatment options include equine heptavalent botulism antitoxin (HBAT) for adults or human botulinum immune globulin (BabyBIG) for infants, with supportive care including mechanical ventilation for respiratory failure. outlook: mortality $< 5\%$ with modern intensive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

119.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 120

Brain Abscess

120.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- A brain abscess is a focal collection of pus within the brain parenchyma, resulting from hematogenous spread, contiguous infection (sinusitis, otitis media, dental infection), or direct inoculation (trauma, neurosurgery)
- Common organisms include *Streptococcus* species, *Staphylococcus aureus*, and anaerobes.

120.2 Causes

This condition happens because of bacterial infection leading to focal suppuration within brain tissue. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

120.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

120.4 Signs and Symptoms

You may experience headache, fever, weakness, numbness, or problems with movement, seizures, and altered consciousness. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

120.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

120.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

120.7 Diagnosis

Doctors diagnose this based on MRI with contrast showing a ring-enhancing lesion with surrounding swelling and mass effect. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

120.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

120.9 Treatment Options

Treatment focuses on targeted intravenous antibiotics (guided by culture when possible) and, for larger abscesses (greater than 2.5 cm), neurosurgical removal by suction or removal. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

120.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

120.11 Outlook

The outlook has improved significantly with modern imaging and treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 121

Brain Tumors in Children

121.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Brain tumors are the most common solid tumors in children and the second most common pediatric malignancy after leukemia. They differ significantly from adult brain tumors in histology and location. Medulloblastoma is an embryonal tumor of the cerebellum, most common in children aged 3–8 years, presenting with symptoms of increased intracranial pressure, loss of coordination, and cranial nerve deficits
- Treatment usually involves surgical removal, craniospinal radiation, and chemotherapy, with 5-year survival of approximately 70–80%. Ependymoma arises from ependymal cells lining the ventricles, most commonly in the posterior fossa
- Treatment typically involves maximal safe surgical removal followed by focal radiation, with 5-year survival of 50–70%. Brainstem glioma, most commonly diffuse intrinsic pontine glioma (DIPG), is a devastating high-grade glioma of the pons in children aged 5–10 years presenting with rapidly progressive cranial neuropathies, loss of coordination, and long-tract signs
- Radiation provides temporary palliation, and median survival is less than 1 year.

121.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

121.3 Risk Factors

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

121.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

121.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Pilocytic astrocytoma is the most common glioma in children, a low-grade (WHO Grade I) tumor typically occurring in the cerebellum, optic pathway, hypothalamus, and brainstem
- Complete surgical removal is curative, with 90–95% survival.

121.6 Complications

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

121.7 Diagnosis

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

121.8 Prevention

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

121.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

121.10 Living with It

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

121.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 122

Breast Cancer

122.1 Overview

Breast cancer is the most commonly diagnosed cancer worldwide (2.3 million new cases annually) and the second leading cause of cancer-related death in women. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

122.2 Causes

The majority of breast cancers (70–80%) are invasive ductal carcinoma (not otherwise specified), with other types including invasive lobular carcinoma (10–15%), inflammatory breast cancer, Paget's disease of the nipple, and male breast cancer (<1%). This condition happens because of cancerous transformation of breast epithelial cells. Screening: mammography is recommended biennially for women ages 50–74 (USPSTF) or annually starting at age 40 (American College of Radiology), with earlier and more intensive screening for high-risk individuals including breast MRI. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

122.3 Risk Factors

Risk factors include female sex, advancing age, family history (particularly BRCA1 and BRCA2 mutations—lifetime risk 45–72% and 26–43% respectively), prior breast cancer, early menarche, late menopause, nulliparity, late first pregnancy, obesity (postmenopausal), alcohol use, and prior chest radiation, with protective factors including breastfeeding, physical activity, and early menopause.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

122.4 Signs and Symptoms

- You may experience a painless breast mass, skin changes, nipple discharge, and axillary lymphadenopathy. Diagnosis: diagnostic mammography with ultrasound
- Core needle biopsy for histopathological evaluation. Molecular subtyping by immunohistochemistry: estrogen receptor (ER), progesterone receptor (PR), HER2, and Ki-67, with four molecular subtypes: luminal A (ER+/PR+, HER2-, low Ki-67—best outlook), luminal B (ER+, HER2+ or high Ki-67), HER2-enriched (ER-, PR-, HER2+), and triple-negative/basal-like (ER-, PR-, HER2—-worst outlook). Staging uses the TNM system.
- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

122.5 Progression and Stages

The right treatment depends on stage, molecular subtype, and patient factors: Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

Radiation therapy: Standard after breast-conserving surgery; post-mastectomy radiation for tumors >5 cm, node-positive disease, or close margins. **Systemic therapy:** Endocrine therapy (tamoxifen for premenopausal; aromatase inhibitors for postmenopausal) for ER+ disease; anti-HER2 therapy (trastuzumab, pertuzumab, T-DM1, T-DXd) for HER2+ disease; chemotherapy (anthracyclines, taxanes) for high-risk disease; CDK4/6 inhibitors for advanced ER+ disease; PARP inhibitors for BRCA-mutated tumors; immune checkpoint inhibitors for triple-negative disease with PD-L1 expression.

- **Surgery:** Breast-conserving surgery (lumpectomy) with sentinel lymph node biopsy, or mastectomy (simple, modified radical, or radical). Oncoplastic techniques optimize cosmetic outcomes.

122.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Infections due to immunosuppression
- Venous thromboembolism

122.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
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- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

122.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

122.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

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- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

122.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

122.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 123

Bronchiectasis

123.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Bronchiectasis is a permanent, abnormal dilation of the bronchi and bronchioles caused by chronic inflammation and recurrent infection
- Etiologies include cystic scarring, primary ciliary dyskinesia, immunodeficiency, prior severe respiratory infections, and idiopathic causes.

123.2 Causes

This condition happens because of impaired mucociliary clearance, mucus stasis, chronic bacterial infection, and progressive airway destruction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

123.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

123.4 Signs and Symptoms

You may experience chronic productive cough with copious purulent sputum, recurrent respiratory infections, hemoptysis, and progressive shortness of breath. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

123.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

123.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

123.7 Diagnosis

Doctors diagnose this using high-resolution CT showing tram-track opacities, signet ring sign, and dilated bronchi. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

123.8 Prevention

- Treatment options include airway clearance techniques, targeted antibiotics for exacerbations and chronic suppression, bronchodilators, and Treatment for underlying causes
- Macrolide prophylaxis reduces exacerbation frequency.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

123.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

123.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

123.11 Outlook

Your outlook depends on the underlying cause and frequency of exacerbations. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 124

Bronchiolitis (RSV)

124.1 Overview

Bronchiolitis is the most common lower respiratory tract infection in infants, caused most frequently by respiratory syncytial virus (RSV). It affects nearly all children by age 2, with peak incidence at 2–6 months. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

124.2 Causes

This condition happens because of inflammation, swelling, and tissue death of the bronchiolar epithelium, leading to airway obstruction, air trapping, and atelectasis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

124.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

124.4 Signs and Symptoms

- You may experience coryza, cough, tachypnea, wheezing, crackles, and respiratory distress
- Apnea is a presenting sign in young infants (<2 months).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

124.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

124.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

124.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- RSV rapid antigen testing or PCR confirms the cause.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

124.8 Prevention

Palivizumab (RSV monoclonal antibody) is used for prophylaxis in high-risk infants.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

124.9 Treatment Options

Treatment typically involves primarily supportive: supplemental oxygen, nasal suctioning, and hydration. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

124.10 Living with It

Most people recover well with supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

124.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 125

Bronchopulmonary Dysplasia (BPD)

125.1 Overview

Bronchopulmonary dysplasia (BPD) is a chronic lung disease of prematurity, defined as the need for supplemental oxygen or respiratory support at 36 weeks postmenstrual age. Incidence is 30–50% in extremely preterm infants. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

125.2 Causes

This condition happens because of the combined effects of lung immaturity, mechanical ventilation, oxygen toxicity, and inflammation, leading to impaired alveolarization and dysregulated vascular development. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

125.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

125.4 Signs and Symptoms

You may experience persistent oxygen requirement, tachypnea, retractions, and wheezing. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

125.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

125.6 Complications

- The outlook varies from person to person
- Long-term complications include reactive airway disease, lung high blood pressure, and increased risk of respiratory infections.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

125.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

125.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

125.9 Treatment Options

Treatment options include optimizing nutrition (high-calorie feeds), fluid restriction, diuretics, bronchodilators, and systemic corticosteroids in severe cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

125.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

125.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 126

Brucellosis

126.1 Overview

Brucellosis is a zoonotic infection caused by gram-negative coccobacilli of the genus *Brucella*, endemic in the Mediterranean basin, Middle East, Central Asia, Africa, and Latin America, with 500,000 new cases annually, and transmission through ingestion of unpasteurized dairy products or direct contact with infected animal tissues. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

126.2 Causes

This condition happens because of *Brucella* surviving and replicating within macrophages by inhibiting phagolysosome fusion and producing a type IV secretion system that modulates the host immune response. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

126.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

126.4 Signs and Symptoms

You may experience acute brucellosis (80%): undulant fever (evening fever spikes that wax and wane), profuse sweating, chills, malaise, joint pain, muscle pain, and back pain, with focal disease including spondylitis, sacroiliitis, peripheral arthritis, epididymo-orchitis, neurobrucellosis, and endocarditis (leading cause of mortality). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

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126.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

126.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

126.7 Diagnosis

Doctors diagnose this using blood culture, bone marrow culture, serology, and PCR. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

126.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

126.9 Treatment Options

Treatment options include doxycycline plus rifampin for 6 weeks (standard), with longer therapy for focal disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

126.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

126.11 Outlook

- Most people recover well for most cases
- Mortality $< 1\%$ except for endocarditis (up to 20%).

Chapter 127

Brugada Syndrome

127.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Brugada syndrome is an inherited sodium channelopathy (SCN5A mutation most common) characterized by an ECG pattern of coved ST-segment elevation in the right precordial leads (V1–V2) and an increased risk of ventricular fibrillation and sudden cardiac death, particularly during sleep or at rest
- It is more common in men and in Southeast Asian populations.

127.2 Causes

This condition happens because of mutations in cardiac sodium channels leading to abnormal repolarization. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

127.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

127.4 Signs and Symptoms

- You may experience syncope, nocturnal agitated resuscitation, or sudden cardiac death
- Patients may be asymptomatic.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

127.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

127.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

127.7 Diagnosis

Doctors diagnose this based on ECG findings, with provocation testing using sodium channel blockers (ajmaline, flecainide) to unmask the pattern when necessary.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

127.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

127.9 Treatment Options

- Treatment focuses on implantable cardioverter-defibrillator placement for high-risk patients
- Fever and certain drugs should be avoided as they can exacerbate the ECG abnormality.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

127.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

127.11 Outlook

Most people recover well in low-risk patients and improved with implantable cardioverter-defibrillator in high-risk patients.

Chapter 128

Budd-Chiari Syndrome

128.1 Overview

Budd-Chiari syndrome is a liver venous outflow obstruction at any level from the small liver veins to the junction of the inferior vena cava and right atrium. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

128.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common cause is myeloproliferative tumors (particularly polycythemia vera)
- Other causes include antiphospholipid syndrome, paroxysmal nocturnal hemoglobinuria, and oral contraceptive use. This condition happens because of obstruction of liver venous outflow, leading to liver congestion, sinusoidal high blood pressure, and hepatocyte reduced blood flow.

128.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

128.4 Signs and Symptoms

- Symptoms can range from asymptomatic to fulminant liver failure
- Classic features include acute abdominal pain, hepatomegaly, and ascites.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

128.5 Progression and Stages

Your outlook depends on the cause and severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

128.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

128.7 Diagnosis

Doctors diagnose this using Doppler ultrasound, CT, or MR venography showing absent or abnormal liver venous flow.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

128.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

128.9 Treatment Options

Treatment options include anticoagulation, catheter-directed thrombolysis, TIPS for decompression, and liver transplantation for progressive liver failure. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

128.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 129

Bulimia Nervosa

129.1 Overview

Bulimia nervosa has a lifetime prevalence of 1–1.5% in women and 0.5% in men, with an age of onset in late adolescence to early adulthood. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

129.2 Causes

This condition happens because of altered reward circuitry, deficits in inhibitory control, and interoceptive dysfunction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

129.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

129.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

129.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

129.6 Complications

Medical complications include electrolyte disturbances (hypokalemic hypochloremic metabolic alkalosis), Russell's sign, parotid gland enlargement, dental erosion, Mallory-Weiss tears, esophageal rupture, and ipecac cardiomyopathy.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

129.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

129.8 Prevention

You may experience recurrent episodes of binge eating (eating a large amount of food in a discrete period with a sense of loss of control) followed by recurrent inappropriate compensatory behaviors to prevent weight gain (self-induced vomiting, laxative/diuretic misuse, fasting, excessive exercise), occurring at least once a week for 3 months, with self-evaluation unduly influenced by body shape and weight.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

129.9 Treatment Options

- Treatment options include CBT-E (first-line), IPT (second-line), and SSRI pharmacotherapy (high-dose fluoxetine 60 mg/day is FDA-approved). outlook: 50–60% achieve recovery
- Relapse rates are moderate.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

129.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

129.11 Outlook

With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 130

Bullous Pemphigoid

130.1 Overview

Bullous pemphigoid (BP) is the most common autoimmune blistering disease, predominantly affecting the elderly (peak incidence 70–80 years). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

130.2 Causes

This condition happens because of IgG autoantibodies targeting BP180 and BP230, components of hemidesmosomes at the dermal-epidermal junction, causing subepidermal blister formation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

130.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

130.4 Signs and Symptoms

- You may experience tense blisters on an erythematous or urticarial base, itching, and predominance on the trunk and flexural surfaces
- Mucosal involvement is uncommon (10–30%).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

130.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

130.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

130.7 Diagnosis

Diagnosis typically involves skin biopsy showing subepidermal split with eosinophilic infiltrate, direct immunofluorescence showing linear IgG and C3 at the basement membrane zone, and anti-BP180/BP230 antibodies by ELISA. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

130.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

130.9 Treatment Options

Treatment options include potent topical corticosteroids (clobetasol, first-line for localized and generalized disease), systemic corticosteroids for extensive disease, and steroid-sparing agents (doxycycline, azathioprine, mycophenolate, rituximab). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

130.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

130.11 Outlook

In most cases, the outlook is favorable but may be chronic. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 131

Bursitis

131.1 Overview

Bursitis is inflammation of a bursa, a synovial fluid–filled sac that reduces friction between bones and overlying soft tissues (tendons, muscles, skin). The most commonly affected bursae include the subacromial (shoulder), olecranon (elbow), prepatellar (knee—"housemaid's knee"), and trochanteric (hip—"weaver's bottom"). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

131.2 Causes

This condition happens because of mechanical irritation or infection of the bursa. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

131.3 Risk Factors

Risk factors include repetitive motion or prolonged pressure, overuse, trauma, inflammatory conditions (gout, rheumatoid arthritis), and infection (septic bursitis). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

131.4 Signs and Symptoms

You may experience localized pain, swelling, tenderness, and sometimes redness of the skin and warmth over the affected bursa

- The overlying joint typically has a full range of motion, distinguishing bursitis from arthritis. Septic bursitis is more common in the olecranon and prepatellar bursae, often caused by *S. aureus*, presenting with more intense inflammation, fever, and systemic symptoms. Diagnosis is primarily clinical
- Ultrasound can confirm bursal fluid collection. removal by suction of bursal fluid is indicated when septic bursitis is suspected. Treatment for aseptic bursitis includes rest, ice, NSAIDs, and corticosteroid injection
- Septic bursitis requires bursal removal by suction, incision and removal of fluid if needed, and antibiotics.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

131.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

131.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

131.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

131.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

131.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

131.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

131.11 Outlook

Most people recover well with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 132

Cadmium Toxicity

132.1 Overview

Cadmium is a heavy metal used in batteries, pigments, electroplating, and fertilizers, with a biological half-life of 15–30 years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

132.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of cadmium accumulating in the kidneys and bone, impairing kidney tubular reabsorption and causing osteomalacia. Acute inhalation exposure causes chemical pneumonitis and lung swelling. Chronic exposure causes kidney tubular damage, Itai-Itai disease (severe osteomalacia, bone pain, and fractures with kidney tubular dysfunction), high blood pressure, and lung cancer. Diagnosis: 24-hour urine and blood cadmium levels. Treatment: removal from exposure
- Chelation is generally ineffective and BAL is contraindicated.

132.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

132.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

132.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

132.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

132.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

132.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

132.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

132.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

132.11 Outlook

Your outlook depends on chronicity of exposure. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 133

Calcium and Phosphorus Disorders

133.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Calcium and phosphorus disorders encompass a group of mineral disorders resulting from dysregulation of calcium and phosphorus homeostasis, which is tightly regulated by parathyroid hormone (PTH), vitamin D, and calcitonin. Hypocalcemia (total calcium less than 8.5 mg/dL or ionized calcium less than 4.2 mg/dL) results from vitamin D deficiency, hypoparathyroidism, chronic kidney disease, hypomagnesemia, and medications
- You may experience neuromuscular irritability (perioral paresthesias, Chvostek sign, Trousseau sign, carpopedal spasm, tetany), prolonged QT interval, seizures, and laryngospasm. Hypercalcemia (total calcium greater than 10.5 mg/dL) is most commonly due to primary hyperparathyroidism or malignancy
- You may experience "bones, stones, groans, and psychiatric overtones"—bone pain, nephrolithiasis, polyuria, polydipsia, constipation, nausea, confusion, and shortened QT interval
- Severe hypercalcemia (greater than 13 mg/dL) is a medical emergency requiring IV fluids, calcitonin, bisphosphonates, and treatment of the underlying cause. Hypophosphatemia (phosphate less than 2.5 mg/dL) causes muscle weakness, rhabdomyolysis, and impaired bone mineralization
- Hyperphosphatemia (phosphate greater than 4.5 mg/dL) is most common in chronic kidney disease, causing secondary hyperparathyroidism, vascular calcium buildup, and kidney osteodystrophy.

133.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

133.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

133.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

133.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

133.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

133.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

133.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

133.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

133.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

133.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 134

Calcium Channel Blocker Overdose

134.1 Overview

Calcium channel blocker (CCB) overdose (verapamil, diltiazem, nifedipine, amlodipine, and other dihydropyridines) is a highly lethal pharmaceutical overdose. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

134.2 Causes

This condition happens because of CCBs inhibiting L-type calcium channels in cardiac myocytes, vascular smooth muscle, and pancreatic beta cells, leading to negative chronotropy, dromotropy, and inotropy with vasodilation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

134.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

134.4 Signs and Symptoms

You may experience hypotension, bradycardia (or reflex tachycardia with dihydropyridines), atrioventricular block, junctional rhythms, asystole, hyperglycemia, metabolic acidosis, and altered mental status. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

134.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

134.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

134.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

134.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

134.9 Treatment Options

Treatment options include aggressive fluid resuscitation, calcium salts to overcome calcium channel blockade, and high-dose insulin euglycemia therapy (the most effective treatment). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

134.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

134.11 Outlook

- Outlook is guarded
- Vasopressors and ECMO may be needed.

Chapter 135

Calcium Pyrophosphate Deposition Disease

135.1 Overview

Calcium pyrophosphate deposition disease (CPPD, formerly pseudogout) is a crystal-induced arthropathy resulting from the deposition of calcium pyrophosphate dihydrate (CPP) crystals in articular cartilage (chondrocalcinosis) and synovium. The condition occurs predominantly in elderly individuals ($>$ 60 years) and is linked to hyperparathyroidism, hemochromatosis, hypomagnesemia, and hypothyroidism. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

135.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

135.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

135.4 Signs and Symptoms

You may experience acute monoarthritis or oligoarthritis resembling gout, most commonly affecting the knee, wrist, or symphysis pubis, with swelling, warmth, severe pain, and redness of the skin. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

135.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

135.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

135.7 Diagnosis

- Your doctor confirms the diagnosis with synovial fluid removal by suction showing weakly positively birefringent, rhomboid-shaped CPP crystals under polarized light microscopy
- Radiographs demonstrate chondrocalcinosis (linear calcium buildup of articular cartilage). Treatment for acute attacks includes intra-articular corticosteroid injection, oral NSAIDs, or colchicine.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

135.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

135.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

135.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

135.11 Outlook

The outlook is good for acute control. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 136

Candida auris

136.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

136.2 Causes

This condition happens because of *C. auris* forming biofilms on medical devices, with mechanisms of antifungal resistance including mutations in ERG11 (azoles), FKS1 (echinocandins), and overexpression of efflux pumps. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

136.3 Risk Factors

Candida auris is an emerging multidrug-resistant yeast first identified in 2009, causing invasive infections in healthcare settings, with cases reported on six continents, and risk factors including ICU stay, central venous catheters, mechanical ventilation, total parenteral nutrition, recent surgery, and diabetes. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

136.4 Signs and Symptoms

You may experience colonization often being asymptomatic, with invasive disease including candidemia, wound infections, and deep-seated candidiasis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

136.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

136.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

136.7 Diagnosis

To make the diagnosis, doctors look for MALDI-TOF mass spectrometry or DNA sequencing, as standard biochemical methods misidentify *C. auris*. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

136.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

136.9 Treatment Options

Treatment options include echinocandins as first-line, with central line removal essential. outlook: mortality 30–60%. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

136.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

136.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 137

Candidiasis (Invasive / Systemic)

137.1 Overview

Invasive candidiasis encompasses candidemia (bloodstream infection) and deep-seated candidiasis (abscesses, endocarditis, endophthalmitis, osteomyelitis, meningitis), and is the most common healthcare-associated fungal infection, with Candida species being the fourth most common cause of bloodstream infections in hospitalized patients. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

137.2 Causes

This condition happens because of *Candida* invading through breach of mucocutaneous barriers, adhering to endothelium, and forming biofilms on medical devices. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

137.3 Risk Factors

Risk factors include ICU stay, central venous catheters, total parenteral nutrition, broad-spectrum antibiotics, neutropenia, immunosuppression, surgery, and diabetes. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

137.4 Signs and Symptoms

You may experience candidemia: fever refractory to broad-spectrum antibiotics, hypotension, and sepsis, often without localizing signs. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

137.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

137.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

137.7 Diagnosis

Doctors diagnose this using blood culture and (1,3)-β-D-glucan (BDG) assay. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

137.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

137.9 Treatment Options

Treatment options include echinocandins (caspofungin, micafungin, anidulafungin) as first-line, with central line removal being essential. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

137.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

137.11 Outlook

outlook is guarded, with significant mortality in ICU patients. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 138

Cannabis Toxicity and Cannabinoid Hyperemesis Syndrome

138.1 Overview

Cannabis is the most commonly used illicit substance worldwide. Acute intoxication produces euphoria, relaxation, impaired short-term memory, altered judgment, conjunctival injection, tachycardia, and increased appetite. Severe effects include acute psychosis, panic attacks, and in high doses (especially edibles), severe sedation, loss of coordination, and hypotension. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

138.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

138.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

138.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

138.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

138.6 Complications

Cannabis hyperemesis syndrome (CHS) is a complication of chronic heavy cannabis use, characterized by cyclic episodes of severe nausea, vomiting, and abdominal pain, relieved by compulsive hot showers or baths.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

138.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Treatment: cessation of cannabis is curative
- For acute episodes, hot showers, topical capsaicin cream, benzodiazepines, haloperidol, metoclopramide, and ondansetron.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

138.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

138.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

138.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

138.11 Outlook

The outlook is generally positive with cannabis abstinence.

Chapter 139

Carbon Monoxide Poisoning

139.1 Overview

Carbon monoxide (CO) is an odorless, colorless gas produced by incomplete combustion of carbon-based fuels. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

139.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of CO binding to hemoglobin with 240 times greater affinity than oxygen, forming carboxyhemoglobin (COHb), which shifts the oxygen dissociation curve to the left, impairing oxygen delivery and utilization
- CO also binds to myoglobin and cytochrome c oxidase.

139.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

139.4 Signs and Symptoms

- Clinical features: headache (most common), nausea, dizziness, confusion, shortness of breath, blurred vision, loss of consciousness, and seizures
- Severe poisoning causes coma, respiratory failure, cardiovascular collapse, and death. nervous system sequelae include acute encephalopathy and delayed neuropsychiatric syndrome (DNS).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

139.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

139.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

139.7 Diagnosis

Doctors diagnose this using COHb level. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

139.8 Prevention

- Treatment: 100% normobaric oxygen
- Hyperbaric oxygen therapy (HBOT) is indicated for severe poisoning and may prevent DNS.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

139.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

139.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

139.11 Outlook

The outlook is generally positive with prompt treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 140

Cardiac Tamponade

140.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Cardiac tamponade is the accumulation of pericardial fluid under pressure, causing compression of the heart and impaired cardiac filling, leading to reduced cardiac output and obstructive shock
- Causes include malignancy (most common in developed countries), trauma, uremia, infection, aortic dissection, and post-procedural causes.

140.2 Causes

This condition happens because of rapid or excessive pericardial fluid accumulation. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

140.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

140.4 Signs and Symptoms

You may experience Beck's triad (hypotension, muffled heart sounds, and jugular venous distension) and pulsus paradoxus. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

140.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

140.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

140.7 Diagnosis

Doctors diagnose this using echocardiography, which shows pericardial effusion with diastolic collapse of the right atrium and right ventricle. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

140.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

140.9 Treatment Options

Treatment typically involves emergent pericardiocentesis or surgical pericardiotomy. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

140.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

140.11 Outlook

Your outlook depends on the underlying cause and timeliness of intervention. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 141

Carpal Tunnel Syndrome

141.1 Overview

Carpal tunnel syndrome is the most common entrapment neuropathy, caused by compression of the median nerve as it passes through the carpal tunnel in the wrist. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

141.2 Causes

This condition happens because of mechanical compression of the median nerve within the carpal tunnel. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

141.3 Risk Factors

Risk factors include repetitive wrist movements, pregnancy, hypothyroidism, diabetes, rheumatoid arthritis, and space-occupying lesions.

- Family history of the genetic condition
- Consanguineous parents (for recessive disorders)
- Advanced parental age (for new mutations)
- Carrier status in parents
- Ethnic background (some conditions)

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

141.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

141.5 Progression and Stages

You may experience numbness and tingling in the median nerve distribution (thumb, index, middle, and radial half of the ring finger), nocturnal pain, and weakness of the thenar muscles in advanced cases. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

141.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

141.7 Diagnosis

- Doctors diagnose this based on clinical examination (Phalen's test, Tinel's sign) and nerve conduction studies confirming delayed median nerve conduction across the wrist. Management begins with wrist splinting (especially at night), ergonomic modifications, and corticosteroid injection
- Surgical carpal tunnel release is indicated for severe or refractory cases.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

141.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

141.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

141.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

141.11 Outlook

Most people recover well with appropriate treatment.

Chapter 142

Celiac Disease

142.1 Overview

Celiac disease is an autoimmune enteropathy triggered by ingestion of gluten (gliadin and glutenin proteins found in wheat, barley, and rye) in genetically susceptible individuals (HLA-DQ2 or HLA-DQ8). Prevalence is approximately 1% worldwide. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

142.2 Causes

This condition happens because of an immune response causing villous shrinkage or wasting, crypt abnormal increase in cells, and intraepithelial lymphocytosis in the small intestinal mucosa, primarily affecting the duodenum and jejunum. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

142.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

142.4 Signs and Symptoms

- Clinical features vary widely: classical symptoms in children include chronic diarrhea, failure to thrive, and abdominal distension
- In adults, diarrhea, weight loss, iron deficiency anemia, and osteoporosis. Dermatitis herpetiformis is the cutaneous manifestation. Many patients are asymptomatic or have extraintestinal manifestations (fatigue, neurological symptoms, liver disease, infertility).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

142.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

142.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

142.7 Diagnosis

- To make the diagnosis, doctors look for serological testing (anti-tissue transglutaminase IgA, anti-endomysial antibodies) and duodenal biopsy showing villous shrinkage or wasting, crypt abnormal increase in cells, and increased intraepithelial lymphocytes
- HLA testing helps exclude the diagnosis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

142.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

142.9 Treatment Options

Treatment typically involves a strict lifelong gluten-free diet, which leads to mucosal recovery and symptom resolution. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

142.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

142.11 Outlook

Most people recover well with dietary compliance. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 143

Cellular Senescence

143.1 Overview

Cellular senescence is a state of irreversible cell cycle arrest in response to various stressors including telomere shortening, DNA damage, oncogene activation, and oxidative stress. Senescent cells accumulate in multiple tissues with age and secrete a complex mixture of pro-inflammatory cytokines, chemokines, growth factors, and matrix metalloproteinases—the senescence-associated secretory phenotype (SASP). The SASP drives chronic inflammation, disrupts tissue architecture, and contributes to age-related diseases. Key pathways include p53/p21^CIP1 and p16^INK4a/Rb. Mouse studies demonstrate that selective elimination of senescent cells (using senolytic agents such as dasatinib plus quercetin, or fisetin) attenuates age-related pathology and extends healthspan. Senolytics are being actively investigated in human clinical trials for frailty, osteoarthritis, and other conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

143.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

143.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

143.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

143.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

143.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

143.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

143.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

143.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

143.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

143.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 144

Cellulitis

144.1 Overview

Cellulitis is a bacterial infection of the dermis and subcutaneous tissue, most commonly caused by *Streptococcus pyogenes* (group A strep) and *Staphylococcus aureus*. Predisposing factors include skin barrier disruption (fissures, insect bites, surgical wounds), swelling, chronic venous insufficiency, and immunosuppression. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

144.2 Causes

This condition happens because of bacterial entry through disrupted skin barriers. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

144.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

144.4 Signs and Symptoms

- You may experience a well-demarcated area of redness of the skin, warmth, swelling, tenderness, and sometimes lymphangitis and regional lymphadenopathy
- The lower extremities are most commonly affected.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

144.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

144.6 Complications

Complications include abscess formation, necrotizing fasciitis (a surgical emergency), bacteremia, and sepsis.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)

- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

144.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Wound cultures and blood cultures may guide therapy in severe cases.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

144.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

144.9 Treatment Options

Treatment options include oral antibiotics (dicloxacillin, cephalexin, or clindamycin for MRSA coverage) for mild cases and IV antibiotics (ceftriaxone, vancomycin, or daptomycin) for severe disease. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

144.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

144.11 Outlook

- Most people recover well with appropriate antibiotic therapy
- Recurrent cellulitis may require prophylactic antibiotics.

Chapter 145

Central Pontine Myelinolysis

145.1 Overview

Central pontine myelinolysis, also known as osmotic demyelination syndrome, is a non-inflammatory demyelinating condition of the pons caused by overly rapid correction of hyponatremia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

145.2 Causes

This condition happens because of osmotic stress leading to oligodendrocyte damage and demyelination. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

145.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

145.4 Signs and Symptoms

You may experience spastic quadriparesis, pseudobulbar palsy (difficulty swallowing, slurred speech, emotional lability), and in severe cases, locked-in syndrome. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

145.5 Progression and Stages

The outlook varies from person to person, with recovery being partial or absent depending on severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

145.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

145.7 Diagnosis

Doctors diagnose this based on MRI showing a characteristic symmetric T2-hyperintense lesion in the central basis pontis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

145.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

145.9 Treatment Options

- Treatment typically involves primarily preventive: serum sodium should be corrected at a rate not exceeding 8–10 mmol/L in 24 hours
- Treatment typically involves supportive once established.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

145.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

145.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 146

Central Post-Stroke Pain

146.1 Overview

Central post-stroke pain (CPSP, Dejerine-Roussy syndrome) is a central neuropathic pain condition arising from a cerebrovascular lesion (ischemic or involving bleeding) affecting the spinothalamic tract or thalamocortical projections. It occurs in 1–12% of stroke survivors, with onset typically within 1–6 months after the stroke. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

146.2 Causes

This condition happens because of the lesion in the somatosensory pathway. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

146.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

146.4 Signs and Symptoms

You may experience burning, aching, pricking, and freezing sensations contralateral to the lesion, with prominent allodynia and hyperalgesia to cold stimuli. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

146.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

146.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

146.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam with confirmatory neuroimaging identifying the lesion in the appropriate somatosensory pathway. First-line pharmacotherapy includes amitriptyline and lamotrigine
- Gabapentin, pregabalin, and duloxetine are also used.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

146.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

146.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

146.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

146.11 Outlook

- The outlook varies from person to person
- Opioids are generally ineffective for central neuropathic pain.

Chapter 147

Central Sleep Apnea

147.1 Overview

Central sleep apnea (CSA) is a sleep-related breathing disorder characterized by recurrent episodes of absent or reduced respiratory effort during sleep due to instability of the central respiratory control system, without airway obstruction. CSA affects 5–10% of patients referred for sleep studies and is frequently comorbid with heart failure and neurological conditions. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

147.2 Causes

This condition happens because of increased chemosensitivity (high loop gain), prolonged circulation time (Cheyne-Stokes respiration in heart failure), hypobaric hypoxia-induced hyperventilation and hypocapnia (high-altitude periodic breathing), opioid-induced respiratory depression, or brainstem lesions. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

147.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

147.4 Signs and Symptoms

You may experience fragmented sleep, nocturnal shortness of breath, and excessive daytime sleepiness. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

147.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

147.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

147.7 Diagnosis

To make the diagnosis, doctors look for polysomnography demonstrating five or more central apneas or hypopneas per hour with no evidence of obstruction. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

147.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

147.9 Treatment Options

Treatment focuses on treatment of underlying conditions, supplemental oxygen, positive airway pressure (CPAP, bilevel, or adaptive servo-ventilation), acetazolamide, and gradual acclimatization for high-altitude periodic breathing. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

147.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

147.11 Outlook

- Your outlook depends on the underlying cause
- Optimization of cardiac function is essential in heart failure patients.

Chapter 148

Cerebral Aneurysm

148.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- A cerebral aneurysm is a focal dilation of a cerebral artery, most commonly at bifurcation sites in the Circle of Willis
- Saccular (berry) aneurysms are the most prevalent type.

148.2 Causes

This condition happens because of weakness in the arterial wall. Most are asymptomatic and discovered incidentally, but rupture causes subarachnoid bleeding with high mortality (approximately 40%). Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

148.3 Risk Factors

Risk factors include high blood pressure, smoking, family history, connective tissue disorders (eg., Ehlers-Danlos syndrome type II, autosomal dominant polycystic kidney disease), and female sex.

- Advanced age
- Cigarette smoking

- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease
- Obesity and physical inactivity
- Family history and genetic predisposition
- Head trauma or brain injury
- Vascular risk factors (hypertension, diabetes)
- Exposure to environmental toxins
- Chronic stress
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

148.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

148.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

148.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

148.7 Diagnosis

- Doctors diagnose this using CT angiography, MR angiography, or digital subtraction angiography. Treatment for unruptured aneurysms may involve observation or prophylactic treatment depending on size, location, and patient factors
- Ruptured aneurysms require urgent surgical clipping or endovascular coiling.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

148.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

148.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

148.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

148.11 Outlook

The outlook is good for unruptured aneurysms but poor following rupture. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 149

Cerebral Venous Thrombosis

149.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Cerebral venous blood clot formation is a blood clot formation of the dural venous sinuses or cerebral veins that leads to impaired venous removal of fluid, increased intracranial pressure, and potential venous tissue death due to lack of blood supply
- It accounts for less than 1% of all strokes but disproportionately affects young adults and women, particularly those using oral contraceptives or during pregnancy and the puerperium.

149.2 Causes

This condition happens because of thrombotic occlusion of cerebral veins or dural sinuses. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

149.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

149.4 Signs and Symptoms

Clinical features vary widely and may include headache, seizures, weakness, numbness, or problems with movement, papilledema, and altered consciousness. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

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- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

149.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

149.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia

- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

149.7 Diagnosis

To make the diagnosis, doctors look for CT or MR venography showing a filling defect in the venous sinuses. Management consists of anticoagulation (even in the presence of involving bleeding venous tissue death due to lack of blood supply), Treatment for raised intracranial pressure, and treatment of seizures. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

149.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

149.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel)

Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

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149.10 Living with It

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- Monitor your blood pressure and weight regularly
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- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

149.11 Outlook

In most cases, the outlook is favorable with prompt diagnosis and treatment. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 150

Cerumen Impaction

150.1 Overview

Cerumen impaction is an accumulation of earwax that causes symptoms or obscures the tympanic membrane, affecting approximately 6% of the general population and up to 30% of the elderly. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

150.2 Causes

This condition happens because of overproduction of cerumen or from use of cotton swabs and hearing aids that push cerumen deeper into the canal. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

150.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

150.4 Signs and Symptoms

You may experience hearing loss, aural fullness, ringing in the ears, and otalgia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

150.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

150.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

150.7 Diagnosis

Doctors diagnose this based on otoscopic examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

150.8 Prevention

- Most people recover well with proper removal
- Avoidance of cotton swabs is recommended for prevention.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

150.9 Treatment Options

Treatment focuses on manual extraction with a curette, irrigation with warm water, or cerumenolytic agents (dilute hydrogen peroxide, mineral oil, glycerin). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

150.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

150.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 151

Cervical Cancer

151.1 Overview

Treatment for abnormal screening results follows the ASCCP risk-based management consensus guidelines. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

151.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

151.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity

- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

151.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

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- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

151.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

151.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
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- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

151.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

151.8 Prevention

Cervical cancer screening with Pap smear and HPV co-testing remains the cornerstone of prevention. HPV vaccination (targeting ages 11–12, with catch-up through age 26) has the potential to nearly eliminate cervical cancer globally. Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis

B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

151.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

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- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

151.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

151.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 152

Cervical Cancer

152.1 Overview

Cervical cancer is the fourth most common cancer in women worldwide, with approximately 600,000 new cases and 340,000 deaths annually, disproportionately affecting low- and middle-income countries. Nearly all cases (99%) are caused by persistent infection with high-risk human papillomavirus (HPV) types 16 and 18 (responsible for 70% of cases). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

152.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

152.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)

- Excessive alcohol consumption
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- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

152.4 Signs and Symptoms

You may experience abnormal vaginal bleeding, postcoital bleeding, and pelvic pain. Cervical screening involves Pap smear (cytology) and/or HPV DNA testing, with abnormal results evaluated by colposcopy and biopsy. Pre-invasive lesions (cervical intraepithelial neoplasia, CIN) are treated by excisional procedures (LEEP, cold knife conization). Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

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152.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for early-stage disease is radical hysterectomy with pelvic lymph node dissection or chemoradiation
- Advanced disease doctors typically treat it with concurrent chemoradiation and adjuvant chemotherapy.

152.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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152.8 Prevention

Most people recover well with early detection, and prevention through HPV vaccination and screening has dramatically reduced incidence. Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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152.10 Living with It

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(acupuncture, massage, meditation) Build a strong support network of family and friends

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152.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 153

Cervicitis (Chlamydia, Gonorrhea)

153.1 Overview

Cervicitis is inflammation of the uterine cervix, most commonly caused by *Chlamydia trachomatis* and *Neisseria gonorrhoeae*. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

153.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

153.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water

- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

153.4 Signs and Symptoms

You may experience mucopurulent endocervical discharge, cervical friability, and intermenstrual or postcoital bleeding

- Many infections are asymptomatic. Untreated cervicitis can ascend to cause PID, tubal factor infertility, ectopic pregnancy, and neonatal infection. Screening with NAAT is recommended for sexually active women under 25. Treatment for chlamydia: azithromycin 1 g single dose or doxycycline 100 mg BID for 7 days
- Treatment for gonorrhea: ceftriaxone 500 mg IM single dose. Partner treatment and test-of-cure are indicated.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

153.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

153.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts

- Multi-organ failure in severe cases

153.7 Diagnosis

Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

153.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

153.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

153.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

153.11 Outlook

The outlook is generally positive with treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 154

Cervicogenic Headache

154.1 Overview

Cervicogenic headache is a secondary headache originating from bony or soft tissue structures of the cervical spine. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

154.2 Causes

This condition happens because of convergence of cervical afferents (C1–C3) with trigeminal afferents in the trigeminocervical complex. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

154.3 Risk Factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

154.4 Signs and Symptoms

You may experience unilateral pain starting in the neck or occiput and spreading to the frontotemporal region, precipitated or worsened by specific neck movements or sustained head postures, with reduced range of motion and tenderness over upper cervical facets or occipital nerve. Diagnosis is supported by temporary relief with anesthetic block of the C2–3 nerve root or greater occipital nerve. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

154.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

154.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

154.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

154.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

154.9 Treatment Options

Treatment options include physical therapy, occipital nerve block, radiofrequency ablation, and optimization of ergonomics. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

154.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

154.11 Outlook

The outlook varies from person to person and depends on the underlying pathology. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 155

Chagas Disease (American Trypanosomiasis)

155.1 Overview

Chagas disease is a protozoan infection caused by *Trypanosoma cruzi*, transmitted by reduviid (triatomine) bugs ("kissing bugs"), with 6–8 million infected, primarily in Latin America, with increasing cases in the US, Europe, and Japan due to migration. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

155.2 Causes

This condition happens because of metacyclic trypomastigotes entering the skin or mucous membranes, invading macrophages and other nucleated cells, and transforming into amastigotes, with the parasite having a tropism for cardiac muscle, smooth muscle, and autonomic ganglia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

155.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

155.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

155.5 Progression and Stages

You may experience acute phase (6–8 weeks, often asymptomatic): Romãna's sign (unilateral, painless palpebral swelling and conjunctivitis), fever, lymphadenopathy, hepatosplenomegaly, and occasionally acute myocarditis The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Indeterminate phase: asymptomatic, positive serology
- Chronic phase (10–30 years after infection, 30% progress): Chagas cardiomyopathy—the most common and serious manifestation, megaesophagus, and megacolon.

155.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

155.7 Diagnosis

Doctors diagnose this using microscopic identification of trypomastigotes in blood (acute) or serology (chronic). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

155.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

155.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

155.10 Living with It

Treatment options include benznidazole or nifurtimox for acute and congenital disease, with supportive care for chronic cardiomyopathy.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

155.11 Outlook

- The outlook is good for acute disease
- Chronic cardiomyopathy has significant morbidity.

Chapter 156

Chancroid

156.1 Overview

Chancroid is a sexually transmitted infection caused by *Haemophilus ducreyi*, a fastidious gram-negative coccobacillus, most common in resource-limited settings, and a cofactor for HIV transmission due to genital ulceration. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

156.2 Causes

This condition happens because of *H. ducreyi* entering through microabrasions during sexual contact and producing a cytolethal distending toxin that causes epithelial cell tissue death. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

156.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

156.4 Signs and Symptoms

You may experience an incubation of 4–10 days, with painful, non-indurated genital ulcers with ragged, undermined borders and a purulent, necrotic base, and painful inguinal lymphadenopathy occurring in 50% of cases. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

156.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

156.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

156.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with Gram stain showing gram-negative coccobacilli in "school of fish" arrangement. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

156.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

156.9 Treatment Options

Treatment options include azithromycin, ceftriaxone, ciprofloxacin, or erythromycin. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

156.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

156.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 157

Charcot-Marie-Tooth Disease

157.1 Overview

Charcot-Marie-Tooth disease (CMT) is the most common hereditary peripheral neuropathy, affecting 1 in 2,500 individuals. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

157.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

157.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

157.4 Signs and Symptoms

You may experience slowly progressive distal muscle weakness and shrinkage or wasting (beginning in peroneal muscles, leading to foot drop and steppage gait), pes cavus, hammer toes, distal sensory loss, and diminished or absent deep tendon reflexes. Upper extremity involvement occurs later, producing intrinsic hand muscle weakness and claw hand deformity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

157.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- It comprises a genetically heterogeneous group of disorders classified by nerve conduction velocity and inheritance pattern: CMT1 (demyelinating, NCV <38 m/s, autosomal dominant, most commonly PMP22 duplication), CMT2 (axonal, NCV >38 m/s, autosomal dominant, MFN2 most common), CMTX (X-linked, GJB1 mutations), and CMT4 (autosomal recessive). In most cases, the outlook is good
- Most patients remain ambulatory with normal life expectancy, though the severity varies widely by subtype.

157.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

157.7 Diagnosis

Doctors diagnose this based on clinical examination, nerve conduction studies, and genetic testing (next-generation sequencing panels). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

157.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

157.9 Treatment Options

Treatment typically involves supportive: physical and occupational therapy, ankle-foot orthoses for foot drop, surgical correction of foot deformities, and pain Treatment for neuropathic pain. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

157.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

157.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 158

Chickenpox

158.1 Overview

Chickenpox (varicella) is a highly contagious primary infection caused by varicella-zoster virus (VZV, human herpesvirus 3), predominantly affecting unvaccinated children. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

158.2 Causes

This condition happens because of airborne respiratory droplet transmission or direct contact with vesicular fluid, with viral replication in nasopharyngeal lymph nodes followed by viremic dissemination to skin and viscera. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

158.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

158.4 Signs and Symptoms

You may experience a brief viral prodrome (fever, malaise, pharyngitis) followed by a characteristic intensely pruritic maculopapular rash that rapidly evolves into clear vesicles on an erythematous base ('dewdrops on a rose petal'), which rupture and crust over 48 hours. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

158.5 Progression and Stages

Lesions appear in successive crops over 3–5 days, resulting in lesions at all stages of development simultaneously across the body. The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

158.6 Complications

- Complications include secondary bacterial skin infections (Staphylococcus aureus, Streptococcus pyogenes), cerebellar loss of coordination, pneumonia (more severe in adults), and encephalitis
- VZV remains dormant in dorsal root ganglia and may reactivate as herpes zoster later in life.
- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

158.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by Tzanck smear (multinucleated giant cells) or PCR of lesion fluid. Management in healthy children is supportive (antipruritics, cool compresses
- Aspirin is strictly contraindicated due to Reye syndrome)
- Oral acyclovir or valacyclovir is indicated for adolescents, adults, and immunocompromised patients.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

158.8 Prevention

Live attenuated varicella vaccine provides strong protection. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

158.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

158.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

158.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 159

Chikungunya

159.1 Overview

Chikungunya is a mosquito-borne viral illness caused by chikungunya virus (CHIKV), an alphavirus transmitted by *Aedes aegypti* and *A. albopictus*, with outbreaks in Africa, Asia, the Indian Ocean islands, Europe, and the Americas. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

159.2 Causes

This condition happens because of CHIKV infecting fibroblasts, epithelial cells, and monocytes, activating immune cells that produce cytokines mediating acute fever and severe joint pain, with the virus persisting in macrophages and synovial tissue, contributing to chronic arthropathy. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

159.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

159.4 Signs and Symptoms

You may experience an incubation of 2–6 days, with abrupt onset of high fever, severe polyarthralgia (symmetric, involving hands, wrists, ankles, knees, and spine), maculopapular rash on trunk and limbs, and muscle pain, with chronic joint pain (> 3 months) occurring in 30–60% of patients. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

159.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

159.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

159.7 Diagnosis

Doctors diagnose this using RT-PCR (first 5 days of illness) and serology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

159.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

159.9 Treatment Options

Treatment typically involves supportive (rest, hydration, acetaminophen for fever, NSAIDs for joint pain after dengue is excluded). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

159.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

159.11 Outlook

- The outlook is good for acute disease
- Chronic joint pain can be debilitating.

Chapter 160

Choanal Atresia

160.1 Overview

Choanal atresia is a congenital obstruction of the posterior nasal airway caused by a bony (most common), membranous, or mixed atresia plate, and may be associated with CHARGE syndrome. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

160.2 Causes

This condition happens because of failure of the nasobuccal membrane to rupture during embryonic development. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

160.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

160.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

160.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Bilateral choanal atresia presents at birth with cyclic cyanosis (worsening with feeding, improving with crying) and severe respiratory distress
- Unilateral choanal atresia may present later with unilateral nasal obstruction and rhinorrhea.

160.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

160.7 Diagnosis

Your doctor confirms the diagnosis with inability to pass a nasogastric tube and by CT imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

160.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

160.9 Treatment Options

Treatment typically involves surgical transnasal or transpalatal repair, ideally in the neonatal period for bilateral disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

160.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

160.11 Outlook

The outlook is good following surgical repair. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 161

Cholecystitis

161.1 Overview

Acute calculous cholecystitis results from persistent impaction of a gallstone in the cystic duct, causing bile stasis, mucosal reduced blood flow, chemical irritation, and secondary bacterial superinfection (Escherichia coli, Klebsiella, Enterococcus). Acalculous cholecystitis occurs in critically ill patients (burns, trauma, major surgery, prolonged parenteral nutrition). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

161.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

161.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

161.4 Signs and Symptoms

- You may experience severe steady right upper quadrant (RUQ) or epigastric pain radiating to the right scapula/shoulder, fever, leukocytosis, and a positive Murphy sign (inspiratory arrest on deep palpation of the RUQ). Diagnosis is supported by abdominal ultrasonography showing gallbladder wall thickening (> 4 mm), pericholecystic fluid, and sonographic Murphy sign
- HIDA scan confirms cystic duct obstruction if ultrasound is equivocal.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

161.5 Progression and Stages

Cholecystitis is inflammation of the gallbladder wall, classified as acute calculous (90% of cases), acalculous (10%), or chronic. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

161.6 Complications

Complications include gallbladder gangrene, perforation, and emphysematous cholecystitis.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

161.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

161.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

161.9 Treatment Options

Treatment options include NPO status, IV fluids, IV broad-spectrum antibiotics, and early using small incisions and a camera cholecystectomy (within 24–72 hours). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

161.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

161.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 162

Cholelithiasis

162.1 Overview

Cholelithiasis is the presence of solid concretions (gallstones) within the gallbladder lumen, affecting 10–15% of the adult population. Most gallstones ($> 80\%$) are asymptomatic. Symptomatic cholelithiasis presents as biliary colic: episodic right upper quadrant or epigastric dull pain precipitated by fatty meal ingestion, lasting 1–5 hours, without fever or peritoneal signs. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

162.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

162.3 Risk Factors

Risk factors for cholesterol stones include the classic '4 Fs': Female, Fat (obesity), Fertile (multiparity/estrogen therapy), and Forty (> 40 years). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

162.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

162.5 Progression and Stages

Gallstones are classified as cholesterol stones (80%, radiolucent, resulting from bile supersaturated with cholesterol, gallbladder hypomotility, and excess bilirubin) or pigment stones (20%, radiopaque, divided into black stones from chronic hemolysis/cirrhosis and brown stones from biliary tract infection). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

162.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

162.7 Diagnosis

- Doctors diagnose this using abdominal ultrasound. Asymptomatic stones require no treatment
- Symptomatic biliary colic is managed with elective using small incisions and a camera cholecystectomy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

162.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

162.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

162.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

162.11 Outlook

Most people recover well following surgery. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 163

Cholera (Vibrio cholerae)

163.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

163.2 Causes

This condition happens because of colonization of the small intestine and secretion of cholera toxin (CT), an A-B subunit enterotoxin that ADP-ribosylates Gs protein, constitutively activating adenylate cyclase, increasing cAMP, and causing massive chloride and water secretion into the intestinal lumen. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

163.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

163.4 Signs and Symptoms

You may experience sudden onset of profuse, painless, watery diarrhea (rice-water stools), vomiting, and rapidly progressing dehydration leading to hypovolemic shock, metabolic acidosis, and kidney failure within hours in severe cases. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

163.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

163.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage

- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

163.7 Diagnosis

Doctors diagnose this based on stool culture (TCBS agar) or rapid dipstick test in endemic settings. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

163.8 Prevention

Cholera is an acute diarrheal disease caused by *Vibrio cholerae* serogroups O1 and O139, with 1.3–4 million cases and 21,000–143,000 deaths annually worldwide, primarily in developing countries with poor water sanitation. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings

- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

163.9 Treatment Options

Treatment focuses on aggressive rehydration with oral rehydration solution (WHO-ORS) for mild to moderate dehydration, intravenous Ringer's lactate for severe dehydration, and antibiotics (doxycycline, azithromycin, or ciprofloxacin) to reduce duration and volume of diarrhea. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

163.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

163.11 Outlook

- Most people recover well with prompt rehydration (mortality $< 1\%$)
- Without treatment, mortality exceeds 50%.

Chapter 164

Chondrodermatitis Nodularis Helicis

164.1 Overview

Chondrodermatitis nodularis helicis is a painful, inflammatory condition affecting the cartilage of the pinna, most commonly the helix or antihelix. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

164.2 Causes

This condition happens because of necrotizing chondritis with overlying epidermal changes, often triggered by chronic pressure during sleep. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

164.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

164.4 Signs and Symptoms

You may experience a small, exquisitely tender nodule with overlying crusting or ulceration, frequently on the side most affected by sleeping pressure. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

164.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

164.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

164.7 Diagnosis

Doctors diagnose this based on clinical examination and histopathology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

164.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

164.9 Treatment Options

- Treatment options include pressure relief (using a donut-shaped pillow), topical corticosteroids, and intralesional corticosteroid injection
- Refractory cases may require surgical removal.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

164.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

164.11 Outlook

In most cases, the outlook is good with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 165

Chronic Fatigue Syndrome

165.1 Overview

Chronic fatigue syndrome/myalgic encephalomyelitis (CFS/ME) is a complex, debilitating multisystem illness affecting 0.2–0.4% of the population, with a female predominance. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

165.2 Causes

The cause is unknown but is likely multifactorial, involving post-infectious triggers (EBV, enterovirus, Q fever, Ross River virus), immune dysregulation, neuroendocrine dysfunction, and genetic predisposition, with proposed mechanisms including chronic immune activation, hypothalamic-pituitary-adrenal axis hypofunction, mitochondrial dysfunction, and central sensitization. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

165.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

165.4 Signs and Symptoms

Clinical features (CDC/Institute of Medicine 2015 diagnostic criteria) include substantial reduction in previous activity level persisting >6 months, post-exertional malaise (PEM), unrefreshing sleep, thinking and memory impairment (brain fog), and orthostatic intolerance, along with additional symptoms such as chronic pain, sore throat, tender lymphadenopathy, and flu-like malaise.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

165.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

165.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

165.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (no biomarkers), requiring ruling out other causes.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

165.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

165.9 Treatment Options

Treatment options include symptom management with pacing/activity management (avoiding PEM), and symptom-specific treatments such as sleep hygiene, low-dose naltrexone, and SSRIs. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

165.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

165.11 Outlook

The outlook varies from person to person, with approximately 5% recovering fully and most patients experiencing a fluctuating course.

Chapter 166

Chronic Gastritis

166.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Chronic gastritis is marked by chronic inflammatory infiltrate in the gastric mucosa. Two main types exist: Type A (autoimmune) gastritis involves the fundus and body and is linked to pernicious anemia and increased risk of gastric carcinoid tumors
- Type B (atrophic) gastritis involves the antrum and is the most common form worldwide.

166.2 Causes

This condition happens because of autoimmune destruction of parietal cells with anti-parietal cell and anti-intrinsic factor antibodies (Type A) or *H. pylori* infection (Type B). Long-standing *H. pylori* gastritis can progress through atrophic gastritis, intestinal metaplasia, dysplasia, and gastric adenocarcinoma (Correa cascade). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

166.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

166.4 Signs and Symptoms

You may experience dyspepsia and potential symptoms of vitamin B12 deficiency in Type A. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

166.5 Progression and Stages

Doctors diagnose this based on endoscopy with biopsy using the Sydney system for grading. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

166.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

166.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

166.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

166.9 Treatment Options

Treatment focuses on *H. pylori* eradication therapy for Type B and vitamin B12 supplementation for Type A. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

166.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

166.11 Outlook

Your outlook depends on the subtype and the presence of precancerous changes. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 167

Chronic Hypertension in Pregnancy

167.1 Overview

Chronic high blood pressure in pregnancy is defined as high blood pressure predating pregnancy or diagnosed before 20 weeks of gestation. It affects 1–5% of pregnancies. Women with chronic high blood pressure are at increased risk of superimposed preeclampsia (25%), placental abruption, preterm birth, and fetal growth restriction. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

167.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

167.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus

- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

167.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

167.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

167.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

167.7 Diagnosis

History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

167.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

167.9 Treatment Options

- Treatment options include antihypertensives (labetalol, nifedipine, methyldopa) for BP $\geq$160/100 mmHg
- ACE inhibitors and ARBs are contraindicated in pregnancy due to fetotoxicity.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

167.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

167.11 Outlook

In most cases, the outlook is favorable with appropriate management. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 168

Chronic Insomnia Disorder

168.1 Overview

Chronic insomnia disorder is a sleep disorder defined by persistent difficulty initiating sleep, maintaining sleep, or early morning awakening with inability to return to sleep, despite adequate opportunity for sleep, occurring at least three nights per week for at least three months, with significant daytime impairment (fatigue, mood disturbance, thinking and memory dysfunction, or reduced performance). The prevalence is 6–10% in the general population, with females affected 1.5–2 times more frequently than males. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

168.2 Causes

This condition happens because of hyperarousal of the hypothalamic-pituitary-adrenal axis (elevated cortisol, ACTH), increased high-frequency EEG activity during sleep, activation of the sympathetic nervous system, and maladaptive conditioning (negative associations with the sleep environment). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

168.3 Risk Factors

Risk factors include advanced age, female sex, shift work, medical and psychiatric comorbidities, and genetic predisposition.

- Family history of mental illness
- Trauma or adverse childhood experiences
- Chronic stress
- Substance use
- Social isolation
- Underlying medical conditions
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Medication use

168.4 Signs and Symptoms

You may experience difficulty falling asleep, frequent awakenings, early morning awakening, and daytime impairment.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

168.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

168.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

168.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam based on ICSD-3 and DSM-5 criteria, supplemented by sleep diaries and actigraphy
- Polysomnography is not routinely indicated but may evaluate comorbid sleep disorders.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

168.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

168.9 Treatment Options

Treatment focuses on thinking and memory behavioral therapy for insomnia (CBT-I) as first-line therapy—stimulus control, sleep restriction, thinking and memory restructuring, and sleep hygiene education—with pharmacotherapy (melatonin, ramelteon, orexin receptor antagonists, benzodiazepine receptor agonists, trazodone, doxepin) as adjunctive treatment. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

168.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups

- Stay informed about your condition

168.11 Outlook

- The outlook is generally positive with CBT-I
- Long-term outcomes favor CBT-I over medication alone.

Chapter 169

Chronic Kidney Disease

169.1 Overview

Chronic kidney disease (CKD) is abnormalities of kidney structure or function present for more than 3 months, with health implications. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

169.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common causes are diabetes mellitus (leading cause in developed countries) and high blood pressure
- Other causes include glomerulonephritis, polycystic kidney disease, and chronic obstruction. This condition happens because of progressive loss of nephrons leading to declining GFR.

169.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

169.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

169.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- CKD is staged by the glomerular filtration rate (GFR): Stage 1 (GFR $\geq$90 with kidney damage markers), Stage 2 (GFR 60–89), Stage 3a (GFR 45–59), Stage 3b (GFR 30–44), Stage 4 (GFR 15–29), and Stage 5 (GFR <15, end-stage kidney disease, ESRD). Treatment options include BP control (target <130/80 with ACE inhibitors/ARBs), glycemic control, SGLT2 inhibitors (dapagliflozin, empagliflozin), dietary modification, and avoidance of nephrotoxins
- Kidney replacement therapy is required for Stage 5 CKD. Your outlook depends on stage and progression rate
- Cardiovascular disease is the leading cause of death in CKD.

169.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

169.7 Diagnosis

Doctors diagnose this based on GFR estimation and markers of kidney damage (albuminuria, abnormal imaging, abnormal histology). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

169.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

169.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

169.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

169.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 170

Chronic Low Back Pain

170.1 Overview

Chronic low back pain (cLBP) is pain in the lower back persisting for $\geq$3 months. It is the leading cause of years lived with disability worldwide, with a lifetime prevalence of 60–80%. Most cases are non-specific (no identifiable structural pathology). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

170.2 Causes

This condition happens because of a biopsychosocial interaction of biological factors (structural pathology, inflammation), psychological factors (catastrophizing, fear-avoidance beliefs, depression), and social factors (occupational demands, disability compensation). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

170.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

170.4 Signs and Symptoms

You may experience pain in the lower back that may radiate to the legs (sciatica), with or without neurological deficits. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

170.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

170.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

170.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Red flags for serious pathology include age >50 or <20, recent trauma, unexplained weight loss, fever, night pain, history of cancer, and progressive neurological deficits. Initial Treatment options include education, activity modification, NSAIDs, and muscle relaxants
- First-line pharmacotherapy is NSAIDs, with TCAs, SNRIs, and gabapentinoids as adjuncts
- Non-pharmacological treatments include physical therapy, manual therapy, acupuncture, thinking and memory behavioral therapy, and graded activity programs
- Interventional procedures include epidural steroid injections, facet joint injections, and spinal cord stimulation.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

170.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

170.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

170.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

170.11 Outlook

- The outlook varies from person to person
- Most cases improve with conservative management.

Chapter 171

Chronic Lymphocytic Leukemia

171.1 Overview

Chronic lymphocytic leukemia is the most common leukemia in adults in Western countries, with a median age at To diagnose 70 years, characterized by accumulation of mature-appearing but functionally incompetent B lymphocytes in the blood, bone marrow, lymph nodes, and spleen. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

171.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

171.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

171.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

171.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

171.6 Complications

You may experience generalized lymphadenopathy, hepatosplenomegaly, fatigue, weight loss, night sweats, and autoimmune complications (autoimmune hemolytic anemia, immune thrombocytopenia), with many patients being asymptomatic and diagnosed incidentally on routine CBC showing lymphocytosis (>5000/μL B lymphocytes). Treatment typically involves indicated for symptomatic or bulky disease, progressive cytopenias, or autoimmune complications, with options including BTK inhibitors (ibrutinib, acalabrutinib, zanubrutinib), BCL-2 inhibitor (venetoclax) plus obinutuzumab, and FCR (fludarabine, cyclophosphamide, rituximab) for young patients with mutated IGHV. Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

171.7 Diagnosis

Doctors diagnose this using flow cytometry showing clonal B cells co-expressing CD5, CD19, CD20 (dim), CD23, with prognostic markers including IGHV mutation status (mutated = favorable, unmutated = unfavorable), TP53 mutations/deletions (very unfavorable), del(13q) (favorable), trisomy 12, and del(11q). Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel

- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

171.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

171.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support

- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

171.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

171.11 Outlook

In most cases, the outlook is favorable, with CLL being incurable but highly treatable and many patients having prolonged survival. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 172

Chronic Neck Pain

172.1 Overview

Chronic neck pain has a point prevalence of 10–15% and a lifetime prevalence of 50–70%. It is more common in women and office workers. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

172.2 Causes

This condition happens because of mechanical neck pain (poor posture, prolonged computer use), cervical spondylosis, cervical radiculopathy, or myofascial pain. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

172.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

172.4 Signs and Symptoms

You may experience pain in the neck that may radiate to the shoulders or arms. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

172.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

172.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

172.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Management parallels chronic low back pain: physical therapy, manual therapy, exercise, NSAIDs, and muscle relaxants
- Cervical epidural steroid injections and selective nerve root blocks are used for radicular pain.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

172.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

172.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

172.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

172.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 173

Chronic Obstructive Pulmonary Disease

173.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

173.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of chronic inflammation leading to airway remodeling and alveolar destruction
- COPD encompasses chronic bronchitis (persistent cough and sputum production) and emphysema (destruction of alveolar walls).

173.3 Risk Factors

- Chronic obstructive lung disease is a common, preventable, and treatable disease characterized by persistent respiratory symptoms and airflow limitation due to airway or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases
- Cigarette smoking is the predominant risk factor (approximately 80% of cases).
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

173.4 Signs and Symptoms

You may experience progressive shortness of breath, chronic cough, sputum production, wheezing, and chest tightness. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

173.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Doctors diagnose this using spirometry showing post-bronchodilator FEV_1/FVC ratio less than 0.70
- GOLD classification stages severity by FEV_1. Treatment options include smoking cessation (most important intervention), bronchodilators, inhaled corticosteroids for frequent exacerbators, lung rehabilitation, long-term oxygen therapy for chronic hypoxemic failure, and lung volume reduction surgery or lung transplantation for advanced disease. Your outlook depends on disease severity, exacerbation frequency, and smoking cessation.

173.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

173.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

173.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

173.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

173.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

173.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 174

Chronic Otitis Media

174.1 Overview

Chronic otitis media is a chronic or recurrent perforation of the tympanic membrane with or without active middle ear infection. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

174.2 Causes

This condition happens because of persistent mucosal inflammation or squamous disease (cholesteatoma), which is an abnormal accumulation of keratinizing squamous epithelium that can erode the ossicles, facial nerve, and bony labyrinth. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

174.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

174.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

174.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

174.6 Complications

- You may experience persistent or recurrent otorrhea through a perforated tympanic membrane, hearing loss, and occasional feeling of spinning
- Cholesteatoma may cause conductive hearing loss, facial nerve palsy, and intracranial complications.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

174.7 Diagnosis

Doctors diagnose this based on otoscopic examination and temporal bone CT. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

174.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

174.9 Treatment Options

Treatment typically involves primarily surgical (tympanoplasty, mastoidectomy). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

174.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

174.11 Outlook

Your outlook depends on the extent of disease and success of surgical intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 175

Chronic Pain vs Acute Pain

175.1 Overview

Acute pain is a direct physiological response to a noxious stimulus or injury, with a reversible time course proportional to tissue healing (typically days to weeks). It serves an adaptive, protective function and resolves with treatment of the underlying cause. Chronic pain is pain persisting beyond normal tissue healing time (typically >3 months) or pain associated with a chronic condition. It is maladaptive, often lacks a clear protective function, and involves neuroplastic changes in the peripheral and central nervous systems. Chronic pain affects approximately 20–30% of the adult population worldwide. Management requires a biopsychosocial approach rather than a purely biomedical one. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

175.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

175.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

175.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

175.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

175.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

175.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

175.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

175.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

175.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

175.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 176

Chronic Pancreatitis

176.1 Overview

Chronic pancreatitis is progressive, irreversible morphological changes of the pancreatic parenchyma leading to impairment of exocrine and endocrine function. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

176.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common cause in Western countries is chronic alcohol use (70–80%)
- Other causes include hereditary pancreatitis (PRSS1 mutations), cystic scarring (CFTR mutations), autoimmune pancreatitis, tropical pancreatitis, and idiopathic. This condition happens because of repeated episodes of pancreatic injury leading to scarring, parenchymal shrinkage or wasting, ductal strictures, and calcifications.

176.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

176.4 Signs and Symptoms

You may experience chronic epigastric pain (often worse after meals), steatorrhea (exocrine insufficiency), diabetes mellitus (endocrine insufficiency), and weight loss. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

176.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

176.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

176.7 Diagnosis

Doctors diagnose this based on CT scan showing pancreatic calcifications, ductal dilatation, and parenchymal shrinkage or wasting

- MRCP for ductal morphology
- And functional testing (fecal elastase, secretin stimulation).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

176.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

176.9 Treatment Options

Treatment options include pancreatic enzyme replacement therapy (PERT), pain management (analgesics, using a thin camera tube removal of fluid of ductal strictures, celiac plexus block), insulin for diabetes, and total pancreatectomy with islet autotransplantation for refractory cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

176.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

176.11 Outlook

The outlook varies from person to person with progressive functional decline. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 177

Chronic Pelvic Pain

177.1 Overview

Chronic pelvic pain (CPP) is persistent or recurrent pain in the pelvic region lasting $\geq$6 months in women (and $\geq$3–6 months in men). In women, estimated prevalence is 15–25%. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

177.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of various causes: in women, endometriosis, interstitial cystitis, chronic pelvic inflammatory disease, pelvic adhesions, adenomyosis, and pelvic floor myofascial pain
- In men, chronic prostatitis and pudendal neuralgia.

177.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

177.4 Signs and Symptoms

You may experience persistent or recurrent pelvic pain. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

177.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

177.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

177.7 Diagnosis

To make the diagnosis, doctors look for a multidisciplinary approach including gynecological, urological, gastroenterological, and pain medicine evaluation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

177.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

177.9 Treatment Options

Treatment typically involves directed at the identified cause

- Pelvic floor physical therapy is highly effective for pelvic floor myofascial dysfunction
- Pharmacotherapy includes NSAIDs, gabapentinoids, TCAs, and hormonal therapy for endometriosis.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

177.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

177.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 178

Chronic Rhinosinusitis

178.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

178.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of persistent mucosal inflammation, often with bacterial biofilms
- CRSwNP involves type 2 inflammation with eosinophilic infiltration and IL-5/IL-13 pathways.

178.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

178.4 Signs and Symptoms

You may experience nasal obstruction, facial pain or pressure, purulent nasal discharge, and hyposmia. Symptoms vary depending on the severity and extent of the condition

Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

178.5 Progression and Stages

Chronic rhinosinusitis is a symptomatic inflammation of the paranasal sinuses persisting for 12 weeks or longer, classified into CRS with nasal polyps (CRSwNP) and CRS without nasal polyps (CRSsNP). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

178.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

178.7 Diagnosis

To make the diagnosis, doctors look for using a thin camera tube or CT evidence of sinus inflammation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

178.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

178.9 Treatment Options

Treatment options include nasal saline irrigation, intranasal corticosteroids, and antibiotics for acute exacerbations

- Functional using a thin camera tube sinus surgery (FESS) is indicated for refractory disease
- Biologic therapies (dupilumab, omalizumab) have shown efficacy for CRSwNP.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

178.10 Living with It

The outlook varies from person to person, with many patients requiring long-term management.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

178.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 179

Chronic Traumatic Encephalopathy

179.1 Overview

Chronic traumatic encephalopathy is a progressive neurodegenerative disease associated with repetitive head impacts, found predominantly in contact sport athletes, military personnel exposed to blast injuries, and victims of repeated physical abuse. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

179.2 Causes

This condition happens because of accumulation of hyperphosphorylated tau protein in neurons and astrocytes, typically in a perivascular distribution at the depths of cortical sulci. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

179.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

179.4 Signs and Symptoms

- You may experience behavioral and mood changes (depression, apathy, irritability, impulsivity), thinking and memory impairment (progressive dementia), and motor abnormalities (parkinsonism)
- Symptoms typically manifest years to decades after the repetitive head trauma. Diagnosis is currently only confirmed postmortem by neuropathological examination.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

179.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

179.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

179.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

179.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

179.9 Treatment Options

Treatment typically involves supportive, as no disease-modifying treatment exists. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

179.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

179.11 Outlook

- Outlook is progressive and incurable
- The discovery of chronic traumatic encephalopathy has led to significant changes in sports safety protocols.

Chapter 180

Chronic Tubulointerstitial Nephritis

180.1 Overview

Chronic tubulointerstitial nephritis is gradual tubular shrinkage or wasting and interstitial scarring. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

180.2 Causes

Causes include chronic analgesic use (NSAIDs, combined analgesics), lithium therapy, heavy metals (lead, cadmium), hyperuricemia, sarcoidosis, and chronic infections. This condition happens because of prolonged exposure to nephrotoxic agents or chronic inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

180.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

180.4 Signs and Symptoms

You may experience slowly progressive CKD, often with inability to concentrate urine (polyuria, nocturia) and mild proteinuria. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

180.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- The outlook varies from person to person
- Progression to ESRD may occur with continued exposure.

180.6 Complications

Treatment focuses on removal of the offending agent and Treatment for CKD complications.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

180.7 Diagnosis

Doctors diagnose this using kidney biopsy showing tubular shrinkage or wasting, interstitial scarring, and chronic inflammatory infiltrate. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

180.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

180.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

180.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

180.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 181

Circadian Rhythm Regulation

181.1 Overview

Circadian rhythms are endogenous near-24-hour biological cycles regulated by the suprachiasmatic nucleus (SCN) of the anterior hypothalamus. The SCN receives photic input from intrinsically photosensitive retinal ganglion cells (ipRGCs containing melanopsin) via the retinohypothalamic tract and coordinates peripheral clocks throughout the body via neuroendocrine (melatonin, cortisol) and autonomic outputs. Cortisol exhibits a reciprocal pattern, with nadir around midnight and peak upon awakening (cortisol awakening response). The circadian system interacts homeostatically with the sleep-wake drive (Process C and Process S): adenosine accumulation during wakefulness drives homeostatic sleep pressure (Process S), while circadian alerting signals (Process C) counterbalance this pressure to maintain consolidated wakefulness. The two-process model explains the timing and intensity of sleep under normal and sleep-deprived conditions. Circadian misalignment occurs when the endogenous clock is desynchronized from the external light-dark cycle or social schedule. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

181.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

181.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

181.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

181.5 Progression and Stages

Melatonin, synthesized and released by the pineal gland during darkness, is the principal hormonal signal for sleep onset and phase shifting. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

181.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

181.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

181.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

181.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

181.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

181.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 182

Cirrhosis

182.1 Overview

This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

182.2 Causes

This condition happens because of chronic liver injury due to viral hepatitis (B, C), alcohol, NAFLD/NASH, autoimmune hepatitis, primary biliary cholangitis, primary sclerosing cholangitis, or hereditary hemochromatosis. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

182.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

182.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

182.5 Progression and Stages

Cirrhosis is the end stage of chronic liver disease characterized by diffuse scarring and formation of regenerative nodules that distort the liver architecture and vasculature, and is the 12th leading cause of death worldwide. Your symptoms depend on the stage and complications. Your outlook depends on the stage and complications, with transplantation offering The main treatment. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

182.6 Complications

Cirrhosis may be compensated (no complications) or decompensated (development of ascites, variceal bleeding, liver encephalopathy, or jaundice). Portal high blood pressure drives many complications including esophageal and gastric varices, ascites, hepatorenal syndrome, and splenomegaly with hypersplenism.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

182.7 Diagnosis

- Doctors diagnose this using liver biopsy, imaging (ultrasound, CT, MRI), and elastography (FibroScan)
- The MELD score predicts mortality and guides transplant listing.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

182.8 Prevention

Treatment options include treatment of the underlying cause (antivirals, alcohol abstinence, weight loss), diuretics for ascites, beta-blockers or band ligation for variceal prophylaxis, lactulose and rifaximin for liver encephalopathy, and liver transplantation for end-stage disease.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

182.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

182.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

182.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 183

Cleft Lip and Palate

183.1 Overview

Cleft lip and palate are among the most common congenital anomalies (1 per 700 live births), resulting from failure of fusion of the maxillary and medial nasal processes (cleft lip) or the palatal shelves (cleft palate). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

183.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

183.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

183.4 Signs and Symptoms

You may experience feeding difficulties, speech abnormalities, hearing loss, and dental anomalies. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

183.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

183.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

183.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam at birth. Treatment requires a multidisciplinary team approach: surgical repair (cheiloplasty at 3–6 months, palatoplasty at 9–18 months), speech therapy, orthodontic intervention, and Treatment for associated ear issues. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

183.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

183.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

183.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

183.11 Outlook

Most people recover well with appropriate multidisciplinary care. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 184

Cluster Headache

184.1 Overview

Cluster headache is a trigeminal autonomic cephalalgia (TAC) characterized by strictly unilateral, severe to very severe pain in the orbital, supraorbital, temporal, or any combination of these areas, lasting 15–180 minutes if untreated, with attacks occurring from once every other day to 8 times per day. Prevalence is approximately 0.1%, with a male predominance (3:1). This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

184.2 Causes

This condition happens because of activation of the posterior hypothalamic gray matter and the trigeminovascular system. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

184.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

184.4 Signs and Symptoms

- You may experience unilateral burning, boring, or stabbing pain with ipsilateral autonomic features (conjunctival injection, lacrimation, nasal congestion, rhinorrhea, miosis, ptosis, eyelid swelling) and restlessness or agitation during attacks. Episodic cluster headache occurs in bouts with remission periods
- Chronic cluster headache has attacks for >1 year without remission.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

184.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

184.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

184.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam based on ICHD-3 criteria. Acute Treatment options include subcutaneous sumatriptan, intranasal zolmitriptan, and high-flow oxygen. Transitional therapy uses short-course oral corticosteroids or greater occipital nerve block. Preventive therapy includes verapamil, lithium, topiramate, and valproic acid. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

184.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

184.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

184.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

184.11 Outlook

outlook is chronic but attacks can be prevented during bouts with appropriate therapy. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 185

Coarctation of the Aorta

185.1 Overview

Coarctation of the Aorta is a congenital cardiac anomaly characterized by a discrete constriction of the aortic lumen, most commonly located distal to the origin of the left subclavian artery near the insertion of the ductus arteriosus (juxtaductal). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

185.2 Causes

This condition happens because of ectopic ductal tissue tissue contraction or impaired embryonic aortic arch development. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

185.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

185.4 Signs and Symptoms

- You may experience upper extremity high blood pressure with delayed, weak, or absent femoral pulses and a lower extremity blood pressure gradient ($> 10 - -20$ mmHg lower in legs)
- Patients may present with headache, cold feet, leg claudication, or heart failure. A continuous systolic murmur is best heard over the back. Chest radiograph demonstrates rib notching (due to collateral intercostal arterial enlargement) and the 'figure-3' sign on the aorta.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

185.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

185.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

185.7 Diagnosis

Your doctor confirms the diagnosis with echocardiography or cardiac MRI. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

185.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

185.9 Treatment Options

Treatment focuses on surgical removal with end-to-end anastomosis or balloon angioplasty with stenting. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

185.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

185.11 Outlook

The outlook is good following early intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 186

Coccidioidomycosis (Valley Fever)

186.1 Overview

Coccidioidomycosis (Valley fever) is a systemic fungal infection caused by *Coccidioides immitis* and *C. posadasii*, dimorphic fungi endemic to the Sonoran Desert (southwestern US, northern Mexico, and parts of Central and South America), with 150,000 infections annually in the US, and incidence increasing due to climate change. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

186.2 Causes

This condition happens because of inhalation of arthroconidia leading to conversion into spherules containing endospores in the lung, with a robust Th1 response controlling infection. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

186.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

186.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

186.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

186.6 Complications

You may experience acute lung: 60% asymptomatic, 40% develop Valley fever (influenza-like illness with fever, cough, chest pain, joint pain, and redness of the skin nodosum)

- Chronic lung: persistent cough, hemoptysis, cavities
- Disseminated coccidioidomycosis: skin, bones/joints, and CNS (coccidioidal meningitis—the most feared complication).
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

186.7 Diagnosis

Doctors diagnose this using culture, serology, and histopathology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

186.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

186.9 Treatment Options

Treatment options include fluconazole for mild-moderate disease and liposomal amphotericin B for severe disseminated disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

186.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

186.11 Outlook

- Most people recover well for acute lung disease
- Meningitis requires lifelong therapy.

Chapter 187

Cogan's Syndrome

187.1 Overview

Cogan's syndrome is a rare, chronic inflammatory disorder characterized by non-syphilitic interstitial keratitis and vestibuloauditory symptoms reminiscent of Ménière's disease/hearing loss, associated with systemic vasculitis, typically affecting young adults (median age 25–40 years), with equal sex distribution. The Disease mechanism is poorly understood but likely involves a T-cell-mediated and humoral autoimmune response directed against a common antigen in the cornea and inner ear, with approximately 50–70% of patients having systemic vasculitis that can involve large, medium, or small vessels. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

187.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

187.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

187.4 Signs and Symptoms

You may experience ocular involvement (interstitial keratitis—the hallmark, presenting with eye pain, photophobia, blurred vision, conjunctival injection), vestibuloauditory involvement (acute-onset Ménière-like attacks of feeling of spinning, nausea, vomiting, ringing in the ears, and fluctuating sensorineural hearing loss that can progress to bilateral profound deafness), and systemic vasculitis (constitutional symptoms, skin nodules or purple spots on the skin, neuropathy, arthritis).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

187.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

187.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

187.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, based on ocular and vestibuloauditory manifestations in the absence of syphilis or other infection.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

187.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

187.9 Treatment Options

Treatment typically involves with corticosteroids (high-dose systemic and topical eye), with methotrexate, azathioprine, cyclophosphamide, rituximab, and TNF inhibitors used for refractory or severe disease. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

187.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

187.11 Outlook

The outlook varies from person to person: up to 50% of patients develop significant visual or hearing impairment.

Chapter 188

Colorectal Adenoma

188.1 Overview

Colorectal adenoma is a non-cancerous epithelial tumor that is a precursor to colorectal cancer through the adenoma-carcinoma sequence. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

188.2 Causes

Most adenomas are asymptomatic and detected through screening colonoscopy. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

188.3 Risk Factors

Risk factors include age (>50), male sex, family history, hereditary polyposis syndromes (FAP, Lynch syndrome), inflammatory bowel disease, obesity, smoking, and high red meat consumption.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

188.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

188.5 Progression and Stages

It is classified as tubular, tubulovillous, or villous based on histology, and by the degree of dysplasia (low-grade or high-grade). Larger adenomas (>1 cm), villous histology, and high-grade dysplasia carry higher cancerous potential. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

188.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

188.7 Diagnosis

Doctors diagnose this based on colonoscopy with histopathological evaluation. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

188.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

188.9 Treatment Options

Treatment focuses on using a thin camera tube removal (polypectomy) with surveillance colonoscopy intervals based on polyp number, size, and histology. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

188.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

188.11 Outlook

Most people recover well with complete removal. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment.

Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 189

Colorectal Cancer

189.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Colorectal cancer (CRC) is the third most commonly diagnosed cancer worldwide and the second leading cause of cancer-related death. Approximately 70% of CRCs arise from adenomatous polyps via the chromosomal instability pathway (APC/KRAS/TP53 mutations)
- Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC) accounts for 2–4% of cases due to mismatch repair gene mutations.

189.2 Causes

This condition happens because of accumulated genetic and epigenetic alterations driving cancerous transformation. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

189.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal

factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

189.4 Signs and Symptoms

- Symptoms vary depending on location
- Right-sided colon cancers tend to present with iron deficiency anemia and occult GI bleeding, while left-sided cancers present with altered bowel habits, obstruction, and hematochezia.
- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

189.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for localized disease is surgical removal with lymphadenectomy, with adjuvant chemotherapy (FOLFOX, CapeOX) for stage III disease
- Neoadjuvant chemoradiation is standard for locally advanced rectal cancer. Your outlook depends on stage at diagnosis.

189.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

189.7 Diagnosis

- Doctors diagnose this based on colonoscopy with biopsy
- Screening methods include colonoscopy (gold standard), fecal occult blood testing (FIT), and CT colonography. Staging uses TNM classification with CT, MRI (rectal cancer), and PET. Metastatic CRC doctors typically treat it with systemic chemotherapy, targeted therapy (bevacizumab [anti-VEGF], cetuximab/panitumumab [anti-EGFR] for RAS wild-type tumors), immunotherapy (pembrolizumab for MSI-high/dMMR tumors), and metastasectomy for selected patients.
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

189.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

189.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

189.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications

- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

189.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 190

Coma

190.1 Overview

Coma is a state of profound unconsciousness in which the patient cannot be aroused and does not respond to external stimuli. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

190.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of dysfunction of the ascending reticular activating system in the brainstem or bilateral cerebral hemispheres
- Common causes include traumatic brain injury, stroke, intracranial bleeding, status epilepticus, metabolic encephalopathy, drug intoxication, infections, and hypoxic-ischemic brain injury.

190.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

190.4 Signs and Symptoms

You may experience unresponsiveness to external stimuli with eyes closed. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

190.5 Progression and Stages

Recovery varies widely depending on the cause and severity of injury. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

190.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

190.7 Diagnosis

Doctors diagnose this based on clinical assessment using the Glasgow Coma Scale (score 3–15), urgent neuroimaging, and laboratory studies. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

190.8 Prevention

Management targets the underlying cause, with supportive measures including airway management, intracranial pressure monitoring, and prevention of secondary brain injury.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

190.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

190.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

190.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 191

Common Variable Immunodeficiency

191.1 Overview

Common variable immunodeficiency is the most common symptomatic primary antibody deficiency, affecting approximately 1 in 25,000–50,000. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

191.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

191.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

191.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities

- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

191.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

191.6 Complications

Complications include bronchiectasis (from recurrent infections), autoimmune cytopenias, granulomatous disease, lymphoproliferation, and increased risk of digestive tract and lymphoid malignancies.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

191.7 Diagnosis

To make the diagnosis, doctors look for exclusion of other causes of hypogammaglobulinemia and is typically made in the second to fourth decade of life. You may experience recurrent infections.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

191.8 Prevention

The condition is marked by hypogammaglobulinemia (decreased IgG and IgA, with or without decreased IgM), impaired vaccine antibody responses, and increased susceptibility to bacterial infections, particularly sinopulmonary infections (pneumonia, sinusitis, bronchitis).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

191.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

191.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

191.11 Outlook

The outlook varies from person to person but improved with immunoglobulin replacement.

Chapter 192

Community-Acquired Pneumonia

192.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- Community-acquired pneumonia is an acute infection of the lung parenchyma acquired outside of a hospital setting
- Common pathogens include *Streptococcus pneumoniae* (most common typical bacterial cause), *Haemophilus influenzae*, atypical organisms (*Mycoplasma pneumoniae*, *Chlamydophila pneumoniae*, *Legionella pneumophila*), and respiratory viruses.

192.2 Causes

This condition happens because of microbial infection of the alveolar spaces leading to consolidation. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

192.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

192.4 Signs and Symptoms

You may experience fever, productive cough, shortness of breath, pleuritic chest pain, and malaise. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

192.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Doctors diagnose this based on chest X-ray showing consolidation
- The CURB-65 score assesses severity and guides disposition. Your outlook depends on severity and patient factors.

192.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

192.7 Diagnosis

Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

192.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)

- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

192.9 Treatment Options

Treatment options include empiric antibiotics: amoxicillin or macrolides for outpatient mild disease

- Respiratory fluoroquinolones or beta-lactam plus macrolide for inpatient disease
- Severe disease requires ICU admission and combination therapy.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

192.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

192.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 193

Compartment Syndrome

193.1 Overview

Compartment syndrome is elevated pressure within a closed fascial compartment, compromising tissue perfusion and leading to reduced blood flow and tissue death of muscle and nerves. Acute compartment syndrome is a surgical emergency most commonly caused by fractures (tibial fractures most common), crush injuries, reperfusion injury, or tight casts. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

193.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

193.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

193.4 Signs and Symptoms

- You may experience pain out of proportion to the injury, worsened by passive stretching of the affected muscles, with tense, swollen compartments
- The five P's (pulselessness, tingling or numbness, paralysis, pallor, poikilothermia) are late signs.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

193.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

193.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

193.7 Diagnosis

Doctors diagnose this based on clinical examination and compartment pressure measurement (pressures >30 mmHg or within 30 mmHg of diastolic pressure indicate need for fasciotomy).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

193.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

193.9 Treatment Options

- Treatment typically involves emergent fasciotomy to decompress all affected compartments
- Delayed treatment leads to irreversible muscle tissue death, contracture (Volkmann's ischemic contracture), and potentially limb loss.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

193.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

193.11 Outlook

Most people recover well with early fasciotomy but poor if Treatment typically involves delayed.

Chapter 194

Complex Regional Pain Syndrome

194.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Complex regional pain syndrome (CRPS) is a painful condition characterized by disproportionate pain, sensory changes, vasomotor dysfunction, sudomotor dysfunction, and trophic changes in a limb. CRPS Type I occurs without a definable nerve lesion, typically after trauma or immobilization
- CRPS Type II occurs with a definable major nerve injury. The incidence is 5–26 per 100,000 person-years, with a female-to-male ratio of 3–4:1.

194.2 Causes

This condition happens because of multifactorial Disease mechanism including peripheral and central sensitization, sympathetic nervous system dysregulation, neurogenic inflammation, and altered somatosensory processing. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

194.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

194.4 Signs and Symptoms

Clinical features are organized into four domains: sensory (allodynia, hyperalgesia), vasomotor (temperature asymmetry, skin color changes), sudomotor/swelling, and motor/trophic (decreased range of motion, weakness, tremor, dystonia).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

194.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

194.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

194.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam using the Budapest diagnostic criteria.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

194.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

194.9 Treatment Options

Treatment typically involves most effective when initiated early and includes a multidisciplinary approach with physical therapy, pharmacotherapy (gabapentinoids, TCAs, bisphosphonates), and interventional procedures (sympathetic nerve blocks, spinal cord stimulation). Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

194.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

194.11 Outlook

- The outlook varies from person to person
- Early treatment improves outcomes but up to 20% of patients have persistent disability.

Chapter 195

Complex Sleep Apnea

195.1 Overview

Complex sleep apnea (treatment-emergent central sleep apnea) is a sleep-related breathing disorder defined by the emergence or persistence of CSA during PAP therapy in patients initially diagnosed with OSA. It occurs in 5–15% of patients initiating CPAP therapy, particularly those with cardiovascular disease, opioid use, or high loop gain physiology. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

195.2 Causes

This condition happens because of PAP-induced elimination of the obstructive component, unmasking a latent central ventilatory instability. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

195.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

195.4 Signs and Symptoms

You may experience persistent respiratory events despite adequate PAP therapy for OSA. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

195.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

195.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

195.7 Diagnosis

To make the diagnosis, doctors look for polysomnography on PAP demonstrating central events. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

195.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

195.9 Treatment Options

- Treatment focuses on continued PAP adherence, as many cases resolve spontaneously within 3–6 months
- Persistent cases may require switching to adaptive servo-ventilation or bilevel PAP with a backup rate.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

195.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

195.11 Outlook

In most cases, the outlook is favorable. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 196

Concussion

196.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Concussion is a mild traumatic brain injury caused by a direct blow to the head or a biomechanical force transmitted to the brain, resulting in a transient disturbance of neurological function
- It is the most common form of traumatic brain injury.

196.2 Causes

This condition happens because of a complex neurometabolic cascade involving ionic flux, neurotransmitter release, impaired cerebral blood flow regulation, and axonal injury, without macrostructural damage visible on conventional neuroimaging. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

196.3 Risk Factors

- Doctors usually diagnose this based on your symptoms and a physical exam
- CT head is indicated to exclude intracranial bleeding in patients with risk factors (age greater than 65, anticoagulant use, focal neurological signs, persistent vomiting, worsening headache).

- Advanced age
- Family history and genetic predisposition
- Head trauma or brain injury
- Vascular risk factors (hypertension, diabetes)
- Exposure to environmental toxins
- Chronic stress
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

196.4 Signs and Symptoms

- You may experience headache, dizziness, confusion, amnesia, visual disturbances, sensitivity to light and noise, difficulty concentrating, and emotional lability
- Loss of consciousness occurs in fewer than 10% of cases.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

196.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

196.6 Complications

- The outlook is generally positive, although post-concussion syndrome (persistent symptoms beyond 4 weeks) occurs in 10–30% of patients
- Second impact syndrome is a rare but potentially fatal complication.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

196.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

196.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

196.9 Treatment Options

Treatment focuses on physical and thinking and memory rest for 24–48 hours, followed by a graduated return-to-activity protocol. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

196.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

196.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 197

Congenital Cataract

197.1 Overview

Congenital cataract is an opacification of the lens present at or shortly after birth, affecting 1–6 per 10,000 live births. The condition may be idiopathic, hereditary (autosomal dominant most common), or associated with systemic conditions (Down syndrome, congenital rubella, galactosemia, Lowe syndrome, persistent fetal vasculature). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

197.2 Causes

This condition happens because of abnormal lens development or metabolic insult in utero. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

197.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

197.4 Signs and Symptoms

You may experience lens opacity present at birth or in early childhood.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

197.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

197.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

197.7 Diagnosis

Doctors diagnose this using slit-lamp examination.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

197.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

197.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

197.10 Living with It

Treatment focuses on urgent surgical removal, ideally within the first few weeks of life for unilateral cases, followed by postoperative visual rehabilitation with patching and corrective lenses. Your outlook depends on timing of intervention and compliance with visual rehabilitation.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 198

Congenital Diaphragmatic Hernia

198.1 Overview

Congenital diaphragmatic hernia (CDH) is the herniation of abdominal contents into the thoracic cavity through a defect in the diaphragm, most commonly the left posterolateral foramen of Bochdalek. Incidence is 1 per 2500 live births. CDH causes lung underdevelopment and lung high blood pressure, which determine outlook. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

198.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

198.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

198.4 Signs and Symptoms

Clinical features at birth include respiratory distress, scaphoid abdomen, and decreased breath sounds on the affected side.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

198.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

198.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

198.7 Diagnosis

Your doctor confirms the diagnosis with a nasogastric tube in the chest on radiograph.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

198.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

198.9 Treatment Options

- Treatment options include immediate endotracheal intubation (avoid bag-mask ventilation), nasogastric decompression, and gentle ventilation strategies
- Surgical repair is performed after physiologic stabilization. Overall survival is approximately 70–80%.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

198.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 199

Congenital Heart Disease (Overview)

199.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Congenital heart disease (CHD) is the most common congenital anomaly, affecting approximately 1% of live births. CHD encompasses a wide spectrum of structural cardiac defects (see Chapter 12, Diseases of the Heart for comprehensive coverage). Septal defects (ASD, VSD, AVSD) allow left-to-right shunting, leading to lung overcirculation and, over time, lung high blood pressure and Eisenmenger syndrome
- Small defects may close spontaneously, while larger defects require surgical or transcatheter closure. Tetralogy of Fallot is the most common cyanotic CHD, consisting of VSD, overriding aorta, right ventricular outflow tract obstruction, and right ventricular enlargement
- Surgical repair is performed in infancy. Transposition of the great arteries creates parallel circulations incompatible with life without mixing
- The arterial switch operation is performed in the first week of life. Coarctation of the aorta presents with high blood pressure in the upper extremities and diminished femoral pulses
- Treatment typically involves surgical removal or balloon angioplasty. Hypoplastic left heart syndrome requires staged surgical palliation (Norwood, Glenn, Fontan procedures) or heart transplantation.

199.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

199.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

199.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

199.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

199.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

199.7 Diagnosis

History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

199.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

199.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

199.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

199.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 200

Congenital Hip Dysplasia

200.1 Overview

Developmental dysplasia of the hip (DDH) encompasses a spectrum from mild acetabular dysplasia to frank dislocation of the femoral head from the acetabulum. Incidence is 1–3 per 1000 live births. Clinical screening includes the Ortolani and Barlow maneuvers. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

200.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

200.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

200.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

200.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

200.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

200.7 Diagnosis

- Your doctor confirms the diagnosis with ultrasound (for infants <6 months) or radiography (for older infants). Treatment for infants under 6 months is a Pavlik harness to maintain hip flexion and abduction
- For older infants or failed Pavlik treatment, closed or open reduction with a spica cast is required.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

200.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

200.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

200.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

200.11 Outlook

- Most people recover well with early treatment
- Untreated DDH leads to osteoarthritis and limb-length discrepancy.

Chapter 201

Congenital Talipes Equinovarus (Clubfoot)

201.1 Overview

Congenital talipes equinovarus (clubfoot) is a congenital deformity of the foot characterized by four components: equinus, varus, adductus, and cavus. Incidence is 1 per 1000 live births, more common in males (2:1), and 50% are bilateral. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

201.2 Causes

The cause is multifactorial. Diagnosis is primarily clinical at birth. The Ponseti method is the standard of care: serial manipulation and long-leg casting for 6–8 weeks, followed by through the skin Achilles tenotomy, and then a foot abduction brace. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

201.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

201.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

201.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

201.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

201.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

201.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

201.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

201.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

201.11 Outlook

- Most people recover well with adherence to bracing
- Surgical release is reserved for resistant or neglected cases.

Chapter 202

Conjunctivitis

202.1 Overview

Viral conjunctivitis, most commonly caused by adenovirus, results from viral infection of the conjunctiva. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

202.2 Causes

Bacterial conjunctivitis results from bacterial infection (S. aureus, S. pneumoniae, H. influenzae, N. gonorrhoeae, C. trachomatis). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

202.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

202.4 Signs and Symptoms

Symptoms of viral conjunctivitis include acute onset, watery discharge, and preauricular lymphadenopathy

- Bacterial conjunctivitis presents with purulent discharge and eyelids matted shut
- Allergic conjunctivitis features bilateral itching, tearing, and stringy mucus.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

202.5 Progression and Stages

Conjunctivitis is an inflammation of the conjunctiva and the most common eye disease worldwide, classified as infectious (viral, bacterial), allergic, or chemical or irritant. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

202.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

202.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with Gram stain and culture used for hyperacute or bacterial cases. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

202.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

202.9 Treatment Options

- Treatment typically involves etiologic: supportive for viral, topical antibiotics for bacterial, and topical antihistamine or mast cell stabilizers for allergic
- Gonococcal conjunctivitis requires intramuscular ceftriaxone.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

202.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

202.11 Outlook

- Most people recover well with appropriate treatment
- Untreated gonococcal conjunctivitis can lead to corneal perforation within 24 hours.

Chapter 203

Constrictive Pericarditis

203.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Constrictive pericarditis is a condition in which the pericardium becomes thickened, fibrotic, and calcified, restricting diastolic filling of the heart
- Common causes include prior cardiac surgery, radiation therapy, tuberculosis, idiopathic pericarditis, and autoimmune disease.

203.2 Causes

This condition happens because of chronic inflammation leading to pericardial scarring and restriction. The Disease mechanism mimics restrictive cardiomyopathy but with ventricular interdependence and exaggerated respiratory variation in filling. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

203.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

203.4 Signs and Symptoms

You may experience signs of right heart failure (swelling, ascites, hepatomegaly, elevated jugular venous pressure with Kussmaul's sign). Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

203.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

203.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

203.7 Diagnosis

Diagnosis typically involves echocardiography, cardiac MRI (pericardial thickening and late enhancement), and cardiac catheterization (equalization of diastolic pressures). Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

203.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

203.9 Treatment Options

Treatment typically involves pericardiectomy, which is curative but carries significant surgical morbidity. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

203.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

203.11 Outlook

Most people recover well after successful pericardiectomy. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 204

Contact Dermatitis

204.1 Overview

Contact dermatitis is inflammation of the skin caused by direct contact with an irritant or allergen. Irritant contact dermatitis (ICD) accounts for 80% of cases and is caused by direct chemical damage to the skin (soaps, detergents, solvents, acids). Allergic contact dermatitis (ACD) is a type IV delayed hypersensitivity reaction requiring prior sensitization to the allergen. Common allergens include nickel, poison ivy/oak/urushiol, fragrances, preservatives, rubber chemicals, and hair dyes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

204.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

204.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

204.4 Signs and Symptoms

You may experience pruritic, erythematous, vesicular or bullous dermatitis in the distribution of contact. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

204.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

204.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

204.7 Diagnosis

Doctors diagnose this based on history and patch testing for ACD. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

204.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

204.9 Treatment Options

Treatment options include identification and avoidance of the causative agent, topical corticosteroids, and in severe cases, systemic corticosteroids. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

204.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

204.11 Outlook

Most people recover well with avoidance of the offending agent. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 205

Copper Deficiency and Excess

205.1 Overview

Copper deficiency is a rare mineral deficiency disorder that occurs after gastric bypass surgery, in malabsorption, prolonged parenteral nutrition, excessive zinc intake (zinc induces metallothionein which blocks intestinal copper absorption), and in Menkes disease. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

205.2 Causes

This condition happens because of deficiency of copper, an essential trace element serving as a cofactor for cytochrome c oxidase, superoxide dismutase, lysyl oxidase, dopamine β-hydroxylase, and ceruloplasmin. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

205.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

205.4 Signs and Symptoms

You may experience microcytic hypochromic anemia (refractory to iron therapy), neutropenia, myeloneuropathy (spastic gait, sensory loss of coordination, peripheral neuropathy mimicking B12 deficiency), and pancytopenia.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

205.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

205.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

205.7 Diagnosis

Doctors diagnose this using low serum copper and low ceruloplasmin.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

205.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

205.9 Treatment Options

- Treatment focuses on oral copper 2–5 mg/kg/day as copper sulfate. Menkes disease (kinky hair disease) is an X-linked recessive disorder of ATP7A causing impaired intestinal copper absorption, presenting in infancy with progressive neurodegeneration, seizures, hypotonia, failure to thrive, characteristic brittle, depigmented, kinky hair (pili torti), and tortuous cerebral arteries
- Subcutaneous copper-histidine may improve outcomes if started early, but Unfortunately, the outlook is often poor. Wilson disease (copper excess) is covered in Chapter 8.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

205.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 206

COVID-19

206.1 Overview

Coronavirus disease 2019 (COVID-19) is an infectious disease caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), first identified in Wuhan, China in December 2019 and declared a pandemic by the WHO in March 2020. SARS-CoV-2 is a betacoronavirus that uses the angiotensin-converting enzyme 2 receptor for cell entry. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

206.2 Causes

This condition happens because of viral infection causing a spectrum of illness from asymptomatic infection to severe pneumonia with acute respiratory distress syndrome, multi-organ failure, and death. Transmission occurs primarily through respiratory droplets and aerosols, with an incubation period of 2–14 days. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

206.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

206.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

206.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

206.6 Complications

You may experience fever, cough, fatigue, muscle pain, loss of smell or ageusia, headache, and shortness of breath

- Severe disease features hypoxemia, bilateral lung infiltrates, lymphopenia, elevated inflammatory markers, and cytokine storm
- Complications include ARDS, thromboembolism, myocarditis, acute kidney injury, neurological manifestations, and multisystem inflammatory syndrome in children.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

206.7 Diagnosis

Doctors diagnose this using RT-PCR testing or rapid antigen tests. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

206.8 Prevention

- Your outlook depends on age, comorbidities, and disease severity
- Vaccination has significantly reduced severe disease and death.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

206.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

206.10 Living with It

- The right treatment depends on severity: supportive care for mild disease
- Dexamethasone, remdesivir, baricitinib, tocilizumab, and anticoagulation for hospitalized patients
- Long COVID is marked by persistent symptoms lasting weeks to months after acute infection.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

206.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 207

Creutzfeldt-Jakob Disease (CJD)

207.1 Overview

Creutzfeldt-Jakob disease is a rapidly progressive, fatal neurodegenerative disease caused by prions—misfolded, protease-resistant isoforms (PrP^{Sc}) of the normal cellular prion protein (PrP^{C}), with an incidence of 1–2 per million annually worldwide, median age at onset 60–65 years, with sporadic CJD (sCJD) accounting for 85–90% of cases, familial CJD (10%) linked to PRNP mutations, and iatrogenic CJD (<1%) from contaminated surgical instruments. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

207.2 Causes

This condition happens because of PrP^{Sc} converting native PrP^{C} into the misfolded conformation through a template-directed process, leading to exponential accumulation of PrP^{Sc} aggregates, causing vacuolation (spongiform change), neuronal loss, and astrocytosis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

207.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

207.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

207.5 Progression and Stages

You may experience rapidly progressive dementia (weeks to months), myoclonus (startle-sensitive, characteristic), cerebellar loss of coordination, pyramidal and extrapyramidal signs, visual disturbances, and akinetic mutism in the terminal stage, with death typically occurring within 1 year of symptom onset. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

207.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

207.7 Diagnosis

Doctors diagnose this using clinical criteria, EEG showing periodic sharp wave complexes (PSWCs), CSF RT-QuIC (specific and sensitive), and MRI showing T2/FLAIR hyperintensity in the basal ganglia and thalamus. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

207.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

207.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

207.10 Living with It

Treatment typically involves palliative.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

207.11 Outlook

outlook is uniformly fatal. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 208

Cri-du-Chat Syndrome

208.1 Overview

Cri-du-chat syndrome is a chromosomal disorder caused by deletion of the short arm of chromosome 5 (5p-), with an incidence of 1 in 15,000–50,000 live births. The name derives from the characteristic high-pitched, cat-like cry in infancy due to laryngeal underdevelopment. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

208.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

208.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

208.4 Signs and Symptoms

You may experience microcephaly, round face, widely spaced eyes, epicanthal folds, low-set ears, hypotonia, severe intellectual disability, and congenital heart disease. Most cases are de novo deletions.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

208.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

208.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

208.7 Diagnosis

Doctors diagnose this using karyotype or chromosomal microarray.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

208.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

208.9 Treatment Options

Treatment typically involves supportive with early intervention therapies. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

208.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 209

Crohn's Disease

209.1 Overview

Crohn's disease (CD) is a chronic, relapsing inflammatory bowel disease that can affect any part of the digestive tract tract from mouth to anus, though the terminal ileum and colon are most commonly involved. Peak incidence occurs in the second to fourth decades of life, with a bimodal distribution. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

209.2 Causes

This condition happens because of an interplay of genetic susceptibility (over 240 identified loci, with NOD2/CARD15 being the most significant), environmental factors (smoking doubles the risk), and dysregulated immune response, characterized by transmural inflammation, skip lesions, and non-caseating granulomas. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

209.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease

- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

209.4 Signs and Symptoms

- You may experience chronic diarrhea (often non-bloody), abdominal pain (typically right lower quadrant), weight loss, fatigue, and perianal disease (fistulas, skin tags, abscesses)
- Extraintestinal manifestations include arthritis, redness of the skin nodosum, pyoderma gangrenosum, uveitis, and primary sclerosing cholangitis.
- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

209.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

209.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

209.7 Diagnosis

Diagnosis typically involves colonoscopy with biopsies, MR enterography or CT enterography, and serological markers (CRP, fecal calprotectin). Management follows a step-up or top-down approach: 5-aminosalicylates for mild colonic disease, corticosteroids for moderate flares (not for maintenance), immunomodulators (azathioprine, methotrexate), and biologic therapies (anti-TNF [infliximab, adalimumab], anti-integrin [vedolizumab], anti-IL12/23 [ustekinumab], anti-IL23 [risankizumab]). Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

209.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

209.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

209.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

209.11 Outlook

- The outlook varies from person to person
- Surgery is required in approximately 50% of patients within 10 years but is not curative.

Chapter 210

Croup (Laryngotracheobronchitis)

210.1 Overview

Croup is a viral infection of the upper airway causing subglottic swelling and obstruction, most commonly caused by parainfluenza virus types 1 and 3. It predominantly affects children 6 months to 3 years of age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

210.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

210.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

210.4 Signs and Symptoms

You may experience abrupt onset of a barking “seal-like” cough, inspiratory stridor, hoarseness, and respiratory distress, often worse at night. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

210.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

210.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

210.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Neck radiograph may show the “steeple sign” (subglottic narrowing).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

210.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

210.9 Treatment Options

Treatment options include a single dose of oral or intramuscular dexamethasone (0.6 mg/kg) for mild to moderate croup and nebulized epinephrine for moderate to severe croup. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

210.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

210.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 211

Cryoglobulinemic Vasculitis

211.1 Overview

Mixed cryoglobulinemia (types II and III) represents 80% of cases and is predominantly HCV-associated. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

211.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

211.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

211.4 Signs and Symptoms

You may experience palpable purple spots on the skin (the most common feature, recurrent, involving lower extremities), arthralgias, peripheral neuropathy (symmetric, often painful), membranoproliferative glomerulonephritis (proteinuria, hematuria, high blood pressure, progressive kidney dysfunction), and less commonly, Raynaud's phenomenon, skin ulcers, and digital gangrene. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

211.5 Progression and Stages

Cryoglobulinemic vasculitis is a small-vessel vasculitis caused by immune complexes of cryoglobulins (immunoglobulins that precipitate at temperatures <37°C and dissolve on rewarming) deposited in vessel walls, classified into three types: type I (monoclonal, associated with lymphoproliferative disorders), type II (mixed monoclonal and polyclonal, most commonly associated with hepatitis C virus—HCV), and type III (polyclonal, associated with autoimmune diseases such as SLE or Sjögren's syndrome). Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

211.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use

- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

211.7 Diagnosis

To make the diagnosis, doctors look for detection of serum cryoglobulins (blood collected and transported at 37°C), low complement levels (C4 markedly decreased), positive rheumatoid factor, and evidence of HCV infection. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

211.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

211.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

211.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

211.11 Outlook

Your outlook depends on the underlying cause and organ involvement. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 212

Cryptococcosis

212.1 Overview

Cryptococcosis is a systemic fungal infection caused by *Cryptococcus neoformans* (primarily in immunocompromised) and *Cryptococcus gattii* (emerging, affects immunocompetent individuals), with 250,000 cases annually worldwide and 180,000 deaths, mostly in HIV/AIDS (CD4 < 100), and cryptococcal meningitis being the leading cause of meningitis in HIV-infected adults in sub-Saharan Africa. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

212.2 Causes

This condition happens because of inhalation of basidiospores, with the polysaccharide capsule being the major virulence factor, inhibiting phagocytosis and suppressing T cell responses. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

212.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

212.4 Signs and Symptoms

You may experience cryptococcal meningitis: subacute headache, fever, nausea, and altered mental status, with elevated intracranial pressure being a major cause of morbidity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

212.5 Progression and Stages

The outlook has improved with modern antifungal therapy but remains poor in advanced HIV/AIDS. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

212.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

212.7 Diagnosis

Doctors diagnose this using serum cryptococcal antigen (CrAg) and CSF analysis (India ink stain, CrAg). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

212.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

212.9 Treatment Options

Treatment options include induction therapy with liposomal amphotericin B plus flucytosine, followed by consolidation and maintenance with fluconazole. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

212.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

212.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 213

Cryptorchidism (Undescended Testis)

213.1 Overview

Cryptorchidism is the absence of one or both testes from the scrotum, occurring in 3% of full-term and 30% of preterm male newborns. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

213.2 Causes

This condition happens because of failure of testicular descent. Spontaneous descent occurs in most cases within the first 3–6 months of life. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

213.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

213.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

213.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

213.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

213.7 Diagnosis

- Doctors diagnose this using physical examination
- Ultrasound or laparoscopy may be used for nonpalpable testes.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

213.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

213.9 Treatment Options

- Treatment typically involves surgical orchiopexy, optimally performed between 6–18 months of age, to preserve fertility, reduce the risk of testicular torsion, and allow for scrotal examination. Undescended testes are associated with impaired spermatogenesis and increased risk of testicular malignancy
- Early orchiopexy does not eliminate the cancer risk but facilitates self-examination.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

213.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

213.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 214

Cushing Syndrome

214.1 Overview

Cushing syndrome is a clinical state resulting from prolonged exposure to excess systemic glucocorticoids. Exogenous administration of glucocorticoids is the most common cause. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

214.2 Causes

Endogenous Cushing syndrome results from Cushing disease (ACTH-secreting pituitary adenoma, 70% of endogenous cases), ectopic ACTH secretion (eg., small cell lung carcinoma), or primary adrenal adenoma/carcinoma. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

214.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

214.4 Signs and Symptoms

You may experience central obesity, moon facies, buffalo hump, abdominal purple striae, proximal muscle weakness, skin fragility, high blood pressure, hyperglycemia, and osteopenia.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

214.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

214.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

214.7 Diagnosis

Diagnosis typically involves initial screening with 24-hour urinary free cortisol, late-night salivary cortisol, or 1-mg overnight dexamethasone suppression test, followed by plasma ACTH measurement to differentiate ACTH-dependent from ACTH-independent causes.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

214.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

214.9 Treatment Options

Treatment typically involves targeted: surgical removal of pituitary or adrenal tumors, discontinuation of exogenous steroids, or medical therapy (ketoconazole, metyrapone). Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

214.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

214.11 Outlook

The outlook is good following successful surgical cure.

Chapter 215

Cushing's Syndrome

215.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Cushing's syndrome results from chronic exposure to excess cortisol, regardless of the cause, with exogenous glucocorticoid use (iatrogenic) being the most common cause overall. Endogenous causes are divided into ACTH-dependent (70–80%): Cushing's disease (pituitary adenoma secreting ACTH, 70%) and ectopic ACTH syndrome (small cell lung cancer, bronchial carcinoid)
- And ACTH-independent (20–30%): adrenal adenoma, adrenal carcinoma, and bilateral adrenal abnormal increase in cells.

215.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

215.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

215.4 Signs and Symptoms

You may experience central (truncal) obesity, moon facies, dorsocervical fat pad ("buffalo hump"), supraclavicular fat pads, thin extremities, purple striae (>1 cm wide), easy bruising, proximal muscle weakness, high blood pressure, hyperglycemia/diabetes, osteoporosis, menstrual irregularity, hirsutism, and psychiatric disturbances.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

215.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

215.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

215.7 Diagnosis

Diagnosis typically involves confirmation of hypercortisolism (24-hour urinary free cortisol, late-night salivary cortisol, or 1 mg overnight dexamethasone suppression test) followed by determination of the cause (plasma ACTH, high-dose dexamethasone suppression test, CRH stimulation test, MRI pituitary, CT adrenals, inferior petrosal sinus sampling).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

215.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

215.9 Treatment Options

The right treatment depends on the cause: transsphenoidal surgery for Cushing's disease, surgical removal of adrenal tumors, chemotherapy for adrenal carcinoma, and ketoconazole or metyrapone for medical management. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

215.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

215.11 Outlook

The outlook is generally positive with successful treatment.

Chapter 216

Cyanide Toxicity

216.1 Overview

Cyanide is a rapidly acting poison from industrial sources, smoke inhalation, and certain plants. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

216.2 Causes

This condition happens because of cyanide binding to cytochrome c oxidase (complex IV of the mitochondrial electron transport chain), inhibiting cellular respiration and causing histotoxic hypoxia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

216.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

216.4 Signs and Symptoms

- Clinical features: rapid onset (seconds to minutes) of altered mental status, seizures, coma, trismus, hyperventilation, hypotension, bradycardia, ventricular arrhythmias, and cardiovascular collapse
- Classic sign is bright red venous blood.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

216.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

216.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

216.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam, especially in smoke inhalation victims
- Elevated lactate suggests cyanide toxicity. Management: hydroxocobalamin (5 g IV) is first-line, binding cyanide to form nontoxic cyanocobalamin
- Sodium thiosulfate provides substrate for rhodanese to convert cyanide to thiocyanate.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

216.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

216.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

216.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

216.11 Outlook

The outlook is generally positive with rapid treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 217

Cyclospora cayetanensis

217.1 Overview

Cyclosporiasis is a diarrheal illness caused by *Cyclospora cayetanensis*, a coccidian protozoan parasite, with outbreaks associated with imported fresh produce from endemic regions (Central and South America, the Caribbean, Southeast Asia). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

217.2 Causes

This condition happens because of ingested sporulated oocysts releasing sporozoites in the small intestine that invade enterocytes, causing villous blunting, crypt abnormal increase in cells, and inflammation of the duodenum and jejunum. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

217.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

217.4 Signs and Symptoms

You may experience an incubation of 1–14 days, with prolonged, relapsing watery diarrhea (explosive, non-bloody), nausea, anorexia, weight loss, abdominal cramps, and profound fatigue, with average illness duration of 5–7 weeks if untreated. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

217.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

217.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

217.7 Diagnosis

Doctors diagnose this using modified Ziehl-Neelsen acid-fast stain of stool or PCR. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

217.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

217.9 Treatment Options

Treatment options include trimethoprim-sulfamethoxazole (TMP-SMX). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

217.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

217.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 218

Cyclothymic Disorder

218.1 Overview

Cyclothymic disorder has a lifetime prevalence of 0.5–1%, characterized by at least 2 years (1 year in children and adolescents) of numerous periods of hypomanic symptoms that do not meet full criteria for a hypomanic episode and numerous periods of depressive symptoms that do not meet full criteria for a major depressive episode, with symptoms present at least half the time. The condition is often described as a temperamental or personality disposition with mood instability. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

218.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

218.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

218.4 Signs and Symptoms

You may experience chronic mood instability without meeting full criteria for hypomanic or depressive episodes. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

218.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

218.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

218.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

218.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

218.9 Treatment Options

Treatment options include mood stabilizers (lithium, lamotrigine) and psychotherapy (IPSRT, CBT). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

218.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

218.11 Outlook

The outlook varies from person to person, with up to 30% of patients eventually developing Bipolar I or Bipolar II disorder. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 219

Cystic Fibrosis

219.1 Overview

Cystic scarring is an autosomal recessive disorder caused by mutations in the CFTR gene on chromosome 7, affecting approximately 1 in 2,500 Caucasian individuals. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

219.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

219.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

219.4 Signs and Symptoms

You may experience chronic cough, recurrent infections, and growth failure.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

219.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

219.6 Complications

This condition happens because of a defective chloride channel leading to thick, viscous secretions in multiple organs. lung manifestations are the primary cause of morbidity and mortality, including chronic obstructive lung disease with bronchiectasis, chronic *Pseudomonas aeruginosa* colonization, and progressive respiratory failure

- Pancreatic insufficiency causes malabsorption and failure to thrive
- Other complications include CF-related diabetes, liver disease, sinusitis, and male infertility.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

219.7 Diagnosis

Doctors diagnose this using newborn screening, sweat chloride testing (greater than 60 mmol/L), and CFTR genotyping.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

219.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

219.9 Treatment Options

Treatment options include CFTR modulators that target the underlying protein defect, airway clearance, inhaled mucolytics, and antimicrobials. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

219.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

219.11 Outlook

The outlook has improved dramatically, with median survival of approximately 50 years in developed countries.

Chapter 220

Cystitis

220.1 Overview

Cystitis (lower urinary tract infection) is one of the most common bacterial infections, particularly in women (50–60% of women experience at least one episode). *Escherichia coli* is the causative organism in 80–90% of cases. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

220.2 Causes

This condition happens because of bacterial colonization and inflammation of the bladder mucosa. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

220.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

220.4 Signs and Symptoms

You may experience dysuria, urinary frequency, urgency, suprapubic pain, and hematuria. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

220.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

220.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

220.7 Diagnosis

Doctors diagnose this based on urinalysis and urine culture. Treatment for uncomplicated cystitis in non-pregnant women includes nitrofurantoin, trimethoprim-sulfamethoxazole, or fosfomycin

- Recurrent cystitis may require prophylactic antibiotics, behavioral modifications, and cranberry products
- Complicated cystitis (in males, pregnant women, or with structural abnormalities) requires broader-spectrum antibiotics and evaluation for underlying causes.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

220.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

220.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

220.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

220.11 Outlook

Most people recover well with appropriate antibiotic therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 221

Cytomegalovirus (CMV)

221.1 Overview

Cytomegalovirus is a beta-herpesvirus that infects 60–90% of the adult population worldwide, generally asymptomatic in immunocompetent hosts but causes significant disease in immunocompromised patients and congenitally infected neonates, and is the most common congenital infection (0.5–1% of live births). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

221.2 Causes

This condition happens because of CMV infecting epithelial cells, endothelial cells, fibroblasts, and monocytes, with primary infection followed by lifelong latency and reactivation occurring with immunosuppression. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

221.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

221.4 Signs and Symptoms

- Your symptoms depend on the host: immunocompetent host presents with mononucleosis-like syndrome
- Immunocompromised host (HIV with CD4 < 50, transplant recipients) presents with CMV retinitis, CMV pneumonitis, CMV colitis, and CMV esophagitis
- Congenital CMV presents with sensorineural hearing loss (most common sequela), microcephaly, intracranial calcifications, chorioretinitis, and hepatosplenomegaly.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

221.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

221.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

221.7 Diagnosis

Doctors diagnose this using PCR (quantitative) of blood, urine, CSF, or BAL fluid. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

221.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

221.9 Treatment Options

Treatment options include ganciclovir or valganciclovir for immunocompromised patients and congenital CMV. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

221.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

221.11 Outlook

Your outlook depends on the host immune status. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 222

Deep Vein Thrombosis

222.1 Overview

This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

222.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

222.3 Risk Factors

Risk factors are described by Virchow's triad: venous stasis (immobilization, prolonged travel, paralysis, heart failure), endothelial injury (surgery, trauma, indwelling catheters), and hypercoagulability (malignancy, oral contraceptives, pregnancy, inherited thrombophilia [Factor V Leiden, prothrombin gene mutation], antiphospholipid syndrome).

- Advanced age
- Cigarette smoking
- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease
- Obesity and physical inactivity
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol

- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

222.4 Signs and Symptoms

You may experience unilateral leg swelling, pain (often described as a dull ache or cramping), warmth, redness of the skin, and palpable cord along the affected vein, though many DVTs are asymptomatic. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

222.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

222.6 Complications

The outlook is good with treatment, though complications include post-thrombotic syndrome (chronic venous insufficiency in 20–50% of patients) and chronic thromboembolic lung high blood pressure.

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia

- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

222.7 Diagnosis

Diagnosis typically involves clinical probability assessment (Wells score), D-dimer testing (high negative predictive value), and compression ultrasonography (first-line imaging, showing non-compressibility of the affected vein). Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

222.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

222.9 Treatment Options

Treatment typically involves anticoagulation: initial therapy with low-molecular-weight heparin (enoxaparin), fondaparinux, or direct oral anticoagulants (DOACs—rivaroxaban, apixaban), followed by long-term anticoagulation for 3–6 months (first unprovoked DVT) or longer depending on risk of recurrence. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

222.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

222.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 223

Delayed Sleep-Wake Phase Disorder

223.1 Overview

Prevalence is 1–3% in adults and 7–16% in adolescents, making it the most common circadian rhythm disorder. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

223.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

223.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

223.4 Signs and Symptoms

You may experience severe difficulty falling asleep at conventional times but normal sleep once initiated when allowed to follow the endogenous schedule. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

223.5 Progression and Stages

Delayed sleep-wake phase disorder (DSWPD) is a circadian rhythm sleep-wake disorder characterized by a stable delay of the sleep-wake phase relative to the desired or required schedule, with sleep onset typically after 1:00–3:00 AM and wake times in the late morning or early afternoon. This condition happens because of an abnormally long intrinsic circadian period, reduced phase-advance capacity, or a hypo-responsive phase-advance mechanism to morning light. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

223.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

223.7 Diagnosis

Doctors diagnose this using clinical history, sleep diaries, actigraphy, and dim-light melatonin onset measurement. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

223.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

223.9 Treatment Options

- Treatment options include timed bright light therapy in the morning, low-dose melatonin (0.5–3 mg) administered 5–6 hours before target bedtime, and consistent sleep-wake scheduling
- Chronotherapy may be considered in refractory cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

223.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

223.11 Outlook

The outlook is generally positive with treatment compliance. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 224

Delirium

224.1 Overview

Delirium is an acute, fluctuating disturbance in attention, awareness, and thinking, caused by an underlying medical condition, intoxication, or medication side effect. It is a medical emergency affecting 10–30% of hospitalized older adults and up to 50% of postoperative patients. Delirium is linked to increased mortality, longer hospital stays, functional decline, and incident dementia. Three subtypes exist: hyperactive (agitation, restlessness), hypoactive (withdrawal, lethargy, reduced speech—often missed or misdiagnosed as depression), and mixed. The Confusion Assessment Method (CAM) is the most widely used diagnostic tool: acute onset and fluctuating course, plus inattention, plus either disorganized thinking or altered level of consciousness. Precipitating factors include infection, electrolyte disturbance, surgery, psychoactive medications, pain, sleep deprivation, and urinary retention. Pharmacologic treatment (haloperidol or atypical antipsychotics) is reserved for severe agitation that threatens safety. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

224.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

224.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

224.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

224.5 Progression and Stages

Predisposing factors include advanced age, dementia, sensory impairment, polypharmacy, and multiple comorbidities. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

224.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

224.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

224.8 Prevention

Prevention is paramount (the Hospital Elder Life Program—HELP).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

224.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

224.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

224.11 Outlook

- The outlook varies from person to person
- Delirium is linked to increased mortality and functional decline.

Chapter 225

Delusional Disorder

225.1 Overview

Delusional disorder has a lifetime prevalence of 0.2%, with a later age of onset (typically 40–60 years) compared to schizophrenia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

225.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

225.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

225.4 Signs and Symptoms

You may experience one or more delusions that persist for at least 1 month, without meeting full criteria for schizophrenia (hallucinations, if present, are not prominent and are related to the delusional theme

- Negative symptoms are absent
- Functioning is not markedly impaired). Common subtypes include persecutory (most common), jealous, erotomanic (de Clérambault syndrome), grandiose, and somatic.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

225.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

225.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

225.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

225.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

225.9 Treatment Options

Treatment options include antipsychotics (low-dose). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

225.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

225.11 Outlook

outlook is better overall than schizophrenia. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 226

Dementia with Lewy Bodies

226.1 Overview

Dementia with Lewy bodies (DLB) is a progressive neurodegenerative disorder accounting for 5–15% of dementia cases. Core You may experience fluctuating thinking with pronounced variations in attention and alertness, recurrent well-formed visual hallucinations (often detailed, involving people or animals), and parkinsonism (bradykinesia, rigidity, gait disorder). REM sleep behavior disorder (RBD) is a supportive feature, often predating dementia onset by years. In the geriatric population, DLB patients are extremely sensitive to neuroleptics (neuroleptic sensitivity reaction—severe parkinsonism, sedation, autonomic instability, increased mortality), which should be avoided. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

226.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

226.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

226.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

226.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

226.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

226.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

226.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

226.9 Treatment Options

Treatment options include cholinesterase inhibitors (rivastigmine is FDA-approved for DLB), avoidance of anticholinergic medications, and low-dose levodopa for parkinsonism. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

226.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

226.11 Outlook

outlook is progressive. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 227

Dengue Fever

227.1 Overview

Dengue fever is a mosquito-borne viral illness caused by dengue virus (DENV 1–4), a flavivirus transmitted by *Aedes aegypti* mosquitoes, with 400 million infections and 100 million symptomatic cases annually, and 40,000 deaths, endemic in tropical and subtropical regions. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

227.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

227.3 Risk Factors

This condition happens because of the virus infecting dendritic cells, keratinocytes, and macrophages, with a secondary infection with a different serotype being the primary risk factor for severe dengue (antibody-dependent enhancement, ADE).

- Close contact with infected individuals
- Compromised immune system
- Travel to endemic areas
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

227.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

227.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- You may experience incubation of 3–14 days, with classic dengue fever presenting with sudden onset high fever, severe headache (retro-orbital), joint pain and muscle pain (“breakbone fever”), maculopapular rash, lymphadenopathy, and leukopenia
- Severe dengue (dengue involving bleeding fever/dengue shock syndrome) presents with plasma leakage, severe thrombocytopenia, mucosal bleeding, and shock during the critical phase at defervescence.

227.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

227.7 Diagnosis

Doctors diagnose this using NS1 antigen test, IgM and IgG serology, and RT-PCR. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

227.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)

- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

227.9 Treatment Options

Treatment typically involves supportive (acetaminophen for fever, avoiding NSAIDs and aspirin due to bleeding risk, fluid resuscitation). Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

227.10 Living with It

- Most people recover well for classic dengue
- Severe dengue has significant mortality without prompt supportive care.
- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

227.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 228

Dermatitis Herpetiformis

228.1 Overview

Dermatitis herpetiformis (DH) is the cutaneous manifestation of celiac disease, characterized by intensely pruritic, grouped vesicles and papules on an erythematous base, symmetrically distributed on extensor surfaces (elbows, knees, buttocks, back). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

228.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of IgA antibody deposition at the dermal papillae, triggered by gluten ingestion
- Nearly all patients have underlying celiac disease, though digestive tract symptoms may be absent.

228.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

228.4 Signs and Symptoms

You may experience intensely pruritic, grouped vesicles on extensor surfaces. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

228.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

228.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

228.7 Diagnosis

Doctors diagnose this based on skin biopsy showing granular IgA deposition at the dermal papillae. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

228.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

228.9 Treatment Options

- Treatment options include a strict gluten-free diet and dapsone (which rapidly controls skin symptoms)
- Sulfapyridine is an alternative for patients intolerant to dapsone.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

228.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

228.11 Outlook

Most people recover well with dietary compliance. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 229

Dermatomyositis

229.1 Overview

Dermatomyositis is an idiopathic inflammatory myopathy with characteristic skin manifestations, discussed in Chapter ??. In addition to proximal muscle weakness, it presents with heliotrope rash, Gottron's papules, V-sign, shawl sign, and mechanic's hands. It is mediated by complement attack on intramuscular blood vessels (perifascicular shrinkage or wasting on biopsy). Adults with dermatomyositis have an increased risk of underlying malignancy, necessitating age-appropriate cancer screening. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

229.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

229.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

229.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

229.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

229.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

229.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

229.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

229.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

229.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

229.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 230

Dermatomyositis

230.1 Overview

Dermatomyositis is an idiopathic inflammatory myopathy with characteristic skin manifestations, affecting both children and adults. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

230.2 Causes

This condition happens because of autoimmune-mediated inflammation of skin and muscle. Cutaneous features include heliotrope rash (violaceous discoloration of the eyelids), Gottron's papules (erythematous to violaceous papules overlying the knuckles), V-sign (erythematous rash on the anterior neck and chest), shawl sign (erythematous rash over shoulders and upper back), mechanic's hands (rough, cracked skin on the fingers), and nail fold capillary changes. Proximal muscle weakness is the hallmark of myopathy. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

230.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

230.4 Signs and Symptoms

- You may experience both skin and muscle manifestations
- Dermatomyositis in adults carries an increased risk of underlying malignancy (ovarian, lung, GI), necessitating age-appropriate cancer screening.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

230.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

230.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

230.7 Diagnosis

Diagnosis typically involves CK levels, myositis-specific antibodies (anti-Mi-2, anti-MDA5, anti-TIF1-γ), muscle MRI, EMG, and muscle biopsy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

230.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

230.9 Treatment Options

Treatment options include corticosteroids, steroid-sparing agents (methotrexate, azathioprine, mycophenolate), IVIG, and rituximab. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

230.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

230.11 Outlook

- The outlook varies from person to person
- Malignancy screening is essential in adults.

Chapter 231

Deviated Nasal Septum

231.1 Overview

Nasal septal deviation is a misalignment of the cartilaginous or bony septum from the midline that is present to some degree in most individuals. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

231.2 Causes

This condition happens because of developmental or traumatic factors. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

231.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

231.4 Signs and Symptoms

- You may experience unilateral or bilateral nasal obstruction, which may be worsened by mucosal swelling during allergic rhinitis or upper respiratory infections
- Deviation may cause recurrent epistaxis, facial pain, and contribute to obstructive sleep apnea.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

231.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

231.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

231.7 Diagnosis

Doctors diagnose this using anterior rhinoscopy and CT imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

231.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

231.9 Treatment Options

Treatment focuses on septoplasty for severe symptomatic deviation, which may be combined with turbinate reduction for concurrent turbinate enlargement. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

231.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

231.11 Outlook

Most people recover well following surgical correction. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 232

Diabetes Insipidus

232.1 Overview

Diabetes insipidus is a condition of impaired water homeostasis caused by deficient production (central DI) or action (nephrogenic DI) of antidiuretic hormone (ADH/vasopressin). This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

232.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Central DI results from decreased ADH secretion due to pituitary/hypothalamic disease (tumors, surgery, trauma, infiltrative disease, idiopathic), while nephrogenic DI results from kidney resistance to ADH (lithium use is the most common acquired cause
- Genetic forms include X-linked aquaporin-2 mutations).

232.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

232.4 Signs and Symptoms

You may experience polyuria (output >3 L/day, often 5–20 L/day), polydipsia, nocturia, and dilute urine (specific gravity <1.005, osmolality <300 mOsm/kg), with severe dehydration and hypernatremia possibly occurring if water intake is restricted.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

232.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

232.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

232.7 Diagnosis

Diagnosis typically involves the water deprivation test, with desmopressin administration differentiating central DI (improved urine concentration) from nephrogenic DI (no response). Treatment for central DI is desmopressin (DDAVP) replacement, while Treatment for nephrogenic DI includes thiazide diuretics, low-salt diet, and avoidance of nephrotoxic agents.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

232.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

232.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

232.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

232.11 Outlook

The outlook is good with appropriate management.

Chapter 233

Diabetic Ketoacidosis

233.1 Overview

Diabetic ketoacidosis (DKA) is a life-threatening metabolic emergency characterized by hyperglycemia (glucose >250 mg/dL), metabolic acidosis (pH <7.3, bicarbonate <18 mEq/L), and ketosis (elevated ketones in blood and urine). It occurs predominantly in T1DM but can occur in T2DM under significant physiological stress. Precipitants include insulin non-compliance, infections, new-onset diabetes, heart attack, and medications. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

233.2 Causes

This condition happens because of absolute or relative insulin deficiency counteracted by counter-regulatory hormones (glucagon, cortisol, catecholamines), leading to lipolysis, ketone body production (beta-hydroxybutyrate, acetoacetate), and osmotic diuresis causing dehydration and electrolyte depletion. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

233.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

233.4 Signs and Symptoms

You may experience polyuria, polydipsia, nausea, vomiting, abdominal pain, Kussmaul respirations (deep, rapid breathing), and a fruity breath odor. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

233.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

233.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

233.7 Diagnosis

Doctors diagnose this based on blood glucose, arterial blood gas, serum ketones, and anion gap calculation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

233.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

233.9 Treatment Options

Treatment focuses on aggressive IV fluid resuscitation, continuous IV insulin infusion, and careful potassium replacement. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

233.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

233.11 Outlook

- The outlook is generally positive with prompt treatment
- Mortality is low in well-managed cases.

Chapter 234

Diabetic Nephropathy

234.1 Overview

Diabetic nephropathy is the most common cause of ESRD worldwide, affecting 30–40% of patients with diabetes mellitus (both T1DM and T2DM). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

234.2 Causes

This condition happens because of hyperglycemia-induced metabolic and hemodynamic changes leading to glomerular hyperfiltration, mesangial expansion, basement membrane thickening, and ultimately nodular glomerulosclerosis (Kimmelstiel-Wilson nodules). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

234.3 Risk Factors

outlook is improved with early intervention and control of risk factors. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

234.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

234.5 Progression and Stages

Clinical progression includes initial hyperfiltration (increased GFR), microalbuminuria (moderately increased albuminuria), macroalbuminuria (heavily increased albuminuria), declining GFR, and ESRD. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

234.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

234.7 Diagnosis

Doctors diagnose this based on persistent albuminuria in the setting of diabetes. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

234.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

234.9 Treatment Options

Treatment options include optimal glycemic control, RAAS blockade (ACE inhibitors or ARBs), SGLT2 inhibitors, BP control (target <130/80), and finerenone (non-steroidal mineralocorticoid receptor antagonist). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

234.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

234.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 235

Diabetic Peripheral Neuropathic Pain

235.1 Overview

Diabetic peripheral neuropathy (DPN) is a neuropathic pain condition affecting approximately 50% of patients with diabetes mellitus, with 15–25% of these patients experiencing neuropathic pain. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

235.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

235.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

235.4 Signs and Symptoms

You may experience bilateral, symmetric, stocking-glove distribution of burning, tingling, shooting, and electric sensations, often worse at night, with reduced sensation, absent ankle reflexes, and allodynia. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

235.5 Progression and Stages

This condition happens because of hyperglycemia-induced metabolic pathways (polyol pathway flux, advanced glycation end products, oxidative stress, protein kinase C activation) leading to axonal deterioration, demyelination, and loss of small and large nerve fibers. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

235.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

235.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. First-line pharmacotherapy includes duloxetine, pregabalin, gabapentin, and TCAs.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

235.8 Prevention

- Outlook is chronic
- Pain management does not reverse the underlying neuropathy but improves quality of life
- Glycemic control is essential for prevention and slowing progression.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

235.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

235.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

235.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 236

Diabetic Retinopathy

236.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

236.2 Causes

This condition happens because of chronic hyperglycemia damaging retinal capillaries, causing vascular occlusion, non-perfusion, microaneurysms, hemorrhages, hard exudates, and cotton-wool spots (non-proliferative DR). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

236.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

236.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

236.5 Progression and Stages

Your outlook depends on glycemic control and stage at diagnosis. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

236.6 Complications

Diabetic retinopathy is a microvascular complication of diabetes mellitus and a leading cause of blindness in working-age adults.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

236.7 Diagnosis

Diagnosis typically involves dilated fundus examination and fundus photography. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

236.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

236.9 Treatment Options

- Treatment options include optimal glycemic control, anti-VEGF intravitreal injections for diabetic macular swelling, focal or grid laser photocoagulation, and panretinal photocoagulation for proliferative disease
- Regular screening is recommended for all diabetic patients.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

236.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

236.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 237

DiGeorge Syndrome

237.1 Overview

DiGeorge syndrome (22q11.2 deletion syndrome, velocardiofacial syndrome) is a developmental disorder caused by a microdeletion on chromosome 22q11.2, resulting in developmental abnormalities derived from the third and fourth pharyngeal arches. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

237.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

237.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

237.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

237.5 Progression and Stages

You may experience thymic underdevelopment or aplasia (T-cell immunodeficiency of varying severity), hypoparathyroidism (hypocalcemia), congenital heart defects (tetralogy of Fallot, interrupted aortic arch, truncus arteriosus, ventricular septal defect), characteristic facies (broad nasal bridge, small ears, micrognathia), palatal abnormalities (velopharyngeal insufficiency, cleft palate), and learning difficulties, with the severity of immunodeficiency varying widely. Your outlook depends on the severity of associated anomalies. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

237.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

237.7 Diagnosis

Your doctor confirms the diagnosis with FISH or chromosomal microarray showing 22q11.2 deletion. Management addresses individual manifestations: cardiac surgery, calcium/vitamin D supplementation for hypocalcemia, thymic transplantation for complete aplasia, and developmental/educational support.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

237.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

237.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

237.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 238

Digoxin Toxicity

238.1 Overview

Digoxin toxicity is a pharmaceutical overdose involving a cardiac glycoside that inhibits Na+/K+-ATPase, increasing intracellular calcium and enhancing cardiac contractility. Toxicity occurs with therapeutic excess (most common, especially with kidney impairment, hypokalemia, hypomagnesemia, hypercalcemia, or drug interactions) or acute overdose. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

238.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

238.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

238.4 Signs and Symptoms

- You may experience digestive tract (nausea, vomiting, anorexia), neurological (visual disturbances—yellow-green halos, blurred vision, confusion), and cardiac manifestations including any bradyarrhythmia, atrial tachycardia with block, accelerated junctional rhythm, and bidirectional ventricular tachycardia (pathognomonic)
- Hyperkalemia is a marker of severe acute toxicity.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

238.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

238.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

238.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with ECG findings and serum digoxin level. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

238.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

238.9 Treatment Options

Treatment options include digoxin immune Fab (Digibind) as the definitive antidote for life-threatening arrhythmias or hyperkalemia. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

238.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

238.11 Outlook

The outlook is generally positive with timely administration of Fab fragments. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 239

Dilated Cardiomyopathy

239.1 Overview

Dilated cardiomyopathy is a condition characterized by ventricular chamber dilation and systolic dysfunction in the absence of coronary artery disease, valvular disease, or hypertensive heart disease. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

239.2 Causes

Causes include familial or genetic factors (mutations in titin, lamin A/C, and other cytoskeletal genes in 30–50%), viral myocarditis, alcohol, toxins (doxorubicin), peripartum cardiomyopathy, and tachycardia-mediated cardiomyopathy. This condition happens because of impaired myocardial contractility leading to ventricular dilation. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

239.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age

- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

239.4 Signs and Symptoms

You may experience heart failure symptoms. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

239.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

239.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

239.7 Diagnosis

- Doctors diagnose this based on echocardiography showing a dilated left ventricle with reduced ejection fraction
- Endomyocardial biopsy may be indicated for suspected myocarditis.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

239.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

239.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

239.10 Living with It

- Management follows guideline-directed medical therapy for heart failure with reduced ejection fraction
- Anticoagulation may be considered for high thromboembolic risk
- Genetic counseling and family screening are recommended.
- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

239.11 Outlook

Your outlook depends on the underlying cause and response to therapy. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 240

Diphtheria

240.1 Overview

Diphtheria is a serious bacterial infection caused by *Corynebacterium diphtheriae* that produces an exotoxin causing tissue tissue death and systemic toxicity. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

240.2 Causes

This condition happens because of infection with toxigenic strains of *C. diphtheriae*. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

240.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

240.4 Signs and Symptoms

- You may experience a tough, gray-white pseudomembrane on the pharynx and tonsils, neck swelling (“bull neck”), fever, and toxic appearance
- The exotoxin can cause myocarditis and neuropathy.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

240.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

240.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts

- Multi-organ failure in severe cases

240.7 Diagnosis

Doctors diagnose this using culture on tellurite agar and toxin detection. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

240.8 Prevention

The outlook has been dramatically improved by vaccination with diphtheria toxoid (as part of DTaP/Tdap), which has made the disease rare in developed countries. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

240.9 Treatment Options

Treatment options include diphtheria antitoxin and antibiotics (penicillin or erythromycin). Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

240.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

240.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 241

Diphtheria (Corynebacterium diphtheriae)

241.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

241.2 Causes

This condition happens because of *C. diphtheriae* colonizing the nasopharynx or skin and producing diphtheria toxin, a potent ADP-ribosyltransferase that inactivates elongation factor 2 (EF-2), inhibiting protein synthesis and causing cell tissue death, with toxin causing local tissue destruction (pseudomembrane formation) and systemic effects (myocarditis, neuritis). Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

241.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational

exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

241.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

241.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

241.6 Complications

You may experience respiratory diphtheria: sore throat, difficulty swallowing, low-grade fever, and a characteristic grayish-white, thick, adherent pseudomembrane over the tonsils, pharynx, or larynx, with airway obstruction being the primary local risk, and systemic complications including myocarditis and neuropathy.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

241.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with Gram stain and culture confirmation. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

241.8 Prevention

Diphtheria is a toxin-mediated disease caused by *Corynebacterium diphtheriae*, a gram-positive bacillus, with sporadic cases in high-income countries and outbreaks persisting in regions with low vaccination coverage, transmission by respiratory droplets or contact with infected skin lesions. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

241.9 Treatment Options

Treatment options include diphtheria antitoxin (equine) administered immediately and antibiotics (erythromycin or penicillin G). outlook: mortality 5–10%, lower with antitoxin. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

241.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

241.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 242

Disseminated Intravascular Coagulation

242.1 Overview

Disseminated intravascular coagulation is an acquired, life-threatening coagulation disorder characterized by systemic activation of coagulation, leading to widespread microvascular blood clot formation and consumption of platelets and clotting factors, resulting in a paradoxical bleeding tendency. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

242.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

242.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

242.4 Signs and Symptoms

You may experience widespread bleeding (from IV sites, surgical wounds, mucosal surfaces), microvascular blood clot formation (organ dysfunction), and hemodynamic instability. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

242.5 Progression and Stages

Your outlook depends on the underlying cause and severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

242.6 Complications

The condition is always secondary to an underlying condition: sepsis (most common cause), trauma, malignancy, obstetric complications (placental abruption, amniotic fluid blockage by a blood clot), transfusion reactions, and severe liver disease.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

242.7 Diagnosis

Doctors diagnose this based on clinical findings and laboratory evidence: thrombocytopenia, prolonged PT and aPTT, low fibrinogen, elevated D-dimer and fibrin degradation products, and schistocytes on peripheral smear, with the ISTH DIC score aiding diagnosis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

242.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

242.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

242.10 Living with It

Management targets the underlying cause, with supportive care including platelet and FFP transfusions for active bleeding, cryoprecipitate for fibrinogen <100 mg/dL, and heparin for thrombotic predominant DIC.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

242.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 243

Dissociative Amnesia

243.1 Overview

Dissociative amnesia has a prevalence of 1–3% in the general population. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

243.2 Causes

This condition happens because of stress-induced disruption of hippocampal and prefrontal memory encoding and retrieval processes, typically associated with trauma (sexual assault, combat, natural disaster, interpersonal violence). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

243.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

243.4 Signs and Symptoms

You may experience inability to recall important autobiographical information, usually of a traumatic or stressful nature, that is inconsistent with ordinary forgetting, with the amnesia possibly being localized, selective, generalized, or systematized. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

243.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

243.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

243.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with neuroimaging and EEG to rule out organic amnesia. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

243.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

243.9 Treatment Options

Treatment options include psychotherapy (psychodynamic, CBT, hypnotherapy) focused on recovery from trauma and integration. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

243.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

243.11 Outlook

The outlook is good for localized and selective amnesia—memory typically returns spontaneously or with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 244

Dissociative Identity Disorder

244.1 Overview

Dissociative identity disorder has a prevalence of 1–1.5% in the general population, though it is frequently underdiagnosed. The condition typically arises from severe, repetitive, early childhood trauma before age 6–9, when the child's personality is still developing and dissociation serves as a psychological defense, with evidence of functional differences between identities on neuroimaging. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

244.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

244.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

244.4 Signs and Symptoms

You may experience disruption of identity involving two or more distinct personality states (alter identities), each with its own relatively enduring pattern of perceiving, relating to, and thinking about the environment and self, accompanied by recurrent gaps in the recall of everyday events, important personal information, and/or traumatic events. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

244.5 Progression and Stages

Treatment typically involves long-term, phase-oriented treatment focusing on safety/stabilization, trauma processing, and identity integration. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

244.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

244.7 Diagnosis

Doctors diagnose this using clinical interview (SCID-D). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

244.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

244.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

244.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

244.11 Outlook

- The outlook varies from person to person
- Treatment aims for improved cooperation between identities or integration.

Chapter 245

Diverticulitis

245.1 Overview

Diverticulitis is inflammation or infection of a diverticulum, occurring when a diverticulum becomes obstructed by fecal material, leading to microperforation and localized inflammation. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

245.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

245.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

245.4 Signs and Symptoms

- Symptoms of uncomplicated diverticulitis include left lower quadrant pain, fever, leukocytosis, and altered bowel habits

- Complicated diverticulitis may involve abscess formation, perforation with peritonitis, fistula formation (colovesical fistula most common), or stricture.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

245.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

245.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

245.7 Diagnosis

- Doctors diagnose this based on CT abdomen/pelvis with contrast, which demonstrates pericolonic fat stranding, wall thickening, and abscess. Treatment for uncomplicated diverticulitis includes antibiotics (fluoroquinolones + metronidazole, or amoxicillin-clavulanate) and a liquid diet progressing to a high-fiber diet
- Complicated diverticulitis requires hospitalization, IV antibiotics, and through the skin removal of fluid for abscesses.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

245.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

245.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

245.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

245.11 Outlook

- The outlook is good with appropriate treatment
- Elective colectomy is considered after two or more episodes of complicated diverticulitis or for immunocompromised patients.

Chapter 246

Diverticulosis

246.1 Overview

Diverticulosis is the presence of small, bulging pouches (diverticula) in the wall of the colon, most commonly in the sigmoid colon. It affects over 50% of individuals over 60 in Western countries. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

246.2 Causes

This condition happens because of increased intraluminal pressure at weak points in the colonic wall where vasa recta penetrate the circular muscle layer, and is strongly associated with low-fiber diets, obesity, and aging. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

246.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

246.4 Signs and Symptoms

You may experience mostly asymptomatic presentation, discovered incidentally on colonoscopy or CT. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

246.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

246.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

246.7 Diagnosis

Doctors diagnose this based on colonoscopy or imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

246.8 Prevention

Treatment options include high-fiber diets and stool softeners for prevention of complications.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

246.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

246.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

246.11 Outlook

- Most people recover well
- Most patients remain asymptomatic.

Chapter 247

Down Syndrome (Trisomy 21)

247.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Down syndrome is the most common chromosomal disorder, with an incidence of 1 in 700 live births. It is caused by trisomy 21 (94%), Robertsonian translocation (4%), or mosaic trisomy 21 (2%)
- Risk increases with maternal age.

247.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

247.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

247.4 Signs and Symptoms

You may experience flat facial profile, upslanting palpebral fissures, epicanthal folds, Brushfield spots, single transverse palmar crease, hypotonia, joint hyperlaxity, short stature, intellectual disability (mild to moderate), and congenital heart disease (50%). Associated conditions include hearing loss, hypothyroidism, celiac disease, duodenal atresia, and increased risk of leukemia. Prenatal screening includes first-trimester nuchal translucency and maternal serum markers.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

247.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

247.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

247.7 Diagnosis

- Your doctor confirms the diagnosis with karyotype or FISH. Management requires lifelong multidisciplinary care
- Life expectancy has increased to approximately 60 years.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

247.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

247.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

247.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 248

Drowning (Submersion Injury)

248.1 Overview

Drowning is respiratory impairment from submersion or immersion in a liquid medium. It is a leading cause of unintentional injury death worldwide, particularly in children under 5 years. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

248.2 Causes

This condition happens because of initial breath-holding and panic, laryngospasm (10–15% dry drowning—no removal by suction), then removal by suction of water, surfactant washout, lung swelling, and hypoxemia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

248.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

248.4 Signs and Symptoms

- Clinical features: respiratory distress, coughing, tachypnea, cyanosis, foam from mouth/nose, altered mental status, seizures, cardiac arrest
- The primary determinant of the outcome is duration of anoxia.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

248.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

248.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

248.7 Diagnosis

Doctors diagnose this using history of submersion, clinical presentation, chest X-ray, and arterial blood gas. Management: immediate rescue, CPR if pulseless, airway management, positive pressure ventilation, supplemental oxygen, and rewarming for hypothermic victims. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

248.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

248.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

248.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

248.11 Outlook

outlook is better in children and those with submersion time <10 minutes and rapid resuscitation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 249

Drug Hypersensitivity

249.1 Overview

Drug hypersensitivity reactions are adverse drug reactions with an immunologic basis, representing approximately 5–10% of all adverse drug reactions. The condition may be immediate (IgE-mediated, occurring within 1–6 hours: urticaria, angioedema, anaphylaxis) or delayed (T-cell mediated, occurring hours to days after exposure: maculopapular exanthema, DRESS syndrome, Stevens-Johnson syndrome, toxic epidermal necrolysis, acute generalized exanthematous pustulosis). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

249.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

249.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

249.4 Signs and Symptoms

Common culprits include beta-lactam antibiotics, sulfonamides, anticonvulsants, and NSAIDs. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

249.5 Progression and Stages

Your symptoms depend on the type and severity of reaction. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

249.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

249.7 Diagnosis

Diagnosis typically involves detailed clinical history, skin testing (for IgE-mediated reactions), drug provocation tests, and lymphocyte transformation tests (for delayed reactions). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

249.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

249.9 Treatment Options

Treatment options include drug avoidance, acute treatment of reactions, and in some cases, desensitization protocols for essential medications. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

249.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

249.11 Outlook

In most cases, the outlook is favorable with avoidance of the offending drug. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 250

Drug Interactions in the Elderly

250.1 Overview

Age-related physiological changes alter pharmacokinetics (absorption, distribution—increased body fat, decreased total body water, decreased albumin—metabolism, and elimination—reduced kidney function) and pharmacodynamics (increased sensitivity to CNS-active drugs, anticholinergics, anticoagulants). Common drug-drug interactions in older adults include warfarin—NSAIDs (increased bleeding risk), warfarin—amiodarone (increased INR), ACE inhibitors—spironolactone (hyperkalemia), SSRIs—NSAIDs (GI bleeding), and opioids—benzodiazepines (respiratory depression). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

250.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

250.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

250.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

250.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

250.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

250.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

250.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

250.9 Treatment Options

Treatment options include maintaining an up-to-date medication list, using a single pharmacy and prescriber when possible, prioritizing drug-drug interaction checking, and avoiding high-risk combinations. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

250.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

250.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 251

Dry Eye Disease

251.1 Overview

Dry eye disease is a multifactorial disorder of the ocular surface characterized by loss of homeostasis of the tear film, affecting approximately 5–30% of the global population, with higher prevalence in the elderly, women, and those in arid or windy environments. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

251.2 Causes

This condition happens because of tear film instability due to aqueous-deficient (reduced lacrimal gland production, as in Sjögren's syndrome, aging, or medication side effects) or evaporative causes (meibomian gland dysfunction, lid abnormalities, or environmental factors). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

251.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

251.4 Signs and Symptoms

- You may experience dryness, grittiness, burning, foreign body sensation, photophobia, and blurred vision
- Signs include reduced tear break-up time (less than 10 seconds), Schirmer's test less than 5 mm in 5 minutes, corneal staining, and meibomian gland dropout.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

251.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

251.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

251.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by tear osmolarity testing and inflammatory markers. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

251.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

251.9 Treatment Options

Treatment typically involves stepwise: artificial tears, punctal plugs, topical cyclosporine A or lifitegrast, secretagogues, warm compresses and lid hygiene, and environmental modifications. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

251.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

251.11 Outlook

The outlook varies from person to person, with most patients achieving symptom control with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 252

Duchenne Muscular Dystrophy

252.1 Overview

Duchenne muscular dystrophy (DMD) is the most common and severe childhood muscular dystrophy, affecting approximately 1 in 3500–5000 live-born males. It is an X-linked recessive disorder caused by mutations in the dystrophin gene (DMD) on Xp21, leading to absence of the dystrophin protein essential for sarcolemmal integrity. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

252.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

252.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

252.4 Signs and Symptoms

- You may experience progressive proximal muscle weakness beginning between ages 2–5, with Gowers' sign (using hands to "climb up" the thighs when rising from the floor), waddling gait, toe walking, and calf pseudohypertrophy
- By age 10–12, most patients lose ambulation. Cardiomyopathy and respiratory failure develop in adolescence.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

252.5 Progression and Stages

Treatment options include corticosteroids (deflazacort, prednisone) to slow progression, cardiac management (ACE inhibitors, beta-blockers), respiratory support, and newer therapies including exon-skipping agents (eteplirsen, golodirsen, viltolarsen, casimersen for specific mutations) and gene therapy (delandistrogene moxeparvovec). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

252.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

252.7 Diagnosis

Doctors diagnose this based on elevated CK (10–100 times normal), genetic testing showing dystrophin mutations, and muscle biopsy showing dystrophic changes with absent dystrophin on immunostaining.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

252.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

252.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

252.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

252.11 Outlook

outlook is severe, with most patients losing ambulation by age 10–12 and survival into the third decade with respiratory support.

Chapter 253

Ductal Carcinoma In Situ (DCIS)

253.1 Overview

Ductal carcinoma in situ (DCIS) is a non-invasive neoplastic growth of epithelial cells confined within the mammary ductal-lobular system without penetration of the basement membrane, accounting for 20–25% of screen-detected breast malignancies. Driven by localized genetic mutations and hormonal stimulation, DCIS is categorized by architectural patterns (comedo, cribriform, papillary, solid), with comedo DCIS presenting central high-grade tissue death and dystrophic calcium buildup. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

253.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

253.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age

- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

253.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

253.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

253.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

253.7 Diagnosis

Diagnosis is established via stereotactic or ultrasound-guided core needle biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

253.8 Prevention

The condition is overwhelmingly asymptomatic and detected on routine screening mammography as microcalcifications (clustered, pleomorphic, or linear branching). Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

253.9 Treatment Options

- Treatment focuses on breast-conserving surgery (lumpectomy) with negative margins (> 2 mm) followed by radiation therapy and adjuvant tamoxifen/aromatase inhibitors for ER-positive lesions
- Total mastectomy is indicated for widespread or multicentric disease.
- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

253.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies

(acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

253.11 Outlook

Most people recover well, with 20-year disease-specific survival exceeding 98%. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 254

Duodenal Ulcer

254.1 Overview

Duodenal ulcer is the most common type of peptic ulcer, occurring predominantly in the first part of the duodenum (duodenal bulb). *Helicobacter pylori* is the primary cause in 90–95% of cases, with NSAIDs as the second most common cause. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

254.2 Causes

This condition happens because of acid hypersecretion due to loss of negative feedback from somatostatin-producing D-cells infected by *H. pylori*. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

254.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

254.4 Signs and Symptoms

You may experience epigastric pain relieved by food and antacids ("hunger pain"), nocturnal pain, and bloating. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

254.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

254.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

254.7 Diagnosis

Doctors diagnose this based on upper endoscopy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

254.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

254.9 Treatment Options

Treatment focuses on *H. pylori* eradication (triple or quadruple therapy for 14 days), PPI therapy, and smoking cessation. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

254.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

254.11 Outlook

- The outlook is generally positive
- Unlike gastric ulcers, duodenal ulcers rarely undergo cancerous transformation.

Chapter 255

Dupuytren Contracture

255.1 Overview

Dupuytren contracture is a progressive fibroproliferative disorder of the palmar fascia resulting in nodule formation and fixed flexion deformities of the digits. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

255.2 Causes

This condition happens because of abnormal myofibroblast growth and excessive collagen type III deposition, leading to thickening and shortening of the palmar aponeurosis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

255.3 Risk Factors

Risk factors include Northern European ancestry, male sex, advanced age, positive family history, diabetes mellitus, heavy alcohol consumption, and smoking. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

255.4 Signs and Symptoms

Clinical features begin with painless palmar skin pitting and subcutaneous nodules (most commonly over the 4th and 5th rays), progressing to dense fascial cords and fixed flexion contractures at the metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints (positive Tabletop test). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

255.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

255.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

255.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

255.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

255.9 Treatment Options

Treatment options include observation for mild non-functional cases, collagenase Clostridium histolyticum injections, needle aponeurotomy, or subtotal palmar fasciectomy for functional impairment ($> 30°$ MCP contracture). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

255.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

255.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 256

Dysbetalipoproteinemia

256.1 Overview

Dysbetalipoproteinemia (type III hyperlipoproteinemia, remnant removal disease) is a disorder of chylomicron and VLDL remnant clearance caused by homozygosity for apolipoprotein E2 (apoE2/E2 genotype) in conjunction with a second metabolic hit (obesity, diabetes, hypothyroidism, or estrogen deficiency). Only 1–5% of individuals with the apoE2/E2 genotype develop the full phenotype, which is otherwise rare (prevalence approximately 1 in 10,000). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

256.2 Causes

This condition happens because of impaired liver clearance of remnant lipoproteins due to poor binding of apoE2 to the LDL receptor. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

256.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

256.4 Signs and Symptoms

You may experience both cholesterol and triglyceride elevation (typically in equal proportion, greater than 300 mg/dL), tuberoeruptive xanthomas, and palmar crease xanthomas (pathognomonic). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

256.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

256.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

256.7 Diagnosis

Doctors diagnose this using ultracentrifugation showing a broad β-band, VLDL cholesterol-to-total triglyceride ratio greater than 0.30, and apoE genotyping (E2/E2). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

256.8 Prevention

- Treatment options include lifestyle modification
- Statins and fibrates are synergistic and often normalize lipids.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

256.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

256.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

256.11 Outlook

The outlook is generally positive with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 257

Dysmenorrhea

257.1 Overview

Dysmenorrhea is painful menstrual cramps, the most common gynecologic complaint, affecting 50–80% of reproductive-age women. Primary dysmenorrhea occurs without organic pelvic pathology, typically beginning within a few years of menarche, caused by excessive endometrial prostaglandin production leading to myometrial hypercontractility and uterine reduced blood flow. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

257.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

257.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

257.4 Signs and Symptoms

You may experience cramping lower abdominal pain radiating to the back and thighs, beginning just before or at the onset of menstruation and lasting 1–3 days. Secondary dysmenorrhea is due to underlying pathology (endometriosis, adenomyosis, uterine fibroids, pelvic inflammatory disease). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

257.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

257.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

257.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Secondary dysmenorrhea requires pelvic ultrasound to identify underlying causes. Treatment of primary dysmenorrhea includes NSAIDs and combined oral contraceptives.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

257.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

257.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

257.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

257.11 Outlook

The outlook is generally positive. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 258

Dystonia

258.1 Overview

Dystonia is a movement disorder characterized by sustained or intermittent muscle contractions causing abnormal, often repetitive, movements, postures, or both, with a prevalence of 1 in 3,000. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

258.2 Causes

This condition happens because of dysfunction of the basal ganglia-thalamo-cortical motor circuit, with genetic forms caused by mutations in genes including TOR1A (DYT1, early-onset generalized), THAP1 (DYT6, mixed phenotype), GCH1 (DYT5, dopa-responsive dystonia), and SGCE (DYT11, myoclonus-dystonia). Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

258.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

258.4 Signs and Symptoms

Symptoms vary depending on distribution: focal (most common, eg., cervical dystonia/spasmodic torticollis, blepharospasm, oromandibular dystonia, laryngeal dystonia/spasmodic dysphonia, writer's cramp), segmental (involving two contiguous body regions), multifocal, hemidystonia, or generalized. Dystonic movements are typically directional, patterned, and often action-specific, worsened by stress and fatigue, and may be alleviated by sensory tricks (geste antagoniste). Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

258.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

258.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

258.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with genetic testing for early-onset or familial cases. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

258.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

258.9 Treatment Options

- Treatment options include oral medications (trihexyphenidyl, benzodiazepines, baclofen, tetrabenazine), botulinum toxin injections (first-line for focal dystonia), deep brain stimulation of the globus pallidus internus (GPi-DBS, most effective for generalized and cervical dystonia), and levodopa trial for suspected dopa-responsive dystonia. outlook: focal dystonia is usually non-progressive after initial spread
- Generalized dystonia may progress over years but stabilize with treatment.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

258.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

258.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 259

Eastern Equine Encephalitis

259.1 Overview

Eastern equine encephalitis is a rare but severe mosquito-borne viral disease caused by Eastern equine encephalitis virus (EEEV), an alphavirus, and is the most severe arboviral encephalitis in North America, with 5–10 human cases annually in the United States and a case fatality rate of 30–50%. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

259.2 Causes

This condition happens because of EEEV crossing the blood-brain barrier and infecting neurons, causing neuronal tissue death, perivascular cuffing, and intense neutrophilic inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

259.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

259.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

259.5 Progression and Stages

You may experience an incubation of 4–10 days, with a systemic phase (sudden fever, chills, headache, muscle pain, malaise) followed by an encephalitic phase (altered mental status, seizures, cranial nerve palsies, and focal deficits), with rapid progression to coma and respiratory failure. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

259.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

259.7 Diagnosis

Doctors diagnose this using EEEV-specific IgM in CSF or serum by ELISA. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

259.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

259.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

259.10 Living with It

- Treatment typically involves supportive care. outlook: 30–50% mortality
- Survivors have a high rate (50–80%) of severe nervous system sequelae.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

259.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 260

Ebola Virus Disease

260.1 Overview

Ebola virus disease is a severe, often fatal involving bleeding fever caused by Ebola virus, with five species: Zaire (most virulent), Sudan, Bundibugyo, Taï Forest, and Reston, with the 2014–2016 West Africa outbreak being the largest (28,646 cases, 11,323 deaths), and transmission through direct contact with blood, secretions, or bodily fluids of infected humans or animals. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

260.2 Causes

This condition happens because of the virus entering through mucosal surfaces or breaks in skin, infecting dendritic cells, macrophages, and monocytes, spreading to lymph nodes, liver, spleen, and adrenal glands, causing profound immunosuppression and release of pro-inflammatory cytokines (cytokine storm), with endothelial cell dysfunction, DIC, and multi-organ failure. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

260.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to

endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

260.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

260.5 Progression and Stages

You may experience an incubation of 2–21 days, with prodrome (sudden fever, severe headache, muscle pain, malaise), digestive tract phase (severe watery diarrhea, nausea, vomiting, abdominal pain), and systemic phase (multi-organ failure, coagulopathy, hypotension/shock, and nervous system involvement). The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

260.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

260.7 Diagnosis

Doctors diagnose this using RT-PCR of blood. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

260.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer

- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

260.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

260.10 Living with It

- Treatment options include supportive care and two FDA-approved monoclonal antibody therapies: mAb114 (ansuvimab) and REGN-EB3. outlook: Zaire species 50–80% mortality without treatment
- 33–50% with supportive care and monoclonal antibodies.
- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

260.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 261

Eclampsia

261.1 Overview

Eclampsia is a hypertensive disorder of pregnancy defined as the occurrence of generalized tonic-clonic seizures in a woman with preeclampsia that cannot be attributed to other causes. It occurs in 1 in 2000 deliveries in developed countries and more frequently in resource-limited settings. Seizures may occur antepartum (38%), intrapartum (18%), or postpartum (44%). This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

261.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

261.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

261.4 Signs and Symptoms

You may experience generalized tonic-clonic seizures in the setting of preeclampsia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

261.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

261.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

261.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

261.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

261.9 Treatment Options

- Treatment typically involves a medical emergency: magnesium sulfate (4–6 g IV loading dose followed by 1–2 g/hour infusion) is the first-line anticonvulsant, superior to diazepam or phenytoin
- Airway protection, seizure precautions, and antihypertensive therapy are essential. The main treatment typically involves delivery after maternal stabilization.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

261.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

261.11 Outlook

- The outlook is generally positive with prompt treatment
- Magnesium toxicity doctors typically treat it with calcium gluconate.

Chapter 262

Ectopic Pregnancy

262.1 Overview

Ectopic pregnancy occurs when a fertilized ovum implants outside the uterine cavity, most commonly in the fallopian tube (95%), but also in the cervix, ovary, abdominal cavity, or previous cesarean scar. The incidence is 1–2% of all pregnancies. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

262.2 Causes

This condition happens because of impaired tubal function or anatomy preventing normal embryo transport. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

262.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

262.4 Signs and Symptoms

- You may experience amenorrhea, vaginal bleeding, and unilateral lower abdominal pain
- Ruptured ectopic pregnancy presents with sudden severe pain, syncope, involving bleeding shock, and peritonitis.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

262.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

262.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

262.7 Diagnosis

- Doctors diagnose this using serum β-hCG and transvaginal ultrasound (empty uterine cavity, adnexal mass, free fluid)
- A discriminatory zone of β-hCG >2000 IU/L with no intrauterine pregnancy suggests ectopic pregnancy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

262.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

262.9 Treatment Options

- The right treatment depends on stability: stable patients with small ectopic may be treated with methotrexate
- Unstable patients require using small incisions and a camera salpingectomy or salpingotomy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

262.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

262.11 Outlook

- The outlook is generally positive with early diagnosis
- Ruptured ectopic pregnancy is life-threatening.

Chapter 263

Edwards Syndrome (Trisomy 18)

263.1 Overview

Edwards syndrome is a chromosomal disorder caused by trisomy 18, with an incidence of 1 in 5000 live births. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

263.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

263.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

263.4 Signs and Symptoms

- You may experience severe intellectual disability, growth restriction, low-set malformed ears, micrognathia, prominent occiput, clenched fists with overlapping fingers, rocker-bottom feet, and congenital heart disease. Median survival is less than 1 year
- Only 5–10% survive to age 1 year.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

263.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

263.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

263.7 Diagnosis

Doctors diagnose this using karyotype.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

263.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

263.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

263.10 Living with It

Treatment typically involves supportive and palliative.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 264

Ehlers-Danlos Syndrome

264.1 Overview

Ehlers-Danlos syndrome (EDS) is a heterogeneous group of connective tissue disorders caused by defects in collagen synthesis and processing, with a combined prevalence of 1 in 5,000. The most common subtype, hypermobile EDS (hEDS, 90%), has unknown genetic basis. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

264.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Classical EDS (cEDS) results from COL5A1/A2 mutations causing abnormal type V collagen
- Vascular EDS (vEDS) results from COL3A1 mutations causing abnormal type III collagen.

264.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

264.4 Signs and Symptoms

Symptoms vary depending on subtype: hEDS presents with joint hypermobility, skin hyperextensibility, easy bruising, chronic pain, and recurrent dislocations

- CEDS features velvety, hyperextensible skin, atrophic scars, and easy bruising
- VEDS presents with thin, translucent skin, characteristic facial appearance (prominent eyes, thin nose, lobeless ears), and life-threatening arterial, intestinal, or uterine rupture.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

264.5 Progression and Stages

Doctors diagnose this based on Villefranche classification and genetic testing. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

264.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

264.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

264.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

264.9 Treatment Options

Treatment focuses on symptomatic treatment: physical therapy for joint instability, pain management, wound care, and cardiovascular surveillance in vEDS. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

264.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

264.11 Outlook

- Outlook is normal for hEDS and cEDS
- VEDS has a median survival of 50 years, primarily due to arterial rupture.

Chapter 265

Ehrlichiosis

265.1 Overview

Ehrlichiosis is a tick-borne illness caused by obligate intracellular bacteria of the family Anaplasmataceae, with human monocytic ehrlichiosis (HME) caused by *Ehrlichia chaffeensis*, transmitted by the Lone Star tick (*Amblyomma americanum*), with over 1500 cases annually in the United States. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

265.2 Causes

This condition happens because of *E. chaffeensis* infecting monocytes and macrophages, forming intracytoplasmic microcolonies (morulae), and inhibiting phagolysosomal fusion. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

265.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

265.4 Signs and Symptoms

You may experience an incubation of 5–14 days, with fever, severe headache, muscle pain, malaise, nausea, and rigors, with rash being less common (30%) than in RMSF, and laboratory findings of leukopenia, thrombocytopenia, and elevated liver transaminases being characteristic. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

265.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

265.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

265.7 Diagnosis

Doctors diagnose this using PCR of blood (high sensitivity in the first week). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

265.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

265.9 Treatment Options

Treatment options include doxycycline. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

265.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

265.11 Outlook

- Most people recover well with treatment
- Mortality 1–3%.

Chapter 266

Electrical Injury and Lightning Strike

266.1 Overview

Electrical injuries from high-voltage (>1,000 V) and low-voltage (<1,000 V, typically household 110–240 V) sources cause damage via thermal burns, direct cellular injury, and electroporation of cell membranes. Current follows the path of least resistance: nerves and blood vessels are most affected, followed by muscle, then skin and bone. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

266.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

266.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

266.4 Signs and Symptoms

Clinical features: cutaneous burns (entry/exit wounds—may be deceptively small), muscle tissue death, rhabdomyolysis, compartment syndrome, irregular heartbeats, neurological injury, and fractures. Lightning strike produces a massive direct current (DC) shock of up to 30 million volts and 30,000 amps. Clinical features: cardiac arrest, respiratory arrest, confusion, amnesia, kerauonographic markings, tympanic membrane rupture, and cataracts. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

266.5 Progression and Stages

Management: advanced cardiac life support, fluid resuscitation, surgical fasciotomy for compartment syndrome, and wound care. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

266.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

266.7 Diagnosis

Doctors diagnose this using history and physical examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

266.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

266.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

266.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

266.11 Outlook

- The outlook varies from person to person
- Lightning strike patients in cardiac arrest may be salvageable with prolonged resuscitation.

Chapter 267

Empyema

267.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

267.2 Causes

This condition happens because of bacterial infection spreading to the pleural space. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

267.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

267.4 Signs and Symptoms

You may experience persistent fever despite appropriate antibiotic therapy for pneumonia, pleuritic chest pain, and shortness of breath. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

267.5 Progression and Stages

Treatment options include chest tube removal of fluid, intrapleural fibrinolytics (alteplase and DNase), VATS decortication for advanced disease, and appropriate antibiotics. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

267.6 Complications

- Empyema is the accumulation of purulent fluid within the pleural space, most commonly as a complication of pneumonia, thoracic surgery, or trauma
- It progresses through exudative, fibrinopurulent, and organizational stages.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

267.7 Diagnosis

Doctors diagnose this using chest X-ray, ultrasound, and diagnostic thoracentesis showing purulent fluid with low pH, low glucose, high LDH, and positive Gram stain or culture. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

267.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

267.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

267.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

267.11 Outlook

The outlook is good with prompt removal of fluid and appropriate antibiotics. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 268

End-Stage Renal Disease

268.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

268.2 Causes

This condition happens because of progressive nephron loss from any cause of CKD. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

268.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

268.4 Signs and Symptoms

You may experience uremic syndrome: nausea, vomiting, anorexia, itching, pericarditis, encephalopathy, peripheral neuropathy, and restless legs syndrome. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected

area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

268.5 Progression and Stages

End-stage kidney disease (ESRD) is the final stage of CKD (GFR <15 mL/min/1.73 m^2) where the kidneys can no longer sustain life without kidney replacement therapy. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

268.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

268.7 Diagnosis

Doctors diagnose this based on GFR criteria and clinical manifestations. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

268.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

268.9 Treatment Options

Post-transplant immunosuppression (tacrolimus, mycophenolate, prednisone) carries risks of infection and malignancy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

268.10 Living with It

- Treatment focuses on dialysis (hemodialysis or peritoneal dialysis) or kidney transplantation
- Kidney transplantation is the treatment of choice for eligible patients, offering superior survival and quality of life.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

268.11 Outlook

- The outlook varies from person to person
- Living donor transplantation provides the best outcomes, while dialysis has a 5-year survival of approximately 40–50%.

Chapter 269

Endometrial / Uterine Cancer

269.1 Overview

Endometrial cancer is the most common gynecologic malignancy in developed countries. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

269.2 Causes

Two broad histologic types exist: type I (endometrioid adenocarcinoma, 80%, estrogen-driven, good outlook) and type II (serous carcinoma, clear cell carcinoma, not estrogen-driven, poor outlook). Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

269.3 Risk Factors

Risk factors include unopposed estrogen exposure, Lynch syndrome, advanced age, and nulliparity.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption

- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

269.4 Signs and Symptoms

Clinical The condition usually appears as postmenopausal bleeding (90% present with bleeding). Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

269.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for early-stage disease is total hysterectomy with bilateral salpingo-oophorectomy
- Progestin therapy may be used for fertility-sparing management. In most cases, the

outlook is good: 5-year survival is 95% for stage I disease.

269.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

269.7 Diagnosis

Doctors diagnose this using endometrial biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

269.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

269.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

269.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

269.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 270

Endometriosis

270.1 Overview

Endometriosis is the most common cause of chronic pelvic pain and a leading cause of infertility. Beyond first-line therapies (NSAIDs, combined oral contraceptives), surgical removal of deeply infiltrating endometriosis may be necessary for pain refractory to medical management. GnRH agonists or antagonists with add-back therapy are used for moderate to severe disease. Dienogest and norethindrone acetate are effective progestin-only options. Treatment of endometriosis-associated infertility includes using small incisions and a camera removal followed by IVF. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

270.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

270.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

270.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

270.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

270.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

270.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

270.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

270.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

270.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

270.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 271

Endometriosis

271.1 Overview

Endometriosis is defined by the presence of endometrial-like tissue outside the uterus, affecting approximately 10% of reproductive-age women, with the most common sites being the ovaries (endometriomas—“chocolate cysts”), uterosacral ligaments, pelvic peritoneum, and rectovaginal septum. The pathogenesis is debated, with Sampson’s theory of retrograde menstruation being the most widely accepted. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

271.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

271.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

271.4 Signs and Symptoms

You may experience chronic pelvic pain (dysmenorrhea, dyspareunia, chronic pelvic pain), infertility (present in 30–50% of women with endometriosis), and cyclic symptoms. Diagnosis is typically made by laparoscopy with visualization of endometriotic implants and histological confirmation, with MRI useful for deep infiltrating endometriosis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

271.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

271.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

271.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

271.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

271.9 Treatment Options

Treatment options include NSAIDs for pain, combined oral contraceptives, progestins, GnRH agonists/antagonists with add-back therapy, and surgical removal/ablation of endometriotic lesions. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

271.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

271.11 Outlook

The outlook varies from person to person, with recurrence possible after treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 272

Eosinophilic Fasciitis (Shulman Syndrome)

272.1 Overview

Eosinophilic fasciitis is a rare, fibrosing disorder of the fascia characterized by symmetrical, painful swelling and induration of the skin and subcutaneous tissue, with peripheral eosinophilia, with an incidence of approximately 1 per million, equal sex distribution, and typically affecting adults 40–60 years of age. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

272.2 Causes

The cause is unknown, but triggers include vigorous exercise or trauma (in up to 50%), infections, and medications (statins, antitubercular drugs), with Disease mechanism involving inflammation and thickening of the deep fascia, infiltration by lymphocytes, eosinophils, macrophages, and plasma cells, leading to collagen deposition and scarring. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

272.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

272.4 Signs and Symptoms

You may experience acute or subacute onset of symmetrical, painful swelling and induration of the arms and legs (sparing the hands, feet, and face—a distinguishing feature from systemic sclerosis), with a characteristic "peau d'orange" or cobblestone appearance, and the groove sign (linear depression overlying superficial veins) being a pathognomonic finding, with joint contractures developing due to fascial scarring.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

272.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

272.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

272.7 Diagnosis

To make the diagnosis, doctors look for a deep incisional biopsy including skin, subcutaneous tissue, fascia, and muscle, showing fasciitis with a mixed inflammatory infiltrate and often eosinophils, with peripheral eosinophilia ($>5\%$ or absolute count $>500/\mu$L) present in 70–90%.

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

272.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

272.9 Treatment Options

Treatment typically involves with corticosteroids (prednisone 0.5–1 mg/kg/day), with methotrexate, mycophenolate, and azathioprine used as steroid-sparing agents. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

272.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

272.11 Outlook

- The outlook varies from person to person
- Most patients improve with corticosteroids, but joint contractures may persist.

Chapter 273

Eosinophilic Granulomatosis with Polyangiitis

273.1 Overview

Eosinophilic granulomatosis with polyangiitis is a rare ANCA-associated vasculitis characterized by asthma, eosinophilia, and necrotizing vasculitis of small and medium-sized vessels with extravascular eosinophilic granulomas, with an incidence of 1–3 per million, equal sex distribution, and typically presenting in the fourth to fifth decade. The condition involves a Th2-driven immune response with prominent eosinophil activation and degranulation, ANCA (typically MPO-ANCA, positive in 30–50%, more commonly in patients with kidney and nervous system involvement), and eosinophilic infiltration of tissues. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

273.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

273.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

273.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

273.5 Progression and Stages

Clinical features evolve through three phases: prodromal (asthma, allergic rhinitis, often years before systemic disease), eosinophilic (peripheral eosinophilia $>1500/\mu$L, eosinophilic pneumonia, gastroenteritis), and vasculitic (systemic vasculitis with mononeuritis multiplex—the most common vasculitic manifestation, purple spots on the skin, glomerulonephritis, cardiomyopathy, and skin nodules), with cardiac involvement (eosinophilic myocarditis, coronary arteritis, heart failure) being the leading cause of death. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

273.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

273.7 Diagnosis

Doctors diagnose this based on the 2022 ACR/EULAR criteria. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

273.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

273.9 Treatment Options

- Treatment typically involves stratified: patients without poor prognostic factors (FFS five-factor score=0) receive corticosteroids alone
- Patients with poor prognostic factors (FFS $\geq$1) receive corticosteroids plus cyclophosphamide or rituximab.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

273.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

273.11 Outlook

- Your outlook depends on cardiac involvement and kidney function
- 5-year survival exceeds 90%.

Chapter 274

Epidermolysis Bullosa

274.1 Overview

Epidermolysis bullosa (EB) is a group of inherited blistering disorders characterized by mechanical fragility of the skin and mucous membranes, with an overall prevalence of 1 in 50,000. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

274.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

274.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

274.4 Signs and Symptoms

You may experience blister formation at sites of minor friction or trauma, ranging from mild acral blistering (EBS) to widespread blistering, scarring, milia, nail dystrophy, and mitten-hand deformities from repeated scarring (severe DEB). Extracutaneous involvement includes oral, esophageal, and laryngeal blistering, corneal erosions, anemia, growth retardation, and increased risk of aggressive squamous cell carcinoma in severe DEB. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

274.5 Progression and Stages

This condition happens because of mutations in genes encoding structural proteins of the skin, classified by the level of blister formation within the basement membrane zone: EB simplex (EBS, keratin 5/14 mutations, intraepidermal cleavage), junctional EB (JEB, laminin 332/collagen XVII mutations, lamina lucida cleavage), and dystrophic EB (DEB, collagen VII mutations, sublamina densa cleavage). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

274.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

274.7 Diagnosis

Doctors diagnose this based on clinical features, skin biopsy for immunofluorescence mapping, transmission electron microscopy, and genetic testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

274.8 Prevention

Treatment typically involves multidisciplinary: gentle skin care, wound dressings (non-adherent), prevention and treatment of infection, nutritional support, esophageal dilation for strictures, physical therapy for contractures, and surveillance for squamous cell carcinoma.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

274.9 Treatment Options

Emerging therapies include gene therapy (beremagene geperpavec for DEB), protein replacement therapy, and cell-based therapies. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

274.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

274.11 Outlook

- Recovery varies widely: EBS has normal life expectancy
- Severe JEB and recessive DEB cause significant morbidity and mortality from infection, malnutrition, and aggressive squamous cell carcinoma.

Chapter 275

Epidural Hematoma

275.1 Overview

An epidural hematoma is a traumatic intracranial bleeding accumulating between the inner skull table and the dura mater, most commonly resulting from skull fracture crossing the pterion with laceration of the middle meningeal artery. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

275.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

275.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

275.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance

or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

275.5 Progression and Stages

You may experience transient loss of consciousness at injury followed by a lucid interval (lasting hours), after which neurological deterioration ensues with severe headache, contralateral hemiparesis, ipsilateral dilated and unreactive pupil (due to uncal herniation with CN III compression), and Cushing triad (high blood pressure, bradycardia, irregular respiration). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

275.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

275.7 Diagnosis

Doctors diagnose this using non-contrast head CT showing a convex, biconvex (lens-shaped) hyperdense collection that does not cross cranial suture lines. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

275.8 Prevention

Management requires urgent neurosurgical surgical opening of the skull and hematoma evacuation to prevent fatal brain herniation.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

275.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

275.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

275.11 Outlook

Most people recover well if evacuated promptly before herniation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 276

Epiglottitis

276.1 Overview

Epiglottitis is a rapidly progressive, life-threatening inflammation of the epiglottis and supraglottic structures. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

276.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

276.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

276.4 Signs and Symptoms

You may experience abrupt onset of high fever, severe sore throat, drooling, muffled voice (“hot potato” voice), tripod positioning, and inspiratory stridor. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected

area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

276.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

276.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

276.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Lateral neck radiograph shows the “thumbprint sign” (enlarged epiglottis). Airway Treatment typically involves the priority: immediate establishment of a secured airway (endotracheal intubation in the operating room). Blood and epiglottic cultures are obtained, and intravenous antibiotics (ceftriaxone or cefotaxime) are initiated.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

276.8 Prevention

Historically caused by Haemophilus influenzae type b (Hib), its incidence has declined dramatically due to the Hib vaccine.

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

276.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

276.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

276.11 Outlook

Most people recover well with prompt airway management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 277

Epiglottitis

277.1 Overview

Epiglottitis (supraglottitis) is a rapidly progressive, life-threatening cellulitis and swelling of the epiglottis and surrounding supraglottic structures (arytenoids, aryepiglottic folds), leading to catastrophic upper airway obstruction. Primary bacterial invasion causes massive supraglottic swelling, narrowing the laryngeal inlet. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

277.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

277.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

277.4 Signs and Symptoms

Clinical features present with acute onset of high fever, severe sore throat, odynophagia, muffled 'hot potato' voice, drooling, and respiratory distress. Patients characteristically assume a 'tripod position' (sitting upright, leaning forward, neck hyperextended, chin thrust forward) to optimize airway caliber. Direct visualization with a tongue depressor is strictly contraindicated due to risk of precipitating complete laryngospasm. Lateral neck X-ray reveals a swollen epiglottis ('thumbprint sign'). Management mandates immediate airway stabilization in an operating room via endotracheal intubation or tracheostomy, followed by parenteral broad-spectrum antibiotics (ceftriaxone plus vancomycin). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

277.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

277.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

277.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

277.8 Prevention

Historically caused by Haemophilus influenzae type b (Hib) in young children, widespread Hib vaccination has shifted epidemiology toward older adults and non-typeable H. influenzae, Streptococcus pneumoniae, Staphylococcus aureus, and Beta-hemolytic streptococci.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

277.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

277.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

277.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 278

Epilepsy

278.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures resulting from abnormal excessive or synchronous neuronal activity in the brain
- It is defined by two or more unprovoked seizures occurring more than 24 hours apart, or one unprovoked seizure with a high recurrence risk.

278.2 Causes

This condition happens because of genetic, structural (eg., cortical dysplasia, tumor, stroke), infectious, metabolic, or immune causes. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

278.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

278.4 Signs and Symptoms

Your symptoms depend on seizure type. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

278.5 Progression and Stages

Seizure classification distinguishes focal (partial) from generalized seizures, with further subtypes including absence, tonic-clonic, myoclonic, atonic, and tonic seizures. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

278.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

278.7 Diagnosis

Diagnosis relies on clinical history, EEG showing epileptiform discharges, and neuroimaging. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

278.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

278.9 Treatment Options

- Treatment focuses on antiseizure medications selected based on seizure type and syndrome classification
- Drug-resistant epilepsy may be treated with epilepsy surgery, vagus nerve stimulation, or responsive neurostimulation.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

278.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

278.11 Outlook

The outlook varies from person to person, with many patients achieving seizure control with appropriate therapy. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 279

Epistaxis

279.1 Overview

Epistaxis is a nosebleed that affects approximately 60% of the population at some point, with most bleeds originating from Kiesselbach's plexus on the anterior nasal septum (anterior epistaxis). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

279.2 Causes

This condition happens because of local trauma or systemic factors such as dry air, digital trauma, allergic rhinitis, nasal foreign bodies, anticoagulant use, high blood pressure, coagulopathy, and hereditary involving bleeding telangiectasia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

279.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

279.4 Signs and Symptoms

- You may experience bleeding from the anterior or posterior nasal cavity
- Posterior bleeds from the sphenopalatine artery are less common but more severe.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

279.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

279.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

279.7 Diagnosis

- Doctors diagnose this based on clinical examination. Anterior epistaxis is managed with direct pressure, chemical burning to seal tissue (silver nitrate), or anterior nasal packing
- Posterior epistaxis requires posterior packing, using a thin camera tube sphenopalatine artery ligation, or embolization.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

279.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

279.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

279.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

279.11 Outlook

The outlook is good, with most cases resolving with simple measures. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 280

Epstein-Barr Virus / Infectious Mononucleosis (Systemic Aspects)

280.1 Overview

Epstein-Barr virus is a gamma-herpesvirus that infects > 90% of the global population, with primary infection causing infectious mononucleosis (IM) in adolescents and young adults, and EBV being associated with several malignancies: Burkitt lymphoma, Hodgkin lymphoma, post-transplant lymphoproliferative disorder, nasopharyngeal carcinoma, and gastric carcinoma. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

280.2 Causes

This condition happens because of EBV infecting B lymphocytes via CD21 (CR2) receptor, leading to polyclonal B cell growth, with the host T cell response producing the atypical lymphocytosis characteristic of IM. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

280.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to

endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

280.4 Signs and Symptoms

You may experience an incubation of 30–50 days, with the classical triad of fever, pharyngitis (severe, exudative), and cervical lymphadenopathy, along with profound fatigue, splenomegaly, hepatomegaly, and rash (especially with ampicillin/amoxicillin administration). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

280.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

280.6 Complications

Systemic complications include airway obstruction, splenic rupture, hemolytic anemia, thrombocytopenia, myocarditis, and encephalitis.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

280.7 Diagnosis

Doctors diagnose this using atypical lymphocytosis, Monospot test, and EBV-specific antibodies. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

280.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification

- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

280.9 Treatment Options

Treatment typically involves supportive. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

280.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

280.11 Outlook

Most people recover well for acute IM. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 281

Erectile Dysfunction

281.1 Overview

Erectile dysfunction is the persistent inability to achieve or maintain an erection sufficient for satisfactory sexual performance, affecting approximately 50% of men between ages 40 and 70 to varying degrees. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

281.2 Causes

This condition happens because of impaired vascular, neurological, or endocrine function. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

281.3 Risk Factors

Risk factors include age, cardiovascular disease, diabetes mellitus, high blood pressure, dyslipidemia, smoking, obesity, depression, and medications, with the most common cause being vasculogenic (arterial insufficiency or veno-occlusive dysfunction), often reflecting subclinical atherosclerosis. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

281.4 Signs and Symptoms

You may experience difficulty achieving or maintaining erections. Evaluation includes sexual history, hormonal testing (testosterone, prolactin, TSH), and cardiovascular risk assessment. The first-choice treatment typically involves PDE5 inhibitors (sildenafil, tadalafil, vardenafil, avanafil), with second-line options including vacuum erection devices and intracavernosal injection therapy, and third-line being penile prosthesis implantation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

281.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

281.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

281.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

281.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

281.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

281.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

281.11 Outlook

The outlook is good with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 282

Esophageal Atresia / Tracheoesophageal Fistula

282.1 Overview

Esophageal atresia (EA) with or without tracheoesophageal fistula (TEF) is a congenital anomaly with incidence of 1 per 3500 live births. The most common type (86%) is EA with distal TEF (Gross Type C). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

282.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

282.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

282.4 Signs and Symptoms

Clinical presentation includes polyhydramnios (prenatal), excessive drooling, coughing, choking, and cyanosis with feeding. Failure to pass a nasogastric tube beyond 8–10 cm from the gums is diagnostic. Radiography shows a coiled nasogastric tube in the proximal esophageal pouch and gas in the stomach (if distal TEF is present). Associated anomalies include the VACTERL association. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

282.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

282.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

282.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

282.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

282.9 Treatment Options

Treatment typically involves surgical: primary repair with division of the fistula and end-to-end esophageal anastomosis. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

282.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

282.11 Outlook

Most people recover well for isolated EA/TEF. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 283

Esophageal Varices

283.1 Overview

Esophageal varices are dilated submucosal veins in the lower third of the esophagus resulting from portal high blood pressure, most commonly caused by liver cirrhosis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

283.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of collateral venous circulation routing portal blood flow from the left gastric (coronary) vein into the systemic azygos vein due to elevated intrahepatic vascular resistance. Unruptured varices are asymptomatic
- Rupture causes catastrophic upper digestive tract bleeding presenting with painless hematemesis, melena, hypovolemic shock, and altered consciousness.

283.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

283.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

283.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

283.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

283.7 Diagnosis

- Doctors diagnose this using upper digestive tract endoscopy showing tortuous blue submucosal vessels. Treatment for acute variceal bleeding requires hemodynamic resuscitation, IV octreotide/somatostatin (to reduce splanchnic blood flow), prophylactic IV ceftriaxone, and urgent using a thin camera tube band ligation (EBL)
- Transjugular intrahepatic portosystemic shunt (TIPS) is indicated for refractory bleeding. Mortality per bleeding episode is 15–20%.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

283.8 Prevention

Primary prophylaxis includes non-selective beta-blockers (nadolol, propranolol) or elective using a thin camera tube band ligation.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

283.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

283.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

283.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 284

Essential Thrombocythemia

284.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Essential thrombocythemia is a myeloproliferative tumor characterized by sustained thrombocytosis (platelet count >450,000/μL) with increased megakaryocyte growth, in the absence of other myeloproliferative tumors. The condition is driven by JAK2 V617F (50%), CALR (25–30%), or MPL (3–5%) mutations
- Approximately 10% are triple-negative.

284.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

284.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

284.4 Signs and Symptoms

Clinical features may include microvascular symptoms (erythromelalgia, headache, visual disturbances), blood clot formation (arterial or venous), or bleeding (with very high platelet counts >1,000,000/μL due to acquired von Willebrand syndrome), though many patients are asymptomatic. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

284.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

284.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

284.7 Diagnosis

- Doctors diagnose this based on sustained thrombocytosis with the characteristic mutation profile. Treatment for low-risk ET involves observation with low-dose aspirin
- High-risk ET (age >60 or history of blood clot formation) requires cytoreduction with hydroxyurea or anagrelide.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

284.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

284.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

284.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

284.11 Outlook

The outlook is generally positive with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 285

Essential Tremor

285.1 Overview

Essential tremor is one of the most common movement disorders, affecting 1% of the general population and 5% of individuals over 65 years. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

285.2 Causes

This condition happens because of abnormal oscillatory activity in the cerebello-thalamo-cortical circuit, with genetic heterogeneity (autosomal dominant inheritance in familial cases, ETM1/ETM2 loci). Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

285.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

285.4 Signs and Symptoms

You may experience bilateral, symmetric postural and kinetic tremor of the upper limbs (6–12 Hz), which may also involve the head (titubation), voice, chin, or trunk. Tremor is typically absent at rest, worsens with goal-directed movement, and is exacerbated by stress, fatigue, and caffeine. Alcohol consumption temporarily reduces tremor amplitude in 50–70% of patients. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

285.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

285.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

285.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, based on the Movement Disorder Society consensus criteria and exclusion of other causes (Parkinson disease, dystonic tremor, enhanced physiologic tremor, drug-induced tremor). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

285.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

285.9 Treatment Options

Treatment options include The first-choice treatment with propranolol or primidone, alternative agents (topiramate, gabapentin, benzodiazepines), and for medication-refractory cases, deep brain stimulation of the ventral intermediate nucleus (VIM) of the thalamus or focused ultrasound thalamotomy. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

285.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

285.11 Outlook

- Outlook is non-cancerous but slowly progressive
- Disability from tremor may be significant.

Chapter 286

Esthesioneuroblastoma

286.1 Overview

Esthesioneuroblastoma (olfactory neuroblastoma) is a rare, cancerous neuroectodermal tumor originating from the specialized olfactory neuroepithelium lining the cribriform plate, superior nasal cavity, and upper nasal septum. The tumor exhibits a bimodal age distribution (10–20 and 50–60 years) and is marked by slow, locally invasive growth with erosion into the anterior cranial fossa, orbits, and paranasal sinuses. Histopathology shows small round blue cells forming Homer-Wright pseudorosettes with positive neuroendocrine markers (synaptophysin, chromogranin, S100 at tumor margins). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

286.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

286.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

286.4 Signs and Symptoms

Staging uses the Kadish system. Management requires multimodality therapy comprising craniofacial or using a thin camera tube endonasal removal combined with adjuvant radiotherapy and chemotherapy. 5-year survival ranges from 60 to 80%. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

286.5 Progression and Stages

Symptoms often appear as unilateral nasal obstruction, recurrent epistaxis, loss of smell, and advanced orbital symptoms (proptosis, diplopia). Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II:

Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

286.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

286.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

286.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

286.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

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- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
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- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

286.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

286.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 287

Ethical Issues

287.1 Overview

Geriatric care raises complex ethical challenges: This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

Informed consent: Must be obtained from patients with capacity; for those lacking capacity, consent is obtained from the legally authorized representative.

- **Decision-making capacity:** Capacity is task-specific and can be assessed using the Aid to Capacity Evaluation (ACE) tool. A patient with capacity can understand relevant information, appreciate the situation and its consequences, reason about treatment options, and communicate a choice. Incapacity requires surrogate decision-making based on substituted judgment or best interests.
- **Truth-telling:** Geriatric patients should be informed of their diagnosis and outlook in a sensitive, individualized manner. Withholding information (therapeutic privilege) is rarely justified.
- **Euthanasia and assisted dying:** Medical assistance in dying (MAiD) is legal in several jurisdictions. In Canada, MAiD is available to adults with a grievous and irremediable medical condition causing enduring suffering, with a reasonably foreseeable natural death. Clinicians have an ethical and legal duty to report suspected abuse to adult protective services.

287.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

287.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

287.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

287.5 Progression and Stages

Advance requests for MAiD in dementia require careful ethical and legal consideration.

Elder abuse and neglect: Affects 10–15% of older adults, including physical, emotional, financial abuse, and neglect. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

287.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

287.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

287.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

287.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

287.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

287.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 288

Ewing Sarcoma

288.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Ewing sarcoma is the second most common primary bone malignancy in children and adolescents, arising from primitive neuroectodermal cells. The classic translocation t(11
- 22)(q24
- Q12) results in the EWS-FLI1 fusion gene. It typically affects the diaphysis of long bones (femur, tibia, fibula) and the pelvis.

288.2 Causes

This condition happens because of the EWS-FLI1 fusion gene driving oncogenesis. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

288.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
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- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

288.4 Signs and Symptoms

You may experience bone pain, swelling, and a mass, with systemic symptoms (fever, weight loss) that may mimic infection. X-rays show a permeative lytic lesion with "onion skin" periosteal reaction. MRI and CT chest are essential for staging. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

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- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

288.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

288.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
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- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

288.7 Diagnosis

Your doctor confirms the diagnosis with biopsy with molecular testing for the translocation. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

288.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

288.9 Treatment Options

Treatment options include neoadjuvant chemotherapy (vincristine, doxorubicin, cyclophosphamide alternating with ifosfamide/etoposide), local control with surgery and/or radiation therapy, and adjuvant chemotherapy. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
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288.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

288.11 Outlook

- The outlook is generally positive for localized disease
- The 5-year survival rate is approximately 70%.

Chapter 289

Ewing Sarcoma in Children

289.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Ewing sarcoma is the second most common cancerous bone tumor in children and adolescents (see Chapter 26, Diseases of the Musculoskeletal System for comprehensive coverage). Peak incidence is 10–20 years. It arises in the diaphysis of long bones and flat bones (pelvis, ribs). The hallmark is a t(11
- 22)(q24
- Q12) translocation producing the EWSR1-FLI1 fusion gene.

289.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

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- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

289.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

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289.8 Prevention

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- Avoid exposure to known carcinogens in the workplace and environment

289.9 Treatment Options

Treatment options include chemotherapy, surgical removal, and/or radiation therapy. Survival for localized disease is 70–75%. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

289.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

289.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 290

Facioscapulohumeral Muscular Dystrophy

290.1 Overview

Facioscapulohumeral muscular dystrophy (FSHD) is an autosomal dominant myopathy caused by contraction of D4Z4 repeats on chromosome 4q35, leading to inappropriate expression of the DUX4 gene. It has a prevalence of approximately 1 in 8000–20,000. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

290.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

290.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

290.4 Signs and Symptoms

- You may experience initial involvement of facial muscles (inability to fully close the eyes, difficulty whistling, transverse smile) and scapular stabilizers ("scapular winging"), followed by humeral and pelvic girdle muscles
- Asymmetric involvement is common. Hearing loss and retinal telangiectasias (Coats disease) may occur.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

290.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

290.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

290.7 Diagnosis

Doctors diagnose this based on genetic testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

290.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

290.9 Treatment Options

- Treatment options include physical therapy and orthopedic interventions (scapular bracing/fusion) and respiratory monitoring
- There is no curative treatment.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

290.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

290.11 Outlook

The outlook varies from person to person, with slowly progressive weakness but usually normal life expectancy.

Chapter 291

Failure to Thrive

291.1 Overview

Failure to thrive (FTT) in older adults is a geriatric syndrome of unintentional weight loss (>5% in 6–12 months), decreased appetite, poor nutrition, and functional decline, often accompanied by social withdrawal and depression. It is multifactorial, resulting from medical (malignancy, dementia, depression, infection, hyperthyroidism, malabsorption, difficulty swallowing), social (isolation, poverty, caregiver loss), and psychological (bereavement, delirium) causes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

291.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

291.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

291.4 Signs and Symptoms

You may experience weight loss, poor appetite, functional decline, and social withdrawal. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

291.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

291.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

291.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with identification of underlying causes. Treatment targets underlying causes, nutritional supplementation, appetite stimulants (megestrol acetate, mirtazapine), and social support. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

291.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

291.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

291.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

291.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 292

Falls in the Elderly

292.1 Overview

Falls are the leading cause of fatal and non-fatal injuries in adults aged 65 and older. Approximately 30% of community-dwelling older adults fall each year, increasing to 50% in those over 80. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

292.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

292.3 Risk Factors

Risk factors are classified as intrinsic (muscle weakness, gait/balance impairment, visual impairment, thinking and memory impairment, orthostatic hypotension, polypharmacy) and extrinsic (environmental hazards). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

292.4 Signs and Symptoms

You may experience fall-related injuries, fear of falling, and functional decline. The STEADI toolkit recommends annual fall screening. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

292.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

292.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

292.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

292.8 Prevention

Prevention is multifactorial: strength and balance exercises (Tai Chi, Otago Exercise Programme), medication review and deprescribing, vitamin D supplementation, vision optimization, home hazard modification, and hip protectors for high-risk individuals.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

292.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

292.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

292.11 Outlook

- The outlook varies from person to person
- Multifactorial intervention reduces falls by 20–30%.

Chapter 293

Familial Adenomatous Polyposis

293.1 Overview

Familial adenomatous polyposis (FAP) is an autosomal dominant condition caused by germline mutations in the APC gene on chromosome 5q21, characterized by the development of hundreds to thousands of adenomatous polyps throughout the colon and rectum, typically beginning in adolescence. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

293.2 Causes

This condition happens because of APC gene mutation leading to uncontrolled cell growth. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

293.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

293.4 Signs and Symptoms

- You may experience the development of polyps in adolescence
- Without treatment, CRC develops in virtually 100% of patients by age 40.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

293.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

293.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

293.7 Diagnosis

Doctors diagnose this based on colonoscopy showing innumerable polyps and genetic testing. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

293.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

293.9 Treatment Options

Treatment focuses on prophylactic proctocolectomy with ileal pouch-anal anastomosis (IPAA) recommended in the late teens to early twenties. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

293.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

293.11 Outlook

- Most people recover well with prophylactic surgery
- Attenuated FAP (AFAP) has fewer polyps (10–100) and later-onset CRC. Gardner syndrome is a variant with extraintestinal manifestations (osteomas, desmoid tumors, congenital enlargement of retinal pigment epithelium).

Chapter 294

Familial Combined Hyperlipidemia

294.1 Overview

Familial combined hyperlipidemia (FCHL) is the most common genetic dyslipidemia, with a prevalence of 1–2% in the general population and up to 10–20% in patients with premature coronary artery disease. It is a complex oligogenic disorder characterized by overproduction of apolipoprotein B-100 and very-low-density lipoprotein (VLDL) by the liver. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

294.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

294.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

294.4 Signs and Symptoms

You may experience premature coronary artery disease, obesity, insulin resistance, and the absence of tendon xanthomas (distinguishing it from FH). The lipid phenotype is variable: affected individuals may have elevated total cholesterol and/or elevated triglycerides. Diagnosis is made in individuals with elevated apolipoprotein B (greater than 120 mg/dL), elevated LDL-C and/or triglycerides, and a family history of premature cardiovascular disease.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

294.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

294.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

294.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

294.8 Prevention

Treatment options include lifestyle modification, statins as first-line therapy, with the addition of fibrates for hypertriglyceridemia.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

294.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

294.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

294.11 Outlook

The outlook is generally positive with treatment.

Chapter 295

Familial Hypercholesterolemia

295.1 Overview

Familial hypercholesterolemia (FH) is an autosomal codominant disorder of LDL metabolism caused by loss-of-function mutations in the LDLR gene (LDL receptor, 90% of cases), APOB (apolipoprotein B-100), or gain-of-function mutations in PCSK9. It is the most common monogenic lipid disorder, with a prevalence of 1 in 250 for heterozygous FH (HeFH) and 1 in 300,000 for homozygous FH (HoFH). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

295.2 Causes

This condition happens because of impaired liver clearance of LDL particles. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

295.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

295.4 Signs and Symptoms

- You may experience tendon xanthomas (Achilles, extensor tendons of the hands), tuberous xanthomas, xanthelasmas, corneal arcus before age 45, and premature atherosclerotic cardiovascular disease
- Homozygous FH presents in childhood with LDL-C greater than 500 mg/dL, widespread xanthomas, and coronary atherosclerosis before age 20.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

295.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

295.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

295.7 Diagnosis

Doctors diagnose this using clinical criteria (Dutch Lipid Clinic Network or Simon Broome criteria) supported by genetic testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

295.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

295.9 Treatment Options

Treatment options include high-dose statins with ezetimibe as first-line for HeFH

- PCSK9 inhibitors for inadequate response
- For HoFH, all available agents and LDL apheresis are needed.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

295.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

295.11 Outlook

Most people recover well with early diagnosis and treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 296

Fanconi Syndrome

296.1 Overview

Fanconi syndrome is generalized dysfunction of the proximal kidney tubule, leading to impaired reabsorption of glucose, amino acids, phosphate, bicarbonate, uric acid, and other solutes. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

296.2 Causes

Causes include hereditary disorders (cystinosis, Wilson's disease, galactosemia), drugs (ifosfamide, outdated tetracycline, aminoglycosides), multiple myeloma, and heavy metal poisoning. This condition happens because of impaired proximal tubular transport mechanisms. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

296.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

296.4 Signs and Symptoms

You may experience metabolic acidosis (proximal RTA), hypophosphatemic rickets/osteomalacia, polyuria, and growth retardation in children.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

296.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

296.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

296.7 Diagnosis

Doctors diagnose this based on laboratory findings of generalized proximal tubular dysfunction. Management addresses the underlying cause and provides supplementation of lost solutes (phosphate, vitamin D, bicarbonate).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

296.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

296.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

296.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

296.11 Outlook

Your outlook depends on the underlying cause.

Chapter 297

Fat Necrosis of the Breast

297.1 Overview

Fat tissue death of the breast is a non-cancerous, non-enzymatic inflammatory reaction resulting from focal adipocyte destruction following blunt trauma, breast surgery, core needle biopsy, or radiation therapy. Disrupted adipocytes release neutral triglycerides that undergo enzymatic saponification by tissue lipases, precipitating a foreign-body granulomatous response dominated by lipid-laden macrophages (foam cells), multinucleated giant cells, and peripheral fibrous tissue scarring. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

297.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

297.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

297.4 Signs and Symptoms

You may experience a solitary, firm, irregular, tethered breast mass occasionally accompanied by skin dimpling, nipple retraction, or bruising, closely mimicking invasive ductal carcinoma. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

297.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

297.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

297.7 Diagnosis

Your doctor confirms the diagnosis with mammography (showing lipid cysts with eggshell calcifications or spiculated masses) and core needle biopsy when cancerous features cannot be excluded. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

297.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

297.9 Treatment Options

- Treatment typically involves conservative with patient reassurance following definitive diagnosis
- Symptomatic oil cysts may be aspirated.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

297.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

297.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 298

Fear of Falling

298.1 Overview

Fear of falling is a common psychological consequence occurring in 20–60% of older adults after a fall, and can occur even without a history of falls. It leads to activity restriction, social isolation, and deconditioning (the "post-fall syndrome"), paradoxically increasing the risk of future falls and functional decline. Assessment uses the Falls Efficacy Scale—International (FES-I) or the Activities-specific Balance Confidence (ABC) Scale. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

298.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

298.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

298.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

298.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

298.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

298.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

298.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

298.9 Treatment Options

Treatment options include thinking and memory-behavioral therapy, graded exposure to feared activities, balance training (Tai Chi, yoga), and addressing the underlying physical deficits that perpetuate fear. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

298.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

298.11 Outlook

The outlook is generally positive with appropriate intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 299

Febrile Seizures

299.1 Overview

Febrile seizures are the most common seizures in childhood, occurring in 2–5% of children between 6 months and 5 years of age. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

299.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of fever (temperature greater than 38 degrees Celsius) associated with an extracranial infection, in the absence of central nervous system infection or defined electrolyte imbalance. Simple febrile seizures are generalized, last less than 15 minutes, and do not recur within 24 hours
- Complex febrile seizures are focal, prolonged (greater than 15 minutes), or recurrent.

299.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

299.4 Signs and Symptoms

You may experience generalized or focal seizure activity triggered by fever. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

299.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

299.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

299.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, after excluding central nervous system infection. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

299.8 Prevention

- Treatment typically involves supportive
- Antipyretics prevent recurrence only partially, and prophylactic anticonvulsants are generally not recommended.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

299.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

299.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

299.11 Outlook

Most people recover well, with most children outgrowing febrile seizures by age 6. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 300

Fecal Incontinence

300.1 Overview

Fecal incontinence (FI)—involuntary loss of solid or liquid stool—affects 10–20% of community-dwelling older adults and up to 50% of nursing home residents. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

300.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

300.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

300.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight

changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

300.5 Progression and Stages

Causes include impaired anal sphincter function (obstetric trauma, surgical injury, age-related deterioration), reduced rectal sensation (diabetes, spinal cord injury, fecal impaction with overflow), diarrhea, and thinking and memory/mobility impairment. Mark's (Vaizey) score to quantify severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

300.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

300.7 Diagnosis

Doctors use the St. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

300.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

300.9 Treatment Options

Addressing constipation and fecal impaction in institutionalized patients is critical. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

300.10 Living with It

Treatment options include dietary modification (fiber supplementation for formed stool, avoidance of dietary triggers), scheduled toilet routines, pelvic floor rehabilitation, antidiarrheal medications, anal plugs, and surgical options for severe refractory cases.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

300.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 301

Fibroadenoma

301.1 Overview

Fibroadenomas are the most common non-cancerous breast tumors in young women (ages 15–35). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

301.2 Causes

This condition happens because of estrogen-dependent non-cancerous tumors composed of fibrous and glandular tissue. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

301.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

301.4 Signs and Symptoms

Clinical features typically include a solitary, firm, mobile, well-circumscribed, rubbery mass—often described as a "breast mouse" due to its mobility—which can grow during pregnancy or hormonal stimulation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

301.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

301.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

301.7 Diagnosis

Doctors diagnose this using ultrasound (well-circumscribed, hypoechoic, oval mass with horizontal orientation) and core needle biopsy to confirm. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

301.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

301.9 Treatment Options

- Treatment options include surgical removal for giant fibroadenomas or those causing symptoms
- Small, asymptomatic fibroadenomas may be observed.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

301.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

301.11 Outlook

Most people recover well. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 302

Fibrocystic Changes

302.1 Overview

Fibrocystic changes are the most common non-cancerous breast condition, affecting up to 60% of women at some point. Histologically, it encompasses a spectrum of findings: cysts (microcystic and macrocystic), scarring, adenosis, and epithelial abnormal increase in cells. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

302.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

302.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

302.4 Signs and Symptoms

You may experience breast pain and nodularity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

302.5 Progression and Stages

The condition is marked by breast pain (mastalgia), tenderness, nodularity, and thickening, often fluctuating with the menstrual cycle, and is hormone-dependent, typically worsening in the luteal phase. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

302.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

302.7 Diagnosis

Doctors diagnose this using clinical breast examination, mammography (for women over 40), and ultrasound (to evaluate dominant cysts). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

302.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

302.9 Treatment Options

Treatment options include reassurance, well-fitting support bras, NSAIDs, and for severe symptoms, oral contraceptives or tamoxifen. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

302.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

302.11 Outlook

Most people recover well, with no cancerous potential. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 303

Fibromyalgia

303.1 Overview

Fibromyalgia is a chronic pain condition characterized by widespread musculoskeletal pain, fatigue, sleep disturbance, thinking and memory dysfunction ("fibro fog"), and mood disorders, affecting 2–4% of the population, predominantly women. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

303.2 Causes

This condition happens because of central sensitization (amplified pain processing in the CNS), altered neurotransmitter levels (substance P, serotonin), and abnormal functional connectivity in brain pain networks. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

303.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

303.4 Signs and Symptoms

You may experience chronic widespread pain, tender points (18 defined anatomic sites historically), fatigue, and sleep disturbance. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

303.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Doctors diagnose this based on the 2010 ACR criteria using a widespread pain index and symptom severity scale
- It is a clinical To diagnose exclusion with no specific laboratory tests.

303.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

303.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

303.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

303.9 Treatment Options

Treatment typically involves multimodal: exercise (most evidence-based intervention), thinking and memory-behavioral therapy, pharmacotherapy (pregabalin, duloxetine, milnacipran, amitriptyline, tramadol), and patient education. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

303.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

303.11 Outlook

- The outlook varies from person to person
- Symptoms are chronic but functional status can improve with multimodal management.

Chapter 304

Fibromyalgia

304.1 Overview

See Chapter ??, Section 303. From a pain medicine perspective, fibromyalgia is the prototypical nociplastic pain condition involving central sensitization, dysfunctional descending pain modulation, and aberrant connectivity within the default mode network and salience network. Management emphasizes a multimodal approach: pharmacotherapy (duloxetine 60–120 mg/day, milnacipran 100–200 mg/day, pregabalin 150–600 mg/day—the three FDA-approved agents), graded aerobic exercise, thinking and memory behavioral therapy, sleep hygiene, and patient education. Opioids are ineffective and should be avoided. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

304.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

304.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

304.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

304.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

304.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

304.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

304.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

304.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

304.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

304.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 305

Fibrous Dysplasia

305.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Fibrous dysplasia is a non-cancerous bone disorder where normal bone is replaced by fibrous tissue and immature woven bone, caused by postzygotic activating mutations in the GNAS gene. It may be monostotic (single bone, 70–80%) or polyostotic (multiple bones)
- The ribs, femur, tibia, and skull are commonly affected.

305.2 Causes

This condition happens because of GNAS mutation leading to abnormal bone formation. McCune-Albright syndrome is a variant combining polyostotic fibrous dysplasia with café-au-lait spots and endocrine hyperfunction (precocious puberty). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

305.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

305.4 Signs and Symptoms

You may experience bone pain, deformity, or pathologic fracture. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

305.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

305.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

305.7 Diagnosis

Doctors diagnose this using X-ray showing ground-glass appearance and thinning of the cortex. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

305.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

305.9 Treatment Options

Treatment typically involves symptomatic: observation, bisphosphonates for pain, and surgical correction of deformities or fractures. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

305.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

305.11 Outlook

- In most cases, the outlook is good
- Cancerous transformation is rare.

Chapter 306

Filariasis (Lymphatic Filariasis)

306.1 Overview

Lymphatic filariasis (elephantiasis) is a mosquito-borne nematode infection caused by *Wuchereria bancrofti* (90% of cases), *Brugia malayi*, and *B. timori*, with 120 million infected, primarily in tropical and subtropical regions, and is the second leading cause of permanent disability worldwide. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

306.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

306.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

306.4 Signs and Symptoms

You may experience asymptomatic microfilaremia (most infected individuals), acute filarial fevers (adenolymphangitis with painful lymphadenitis, fever, chills), and chronic manifestations (lymphedema of the leg, scrotal hydrocele, elephantiasis). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

306.5 Progression and Stages

This condition happens because of third-stage filariform larvae (L3) being inoculated by mosquito bite, migrating to the lymphatics, and maturing into adult worms in lymphatic vessels and lymph nodes, with the host immune response to dying adult worms causing granulomatous inflammation and lymphatic damage. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

306.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

306.7 Diagnosis

Doctors diagnose this using microscopic detection of microfilariae on thick blood smear or antigen detection. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

306.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

306.9 Treatment Options

Treatment options include diethylcarbamazine (DEC) and albendazole. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

306.10 Living with It

- The outlook is good with early treatment
- Chronic lymphedema requires long-term care.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

306.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 307

Fluoride Excess — Fluorosis

307.1 Overview

Fluorosis is a mineral excess disorder caused by chronic excessive intake of fluoride, typically from drinking water with high fluoride content (greater than 1.5 ppm), but also from dental products, fluoride supplements, and industrial exposure. It is endemic in regions of India, China, Africa, and the Middle East. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

307.2 Causes

This condition happens because of fluoride incorporation into hydroxyapatite crystals of bones and teeth. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

307.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

307.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

307.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- You may experience dental fluorosis (chalky white or yellow-brown mottling, pitting, and staining of teeth, occurring when excess fluoride is ingested during tooth development) and skeletal fluorosis (osteosclerosis, bone and joint pain, ligamentous calcium buildup, periosteal bone formation, and spinal rigidity mimicking ankylosing spondylitis
- Advanced stages lead to crippling deformities, kyphosis, and nervous system complications from spinal canal stenosis).

307.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

307.7 Diagnosis

Doctors diagnose this based on history of high fluoride exposure, radiographic findings (osteosclerosis, periosteal bone apposition, interosseous membrane calcium buildup), and elevated urinary fluoride. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

307.8 Prevention

- Treatment focuses on prevention: defluoridation of drinking water, use of alternative water sources, and avoidance of fluoride-containing products in endemic areas. The outlook is good with prevention
- No specific treatment exists for established disease.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

307.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

307.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

307.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 308

Focal Segmental Glomerulosclerosis

308.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Focal segmental glomerulosclerosis (FSGS) is a pattern of glomerular injury characterized by segmental sclerosis in some (focal) glomeruli. It is a common cause of nephrotic syndrome in adults. The condition may be primary (idiopathic) or secondary (to obesity, HIV infection, sickle cell disease, drug abuse, or adaptive hyperfiltration)
- Podocyte injury is the central pathological event.

308.2 Causes

Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

308.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

308.4 Signs and Symptoms

You may experience proteinuria (often nephrotic-range), hematuria, high blood pressure, and progressive kidney insufficiency.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

308.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

308.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

308.7 Diagnosis

- Doctors diagnose this using kidney biopsy, which must include sufficient glomeruli to demonstrate the focal pattern. Treatment for primary FSGS includes corticosteroids (6 months), calcineurin inhibitors for steroid-resistant cases, and rituximab
- Secondary FSGS requires treatment of the underlying cause.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

308.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

308.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

308.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

308.11 Outlook

- The outlook varies from person to person
- APOL1 gene variants are associated with increased risk, particularly in individuals of African ancestry.

Chapter 309

Folate Deficiency Anemia

309.1 Overview

Folate deficiency anemia is a megaloblastic anemia that is clinically indistinguishable from B12 deficiency anemia but without the neurological manifestations. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

309.2 Causes

This condition happens because of folate (vitamin B9) deficiency, with causes including poor dietary intake (alcoholism, inadequate fruit/vegetable consumption), malabsorption (celiac disease, tropical sprue), increased demand (pregnancy, hemolytic anemias), and medications (methotrexate, trimethoprim, phenytoin). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

309.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

309.4 Signs and Symptoms

You may experience fatigue, pallor, and glossitis.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

309.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

309.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

309.7 Diagnosis

Doctors diagnose this using macrocytic anemia, low serum folate, and elevated homocysteine (with normal methylmalonic acid, distinguishing it from B12 deficiency).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

309.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

309.9 Treatment Options

Treatment focuses on oral folic acid supplementation (1 mg daily) and addressing the underlying cause. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

309.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

309.11 Outlook

Most people recover well with treatment.

Chapter 310

Fractures

310.1 Overview

A fracture is a break in the continuity of bone. Common fracture types include distal radius (Colles' fracture), hip (intertrochanteric or subcapital femoral neck), vertebral compression, scaphoid, and clavicle fractures. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

310.2 Causes

This condition happens because of mechanical force exceeding bone strength. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

310.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

310.4 Signs and Symptoms

You may experience pain, deformity, swelling, and loss of function. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

310.5 Progression and Stages

Fractures are classified as open (compound, with skin disruption) or closed (simple), and by pattern (transverse, oblique, spiral, comminuted, greenstick, impacted, avulsion). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

310.6 Complications

- In most cases, the outlook is good
- Hip fractures in the elderly are orthopedic emergencies requiring surgical fixation within 24–48 hours to reduce mortality and complications.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

310.7 Diagnosis

- Doctors diagnose this using X-rays (at least two views)
- CT for complex fractures, MRI for occult fractures and ligamentous injury.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

310.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

310.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

310.10 Living with It

Treatment aims for reduction (realigning fracture fragments), fixation (maintaining alignment during healing), and rehabilitation

- Closed reduction with casting is appropriate for stable, minimally displaced fractures
- Open reduction with internal fixation (ORIF) is indicated for displaced, unstable, or intra-articular fractures.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

310.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 311

Fragile X Syndrome

311.1 Overview

Fragile X syndrome is the most common inherited cause of intellectual disability and the most common monogenic cause of autism spectrum disorder. It is caused by a CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 gene on the X chromosome, leading to hypermethylation and gene silencing. Incidence is 1 in 4000 males and 1 in 8000 females. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

311.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

311.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

311.4 Signs and Symptoms

Clinical features in males include intellectual disability, long narrow face, prominent ears, large testes (macroorchidism, postpubertal), joint hypermobility, autistic behaviors, attention deficits, and anxiety. Females may have mild intellectual disability or learning difficulties. Premutation carriers (55–200 repeats) are at risk of FXTAS and FXPOI.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

311.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

311.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

311.7 Diagnosis

Doctors diagnose this using FMR1 DNA testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

311.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

311.9 Treatment Options

Treatment typically involves supportive and includes behavioral therapies, educational interventions, and pharmacotherapy for associated conditions. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

311.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 312

Frailty

312.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Frailty is a geriatric syndrome of increased vulnerability to stressors resulting from age-related decline in physiological reserve across multiple organ systems. It affects a significant proportion of older adults. Two major conceptual frameworks guide frailty assessment. The **Fried Frailty Phenotype** (2001) defines frailty by the presence of three or more of five criteria: unintentional weight loss (>10 lb in one year), self-reported exhaustion, weakness (low grip strength), slow gait speed, and low physical activity
- Pre-frailty is defined as one or two criteria. The **Frailty Index** (Rockwood and Mitnitski) accumulates deficits across multiple domains as a ratio of deficits present to total deficits assessed (typically 30–70 items).

312.2 Causes

This condition happens because of age-related decline in physiologic reserve, with sarcopenia as a core biological substrate. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

312.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

312.4 Signs and Symptoms

You may experience unintentional weight loss, exhaustion, weakness, slow gait speed, and low physical activity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

312.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- The outlook varies from person to person
- Frailty is a dynamic process that can improve or worsen over time.

312.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

312.7 Diagnosis

Doctors use the Fried Frailty Phenotype (three or more of five criteria) or the Frailty Index (>0.25 indicates frailty). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

312.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

312.9 Treatment Options

Treatment options include resistance and aerobic exercise, protein-calorie supplementation, vitamin D, reduction of polypharmacy, and comprehensive geriatric assessment (CGA). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

312.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

312.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 313

Frailty as a Syndrome

313.1 Overview

Frailty is a clinically recognizable state of increased vulnerability to stressors resulting from age-related decline in physiological reserve across multiple organ systems. It predicts adverse outcomes including falls, hospitalization, institutionalization, and death, independent of chronological age and comorbidity. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

313.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

313.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

313.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

313.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

313.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

313.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

313.8 Prevention

Frailty is a dynamic process that can improve or worsen over time, making early detection and intervention clinically important.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

313.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

313.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 314

Friedreich's Ataxia

314.1 Overview

Friedreich's loss of coordination is the most common hereditary loss of coordination, caused by homozygous GAA trinucleotide repeat expansion in the FXN gene encoding frataxin, with a prevalence of 1 in 50,000. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

314.2 Causes

This condition happens because of frataxin deficiency, leading to mitochondrial iron accumulation, oxidative stress, and impaired energy metabolism, primarily affecting the dorsal root ganglia, posterior columns, corticospinal tracts, and cerebellum. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

314.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

314.4 Signs and Symptoms

You may experience progressive gait and limb loss of coordination, slurred speech, areflexia, loss of vibratory and proprioceptive sensation, extensor plantar responses, scoliosis, pes cavus, hypertrophic cardiomyopathy, and increased risk of diabetes mellitus. Onset is typically in childhood or adolescence (mean age 10–15 years). Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

314.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

314.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

314.7 Diagnosis

Doctors diagnose this based on clinical criteria and confirmed by FXN genetic testing. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

314.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

314.9 Treatment Options

Treatment focuses on symptomatic treatment: physical therapy for balance and mobility, occupational therapy, speech therapy, and orthopedic Treatment for scoliosis. Cardiac monitoring is essential (echocardiography, ECG). Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

314.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider

- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

314.11 Outlook

- Outlook is progressive
- Most patients lose ambulation 10–15 years after symptom onset, with mean survival to age 40–50, primarily due to cardiomyopathy.

Chapter 315

Frontotemporal Dementia

315.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Frontotemporal dementia encompasses a group of neurodegenerative disorders selectively affecting the frontal and temporal lobes
- Onset typically occurs between ages 40 and 65, making it a common cause of early-onset dementia.

315.2 Causes

This condition happens because of accumulation of tau, TDP-43, or FUS proteins. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

315.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

315.4 Signs and Symptoms

- You may experience behavioral variant FTD with personality changes, disinhibition, apathy, and executive dysfunction
- Language variants include semantic dementia (progressive loss of word meaning) and non-fluent or agrammatic primary progressive aphasia.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

315.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

315.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

315.7 Diagnosis

Doctors diagnose this based on MRI showing disproportionate frontal and temporal lobe shrinkage or wasting. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

315.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

315.9 Treatment Options

Treatment typically involves supportive, focusing on behavioral management, speech therapy, and caregiver support. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

315.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

315.11 Outlook

outlook is progressive, with no disease-modifying treatment available. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 316

Frontotemporal Dementia

316.1 Overview

Frontotemporal dementia (FTD) is a group of neurodegenerative disorders characterized by progressive dysfunction of the frontal and temporal lobes. Although typically presenting at a younger age (45–65 years), FTD can occur in older adults. The behavioral variant (bvFTD) presents with early and progressive changes in personality, social conduct, and executive function—disinhibition, apathy, loss of empathy, compulsive behaviors, and hyperorality. Language variants include non-fluent/agrammatic primary progressive aphasia (PPA) and semantic variant PPA. Approximately 30–50% of FTD is familial, with mutations in MAPT (tau), GRN (progranulin), and C9orf72. There is no disease-modifying therapy. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

316.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

316.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

316.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

316.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

316.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

316.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

316.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

316.9 Treatment Options

Treatment typically involves symptomatic and supportive, focusing on behavior management (SSRIs for disinhibition), caregiver support, and addressing safety and legal concerns. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

316.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

316.11 Outlook

outlook is progressive. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 317

Fuchs' Endothelial Dystrophy

317.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Fuchs' endothelial corneal dystrophy is a bilateral, progressive disorder of the corneal endothelium characterized by guttae, endothelial cell loss, and corneal swelling
- It is autosomal dominant and more common in women.

317.2 Causes

This condition happens because of progressive endothelial cell dysfunction and loss. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

317.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

317.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

317.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Early disease is asymptomatic
- As the disease progresses, You may experience morning blurring that improves through the day, glare, and reduced visual acuity
- Advanced disease causes persistent corneal swelling and bullous keratopathy.

317.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

317.7 Diagnosis

Doctors diagnose this using slit-lamp examination and specular microscopy showing reduced endothelial cell density. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

317.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

317.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

317.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

317.11 Outlook

Most people recover well with surgical intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 318

Fungal Sinusitis

318.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- Fungal sinusitis encompasses a spectrum of conditions ranging from non-invasive forms (fungal ball, allergic fungal sinusitis) to life-threatening invasive forms (acute invasive fungal rhinosinusitis). Non-invasive forms result from fungal colonization or allergic reaction
- Invasive forms occur in immunocompromised patients (diabetes with ketoacidosis, neutropenia, transplant recipients) and are caused by Mucorales or *Aspergillus* species.

318.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

318.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)

- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

318.4 Signs and Symptoms

Symptoms of acute invasive disease include rapidly progressive facial pain, nasal congestion, black necrotic eschar, and extension to the orbit and brain. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

318.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

318.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)

- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

318.7 Diagnosis

To diagnose invasive disease requires tissue biopsy. Treatment for invasive disease involves aggressive surgical removal of dead tissue, systemic antifungal therapy (amphotericin B), and reversal of immunosuppression. outlook for invasive disease is poor with high mortality. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

318.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)

- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

318.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

318.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

318.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 319

Gait Disorders

319.1 Overview

Gait disorders affect 15% of adults aged 60 and over and 80% of those over 85. Common patterns include: This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- **Parkinsonian gait:** Shuffling, reduced step length, decreased arm swing, festination (accelerating steps), freezing, and difficulty initiating gait. Associated with Parkinson disease and atypical parkinsonian disorders.
- **Hemiparetic gait:** Unilateral leg weakness with circumduction (leg swings in a semicircle), foot drop, and arm held in flexion. Seen in stroke and other unilateral brain lesions.
- **Ataxic gait:** Wide-based, unsteady, staggering, with inconsistent step width. Cerebellar cause (stroke, tumor, alcohol-related, multiple system shrinkage or wasting). Also seen in sensory loss of coordination (impaired proprioception from peripheral neuropathy or dorsal column disease) where vision partially compensates (positive Romberg sign).
- **Neuropathic gait** (steppage gait): Foot drop with high-stepping to clear the toe, often slapping of the foot. Caused by peripheral neuropathy (diabetic, alcoholic, Charcot-Marie-Tooth), L5 radiculopathy, or common peroneal nerve palsy.
- **Senile gait** (cautious gait): Slightly wide-based, shorter steps, reduced arm swing, mild stooped posture, and general caution. It may represent normal aging or a prodrome of neurological disease.

319.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

319.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

319.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

319.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

319.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

319.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

319.8 Prevention

Management addresses underlying cause, assistive devices (canes, walkers), physical therapy (gait training, balance exercises), and fall prevention strategies.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

319.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

319.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

319.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 320

Galactocele

320.1 Overview

A galactocele is a non-cancerous retention cyst filled with milk or milky fluid, occurring predominantly in lactating women during or immediately following weaning. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

320.2 Causes

This condition happens because of obstruction of a lactiferous duct (secondary to inflammation, inspissated milk secretions, or external compression), leading to proximal ductal dilation and accumulation of fat-rich milk elements. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

320.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

320.4 Signs and Symptoms

You may experience a painless, smooth, mobile, well-circumscribed subareolar mass that may fluctuate in size with breastfeeding. Ultrasound demonstrates a well-marginated oval lesion with acoustic enhancement and a characteristic fat-fluid level. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

320.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

320.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

320.7 Diagnosis

Doctors confirm the diagnosis through clinical examination, sonography, and fine-needle removal by suction yielding milky fluid, which is both diagnostic and therapeutic. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

320.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

320.9 Treatment Options

- Treatment focuses on needle removal by suction for symptomatic relief
- Infected galactoceles require antibiotic coverage and removal of fluid.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

320.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

320.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 321

Galactosemia

321.1 Overview

Galactosemia is an autosomal recessive disorder of galactose metabolism caused most commonly by deficiency of galactose-1-phosphate uridylyltransferase (GALT), with an incidence of approximately 1 in 30,000–60,000 live births. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

321.2 Causes

This condition happens because of accumulation of galactose-1-phosphate and galactitol when galactose from dietary lactose cannot be converted to glucose. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

321.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

321.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

321.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

321.6 Complications

- Clinical features present after the initiation of milk feeding with poor feeding, vomiting, diarrhea, failure to thrive, hypoglycemia, jaundice, hepatomegaly, ascites, and liver dysfunction
- Escherichia coli neonatal sepsis is a frequent and life-threatening complication
- Cataracts develop due to galactitol accumulation in the lens. outlook is guarded despite treatment
- Long-term complications including speech apraxia, thinking and memory deficits, loss of coordination, tremor, and premature ovarian insufficiency are common.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

321.7 Diagnosis

Doctors diagnose this using newborn screening (elevated total galactose), red blood cell GALT enzyme activity (absent or severely reduced), and genetic testing of the GALT gene. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

321.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

321.9 Treatment Options

Treatment focuses on immediate elimination of dietary lactose and galactose (soy-based or elemental formula) for life. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

321.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

321.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 322

Gas Gangrene (Clostridial Myonecrosis)

322.1 Overview

Gas gangrene is a rapidly progressive, life-threatening infection of skeletal muscle caused by *Clostridium perfringens* (80% of cases) and other clostridial species, with low incidence but high mortality (20–60%), occurring post-traumatically or spontaneously (associated with colorectal carcinoma, neutropenic colitis). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

322.2 Causes

This condition happens because of spores germinating in devitalized, hypoxic tissue and producing potent exotoxins, including alpha-toxin (lecithinase, phospholipase C) causing hemolysis, platelet aggregation, increased vascular permeability, and direct myonecrosis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

322.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

322.4 Signs and Symptoms

You may experience sudden onset of severe pain at the wound site, marked swelling, tenderness, and crepitus on palpation, with skin progressing from pale to bronze to purplish-black with involving bleeding bullae, and systemic toxicity including tachycardia, tachypnea, hypotension, fever, and kidney failure. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

322.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

322.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

322.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with Gram stain showing gram-positive rods and radiographs showing gas in soft tissues. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

322.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

322.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

322.10 Living with It

- Management requires urgent surgical exploration with extensive removal of dead tissue, high-dose penicillin G plus clindamycin, and aggressive supportive care. outlook: mortality is 100% without treatment
- With treatment, 20–60%.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

322.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 323

Gastric Adenocarcinoma

323.1 Overview

Gastric adenocarcinoma is the fifth most common cancer worldwide and the fourth leading cause of cancer-related death. The two histological subtypes are intestinal (associated with *H. pylori*, chronic atrophic gastritis, intestinal metaplasia, and dietary factors) and diffuse (associated with CDH1 mutations, affecting younger patients with worse outlook). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

323.2 Causes

This condition happens because of multifactorial carcinogenesis involving *H. pylori* infection, high-salt diet, smoking, family history, and pernicious anemia. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

323.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity

Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

323.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

323.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment focuses on using a thin camera tube mucosal removal or surgical gastrectomy with lymphadenectomy for early-stage disease
- Perioperative chemotherapy (FLOT regimen), targeted therapy (trastuzumab for

HER2-positive disease), and immunotherapy (nivolumab for PD-L1-positive tumors) for advanced disease. Unfortunately, the outlook is often poor for advanced disease but favorable for early-stage disease with appropriate treatment.

323.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

323.7 Diagnosis

Doctors diagnose this based on upper endoscopy with biopsy, with staging using TNM classification, CT imaging, and laparoscopy for peritoneal assessment. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status

- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

323.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

323.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

323.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

323.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 324

Gastric Lymphoma

324.1 Overview

Gastric lymphoma is the most common extranodal lymphoma. Mucosa-associated lymphoid tissue (MALT) lymphoma accounts for the majority and is strongly associated with *H. pylori* infection. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

324.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

- This condition happens because of *H. pylori*-driven lymphoproliferation
- High-grade (diffuse large B-cell) lymphoma may arise from low-grade MALT lymphoma or develop de novo.

324.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age

- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

324.4 Signs and Symptoms

You may experience epigastric pain, weight loss, and GI bleeding. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

324.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment focuses on *H. pylori* eradication therapy for low-grade MALT lymphoma
- Antibiotic-refractory or high-grade disease requires chemotherapy (R-CHOP) and/or radiation.

324.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

324.7 Diagnosis

Doctors diagnose this based on endoscopy with deep biopsy and immunohistochemistry. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

324.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

324.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
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- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

324.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

324.11 Outlook

The outlook is generally positive for low-grade MALT lymphoma, which may regress completely with eradication therapy alone. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 325

Gastric Ulcer

325.1 Overview

Gastric ulcer is a mucosal defect extending through the muscularis mucosae into the deeper layers of the gastric wall, most commonly located along the lesser curvature and antrum. *Helicobacter pylori* infection is present in 60–80% of cases and NSAID use (including aspirin and low-dose aspirin) in most others. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

325.2 Causes

This condition happens because of mucosal injury caused by *H. pylori* infection, NSAID-induced prostaglandin inhibition, and less commonly Zollinger-Ellison syndrome, Crohn's disease, or CMV infection in immunocompromised patients. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

325.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

325.4 Signs and Symptoms

You may experience epigastric pain aggravated by food (distinguishing it from duodenal ulcers, which improve with food), nausea, vomiting, and early satiety. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

325.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

325.6 Complications

- In most cases, the outlook is good with treatment
- Complications include bleeding (most common), perforation, gastric outlet obstruction, and rarely cancerous transformation.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

325.7 Diagnosis

Doctors diagnose this based on upper digestive tract endoscopy with biopsy to exclude malignancy (all gastric ulcers should be biopsied). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

325.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

325.9 Treatment Options

Treatment focuses on *H. pylori* eradication when positive, cessation of NSAIDs, and proton pump inhibitors (PPIs) for 8 weeks with repeat endoscopy to confirm healing. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

325.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

325.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 326

Gastroesophageal Reflux Disease

326.1 Overview

Gastroesophageal reflux disease (GERD) is a chronic condition characterized by symptoms of heartburn and regurgitation caused by reflux of gastric contents into the esophagus, occurring when the lower esophageal sphincter (LES) relaxes inappropriately or transiently. It affects approximately 20–30% of adults in Western countries weekly. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

326.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

326.3 Risk Factors

This condition happens because of transient LES relaxations, reduced basal LES pressure, impaired esophageal acid clearance, and hiatal hernia, with risk factors including obesity, pregnancy, smoking, and certain foods (fatty foods, chocolate, caffeine, alcohol). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

326.4 Signs and Symptoms

- Diagnosis is primarily clinical
- A therapeutic trial of proton pump inhibitors (PPIs) is both diagnostic and therapeutic. Upper endoscopy is indicated for alarm features (difficulty swallowing, weight loss, GI bleeding, anemia) or symptoms persisting despite PPI therapy. Ambulatory pH monitoring (with impedance) is the gold standard for confirming abnormal acid exposure.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

326.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

326.6 Complications

- You may experience pyrosis (heartburn), regurgitation of acid or food, difficulty swallowing, odynophagia, chronic cough, laryngitis, and asthma-like symptoms
- Complications include erosive esophagitis, peptic stricture, Barrett's esophagus (intestinal metaplasia of the esophageal mucosa, a precursor to esophageal adenocarcinoma), and esophageal adenocarcinoma.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

326.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

326.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

326.9 Treatment Options

Treatment focuses on lifestyle modifications (weight loss, elevation of the head of the bed, avoidance of trigger foods and late meals), antacids and alginate agents for mild symptoms, H2 receptor antagonists for nocturnal symptoms, and PPIs (omeprazole, lansoprazole, esomeprazole) as the mainstay of therapy for moderate-to-severe disease. using small incisions and a camera Nissen fundoplication or magnetic sphincter augmentation (LINX device) are surgical options for refractory disease or patient preference to discontinue medical therapy. using a thin camera tube therapies (transoral incisionless fundoplication, radiofrequency ablation) are alternative options. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

326.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

326.11 Outlook

In most cases, the outlook is good with appropriate medical or surgical management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 327

Gastrointestinal Stromal Tumor

327.1 Overview

digestive tract stromal tumor (GIST) is the most common mesenchymal tumor of the GI tract, arising from interstitial cells of Cajal. The stomach is the most common location (60–70% of GISTs). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

327.2 Causes

This condition happens because of activating mutations in the KIT or PDGFRA receptor tyrosine kinase genes. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

327.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

327.4 Signs and Symptoms

Symptoms can range from incidental discovery to GI bleeding, abdominal pain, or a palpable mass. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

327.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment focuses on surgical removal for localized disease

- Imatinib (a tyrosine kinase inhibitor) is used for advanced, unresectable, or metastatic disease and as adjuvant therapy for high-risk removed tumors.

327.6 Complications

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

327.7 Diagnosis

Doctors diagnose this based on endoscopy or imaging, confirmed by biopsy showing KIT (CD117) or DOG1 positivity. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

327.8 Prevention

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)

- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

327.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

327.10 Living with It

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

327.11 Outlook

The outlook has been significantly improved with targeted tyrosine kinase inhibitor therapy. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 328

Gastroparesis

328.1 Overview

Gastroparesis is delayed gastric emptying in the absence of mechanical obstruction. It affects up to 5% of diabetic patients, and also occurs post-surgically (particularly after vagotomy or fundoplication), idiopathically, or due to medications (opioids, anticholinergics) and connective tissue disorders (scleroderma). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

328.2 Causes

This condition happens because of impaired gastric motility due to autonomic neuropathy (diabetes), vagal nerve injury, or medication effects. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

328.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

328.4 Signs and Symptoms

You may experience nausea, vomiting of undigested food hours after eating, early satiety, postprandial fullness, and weight loss. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

328.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

328.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

328.7 Diagnosis

Doctors diagnose this based on gastric emptying scintigraphy (4-hour study showing >10% retention at 4 hours). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

328.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

328.9 Treatment Options

- Treatment focuses on dietary modification (small, frequent, low-fat, low-fiber meals), prokinetic agents (metoclopramide or domperidone), erythromycin (motilin receptor agonist), and gastric electrical stimulation (Enterra therapy) for refractory cases
- Metoclopramide carries a black box warning for tardive dyskinesia.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

328.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

328.11 Outlook

The outlook varies from person to person, with symptomatic improvement achievable but cure uncommon. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 329

Generalized Anxiety Disorder (GAD)

329.1 Overview

Generalized anxiety disorder has a lifetime prevalence of 5–9% and is twice as common in women. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

329.2 Causes

This condition happens because of hyperactivation of the amygdala, prefrontal cortex dysfunction, reduced GABAergic tone, elevated norepinephrine activity, and serotonergic dysregulation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

329.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

329.4 Signs and Symptoms

You may experience excessive, uncontrollable worry about multiple domains for at least 6 months, occurring more days than not, with associated symptoms including restlessness, easy fatiguability, difficulty concentrating, irritability, muscle tension, and sleep disturbance, with at least three of these required for diagnosis.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

329.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

329.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

329.7 Diagnosis

- Doctors diagnose this using clinical interview
- The GAD-7 scale is a validated screening tool.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

329.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

329.9 Treatment Options

Treatment options include pharmacotherapy (SSRIs and SNRIs are first-line, with buspirone as an alternative) and psychotherapy (CBT is first-line). Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

329.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

329.11 Outlook

- The outlook varies from person to person
- Many patients respond to treatment but symptoms may be chronic.

Chapter 330

Genital Herpes (HSV)

330.1 Overview

Genital herpes is caused by herpes simplex virus type 2 (HSV-2, most cases) and type 1 (HSV-1, increasing proportion). In the obstetric context, the primary concern is neonatal herpes transmission during vaginal delivery in women with active primary or recurrent genital lesions. The risk of neonatal transmission is 25–60% with primary infection during the third trimester and less than 2% with recurrent infection. Management at onset of labor: if active lesions or prodromal symptoms are present, cesarean delivery is recommended. For women with a history of recurrent genital herpes, suppressive acyclovir or valacyclovir is given from 36 weeks to delivery. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

330.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

330.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

330.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

330.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

330.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)

- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

330.7 Diagnosis

Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

330.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

330.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

330.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

330.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 331

Genital Warts (HPV)

331.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Genital warts are caused by low-risk human papillomavirus types 6 and 11. In the obstetric context, genital warts may increase in size and number during pregnancy. Treatment options in pregnancy include freezing treatment, trichloroacetic acid (TCA), or surgical removal
- Podophyllin, podofilox, and imiquimod are contraindicated. Cesarean delivery is reserved for cases where warts cause mechanical obstruction of the birth canal.

331.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

331.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

331.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

331.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

331.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

331.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

331.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

331.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

331.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

331.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 332

Gestational Diabetes Mellitus

332.1 Overview

Gestational diabetes mellitus (GDM) is glucose intolerance first recognized during pregnancy, affecting 2–10% of pregnancies. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

332.2 Causes

This condition happens because of insufficient insulin secretion to overcome pregnancy-related insulin resistance mediated by placental hormones (human placental lactogen, cortisol, progesterone). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

332.3 Risk Factors

Risk factors include obesity, advanced maternal age, family history of diabetes, prior GDM, and polycystic ovary syndrome.

- Family history
- Obesity and sedentary lifestyle
- Poor dietary habits

- Advancing age
- Hormonal imbalances
- Certain medications
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

332.4 Signs and Symptoms

- Clinical features are typically asymptomatic
- Screening is performed at 24–28 weeks gestation using the 75-g oral glucose tolerance test.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

332.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

332.6 Complications

- Complications for the mother include preeclampsia and increased cesarean delivery rate
- Fetal/neonatal complications include macrosomia, shoulder dystocia, neonatal hypoglycemia, and long-term risk of obesity and T2DM.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

332.7 Diagnosis

Doctors diagnose this based on the oral glucose tolerance test.

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

332.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

332.9 Treatment Options

- Treatment options include medical nutrition therapy, exercise, and insulin therapy when glycemic targets are not met with lifestyle measures
- Metformin may be used in select cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

332.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

332.11 Outlook

- In most cases, the outlook is favorable
- GDM usually resolves postpartum, but women with GDM have a 50% lifetime risk of developing T2DM and should undergo periodic screening.

Chapter 333

Gestational Diabetes Mellitus (Obstetric Focus)

333.1 Overview

Gestational diabetes mellitus (GDM) is a pregnancy-related condition requiring coordinated obstetric care given its implications for pregnancy outcomes. Universal screening is performed at 24–28 weeks using a 75-g oral glucose tolerance test (OGTT). Poor glycemic control in GDM is linked to an increased risk of preeclampsia, polyhydramnios, preterm labor, and cesarean delivery. Fetal surveillance includes ultrasound monitoring for macrosomia and amniotic fluid index. Intrapartum Treatment options include close monitoring of maternal glucose to maintain normoglycemia, reducing the risk of neonatal hypoglycemia. Postpartum testing (75-g OGTT at 6–12 weeks) is essential as up to 50% of women with GDM develop type 2 diabetes within 10 years. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

333.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

333.3 Risk Factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

333.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

333.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

333.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

333.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

333.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

333.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

333.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 334

Gestational Hypertension

334.1 Overview

Gestational high blood pressure is a hypertensive disorder of pregnancy defined as new-onset high blood pressure (systolic BP $\geq$140 mmHg or diastolic BP $\geq$90 mmHg) after 20 weeks of gestation in the absence of proteinuria or other features of preeclampsia. It affects 6–8% of pregnancies. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

334.2 Causes

This condition happens because of pregnancy-induced vasospasm and endothelial dysfunction. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

334.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age

- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

334.4 Signs and Symptoms

You may experience elevated blood pressure after 20 weeks without proteinuria. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

334.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

334.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

334.7 Diagnosis

To make the diagnosis, doctors look for new-onset high blood pressure after 20 weeks in the absence of proteinuria or end-organ dysfunction. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

334.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

334.9 Treatment Options

- Treatment options include close monitoring of blood pressure and fetal surveillance
- Antihypertensive therapy is initiated for persistent BP $\geq$160/105 mmHg using labetalol, nifedipine, or methyldopa.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)

- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

334.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

334.11 Outlook

- The outlook is generally positive
- Blood pressure typically normalizes by 12 weeks postpartum, though approximately 25% of women progress to preeclampsia.

Chapter 335

Gestational Trophoblastic Disease

335.1 Overview

Gestational trophoblastic disease (GTD) encompasses a spectrum of disorders arising from abnormal trophoblast growth, ranging from non-cancerous hydatidiform mole to cancerous choriocarcinoma. Incidence is 1 in 1000 pregnancies in Western countries. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

335.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

335.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

335.4 Signs and Symptoms

You may experience vaginal bleeding, exaggerated uterine size for dates, hyperemesis gravidarum, and elevated β-hCG. Ultrasonography shows a characteristic "snowstorm" pattern in complete mole. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

335.5 Progression and Stages

Hydatidiform mole is classified as complete (46,XX, entirely paternal genome, risk of cancerous transformation 15–20%) or partial (69,XXY, triploid, risk 1–5%). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

335.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

335.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

335.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

335.9 Treatment Options

Treatment typically involves suction dilation and curettage with serial β-hCG monitoring. Gestational trophoblastic neoplasia (GTN) is diagnosed when β-hCG plateaus or rises after molar evacuation

- Low-risk GTN doctors typically treat it with single-agent methotrexate or actinomycin D
- High-risk GTN requires multi-agent chemotherapy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

335.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

335.11 Outlook

Most people recover well, with cure rates exceeding 90% for low-risk and 80–85% for high-risk disease. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 336

Giant Cell Arteritis

336.1 Overview

Giant cell arteritis is the most common primary systemic vasculitis, characterized by granulomatous inflammation of large and medium-sized arteries, with a predilection for the cranial branches of the aorta, particularly the temporal artery, exclusively affecting individuals over 50 years of age (peak incidence 70–80 years), with a female predominance of 2–3:1, and being more common in Northern European descendants. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

336.2 Causes

This condition happens because of a Th1- and Th17-driven inflammatory response targeting the vessel wall, with CD4+ T-cell infiltration, multinucleated giant cells, intimal abnormal increase in cells, and luminal occlusion, with IL-6 being a key driver. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

336.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

336.4 Signs and Symptoms

You may experience new-onset severe headache (the most common symptom, often temporal), scalp tenderness, jaw claudication (pathognomonic), visual disturbances (amaurosis fugax, diplopia, anterior ischemic optic neuropathy leading to irreversible blindness in 15–20% if untreated), polymyalgia rheumatica symptoms in 40–60%, fever, weight loss, and an elevated ESR, with aortic involvement (aortitis, aortic aneurysm, aortic dissection) occurring in 10–20%. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

336.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

336.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

336.7 Diagnosis

To make the diagnosis, doctors look for a high index of suspicion: ESR is almost always markedly elevated (>50–100 mm/hr), CRP is elevated, and temporal artery biopsy (TAB, at least 3–5 cm in length) shows granulomatous inflammation with multinucleated giant cells. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

336.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

336.9 Treatment Options

Treatment typically involves with high-dose corticosteroids (prednisone 40–60 mg/day or IV methylprednisolone if vision threatened), with tocilizumab (anti-IL-6 receptor) approved for GCA and used as a steroid-sparing agent. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

336.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

336.11 Outlook

The outlook is good with prompt treatment, though delayed treatment risks irreversible blindness. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 337

Giardiasis

337.1 Overview

Giardiasis is a protozoan diarrheal illness caused by the flagellate *Giardia lamblia*, the most common intestinal parasitic infection in the United States and a leading cause of waterborne outbreaks globally, with 300 million cases annually worldwide. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

337.2 Causes

This condition happens because of cysts excysting in the duodenum, releasing trophozoites that adhere to enterocytes via a ventral sucking disk, causing enterocyte damage, microvillus shrinkage or wasting, and increased intestinal permeability. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

337.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

337.4 Signs and Symptoms

You may experience an incubation of 1–3 weeks, with acute giardiasis presenting with watery, foul-smelling diarrhea (sulfuriferous), abdominal cramps, bloating, flatulence, and steatorrhea, and chronic giardiasis presenting with persistent diarrhea, weight loss, malabsorption, and fatigue. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

337.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

337.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

337.7 Diagnosis

Doctors diagnose this using microscopy of stool for cysts and trophozoites or antigen detection. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

337.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

337.9 Treatment Options

Treatment options include tinidazole or metronidazole. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

337.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

337.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 338

Gingivitis and Periodontitis

338.1 Overview

Gingivitis is an inflammation of the gingiva without attachment loss, representing the earliest and most common form of periodontal disease, caused by dental plaque biofilm accumulation. Periodontitis extends beyond the gingiva with loss of connective tissue attachment and alveolar bone, leading to pocket formation, gingival recession, tooth mobility, and eventual tooth loss. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

338.2 Causes

This condition happens because of bacterial plaque accumulation and the host inflammatory response. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

338.3 Risk Factors

Risk factors include smoking, diabetes mellitus, genetic susceptibility, and immunodeficiency. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

338.4 Signs and Symptoms

- You may experience red, swollen, bleeding gingiva for gingivitis
- Periodontitis additionally presents with pocket formation, gingival recession, and tooth mobility.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

338.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

338.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

338.7 Diagnosis

- Doctors diagnose this based on clinical examination and probing depths. Treatment for gingivitis involves oral hygiene instruction and professional cleaning
- Periodontitis Treatment usually involves scaling and root planing, surgical intervention, and systemic antibiotics for aggressive disease.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

338.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

338.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

338.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

338.11 Outlook

- Most people recover well for gingivitis with proper hygiene
- Periodontitis requires long-term maintenance.

Chapter 339

Glaucoma in Children

339.1 Overview

Primary congenital glaucoma is a condition resulting from developmental abnormality of the trabecular meshwork (goniodysgenesis), causing elevated intraocular pressure from birth or early childhood. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

339.2 Causes

This condition happens because of abnormal development of the aqueous outflow pathways. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

339.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

339.4 Signs and Symptoms

- You may experience epiphora, blepharospasm, photophobia, buphthalmos, and corneal swelling
- Juvenile open-angle glaucoma presents in older children and adolescents.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

339.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

339.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

339.7 Diagnosis

Doctors diagnose this based on examination under anesthesia and tonometry. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

339.8 Prevention

Your outlook depends on early intervention to prevent irreversible optic nerve damage and amblyopia.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

339.9 Treatment Options

- Treatment focuses on early surgical intervention (goniotomy or trabeculotomy)
- Secondary childhood glaucoma may result from aphakia, trauma, uveitis, or steroids.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

339.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

339.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 340

Glioblastoma

340.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

340.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

- This condition happens because of rapid infiltrative growth with microvascular growth and areas of tissue death
- The IDH-wildtype molecular profile is characteristic.

340.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

340.4 Signs and Symptoms

Your symptoms depend on tumor location but typically include progressive headache, seizures, weakness, numbness, or problems with movement, and thinking and memory changes. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

340.5 Progression and Stages

Glioblastoma is a WHO grade 4 astrocytoma and the most common and aggressive primary cancerous brain tumor in adults. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

340.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

340.7 Diagnosis

Doctors diagnose this based on MRI showing a ring-enhancing lesion with surrounding swelling and mass effect. Management follows the Stupp protocol: maximal safe surgical removal followed by concurrent temozolomide chemotherapy and radiotherapy, then adjuvant temozolomide

- MGMT promoter methylation status predicts response to temozolomide
- Tumor-treating fields (TTFields) have shown incremental survival benefit.
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

340.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs

Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

340.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

340.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian

- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

340.11 Outlook

Unfortunately, the outlook is often poor despite aggressive treatment, with median survival of approximately 15 months. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 341

Glossopharyngeal Neuralgia

341.1 Overview

Glossopharyngeal neuralgia is a rare neuropathic pain disorder characterized by paroxysmal, severe, stabbing pain in the posterior pharynx, tonsillar fossa, base of tongue, and deep ear (cranial nerves IX and X). It may be caused by neurovascular compression of the glossopharyngeal nerve. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

341.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

341.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

341.4 Signs and Symptoms

You may experience attacks triggered by swallowing, coughing, talking, or yawning, which can be associated with bradycardia, syncope, or asystole due to vagal activation. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

341.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

341.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

341.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

341.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

341.9 Treatment Options

- Treatment options include carbamazepine or oxcarbazepine as first-line
- Surgical options include microvascular decompression or through the skin radiofrequency rhizotomy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

341.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

341.11 Outlook

The outlook is generally positive with treatment. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 342

Glycogen Storage Diseases

342.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Glycogen storage diseases (GSDs) are a group of autosomal recessive disorders (except GSD IX, X-linked) resulting from defects in glycogen synthesis, degradation, or glycolysis, leading to abnormal glycogen accumulation. Von Gierke disease (GSD I) is caused by glucose-6-phosphatase deficiency, presenting with severe fasting hypoglycemia, hepatomegaly, growth retardation, lactic acidosis, hyperuricemia, hypertriglyceridemia, and a rounded, doll-like facies
- Treatment usually involves continuous nasogastric infusion of uncooked cornstarch. Pompe disease (GSD II) is caused by acid maltase deficiency, a lysosomal storage disorder presenting in infants with hypertrophic cardiomyopathy, macroglossia, hypotonia (floppy baby), and rapidly progressive weakness
- Enzyme replacement therapy (alglucosidase alfa) improves survival. Cori disease (GSD III) is caused by debranching enzyme deficiency, causing hypoglycemia, hepatomegaly, myopathy, and possible cardiomyopathy. McArdle disease (GSD V) is myophosphorylase deficiency (see Chapter 16, Muscles).

342.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

342.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

342.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

342.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

342.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

342.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

342.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

342.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

342.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

342.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 343

Gout

343.1 Overview

Gout is the most common inflammatory arthropathy, caused by deposition of monosodium urate crystals in joints and soft tissues due to chronic hyperuricemia (serum urate >6.8 mg/dL, the saturation point). It predominantly affects middle-aged men and postmenopausal women. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

343.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Acute gouty arthritis classically presents as sudden, severe monoarthritis of the first metatarsophalangeal joint (podagra), with intense pain, redness of the skin, warmth, and swelling
- Recurrent attacks may involve multiple joints. Chronic tophaceous gout causes tophi (urate deposits) in joints, cartilage, and soft tissues.

343.3 Risk Factors

This condition happens because of chronic hyperuricemia with risk factors including high-purine diet, alcohol (especially beer), obesity, chronic kidney disease, and certain medications (thiazide diuretics).

- Family history
- Obesity and sedentary lifestyle

- Poor dietary habits
- Advancing age
- Hormonal imbalances
- Certain medications
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

343.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

343.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

343.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

343.7 Diagnosis

- Doctors diagnose this based on synovial fluid analysis showing needle-shaped, negatively birefringent crystals under polarized light microscopy. Treatment for acute attacks includes NSAIDs, colchicine (most effective within 12 hours of onset), or corticosteroids
- Long-term urate-lowering therapy (ULT) with allopurinol or febuxostat (xanthine oxidase inhibitors) or probenecid (uricosuric) is indicated for recurrent gout, tophi, or joint damage, targeting serum urate <6 mg/dL.
- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

343.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

343.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

343.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

343.11 Outlook

Most people recover well with effective urate-lowering therapy.

Chapter 344

Granulomatosis with Polyangiitis

344.1 Overview

Granulomatosis with polyangiitis is an ANCA-associated necrotizing granulomatous vasculitis affecting small and medium-sized vessels, with a predilection for the respiratory tract and kidneys, with an incidence of approximately 2–12 per million, equal sex distribution, and typically presenting in the fourth to sixth decade. The condition involves c-ANCA (cytoplasmic) directed against proteinase 3 (PR3), leading to neutrophil activation, degranulation, and respiratory burst, causing endothelial injury and necrotizing inflammation. The classic triad involves the upper respiratory tract (sinusitis, epistaxis, nasal crusting, saddle nose deformity, subglottic stenosis), lower respiratory tract (cough, hemoptysis, lung nodules and cavitary lesions, alveolar bleeding), and kidney (rapidly progressive glomerulonephritis with crescents—pauci-immune on immunofluorescence), with other features including ocular involvement (scleritis, episcleritis, orbital pseudotumor), cutaneous purple spots on the skin, mononeuritis multiplex, and arthralgias. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

344.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

344.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

344.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
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- Symptoms may develop gradually or appear suddenly

344.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

344.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

344.7 Diagnosis

To make the diagnosis, doctors look for histologic confirmation (nasal, endobronchial, or kidney biopsy showing granulomatous inflammation, multinucleated giant cells, and necrotizing vasculitis) or serologic evidence of PR3-ANCA (positive in >90% of active generalized GPA). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

344.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

344.9 Treatment Options

Treatment typically involves with high-dose corticosteroids combined with rituximab or cyclophosphamide for induction of remission, with maintenance therapy using rituximab (scheduled infusions every 6 months) or methotrexate/azathioprine. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

344.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

344.11 Outlook

The outlook has improved dramatically with modern therapy, with 5-year survival exceeding 80%. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 345

Graves Disease

345.1 Overview

Graves disease is an autoimmune disorder and the most common cause of hyperthyroidism (60–80% of cases), occurring predominantly in women aged 20–50. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

345.2 Causes

This condition happens because of thyroid-stimulating immunoglobulin (TSI/TSHR autoantibodies) binding to and activating the TSH receptor, stimulating unregulated thyroid hormone synthesis and diffuse thyroid abnormal increase in cells. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

345.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

345.4 Signs and Symptoms

You may experience heat intolerance, weight loss despite increased appetite, tachycardia, palpitations, fine tremor, diffuse non-tender goiter with bruit, and specific extrathyroidal manifestations: Graves ophthalmopathy (exophthalmos, proptosis, lid lag due to orbital fibroblast TSHR expression) and pretibial myxedema. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

345.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

345.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

345.7 Diagnosis

Your doctor confirms the diagnosis with suppressed TSH, elevated free T4 and T3, positive TSI antibodies, and diffuse increased radioactive iodine uptake (RAIU). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

345.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

345.9 Treatment Options

Treatment options include antithyroid drugs (methimazole, propylthiouracil), radioactive iodine ablation, or thyroidectomy, along with beta-blockers (propranolol) for symptom control. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

345.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

345.11 Outlook

In most cases, the outlook is favorable. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 346

Guillain-Barré Syndrome

346.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Guillain-Barré syndrome is an acute, immune-mediated polyneuropathy characterized by rapidly progressive, symmetric limb weakness with areflexia
- It often follows a respiratory or digestive tract infection by 1–4 weeks, with *Campylobacter jejuni*, cytomegalovirus, Epstein-Barr virus, and Zika virus being common triggers.

346.2 Causes

This condition happens because of an autoimmune attack on peripheral nerves triggered by preceding infection. The most common variant is acute inflammatory demyelinating polyneuropathy (AIDP), while acute motor axonal neuropathy (AMAN) is more prevalent in Asia. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

346.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

346.4 Signs and Symptoms

You may experience rapidly progressive symmetric weakness, areflexia, and possible autonomic dysfunction (tachycardia, blood pressure lability) and respiratory muscle weakness requiring ventilatory support.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

346.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

346.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

346.7 Diagnosis

Doctors diagnose this based on clinical features and CSF analysis showing albuminocytological dissociation (elevated protein with normal cell count).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

346.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

346.9 Treatment Options

Treatment options include intravenous immunoglobulin (IVIg) or plasmapheresis. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

346.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

346.11 Outlook

- The outlook varies from person to person, with most patients recovering
- 5–10% die and 10–20% have significant residual disability.

Chapter 347

Gynecomastia

347.1 Overview

Gynecomastia is non-cancerous enlargement of breast tissue in males, caused by an imbalance of estrogen and androgen effects, present in up to 70% of adolescent males (physiologic gynecomastia, typically resolving within 1–2 years) and up to 65% of men over 50. Pathological causes include medications (spironolactone, finasteride, cimetidine, chemotherapy, exogenous testosterone/steroids, cannabinoids), liver disease, hypogonadism, hyperthyroidism, Klinefelter syndrome, and obesity (pseudogynecomastia). This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

347.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

347.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

347.4 Signs and Symptoms

You may experience a palpable, firm, disk-shaped tissue beneath the areola. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

347.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

347.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

347.7 Diagnosis

Doctors diagnose this using clinical examination and evaluation of hormonal levels. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

347.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

347.9 Treatment Options

Treatment options include addressing reversible causes, observation for physiologic or medication-induced gynecomastia, and surgical mastectomy for persistent, symptomatic cases. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

347.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

347.11 Outlook

Most people recover well. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 348

Haff Disease

348.1 Overview

Haff disease is a rare syndrome of rhabdomyolysis occurring within 24 hours of consuming certain freshwater or marine fish or crustaceans. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

348.2 Causes

The cause is an unknown toxin (possibly a heat-stable palytoxin-like compound from algae/dinoflagellates that accumulates in the food chain). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

348.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

348.4 Signs and Symptoms

Clinical features: sudden onset of severe muscle pain, muscle tenderness, weakness, dark urine (myoglobinuria), and elevated serum creatine kinase (CK, often >10,000 IU/L) within 6–24 hours of fish consumption, without fever, nervous system deficits, or rash. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

348.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

348.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

348.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with CK elevation, urinalysis (heme-positive without RBCs), and history of recent fish consumption. Management: aggressive IV fluids (target urine output >200 mL/hour), monitoring for acute kidney injury and compartment syndrome, and correction of electrolyte abnormalities. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

348.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

348.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

348.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

348.11 Outlook

- Most people recover well
- Most patients recover within days without long-term sequelae if treated promptly.

Chapter 349

Hallucinogen Toxicity

349.1 Overview

LSD (lysergic acid diethylamide) and psilocybin are serotonin 5-HT_{2A} receptor agonists producing altered perception, synesthesias, depersonalization, emotional lability, and hallucinations (typically visual). Physiological effects include dilated pupils, tachycardia, high blood pressure, hyperthermia, and diaphoresis. Acute adverse events include bad trips (anxiety, panic, dysphoria), psychosis, flashbacks, and serotonin syndrome. Phencyclidine (PCP) is a dissociative anesthetic acting as an NMDA receptor antagonist. Toxicity produces involuntary eye movements (horizontal, vertical, or rotatory—diagnostic clue), high blood pressure, tachycardia, hyperthermia, psychosis, violent behavior, rhabdomyolysis, and seizures. Large-volume fluid resuscitation for rhabdomyolysis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

349.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

349.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

349.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

349.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

349.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

349.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

349.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

349.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

349.10 Living with It

Treatment: benzodiazepines for agitation, antipsychotics for psychosis (caution: haloperidol may lower seizure threshold), and supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

349.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 350

Hallux Valgus

350.1 Overview

Hallux valgus is a structural deformity of the first ray of the foot characterized by lateral deviation of the hallux (great toe) and medial displacement of the first metatarsal head. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

350.2 Causes

This condition happens because of biomechanical instability of the first metatarsophalangeal (MTP) joint, often exacerbated by constrictive footwear, ligamentous laxity, flat feet (pes planus), or genetic predisposition. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

350.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

350.4 Signs and Symptoms

You may experience a painful exostosis (bunion) over the medial eminence of the first metatarsal head, lateral crowding of second and third toes, metatarsalgia, and difficulty fitting shoes. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

350.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

350.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

350.7 Diagnosis

- Your doctor confirms the diagnosis with weight-bearing foot radiographs showing a first MTP angle $> 15°$ and an intermetatarsal angle $> 9°$. Management begins with wide-toebox shoes, orthotics, and night splints
- Surgical osteotomy and soft tissue balancing are indicated for severe refractory pain.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

350.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

350.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

350.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

350.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 351

Hand, Foot, and Mouth Disease

351.1 Overview

Hand, foot, and mouth disease (HFMD) is a highly contagious pediatric viral exanthem caused by enteroviruses, most commonly Coxsackievirus A16 and Enterovirus 71. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

351.2 Causes

This condition happens because of fecal-oral or oral-oral transmission with primary viral replication in pharyngeal and digestive tract lymphoid tissue followed by viremic cutaneous and mucosal dissemination. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

351.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

351.4 Signs and Symptoms

You may experience low-grade fever, malaise, and sore throat followed by painful vesiculoulcerative enanthem on the buccal mucosa, tongue, and soft palate, and a non-pruritic, non-tender maculopapular or oval vesicular exanthem on the palms of the hands, soles of the feet, and buttocks. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

351.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

351.6 Complications

- Treatment typically involves supportive with oral fluid hydration and analgesics (aspirin is strictly avoided)
- Severe neurological complications (aseptic meningitis, brainstem encephalitis) may rarely occur with Enterovirus 71.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

351.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam based on characteristic lesion distribution. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

351.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

351.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

351.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

351.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 352

Hantavirus Pulmonary Syndrome

352.1 Overview

Hantavirus lung syndrome is a severe, often fatal respiratory illness caused by hantaviruses, with the primary cause in North America being Sin Nombre virus (SNV), with 800 cases reported in the Americas since 1993, mortality 30–40%, and transmission through inhalation of aerosolized excreta from infected rodents. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

352.2 Causes

This condition happens because of the virus infecting endothelial cells (particularly in the lungs), causing increased vascular permeability without direct cytotoxicity, mediated by CD8+ T cell activation and cytokine release disrupting endothelial barrier function. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

352.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

352.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

352.5 Progression and Stages

You may experience an incubation of 9–35 days, with prodrome (3–5 days): fever, muscle pain, headache, and digestive tract symptoms, followed by rapid onset of cardiopulmonary phase: cough, shortness of breath, and respiratory failure from lung swelling, often within 24 hours, with hypotension and shock being common. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

352.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

352.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam in endemic areas, confirmed by serology.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

352.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

352.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

352.10 Living with It

Treatment typically involves aggressive supportive care (mechanical ventilation, vasopressors, ECMO in severe cases). outlook: mortality 30–40%.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 353

Harmful Algal Blooms

353.1 Overview

Harmful algal blooms (HABs) are caused by toxin-producing cyanobacteria (blue-green algae) and marine algae that accumulate in freshwater lakes, ponds, and coastal waters during warm weather. Cyanobacteria produce multiple toxin classes: hepatotoxins (microcystins, nodularins), neurotoxins (anatoxin-a, saxitoxin), and dermatoxins (lyngbyatoxin). Exposure occurs through recreational water contact, drinking contaminated water, or consumption of toxin-accumulating shellfish. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

353.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

353.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

353.4 Signs and Symptoms

- Clinical features: contact—dermatitis, conjunctivitis, allergic rhinitis
- Ingestion—acute gastroenteritis, hepatotoxicity, neurotoxicity
- Inhalation—asthma-like symptoms.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

353.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

353.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

353.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with history of exposure to water with visible algal blooms. Management: supportive—decontamination, IV fluids for GI losses, respiratory support for neurotoxic effects. No specific antidotes exist. Climate change is increasing HAB frequency, duration, and geographic range. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

353.8 Prevention

Prevention includes avoiding water contact where blooms are visible.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

353.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

353.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

353.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 354

Hashimoto Thyroiditis

354.1 Overview

Hashimoto thyroiditis (chronic lymphocytic thyroiditis) is an autoimmune disease and the most common cause of hypothyroidism in iodine-sufficient regions, affecting women 10 times more frequently than men. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

354.2 Causes

This condition happens because of cell-mediated cytotoxicity (CD8+ T cells) and anti-thyroid peroxidase (anti-TPO) and anti-thyroglobulin autoantibodies, leading to lymphocytic infiltration, germinal center formation, Hurthle cell metaplasia, and progressive destruction of thyroid follicles. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

354.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

354.4 Signs and Symptoms

You may experience cold intolerance, weight gain, fatigue, constipation, dry skin, bradycardia, muscle weakness, and a non-tender firm goiter. Patients may experience transient hyperthyroidism (hashitoxicosis) early in the disease due to follicular destruction. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

354.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

354.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

354.7 Diagnosis

Your doctor confirms the diagnosis with elevated serum TSH, low free T4, and high titers of anti-TPO antibodies. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

354.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

354.9 Treatment Options

Treatment typically involves lifelong hormone replacement with levothyroxine, adjusted to maintain TSH within normal limits. Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

354.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

354.11 Outlook

Most people recover well with proper dosing. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 355

Head Lice (Pediculosis Capitis)

355.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Head lice is infestation of the scalp by *Pediculus humanus capitis*, common in school-age children (6–12 million cases/year in the US). Transmission occurs through direct head-to-head contact
- Fomite transmission is less common.

355.2 Causes

This condition happens because of louse infestation with eggs (nits) laid at the hair base that hatch in 7–10 days and mature in 2–3 weeks. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

355.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

355.4 Signs and Symptoms

You may experience scalp itching (from hypersensitivity to louse saliva), visible nits (0.8 mm, yellow-white, firmly attached to the hair shaft), and less commonly live lice. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

355.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

355.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

355.7 Diagnosis

Doctors diagnose this using visual inspection with a fine-toothed comb

- Wood's lamp shows nits as pale blue. The first-choice treatment typically involves permethrin 1% or pyrethrins (OTC), applied for 10 minutes and repeated in 7–10 days
- Second-line options include malathion 0.5% lotion, benzyl alcohol 5%, spinosad 0.9%, and ivermectin 0.5% lotion
- Oral ivermectin is used for refractory cases. Wet combing is a non-chemical option. All close contacts should be checked
- Environmental measures include washing bedding in hot water.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

355.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

355.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

355.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

355.11 Outlook

- Most people recover well with appropriate treatment
- No-nit policies in schools are not recommended by the AAP/CDC.

Chapter 356

Headache Attributed to Giant Cell Arteritis

See Chapter ??, Section 336.

356.1 Overview

This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

356.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

356.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

356.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

356.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

356.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

356.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

356.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

356.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

356.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

356.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 357

Headache Attributed to Intracranial Hypertension

357.1 Overview

Idiopathic intracranial high blood pressure (IIH, pseudotumor cerebri) is a secondary headache disorder characterized by elevated intracranial pressure without identifiable cause. The annual incidence is 0.5–3 per 100,000, rising to 12–20 per 100,000 in obese women of childbearing age. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

357.2 Causes

This condition happens because of impaired CSF absorption at the arachnoid granulations or elevated cerebral venous sinus pressure. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

357.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

357.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

357.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

357.6 Complications

- You may experience daily, diffuse, pressure-like headache worse in the morning or with Valsalva, transient visual obscurations, pulsatile ringing in the ears, and papilledema
- Visual loss is the most serious complication.
- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

357.7 Diagnosis

To make the diagnosis, doctors look for elevated opening pressure on lumbar puncture (>25 cm H2O) with normal CSF composition, normal neuroimaging, and no alternative cause. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

357.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose

- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

357.9 Treatment Options

- Treatment options include weight loss, acetazolamide, and serial ophthalmological monitoring
- Refractory cases may require optic nerve sheath fenestration or CSF shunting.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

357.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

357.11 Outlook

- The outlook varies from person to person
- Visual loss occurs in 10–25% of patients.

Chapter 358

Headache Attributed to Intracranial Hypotension

358.1 Overview

Spontaneous intracranial hypotension (SIH) is a secondary headache disorder caused by spontaneous CSF leak, typically at the spinal level (thoracic or cervical). This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

358.2 Causes

This condition happens because of CSF leak causing low CSF volume and pressure. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

358.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

358.4 Signs and Symptoms

You may experience orthostatic headache worse within seconds to minutes of upright posture and improving with recumbency, with associated neck pain, nausea, ringing in the ears, and dizziness. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

358.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

358.6 Complications

- Myocardial infarction (heart attack)

- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

358.7 Diagnosis

Doctors diagnose this using clinical presentation and MRI showing diffuse pachymeningeal enhancement, brain sagging, and subdural fluid collections. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

358.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

358.9 Treatment Options

- Treatment typically involves conservative initially (bed rest, caffeine)

- Epidural blood patch is the The main treatment for persistent symptoms, with success rates of 80–90%.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

358.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

358.11 Outlook

The outlook is generally positive with treatment. Post-dural puncture headache (PDPH) is a secondary headache disorder complicating lumbar puncture, spinal anesthesia, or epidural anesthesia, occurring in 1–30% of cases. This condition happens because of persistent CSF leak through the dural puncture site. You may experience orthostatic headache typically fronto-occipital, with onset within 24–72 hours after dural puncture. Doctors usually diagnose this based on your symptoms and a physical exam. Treatment options include conservative measures for mild cases and epidural blood patch for moderate to severe cases. The outlook is generally positive. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 359

Headache Attributed to Sinusitis

See Chapter ??, Section 178.

359.1 Overview

This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

359.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

359.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

359.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

359.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

359.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

359.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

359.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

359.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

359.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

359.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 360

Headache in Brain Tumors

See Chapter ??, Section ??.

360.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

360.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

360.3 Risk Factors

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

360.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

360.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

360.6 Complications

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

360.7 Diagnosis

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

360.8 Prevention

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

360.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

360.10 Living with It

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

360.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 361

Heart Failure with Preserved Ejection Fraction

361.1 Overview

Heart failure with preserved ejection fraction (HFpEF, EF $\geq$50%) is the predominant heart failure phenotype in older adults, especially older women with high blood pressure, diabetes, obesity, and atrial fibrillation. Disease mechanism includes left ventricular diastolic dysfunction, systemic inflammation, lung high blood pressure, and chronotropic incompetence. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

361.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

361.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

361.4 Signs and Symptoms

Clinical The condition usually appears as shortness of breath on exertion, exercise intolerance, and fluid overload. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

361.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

361.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

361.7 Diagnosis

To make the diagnosis, doctors look for clinical signs/symptoms of HF, preserved LVEF, and evidence of diastolic dysfunction or elevated filling pressures. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

361.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

361.9 Treatment Options

Treatment options include diuretics for volume overload, blood pressure control, SGLT2 inhibitors, and Treatment for comorbidities. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

361.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

361.11 Outlook

The outlook varies from person to person. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 362

Heart Failure with Preserved Ejection Fraction

362.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Heart failure with preserved ejection fraction accounts for approximately half of heart failure cases and is marked by symptoms of heart failure with left ventricular ejection fraction of 50% or greater
- It is linked to aging, high blood pressure, obesity, diabetes mellitus, and atrial fibrillation.

362.2 Causes

This condition happens because of diastolic dysfunction (impaired myocardial relaxation and increased stiffness), elevated filling pressures, and systemic inflammation. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

362.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

362.4 Signs and Symptoms

- You may experience exertional shortness of breath, fatigue, and volume overload. Diagnosis is often challenging, requiring integration of symptoms, echocardiography (including diastolic function assessment), elevated natriuretic peptides, and sometimes invasive hemodynamic measurement. Management targets comorbidities, volume management with diuretics, and SGLT2 inhibitors (empagliflozin, dapagliflozin)
- Exercise training and weight management are important non-pharmacological interventions.
- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

362.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

362.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events

- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

362.7 Diagnosis

History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

362.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

362.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)

- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

362.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

362.11 Outlook

The outlook varies from person to person, with similar mortality rates to heart failure with reduced ejection fraction. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 363

Heart Failure with Reduced Ejection Fraction

363.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Heart failure with reduced ejection fraction is a clinical syndrome characterized by symptoms and signs of heart failure with left ventricular ejection fraction of 40% or less
- Common etiologies include ischemic heart disease, high blood pressure, dilated cardiomyopathy, valvular disease, and myocarditis.

363.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

363.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus

- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

363.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

363.5 Progression and Stages

Symptoms can range from shortness of breath and fatigue (NYHA class II–IV) to volume overload with congestion. This condition happens because of neurohormonal activation (sympathetic nervous system, renin-angiotensin-aldosterone system) that drives disease progression. Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

363.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

363.7 Diagnosis

Doctors diagnose this based on clinical assessment and echocardiography demonstrating reduced left ventricular ejection fraction. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

363.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

363.9 Treatment Options

Treatment focuses on four pillars of guideline-directed medical therapy: ACE inhibitor, ARB, or ARNI

- Beta-blocker
- Mineralocorticoid receptor antagonist
- And SGLT2 inhibitor
- Device therapy with implantable cardioverter-defibrillator and cardiac resynchronization therapy reduces sudden death and improves symptoms.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)

- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

363.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

363.11 Outlook

The outlook has improved significantly with modern medical therapy. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 364

Heat Stroke and Hyperthermia

364.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Heat stroke is a life-threatening condition characterized by core body temperature >40°C (104°F) with CNS dysfunction. Two forms: classic (nonexertional)—affects elderly, infants, and those in heat waves
- And exertional—affects athletes, military personnel, and laborers.

364.2 Causes

This condition happens because of failure of thermoregulatory mechanisms with overwhelming heat production or impaired heat dissipation, causing direct cellular injury, endotoxemia, systemic inflammatory response, and multiorgan failure. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

364.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

364.4 Signs and Symptoms

- Clinical features: hyperthermia, anhydrosis in classic heat stroke (skin hot and dry), CNS dysfunction (confusion, loss of coordination, delirium, seizures, coma), tachycardia, hypotension, and circulatory collapse. Diagnosis: core temperature >40°C with CNS dysfunction after ruling out other causes. Management: rapid cooling—evaporative cooling, ice packs, cold IV fluids, and ice water immersion
- Aggressive IV fluids
- Benzodiazepines for uncontrolled shivering or seizures.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

364.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

364.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

364.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

364.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

364.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

364.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

364.11 Outlook

Your outlook depends on promptness of treatment. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 365

Helicobacter Pylori Infection

365.1 Overview

Helicobacter pylori is a gram-negative, microaerophilic, spiral bacterium that colonizes the gastric mucosa of approximately half the world's population. It is the primary cause of chronic gastritis, peptic ulcer disease, gastric mucosa-associated lymphoid tissue (MALT) lymphoma, and gastric adenocarcinoma. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

365.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

365.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices

- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

365.4 Signs and Symptoms

Symptoms can range from asymptomatic colonization to dyspepsia, peptic ulcer disease, and gastric malignancy. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

365.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- This condition happens because of fecal-oral, oral-oral, or gastro-oral transmission, with most infections acquired in childhood
- The bacterium uses urease to neutralize gastric acid, flagella to penetrate the mucus layer, and adhesins to attach to epithelial cells, with virulence factors (CagA, VacA) determining disease severity.

365.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

365.7 Diagnosis

Doctors diagnose this based on urea breath test, stool antigen test, or using a thin camera tube biopsy-based testing (rapid urease test, histology, culture). Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

365.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)

- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

365.9 Treatment Options

Treatment focuses on PPI-based triple therapy (PPI + amoxicillin + clarithromycin) or bismuth quadruple therapy for 14 days, with confirmation of eradication by urea breath test or stool antigen. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

365.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

365.11 Outlook

- Most people recover well with successful eradication
- Untreated infection carries increased risk of gastric adenocarcinoma.

Chapter 366

HELLP Syndrome

366.1 Overview

It occurs in 0.5–0.9% of pregnancies and in 10–20% of women with severe preeclampsia. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

366.2 Causes

This condition happens because of microvascular endothelial damage leading to platelet activation, microangiopathic hemolytic anemia, and hepatocellular tissue death. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

366.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

366.4 Signs and Symptoms

You may experience right upper quadrant or epigastric pain (65%), nausea and vomiting (50%), and headache (30%). The Tennessee classification requires all three criteria: hemolysis (LDH >600 U/L, schistocytes on smear, bilirubin >1.2 mg/dL), elevated liver enzymes (AST $\geq$70 U/L), and low platelets (<100,000/μL).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

366.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

366.6 Complications

HELLP syndrome is a serious complication of severe preeclampsia characterized by Hemolysis, Elevated Liver enzymes, and Low Platelets.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

366.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

366.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

366.9 Treatment Options

Treatment typically involves delivery, which usually results in disease resolution

- Corticosteroids may improve platelet count temporarily
- Plasma exchange may be considered for persistent postpartum HELLP.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

366.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

366.11 Outlook

- The outlook is generally positive with delivery
- Differential diagnosis includes acute fatty liver of pregnancy, TTP, and HUS.

Chapter 367

Hemicrania Continua

367.1 Overview

Hemicrania continua is a trigeminal autonomic cephalalgia presenting as a continuous, strictly unilateral headache of moderate intensity, with superimposed exacerbations of severe pain, present for >3 months without remission. Prevalence is approximately 0.1% of headache clinic populations, with a female predominance. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

367.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

367.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

367.4 Signs and Symptoms

You may experience continuous unilateral headache with ipsilateral cranial autonomic features during exacerbations and a sense of restlessness or agitation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

367.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

367.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

367.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- It is exquisitely responsive to indomethacin (75–150 mg/day), with continuous pain resolving within 24–72 hours of starting therapy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

367.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

367.9 Treatment Options

Treatment focuses on indomethacin with digestive tract and kidney monitoring. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

367.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

367.11 Outlook

The outlook is generally positive with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 368

Hemolytic Anemias

368.1 Overview

Hemolytic anemias are a group of disorders characterized by premature destruction of red blood cells (RBCs), leading to anemia, elevated LDH, indirect hyperbilirubinemia (unconjugated), and reticulocytosis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

368.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

368.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

368.4 Signs and Symptoms

You may experience fatigue, jaundice, dark urine, and splenomegaly. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected

area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

368.5 Progression and Stages

The condition is classified as intrinsic (membrane disorders: hereditary spherocytosis, hereditary elliptocytosis The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Enzyme disorders: G6PD deficiency, pyruvate kinase deficiency
- Hemoglobinopathies: sickle cell disease, thalassemia) or extrinsic (immune: autoimmune hemolytic anemia, hemolytic disease of the newborn
- Mechanical: microangiopathic hemolytic anemia in TTP/HUS, DIC, prosthetic valves
- Infectious: malaria, babesiosis).

368.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

368.7 Diagnosis

Diagnosis typically involves reticulocyte count, peripheral blood smear, direct antiglobulin test (Coombs test), hemoglobin electrophoresis, and specific tests based on suspected cause. Management varies by cause and may include transfusions, corticosteroids (for immune hemolysis), splenectomy (for spherocytosis and some other hemolytic anemias), and disease-specific therapies. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

368.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

368.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

368.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

368.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 369

Hemophagocytic Lymphohistiocytosis and Macrophage Activation Syndrome

369.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Hemophagocytic lymphohistiocytosis is a life-threatening syndrome of pathologic immune activation, characterized by uncontrolled growth of activated lymphocytes and macrophages (histiocytes) producing a massive cytokine storm (hypercytokinemia: IFN-γ, IL-1, IL-6, IL-18, TNF). There are two forms: primary (familial) HLH (fHLH), caused by autosomal recessive mutations in genes critical for perforin-mediated cytotoxic function, typically presenting in infancy or early childhood
- And secondary HLH (sHLH), triggered by infections (EBV is the most common trigger), malignancies, autoimmune diseases (macrophage activation syndrome, MAS, associated with systemic juvenile idiopathic arthritis [sJIA] and adult-onset Still's disease [AOSD]), and immunosuppression. The incidence of HLH is estimated at 1 per 50,000–100,000 in children.

369.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

369.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

369.4 Signs and Symptoms

You may experience persistent high fever, hepatosplenomegaly, lymphadenopathy, cytopenias (at least two lineages: anemia, thrombocytopenia, neutropenia), skin rash, and neurological symptoms (seizures, altered mental status, meningismus).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

369.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

369.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

369.7 Diagnosis

Doctors diagnose this using HLH-2004 criteria (five of eight: fever, splenomegaly, cytopenias, hypertriglyceridemia and/or hypofibrinogenemia, hemophagocytosis in bone marrow/spleen/lymph nodes, low or absent NK-cell activity, ferritin $\geq$500 ng/mL, elevated soluble CD25/IL-2R $\geq$2400 U/mL). Treatment for HLH includes immunosuppression with dexamethasone and etoposide (HLH-94/2004 protocol), with allogeneic hematopoietic stem cell transplantation for familial HLH and refractory cases.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

369.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

369.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

369.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

369.11 Outlook

Unfortunately, the outlook is often poor without treatment but has improved with protocolized therapy.

Chapter 370

Hemophilia A

370.1 Overview

Hemophilia A is an X-linked recessive bleeding disorder caused by deficiency or dysfunction of coagulation factor VIII, affecting approximately 1 in 5000 males. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

370.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

370.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

370.4 Signs and Symptoms

You may experience spontaneous hemarthroses (knee, elbow, ankle—leading to chronic hemophilic arthropathy), deep muscle hematomas, retroperitoneal bleeding, and prolonged bleeding after surgery or trauma, with intracranial bleeding being the leading cause of death.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

370.5 Progression and Stages

This condition happens because of factor VIII deficiency, with severity classified by residual factor VIII activity: severe (<1%), moderate (1–5%), mild (5–40%). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

370.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

370.7 Diagnosis

Doctors diagnose this using prolonged aPTT with normal PT, decreased factor VIII activity, and positive family history.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

370.8 Prevention

Treatment focuses on factor VIII replacement therapy (recombinant or plasma-derived), given prophylactically (for severe hemophilia, typically 3 times weekly) or on-demand for bleeding episodes, with emicizumab (a bispecific antibody bridging factor IXa and factor X) used for prophylaxis.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

370.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

370.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

370.11 Outlook

The outlook has been transformed by factor replacement and gene therapy (valoctocogene roxaparvovec), though inhibitors (neutralizing alloantibodies against factor VIII) develop in 25–30% of severe hemophilia A and require bypassing agents or immune tolerance induction.

Chapter 371

Hemophilia B

371.1 Overview

Hemophilia B (Christmas disease) is an X-linked recessive bleeding disorder caused by deficiency or dysfunction of coagulation factor IX, accounting for approximately 15% of hemophilia cases. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

371.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

371.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

371.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

371.5 Progression and Stages

Clinical features and classification are identical to hemophilia A. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

371.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

371.7 Diagnosis

Doctors diagnose this using prolonged aPTT, normal PT, and decreased factor IX activity.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

371.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

371.9 Treatment Options

Treatment focuses on factor IX replacement therapy (recombinant or plasma-derived), with a longer half-life than factor VIII allowing less frequent dosing, and emicizumab is not effective for hemophilia B. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

371.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

371.11 Outlook

The outlook has improved with gene therapy (etranacogene dezaparvovec), which offers durable factor IX expression and reduction in bleeding episodes.

Chapter 372

Hemorrhagic Stroke

372.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Involving bleeding stroke is a cerebrovascular event resulting from rupture of a blood vessel within the brain, leading to intraparenchymal or subarachnoid bleeding. Intracerebral bleeding (ICH) is often associated with chronic high blood pressure, cerebral amyloid angiopathy, vascular malformations, or coagulopathy
- Subarachnoid bleeding (SAH) most commonly results from rupture of a cerebral aneurysm.

372.2 Causes

This condition happens because of spontaneous rupture of a cerebral blood vessel. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

372.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

372.4 Signs and Symptoms

- You may experience sudden severe headache, vomiting, decreased consciousness, and weakness, numbness, or problems with movement that may progress rapidly
- SAH presents with thunderclap headache, meningismus, and loss of consciousness.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

372.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

372.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

372.7 Diagnosis

Doctors diagnose this based on CT head as the initial diagnostic study. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

372.8 Prevention

- Treatment for ICH focuses on blood pressure control, reversal of anticoagulation, and neurosurgical intervention when indicated
- SAH requires urgent neurosurgical clipping or endovascular coiling of the aneurysm with nimodipine to prevent vasospasm.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

372.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

372.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

372.11 Outlook

Your outlook depends on bleeding location, size, and timeliness of intervention. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 373

Henoch-Schönlein Purpura (IgA Vasculitis)

373.1 Overview

Henoch-Schönlein purple spots on the skin (HSP) is an IgA-mediated small-vessel vasculitis, the most common vasculitis in children (see Chapter 27, Connective Tissue Diseases for comprehensive coverage). Classic tetrad: palpable purple spots on the skin (dependent areas), arthritis/joint pain, abdominal pain, and kidney involvement (hematuria, proteinuria). Mean age is 4–6 years. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

373.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

373.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection

- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

373.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

373.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

373.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

373.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Skin or kidney biopsy shows IgA deposition.
- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

373.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

373.9 Treatment Options

- Treatment typically involves supportive
- Corticosteroids are used for severe abdominal pain or significant kidney disease.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

373.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management

- Monitor for medication side effects
- Plan activities around energy levels during flares

373.11 Outlook

- Most people recover well
- Most cases resolve spontaneously.

Chapter 374

Hepatic Encephalopathy

374.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

374.2 Causes

This condition happens because of impaired liver clearance of neurotoxins, precipitated by infection, digestive tract bleeding, constipation, electrolyte imbalances, dehydration, and hepatorenal syndrome. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

374.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

374.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

374.5 Progression and Stages

Symptoms can range from subtle thinking and memory impairment (grade 1) to confusion (grade 2), somnolence (grade 3), and coma (grade 4). liver encephalopathy is a neuropsychiatric syndrome occurring in patients with advanced liver disease or portosystemic shunting, resulting from accumulation of neurotoxic substances (ammonia being the most important) that are normally cleared by the liver. outlook is related to the severity of underlying liver disease. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

374.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

374.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by elevated ammonia levels and exclusion of other causes of altered mental status. Management targets precipitating factors and reduces ammonia production and absorption with lactulose (titrated to 2–3 loose stools per day) and rifaximin. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

374.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

374.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

374.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

374.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 375

Hepatitis A

375.1 Overview

Hepatitis A is an acute, self-limited viral hepatitis transmitted via the fecal-oral route through contaminated food or water or close contact with infected individuals. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

375.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of infection with hepatitis A virus (HAV), with an incubation period of 15–50 days. Most infections in children are asymptomatic
- Adults present with malaise, anorexia, nausea, vomiting, right upper quadrant pain, jaundice, and dark urine.

375.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

375.4 Signs and Symptoms

You may experience malaise, anorexia, nausea, vomiting, right upper quadrant pain, jaundice, and dark urine. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

375.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

375.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)

- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

375.7 Diagnosis

Doctors diagnose this based on elevated serum aminotransferases (ALT, AST) and confirmed by anti-HAV IgM antibodies. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

375.8 Prevention

- Treatment typically involves supportive, as no specific antiviral treatment exists
- Prevention is through vaccination (inactivated virus) and hygiene measures.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

375.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

375.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

375.11 Outlook

- Most people recover well, as HAV does not cause chronic infection or cirrhosis
- Fulminant liver failure is rare (0.1–0.3%).

Chapter 376

Hepatitis B

376.1 Overview

Hepatitis B is a major global infection caused by hepatitis B virus (HBV) that affects approximately 296 million chronically infected individuals worldwide. Transmission occurs through through the skin exposure (needle sharing, needlestick injuries), sexual contact, and perinatal transmission. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

376.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of HBV infection, with chronic infection defined by persistence of HBsAg for more than 6 months
- Chronic hepatitis B carries risks of cirrhosis (15–40%) and hepatocellular carcinoma (2–5% per year in cirrhotics).

376.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

376.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

376.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Symptoms of acute infection range from subclinical to fulminant liver failure
- The natural history follows a spectrum of immune tolerance, immune active phase, inactive carrier state, and reactivation.

376.6 Complications

- Sepsis and septic shock

- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

376.7 Diagnosis

Diagnosis typically involves serological testing (HBsAg, anti-HBs, HBeAg, anti-HBe, anti-HBc IgM/IgG) and HBV DNA quantification. Management with nucleos(t)ide analogues (entecavir, tenofovir) or pegylated interferon-alpha is indicated for active disease. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

376.8 Prevention

- Outlook is improved with antiviral therapy
- Vaccination is highly effective in preventing infection.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated

- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

376.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

376.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

376.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 377

Hepatitis C

377.1 Overview

Hepatitis C is a viral infection caused by hepatitis C virus (HCV) that affects approximately 58 million people globally and is a leading cause of cirrhosis, liver failure, and hepatocellular carcinoma. Transmission is primarily through blood exposure (intravenous drug use, blood transfusions before screening, needlestick injuries). This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

377.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of HCV infection, with approximately 75–85% of acute cases progressing to chronic infection. Chronic hepatitis C follows an indolent course, with 10–20% developing cirrhosis over 20–30 years
- Extrahepatic manifestations include cryoglobulinemia, membranoproliferative glomerulonephritis, and B-cell lymphoma.

377.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

377.4 Signs and Symptoms

Symptoms of acute infection are usually asymptomatic. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

377.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

377.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage

- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

377.7 Diagnosis

Diagnosis typically involves anti-HCV antibody testing followed by HCV RNA quantification and genotyping. Management with direct-acting antivirals (DAAs) achieves sustained virological response rates exceeding 95% with 8–12 weeks of oral therapy. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

377.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)

- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

377.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

377.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

377.11 Outlook

- Most people recover well with treatment
- The WHO has set a goal of hepatitis C elimination by 2030.

Chapter 378

Hepatitis D

378.1 Overview

Hepatitis D is an infection caused by hepatitis D virus (HDV), a defective RNA virus that requires co-infection or superinfection with HBV (using HBsAg for its envelope), affecting approximately 5% of HBV-infected individuals. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

378.2 Causes

This condition happens because of HDV infection in the presence of HBV. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

378.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

378.4 Signs and Symptoms

You may experience more severe hepatitis compared with HBV monoinfection. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

378.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Co-infection results in more severe acute hepatitis and a higher risk of fulminant liver failure than HBV alone
- Superinfection in chronic HBV carriers accelerates progression to cirrhosis and HCC.

378.6 Complications

- Sepsis and septic shock

- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

378.7 Diagnosis

Doctors diagnose this using anti-HDV antibody testing and HDV RNA quantification. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

378.8 Prevention

outlook is worse than HBV monoinfection, although prevention of HBV through vaccination prevents HDV infection. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings

- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

378.9 Treatment Options

- Treatment focuses on pegylated interferon-alpha with limited efficacy
- Bulevirtide has been approved in Europe as an entry inhibitor.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

378.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

378.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 379

Hepatitis E

379.1 Overview

Hepatitis E is an inflammatory liver disease caused by hepatitis E virus (HEV, Orthohepevirus A), a non-enveloped single-stranded RNA virus in the Hepeviridae family, with four major genotypes: genotypes 1 and 2 cause waterborne epidemics in developing countries, and genotypes 3 and 4 cause zoonotic infection (primarily from swine) in developed countries. An estimated 20 million HEV infections occur annually, with 3.3 million symptomatic cases and 44,000 deaths. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

379.2 Causes

This condition happens because of fecal-oral transmission (genotypes 1, 2 through contaminated water) or consumption of undercooked pork or game meat (genotypes 3, 4). HEV replicates in hepatocytes, causing hepatocellular injury and tissue death. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

379.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to

endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

379.4 Signs and Symptoms

Symptoms of acute hepatitis E include jaundice, malaise, anorexia, abdominal pain, hepatomegaly, fever, and elevated transaminases, and are usually self-limited over 4–6 weeks. A distinctive feature is severe disease in pregnancy (genotypes 1, 2), with mortality of 20–30% in the third trimester, high rates of fulminant liver failure, and adverse fetal outcomes (preterm delivery, stillbirth). Chronic HEV infection occurs in immunocompromised individuals (organ transplant recipients, HIV, hematologic malignancy) with genotypes 3, 4, leading to rapidly progressive cirrhosis. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

379.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

379.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

379.7 Diagnosis

Doctors diagnose this based on serum anti-HEV IgM and HEV RNA detection (PCR). Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

379.8 Prevention

A recombinant HEV vaccine (Hecolin) is licensed in China. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

379.9 Treatment Options

Treatment typically involves supportive for acute hepatitis E

- Ribavirin for chronic HEV (12 weeks)
- Reduction of immunosuppression is first-line for transplant recipients.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

379.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

379.11 Outlook

- Most people recover well for acute HEV in immunocompetent hosts
- Chronic HEV is treatable with ribavirin.

Chapter 380

Hepatoblastoma

380.1 Overview

Hepatoblastoma is the most common cancerous liver tumor in children, primarily affecting children under 3 years of age. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

380.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

380.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity

- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

380.4 Signs and Symptoms

You may experience an asymptomatic abdominal mass, abdominal distension, and occasionally pain, anorexia, and jaundice. Serum alpha-fetoprotein is markedly elevated in >90% of cases. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

380.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

380.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

380.7 Diagnosis

Doctors diagnose this using abdominal ultrasound, CT/MRI, and AFP level. Complete surgical removal is the cornerstone of treatment, with neoadjuvant chemotherapy for initially unresectable tumors. Survival is approximately 70–80% with complete removal and chemotherapy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

380.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

380.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

380.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

380.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 381

Hepatocellular Carcinoma

381.1 Overview

Hepatocellular carcinoma is the most common primary liver cancer and the third leading cause of cancer-related death worldwide, arising predominantly in the setting of chronic liver disease and cirrhosis. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

381.2 Causes

This condition happens because of cancerous transformation of hepatocytes in the setting of chronic liver disease. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

381.3 Risk Factors

Major risk factors include chronic HBV, chronic HCV, alcohol-related liver disease, NAFLD/NASH, and hereditary hemochromatosis.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption

- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

381.4 Signs and Symptoms

Clinical features may include abdominal pain, weight loss, and liver decompensation. Staging uses the Barcelona Clinic Liver Cancer (BCLC) system. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

381.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- The right treatment depends on stage: curative options include surgical removal, liver transplantation (Milan criteria), and radiofrequency ablation for early-stage

disease
- Intermediate-stage disease doctors typically treat it with transarterial chemoembolization (TACE)
- Advanced disease is managed with systemic therapy (sorafenib, lenvatinib, atezolizumab plus bevacizumab). Your outlook depends on stage at diagnosis and treatment received.

381.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

381.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
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- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging

- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

381.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

381.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

381.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
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- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

381.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 382

Hereditary Angioedema

382.1 Overview

Hereditary angioedema (HAE) is an autosomal dominant disorder caused by deficiency (type I, 85%) or dysfunction (type II, 15%) of C1 esterase inhibitor (C1-INH), with a prevalence of 1 in 50,000. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

382.2 Causes

This condition happens because of uncontrolled activation of the contact system, leading to excessive bradykinin production, increased vascular permeability, and recurrent episodes of subcutaneous and submucosal swelling. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

382.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

382.4 Signs and Symptoms

You may experience recurrent episodes of non-pitting, non-pruritic swelling affecting the extremities, face, genitals, trunk, and potentially life-threatening laryngeal swelling leading to airway obstruction. digestive tract involvement causes severe abdominal pain, nausea, vomiting, and diarrhea, often misdiagnosed as acute abdomen. Episodes last 2–5 days without treatment. Unlike allergic angioedema, there is no urticaria, itching, or response to antihistamines or epinephrine.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

382.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

382.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

382.7 Diagnosis

Doctors diagnose this based on low C4 levels during attacks (screening test), low C1-INH antigenic levels (type I), low functional C1-INH activity (type II), and C1-INH genetic testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

382.8 Prevention

Treatment options include on-demand treatment for acute attacks (plasma-derived C1-INH, recombinant C1-INH, icatibant [bradykinin B2 receptor antagonist], ecallantide [kallikrein inhibitor]), short-term prophylaxis prior to procedures (plasma-derived C1-INH or attenuated androgens), and long-term prophylaxis (plasma-derived C1-INH, lanadelumab [monoclonal antibody against plasma kallikrein], berotralstat [oral plasma kallikrein inhibitor], or attenuated androgens: danazol, stanozolol).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

382.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

382.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

382.11 Outlook

- Most people recover well with appropriate management
- Before modern therapy, mortality from laryngeal swelling was 25–30%.

Chapter 383

Hereditary Hemochromatosis

383.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Hereditary hemochromatosis is a group of genetic disorders characterized by excessive intestinal iron absorption and progressive iron deposition in tissues, leading to organ damage
- The most common form is type 1 (HFE-related HH), an autosomal recessive disorder caused by mutations in the HFE gene (predominantly C282Y homozygosity), affecting approximately 1 in 200–300 individuals of Northern European descent, though clinical disease develops in only 10–30% of C282Y homozygotes.

383.2 Causes

This condition happens because of excessive iron absorption leading to iron deposition in the liver, heart, pancreas, joints, skin, and endocrine glands. Clinical manifestations include hepatomegaly, cirrhosis, hepatocellular carcinoma, diabetes mellitus, cardiomyopathy, arthropathy, hypogonadism, and skin hyperpigmentation. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

383.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

383.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

383.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

383.6 Complications

Treatment focuses on regular phlebotomy, which is highly effective in preventing complications if initiated early.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

383.7 Diagnosis

- Diagnosis typically involves elevated serum transferrin saturation (greater than 45%), elevated serum ferritin, and HFE genotyping
- Liver biopsy or MRI T2* quantification assesses iron burden.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

383.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

383.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

383.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

383.11 Outlook

- Most people recover well with early diagnosis and treatment but poor if cirrhosis or cardiomyopathy has developed
- First-degree relatives should be screened.

Chapter 384

Hereditary Hemochromatosis

384.1 Overview

Hereditary hemochromatosis (HH) is an autosomal recessive disorder of iron metabolism characterized by excessive intestinal absorption of dietary iron, leading to progressive iron overload and organ damage. The most common form (HFE-related, type 1) is caused by the C282Y mutation in the HFE gene, with a prevalence of approximately 1 in 200–300 in Northern European populations. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

384.2 Causes

This condition happens because of the mutation impairing hepcidin production, leading to unregulated ferroportin-mediated iron efflux from enterocytes and macrophages. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

384.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

384.4 Signs and Symptoms

Clinical features typically manifest in adulthood and include liver cirrhosis (most common cause of morbidity), hepatocellular carcinoma (30% risk with cirrhosis), skin hyperpigmentation (bronze diabetes), diabetes mellitus, hypogonadism, arthropathy (chondrocalcinosis), and cardiomyopathy.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

384.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

384.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

384.7 Diagnosis

Doctors diagnose this using elevated transferrin saturation (greater than 45%) and ferritin, confirmatory genetic testing for HFE mutations, and liver biopsy when indicated.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

384.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

384.9 Treatment Options

- Treatment typically involves therapeutic phlebotomy (weekly until ferritin less than 50 ng/mL, then maintenance)
- Iron chelation is used if phlebotomy is contraindicated.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

384.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

384.11 Outlook

Most people recover well with early diagnosis and treatment.

Chapter 385

Hereditary Spherocytosis

385.1 Overview

Hereditary spherocytosis is the most common inherited hemolytic anemia in individuals of Northern European descent, affecting approximately 1 in 2000. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

385.2 Causes

This condition happens because of mutations in genes encoding red blood cell membrane proteins (spectrin, ankyrin, band 3, protein 4.2), leading to defective vertical interactions between the membrane and cytoskeleton, producing spherocytic, less deformable RBCs that are trapped and destroyed in the spleen. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

385.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

385.4 Signs and Symptoms

Symptoms can range from asymptomatic (compensated hemolysis) to severe hemolytic anemia with jaundice, splenomegaly, gallstones, and aplastic crisis (parvovirus B19 infection).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

385.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

385.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

385.7 Diagnosis

Doctors diagnose this using osmotic fragility testing (increased fragility), EMA (eosin-5-maleimide) binding test by flow cytometry (reduced fluorescence), and peripheral smear showing spherocytes. Management ranges from observation and folic acid supplementation to splenectomy (for moderate-to-severe cases, ideally deferred until age 5 to reduce OPSI risk).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

385.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

385.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

385.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

385.11 Outlook

The outlook is good with appropriate management.

Chapter 386

Herpes Simplex

386.1 Overview

Herpes simplex virus (HSV) infection causes recurrent, painful vesicular eruptions on an erythematous base. HSV-1 typically causes orolabial herpes (cold sores), while HSV-2 typically causes genital herpes, though there is increasing overlap. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

386.2 Causes

This condition happens because of HSV infection with establishment of latency in sensory ganglia (trigeminal for HSV-1, sacral for HSV-2) and reactivation with stress, illness, UV exposure, or immunosuppression. Primary infection may be asymptomatic or present with fever, malaise, and widespread vesicles (especially in children—gingivostomatitis). Recurrent disease is milder, with clustered vesicles at a consistent site, lasting 7–10 days. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

386.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to

endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

386.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

386.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

386.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

386.7 Diagnosis

Doctors diagnose this based on clinical examination, viral culture, PCR, or direct fluorescent antibody testing. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

386.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules

- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

386.9 Treatment Options

- Treatment options include topical or oral antivirals (acyclovir, valacyclovir, famciclovir), most effective when started early
- Suppressive antiviral therapy reduces recurrence frequency.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

386.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

386.11 Outlook

- The outlook is good
- Recurrences decrease over time in many patients.

Chapter 387

Herpes Zoster

387.1 Overview

Herpes zoster (shingles) results from reactivation of latent varicella-zoster virus (VZV) in dorsal root or cranial nerve ganglia, typically occurring decades after primary chickenpox infection. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

387.2 Causes

This condition happens because of declining cell-mediated immunity allowing VZV reactivation. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

387.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

387.4 Signs and Symptoms

- You may experience a unilateral, dermatomal distribution of painful vesicular eruption, most commonly affecting the thoracic dermatomes
- Prodromal pain may precede the rash by 1–3 days.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

387.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

387.6 Complications

- Complications include postherpetic neuralgia (PHN, persistent pain for >3 months after rash, most common in patients >50), bacterial superinfection, motor neuropathy, and disseminated zoster in immunocompromised patients
- Eye zoster (V1 distribution) requires urgent ophthalmologic evaluation.
- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage

- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

387.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- PCR or DFA testing may confirm atypical cases. Management with antivirals (valacyclovir or famciclovir) should be initiated within 72 hours of rash onset.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

387.8 Prevention

- In most cases, the outlook is favorable
- The Shingrix recombinant zoster vaccine is recommended for adults ≥50 to prevent herpes zoster and PHN.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

387.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed

Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

387.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

387.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 388

Hip Fracture

388.1 Overview

Hip fracture is one of the most devastating injuries in older adults, with a 1-year mortality of 20–30% and up to 50% of survivors losing independent mobility. The incidence increases exponentially with age, doubling every 5–6 years after age 65. Osteoporotic hip fractures typically result from a fall from standing height or less. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

388.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

388.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

388.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

388.5 Progression and Stages

Anatomically, hip fractures are classified as femoral neck (intracapsular) or intertrochanteric (extracapsular). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

388.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

388.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

388.8 Prevention

Management requires urgent orthopedic consultation, surgical repair within 24–48 hours (associated with reduced mortality and complications), and comprehensive perioperative care including pain management, delirium prevention, thromboprophylaxis, infection prophylaxis, early mobilization, nutritional support, and osteoporosis treatment.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

388.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

388.10 Living with It

Post-operative rehabilitation with physical therapy is essential for functional recovery.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

388.11 Outlook

- Outlook is guarded
- 1-year mortality is 20–30%.

Chapter 389

Hirschsprung Disease

389.1 Overview

Hirschsprung's disease is a congenital disorder of enteric nervous system development characterized by absence of parasympathetic ganglion cells in the myenteric and submucosal plexuses of the distal colon, with an incidence of 1 in 5,000 live births. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

389.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of failure of neural crest cell migration during embryogenesis, most commonly involving the rectosigmoid colon (80%)
- Long-segment disease extends more proximally. RET proto-oncogene mutations account for 50% of familial cases.

389.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

389.4 Signs and Symptoms

You may experience failure to pass meconium within 48 hours of birth (in 90% of term infants), abdominal distension, bilious vomiting, and chronic constipation. On physical examination, the anal canal is empty and the rectum is tight on digital examination, with explosive expulsion of stool upon withdrawal of the examining finger. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

389.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

389.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

389.7 Diagnosis

- Doctors diagnose this based on rectal biopsy showing absence of ganglion cells and hypertrophied acetylcholinesterase-positive nerve trunks
- Anorectal manometry shows absence of the rectoanal inhibitory reflex.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

389.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

389.9 Treatment Options

Treatment focuses on surgical removal of the aganglionic segment with pull-through procedure (Swenson, Duhamel, or Soave technique), performed in the neonatal period or after initial colostomy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

389.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

389.11 Outlook

- Most people recover well
- Most children achieve normal bowel function, though some experience enterocolitis, constipation, or fecal incontinence.

Chapter 390

Histoplasmosis

390.1 Overview

Histoplasmosis is a systemic fungal infection caused by *Histoplasma capsulatum*, a dimorphic fungus, endemic to the Ohio and Mississippi River Valleys in the United States, and is the most common endemic mycosis in the US, with infection occurring through inhalation of conidia from soil enriched with bat or bird guano. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

390.2 Causes

This condition happens because of conidia being inhaled, reaching the alveoli, and converting to yeast form at body temperature, with yeasts being phagocytosed by macrophages and surviving within phagolysosomes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

390.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

390.4 Signs and Symptoms

You may experience acute lung histoplasmosis: asymptomatic (90%) or self-limited flu-like illness

- Progressive disseminated histoplasmosis (PDH): fever, weight loss, hepatosplenomegaly, lymphadenopathy, pancytopenia, and mucocutaneous ulcers
- And chronic lung histoplasmosis: cavitary lung disease resembling TB.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

390.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

390.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

390.7 Diagnosis

Doctors diagnose this using culture, histopathology, and *Histoplasma* polysaccharide antigen in urine or serum. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

390.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

390.9 Treatment Options

Treatment options include itraconazole for mild-moderate disease and liposomal amphotericin B for severe disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

390.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

390.11 Outlook

- Most people recover well for mild disease
- Disseminated disease can be fatal without treatment.

Chapter 391

HIV/AIDS

391.1 Overview

Human immunodeficiency virus (HIV) infection is a progressive immune deficiency caused by HIV-1 (global pandemic) or HIV-2 (West Africa), ultimately leading to acquired immunodeficiency syndrome (AIDS) if untreated. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

391.2 Causes

This condition happens because of targeting of CD4+ T lymphocytes, macrophages, and dendritic cells, with transmission occurring through sexual contact, blood-borne exposure, and vertical transmission (mother to child). Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

391.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

391.4 Signs and Symptoms

Clinical features follow a natural history including acute retroviral syndrome (fever, rash, lymphadenopathy, muscle pain 2–4 weeks after infection), clinical latency (chronic infection with asymptomatic period lasting years), and AIDS (CD4 count $<$200 cells/μL or presence of AIDS-defining opportunistic infections and malignancies). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

391.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

391.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

391.7 Diagnosis

Doctors diagnose this using fourth-generation HIV antigen/antibody combination immunoassay, with confirmatory testing using HIV-1/HIV-2 differentiation immunoassay and HIV RNA viral load. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

391.8 Prevention

Treatment typically involves with antiretroviral therapy (ART), typically a combination of three or more drugs from different classes (NRTIs, NNRTIs, integrase inhibitors, protease inhibitors, entry inhibitors), with goals of viral suppression (undetectable viral load), immune reconstitution (CD4 recovery), and prevention of transmission (U=U: Undetectable = Untransmittable). Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

391.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

391.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

391.11 Outlook

The outlook has been transformed by ART, with near-normal life expectancy for adherent patients. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 392

Hoarding Disorder

392.1 Overview

Hoarding disorder has a prevalence of 2–6% in the general population, increasing with age (particularly in those over 55). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

392.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

392.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

392.4 Signs and Symptoms

You may experience persistent difficulty discarding or parting with possessions, regardless of their actual value, due to a perceived need to save them and distress associated with discarding them, leading to accumulation of possessions that congest and clutter active living areas, causing significant distress or functional impairment including health and safety risks. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

392.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

392.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

392.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

392.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

392.9 Treatment Options

Treatment options include CBT specifically adapted for hoarding (skills training in decision-making, organizing, categorizing

- Motivational interviewing
- In-home decluttering sessions).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

392.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

392.11 Outlook

The outlook varies from person to person, with high rates of relapse. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 393

Hodgkin Lymphoma

393.1 Overview

Hodgkin lymphoma is a cancerous lymphoid tumor characterized by Reed-Sternberg cells (large, binucleated or multinucleated cells with prominent nucleoli) in a background of reactive inflammatory cells, with two major types: classical HL (nodular sclerosis, mixed cellularity, lymphocyte-rich, lymphocyte-depleted—accounting for 95% of cases) and nodular lymphocyte-predominant HL (NLPHL), with a bimodal age distribution (peaks at 20–30 and >60 years). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

393.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

393.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age

- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

393.4 Signs and Symptoms

Clinical features typically include painless lymphadenopathy (cervical, supraclavicular, mediastinal), B symptoms (fever >38°C, drenching night sweats, weight loss >10% in 6 months), and itching. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

393.5 Progression and Stages

Treatment for early-stage favorable disease involves combined modality therapy (ABVD chemotherapy [doxorubicin, bleomycin, vinblastine, dacarbazine] + involved-field radiation therapy), while advanced-stage disease uses ABVD or escalated BEACOPP with PET-adapted therapy. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

393.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

393.7 Diagnosis

To make the diagnosis, doctors look for lymph node biopsy showing Reed-Sternberg cells (CD15+, CD30+, CD45-), with PET-CT used for staging (Lugano classification). Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology

- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

393.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

393.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

393.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

393.11 Outlook

- Most people recover well, with cure rates exceeding 80%
- Brentuximab vedotin and checkpoint inhibitors (nivolumab, pembrolizumab) are used for relapsed/refractory disease.

Chapter 394

Homocystinuria

394.1 Overview

Homocystinuria (classic form) is an autosomal recessive disorder caused by deficiency of cystathionine β-synthase (CBS), the enzyme that converts homocysteine to cystathionine in the transsulfuration pathway. The incidence is approximately 1 in 200,000–300,000 live births. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

394.2 Causes

This condition happens because of accumulation of homocysteine and methionine, producing a characteristic clinical phenotype resembling Marfan syndrome. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

394.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

394.4 Signs and Symptoms

You may experience tall, thin build, ectopia lentis (downward lens dislocation), myopia, intellectual disability (variable), skeletal abnormalities (scoliosis, pectus excavatum, arachnodactyly, genu valgum), osteoporosis, and a high risk of thromboembolism (the most life-threatening feature). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

394.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

394.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

394.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

394.8 Prevention

- Doctors diagnose this using elevated total plasma homocysteine (greater than 100 µmol/L), elevated methionine, and low cystine
- Newborn screening enables early detection.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

394.9 Treatment Options

Treatment focuses on pyridoxine for responsive patients

- Pyridoxine-nonresponsive patients require methionine restriction, betaine supplementation, and B12/folate supplementation
- Antiplatelet therapy reduces thromboembolic risk.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

394.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

394.11 Outlook

The outlook is generally positive with early treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 395

Hookworm / Soil-Transmitted Helminths

395.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

395.2 Causes

This condition happens because of hookworm larvae penetrating skin, migrating through the circulation to the lungs, ascending the trachea, and maturing in the small intestine, with adult worms attaching to intestinal mucosa and feeding on blood. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

395.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

395.4 Signs and Symptoms

You may experience hookworm: cutaneous larva currens, iron deficiency anemia (from chronic blood loss), fatigue, pallor, and growth retardation

- *Ascaris*: heavy infection causes abdominal pain, malnutrition, and intestinal obstruction
- *Trichuris*: dysentery syndrome (bloody, mucoid stools, tenesmus, rectal prolapse).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

395.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

395.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

395.7 Diagnosis

Doctors diagnose this using stool microscopy with Kato-Katz method. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

395.8 Prevention

Soil-transmitted helminths include hookworm (*Ancylostoma duodenale*, *Necator americanus*), roundworm (*Ascaris lumbricoides*), and whipworm (*Trichuris trichiura*), constituting the most common infections of humans globally, with 1.5 billion people infected, endemic in tropical and subtropical regions with poor sanitation.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

395.9 Treatment Options

Treatment options include albendazole or mebendazole. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

395.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

395.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 396

Hospice Care

396.1 Overview

The interdisciplinary team includes physicians, nurses, social workers, chaplains, volunteers, and bereavement counselors. Earlier referral improves outcomes and family satisfaction. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

396.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

396.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

396.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight

changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

396.5 Progression and Stages

Eligibility for common geriatric conditions: dementia (Functional Assessment Staging—FAST stage 7), heart failure, COPD, and cancer. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

396.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

396.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

396.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

396.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

396.10 Living with It

- Hospice care is a model of interdisciplinary, palliative care for patients with terminal illness and a life expectancy of 6 months or less (certified by two physicians), focused on comfort and quality of life rather than curative treatment. Hospice is provided in the home, nursing home, or hospice facility. Despite its benefits, hospice is underused in older adults
- Median length of stay is 18 days, and 30% of patients die within 7 days of enrollment.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

396.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 397

Human Metapneumovirus (hMPV)

397.1 Overview

Human metapneumovirus is a single-stranded RNA virus (family Paramyxoviridae) first identified in 2001, causing acute respiratory tract infections across all ages, with the highest burden in young children, older adults, and immunocompromised patients, accounting for 5–15% of hospitalized respiratory illnesses in children. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

397.2 Causes

This condition happens because of hMPV infecting ciliated epithelial cells, causing syncytia formation, ciliary dysfunction, and epithelial sloughing, with the virus inhibiting type I interferon signaling. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

397.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

397.4 Signs and Symptoms

- You may experience an incubation of 3–6 days, with presentation nearly indistinguishable from RSV: infants and young children present with coryza, cough, fever, wheezing, bronchiolitis, and pneumonia
- Older adults and immunocompromised present with severe lower respiratory tract infection.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

397.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

397.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

397.7 Diagnosis

Doctors diagnose this using RT-PCR of nasopharyngeal swab or BAL. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

397.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

397.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

397.10 Living with It

Treatment typically involves supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

397.11 Outlook

Most people recover well for immunocompetent hosts. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 398

Human Metapneumovirus (hMPV)

398.1 Overview

Human metapneumovirus (hMPV) is a paramyxovirus closely related to respiratory syncytial virus (RSV), discovered in 2001, yet ubiquitous—virtually all children seroconvert by age 5. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

398.2 Causes

This condition happens because of hMPV infecting respiratory epithelium causing syncytia formation, ciliary dysfunction, airway obstruction, and inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

398.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

398.4 Signs and Symptoms

- Symptoms can range from mild URI to severe lower respiratory tract disease (bronchiolitis, croup, pneumonia)
- Severe disease is most common in premature infants, those with cardiopulmonary disease, and immunocompromised children.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

398.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

398.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

398.7 Diagnosis

Doctors diagnose this using multiplex PCR. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

398.8 Prevention

- Treatment typically involves supportive care

- No licensed vaccine or specific antiviral therapy is available.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

398.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

398.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

398.11 Outlook

Most people recover well for immunocompetent children. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 399

Huntington's Disease

399.1 Overview

Huntington's disease is an autosomal dominant neurodegenerative disorder caused by a CAG trinucleotide repeat expansion in the huntingtin (HTT) gene on chromosome 4. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

399.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

399.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

399.4 Signs and Symptoms

You may experience a triad of motor dysfunction (chorea, dystonia, bradykinesia), thinking and memory decline (executive dysfunction, dementia), and psychiatric disturbances (depression, irritability, psychosis). Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

399.5 Progression and Stages

This condition happens because of mutant huntingtin protein causing progressive deterioration of neurons, particularly in the caudate nucleus and putamen (striatum). Onset typically occurs between ages 30 and 50, with disease severity correlating inversely with the number of CAG repeats. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

399.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

399.7 Diagnosis

Your doctor confirms the diagnosis with genetic testing. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

399.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

399.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

399.10 Living with It

- Treatment focuses on symptomatic treatment with tetrabenazine or deutetrabenazine for chorea, antipsychotics, and antidepressants
- Genetic counseling is essential.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

399.11 Outlook

outlook is progressive and fatal, with no curative treatment available. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 400

Hydrogen Sulfide Toxicity

400.1 Overview

Hydrogen sulfide (H_2S) is a colorless gas with a characteristic rotten egg odor (rapid olfactory fatigue occurs, rendering odor unreliable), encountered in sewers, manure pits, oil and gas drilling, and pulp mills. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

400.2 Causes

This condition happens because of H_2S inhibiting cytochrome c oxidase (identical mechanism to cyanide), causing histotoxic hypoxia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

400.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

400.4 Signs and Symptoms

- Clinical features: acute exposure causes conjunctivitis, respiratory tract irritation, lung swelling, loss of consciousness (knockdown—immediate collapse), seizures, and death
- Chronic low-level exposure causes headache, fatigue, olfactory dysfunction, and thinking and memory impairment.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

400.5 Progression and Stages

Your outlook depends on severity of exposure. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

400.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

400.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

400.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

400.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

400.10 Living with It

Management: immediate removal from exposure, nitrite therapy to induce methemoglobinemia, hyperbaric oxygen, and supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

400.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 401

Hymenoptera Stings

401.1 Overview

Hymenoptera (bees, wasps, hornets, fire ants) cause local pain, redness of the skin, and swelling from venom components. The major concern is immediate hypersensitivity (IgE-mediated anaphylaxis) in sensitized individuals. Anaphylaxis presents with urticaria, angioedema, bronchospasm, laryngeal swelling, hypotension, and cardiovascular collapse occurring within minutes to 1 hour after sting. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

401.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

401.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

401.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

401.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

401.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

401.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Treatment for anaphylaxis: intramuscular epinephrine (0.3–0.5 mg, 1:1,000, anterolateral thigh) is first-line
- Adjunctive includes antihistamines, corticosteroids, and IV fluids. Venom immunotherapy (VIT) for patients with a history of anaphylaxis provides 80–98% protection from future reactions.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

401.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

401.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

401.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

401.11 Outlook

The outlook is generally positive with prompt treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 402

Hyperemesis Gravidarum

402.1 Overview

Hyperemesis gravidarum (HG) is severe, persistent nausea and vomiting during pregnancy causing weight loss (>5% of prepregnancy weight), dehydration, electrolyte abnormalities, and ketonuria. It affects 0.5–2% of pregnancies, distinguishing it from the more common (70–80%) mild nausea and vomiting of pregnancy. The Disease mechanism involves elevated hCG, estrogen, and progesterone, with a potential genetic predisposition (mutation in GDF15 gene). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

402.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

402.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

402.4 Signs and Symptoms

You may experience severe nausea and vomiting leading to weight loss and dehydration. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

402.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

402.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

402.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

402.8 Prevention

Treatment options include non-pharmacologic measures (dietary modifications, ginger, acupressure), antiemetics (doxylamine/pyridoxine combination as first-line, ondansetron, metoclopramide, promethazine), IV fluid and electrolyte repletion, and thiamine supplementation to prevent Wernicke encephalopathy.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

402.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

402.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

402.11 Outlook

- The outlook is generally positive with treatment
- Severe cases requiring parenteral nutrition are rare.

Chapter 403

Hyperopia

403.1 Overview

Hyperopia is a refractive error that occurs when the axial length of the eye is too short or the refractive power is insufficient, causing light to focus behind the retina. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

403.2 Causes

This condition happens because of insufficient refractive power relative to axial length. Mild hyperopia may be accommodated for, especially in young patients. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

403.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

403.4 Signs and Symptoms

You may experience blurred near vision, eye strain, headaches, and in children, esotropia and amblyopia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

403.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

403.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

403.7 Diagnosis

Doctors diagnose this based on refraction testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

403.8 Prevention

- Treatment focuses on correction with convex (plus) lenses
- Severe hyperopia in childhood requires early correction to prevent amblyopia and strabismus.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

403.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

403.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

403.11 Outlook

Most people recover well with correction. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 404

Hyperosmolar Hyperglycemic State

404.1 Overview

Hyperosmolar hyperglycemic state (HHS) is a metabolic emergency primarily affecting T2DM patients, characterized by extreme hyperglycemia (glucose >600 mg/dL), profound dehydration, hyperosmolality (serum osmolality >320 mOsm/kg), and minimal or absent ketosis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

404.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

404.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

404.4 Signs and Symptoms

You may experience severe dehydration, altered mental status, and neurological deficits (seizures, focal deficits). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

404.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

404.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

404.7 Diagnosis

Doctors diagnose this based on blood glucose, serum osmolality, and absence of significant ketosis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

404.8 Prevention

This condition happens because of sufficient residual insulin to prevent ketogenesis but not to control glucose.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

404.9 Treatment Options

Treatment focuses on aggressive fluid resuscitation (initially normal saline), insulin infusion (at lower doses than DKA), and electrolyte replacement. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

404.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

404.11 Outlook

- Outlook is more guarded than DKA
- Mortality is higher (up to 20%).

Chapter 405

Hyperparathyroidism

405.1 Overview

Hyperparathyroidism is marked by excessive secretion of parathyroid hormone (PTH). Primary hyperparathyroidism (most common) results from a parathyroid adenoma (85%), abnormal increase in cells (10–15%), or rarely carcinoma (<1%), causing hypercalcemia which may be asymptomatic or symptomatic: "bones, stones, abdominal groans, and psychic moans" (bone pain/pathologic fractures, kidney stones, abdominal pain/pancreatitis/constipation, depression/thinking and memory dysfunction). Secondary hyperparathyroidism occurs in CKD due to phosphate retention and decreased vitamin D activation. Tertiary hyperparathyroidism occurs after prolonged secondary hyperparathyroidism with autonomous PTH secretion. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

405.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

405.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

405.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

405.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

405.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

405.7 Diagnosis

To diagnose primary disease shows elevated calcium with elevated or inappropriately normal PTH.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

405.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

405.9 Treatment Options

Treatment typically involves parathyroidectomy for symptomatic patients or those meeting criteria (calcium >1 mg/dL above normal, reduced bone density, kidney stones, age <50, or kidney impairment). Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

405.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

405.11 Outlook

Most people recover well with surgical cure.

Chapter 406

Hyperprolactinemia

406.1 Overview

Hyperprolactinemia is the most common pituitary hormone excess disorder, characterized by elevated serum prolactin levels. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

406.2 Causes

Causes include prolactin-secreting pituitary adenomas (prolactinomas, most common), medications (antipsychotics, metoclopramide, domperidone, verapamil), hypothalamic/pituitary stalk compression, primary hypothyroidism (increased TRH stimulates prolactin), chronic kidney failure, and chest wall trauma. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

406.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

406.4 Signs and Symptoms

Clinical features in women include galactorrhea, oligomenorrhea or amenorrhea, infertility, and decreased libido. In men: decreased libido, erectile dysfunction, infertility, gynecomastia, and galactorrhea (rare). Mass effects from macroprolactinomas (>1 cm) include headache, visual field defects (bitemporal hemianopia), and hypopituitarism. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

406.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

406.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

406.7 Diagnosis

Doctors diagnose this based on confirmed elevated serum prolactin (macroprolactin excluded), MRI pituitary with contrast to identify adenoma, and thyroid function tests to exclude hypothyroidism. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

406.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

406.9 Treatment Options

Treatment options include dopamine agonists (cabergoline preferred over bromocriptine due to better efficacy and tolerability) as first-line therapy for symptomatic hyperprolactinemia and prolactinomas, and transsphenoidal surgery for dopamine agonist-resistant or apoplectic macroprolactinomas. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

406.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

406.11 Outlook

- Most people recover well
- Most patients achieve normal prolactin levels with dopamine agonist therapy.

Chapter 407

Hypersplenism

407.1 Overview

Hypersplenism is a syndrome of splenic overactivity resulting in cytopenias (one or more of anemia, leukopenia, thrombocytopenia) due to excessive sequestration and destruction of blood cells. It is always secondary to an underlying condition causing splenomegaly. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

407.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

407.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

407.4 Signs and Symptoms

You may experience cytopenias with corresponding symptoms (anemia, infection risk, bleeding tendency). Diagnosis is suggested by cytopenias, splenomegaly, and bone marrow showing adequate or hypercellular hematopoiesis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

407.5 Progression and Stages

Your outlook depends on the underlying condition and severity of cytopenias. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

407.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

407.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

407.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

407.9 Treatment Options

- Treatment typically involves directed at the underlying cause
- Splenectomy may be considered for severe, symptomatic cytopenias refractory to medical management, though it increases the risk of overwhelming post-splenectomy infection (OPSI).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

407.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

407.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 408

Hypertension

408.1 Overview

high blood pressure is defined as a sustained elevation of systemic arterial blood pressure (systolic $\geq$130 mmHg or diastolic $\geq$80 mmHg per ACC/AHA guidelines). Essential (primary) high blood pressure accounts for 90–95% of cases and results from complex interactions between genetic factors, kidney sodium retention, sympathetic nervous system hyperreactivity, and vascular smooth muscle enlargement. Secondary high blood pressure (5–10% of cases) results from kidney parenchymal disease, renovascular high blood pressure, primary aldosteronism, obstructive sleep apnea, or endocrine tumors. Most patients are asymptomatic until end-organ damage occurs. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

408.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

408.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

408.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

408.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

408.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

408.7 Diagnosis

To make the diagnosis, doctors look for average blood pressure measurements from ≥2 occasions. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

408.8 Prevention

Treatment options include lifestyle modification (DASH diet, sodium restriction, exercise) and first-line pharmacotherapy (ACE inhibitors/ARBs, thiazide diuretics, calcium channel blockers).

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

408.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation

- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

408.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

408.11 Outlook

Most people recover well with early control. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 409

Hypertension in the Elderly

409.1 Overview

high blood pressure affects over 70% of adults aged 65 and older. The predominant form in older adults is isolated systolic high blood pressure (ISH), defined as systolic BP ≥130 mmHg with diastolic BP <80 mmHg, resulting from age-related arterial stiffening. ISH confers significant risk for stroke, coronary artery disease, heart failure, and thinking and memory decline. The SPRINT trial demonstrated that intensive systolic BP lowering to <120 mmHg reduced cardiovascular events and mortality in adults over 75, though with increased risk of orthostatic hypotension, electrolyte abnormalities, and acute kidney injury. Current guidelines recommend a target of <130/80 mmHg for most older adults, but treatment decisions should be individualized based on frailty, falls risk, polypharmacy, and life expectancy. First-line agents include thiazide diuretics, calcium channel blockers, and ACE inhibitors/ARBs. Initial doses should be low with gradual up-titration. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

409.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

409.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

409.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

409.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

409.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events

- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

409.7 Diagnosis

History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

409.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

409.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)

- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

409.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

409.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 410

Hyperthyroidism

410.1 Overview

Hyperthyroidism (thyrotoxicosis) is a state of excessive thyroid hormone production, affecting approximately 1–2% of the population. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

410.2 Causes

The most common cause is Graves' disease (an autoimmune condition with thyroid-stimulating immunoglobulins [TSI] that activate the TSH receptor, causing diffuse goiter, ophthalmopathy, and dermopathy), with other causes including toxic multinodular goiter, toxic adenoma (Plummer's disease), subacute thyroiditis (painful, viral), silent thyroiditis, postpartum thyroiditis, and exogenous thyroid hormone ingestion. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

410.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

410.4 Signs and Symptoms

You may experience weight loss, heat intolerance, palpitations, anxiety, tremor, diarrhea, menstrual irregularities, warm moist skin, lid retraction, and diffuse goiter. Diagnosis shows suppressed TSH with elevated free T4 and/or T3. Management options include antithyroid drugs (methimazole—preferred, or propylthiouracil [PTU]—used in first trimester of pregnancy and thyroid storm), radioactive iodine (I-131) ablation, and thyroidectomy, with beta-blockers (propranolol) providing symptomatic relief.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

410.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

410.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

410.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

410.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

410.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

410.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

410.11 Outlook

The outlook is good with appropriate treatment.

Chapter 411

Hypertrophic Cardiomyopathy

411.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Hypertrophic cardiomyopathy is the most common inherited cardiovascular disease, affecting 1 in 500 individuals, characterized by asymmetric left ventricular enlargement (typically interventricular septum of 15 mm or greater) in the absence of other loading conditions
- It is a leading cause of sudden cardiac death in young athletes.

411.2 Causes

This condition happens because of mutations in sarcomere proteins (beta-myosin heavy chain, myosin-binding protein C being most common). The Disease mechanism involves diastolic dysfunction, dynamic left ventricular outflow tract obstruction (present in 70%), and myocardial reduced blood flow. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

411.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age

- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

411.4 Signs and Symptoms

You may experience exertional shortness of breath, chest pain, syncope, and palpitations. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

411.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

411.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection

- Chronic kidney disease from reduced perfusion

411.7 Diagnosis

Doctors diagnose this using echocardiography and cardiac MRI. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

411.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

411.9 Treatment Options

Treatment options include beta-blockers, calcium channel blockers, and mavacamten for obstructive disease

- Septal ablation or myectomy is for refractory obstruction
- Implantable cardioverter-defibrillator is placed for high-risk patients.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk

- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

411.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

411.11 Outlook

In most cases, the outlook is good with appropriate management, though sudden cardiac death risk requires stratification. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 412

Hypertrophic Pyloric Stenosis

412.1 Overview

Hypertrophic pyloric stenosis (HPS) is progressive thickening of the pyloric muscle causing gastric outlet obstruction. It occurs in 2–4 per 1000 live births, more commonly in males (4:1) and firstborn infants. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

412.2 Causes

This condition happens because of enlargement and abnormal increase in cells of the pyloric circular muscle. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

412.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

412.4 Signs and Symptoms

- You may experience presentation at 3–6 weeks of age with non-bilious, projectile vomiting after feeds
- Infants remain hungry after vomiting.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

412.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

412.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

412.7 Diagnosis

Doctors diagnose this using physical examination (palpable olive-shaped mass in the right upper quadrant) and ultrasound (thickened pyloric muscle). Metabolic alkalosis with hypochloremia and hypokalemia results from loss of gastric acid. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

412.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

412.9 Treatment Options

Treatment typically involves pyloromyotomy (Ramstedt procedure) after correction of fluid and electrolyte abnormalities. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

412.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

412.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 413

Hypoglycemia

413.1 Overview

Hypoglycemia is a blood glucose level below 70 mg/dL, with clinically significant hypoglycemia occurring below 54 mg/dL (Level 2) and severe hypoglycemia (Level 3) being any event requiring external assistance. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

413.2 Causes

This condition happens because of an imbalance between glucose-lowering therapy and glucose supply. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

413.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

413.4 Signs and Symptoms

You may experience adrenergic symptoms (tremor, palpitations, sweating, anxiety) and neuroglycopenic symptoms (confusion, blurred vision, seizures, loss of consciousness). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

413.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

413.6 Complications

It is the most common acute complication of diabetes treatment (insulin and sulfonylureas).

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

413.7 Diagnosis

- Doctors diagnose this based on the Whipple triad: symptoms of hypoglycemia, low measured glucose, and resolution of symptoms with glucose correction. Treatment for mild-moderate hypoglycemia is with rapid-acting oral glucose (15–20 g)
- Severe hypoglycemia requires IV dextrose or glucagon injection.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

413.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

413.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

413.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

413.11 Outlook

- In most cases, the outlook is excellent with prompt treatment
- Recurrent hypoglycemia may cause hypoglycemia unawareness, requiring adjustment of insulin regimens.

Chapter 414

Hypoparathyroidism

414.1 Overview

Hypoparathyroidism is marked by deficient PTH production, leading to hypocalcemia and hyperphosphatemia. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

414.2 Causes

The most common cause is accidental damage to or removal of the parathyroid glands during thyroid surgery (iatrogenic hypoparathyroidism), with other causes including autoimmune hypoparathyroidism (in isolation or as part of APS-1, AIRE gene mutation) and genetic causes such as DiGeorge syndrome (22q11.2 deletion). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

414.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

414.4 Signs and Symptoms

- Symptoms of hypocalcemia include perioral and digital paresthesias, carpopedal spasm, tetany (Chvostek's sign—facial muscle twitch with facial nerve tapping
- Trousseau's sign—carpopedal spasm with blood pressure cuff inflation), seizures, and irregular heartbeats (prolonged QT).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

414.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

414.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

414.7 Diagnosis

Doctors diagnose this using low PTH with hypocalcemia and hyperphosphatemia.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

414.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

414.9 Treatment Options

Treatment typically involves oral calcium supplementation (calcium carbonate or citrate) and active vitamin D analogs (calcitriol). Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

414.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

414.11 Outlook

The outlook is good with appropriate supplementation.

Chapter 415

Hypopituitarism

415.1 Overview

Hypopituitarism is deficiency in one or more pituitary hormones, with causes including pituitary adenomas (most common), craniopharyngiomas, meningiomas, pituitary surgery/radiation, Sheehan syndrome (postpartum pituitary tissue death from bleeding), traumatic brain injury, infiltrative diseases (sarcoidosis, histiocytosis), infections, and autoimmune hypophysitis, with hormones lost in a characteristic order: GH > LH/FSH > TSH > ACTH > prolactin. Clinical manifestations depend on which hormones are deficient: GH deficiency (decreased muscle mass, increased fat), gonadotropin deficiency (amenorrhea, erectile dysfunction, decreased libido), ACTH deficiency (secondary adrenal insufficiency—without hyperpigmentation and with preserved aldosterone), TSH deficiency (central hypothyroidism), and ADH deficiency (central diabetes insipidus). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

415.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

415.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

415.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

415.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

415.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

415.7 Diagnosis

To make the diagnosis, doctors look for dynamic endocrine testing (insulin tolerance test, ACTH stimulation test, GnRH stimulation test) and MRI of the pituitary. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

415.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

415.9 Treatment Options

Treatment typically involves hormone replacement for each deficient axis, with hydrocortisone for ACTH deficiency (must be replaced before thyroid hormone to avoid adrenal crisis). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

415.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

415.11 Outlook

The outlook is good with appropriate replacement. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 416

Hypothermia

416.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

416.2 Causes

Causes include environmental exposure, cold-water immersion, alcohol intoxication, hypothyroidism, hypoglycemia, and trauma. This condition happens because of initial sympathetic activation (shivering, vasoconstriction, tachycardia) transitioning to progressive depression of all organ systems as temperature drops. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

416.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

416.4 Signs and Symptoms

- Clinical features: mild hypothermia: intact shivering, confusion, loss of coordination, slurred speech
- Moderate hypothermia: cessation of shivering, progressive bradycardia, hyporeflexia, paradoxical undressing, obtundation
- Severe hypothermia: coma, apnea, ventricular arrhythmias, asystole. The classic ECG finding is the Osborn wave (J wave).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

416.5 Progression and Stages

Hypothermia is defined as core temperature below 35°C (95°F), classified as mild (32–35°C), moderate (28–32°C), and severe (<28°C). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

416.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

416.7 Diagnosis

Doctors diagnose this using core temperature measurement. Management: passive external rewarming for mild hypothermia

- Active external rewarming for moderate hypothermia
- Active internal rewarming (warmed IV fluids, ECMO) for severe hypothermia.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

416.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

416.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

416.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

416.11 Outlook

The outlook is generally positive with appropriate rewarming. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 417

Hypothyroidism

417.1 Overview

Hypothyroidism is a state of deficient thyroid hormone production, affecting approximately 2% of the population, with higher prevalence in women, the elderly, and iodine-deficient regions. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

417.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of various causes, with the most common in iodine-sufficient areas being Hashimoto's thyroiditis (chronic autoimmune thyroiditis, characterized by lymphocytic infiltration and destruction of the thyroid gland with positive anti-TPO and anti-thyroglobulin antibodies), while iodine deficiency is the most common cause worldwide
- Other causes include post-thyroidectomy or post-radioiodine therapy, external beam radiation, medications (lithium, amiodarone, tyrosine kinase inhibitors), central (pituitary/hypothalamic) hypothyroidism, and congenital hypothyroidism.

417.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

417.4 Signs and Symptoms

You may experience fatigue, weight gain, cold intolerance, constipation, dry skin, hair loss, bradycardia, delayed relaxation of deep tendon reflexes, and depression, with severe hypothyroidism (myxedema coma) being a medical emergency with hypothermia, altered consciousness, and cardiovascular collapse.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

417.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

417.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

417.7 Diagnosis

Doctors diagnose this using elevated TSH (primary hypothyroidism) and low free T4.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

417.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

417.9 Treatment Options

Treatment typically involves levothyroxine (synthetic T4) replacement, dosed to normalize TSH. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

417.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

417.11 Outlook

Most people recover well with appropriate replacement therapy.

Chapter 418

Idiopathic Granulomatous Mastitis

418.1 Overview

Idiopathic granulomatous mastitis (IGM) is a rare, chronic, non-caseating inflammatory breast disease affecting parous women of childbearing age, typically within five years of lactation. The underlying Disease mechanism involves an autoimmune delayed-type hypersensitivity reaction directed against extravasated proteinaceous luminal secretions from damaged ductal epithelium. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

418.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

418.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

418.4 Signs and Symptoms

Symptoms often appear as a rapidly enlarging, painful, hard, irregular breast mass frequently accompanied by overlying skin redness of the skin, nipple retraction, ulceration, and chronic sterile sinus tract removal of fluid, closely mimicking breast cancer or acute bacterial abscess. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

418.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

418.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

418.7 Diagnosis

To make the diagnosis, doctors look for core needle biopsy demonstrating non-caseating granulomatous inflammation centered on breast lobules, with negative bacterial, fungal, and mycobacterial cultures. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

418.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

418.9 Treatment Options

- Treatment options include systemic corticosteroids (prednisone), immunosuppressive therapy (methotrexate), and conservative local care
- Surgical removal is reserved for refractory cases due to high recurrence rates ($20--50\%$).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

418.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

418.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 419

Idiopathic Hypersomnia

419.1 Overview

Idiopathic hypersomnia is a disorder of excessive daytime sleepiness despite normal or prolonged nighttime sleep, with prolonged sleep inertia (severe difficulty awakening, confusion, grogginess, sometimes referred to as sleep drunkenness) and non-restorative naps. The prevalence is approximately 5–10 per 100,000. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

419.2 Causes

This condition happens because of unknown Disease mechanism, with hypotheses including altered GABA-A receptor function, increased endogenous sleep-promoting substances, or dysfunction of wake-promoting circuits. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

419.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

419.4 Signs and Symptoms

You may experience excessive daytime sleepiness, prolonged sleep inertia, and non-restorative naps. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

419.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

419.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

419.7 Diagnosis

To make the diagnosis, doctors look for MSLT showing mean sleep latency $\leq$8 minutes with fewer than two SOREMs, or total 24-hour sleep time of >660 minutes on polysomnography or actigraphy, with exclusion of narcolepsy, sleep-disordered breathing, insufficient sleep syndrome, and other causes. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

419.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

419.9 Treatment Options

Treatment options include wake-promoting agents (modafinil, armodafinil, pitolisant, solriamfetol), sodium oxybate, and clarithromycin. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

419.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

419.11 Outlook

- Outlook is chronic
- Response to Treatment typically involves often less robust than in narcolepsy.

Chapter 420

Idiopathic Insomnia

420.1 Overview

Idiopathic insomnia is a lifelong form of insomnia with onset in infancy or early childhood, without identifiable triggers, and in the absence of medical, psychiatric, or environmental causes. It is rare, representing less than 5% of insomnia cases. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

420.2 Causes

This condition happens because of a hypothesized primary abnormality in the sleep-wake regulatory system, such as dysfunction of the VLPO or an imbalance between sleep-promoting and arousal-promoting systems. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

420.3 Risk Factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

420.4 Signs and Symptoms

You may experience objectively reduced total sleep time on polysomnography and persistent hyperarousal.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

420.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

420.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

420.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with polysomnography used to rule out other sleep disorders.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

420.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

420.9 Treatment Options

- Treatment focuses on CBT-I as first-line therapy, though efficacy may be limited
- Pharmacotherapy (hypnotics, sedating antidepressants) is often required long-term.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

420.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

420.11 Outlook

outlook is persistence across the lifespan. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 421

Idiopathic Pulmonary Fibrosis

421.1 Overview

Idiopathic lung scarring is a chronic, progressive, fibrosing interstitial pneumonia of unknown cause, occurring predominantly in older males. The pathology is marked by usual interstitial pneumonia pattern: patchy, subpleural, basal-predominant scarring with honeycombing and fibroblastic foci. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

421.2 Causes

This condition happens because of aberrant wound healing and fibroblast growth in the lung parenchyma. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

421.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

421.4 Signs and Symptoms

- You may experience insidious-onset exertional shortness of breath and dry cough
- Velcro-like crackles are heard on auscultation.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

421.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment options include antifibrotic therapies (pirfenidone and nintedanib) that slow disease progression
- Lung transplantation is the only intervention shown to improve survival.

421.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

421.7 Diagnosis

To make the diagnosis, doctors look for HRCT showing UIP pattern and exclusion of known causes

- Lung biopsy may be needed when HRCT is inconclusive
- Lung function tests show a restrictive pattern with reduced DLCO.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

421.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

421.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

421.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

421.11 Outlook

Unfortunately, the outlook is often poor, with median survival of 3–5 years. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 422

IgA Nephropathy

422.1 Overview

IgA nephropathy (Berger's disease) is the most common primary glomerulonephritis worldwide, with higher prevalence in East Asia and Southern Europe. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

422.2 Causes

This condition happens because of deposition of galactose-deficient IgA1 in the mesangium, triggering an immune response, mesangial growth, and glomerular injury. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

422.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

422.4 Signs and Symptoms

- You may experience episodic gross hematuria (often concurrent with upper respiratory infections—"synpharyngitic hematuria"), microscopic hematuria, or proteinuria
- Some present with nephrotic syndrome or rapidly progressive glomerulonephritis.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

422.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

422.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

422.7 Diagnosis

- To make the diagnosis, doctors look for kidney biopsy showing mesangial IgA deposition on immunofluorescence
- The Oxford Classification (MEST-C scoring) guides outlook.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

422.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

422.9 Treatment Options

Neflgentan (sodium-glucose cotransporter 2 inhibitor) is an emerging therapy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

422.10 Living with It

- The right treatment depends on severity: RAAS blockade for proteinuria, SGLT2 inhibitors, and immunosuppression (corticosteroids) for high-risk patients
- Supportive care includes BP control and smoking cessation.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

422.11 Outlook

- The outlook varies from person to person
- Up to 30% progress to ESRD within 20 years.

Chapter 423

IgA Vasculitis

423.1 Overview

IgA vasculitis is a small-vessel vasculitis characterized by IgA-dominant immune complex deposition in vessel walls, affecting the skin, joints, digestive tract tract, and kidneys, and is the most common vasculitis in children, with an incidence of 3–26 per 100,000 in children versus 0.1–1.8 per 100,000 in adults, with a median age at presentation of 4–6 years in children. The condition involves IgA1 deposition in small vessels with complement activation and leukocytoclasia, with triggers including upper respiratory tract infections. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

423.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

423.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection

- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

423.4 Signs and Symptoms

You may experience palpable purple spots on the skin (non-thrombocytopenic, symmetrically distributed over the lower extremities and buttocks—the universal feature), arthritis/arthralgias (oligoarticular, non-erosive, typically of lower extremity joints), digestive tract involvement (colicky abdominal pain, nausea, vomiting, intussusception, digestive tract bleeding), and kidney involvement (IgA nephropathy—hematuria, proteinuria, high blood pressure, more severe in adults). Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

423.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

423.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use

- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

423.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by skin or kidney biopsy showing leukocytoclastic vasculitis with predominant IgA deposition on immunofluorescence. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

423.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

423.9 Treatment Options

- Treatment typically involves supportive for mild cases
- Corticosteroids are used for significant digestive tract involvement, orchitis, or severe arthritis.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)

- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

423.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

423.11 Outlook

Most people recover well in children (complete recovery in 95%), with relapses in 30%. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 424

IgG4-Related Disease

424.1 Overview

IgG4-related disease is a fibroinflammatory condition characterized by tumefactive (mass-like) lesions, lymphoplasmacytic infiltration rich in IgG4-positive plasma cells, storiform scarring, and often elevated serum IgG4 levels, affecting virtually any organ with the most common sites being the pancreas (autoimmune pancreatitis type 1), salivary glands (sialadenitis, Mikulicz disease), orbits (dacryoadenitis, orbital pseudotumor), retroperitoneum (retroperitoneal scarring), thyroid (Riedel thyroiditis), kidneys (tubulointerstitial nephritis), lymph nodes, lungs, and biliary tree, with an incidence of 1–5 per 100,000, a male predominance (3:1), and typically presenting in the sixth decade. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

424.2 Causes

This condition happens because of a complex interplay of genetic susceptibility, adaptive and innate immune activation, with a Th2-skewed immune response, T follicular helper cells, and overproduction of IL-4, IL-5, IL-13, and TGF-β, driving IgG4 class switching, eosinophilia, and scarring. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

424.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

424.4 Signs and Symptoms

Clinical features are organ-dependent but commonly include painless swelling of affected organs, obstructive symptoms (jaundice from pancreatic/biliary involvement, hydronephrosis from retroperitoneal scarring), and constitutional symptoms (fatigue, weight loss, low-grade fever). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

424.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

424.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

424.7 Diagnosis

To make the diagnosis, doctors look for histologic confirmation: characteristic histopathology (dense lymphoplasmacytic infiltrate with >10 IgG4+ plasma cells per high-power field, IgG4+/IgG+ ratio >40%, storiform scarring, obliterative phlebitis). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

424.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

424.9 Treatment Options

Treatment typically involves with corticosteroids (first-line, highly effective), with rituximab, azathioprine, mycophenolate, and methotrexate used for steroid-refractory or relapsing disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

424.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

424.11 Outlook

In most cases, the outlook is good but relapses are common (30–50%) during steroid tapering. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 425

Illness Anxiety Disorder (Hypochondriasis)

425.1 Overview

Illness anxiety disorder has a prevalence of 1–5% in the general population, with equal distribution across sexes. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

425.2 Causes

This condition happens because of interoceptive amplification, catastrophic misinterpretation, and memory biases for threat-related health information. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

425.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

425.4 Signs and Symptoms

You may experience preoccupation with having or acquiring a serious illness, with somatic symptoms being absent or mild, and high levels of anxiety about health, with the care-seeking type seeking repeated medical evaluations and the care-avoidant type avoiding medical care due to fear of discovering serious illness.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

425.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

425.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

425.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

425.8 Prevention

Treatment options include CBT with psychoeducation, thinking and memory restructuring, behavioral experiments, and response prevention. outlook: chronic fluctuating course.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

425.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

425.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

425.11 Outlook

With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 426

Immune Thrombocytopenic Purpura

426.1 Overview

Immune thrombocytopenic purple spots on the skin is an autoimmune disorder characterized by isolated thrombocytopenia (platelet count $<100,000/\mu L$) due to antibody-mediated platelet destruction and impaired megakaryopoiesis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

426.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

426.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

426.4 Signs and Symptoms

You may experience tiny red or purple spots, purple spots on the skin, mucosal bleeding (epistaxis, gingival bleeding, menorrhagia), and in severe cases, intracranial bleeding (rare but life-threatening). Diagnosis is one of exclusion: isolated thrombocytopenia on CBC with normal WBC and hemoglobin, normal peripheral smear (except for thrombocytopenia), and exclusion of other causes. The first-choice treatment options include corticosteroids (prednisone or dexamethasone) and IVIG or anti-D immunoglobulin for rapid platelet increase, with second-line therapies including thrombopoietin receptor agonists (romiplostim, eltrombopag, avatrombopag), rituximab, splenectomy, and fostamatinib (SYK inhibitor). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

426.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

426.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

426.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

426.8 Prevention

The condition may be primary (idiopathic) or secondary (associated with SLE, antiphospholipid syndrome, infections [HIV, H. pylori], drugs, lymphoproliferative disorders, or vaccination).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

426.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

426.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

426.11 Outlook

In most cases, the outlook is good, with TPO receptor agonists having largely replaced splenectomy as the preferred second-line therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 427

Impetigo

427.1 Overview

Impetigo is a highly contagious superficial bacterial skin infection, most common in children aged 2–5 years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

427.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of inoculation of bacteria (*Staphylococcus aureus* or *Streptococcus pyogenes*) into minor skin abrasions, insect bites, or areas of dermatitis. It occurs in two forms: non-bullous impetigo (70% of cases), caused by *S. aureus* or *S. pyogenes*, presenting with honey-colored crusted erosions typically on the face and extremities
- And bullous impetigo, caused by exfoliative toxin-producing *S. aureus*, presenting with flaccid, fluid-filled bullae that rupture leaving thin, brown crusts.

427.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

427.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

427.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

427.6 Complications

Complications are rare but include cellulitis, lymphangitis, and post-streptococcal glomerulonephritis (from *S. pyogenes*).

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

427.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Treatment for localized disease is topical mupirocin or retapamulin ointment
- Oral antibiotics (cephalexin, flucloxacillin, or dicloxacillin) are indicated for widespread disease, poor response to topical therapy, or lymphadenopathy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

427.8 Prevention

- Most people recover well with treatment
- Adequate hygiene is essential to prevent spread.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

427.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

427.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

427.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 428

Inclusion Body Myositis

428.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Inclusion body myositis (IBM) is the most common acquired myopathy in adults over 50 and the most common inflammatory myopathy in men. It is marked by slowly progressive, asymmetric muscle weakness affecting both proximal and distal muscles, with characteristic involvement of wrist and finger flexors (difficulty gripping objects) and quadriceps (difficulty rising from chairs and frequent falls)
- Difficulty swallowing occurs in 30–40%.

428.2 Causes

This condition happens because of a combination of inflammatory and degenerative mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

428.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

428.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

428.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

428.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

428.7 Diagnosis

- Doctors diagnose this based on muscle biopsy showing endomysial inflammation, rimmed vacuoles, and intracellular amyloid deposits
- CK is only mildly elevated.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

428.8 Prevention

- Treatment typically involves largely supportive, with physical therapy to prevent falls and maintain function
- The disease is refractory to immunosuppressive therapy.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

428.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

428.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

428.11 Outlook

outlook is slowly progressive with significant functional impairment over time. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 429

Indeterminate Colitis

429.1 Overview

It accounts for 5–15% of IBD cases. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

429.2 Causes

This condition happens because of overlapping pathological features of both Crohn's disease and ulcerative colitis. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

429.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)

- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

429.4 Signs and Symptoms

Clinical features are those of colonic inflammation without clear distinguishing characteristics. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

429.5 Progression and Stages

Indeterminate colitis is inflammatory bowel disease confined to the colon where features of both Crohn's disease and ulcerative colitis are present, and definitive classification is not possible even after pathological review. Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

429.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

429.7 Diagnosis

Management follows the same principles as UC and CD, with treatment tailored to the predominant clinical phenotype. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

429.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

429.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)

- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

429.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

429.11 Outlook

The outlook varies from person to person and depends on the clinical course. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 430

Infectious Mononucleosis

430.1 Overview

Infectious mononucleosis is an acute lymphoproliferative disorder caused by Epstein-Barr virus (EBV, human herpesvirus 4), primarily affecting adolescents and young adults. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

430.2 Causes

This condition happens because of EBV infecting B lymphocytes, triggering a robust CD8+ T-cell response that produces the characteristic atypical lymphocytosis and many of the clinical manifestations, with transmission through saliva ("kissing disease") and an incubation period of 4–6 weeks. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

430.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

430.4 Signs and Symptoms

You may experience the classic triad of fever, pharyngitis (with severe exudative tonsillitis), and lymphadenopathy (typically posterior cervical), along with fatigue (often profound and prolonged), hepatosplenomegaly, maculopapular rash (10–15%, but up to 90% if amoxicillin or ampicillin is given), periorbital swelling, and palatal tiny red or purple spots. Diagnosis is suggested by lymphocytosis (>50% lymphocytes, with >10% atypical lymphocytes), positive heterophile antibody test (Monospot), and EBV-specific serology (anti-VCA IgM for acute infection, anti-VCA IgG for past infection, anti-EBNA for past infection). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

430.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

430.6 Complications

In most cases, the outlook is excellent, though splenic rupture is a rare but life-threatening complication.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

430.7 Diagnosis

Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

430.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings

- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

430.9 Treatment Options

Treatment typically involves symptomatic: rest, hydration, analgesics, and antipyretics, with corticosteroids considered for impending airway obstruction, severe thrombocytopenia, or hemolytic anemia. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

430.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

430.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 431

Infective Endocarditis

431.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Infective endocarditis is an infection of the endocardial surface of the heart, most commonly the heart valves
- Acute infective endocarditis is caused by virulent organisms (*Staphylococcus aureus*) and progresses rapidly, while subacute disease is caused by less virulent organisms (viridans group streptococci).

431.2 Causes

This condition happens because of microbial infection of damaged or abnormal endocardium. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

431.3 Risk Factors

Risk factors include prosthetic valves, intravenous drug use, structural heart disease, and indwelling catheters.

- Close contact with infected individuals
- Compromised immune system
- Travel to endemic areas
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Advanced age
- Cigarette smoking
- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease
- Obesity and physical inactivity
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

431.4 Signs and Symptoms

You may experience fever, chills, night sweats, malaise, weight loss, new or changing cardiac murmur, splinter hemorrhages, Osler's nodes, Janeway lesions, Roth spots, and splenomegaly. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

431.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

431.6 Complications

- Disease progression leading to worsening symptoms

- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

431.7 Diagnosis

To make the diagnosis, doctors look for modified Duke criteria: major criteria (positive blood cultures, evidence of endocardial involvement on echocardiography) and minor criteria (predisposing condition, fever, vascular phenomena, immunologic phenomena). Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

431.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

431.9 Treatment Options

- Treatment focuses on prolonged intravenous antibiotics (4–6 weeks) based on culture and sensitivity
- Surgical intervention is indicated for heart failure, uncontrolled infection, large vegetations, or perivalvular abscess.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

431.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

431.11 Outlook

Your outlook depends on the causative organism, patient factors, and timeliness of intervention. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 432

Inflammatory Breast Cancer

432.1 Overview

Inflammatory breast cancer is a rare, aggressive form of breast cancer (2–5% of all breast cancers) characterized by rapid onset of breast swelling, redness of the skin, warmth, swelling (“peau d’orange”—orange peel appearance of the skin), and often a palpable mass, resulting from dermal lymphatic invasion by tumor cells. The condition is more common in younger women and African American women, with triple-negative and HER2-positive subtypes being more common than in typical breast cancer. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

432.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

432.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age

- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

432.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

432.5 Progression and Stages

Doctors usually diagnose this based on your symptoms and a physical exam, usually at an advanced stage (Stage IIIB or higher). Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

432.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

432.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

432.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin

from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

432.9 Treatment Options

Treatment typically involves neoadjuvant chemotherapy (anthracycline- and taxane-based) followed by mastectomy and post-mastectomy radiation. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

432.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

432.11 Outlook

The outlook has improved but remains worse than other breast cancers, with 5-year survival approximately 40–50%. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 433

Influenza

433.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

433.2 Causes

This condition happens because of viral infection of the respiratory epithelium. Transmission occurs via respiratory droplets and aerosols, with an incubation period of 1–4 days. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

433.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)

- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

433.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

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- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

433.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Influenza is an acute respiratory infection caused by influenza viruses (types A and B), belonging to the Orthomyxoviridae family
- Influenza A viruses are further classified by hemagglutinin and neuraminidase surface antigens, with antigenic drift causing seasonal epidemics and antigenic shift potentially causing pandemics.

433.6 Complications

- You may experience abrupt onset of fever, chills, muscle pain, headache, malaise, nonproductive cough, sore throat, and nasal congestion
- Complications include viral or secondary bacterial pneumonia, myocarditis, pericarditis, encephalopathy, myositis, and Reye's syndrome (in children given aspirin).
- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

433.7 Diagnosis

Doctors diagnose this using rapid influenza diagnostic tests, molecular assays (RT-PCR), or viral culture. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

433.8 Prevention

- Most people recover well for most patients
- Annual influenza vaccination is the most effective preventive measure.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules

- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

433.9 Treatment Options

Treatment options include neuraminidase inhibitors (oseltamivir, zanamivir) and cap-dependent endonuclease inhibitor (baloxavir marboxil), most effective when initiated within 48 hours of symptom onset. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

433.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

433.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 434

Initial Stabilization and Decontamination

434.1 Overview

The poisoned patient is managed using an ABCDE (Airway, Breathing, Circulation, Disability, Exposure/Decontamination) approach. Airway assessment is critical in patients with decreased level of consciousness (GCS <8 often requires intubation). Breathing support with supplemental oxygen and monitoring of pulse oximetry and end-tidal CO_2 is essential for agents causing respiratory depression. Circulation involves IV access, continuous cardiac monitoring, blood pressure support with IV fluids, and vasopressors if needed. Disability assessment includes mental status (AVPU/GCS), pupil size, and glucose measurement. Decontamination includes removal of contaminated clothing, skin decontamination with soap and water, and digestive tract decontamination with activated charcoal (1 g/kg, within 1–2 hours of ingestion for most toxins). Gastric lavage is rarely indicated (only for life-threatening ingestions within 1 hour). Whole-bowel irrigation may be used for sustained-release preparations, iron, lithium, and body packers. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

434.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

434.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

434.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

434.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

434.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

434.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

434.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

434.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

434.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

434.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 435

Insomnia in the Context of Medical/Psychiatric Disorders

435.1 Overview

Insomnia comorbid with medical or psychiatric conditions is a sleep disorder that represents the most common presentation of insomnia in clinical practice. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

435.2 Causes

This condition happens because of the underlying medical or psychiatric disorder and its effects on sleep regulation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

435.3 Risk Factors

Psychiatric disorders—particularly major depressive disorder, generalized anxiety disorder, panic disorder, and post-traumatic stress disorder—are the strongest risk factors, with bidirectional relationships.

- Family history of mental illness
- Trauma or adverse childhood experiences
- Chronic stress
- Substance use
- Social isolation
- Underlying medical conditions
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Medication use

435.4 Signs and Symptoms

You may experience difficulty initiating or maintaining sleep in the presence of a comorbid condition.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

435.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

435.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

435.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with identification of the underlying medical or psychiatric disorder.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

435.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

435.9 Treatment Options

- Treatment focuses on treatment of the underlying condition, CBT-I adapted for the specific comorbidity, and cautious use of hypnotics mindful of drug-disease and drug-drug interactions
- Depression with comorbid insomnia may benefit from sedating antidepressants (trazodone, mirtazapine, amitriptyline).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

435.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

435.11 Outlook

Your outlook depends on the course of the underlying condition. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 436

Interventional Pain Management

436.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Epidural steroid injections (ESI) deliver corticosteroids into the epidural space to reduce inflammation and provide pain relief for radicular pain from disc herniation, spinal stenosis, or nerve root compression. Transforminal ESI targets specific nerve roots
- Interlaminar ESI provides a more diffuse effect. Fluoroscopic guidance is standard. Peripheral nerve blocks involve injection of local anesthetic (with or without corticosteroid) around a peripheral nerve to interrupt nociceptive transmission. Common blocks include greater occipital nerve (headache), suprascapular nerve (shoulder pain), pudendal nerve (pelvic pain), and intercostal nerve (chest wall pain). Radiofrequency ablation (RFA) applies heat generated by radiofrequency energy to produce a thermocoagulation lesion on a specific nerve, providing sustained pain relief (3–12 months). Applications include facet joint denervation (medial branch RFA for chronic axial low back pain and neck pain), sacroiliac joint denervation (lateral branch RFA), genicular nerve RFA for knee osteoarthritis, and pulsed RFA for occipital neuralgia. Spinal cord stimulation (SCS) involves implanting electrodes in the epidural space to deliver electrical stimulation to the dorsal columns, modulating pain transmission via the gate control theory. Indications include failed back surgery syndrome, complex regional pain syndrome, painful diabetic neuropathy, and refractory angina. Modern SCS uses tingling or numbness-based (traditional) or tingling or numbness-free (high-frequency, burst) stimulation. Candidates undergo a trial period (5–7 days) before permanent implantation. Intrathecal drug delivery (implanted pump) delivers medications directly into the cerebrospinal fluid, bypassing the blood-brain barrier and achieving high analgesic concentrations with lower systemic doses. Indications include cancer pain, severe spasticity (baclofen), and chronic non-cancer pain refractory to other treatments. Medications used include morphine, hydromorphone, ziconotide, clonidine, bupivacaine, and baclofen (for spasticity).

436.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

436.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

436.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

436.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

436.6 Complications

Complications include infection, hematoma, dural puncture, nerve injury, and intravascular injection.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

436.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

436.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

436.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

436.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups

- Stay informed about your condition

436.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 437

Intraductal Papillary Mucinous Neoplasm (IPMN)

437.1 Overview

Intraductal papillary mucinous tumor (IPMN) is a mucin-producing epithelial precursor lesion arising within the main pancreatic duct or branch ducts, characterized by intraductal papillary growth, mucin hypersecretion, and progressive cystic ductal ectasia. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

437.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

437.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

437.4 Signs and Symptoms

Symptoms often appear as recurrent acute pancreatitis (due to mucin plugging of the ampulla of Vater) or incidental pancreatic cysts on cross-sectional imaging. Mucin extruding from a gaping 'fish-mouth' ampulla on esophagogastroduodenoscopy is pathognomonic. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

437.5 Progression and Stages

IPMNs carry variable cancerous potential, progressing through low-, intermediate-, and high-grade dysplasia to invasive IPMN-associated adenocarcinoma. Classified by ductal location into main-duct (high cancerous risk, $> 60\%$), branch-duct (lower risk), and mixed-type. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

437.6 Complications

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

437.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

437.8 Prevention

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated

- Avoid exposure to known carcinogens in the workplace and environment

437.9 Treatment Options

Treatment typically involves dictated by Sendai/Fukuoka guidelines: main-duct IPMNs (> 5 mm duct) undergo surgical removal, whereas branch-duct IPMNs are managed conservatively or removed based on high-risk stigmata (solid nodule, main duct dilation, jaundice). Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

437.10 Living with It

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

437.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 438

Intraductal Papilloma

438.1 Overview

Intraductal papilloma is a non-cancerous, focal neoplastic growth of the lactiferous ductal epithelium lining a fibrovascular core, representing the most common cause of spontaneous pathological nipple discharge. Solitary intraductal papillomas arise within major subareolar lactiferous ducts in women aged 30–50 years, whereas multiple papillomas originate in peripheral terminal duct lobular units and confer an increased risk of subsequent breast carcinoma. Torsion of the delicate fibrovascular stalk causes localized reduced blood flow, bleeding, and epithelial sloughing. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

438.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

438.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

438.4 Signs and Symptoms

Symptoms often appear as unilateral, spontaneous, bloody (sanguineous) or serous single-duct nipple discharge, occasionally accompanied by a small subareolar palpable nodule. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

438.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

438.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

438.7 Diagnosis

Doctors confirm the diagnosis through galactography (ductography), ultrasound, and core needle or microdochectomy biopsy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

438.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

438.9 Treatment Options

Treatment focuses on complete surgical removal of the involved duct (microdochectomy) to rule out concurrent ductal carcinoma in situ. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

438.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

438.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 439

Intrauterine Growth Restriction

439.1 Overview

Intrauterine growth restriction (IUGR), also termed fetal growth restriction (FGR), is defined as an estimated fetal weight below the 10th percentile for gestational age, often subclassified as symmetric (early onset, involving all biometric parameters) and asymmetric (late onset, with head sparing). It affects 5–10% of pregnancies. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

439.2 Causes

Causes include maternal factors (high blood pressure, preeclampsia, chronic kidney disease, malnutrition, smoking, alcohol, illicit drugs), placental factors (abnormal placentation, tissue death due to lack of blood supply, velamentous cord insertion), and fetal factors (aneuploidy, congenital infections, multiple gestation). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

439.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

439.4 Signs and Symptoms

You may experience subnormal fetal growth on serial ultrasound. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

439.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

439.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

439.7 Diagnosis

- Diagnosis relies on accurate gestational dating and serial ultrasound measurement of fetal biometry, estimated fetal weight, and amniotic fluid volume
- Umbilical artery Doppler velocimetry identifies fetuses at highest risk.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

439.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

439.9 Treatment Options

Treatment options include antenatal surveillance and delivery when fetal maturity is achieved or when the risks of continued in utero existence outweigh neonatal risks. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

439.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

439.11 Outlook

Your outlook depends on the underlying cause and gestational age at delivery. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 440

Intraventricular Hemorrhage (IVH)

440.1 Overview

Intraventricular bleeding (IVH) is bleeding into the germinal matrix and ventricular system of the brain, occurring almost exclusively in preterm infants (<32 weeks gestation). Incidence is 15–25% in very low birth weight infants. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

440.2 Causes

This condition happens because of fragility of the highly vascularized germinal matrix, which is susceptible to rupture from fluctuations in cerebral blood flow. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

440.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

440.4 Signs and Symptoms

Clinical features may include apnea, bradycardia, seizures, bulging fontanel, hypotension, and anemia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

440.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

440.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

440.7 Diagnosis

Doctors diagnose this using cranial ultrasound. IVH is graded on a scale of I–IV. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

440.8 Prevention

- Treatment typically involves supportive
- Prevention is paramount and includes antenatal corticosteroids, delayed cord clamping, and avoidance of rapid volume expansion or hypotension.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

440.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

440.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

440.11 Outlook

Your outlook depends on the grade of bleeding. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 441

Intussusception

441.1 Overview

Intussusception is the telescoping of one segment of the intestine into an adjacent segment, most commonly ileocecal. It is the most common cause of intestinal obstruction in children aged 3 months to 3 years. The classic triad (not always present) is colicky abdominal pain, vomiting, and red-currant jelly stools (late sign). A sausage-shaped mass may be palpable in the right upper quadrant. Ultrasound demonstrates a target sign or pseudokidney sign. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

441.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

441.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

441.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

441.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

441.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

441.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

441.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

441.9 Treatment Options

Treatment typically involves non-surgical reduction by air enema under fluoroscopic guidance, which is successful in 80–95% of cases. Surgical reduction is required for perforation, peritonitis, or failed pneumatic reduction. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

441.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

441.11 Outlook

Most people recover well with early treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 442

Invasive Lobular Carcinoma (ILC)

442.1 Overview

Invasive lobular carcinoma (ILC) is the second most common histological subtype of invasive breast cancer, accounting for 10–15% of all cases. Pathophysiologically, ILC is marked by loss of cellular adhesion mediated by somatic mutation or promoter hypermethylation of the E-cadherin gene (CDH1). Histologically, dyscohesive tumor cells infiltrate the stroma in a single-file, linear array ('Indian file' pattern) surrounding non-neoplastic ducts without eliciting a marked desmoplastic stroma. Metastatic dissemination exhibits a unique tropism for the peritoneum, retroperitoneum, digestive tract tract, leptomeninges, and ovaries. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

442.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

442.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal

factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

442.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

442.5 Progression and Stages

Consequently, ILC frequently presents as an ill-defined thickening or subtle induration rather than a discrete palpable mass, leading to delayed mammographic detection and higher rates of multifocality, bilaterality, and advanced stage at presentation. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement.

Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

442.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

442.7 Diagnosis

Doctors diagnose this using core biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging

- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

442.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

442.9 Treatment Options

Treatment options include surgical removal, radiation, and long-term endocrine therapy, as most ILCs are strongly ER/PR-positive. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

442.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

442.11 Outlook

outlook is comparable to invasive ductal carcinoma, though late recurrences are more frequent. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 443

Inverted Papilloma

443.1 Overview

Inverted papilloma (Schneiderian papilloma) is a non-cancerous but locally aggressive sinonasal epithelial tumor characterized by inward growth (endophytic inversion) of neoplastic epithelium into the underlying stroma rather than outward exophytic growth. Originating predominantly from the lateral nasal wall or maxillary sinus ostium in males aged 40–70 years, it is strongly associated with high-risk human papillomavirus (HPV 16/18) infection. Key pathological features include high recurrence rates ($20 - -30\%$) and a 5–10% risk of synchronous or metachronous transformation into squamous cell carcinoma. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

443.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

443.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

443.4 Signs and Symptoms

You may experience unilateral nasal obstruction, epistaxis, and pressure pain. CT reveals a soft tissue mass with focal hyperostosis indicating the bony site of origin. Management requires complete using a thin camera tube endonasal removal with drilling of the attachment site. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

443.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

443.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

443.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

443.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

443.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

443.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

443.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 444

Iodine Deficiency Disorders

444.1 Overview

Iodine deficiency disorders are a group of mineral deficiency disorders affecting nearly 2 billion people, particularly in mountainous regions (Himalayas, Andes, Alps), flood-prone river valleys, and areas with iodine-depleted soils. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

444.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of deficiency of iodine, an essential component of thyroid hormones (thyroxine, T4
- Triiodothyronine, T3) required for normal growth, development, and metabolism.

444.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

444.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

444.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

444.6 Complications

- You may experience goiter (compensatory thyroid enlargement), cretinism (nervous system cretinism with intellectual disability, deaf-mutism, spastic diplegia
- Myxedematous cretinism with hypothyroidism, growth retardation, coarse facial features), hypothyroidism in older children and adults (fatigue, cold intolerance, weight gain, constipation, dry skin, bradycardia), and pregnancy complications including miscarriage, stillbirth, congenital anomalies, and impaired neurocognitive development.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

444.7 Diagnosis

Doctors diagnose this using low urinary iodine concentration (less than 100 µg/L), elevated TSH, low T4, and goiter on palpation or ultrasound.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

444.8 Prevention

- Treatment focuses on universal salt iodization programs for prevention

- Iodine supplementation (oral potassium iodide 100–200 mg daily) for mild deficiency and levothyroxine for hypothyroidism.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

444.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

444.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

444.11 Outlook

Most people recover well with correction of deficiency.

Chapter 445

Iron Deficiency

445.1 Overview

Iron deficiency is the most common nutritional deficiency worldwide and the leading cause of anemia, affecting approximately 25% of the global population. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

445.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of inadequate dietary iron intake, poor bioavailability (non-heme iron from plant sources), increased demands (pregnancy, growth, menstruation), and chronic blood loss (digestive tract bleeding, menorrhagia, hookworm)
- Iron is essential for hemoglobin synthesis, myoglobin, and numerous enzymes involved in electron transport and DNA synthesis.

445.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

445.4 Signs and Symptoms

Clinical features manifest as microcytic hypochromic anemia including fatigue, pallor, weakness, shortness of breath on exertion, tachycardia, headache, and epithelial changes including koilonychia (spoon nails), glossitis, angular stomatitis, and pagophagia (pica for ice).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

445.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

445.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

445.7 Diagnosis

Doctors diagnose this using low serum ferritin (gold standard), low serum iron, low transferrin saturation, elevated total iron-binding capacity (TIBC), elevated soluble transferrin receptor, and peripheral smear showing microcytic, hypochromic RBCs with pencil-shaped cells and target cells.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

445.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

445.9 Treatment Options

Treatment focuses on oral iron therapy (ferrous sulfate 325 mg three times daily, equivalent to 195 mg elemental iron per day) as first-line, with response monitored by reticulocytosis at 3–7 days and hemoglobin rise of 1 g/dL per 2–3 weeks

- Parenteral iron is used for intolerance, severe deficiency, malabsorption, or chronic kidney disease
- Treatment of the underlying cause is mandatory.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

445.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

445.11 Outlook

Most people recover well with treatment.

Chapter 446

Iron Deficiency Anemia

446.1 Overview

Iron deficiency anemia is a nutritional deficiency disorder that affects approximately 1.2 billion people worldwide, making it the most common nutritional deficiency globally. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

446.2 Causes

This condition happens because of insufficient iron for hemoglobin synthesis, leading to microcytic, hypochromic anemia, with causes including chronic blood loss (menstruation, GI bleeding from peptic ulcers, colorectal cancer, or NSAID use), inadequate dietary intake (vegetarians/vegans), and increased demand (pregnancy, growth). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

446.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

446.4 Signs and Symptoms

You may experience fatigue, pallor, shortness of breath on exertion, pica (craving for non-food items like ice or clay), koilonychia (spoon nails), angular cheilitis, and glossitis.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

446.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

446.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

446.7 Diagnosis

Doctors diagnose this using CBC showing microcytic anemia (low MCV, MCH, MCHC), low serum ferritin (most specific for iron deficiency), low serum iron, elevated TIBC, and low transferrin saturation, with peripheral smear showing hypochromic microcytes, pencil cells, and target cells.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

446.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

446.9 Treatment Options

Treatment focuses on identifying and treating the underlying cause, and iron supplementation (oral ferrous sulfate 325 mg daily, taken with vitamin C to enhance absorption), with IV iron used for intolerance, malabsorption, or chronic kidney disease. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

446.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

446.11 Outlook

Most people recover well with appropriate treatment.

Chapter 447

Iron Overdose

447.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Iron overdose is a leading cause of poisoning death in children under 6 years due to accidental ingestion of prenatal iron supplements. Toxic dose: elemental iron >20 mg/kg
- Severe toxicity >60 mg/kg.

447.2 Causes

This condition happens because of free iron catalyzing hydroxyl radical formation via the Fenton reaction, causing lipid peroxidation, mitochondrial dysfunction, and metabolic acidosis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

447.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

447.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

447.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Clinical stages: Stage 1 (0–6 hours): nausea, vomiting, abdominal pain, diarrhea, hematemesis, hematochezia
- Stage 2 (6–24 hours): apparent recovery
- Stage 3 (12–48 hours): metabolic acidosis, seizures, coma, shock, coagulopathy, liver tissue death, lung swelling
- Stage 4 (2–4 weeks): digestive tract obstruction from strictures.

447.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

447.7 Diagnosis

Doctors diagnose this using serum iron level and abdominal X-ray. Treatment: aggressive IV fluids, deferoxamine (IV iron chelator) for metabolic acidosis, shock, or iron level >500 mcg/dL, and whole-bowel irrigation for retained tablets. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

447.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

447.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

447.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

447.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 448

Irregular Sleep-Wake Rhythm Disorder

448.1 Overview

Irregular sleep-wake rhythm disorder (ISWRD) is a circadian rhythm sleep-wake disorder presenting as a fragmented, disorganized sleep-wake pattern with at least three distinct sleep episodes within a 24-hour period, without a clear circadian rhythm. It is most common in patients with neurodegenerative disorders (Alzheimer disease, dementia), traumatic brain injury, intellectual disability, and in institutionalized elderly populations. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

448.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

448.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

448.4 Signs and Symptoms

You may experience multiple sleep episodes throughout the day and night with no dominant sleep period. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

448.5 Progression and Stages

This condition happens because of deterioration of the SCN or reduced photic input. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

448.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

448.7 Diagnosis

To make the diagnosis, doctors look for actigraphy or sleep diaries over at least 7 days. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

448.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

448.9 Treatment Options

- Treatment focuses on increasing daytime bright light exposure, structured daytime activities and social schedules, evening melatonin, and avoidance of nighttime light
- Combination therapy (bright light plus melatonin plus structured activity) is most effective.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

448.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

448.11 Outlook

The outlook varies from person to person and depends on the underlying condition. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 449

Irritable Bowel Syndrome

449.1 Overview

Irritable bowel syndrome (IBS) is a chronic functional digestive tract disorder characterized by recurrent abdominal pain associated with altered bowel habits, in the absence of structural or biochemical abnormalities. It affects approximately 10–15% of the global population and is more common in women. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

449.2 Causes

This condition happens because of visceral hypersensitivity, altered digestive tract motility, gut-brain axis dysfunction, intestinal dysbiosis, low-grade mucosal inflammation, and post-infectious triggers. IBS is subtyped by predominant stool pattern: IBS with constipation (IBS-C), IBS with diarrhea (IBS-D), IBS with mixed bowel habits (IBS-M), and unsubtyped IBS (IBS-U). Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

449.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

449.4 Signs and Symptoms

You may experience recurrent abdominal pain, on average at least 1 day per week in the last 3 months, associated with defecation, change in stool frequency, or change in stool form.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

449.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

449.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

449.7 Diagnosis

- Doctors diagnose this based on the Rome IV criteria
- Alarm features (rectal bleeding, nocturnal symptoms, unintentional weight loss, family history of colorectal cancer, iron deficiency anemia, onset after age 50) warrant further investigation to exclude organic disease.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

449.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

449.9 Treatment Options

Treatment typically involves multimodal: dietary modification (low FODMAP diet, soluble fiber supplementation), antispasmodics (hyoscine, dicyclomine, peppermint oil), laxatives for IBS-C (linaclotide, lubiprostone, plecanatide), antidiarrheals for IBS-D (loperamide, eluxadoline, rifaximin), neuromodulators (tricyclic antidepressants, SSRIs), and gut-directed psychological therapies (hypnotherapy, thinking and memory-behavioral therapy). Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

449.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

449.11 Outlook

- The outlook varies from person to person
- Symptoms are chronic but can be managed with multimodal therapy.

Chapter 450

Ischemic Cardiomyopathy

450.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Ischemic cardiomyopathy is myocardial dysfunction resulting from chronic reduced blood flow or prior heart attack, leading to progressive ventricular dilation and heart failure
- It is the most common cause of heart failure worldwide.

450.2 Causes

This condition happens because of myocyte loss, scarring, and adverse ventricular remodeling following coronary artery disease. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

450.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus

- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

450.4 Signs and Symptoms

You may experience symptoms of heart failure: shortness of breath on exertion, orthopnea, paroxysmal nocturnal shortness of breath, fatigue, and peripheral swelling. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

450.5 Progression and Stages

Treatment options include guideline-directed medical therapy (ACE inhibitors, ARBs, ARNI, beta-blockers, mineralocorticoid receptor antagonists, SGLT2 inhibitors, and diuretics) Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

- Coronary revascularization may improve outcomes in selected patients
- Advanced therapies include cardiac resynchronization therapy, implantable cardioverter-defibrillators, and cardiac transplantation.

450.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events

- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

450.7 Diagnosis

Doctors diagnose this based on echocardiography showing reduced left ventricular ejection fraction (less than 40%), regional wall motion abnormalities, and mitral regurgitation. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

450.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

450.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

450.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

450.11 Outlook

Your outlook depends on the extent of myocardial dysfunction and response to therapy. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 451

Ischemic Stroke

451.1 Overview

Ischemic stroke is a focal brain tissue death due to lack of blood supply caused by occlusion of a cerebral artery by a blood clot or blood clot that travels, accounting for approximately 85% of all strokes. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

451.2 Causes

This condition happens because of thrombotic or embolic arterial occlusion leading to focal brain reduced blood flow and tissue death due to lack of blood supply. Common etiologies include atherosclerotic large-vessel disease, cardioembolism (eg., atrial fibrillation), small-vessel occlusive disease (lacunar stroke), and cryptogenic causes. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

451.3 Risk Factors

- Treatment options include intravenous thrombolysis with alteplase (within 4.5 hours) and mechanical thrombectomy for large-vessel occlusions (up to 24 hours in selected patients)
- Secondary prevention includes antithrombotic therapy, statins, and risk factor modification.
- Advanced age
- Cigarette smoking
- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease
- Obesity and physical inactivity
- Family history and genetic predisposition
- Head trauma or brain injury
- Vascular risk factors (hypertension, diabetes)
- Exposure to environmental toxins
- Chronic stress
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

451.4 Signs and Symptoms

- Your symptoms depend on the vascular territory affected: middle cerebral artery occlusion causes contralateral hemiparesis, hemisensory loss, and aphasia (if dominant hemisphere)
- Posterior circulation strokes produce visual field defects, loss of coordination, and crossed deficits.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

451.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

451.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

451.7 Diagnosis

Diagnosis typically involves urgent non-contrast CT to exclude bleeding, followed by CT angiography and perfusion imaging. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

451.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

451.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

451.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

451.11 Outlook

Your outlook depends on infarct size, location, and timeliness of intervention. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 452

Jet Lag Disorder

452.1 Overview

Jet lag disorder is a temporary circadian misalignment caused by rapid travel across multiple time zones. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

452.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

452.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

452.4 Signs and Symptoms

Symptoms include difficulty initiating or maintaining sleep at the destination bedtime, daytime fatigue, impaired thinking and memory performance, digestive tract disturbances, and general malaise. This condition happens because of the circadian system adapting at a rate of approximately one time zone per day. You may experience sleep disturbance and daytime impairment following rapid transmeridian travel. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

452.5 Progression and Stages

Severity correlates with the number of time zones crossed (typically three or more) and direction of travel (eastward travel is more difficult because it requires phase-advancing, which is harder than phase-delaying). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

452.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

452.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

452.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

452.9 Treatment Options

Treatment options include controlled light exposure (seek morning light when traveling east, evening light when traveling west), melatonin (0.5–5 mg at destination bedtime), and short-term use of hypnotics for sleep initiation. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

452.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

452.11 Outlook

- The outlook is generally positive
- Symptoms resolve as the circadian system adapts.

Chapter 453

Juvenile Nasopharyngeal Angiofibroma (JNA)

453.1 Overview

Juvenile nasopharyngeal angiofibroma (JNA) is a non-cancerous, non-encapsulated, highly vascular, locally invasive tumor arising almost exclusively in adolescent males (ages 10–25 years). Originating near the sphenopalatine foramen, JNA consists of thin-walled blood vessels lacking smooth muscle coats embedded in a fibrous stroma, causing intense bleeding upon minimal manipulation. The tumor expands locally into the nasopharynx, pterygopalatine fossa, orbits, and middle cranial fossa. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

453.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

453.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

453.4 Signs and Symptoms

Biopsy is strictly contraindicated due to catastrophic bleeding risk. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

453.5 Progression and Stages

Symptoms often appear as the classic triad of recurrent painless unilateral epistaxis, unyielding nasal obstruction, and a nasopharyngeal mass, with advanced cases causing facial asymmetry ('frog-face' deformity). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

453.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

453.7 Diagnosis

Doctors confirm the diagnosis through contrast CT, MRI, and digital subtraction angiography. Management requires preoperative transarterial embolization followed by complete surgical removal. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

453.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

453.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

453.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

453.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 454

Kawasaki Disease

454.1 Overview

Kawasaki disease is an acute, self-limited, systemic vasculitis of medium-sized arteries, primarily affecting children under 5 years of age, with a predilection for the coronary arteries, with the highest incidence in Japan (300 per 100,000 children <5 years), and approximately 20–25 per 100,000 in North America and Europe. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

454.2 Causes

The cause is unknown but suspected to involve an infectious trigger in genetically susceptible individuals, leading to an intense innate and adaptive immune response, with Disease mechanism involving systemic inflammation, endothelial activation, and infiltration of the vessel wall by neutrophils, lymphocytes, and macrophages, with consequent arteritis that can lead to coronary artery aneurysm formation and blood clot formation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

454.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

454.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

454.5 Progression and Stages

Clinical To make the diagnosis, doctors look for fever for at least 5 days plus four of five principal features: bilateral conjunctival injection (non-exudative), oral mucosal changes (strawberry tongue, erythematous lips, pharyngeal redness of the skin), polymorphous rash (usually truncal), extremity changes (redness of the skin and swelling of hands/feet in acute phase, periungual desquamation in subacute phase), and cervical lymphadenopathy (>1.5 cm, typically unilateral). Treatment typically involves intravenous immunoglobulin (IVIG, 2 g/kg single dose) plus high-dose aspirin in the acute phase, reduced to low-dose antiplatelet dose after defervescence. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

454.6 Complications

Complications of coronary arteritis include coronary artery aneurysms (occur in 25% of untreated KD, reduced to <5% with treatment), heart attack, arrhythmias, and sudden death.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

454.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

454.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

454.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

454.10 Living with It

Echocardiography is essential at diagnosis and follow-up.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

454.11 Outlook

Most people recover well with treatment, with coronary artery aneurysm rates reduced to <5%. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 455

Kawasaki Disease

455.1 Overview

Kawasaki disease is an acute, self-limited vasculitis of unknown cause that predominantly affects children under 5 years (see Chapter 27, Connective Tissue Diseases for comprehensive coverage). Diagnostic criteria include fever for ≥5 days plus four of five features: bilateral conjunctival injection, oral mucosal changes, polymorphous rash, extremity changes, and cervical lymphadenopathy (≥1.5 cm). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

455.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

455.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

455.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

455.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

455.6 Complications

The most serious complication is coronary artery aneurysms, which develop in 20–25% of untreated cases.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

455.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

455.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

455.9 Treatment Options

Treatment typically involves intravenous immunoglobulin (IVIG, 2 g/kg) and high-dose aspirin. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

455.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

455.11 Outlook

- Most people recover well with prompt treatment
- The risk of coronary aneurysms is reduced to less than 5%.

Chapter 456

Keratitis

456.1 Overview

Keratitis is an inflammation of the cornea that may be infectious or non-infectious. Infectious keratitis can be bacterial, viral (herpes simplex), fungal, or amoebic (*Acanthamoeba*). Bacterial keratitis, the most common type, results from corneal infection, often associated with contact lens use or trauma. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

456.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

456.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

456.4 Signs and Symptoms

- You may experience pain, redness, tearing, decreased vision, and a corneal infiltrate or ulcer
- Herpes simplex keratitis may present as epithelial (dendritic ulcer) or stromal disease.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

456.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

456.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

456.7 Diagnosis

Diagnosis typically involves corneal scraping for Gram stain, culture, and sensitivity. Management requires frequent topical antimicrobial agents appropriate for the suspected organism. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

456.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

456.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

456.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

456.11 Outlook

- Your outlook depends on the causative organism and timeliness of treatment
- Corneal scarring may require transplantation.

Chapter 457

Keratoconus

457.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Keratoconus is a progressive, non-inflammatory ectatic disorder of the cornea characterized by progressive thinning and steepening, resulting in irregular astigmatism and reduced visual acuity
- Onset is typically in adolescence or early adulthood.

457.2 Causes

This condition happens because of structural weakening of the corneal stroma. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

457.3 Risk Factors

Risk factors include eye rubbing, family history, and associated systemic conditions (Ehlers-Danlos syndrome, Marfan syndrome, Down syndrome). Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

457.4 Signs and Symptoms

- You may experience progressive visual distortion and decreased visual acuity
- Slit-lamp findings include Munson's sign, Vogt's striae, and corneal iron deposition.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

457.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Management progresses from spectacles and rigid gas-permeable contact lenses to corneal crosslinking to halt progression, intrastromal corneal ring segments, and ultimately keratoplasty in advanced cases. The outlook varies from person to person
- Corneal crosslinking effectively halts progression.

457.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

457.7 Diagnosis

Doctors diagnose this based on corneal topography and tomography. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

457.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

457.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

457.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

457.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 458

Kidney Cancer (Renal Cell Carcinoma)

458.1 Overview

kidney cell carcinoma (RCC) accounts for 2–3% of adult malignancies (400,000 cases/year worldwide). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

458.2 Causes

Clear cell RCC (75%) is the most common subtype, followed by papillary (15%), chromophobe (5%), and collecting duct carcinoma. This condition happens because of VHL inactivation leading to HIF accumulation, VEGF overexpression, and angiogenesis in clear cell RCC. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

458.3 Risk Factors

Risk factors include smoking, obesity, high blood pressure, acquired cystic kidney disease, and hereditary syndromes (VHL, hereditary papillary RCC, Birt-Hogg-Dubé, tuberous sclerosis, hereditary leiomyomatosis).

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

458.4 Signs and Symptoms

You may experience the triad of hematuria, flank pain, and palpable abdominal mass (now uncommon

- 60% are found incidentally on imaging)
- Paraneoplastic syndromes include erythrocytosis (EPO), hypercalcemia (PTHrP), high blood pressure (renin), and Stauffer syndrome (reversible liver dysfunction).
- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

458.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment options include partial nephrectomy (nephron-sparing for T1a, T1b tumors), radical nephrectomy for larger tumors, and ablative therapies for small tumors in elderly patients
- Advanced disease doctors typically treat it with tyrosine kinase inhibitors (sunitinib, pazopanib, cabozantinib), immunotherapy (nivolumab + ipilimumab, pembrolizumab + axitinib), mTOR inhibitors (everolimus, temsirolimus), and HIF-2α inhibitors (belzutifan for VHL-associated RCC). Recovery varies by stage: 5-year survival is 93% for localized, 72% regional, and 14% metastatic.

458.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

458.7 Diagnosis

- Doctors diagnose this based on CT abdomen with contrast (heterogeneous, enhancing kidney mass), MRI for cystic/complex lesions, and kidney mass biopsy for small tumors
- Staging uses the TNM system.
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

458.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

458.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

458.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

458.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 459

Kikuchi-Fujimoto Disease

459.1 Overview

Kikuchi-Fujimoto disease (histiocytic necrotizing lymphadenitis) is a rare, non-cancerous, self-limiting inflammatory disorder of lymph nodes, affecting predominantly young Asian females under 40 years of age. Driven by an aberrant T-cell-mediated immune response to viral triggers (EBV, CMV, HHV-6, HIV) or autoimmune stimuli, histopathology reveals localized cortical and paracortical apoptotic tissue death surrounded by abundant histiocytes, plasmacytoid monocytes, and CD8+ T-lymphocytes, with a characteristic absence of neutrophils. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

459.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

459.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

459.4 Signs and Symptoms

Symptoms often appear as painful or tender cervical lymphadenopathy (most commonly posterior cervical chain), accompanied by low-grade fever, night sweats, fatigue, weight loss, and occasional erythematous skin rashes, closely mimicking lymphoma, tuberculosis, or systemic lupus erythematosus. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

459.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

459.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

459.7 Diagnosis

To make the diagnosis, doctors look for excisional lymph node biopsy to distinguish from cancerous lymphoma. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

459.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

459.9 Treatment Options

- Treatment typically involves supportive with analgesics and NSAIDs
- Severe systemic symptoms respond rapidly to a short course of corticosteroids. Spontaneous resolution occurs within 1 to 4 months with minimal recurrence risk ($< 5\%$).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

459.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

459.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 460

Kleine-Levin Syndrome

460.1 Overview

Kleine-Levin syndrome (recurrent hypersomnia) is a rare disorder characterized by recurrent episodes of severe hypersomnia (16–20 hours of sleep per day), lasting days to weeks, with onset typically during adolescence. Prevalence is estimated at 1–2 per million, with a male predominance (2:1). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

460.2 Causes

This condition happens because of unknown Disease mechanism, with proposed mechanisms including autoimmune or parainfectious inflammation in the hypothalamus. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

460.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

460.4 Signs and Symptoms

- You may experience recurrent episodes of severe hypersomnia accompanied by hyperphagia, hypersexuality, mood changes, thinking and memory impairment, and sometimes psychotic features
- Between episodes, patients have normal sleep, thinking, and function.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

460.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

460.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

460.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam using ICSD-3 criteria.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

460.8 Prevention

Management during episodes includes stimulants (modafinil, methylphenidate) for sleepiness, lithium for episode prevention, and mood stabilizers (valproate, lamotrigine).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

460.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

460.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

460.11 Outlook

- The outlook is generally positive
- The disease typically resolves spontaneously over 8–14 years.

Chapter 461

Klinefelter Syndrome (47,XXY)

461.1 Overview

Klinefelter syndrome is a chromosomal disorder caused by the presence of an extra X chromosome (47,XXY), with an incidence of 1 in 650 live-born males. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

461.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

461.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

461.4 Signs and Symptoms

You may experience tall, eunuchoid habitus, gynecomastia, small testes, azoospermia (infertility), sparse body hair, and low testosterone. Intellectual development is usually normal but verbal and language delays are common.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

461.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

461.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

461.7 Diagnosis

Doctors diagnose this using karyotype.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

461.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

461.9 Treatment Options

Treatment options include testosterone replacement therapy beginning at puberty, breast reduction surgery for gynecomastia, and assisted reproductive technology for infertility. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

461.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 462

Kuru

462.1 Overview

Kuru is a prion disease historically endemic among the Fore people of Papua New Guinea, transmitted through ritualistic endocannibalism, with peak incidence in the 1950s–1960s, and the last known case reported in 2009. The disease was the first human prion disease shown to be transmissible (to chimpanzees, by Carleton Gajdusek, 1966). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

462.2 Causes

This condition happens because of the same mechanism as other prion diseases, with the incubation period being inversely related to the dose of infectious material and age at exposure. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

462.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

462.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

462.5 Progression and Stages

You may experience three stages: ambulant (tremor and progressive loss of coordination, "kuru" means "to shiver" in Fore), sedentary (severe loss of coordination, slurred speech, loss of independent ambulation), and terminal (complete motor incapacity, primitive reflexes, incontinence, and mutism), with emotional lability and inappropriate euphoria ("kuru laughter") being characteristic. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

462.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

462.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (historical). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

462.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

462.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

462.10 Living with It

Treatment typically involves palliative. outlook: the disease was eliminated by the cessation of endocannibalism.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

462.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 463

Kwashiorkor

463.1 Overview

Kwashiorkor is a form of severe protein-energy malnutrition resulting from inadequate protein intake despite relatively preserved caloric intake. It typically occurs in children aged 1–3 years after weaning, particularly when the diet consists predominantly of starchy, low-protein foods. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

463.2 Causes

This condition happens because of hypoalbuminemia due to insufficient dietary protein, leading to decreased plasma oncotic pressure and peripheral swelling. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

463.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

463.4 Signs and Symptoms

- You may experience bilateral pitting swelling (beginning in the feet and ascending), hepatomegaly (fatty liver), thinning and depigmentation of hair (flag sign), a characteristic "flaky paint" dermatitis, cheilosis, and stomatitis
- Affected children appear miserable and apathetic. The presence of swelling distinguishes kwashiorkor from marasmus.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

463.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

463.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

463.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam, supplemented by low serum albumin (less than 25 g/L). Management follows the WHO protocol: treatment of hypoglycemia and hypothermia, empiric antibiotics, correction of electrolyte imbalances, and staged refeeding
- Transition from F-75 to F-100 formula is guided by clinical response.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

463.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

463.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

463.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

463.11 Outlook

- The outlook is good with treatment
- Swelling usually resolves within 1–2 weeks, but long-term thinking and memory deficits may persist.

Chapter 464

Kyphosis

464.1 Overview

Kyphosis is excessive anterior curvature of the thoracic spine (normal kyphosis 20–45°). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

464.2 Causes

Causes include Scheuermann's disease (adolescent, with vertebral wedging), osteoporotic compression fractures (elderly women—"dowager's hump"), ankylosing spondylitis, congenital anomalies, and postural causes. This condition happens because of the specific underlying cause leading to vertebral wedging or collapse. Scheuermann's kyphosis presents in adolescence with rigid thoracic kyphosis and your doctor will diagnose it by the presence of >5° of wedging in three or more consecutive vertebrae on X-ray. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

464.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

464.4 Signs and Symptoms

You may experience visible curvature, back pain, and stiffness. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

464.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

464.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

464.7 Diagnosis

Doctors diagnose this based on physical examination and X-ray. Management ranges from observation and physical therapy for mild cases to bracing and surgery for severe deformities. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

464.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

464.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

464.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

464.11 Outlook

- In most cases, the outlook is good
- Most patients function well with conservative management.

Chapter 465

Labyrinthitis

465.1 Overview

Labyrinthitis is an inflammation of the membranous labyrinth of the inner ear that causes both vestibular and auditory dysfunction. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

465.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of viral infection, often following an upper respiratory infection
- Bacterial labyrinthitis is more severe and can result from otogenic or meningogenic spread.

465.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

465.4 Signs and Symptoms

You may experience feeling of spinning, nausea, vomiting, hearing loss, and ringing in the ears. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

465.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

465.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

465.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by audiometry and vestibular testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

465.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

465.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

465.10 Living with It

- Treatment for viral labyrinthitis involves supportive care (vestibular suppressants, antiemetics, followed by vestibular rehabilitation)
- Bacterial labyrinthitis requires aggressive antibiotic therapy and may require labyrinthectomy.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

465.11 Outlook

- The outlook is generally positive for viral labyrinthitis
- Bacterial labyrinthitis carries a risk of profound hearing loss.

Chapter 466

Lambert-Eaton Myasthenic Syndrome (LEMS)

466.1 Overview

Lambert-Eaton myasthenic syndrome (LEMS) is a rare autoimmune disorder of the neuromuscular junction characterized by presynaptic impairment of acetylcholine release. Driven by pathogenic autoantibodies directed against voltage-gated P/Q-type calcium channels (VGCC) on presynaptic motor nerve terminals, the condition reduces calcium influx during nerve action potentials, preventing acetylcholine vesicle exocytosis into the synaptic cleft. Strongly associated with underlying malignancy (paraneoplastic LEMS, 50–60% of cases, predominantly small cell lung cancer) or primary autoimmune disease. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

466.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

466.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

466.4 Signs and Symptoms

Symptoms often appear as proximal muscle weakness (affecting lower limbs first, causing difficulty rising from a chair), hyporeflexia or areflexia, and prominent autonomic dysfunction (dry mouth, erectile dysfunction, constipation). A hallmark feature is transient post-exercise facilitation (improvement of strength and reflexes after brief muscle contraction).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

466.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

466.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

466.7 Diagnosis

Your doctor confirms the diagnosis with high-titer serum anti-VGCC antibodies and repetitive nerve stimulation (RNS) demonstrating low baseline compound muscle action potential (CMAP) amplitude with dramatic increment ($> 100\%$) following high-frequency stimulation or exercise.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

466.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

466.9 Treatment Options

Treatment focuses on screening for small cell lung cancer, 3,4-diaminopyridine (amifampridine, enhances ACh release), pyridostigmine, and immunosuppression (prednisone, IVIG). Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

466.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 467

Large Bowel Obstruction

467.1 Overview

Large bowel obstruction (LBO) is a blockage of the large intestine, less common than SBO. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

467.2 Causes

The most common cause in Western countries is colorectal cancer (50–60%), followed by diverticulitis strictures and volvulus (sigmoid volvulus in elderly/bedridden patients, cecal volvulus in younger patients). This condition happens because of mechanical or functional obstruction of the colon. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

467.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

467.4 Signs and Symptoms

You may experience progressive abdominal distension, constipation, and eventually nausea and vomiting (late in the course). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

467.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- The right treatment depends on the cause
- Sigmoid volvulus may be decompressed by sigmoidoscopy
- Cancerous LBO typically requires surgical decompression (colostomy or removal) as a palliative or curative measure depending on disease stage.

467.6 Complications

Your outlook depends on the underlying cause and presence of complications.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

467.7 Diagnosis

- Doctors diagnose this based on abdominal X-ray showing a dilated colon with a cutoff point
- CT abdomen/pelvis is diagnostic.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

467.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

467.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

467.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

467.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 468

Laryngitis

468.1 Overview

Laryngitis is an inflammation of the larynx that presents with hoarseness (dysphonia), most commonly viral in acute cases. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

468.2 Causes

This condition happens because of vocal cord inflammation due to viral infection, voice overuse (vocal cord nodules), gastroesophageal reflux disease (laryngopharyngeal reflux), smoking, allergies, or chronic infections. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

468.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

468.4 Signs and Symptoms

- You may experience hoarseness, sore throat, cough, and sometimes low-grade fever
- Chronic laryngitis lasts more than 3 weeks.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

468.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

468.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

468.7 Diagnosis

Doctors diagnose this using laryngoscopy showing vocal cord redness of the skin and swelling. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

468.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

468.9 Treatment Options

- Treatment options include voice rest, hydration, humidification, and addressing underlying causes
- Persistent hoarseness requires laryngoscopy to exclude vocal cord polyps, cysts, or malignancy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

468.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

468.11 Outlook

Most people recover well for acute laryngitis, with resolution within 1–2 weeks. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 469

Lead Poisoning (Plumbism)

469.1 Overview

Lead poisoning (plumbism) is a cumulative toxicant affecting multiple organ systems. Sources include lead-based paint (most common in children), contaminated water, soil, industrial exposure, traditional cosmetics, and imported products. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

469.2 Causes

This condition happens because of lead impairing heme synthesis, disrupting calcium homeostasis, and causing oxidative stress The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- In children, lead crosses the blood-brain barrier and impairs neuronal migration, synaptogenesis, and myelination. No safe blood lead level exists
- The CDC reference level is 3.5 mcg/dL.

469.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

469.4 Signs and Symptoms

Clinical features in children include asymptomatic (most cases), neurocognitive deficits, developmental delay, and at very high levels: encephalopathy, seizures, coma, and death. In adults: peripheral neuropathy, abdominal pain, arthralgias, fatigue, headache, high blood pressure, nephropathy, and anemia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

469.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

469.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

469.7 Diagnosis

Doctors diagnose this using venous blood lead level. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

469.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

469.9 Treatment Options

- Treatment options include removal from exposure source and chelation therapy: succimer (DMSA, oral) for asymptomatic children with levels ≥45 mcg/dL
- BAL plus calcium disodium EDTA for severe toxicity.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

469.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

469.11 Outlook

The outlook is generally positive with early intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 470

Leishmaniasis

470.1 Overview

Leishmaniasis is a protozoan infection caused by *Leishmania* species, transmitted by sandflies, with a clinical spectrum ranging from localized cutaneous disease to lethal visceral leishmaniasis (Kala-azar), with 1.5 million new cases annually and 20,000–30,000 deaths, endemic in tropical and subtropical regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

470.2 Causes

This condition happens because of the sandfly injecting promastigotes into the skin, which are phagocytosed by macrophages and convert to amastigotes, replicating intracellularly and spreading to the reticuloendothelial system. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

470.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

470.4 Signs and Symptoms

You may experience visceral leishmaniasis: fever (often double quotidian), weight loss, massive splenomegaly, hepatomegaly, lymphadenopathy, and pancytopenia

- Cutaneous leishmaniasis: single or multiple painless ulcer(s) at sandfly bite sites with raised, indurated borders
- Mucocutaneous leishmaniasis: destructive lesions of the nasopharynx.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

470.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

470.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

470.7 Diagnosis

Doctors diagnose this using microscopic identification of amastigotes in tissue aspirates or biopsies. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

470.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

470.9 Treatment Options

Treatment options include liposomal amphotericin B (first-line for visceral), pentavalent antimonials, and miltefosine. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

470.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

470.11 Outlook

Your outlook depends on the clinical form and treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 471

Leprosy (Hansen's Disease)

471.1 Overview

Leprosy is a chronic granulomatous infection caused by *Mycobacterium leprae*, an obligate intracellular, acid-fast bacillus, with approximately 200,000 new cases reported annually, primarily in India, Brazil, and Indonesia, transmitted through prolonged, close contact with untreated patients, and a long incubation period (2–10 years, up to 40 years). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

471.2 Causes

This condition happens because of *M. leprae* invading Schwann cells and macrophages, with the clinical spectrum correlating with the host immune response, ranging from tuberculoid leprosy (strong cell-mediated immunity, few bacilli) to lepromatous leprosy (weak cell-mediated immunity, numerous bacilli). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

471.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

471.4 Signs and Symptoms

You may experience hypopigmented, anesthetic skin patches, peripheral nerve thickening, and neuropathy, with tuberculoid leprosy presenting with 1–3 well-demarcated anesthetic macules/plaques, and lepromatous leprosy presenting with diffuse, symmetric skin infiltration with numerous nodules and plaques (leonine facies), eyebrow loss, and progressive symmetric peripheral neuropathy. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

471.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

471.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

471.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (anesthetic skin patches, nerve enlargement), with skin slit-smear for acid-fast bacilli. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

471.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

471.9 Treatment Options

Treatment options include rifampin, dapsone, and clofazimine (multidrug therapy). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

471.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

471.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 472

Leptospirosis

472.1 Overview

Leptospirosis is a zoonotic spirochetal infection caused by pathogenic serovars of *Leptospira interrogans* sensu lato, with more than 1 million cases annually worldwide, highest incidence in tropical and subtropical regions, and transmission through contact with water or soil contaminated by urine of infected animals. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

472.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

472.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

472.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

472.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- This condition happens because of *Leptospira* entering through mucous membranes or abraded skin, multiplying in the bloodstream (leptospiremic phase), then localizing to tissues including liver, kidney, meninges, and eye. You may experience anicteric leptospirosis (80–90%): biphasic illness with Phase 1 (leptospiremic: abrupt fever, rigors, severe headache, muscle pain, conjunctival suffusion) and Phase 2 (immune: aseptic meningitis, uveitis, rash)
- Weil's disease (severe icteric leptospirosis, 5–10%): jaundice, kidney failure, bleeding, and myocarditis.

472.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

472.7 Diagnosis

Doctors diagnose this using PCR of blood or urine, culture, and serology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

472.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

472.9 Treatment Options

Treatment options include doxycycline or amoxicillin for mild disease, and ceftriaxone or penicillin G for severe disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

472.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

472.11 Outlook

Most people recover well for mild disease, but mortality is higher with Weil's disease. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 473

Lipoprotein(a) Disorders

473.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Lp(a) levels are predominantly genetically determined (by the LPA gene) and are largely unaffected by diet or lifestyle
- The prevalence of elevated Lp(a) is approximately 20% of the population worldwide.

473.2 Causes

This condition happens because of apolipoprotein(a) component sharing homology with plasminogen and having antifibrinolytic properties, also promoting vascular inflammation and foam cell formation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

473.3 Risk Factors

Lipoprotein(a) (Lp(a)) disorder is a lipid disorder defined by elevated Lp(a) levels (greater than 50 mg/dL or greater than 125 nmol/L), which is an independent, causal risk factor for atherosclerotic cardiovascular disease and calcific aortic valve stenosis. Treatment focuses on aggressive control of other cardiovascular risk factors. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

473.4 Signs and Symptoms

You may experience premature cardiovascular disease and a family history of early heart disease or aortic stenosis, with no other specific clinical findings. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

473.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

473.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

473.7 Diagnosis

Doctors diagnose this using measurement of Lp(a) mass or molar concentration (once in a lifetime, as levels are stable). Standard lipid-lowering therapies do not significantly lower Lp(a)

- Niacin and PCSK9 inhibitors produce modest reductions
- Investigational therapies (pelacarsen, olpasiran) reduce Lp(a) by 80–90%.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

473.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

473.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

473.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

473.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 474

Lithium Toxicity

474.1 Overview

Lithium toxicity is a pharmaceutical overdose involving a monovalent cation used for bipolar disorder with a narrow therapeutic index (0.6–1.2 mEq/L). Toxicity occurs from acute ingestion, acute-on-chronic overdose, or chronic accumulation (often due to dehydration, kidney impairment, or drug interactions). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

474.2 Causes

This condition happens because of lithium substituting for sodium in excitable cells, altering neurotransmitter release, and impairing kidney concentrating ability. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

474.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

474.4 Signs and Symptoms

- Clinical features: acute toxicity presents with digestive tract symptoms and mild nervous system symptoms (tremor, loss of coordination, involuntary eye movements)
- Chronic toxicity presents predominantly with neurotoxicity (confusion, slurred speech, loss of coordination, hyperreflexia, clonus, parkinsonism), with severe cases progressing to seizures, coma, and permanent cerebellar injury.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

474.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

474.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

474.7 Diagnosis

Doctors diagnose this using serum lithium level (interpreted with caution—chronic toxicity can occur at therapeutic levels). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

474.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

474.9 Treatment Options

- Treatment options include IV fluids (normal saline) to correct volume depletion and enhance lithium clearance
- Hemodialysis is indicated for severe toxicity.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

474.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

474.11 Outlook

- The outlook is generally positive with prompt treatment
- Chronic toxicity may cause permanent nervous system injury.

Chapter 475

Lobular Carcinoma In Situ (LCIS)

475.1 Overview

Lobular carcinoma in situ (LCIS) is a non-invasive neoplastic growth of small, dyscohesive atypical cells completely filling and expanding the lobules and terminal ductules of the breast. Like invasive lobular carcinoma, LCIS is driven by loss of E-cadherin expression (CDH1 gene inactivation). Unlike DCIS, LCIS does not produce clinical masses or mammographic calcifications and is typically discovered incidentally on core needle biopsy performed for unrelated non-cancerous lesions. LCIS functions primarily as a powerful non-palpable, non-calcified biomarker indicating a 9- to 10-fold increased risk of developing invasive carcinoma in either breast (bilateral risk). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

475.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

475.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal

factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

475.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

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- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

475.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer

prognosis.

475.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

475.7 Diagnosis

To make the diagnosis, doctors look for histological confirmation with E-cadherin immunohistochemical staining. Management options include active surveillance with annual screening mammography and MRI, selective estrogen receptor modulators (tamoxifen, raloxifene) for chemoprevention, or bilateral prophylactic mastectomy for high-risk BRCA or family history carriers. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
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- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

475.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

475.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
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- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

475.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

475.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 476

Long QT Syndrome

476.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Long QT syndrome is a group of inherited cardiac channelopathies characterized by prolongation of the QT interval on ECG and predisposition to torsades de pointes ventricular arrhythmias and sudden death
- Over 17 genes have been identified, with LQT1 (KCNQ1), LQT2 (KCNH2), and LQT3 (SCN5A) accounting for the majority.

476.2 Causes

This condition happens because of mutations in cardiac ion channels affecting ventricular repolarization. Triggers vary by genotype: exercise (LQT1), auditory stimuli and emotional stress (LQT2), and rest or sleep (LQT3). Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

476.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

476.4 Signs and Symptoms

You may experience syncope, seizures, and sudden cardiac death.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

476.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

476.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

476.7 Diagnosis

Doctors diagnose this using ECG (corrected QT interval greater than 470 ms in men, greater than 480 ms in women is highly suggestive), genetic testing, and clinical criteria (Schwartz score).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

476.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

476.9 Treatment Options

- Treatment options include beta-blockers (nadolol preferred), avoidance of QT-prolonging drugs, and implantable cardioverter-defibrillator for high-risk patients
- Left cardiac sympathetic denervation is an option for drug-refractory cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

476.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

476.11 Outlook

Most people recover well with appropriate management and avoidance of triggers.

Chapter 477

Ludwig Angina

477.1 Overview

Ludwig angina is a rapidly spreading, life-threatening bilateral gangrenous cellulitis of the submandibular, sublingual, and submental spaces, carrying a high risk of fatal airway compromise. Pathophysiologically, $> 80\%$ of cases originate from odontogenic infections, most commonly infected second or third lower molar teeth whose roots extend below the mylohyoid muscle line. Polymicrobial dental flora (Streptococcus viridans, anaerobes, Bacteroides, Fusobacterium) cross tissue planes without lymphadenopathy or abscess cavitation, causing massive inflammatory swelling that elevates and displaces the tongue posteriorly against the soft palate and hypopharynx. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

477.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

477.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

477.4 Signs and Symptoms

Symptoms often appear as severe neck pain, fever, trismus, difficulty swallowing, drooling, wood-like ('woody') induration of the submandibular space, and progressive stridor. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

477.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

477.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection

- Chronic kidney disease from reduced perfusion

477.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by neck CT once airway stability is secured. Management prioritizes emergency airway control (fiberoptic nasal intubation or awake tracheostomy) combined with IV broad-spectrum antibiotics (ampicillin-sulbactam or carbapenems) and surgical incision and removal of fluid of deep tissue spaces. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

477.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

477.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy

(pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

477.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

477.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 478

Lung Abscess

478.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- A lung abscess is a necrotizing infection of the lung parenchyma leading to a cavitary lesion containing purulent material
- It may be primary (removal by suction pneumonia in patients with impaired consciousness, poor dentition, or swallowing disorders) or secondary (bronchial obstruction, hematogenous seeding).

478.2 Causes

This condition happens because of microbial infection leading to lung tissue tissue death. Common organisms include anaerobic bacteria, *Staphylococcus aureus*, and *Klebsiella pneumoniae*. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

478.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

478.4 Signs and Symptoms

You may experience fever, productive cough with foul-smelling sputum, weight loss, and night sweats. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

478.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

478.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

478.7 Diagnosis

Doctors diagnose this using chest X-ray showing a thick-walled cavity, often with an air-fluid level. Management requires prolonged intravenous antibiotics (6–8 weeks), bronchoscopic removal of fluid for large or refractory abscesses, and surgical removal as a last resort. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

478.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings

- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

478.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

478.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

478.11 Outlook

The outlook is good with appropriate antibiotic therapy. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 479

Lung Cancer

479.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Lung cancer is the leading cause of cancer-related death worldwide, with approximately 1.8 million deaths annually
- Non-small cell lung cancer accounts for 85% of cases and includes adenocarcinoma (most common subtype, often in non-smokers, with driver mutations in EGFR, ALK, ROS1, KRAS), squamous cell carcinoma (central, strongly smoking-related), and large cell carcinoma
- Small cell lung cancer accounts for 15%, is strongly associated with smoking, and has aggressive behavior with early metastasis.

479.2 Causes

This condition happens because of cancerous transformation of bronchial epithelial cells driven by carcinogen exposure. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

479.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

479.4 Signs and Symptoms

You may experience cough, hemoptysis, shortness of breath, chest pain, weight loss, and post-obstructive pneumonia. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

479.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for early-stage disease is surgical removal with adjuvant chemotherapy
- Advanced disease doctors typically treat it with platinum-based chemotherapy, immunotherapy, and targeted therapy for driver mutations
- Small cell lung cancer doctors typically treat it with chemoradiation and prophylactic cranial irradiation.

479.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

479.7 Diagnosis

Diagnosis typically involves CT-guided biopsy, bronchoscopy, or mediastinoscopy, followed by molecular testing for driver mutations. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)

- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

479.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

479.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies

- Surveillance: Active monitoring after definitive treatment

479.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
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- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

479.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 480

Lyme Disease (Borrelia burgdorferi)

480.1 Overview

Lyme disease is a tick-borne spirochetal infection caused by *Borrelia burgdorferi* sensu stricto (US) and *B. afzelii* and *B. garinii* (Europe, Asia), and is the most common vector-borne disease in the United States with approximately 500,000 cases annually, transmitted by *Ixodes* ticks requiring a 24–48 hour blood meal for transmission. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

480.2 Causes

This condition happens because of *B. burgdorferi* being injected into the skin with tick saliva, replicating locally, and disseminating hematogenously. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

480.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

480.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

480.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Clinical features occur in three stages: Stage 1 (early localized, 3–30 days): redness of the skin migrans—an expanding annular rash with central clearing ("bull's-eye" lesion) at the tick bite site, often with fever, fatigue, muscle pain, and headache
- Stage 2 (early disseminated, weeks to months): multiple secondary redness of the skin migrans lesions, flu-like symptoms, migratory joint pain, cranial nerve palsies, meningitis, and carditis
- Stage 3 (late, months to years): Lyme arthritis, acrodermatitis chronica atrophicans, and neuroborreliosis.

480.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

480.7 Diagnosis

Doctors diagnose this using serology using two-tier testing (EIA followed by Western blot). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

480.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

480.9 Treatment Options

Treatment options include doxycycline for early disease, ceftriaxone for nervous system disease, and oral antibiotics for Lyme arthritis. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

480.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

480.11 Outlook

Most people recover well with appropriate antibiotic treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 481

Lymphogranuloma Venereum (LGV)

481.1 Overview

Lymphogranuloma venereum is a sexually transmitted infection caused by *Chlamydia trachomatis* serovars L1, L2, and L3, which are invasive and disseminate to regional lymph nodes, endemic in Africa, Southeast Asia, and the Caribbean, with outbreaks among MSM in Europe, North America, and Australia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

481.2 Causes

This condition happens because of LGV serovars infecting epithelial cells and macrophages, then disseminating through lymphatics, causing lymphangitis and perilymphangitis with granulomatous inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

481.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

481.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

481.5 Progression and Stages

You may experience an incubation of 3–30 days, with primary stage: painless papule or ulcer at the inoculation site The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Secondary stage (2–6 weeks): painful inguinal lymphadenopathy with the “groove sign” characteristic
- Tertiary stage: genital elephantiasis and rectal strictures.

481.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

481.7 Diagnosis

Doctors diagnose this using *C. trachomatis* PCR from ulcer, rectal swab, or bubo aspirate, with genotyping distinguishing LGV serovars. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

481.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

481.9 Treatment Options

Treatment options include doxycycline 100 mg twice daily for 21 days. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

481.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

481.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 482

Lynch Syndrome

482.1 Overview

Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC) is an autosomal dominant cancer predisposition syndrome caused by germline mutations in DNA mismatch repair genes (MLH1, MSH2, MSH6, PMS2) or EPCAM deletions. It accounts for 2–4% of all CRCs, with a lifetime CRC risk of 40–80%. Associated cancers include endometrial (highest risk after CRC), ovarian, gastric, urinary tract, hepatobiliary, and small bowel cancers. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

482.2 Causes

This condition happens because of defective mismatch repair leading to microsatellite instability (MSI) or mismatch repair deficiency (dMMR). Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

482.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

482.4 Signs and Symptoms

You may experience early-onset colorectal cancer and increased risk of multiple primary cancers.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

482.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

482.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

482.7 Diagnosis

Doctors diagnose this based on Amsterdam criteria, revised Bethesda guidelines, immunohistochemistry of tumor tissue, and germline genetic testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

482.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

482.9 Treatment Options

- Treatment options include surveillance colonoscopy starting at age 20–25 (or 2–5 years before the earliest CRC in the family) every 1–2 years
- Prophylactic surgery may be considered.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

482.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

482.11 Outlook

- Outlook is improved with surveillance
- Immunotherapy with pembrolizumab is particularly effective for MSI-high/dMMR cancers.

Chapter 483

Lysosomal Storage Diseases

483.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Lysosomal storage diseases (LSDs) are a group of over 50 inherited metabolic disorders caused by deficiency of specific lysosomal hydrolases, resulting in progressive accumulation of undegraded substrates within lysosomes. Gaucher disease is caused by glucocerebrosidase deficiency, presenting with hepatosplenomegaly, bone pain (Erlenmeyer flask deformity of the distal femur), anemia, thrombocytopenia, and Gaucher cells
- Enzyme replacement therapy and substrate reduction therapy are effective. Niemann-Pick disease is caused by sphingomyelinase deficiency (types A and B) causing accumulation of sphingomyelin in macrophages
- Type A presents with severe neurovisceral disease and a cherry-red spot on the macula. Tay-Sachs disease is caused by hexosaminidase A deficiency, presenting in infancy with progressive neurodegeneration, hypotonia, exaggerated startle response, cherry-red spot, and blindness. Metachromatic leukodystrophy is caused by arylsulfatase A deficiency, causing accumulation of sulfatides with progressive demyelination.

483.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

483.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

483.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

483.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Fabry disease is an X-linked α-galactosidase A deficiency causing accumulation of globotriaosylceramide, presenting with acroparesthesias, angiokeratomas, corneal verticillata, hypohidrosis, progressive kidney failure, hypertrophic cardiomyopathy, and stroke
- Enzyme replacement therapy and oral chaperone therapy slow disease progression.

483.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

483.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists

Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

483.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

483.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

483.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

483.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 484

Major Depressive Disorder (MDD)

484.1 Overview

Major depressive disorder is a prevalent psychiatric disorder with a lifetime prevalence of approximately 15–20% and a 12-month prevalence of 7%, twice as common in women as in men, with a peak age of onset of 20–40 years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

484.2 Causes

This condition happens because of dysregulation of monoamine neurotransmitter systems (serotonin, norepinephrine, dopamine), hypothalamic-pituitary-adrenal (HPA) axis hyperactivity with elevated cortisol, reduced hippocampal volume due to chronic stress-induced neurotoxicity, decreased brain-derived neurotrophic factor (BDNF), and altered functional connectivity in frontolimbic circuits, with genetic factors accounting for 37% of risk. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

484.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

484.4 Signs and Symptoms

You may experience depressed mood (sadness, emptiness, hopelessness) nearly every day, anhedonia (markedly diminished interest or pleasure in activities), significant weight change, insomnia or hypersomnia, psychomotor agitation or retardation, fatigue, feelings of worthlessness or excessive guilt, diminished concentration, and recurrent thoughts of death or suicidal ideation, with DSM-5-TR criteria requiring at least five of these symptoms (including depressed mood or anhedonia) present during the same two-week period. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

484.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

484.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

484.7 Diagnosis

Doctors diagnose this using clinical interview using DSM-5-TR criteria, with screening tools including the PHQ-9 and BDI-II. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

484.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

484.9 Treatment Options

Treatment typically involves multimodal: pharmacotherapy (SSRIs and SNRIs are first-line), psychotherapy (CBT, IPT), somatic therapies (ECT for severe cases, TMS, ketamine/esketamine for treatment-resistant depression), and lifestyle interventions. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

484.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

484.11 Outlook

- The outlook varies from person to person
- Many patients respond to treatment but recurrence is common.

Chapter 485

Malaria (Plasmodium spp.)

485.1 Overview

Malaria is a protozoan infection caused by *Plasmodium* species, transmitted by female *Anopheles* mosquitoes, with five species causing human disease: *P. falciparum* (most virulent, responsible for nearly all deaths), *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi*, with 240 million cases and 600,000 deaths annually, primarily in sub-Saharan Africa. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

485.2 Causes

This condition happens because of sporozoites being injected with mosquito saliva, traveling to the liver and invading hepatocytes (exoerythrocytic schizogony), with merozoites released from the liver invading RBCs and initiating the erythrocytic cycle, with *P. falciparum* modifying RBC surfaces and causing cytoadherence, microvascular obstruction, and end-organ damage. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

485.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to

endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

485.4 Signs and Symptoms

You may experience an incubation of 7–30 days, with paroxysms of fever, chills, rigor, and diaphoresis every 48–72 hours depending on species, with uncomplicated malaria presenting with cyclic fever, headache, muscle pain, anemia, and splenomegaly, and severe malaria (from *P. falciparum*) presenting with cerebral malaria, severe anemia, acute kidney injury, lung swelling, hypoglycemia, metabolic acidosis, DIC, and shock. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

485.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

485.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

485.7 Diagnosis

Doctors diagnose this using thick and thin blood smears and rapid diagnostic tests. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

485.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

485.9 Treatment Options

Treatment options include artemisinin-based combination therapy (ACT) for uncomplicated *P. falciparum*, artesunate IV for severe malaria, and chloroquine plus primaquine for *P. vivax* and *P. ovale*. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

485.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

485.11 Outlook

- Most people recover well with prompt treatment
- Severe malaria has significant mortality.

Chapter 486

Male Breast Cancer

486.1 Overview

Male breast cancer accounts for less than 1% of all breast cancers but is increasing in incidence. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

486.2 Causes

This condition happens because of cancerous transformation of breast epithelium in males. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

486.3 Risk Factors

Risk factors include BRCA2 mutations (most common genetic predisposition), Klinefelter syndrome, radiation exposure, gynecomastia, and liver disease, with a median age at To diagnose 67 years.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption

- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

486.4 Signs and Symptoms

Clinical features typically include a painless, subareolar mass, with ER-positive disease being more common than in female breast cancer. Management follows the same principles as female breast cancer: mastectomy with sentinel lymph node biopsy, adjuvant radiation, and endocrine therapy (tamoxifen is first-line for male breast cancer). Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

486.5 Progression and Stages

Your outlook depends on stage at diagnosis. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

486.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

486.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging

- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

486.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

486.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

486.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

486.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 487

Male Infertility

487.1 Overview

Male infertility is responsible for approximately 40–50% of couple infertility. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

487.2 Causes

Causes include varicocele (most common correctable cause, present in 40% of infertile men), obstructive azoospermia (congenital absence of the vas deferens—often associated with CFTR mutations, epididymitis, vasectomy), non-obstructive azoospermia (Klinefelter syndrome, Y-chromosome microdeletions, cryptorchidism, prior chemotherapy/radiation), hypogonadotropic hypogonadism (Kallmann syndrome, pituitary tumors), and idiopathic oligozoospermia. This condition happens because of impaired spermatogenesis or sperm transport. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

487.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

487.4 Signs and Symptoms

You may experience inability to conceive after one year of unprotected intercourse. Evaluation includes semen analysis (WHO 2021 reference values), hormonal evaluation (FSH, LH, testosterone), and scrotal ultrasound. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

487.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

487.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

487.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

487.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

487.9 Treatment Options

The right treatment depends on the cause: varicocelectomy, assisted reproductive techniques (IUI, IVF with ICSI), surgical sperm retrieval, and treatment of reversible causes. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

487.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

487.11 Outlook

Recovery varies by cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 488

Malignant Hyperthermia

488.1 Overview

cancerous hyperthermia (MH) is a life-threatening pharmacogenetic skeletal muscle disorder triggered by exposure to volatile halogenated anesthetic agents (isoflurane, sevoflurane, desflurane) or depolarizing muscle relaxants (succinylcholine) in susceptible individuals. Driven by autosomal dominant gain-of-function mutations in the RYR1 gene (encoding the ryanodine receptor 1 calcium release channel of the sarcoplasmic reticulum) or CACNA1S gene, volatile agents trigger uncontrolled, massive calcium efflux from the sarcoplasmic reticulum into the myoplasm. Persistent intracellular calcium accumulation drives sustained, uncoordinated muscle contractions, hypermetabolism, ATP depletion, and rhabdomyolysis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

488.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

488.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

488.4 Signs and Symptoms

Symptoms often appear as rapid masseter muscle spasm (lockjaw), hypercapnia (rising end-tidal CO2, earliest and most sensitive sign), sinus tachycardia, muscle rigidity, severe mixed respiratory/metabolic acidosis, rhabdomyolysis (myoglobinuria), hyperkalemia, and late fulminant hyperthermia ($> 42°C$). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

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- Functional impairment affecting daily activities
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- Symptoms may develop gradually or appear suddenly

488.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

488.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

488.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam during anesthesia
- Susceptibility your doctor confirms it with caffeine-halothane contracture test (CHCT) or genetic testing. Management requires immediate discontinuation of triggering agents, hyperventilation with 100% oxygen, administration of IV dantrolene sodium (ryanodine receptor antagonist), active cooling, and aggressive treatment of hyperkalemia and acidosis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

488.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

488.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

488.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

488.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 489

Malnutrition in the Elderly

489.1 Overview

Malnutrition is a geriatric syndrome affecting 15–60% of hospitalized older adults and 30–50% of nursing home residents. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

489.2 Causes

This condition happens because of physiologic changes (reduced olfaction and taste, delayed gastric emptying), social factors (isolation, poverty, inability to shop/cook), dentition problems, difficulty swallowing, polypharmacy, depression, and dementia. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

489.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

489.4 Signs and Symptoms

You may experience weight loss, poor nutritional intake, and functional decline.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

489.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

489.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

489.7 Diagnosis

Doctors use the Mini Nutritional Assessment (MNA) as a validated screening tool.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

489.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

489.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

489.10 Living with It

Treatment options include dietary counseling, oral nutritional supplements, texture-modified foods for difficulty swallowing, treatment of depression, dental care, and in severe cases, enteral nutrition.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

489.11 Outlook

- The outlook varies from person to person
- Consequences include impaired immune function, delayed wound healing, muscle wasting, falls, and increased mortality.

Chapter 490

Mammalian Bites

490.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Mammalian bites (dog, cat, human) are common and carry risk of infection. Dog bites cause significant crush injury and laceration
- Pathogens include Pasteurella canis, Staphylococcus aureus, Streptococcus species, and anaerobes. Cat bites create deep puncture wounds carrying bacteria into closed spaces
- Pathogens include Pasteurella multocida, Staphylococcus, Streptococcus, and Bartonella henselae. Human bites carry high risk of infection from oral flora including Eikenella corrodens, Streptococcus, Staphylococcus, and anaerobes. Management: wound irrigation, removal of dead tissue of devitalized tissue, assessment of deep structure involvement
- Primary closure is generally avoided for high-risk wounds except facial wounds.

490.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

490.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

490.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

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- Changes in appearance or function of affected tissues
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- Symptoms may develop gradually or appear suddenly

490.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

490.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

490.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

490.8 Prevention

Antibiotic prophylaxis for high-risk wounds. Rabies prophylaxis indicated for bites from unvaccinated or unknown-source wild animals. Tetanus prophylaxis as indicated.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

490.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

490.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

490.11 Outlook

The outlook is generally positive with appropriate wound management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 491

Mammary Duct Ectasia

491.1 Overview

Mammary duct ectasia (periductal mastitis) is a non-cancerous inflammatory condition characterized by progressive dilation, shortening, and periductal scarring of the subareolar lactiferous ducts, predominantly affecting multiparous postmenopausal women. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

491.2 Causes

This condition happens because of involutional accumulation and lipid oxidation of ductal secretions, producing ductal rupture, lipid extravasation, and chronic periductal granulomatous inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

491.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

491.4 Signs and Symptoms

You may experience thick, sticky, pasty green or brownish single- or multiple-duct nipple discharge, subareolar pain, nipple inversion/retraction, and a firm subareolar mass mimicking breast carcinoma. Ultrasound demonstrates dilated, fluid-filled subareolar ducts containing particulate debris. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
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491.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

491.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

491.7 Diagnosis

Doctors confirm the diagnosis through clinical examination, sonography, and biopsy to exclude malignancy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

491.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

491.9 Treatment Options

- Treatment typically involves conservative with warm compresses and antibiotics if infected
- Major duct removal (Hadfield operation) is indicated for persistent, troublesome discharge or severe pain.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

491.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

491.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 492

Maple Syrup Urine Disease

492.1 Overview

Maple syrup urine disease (MSUD) is an autosomal recessive disorder caused by deficiency of the branched-chain α-ketoacid dehydrogenase (BCKD) enzyme complex, leading to accumulation of branched-chain amino acids (BCAAs)—leucine, isoleucine, valine—and their corresponding ketoacids. Incidence is approximately 1 in 185,000 live births worldwide, but is higher in certain populations (Mennonite communities, 1 in 380). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

492.2 Causes

This condition happens because of deficiency of the BCKD enzyme complex. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

492.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

492.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

492.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- You may experience the classic (severe) form presenting in the first week of life with poor feeding, lethargy, vomiting, hypertonicity, opisthotonos, seizures, and a characteristic sweet, maple syrup odor of cerumen and urine
- Progressive nervous system deterioration leads to coma and respiratory failure if untreated.

492.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

492.7 Diagnosis

Doctors diagnose this using newborn screening (elevated leucine/isoleucine) and plasma amino acid analysis showing markedly elevated leucine, isoleucine, valine, and pathognomonic alloisoleucine. Treatment for acute decompensation includes aggressive fluid resuscitation, high-dose dextrose and insulin to promote anabolism, and dialysis for rapid leucine clearance

- Chronic Treatment focuses on dietary restriction of BCAAs, thiamine supplementation (in responsive variants), and vigilant monitoring during intercurrent illness
- Liver transplantation provides a functional cure.
- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

492.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

492.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

492.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

492.11 Outlook

The outlook is generally positive with early treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 493

Marasmic-Kwashiorkor

493.1 Overview

Marasmic-kwashiorkor is a mixed form of severe acute malnutrition combining features of both marasmus (wasting) and kwashiorkor (swelling). It occurs when there is deficiency of both total calories and protein. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

493.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

493.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

493.4 Signs and Symptoms

You may experience severe wasting accompanied by peripheral swelling. Anthropometry shows severe deficits in weight-for-age and weight-for-height, with concurrent pitting swelling and low serum albumin. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

493.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

493.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

493.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with anthropometric confirmation. Management follows the same WHO guidelines with emphasis on careful monitoring for fluid overload, refeeding syndrome, and infections. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

493.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

493.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

493.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

493.11 Outlook

outlook is worse than for either pure form alone. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 494

Marasmus

494.1 Overview

Marasmus is a form of severe protein-energy malnutrition characterized by inadequate intake of both protein and calories. It is most common in infants and young children in low-resource settings, particularly during the first year of life, and is linked to early weaning, repeated infections, and poverty. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

494.2 Causes

This condition happens because of catabolism of muscle and subcutaneous fat to meet energy needs, leading to profound wasting with relative preservation of serum albumin. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

494.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

494.4 Signs and Symptoms

- You may experience severe wasting of muscle and subcutaneous tissue, a “monkey-like” facies with prominent eyes, wrinkled skin, irritability, and growth retardation
- Subcutaneous fat is absent and there is no peripheral swelling.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

494.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

494.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

494.7 Diagnosis

- Doctors diagnose this based on anthropometric criteria: weight-for-age less than 60% of median, weight-for-height less than 70% of median, or mid-upper arm circumference (MUAC) less than 115 mm in children aged 6–59 months. Management follows the WHO protocol: stabilization with correction of dehydration and electrolyte imbalances (using ReSoMal), empiric antibiotics, micronutrient supplementation, and staged refeeding with F-75 formula followed by F-100 formula
- Slow, cautious refeeding is required to avoid refeeding syndrome.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

494.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

494.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

494.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

494.11 Outlook

The outlook is good with timely treatment, but mortality remains high (20–30%) in severe cases without adequate care. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 495

Marburg Virus

495.1 Overview

Marburg virus disease is a severe involving bleeding fever caused by Marburg virus, a negative-sense RNA virus (genus *Marburgvirus*), the only filovirus other than Ebola to cause human disease, with outbreaks in Africa, and the reservoir being the Egyptian fruit bat (*Rousettus aegyptiacus*). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

495.2 Causes

This condition happens because of identical Disease mechanism to Ebola virus—infection of dendritic cells, macrophages, and monocytes, profound immunosuppression, cytokine storm, endothelial dysfunction, and DIC. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

495.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

495.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

495.5 Progression and Stages

You may experience an incubation of 2–21 days, with abrupt onset of fever, chills, severe headache, muscle pain, and prostration, followed by digestive tract phase (severe watery diarrhea, abdominal pain) and involving bleeding phase (tiny red or purple spots, ecchymoses, mucosal bleeding, uncontrolled bleeding, and multi-organ failure). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

495.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

495.7 Diagnosis

Doctors diagnose this using RT-PCR (blood, urine, throat swab). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

495.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

495.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

495.10 Living with It

Treatment typically involves supportive care with no approved specific therapy. outlook: mortality 23–80% depending on outbreak and strain.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

495.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 496

Marfan Syndrome

496.1 Overview

Marfan syndrome is an autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene encoding fibrillin-1, affecting 1 in 5,000 individuals. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

496.2 Causes

This condition happens because of abnormal microfibril assembly, leading to reduced structural integrity of connective tissue and dysregulation of TGF-beta signaling. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

496.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

496.4 Signs and Symptoms

You may experience skeletal manifestations (tall stature, arachnodactyly, pectus excavatum or carinatum, scoliosis, joint hypermobility), ocular manifestations (ectopia lentis, myopia, retinal detachment), and cardiovascular manifestations (aortic root dilation, aortic aneurysm, aortic dissection, mitral valve prolapse).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

496.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

496.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

496.7 Diagnosis

Doctors diagnose this based on the revised Ghent criteria, including systemic score, aortic root diameter Z-score, ectopia lentis, and FBN1 genetic testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

496.8 Prevention

Treatment focuses on beta-blockers (atenolol) or losartan to reduce aortic growth rate, serial echocardiography for aortic surveillance, elective aortic root replacement when diameter reaches 50 mm (or earlier with family history of dissection), and lifestyle modification (avoidance of contact sports, isometric exercise).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

496.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

496.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

496.11 Outlook

- The outlook has improved dramatically
- Life expectancy now approaches 70 years with appropriate surveillance and prophylactic surgery, compared to 45 years in the 1970s.

Chapter 497

Marine Envenomation

497.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Jellyfish envenomation: nematocysts discharge venom containing toxins causing local pain, redness of the skin, and in severe cases (box jellyfish—Chironex fleckeri), cardiovascular collapse and death within minutes. Treatment: rinse with vinegar (5% acetic acid) to inactivate undischarged nematocysts
- Remove tentacles with forceps
- Hot water immersion (45°C, 20 minutes) inactivates venom. For box jellyfish, antivenom is used for severe cases. Stonefish (Synanceia) envenomation produces excruciating pain, local swelling, hypotension, arrhythmias, and rarely death. Treatment: hot water immersion (45°C for 30–90 minutes denatures the heat-labile toxin)
- Antivenom is available. Cone snail (Conus) venom contains conotoxins causing neuromuscular blockade, bulbar palsy, and respiratory failure.

497.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

497.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

497.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

497.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

497.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

497.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

497.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

497.9 Treatment Options

Treatment typically involves supportive with airway support and ventilation as needed. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

497.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

497.11 Outlook

The outlook is generally positive with prompt treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 498

Massive Splenomegaly

498.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Massive splenomegaly is spleen enlargement extending below the umbilicus or crossing the midline. Common causes in developing countries include malaria, visceral leishmaniasis (kala-azar), schistosomiasis, and myelofibrosis
- In Western countries, myeloproliferative tumors (chronic myeloid leukemia, polycythemia vera, primary myelofibrosis), lymphomas (hairy cell leukemia, splenic marginal zone lymphoma), and storage diseases (Gaucher disease) are more common.

498.2 Causes

This condition happens because of the underlying infiltrative, congestive, or neoplastic process. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

498.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

498.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

498.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

498.6 Complications

- You may experience significant symptoms and complications including splenic tissue death due to lack of blood supply, portal high blood pressure, and severe cytopenias from hypersplenism. Treatment typically involves directed at the underlying cause
- Splenectomy may be required for symptomatic relief or life-threatening complications.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

498.7 Diagnosis

Doctors diagnose this based on imaging and identification of the underlying cause. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

498.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

498.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

498.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

498.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 499

Mastitis (Puerperal)

499.1 Overview

It presents with localized breast redness of the skin, warmth, induration, and systemic symptoms (fever, myalgias). This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

499.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

499.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

499.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance

or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

499.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

499.6 Complications

Puerperal mastitis is a common puerperal complication affecting 2–10% of breastfeeding women, usually within the first six weeks postpartum.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

499.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

499.8 Prevention

Treatment options include continued breastfeeding or milk expression to prevent milk stasis, supportive bras, and antibiotics (dicloxacillin or cephalexin).

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

499.9 Treatment Options

Breast abscess (5–10% of cases) requires ultrasound-guided removal by suction or incision and removal of fluid. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

499.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

499.11 Outlook

The outlook is generally positive. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 500

Mastitis and Breast Abscess

500.1 Overview

Mastitis is inflammation of the breast tissue, most commonly lactational (puerperal) mastitis, occurring in 2–10% of breastfeeding women. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

500.2 Causes

This condition happens because of bacterial infection, most commonly *Staphylococcus aureus*, introduced through a cracked nipple or cracked skin. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

500.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

500.4 Signs and Symptoms

You may experience a focal area of redness of the skin, warmth, tenderness, and induration, often accompanied by fever and malaise. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

500.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

500.6 Complications

Breast abscess is a complication of mastitis requiring removal by suction or incision and removal of fluid.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

500.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

500.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)

- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

500.9 Treatment Options

Treatment options include continued breastfeeding or expression, warm compresses, and antibiotics (dicloxacillin or cephalexin). Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

500.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

500.11 Outlook

Most people recover well with appropriate treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 501

McArdle Disease

501.1 Overview

McArdle disease (glycogen storage disease type V) is an autosomal recessive metabolic myopathy caused by deficiency of muscle glycogen phosphorylase, the enzyme that breaks down glycogen to glucose-1-phosphate in skeletal muscle. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

501.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

501.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

501.4 Signs and Symptoms

- You may experience exercise intolerance, muscle pain, and cramping during vigorous activity, often with a "second wind" phenomenon (improvement in symptoms after a

brief rest during exercise)
- Rhabdomyolysis may occur with severe exercise. This condition happens because of PYGM gene mutations leading to inability to mobilize glycogen stores during anaerobic metabolism.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

501.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

501.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

501.7 Diagnosis

- Doctors diagnose this based on elevated serum CK, ischemic forearm exercise test (failure of lactate to rise), and genetic testing (PYGM mutations)
- Muscle biopsy shows glycogen accumulation and absent phosphorylase activity.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

501.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

501.9 Treatment Options

- Treatment options include exercise conditioning, high-protein diet, and avoidance of strenuous exercise
- Oral sucrose before exercise may improve exercise capacity.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

501.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

501.11 Outlook

The outlook varies from person to person but many patients maintain functional independence with lifestyle modifications. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 502

MDMA (Ecstasy) Toxicity

502.1 Overview

MDMA (3,4-methylenedioxymethamphetamine) is a recreational empathogen with stimulant and hallucinogenic properties. It promotes serotonin release and inhibits reuptake. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

502.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

502.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

502.4 Signs and Symptoms

You may experience euphoria, increased energy, empathy, and sensory enhancement, but also hyperthermia (often severe), hyponatremia (from SIADH plus excessive water intake), seizures, rhabdomyolysis, DIC, acute liver failure, and serotonin syndrome. Hyponatremia can be severe (Na <120 mEq/L) and cause cerebral swelling. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

502.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

502.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

502.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

502.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

502.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

502.10 Living with It

Treatment: supportive care—cooling for hyperthermia, hypertonic saline for severe hyponatremia with nervous system symptoms, benzodiazepines for agitation and seizures, cyproheptadine for serotonin syndrome.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

502.11 Outlook

The outlook is generally positive with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 503

Measles (Rubeola)

503.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

503.2 Causes

This condition happens because of the virus entering through respiratory epithelium, replicating in regional lymph nodes, and causing viremia, spreading to the reticuloendothelial system, with infection causing profound lymphopenia and immunosuppression lasting weeks to months. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

503.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

503.4 Signs and Symptoms

You may experience an incubation of 7–14 days, with prodrome (2–4 days): fever, the three Cs—cough, coryza, conjunctivitis, and Koplik spots (pathognomonic: small, white, bluish-white papules on the buccal mucosa), followed by rash (day 3–5): erythematous, maculopapular, confluent rash beginning on the face and spreading cephalocaudally. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

503.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

503.6 Complications

Complications include pneumonia (most common cause of death), otitis media, laryngotracheobronchitis, encephalitis, and subacute sclerosing panencephalitis (SSPE).

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

503.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by RT-PCR or IgM serology. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

503.8 Prevention

- Measles is a highly contagious vaccine-preventable viral illness caused by measles virus, a paramyxovirus, with 140,000 deaths globally in 2018, and one of the most contagious infectious diseases with a basic reproduction number (R_0) of 12–18. Most people recover well with supportive care
- Prevention through MMR vaccine is highly effective.
- Hand hygiene with soap and water or alcohol-based sanitizer

- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

503.9 Treatment Options

Treatment typically involves supportive with vitamin A. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

503.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

503.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 504

Meconium Aspiration Syndrome

504.1 Overview

Meconium removal by suction syndrome (MAS) occurs when meconium-stained amniotic fluid is aspirated into the fetal airways before or during delivery, causing airway obstruction, pneumonitis, surfactant inactivation, and lung vasoconstriction. It occurs in 1–3% of all deliveries with meconium-stained fluid. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

504.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

504.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

504.4 Signs and Symptoms

You may experience respiratory distress at birth, tachypnea, grunting, retractions, cyanosis, and barrel-shaped chest. Chest radiograph shows coarse, patchy infiltrates, hyperinflation, and atelectasis.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

504.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

504.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

504.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

504.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

504.9 Treatment Options

Treatment options include immediate endotracheal suctioning of non-vigorous infants, supplemental oxygen, surfactant therapy, inhaled nitric oxide for lung high blood pressure, and ECMO for severe refractory cases. The routine practice of intrapartum suctioning for all meconium-stained infants is no longer recommended for vigorous infants. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

504.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

504.11 Outlook

The outlook is generally positive with appropriate management.

Chapter 505

Medication Non-Adherence

505.1 Overview

Medication non-adherence affects 40–60% of older adults, contributing to poor disease control, increased hospitalizations, and mortality. Barriers include thinking and memory impairment (forgetting doses), complex regimens (multiple pills, multiple daily doses), cost, transportation difficulties, poor provider-patient communication, lack of understanding of medication purpose, and side effects. Interventions include simplifying regimens (once-daily dosing, combination pills), medication organizers (blister packs, pillboxes), caregiver involvement, clear written instructions, technology-based reminders, and addressing polypharmacy through deprescribing. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

505.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

505.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

505.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

505.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

505.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

505.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

505.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

505.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

505.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

505.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 506

Medication Overuse Headache

506.1 Overview

Medication overuse headache (MOH), also called rebound headache, is a secondary headache disorder resulting from frequent or excessive use of acute headache medications. It is the most common cause of chronic daily headache, affecting 1–2% of the general population. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

506.2 Causes

This condition happens because of frequent use of acute headache medications, particularly combination analgesics, triptans, opioids, and barbiturates. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

506.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

506.4 Signs and Symptoms

You may experience headache on ≥15 days per month with a causal relationship to medication overuse. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

506.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

506.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

506.7 Diagnosis

To make the diagnosis, doctors look for ICHD-3 criteria. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

506.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

506.9 Treatment Options

Treatment focuses on withdrawal of the overused medication (abruptly for triptans and simple analgesics, gradually for opioids and barbiturates), bridging therapy during withdrawal, and concurrent initiation of preventive therapy. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

506.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

506.11 Outlook

- The outlook is generally positive with treatment
- Relapse rates are 30–40% within the first year.

Chapter 507

Medulloblastoma

507.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

507.2 Causes

This condition happens because of cancerous transformation of primitive neuroectodermal cells. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

507.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)

- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

507.4 Signs and Symptoms

You may experience headache, vomiting (especially morning), loss of coordination, and signs of hydrocephalus from fourth ventricle obstruction. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

507.5 Progression and Stages

Medulloblastoma is the most common cancerous brain tumor of childhood, arising in the cerebellum or posterior fossa, and belongs to the group of embryonal tumors classified into molecular subgroups (WNT, SHH, Group 3, Group 4) with distinct prognoses. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

507.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

507.7 Diagnosis

Diagnosis typically involves MRI showing a posterior fossa mass and cerebrospinal fluid cytology for staging. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies

- Genetic counseling and testing for hereditary cancer syndromes

507.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

507.9 Treatment Options

- Treatment options include maximal surgical removal, craniospinal irradiation, and adjuvant chemotherapy
- Risk-adapted therapy aims to reduce long-term neurocognitive and endocrine sequelae in survivors.
- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

507.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies

(acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

507.11 Outlook

Recovery varies by molecular subgroup, with the WNT subgroup having the best outlook. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 508

Melanoma

508.1 Overview

Melanoma is the most lethal skin cancer, arising from melanocytes. Although representing only 1% of skin cancers, it accounts for the majority of skin cancer deaths. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

508.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

- The ABCDE criteria aid in identification: Asymmetry, Border irregularity, Color variation, Diameter >6 mm, and Evolution. The four main subtypes are superficial spreading (most common), nodular (most aggressive, may not follow ABCDE), lentigo maligna (on chronically sun-damaged skin), and acral lentiginous (palms, soles, nail beds
- Most common melanoma in people of color). This condition happens because of cancerous transformation of melanocytes driven by UV-induced and genetic mutations (BRAF V600, NRAS, KIT).

508.3 Risk Factors

Risk factors include UV exposure (intermittent intense exposure and sunburns), fair skin, multiple nevi, atypical nevi, family history, and genetic susceptibility (CDKN2A, CDK4 mutations).

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

508.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

508.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for thin melanoma ($\leq$1 mm) is surgical removal with sentinel lymph node biopsy for those $\geq$0.8 mm or with ulceration
- Immunotherapy (anti-PD-1 [nivolumab, pembrolizumab], anti-CTLA-4 [ipilimumab], combination therapy) and targeted therapy (BRAF/MEK inhibitors for BRAF V600-mutated melanoma) have revolutionized treatment of advanced melanoma.

508.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

508.7 Diagnosis

- To make the diagnosis, doctors look for excisional biopsy with adequate margins
- Staging uses Breslow thickness, ulceration, mitotic rate, and lymph node status.
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

508.8 Prevention

- Your outlook depends on stage at diagnosis
- Early detection is critical.
- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

508.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
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- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

508.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team

- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

508.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 509

Melasma

509.1 Overview

Melasma is a common acquired hyperpigmentation disorder characterized by symmetric, irregular, brown-to-gray-brown macules and patches on sun-exposed areas, predominantly the cheeks, forehead, upper lip, and chin. It predominantly affects women with darker skin phototypes (Fitzpatrick III–VI). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

509.2 Causes

This condition happens because of UV-induced stimulation of melanocytes with increased melanin production and dermal deposition, exacerbated by sun exposure, pregnancy ("chloasma"), oral contraceptives, and hormonal therapies. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

509.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

509.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

509.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Doctors usually diagnose this based on your symptoms and a physical exam
- Wood's lamp examination helps classify depth (epidermal, dermal, mixed).

509.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

509.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

509.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

509.9 Treatment Options

Treatment options include strict sun protection (broad-spectrum sunscreen with iron oxide), topical depigmenting agents (hydroquinone 4%, tretinoin, azelaic acid, tranexamic acid), and chemical peels. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

509.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

509.11 Outlook

- The outlook varies from person to person
- Recurrence is common, and combination therapy yields the best results.

Chapter 510

Membranous Nephropathy

510.1 Overview

Membranous nephropathy (MN) is the most common cause of nephrotic syndrome in non-diabetic adults. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

510.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of subepithelial immune complex deposition and thickening of the glomerular basement membrane
- Approximately 70–80% of primary MN is caused by autoantibodies against the phospholipase A2 receptor (PLA2R) on podocytes. Secondary causes include malignancy (lung, GI), infections (hepatitis B), autoimmune diseases (lupus), and drugs.

510.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

510.4 Signs and Symptoms

You may experience nephrotic-range proteinuria, swelling, and hypoalbuminemia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

510.5 Progression and Stages

Treatment for primary MN includes conservative management (RAAS blockade) for mild disease and immunosuppression (rituximab, cyclophosphamide, calcineurin inhibitors) for moderate-to-severe or progressive disease. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

510.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

510.7 Diagnosis

Doctors diagnose this based on kidney biopsy with PLA2R antibody serology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

510.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

510.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

510.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

510.11 Outlook

- The outlook varies from person to person
- Anti-PLA2R antibody levels monitor disease activity and response to treatment.

Chapter 511

Meningioma

511.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

511.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of neoplastic growth of arachnoid cap cells. Most are asymptomatic and discovered incidentally on imaging
- Symptomatic meningiomas cause headaches, seizures, or focal deficits depending on location.

511.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

511.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance

or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

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- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

511.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Meningioma is the most common primary intracranial tumor, arising from arachnoid cap cells
- It is usually non-cancerous (WHO grade 1), slow-growing, and attached to the dura
- Meningiomas are more common in women and associated with prior radiation, neurofibromatosis type 2, and high progesterone receptor expression. Treatment focuses on observation for small, asymptomatic lesions
- Surgical removal is the mainstay for symptomatic or growing tumors
- Atypical (grade 2) and anaplastic (grade 3) meningiomas require adjuvant radiotherapy.

511.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

511.7 Diagnosis

Doctors diagnose this based on MRI revealing a well-circumscribed, homogeneously enhancing dural-based mass with a dural tail. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

511.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

511.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

511.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

511.11 Outlook

Most people recover well for non-cancerous meningiomas. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 512

Menopause / Perimenopause

512.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Menopause is the permanent cessation of menstruation for 12 consecutive months, occurring at a median age of 51 years. Perimenopause typically begins 4–8 years before menopause and is marked by variable cycle length, vasomotor symptoms (hot flashes, night sweats), sleep disruption, and mood changes. The Disease mechanism involves ovarian follicular depletion with declining inhibin B and anti-Müllerian hormone, followed by rising FSH and declining estradiol. Long-term consequences of estrogen deficiency include genitourinary syndrome of menopause, bone loss, and increased cardiovascular risk. Management: menopausal hormone therapy (MHT) with estrogen alone or estrogen plus progestin is the most effective treatment for vasomotor symptoms
- Non-hormonal options include SSRIs/SNRIs, gabapentin, and fezolinetant.

512.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

512.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

512.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

512.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

512.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

512.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

512.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

512.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

512.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

512.11 Outlook

- The outlook is generally positive
- The benefits and risks of MHT are related to age at initiation, formulation, and duration.

Chapter 513

Mercury Toxicity

513.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Mercury exists in three forms: elemental (metallic), inorganic (mercurous/mercuric salts), and organic (methylmercury). Elemental mercury: inhaled vapor is absorbed (80%)
- Causes acute pneumonitis, lung swelling, and chronic neurotoxicity (tremor, erethism—emotional lability, pathological shyness, memory loss). Inorganic mercury: ingestion of mercuric chloride causes corrosive gastroenteritis, acute tubular tissue death, and nephrotic syndrome. Acrodynia (Pink disease) is a hypersensitivity reaction in children. Organic mercury (methylmercury): bioaccumulates in fish
- Causes Minamata disease (severe neurological syndrome: paresthesias, loss of coordination, slurred speech, visual field constriction, hearing loss) and is a potent teratogen. Diagnosis: blood and urine mercury levels. Treatment: cessation of exposure
- Succimer (DMSA) is the preferred chelator for elemental and inorganic mercury.

513.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

513.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

513.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

513.5 Progression and Stages

Your outlook depends on form and severity of exposure. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

513.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

513.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

513.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

513.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

513.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

513.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 514

Mesothelioma

514.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

514.2 Causes

This condition happens because of asbestos fibers causing chronic inflammation and cancerous transformation of mesothelial cells. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

514.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

514.4 Signs and Symptoms

You may experience progressive shortness of breath, chest pain, weight loss, and pleural effusion. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

514.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Cancerous pleural mesothelioma is a rare, aggressive tumor arising from the mesothelial cells of the pleura, strongly associated with asbestos exposure (latency 20–40 years)
- It is classified into epithelioid (most common, better outlook), sarcomatoid, and biphasic subtypes. Treatment typically involves palliative in most cases: chemotherapy with pemetrexed and cisplatin is first-line
- Immunotherapy has shown benefit
- Surgery may be considered in early-stage disease.

514.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

514.7 Diagnosis

To make the diagnosis, doctors look for CT scan showing circumferential pleural thickening and pleural biopsy. Staging is by the TNM system. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

514.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

514.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

514.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

514.11 Outlook

Unfortunately, the outlook is often poor, with median survival of 12–18 months. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 515

Metabolic Syndrome

515.1 Overview

Metabolic syndrome is a cluster of interconnected metabolic abnormalities that increase the risk of cardiovascular disease, type 2 diabetes, and all-cause mortality. The NCEP ATP III diagnostic criteria require at least three of the following: central obesity (waist circumference greater than 102 cm in men, 88 cm in women), elevated triglycerides (150 mg/dL or greater), reduced HDL cholesterol (less than 40 mg/dL in men, less than 50 mg/dL in women), elevated blood pressure (130/85 mm Hg or greater), and elevated fasting glucose (100 mg/dL or greater). Central obesity and insulin resistance are considered core pathophysiologic features, resulting from adipose tissue dysfunction and ectopic fat deposition driving insulin resistance, endothelial dysfunction, and a prothrombotic/proinflammatory state. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

515.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

515.3 Risk Factors

Treatment focuses on lifestyle modification and pharmacologic treatment of individual risk factors.

- Family history of the genetic condition
- Consanguineous parents (for recessive disorders)
- Advanced parental age (for new mutations)
- Carrier status in parents
- Ethnic background (some conditions)
- Family history
- Obesity and sedentary lifestyle
- Poor dietary habits
- Advancing age
- Hormonal imbalances
- Certain medications
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

515.4 Signs and Symptoms

You may experience the cluster of metabolic abnormalities without specific symptoms.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

515.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

515.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life

- Emotional and psychological impact

515.7 Diagnosis

To make the diagnosis, doctors look for fulfillment of the diagnostic criteria.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

515.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

515.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

515.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

515.11 Outlook

- The outlook varies from person to person

- Metabolic syndrome confers a 2-fold increased risk of cardiovascular events and a 5-fold increased risk of developing type 2 diabetes.

Chapter 516

Metformin Overdose

516.1 Overview

Metformin overdose is an uncommon but highly lethal pharmaceutical overdose due to metformin-associated lactic acidosis (MALA). This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

516.2 Causes

This condition happens because of metformin inhibiting gluconeogenesis and complex I of the mitochondrial electron transport chain, leading to impaired lactate clearance and increased anaerobic metabolism, with predisposing factors including kidney impairment, liver dysfunction, hypoperfusion, sepsis, and concurrent use of contrast dye. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

516.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

516.4 Signs and Symptoms

You may experience nausea, vomiting, abdominal pain, diarrhea, profound lactic acidosis (pH <7.1, lactate >20 mmol/L), hypotension, hypothermia, altered mental status, and acute kidney injury. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

516.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

516.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

516.7 Diagnosis

Doctors diagnose this using elevated serum metformin level and severe lactic acidosis with elevated lactate-to-pyruvate ratio. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

516.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

516.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

516.10 Living with It

Treatment options include supportive care and early hemodialysis or continuous venovenous hemofiltration as The main treatment.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

516.11 Outlook

- Outlook is guarded
- Severe MALA has mortality exceeding 30–50%.

Chapter 517

Methemoglobinemia

517.1 Overview

Methemoglobinemia is a disorder characterized by increased levels of methemoglobin (ferric Hb, Fe^{3+}), which is unable to bind oxygen and causes functional anemia, with a normal methemoglobin level <1% of total hemoglobin. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

517.2 Causes

This condition happens because of either congenital deficiency of cytochrome b5 reductase (NADH-methemoglobin reductase, autosomal recessive, most common congenital cause) or exposure to oxidizing agents (nitrates, nitrites, dapsone, benzocaine, sulfonamides, aniline dyes). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

517.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

517.4 Signs and Symptoms

- Clinical features correlate with methemoglobin levels: 10–20% causes cyanosis without symptoms
- 20–50% causes headache, shortness of breath, tachycardia, and altered mental status
- >50% causes seizures, coma, irregular heartbeats, and death. A characteristic finding is chocolate-brown discoloration of blood that does not turn red when exposed to air, and pulse oximetry showing a saturation of approximately 85% that does not change with oxygen supplementation.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

517.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

517.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

517.7 Diagnosis

Your doctor confirms the diagnosis with co-oximetry (multi-wavelength blood gas analysis) showing elevated methemoglobin. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

517.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

517.9 Treatment Options

- Treatment options include supplemental oxygen, intravenous methylene blue (1–2 mg/kg over 5 minutes) for symptomatic or severe methemoglobinemia (>20–30%), which acts as an electron donor to reduce methemoglobin to hemoglobin. Methylene blue ineffective in G6PD deficiency (risk of hemolysis)
- Alternative is ascorbic acid or exchange transfusion.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

517.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

517.11 Outlook

- Most people recover well with treatment
- Untreated severe methemoglobinemia can be fatal.

Chapter 518

Methyl Parathion and Other Organophosphate Insecticides

518.1 Overview

Like other organophosphates, it irreversibly inhibits acetylcholinesterase, causing cholinergic crisis (SLUDGE syndrome plus miosis, bronchorrhea, bronchospasm, bradycardia, muscle fasciculations, weakness, seizures, coma). It is particularly dangerous because it is rapidly absorbed through skin. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

518.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

518.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

518.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

518.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

518.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

518.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam with low RBC/plasma cholinesterase activity (<50% of normal). Treatment: atropine (to block muscarinic symptoms), pralidoxime (2-PAM, to reactivate AChE), diazepam for seizures, and decontamination. outlook: recovery is usually complete if treated early
- Delayed neurotoxicity (organophosphate-induced delayed neuropathy/OPIDN) can occur weeks later.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

518.8 Prevention

Methyl parathion is a highly toxic organophosphate insecticide (WHO class Ia—extremely hazardous) used in agriculture.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

518.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

518.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

518.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 519

Microscopic Polyangiitis

519.1 Overview

Microscopic polyangiitis is an ANCA-associated necrotizing vasculitis affecting small vessels (capillaries, venules, arterioles), with a predilection for the kidneys and lungs, but without granuloma formation (distinguishing it from GPA), with an incidence of 2–6 per million, a slight male predominance, and typically presenting in the sixth to seventh decade. The condition involves p-ANCA (perinuclear) directed against myeloperoxidase (MPO) in approximately 60–70% of patients (PR3-ANCA in 20–30%), with ANCA-induced neutrophil activation leading to necrotizing inflammation of small vessels. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

519.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

519.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

519.4 Signs and Symptoms

You may experience rapidly progressive glomerulonephritis (crescentic, pauci-immune: the most common and serious manifestation, present in >80%), lung capillaritis causing diffuse alveolar bleeding (hemoptysis, shortness of breath, bilateral infiltrates), cutaneous purple spots on the skin, mononeuritis multiplex, arthralgias, and myalgias. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

519.5 Progression and Stages

Doctors diagnose this based on the 2022 ACR/EULAR classification criteria, which emphasize MPO-ANCA positivity, kidney involvement, and absence of granulomatous features. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

519.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

519.7 Diagnosis

Management follows the same principles as GPA: high-dose corticosteroids plus rituximab or cyclophosphamide for induction, followed by maintenance therapy with rituximab, azathioprine, or methotrexate. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

519.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

519.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

519.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

519.11 Outlook

outlook is similar to GPA, with 5-year survival exceeding 75%. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 520

Migraine

520.1 Overview

Migraine is a primary headache disorder affecting approximately 15% of the global population (1 billion people), with a female-to-male ratio of 3:1. It is the second leading cause of years lived with disability worldwide. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

520.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of cortical spreading depression (CSD)—a wave of neuronal depolarization followed by prolonged suppression of cortical activity—triggering the trigeminovascular system with release of calcitonin gene-related peptide (CGRP), causing vasodilation and neurogenic inflammation
- Genetic factors play a role, with familial hemiplegic migraine linked to mutations in CACNA1A, ATP1A2, and SCN1A.

520.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

520.4 Signs and Symptoms

You may experience recurrent, unilateral (60%), pulsating headache of moderate to severe intensity, lasting 4–72 hours, associated with nausea, vomiting, photophobia, and phonophobia

- Migraine with aura involves transient focal neurological symptoms preceding or accompanying the headache
- Chronic migraine is defined by headache on ≥15 days per month for >3 months.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

520.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

520.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

520.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam based on ICHD-3 criteria. Acute Treatment focuses on NSAIDs or triptans for mild to moderate attacks and triptans, lasmiditan, or gepants for moderate to severe attacks. Preventive therapy is indicated for ≥4 migraine days per month and includes beta-blockers, amitriptyline, topiramate, candesartan, onabotulinumtoxin A for chronic migraine, and CGRP monoclonal antibodies. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

520.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

520.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

520.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

520.11 Outlook

- The outlook varies from person to person
- Migraine is a chronic condition with episodic attacks that can be managed but not cured.

Chapter 521

Mild Cognitive Impairment

521.1 Overview

Mild thinking and memory impairment (MCI) is an intermediate state between normal thinking and memory aging and dementia, characterized by objective thinking and memory decline (typically memory—amnestic MCI—or other domains) that is greater than expected for age and education but does not significantly impair functional independence. Prevalence is 10–20% in adults aged 65 and older. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

521.2 Causes

Annual conversion rate to dementia is 5–15%, with amnestic MCI conferring highest risk for Alzheimer disease. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

521.3 Risk Factors

- This condition happens because of underlying neurodegenerative pathology, with the APOE epsilon-4 allele as a major genetic risk factor. Treatment options include cardiovascular risk factor control, physical exercise, thinking and memory stimulation, treatment of depression and sleep disorders, and avoidance of anticholinergic medications
- No pharmacotherapy is approved for MCI.
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

521.4 Signs and Symptoms

You may experience subjective thinking and memory complaints with objective deficits on testing. Proposed biomarkers include CSF amyloid-beta 42 (decreased), phosphorylated tau (increased), amyloid PET, and tau PET. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

521.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

521.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

521.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

521.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

521.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

521.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

521.11 Outlook

- The outlook varies from person to person
- Annual conversion rate to dementia is 5–15%.

Chapter 522

Minimal Change Disease

522.1 Overview

Minimal change disease (MCD) is the most common cause of nephrotic syndrome in children (70–90% of cases) and a significant cause in adults (10–15%). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

522.2 Causes

This condition happens because of podocyte injury with foot process effacement on electron microscopy, with normal light microscopy and immunofluorescence (hence "minimal change"). Most cases are idiopathic, though MCD can be associated with NSAIDs, lymphoma, and infections. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

522.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

522.4 Signs and Symptoms

You may experience sudden onset of nephrotic-range proteinuria (>3.5 g/day), swelling, hypoalbuminemia, hyperlipidemia, and lipiduria. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

522.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

522.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

522.7 Diagnosis

Doctors diagnose this based on kidney biopsy demonstrating podocyte foot process effacement on electron microscopy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

522.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

522.9 Treatment Options

- Treatment typically involves with corticosteroids (prednisone), to which 90% of patients respond (steroid-sensitive)
- Frequent relapses or steroid dependence may require second-line agents (calcineurin inhibitors, cyclophosphamide, rituximab).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

522.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

522.11 Outlook

- Most people recover well
- Most patients achieve remission with corticosteroids.

Chapter 523

Mitral Regurgitation

523.1 Overview

Mitral regurgitation (MR) is the retrograde flow of blood from the left ventricle into the left atrium during systole. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

523.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

523.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity

- Unhealthy diet (high in saturated fat, salt, processed foods)

523.4 Signs and Symptoms

- You may experience exertional shortness of breath, fatigue, orthopnea, and atrial fibrillation
- Physical examination reveals a high-pitched holosystolic (pansystolic) murmur at the apex radiating to the left axilla, S3 gallop, and displaced hyperdynamic apical impulse.
- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

523.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

- Primary (organic) MR results from intrinsic valve apparatus pathology, including mitral valve prolapse (myxomatous deterioration), infective endocarditis, or rheumatic heart disease
- Secondary (functional) MR results from left ventricular remodeling or dilation (ischemic or dilated cardiomyopathy) displacing papillary muscles.

523.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

523.7 Diagnosis

Your doctor confirms the diagnosis with transthoracic echocardiography assessing valve anatomy, regurgitant jet volume, and left ventricular function. Treatment for severe MR includes medical optimization (afterload reduction with ACE inhibitors/ARBs, diuretics) and timely surgical mitral valve repair or replacement prior to irreversible left ventricular dysfunction. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

523.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

523.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation

- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

523.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

523.11 Outlook

Your outlook depends on baseline LV ejection fraction. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 524

Mitral Stenosis

524.1 Overview

Mitral stenosis is a narrowing of the mitral valve orifice, most commonly caused by rheumatic heart disease, with a female predominance. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

524.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process.

- This condition happens because of progressive scarring and fusion of the mitral commissures, leading to elevated left atrial pressure, lung congestion, and eventually lung high blood pressure and right heart failure. The normal mitral valve area is 4–6 cm^2
- Symptoms typically appear when the area narrows to less than 2 cm^2.

524.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus

- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

524.4 Signs and Symptoms

You may experience shortness of breath, orthopnea, hemoptysis, and atrial fibrillation. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

524.5 Progression and Stages

Your outlook depends on the severity and success of intervention. Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

524.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

524.7 Diagnosis

Doctors diagnose this using echocardiography. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

524.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

524.9 Treatment Options

- Treatment options include through the skin mitral balloon valvulotomy for suitable anatomy, surgical commissurotomy, or mitral valve replacement
- Anticoagulation is indicated for atrial fibrillation or thromboembolism.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

524.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

524.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 525

Mitral Valve Prolapse

525.1 Overview

Mitral valve prolapse is the most common valvular abnormality, affecting 2–3% of the population, and involves billowing of one or both mitral valve leaflets into the left atrium during systole, with or without mitral regurgitation. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

525.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process.

- This condition happens because of structural abnormality of the mitral valve apparatus. Most patients are asymptomatic and discovered incidentally
- Clinical features may include palpitations, chest pain, and dysautonomic symptoms.

525.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol

- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

525.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

525.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

- Most cases are idiopathic or associated with myxomatous deterioration
- Mitral valve prolapse is linked to Marfan syndrome and Ehlers-Danlos syndrome.

525.6 Complications

- Most people recover well in most cases
- Complications include mitral regurgitation, infective endocarditis, and rarely, arrhythmias.
- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia

- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

525.7 Diagnosis

- Doctors diagnose this using echocardiography. Asymptomatic patients with minimal regurgitation require only reassurance
- Severe regurgitation may require surgical repair.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

525.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

525.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease

- Cardiac rehabilitation programs

525.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

525.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 526

Mixed Connective Tissue Disease

526.1 Overview

Mixed connective tissue disease is a distinct systemic autoimmune disorder characterized by overlapping Symptoms of systemic lupus erythematosus (SLE), systemic sclerosis (SSc), and polymyositis/dermatomyositis (PM/DM), in the presence of high-titer anti-U1 ribonucleoprotein (anti-RNP) antibodies, with a prevalence of approximately 1–2 per 100,000, a 9:1 female predominance, and typically presenting in the third to fourth decade. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

526.2 Causes

This condition happens because of autoantibodies against U1-RNP, leading to immune complex formation, complement activation, and tissue injury, with prominent endothelial and microvascular involvement. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

526.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

526.4 Signs and Symptoms

Clinical features evolve over time and commonly include Raynaud's phenomenon (>85%), puffy hands or sclerodactyly, inflammatory arthritis (non-erosive, resembling SLE), myositis (proximal muscle weakness, elevated creatine kinase), esophageal dysmotility, interstitial lung disease (up to 50%), lung high blood pressure (the leading cause of death), serositis, and a characteristic malar rash or discoid lupus-like lesions, with kidney involvement being less common and less severe than in SLE alone. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

526.5 Progression and Stages

Doctors diagnose this based on the presence of anti-U1-RNP antibodies (high-titer) plus clinical features from at least two of the three component diseases (SLE-like, SSc-like, PM/DM-like), with several classification criteria existing (Alarcón-Segovia, Kahn, Sharp). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

526.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

526.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

526.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

526.9 Treatment Options

Treatment typically involves determined by the dominant clinical manifestation: corticosteroids and disease-modifying antirheumatic drugs (DMARDs) for arthritis and serositis

- Immunosuppressants (mycophenolate, cyclophosphamide) or rituximab for interstitial lung disease
- Calcium channel blockers and endothelin receptor antagonists for Raynaud's phenomenon and lung high blood pressure.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

526.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

526.11 Outlook

The outlook varies from person to person, with lung high blood pressure being the leading cause of death. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 527

Molluscum Contagiosum

527.1 Overview

Molluscum contagiosum is a viral skin infection caused by a poxvirus, transmitted by direct contact (including sexual contact) or fomites. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

527.2 Causes

This condition happens because of poxvirus infection of epidermal keratinocytes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

527.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

527.4 Signs and Symptoms

- You may experience 2–5 mm, dome-shaped, flesh-colored papules with central umbilication ("dell"), often with a white waxy core
- Lesions may be solitary or numerous and can be spread by scratching (autoinoculation). In immunocompromised patients (HIV/AIDS), molluscum can be widespread, large, and refractory to treatment.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

527.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

527.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

527.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Dermoscopy shows a central pore with white amorphous material. Management options include freezing treatment, curettage, topical agents (cantharidin, imiquimod, potassium hydroxide), and mechanical extraction.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

527.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

527.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

527.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

527.11 Outlook

- Most people recover well
- Self-resolution occurs in immunocompetent individuals within months to years.

Chapter 528

Morton Neuroma

528.1 Overview

Morton neuroma is a non-cancerous perineural fibrotic enlargement of the interdigital plantar nerve, most commonly affecting the third intermetatarsal space between the third and fourth metatarsal heads. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

528.2 Causes

This condition happens because of chronic mechanical compression and repetitive irritation of the nerve beneath the deep transverse metatarsal ligament, leading to epineural and endoneural sclerosis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

528.3 Risk Factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

528.4 Signs and Symptoms

You may experience sharp, burning forefoot pain radiating to the toes, exacerbated by weight-bearing and narrow footwear, often described as 'walking on a marble.' Physical examination reveals a positive Mulder sign (palpable click with forefoot squeeze) and localized intermetatarsal tenderness. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

528.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

528.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

528.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by foot ultrasound or MRI if equivocal.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

528.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

528.9 Treatment Options

Treatment options include wide footwear, metatarsal pads, corticosteroid injections, and surgical neurectomy for recalcitrant symptoms. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

528.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

528.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 529

Motor Neuron Disease

529.1 Overview

Motor neuron disease (MND) is a group of progressive neurodegenerative disorders affecting upper motor neurons (UMNs), lower motor neurons (LMNs), or both, with a lifetime risk of 1 in 300. The spectrum includes amyotrophic lateral sclerosis (ALS, most common, both UMN and LMN), progressive bulbar palsy (PBP, predominantly bulbar onset), progressive muscular shrinkage or wasting (PMA, predominantly LMN), and primary lateral sclerosis (PLS, predominantly UMN). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

529.2 Causes

This condition happens because of TDP-43 pathology, glutamate excitotoxicity, oxidative stress, mitochondrial dysfunction, and impaired protein degradation, with genetic causes including C9orf72 hexanucleotide repeat expansion, SOD1, TARDBP, FUS, and TBK1 mutations. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

529.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

529.4 Signs and Symptoms

You may experience progressive muscle weakness, wasting, fasciculations, spasticity, hyperreflexia (UMN signs), and bulbar symptoms (slurred speech, difficulty swallowing). ALS presentations include limb-onset (70%), bulbar-onset (25%), and respiratory-onset (5%). thinking and memory impairment is present in 30–50% (frontotemporal dementia in 15%). Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

529.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

529.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

529.7 Diagnosis

Doctors diagnose this based on revised El Escorial criteria, electromyography showing widespread denervation, and exclusion of mimics.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

529.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

529.9 Treatment Options

Treatment focuses on multidisciplinary care, riluzole (extends survival by 2–3 months), edaravone (slows functional decline in early ALS), noninvasive ventilation for respiratory insufficiency, and symptomatic treatments (sialorrhea, spasticity, pain). Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

529.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

529.11 Outlook

outlook is uniformly fatal in ALS: median survival 3–5 years from symptom onset, with respiratory failure as the most common cause of death. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 530

Mpox

530.1 Overview

Mpox is a viral zoonosis caused by mpox virus, a double-stranded DNA orthopoxvirus closely related to variola virus, formerly called monkeypox, endemic in Central and West Africa, with the 2022 global outbreak involving over 90,000 cases across 110 countries. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

530.2 Causes

This condition happens because of the virus entering through broken skin, respiratory tract, or mucous membranes, replicating at the inoculation site, spreading to regional lymph nodes, and causing viremia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

530.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

530.4 Signs and Symptoms

You may experience an incubation of 5–21 days, with prodrome (fever, headache, muscle pain, lymphadenopathy—hallmark distinguishing mpox from smallpox) followed by rash appearing 1–4 days after fever, progressing from macules to papules, vesicles, pustules, and crusts over 2–4 weeks, with a centrifugal distribution, and anogenital lesions common in the 2022 outbreak. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

530.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

530.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

530.7 Diagnosis

Doctors diagnose this using PCR of lesion swabs. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

530.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

530.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

530.10 Living with It

Treatment options include tecovirimat (TPOXX) and supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

530.11 Outlook

- In most cases, the outlook is good
- Clade I (Central Africa) is more virulent than clade II (West Africa).

Chapter 531

Mucopolysaccharidoses

531.1 Overview

Mucopolysaccharidoses (MPS) are a group of lysosomal storage disorders caused by deficiency of enzymes required for the degradation of glycosaminoglycans (GAGs), leading to progressive accumulation in tissues. Hurler syndrome (MPS I-H, severe) is caused by α-L-iduronidase deficiency, presenting with coarse facies, corneal clouding, hepatosplenomegaly, dysostosis multiplex, cardiac valvular disease, and severe neurodegeneration. Hunter syndrome (MPS II, X-linked) is caused by iduronate-2-sulfatase deficiency, similar to MPS I but without corneal clouding. Sanfilippo syndrome (MPS III) is primarily neurodegenerative with severe behavioral disturbance, sleep disturbance, progressive intellectual decline, and seizures. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

531.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

531.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

531.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

531.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

531.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

531.7 Diagnosis

Doctors diagnose this using urinary GAG screening with fractionation, enzyme assay, and genetic testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

531.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

531.9 Treatment Options

Treatment options include hematopoietic stem cell transplantation for MPS I-H before age 2.5 years and enzyme replacement therapy for MPS I, II, and IVA. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

531.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

531.11 Outlook

- Outlook is guarded
- Enzyme replacement improves somatic features but does not cross the blood-brain barrier.

Chapter 532

Mucormycosis

532.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

532.2 Causes

This condition happens because of spores being inhaled (rhinocerebral, lung) or inoculated through skin, with the fungus being angioinvasive, causing blood clot formation, tissue tissue death due to lack of blood supply, and bleeding. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

532.3 Risk Factors

Mucormycosis is a rare, rapidly progressive, life-threatening fungal infection caused by molds of the order Mucorales (*Rhizopus*, *Mucor*, *Lichtheimia*, *Cunninghamella*), with incidence increasing with the rising number of immunocompromised patients, and risk factors including diabetic ketoacidosis (DKA), neutropenia, HSCT/SOT, iron overload, and corticosteroid use. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

532.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

532.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- You may experience rhinocerebral mucormycosis (most common, particularly in DKA): fever, facial or periorbital pain, nasal congestion, black necrotic eschar, proptosis, vision loss, and cranial nerve palsies
- Lung mucormycosis: fever, cough, hemoptysis, and rapid progression.

532.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

532.7 Diagnosis

Doctors diagnose this using tissue biopsy showing broad, ribbon-like, non-septate hyphae with right-angle branching. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

532.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

532.9 Treatment Options

Treatment options include urgent surgical removal of dead tissue and liposomal amphotericin B as first-line antifungal. outlook: rhinocerebral 50% mortality, lung 60%, disseminated 80–100%. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

532.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

532.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 533

Multiple Endocrine Neoplasia

533.1 Overview

Multiple endocrine neoplasia syndromes are autosomal dominant disorders characterized by tumors in two or more endocrine glands. MEN1 (Wermer syndrome) involves the parathyroids (primary hyperparathyroidism, most common), pancreas (gastrinoma, insulinoma, non-functional tumors), and pituitary (prolactinoma most common), caused by menin gene mutations. MEN2A (Sipple syndrome) involves medullary thyroid carcinoma (nearly 100% penetrance), pheochromocytoma, and primary hyperparathyroidism, caused by RET proto-oncogene mutations. MEN2B includes medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, marfanoid habitus, and intestinal ganglioneuromatosis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

533.2 Causes

This condition happens because of germline mutations in tumor suppressor genes or proto-oncogenes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

533.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

533.4 Signs and Symptoms

Your symptoms depend on the glands involved. Genetic testing guides prophylactic thyroidectomy in RET mutation carriers (within the first year of life for MEN2B, by age 5 for MEN2A). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

533.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

533.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

533.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

533.8 Prevention

Your outlook depends on early detection and intervention.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

533.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

533.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

533.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 534

Multiple Gestation Complications

534.1 Overview

Multiple gestation (twins 3%, higher order multiples less common) carries increased risks of preterm birth, IUGR, preeclampsia, gestational diabetes, and placenta previa. Monochorionic diamniotic (MCDA) twins share a single placenta with potential vascular anastomoses, predisposing to twin-twin transfusion syndrome (TTTS, 10–15% of MCDA twins). Without treatment, severe TTTS has 80–90% fetal mortality. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

534.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

534.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

534.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

534.5 Progression and Stages

TTTS your doctor will diagnose it by polyhydramnios (recipient twin) and oligohydramnios (donor twin), staged by the Quintero staging system (I–V). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

534.6 Complications

Zygosity and chorionicity determine specific complications. Selective intrauterine growth restriction and twin reversed arterial perfusion sequence are additional complications.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

534.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

534.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

534.9 Treatment Options

Treatment options include laser photocoagulation of communicating placental vessels or serial amnioreduction. Monoamniotic twins are at risk for cord entanglement and require close surveillance with planned cesarean delivery at 32–34 weeks. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

534.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

534.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 535

Multiple Myeloma

535.1 Overview

Multiple myeloma is a plasma cell tumor that accounts for 1% of all cancers and 13% of hematologic malignancies, with a median age at To diagnose 70 years. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

535.2 Causes

This condition happens because of clonal growth of cancerous plasma cells in the bone marrow, production of monoclonal immunoglobulin (M-protein), and end-organ damage summarized by the CRAB criteria: Calcium elevation (hypercalcemia), kidney insufficiency, Anemia, and Bone lesions (lytic lesions on imaging). Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

535.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity

Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

535.4 Signs and Symptoms

You may experience bone pain (vertebral compression fractures, pathologic fractures), fatigue (anemia), recurrent infections (immunoparesis—suppression of normal immunoglobulins), and kidney failure. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

535.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Risk stratification uses the International Staging System (ISS) and revised ISS

(R-ISS), with del(17p), t(4
- 14), and t(14
- 16) as high-risk cytogenetic abnormalities.

535.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

535.7 Diagnosis

To make the diagnosis, doctors look for ≥10% clonal bone marrow plasma cells plus evidence of end-organ damage (CRAB) or biomarkers of malignancy (bone marrow plasma cells ≥60%, free light chain ratio ≥100, or >1 focal lesion on MRI). Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
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- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

535.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

535.9 Treatment Options

Treatment typically involves risk-adapted and typically involves induction (bortezomib, lenalidomide, dexamethasone—VRd), followed by autologous stem cell transplantation for eligible patients, followed by maintenance therapy (lenalidomide). Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines

- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

535.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

535.11 Outlook

The outlook has improved to a median survival of 7–10 years with modern therapy, which includes novel agents such as proteasome inhibitors, immunomodulatory drugs, monoclonal antibodies (daratumumab), bispecific antibodies, and CAR-T cell therapy. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 536

Multiple Myeloma (Bone Manifestations)

536.1 Overview

Multiple myeloma is discussed in detail in Chapter ??. Bone disease is a hallmark feature, present in over 80% of patients at diagnosis. Myeloma cells produce factors (RANKL, MIP-1α) that activate osteoclasts, leading to lytic bone lesions without osteoblastic repair. The most common sites are the spine, skull, ribs, pelvis, and long bones. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

536.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

536.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

536.4 Signs and Symptoms

You may experience pathologic fractures, pain, vertebral compression fractures with height loss and kyphosis, and spinal cord compression. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
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- Persistent cough or hoarseness
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536.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer

prognosis.

536.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

536.7 Diagnosis

To diagnose bone disease involves skeletal survey (low sensitivity), CT (high sensitivity for lytic lesions), MRI (detects marrow infiltration), and PET-CT. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies

- Genetic counseling and testing for hereditary cancer syndromes

536.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
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- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

536.9 Treatment Options

Treatment options include bisphosphonates (zoledronic acid, pamidronate) or denosumab to reduce skeletal events, radiation for symptomatic lesions, kyphoplasty/vertebroplasty for compression fractures, and myeloma-directed therapy. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
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- Palliative care: Symptom management, pain control, quality of life support
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- Surveillance: Active monitoring after definitive treatment

536.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

536.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 537

Multiple Sclerosis

537.1 Overview

This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

537.2 Causes

This condition happens because of autoimmune-mediated demyelination. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

537.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

537.4 Signs and Symptoms

You may experience episodic neurological deficits such as optic neuritis, transverse myelitis, brainstem syndromes, and sensory or motor disturbances. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

537.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Multiple sclerosis is a chronic autoimmune demyelinating disease of the central nervous system characterized by inflammatory demyelination and axonal injury occurring at different times and locations
- The relapsing-remitting form (RRMS) is most common, while progressive forms (primary progressive, secondary progressive) involve gradual neurological deterioration.

537.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

537.7 Diagnosis

- Doctors diagnose this based on MRI revealing periventricular, juxtacortical, infratentorial, and spinal cord T2-hyperintense lesions, often with gadolinium enhancement in active plaques
- Cerebrospinal fluid may show oligoclonal bands.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

537.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

537.9 Treatment Options

- Treatment focuses on disease-modifying therapies including interferons, glatiramer acetate, natalizumab, fingolimod, ocrelizumab, and others
- Acute relapses are treated with high-dose corticosteroids or plasmapheresis.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

537.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

537.11 Outlook

Recovery varies widely depending on disease course and treatment response. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 538

Mumps

538.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

538.2 Causes

This condition happens because of the virus entering through respiratory epithelium, replicating in the upper respiratory tract and regional lymph nodes, and causing viremia, with tropism for glandular tissue and the CNS. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

538.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

538.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

538.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

538.6 Complications

You may experience an incubation of 16–18 days, with prodrome (fever, headache, muscle pain, malaise) followed by parotitis (bilateral in 70–90%): painful, tender swelling of the parotid glands, with complications including orchitis (post-pubertal males, 20–30%), oophoritis, meningitis, pancreatitis, and sensorineural hearing loss.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

538.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by RT-PCR of buccal swab or IgM serology. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

538.8 Prevention

- Mumps is an acute viral illness caused by mumps virus, a paramyxovirus, with 500,000 cases annually worldwide before widespread vaccination, now less common in vaccinated populations but with outbreaks in close-contact settings. Most people recover well
- Prevention through MMR vaccine is highly effective.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use

- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

538.9 Treatment Options

Treatment typically involves supportive. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

538.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

538.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 539

Myasthenia Gravis

539.1 Overview

Myasthenia gravis (MG) is a neuromuscular junction disorder caused by autoantibodies against components of the postsynaptic acetylcholine receptor (AChR) or, less commonly, muscle-specific kinase (MuSK) or lipoprotein-related protein 4 (LRP4). Prevalence is approximately 20 per 100,000. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

539.2 Causes

This condition happens because of impaired neuromuscular transmission due to antibody-mediated blockade or destruction of AChR. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

539.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes

- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

539.4 Signs and Symptoms

- You may experience fluctuating, fatigable weakness that worsens with activity and improves with rest
- Ocular muscles are most commonly affected first (ptosis, diplopia), followed by bulbar muscles (difficulty swallowing, slurred speech, facial weakness), and limb and respiratory muscles. Myasthenic crisis (acute respiratory failure) occurs in 15–20% of patients.
- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

539.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

539.6 Complications

In most cases, the outlook is good with treatment, though myasthenic crisis remains a life-threatening complication.

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy

- Lymphoma risk in some autoimmune conditions

539.7 Diagnosis

Doctors diagnose this based on AChR or MuSK antibody testing, edrophonium (Tensilon) test (transient improvement), repetitive nerve stimulation (decremental response), and single-fiber EMG (increased jitter). Thymoma is present in 10–15% of patients. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

539.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

539.9 Treatment Options

Treatment options include acetylcholinesterase inhibitors (pyridostigmine), immunosuppression (prednisone, azathioprine, mycophenolate), thymectomy, and targeted therapies (eculizumab, ravulizumab [complement inhibitors], efgartigimod, rozanolixizumab [FcRn inhibitors]). Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

539.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

539.11 Outlook

Autoimmune diseases are generally chronic conditions requiring long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 540

Myelodysplastic Syndromes

540.1 Overview

Myelodysplastic syndromes are a heterogeneous group of clonal hematopoietic stem cell disorders that predominantly affect the elderly (median age 70–75). The condition is marked by dysplastic and ineffective hematopoiesis, peripheral cytopenias, and a risk of transformation to acute myeloid leukemia (AML). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

540.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

540.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

540.4 Signs and Symptoms

You may experience symptoms of cytopenias: fatigue (anemia), infections (neutropenia), and bleeding (thrombocytopenia).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

540.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

540.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

540.7 Diagnosis

To make the diagnosis, doctors look for bone marrow biopsy showing dysplasia (≥10% of cells in one or more myeloid lineages), blast percentage (<20%—above 20% classifies as AML), and cytogenetic/molecular abnormalities (del(5q), del(7q), del(20q), SF3B1, TET2, ASXL1 mutations). Risk stratification uses the International Prognostic Scoring System (IPSS-R, now IPSS-M).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

540.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

540.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

540.10 Living with It

Treatment for low-risk MDS includes supportive care, erythropoiesis-stimulating agents ($\pm$ lenalidomide for del(5q) MDS), and luspatercept, while higher-risk MDS doctors typically treat it with hypomethylating agents (azacitidine, decitabine).

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

540.11 Outlook

Your outlook depends on risk category, with allogeneic stem cell transplantation being the only curative option, typically for patients <75 with a donor.

Chapter 541

Myelofibrosis

541.1 Overview

Myelofibrosis is a myeloproliferative tumor characterized by clonal megakaryocyte growth, bone marrow scarring (reticulin and/or collagen deposition), extramedullary hematopoiesis (splenomegaly, hepatomegaly), and progressive cytopenias, which may arise de novo (primary MF) or evolve from PV or ET (post-PV/post-ET MF). The condition is driven by JAK2, CALR, or MPL mutations in over 90% of cases, with ASXL1, EZH2, SRSF2 mutations associated with worse outlook. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

541.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

541.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

541.4 Signs and Symptoms

You may experience massive splenomegaly (early satiety, abdominal discomfort), constitutional symptoms (fatigue, weight loss, night sweats, fever), bone pain, and progressive anemia requiring transfusions. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

541.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

541.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

541.7 Diagnosis

To make the diagnosis, doctors look for bone marrow biopsy showing megakaryocyte growth and atypia, reticulin/collagen scarring, and exclusion of other causes. The Dynamic International Prognostic Scoring System (DIPSS) and molecular-enhanced versions (MIPSS70) guide treatment. Management options include ruxolitinib (JAK1/2 inhibitor, improves splenomegaly and symptoms), fedratinib (JAK2 inhibitor), momelotinib (JAK1/2 inhibitor with anemia benefit), and allogeneic stem cell transplantation (the only curative option). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

541.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

541.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

541.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

541.11 Outlook

The outlook varies from person to person depending on risk category. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 542

Myocardial Infarction

542.1 Overview

heart attack is irreversible myocardial tissue death resulting from prolonged reduced blood flow, most commonly caused by acute thrombotic occlusion of a coronary artery at the site of a ruptured or eroded atherosclerotic plaque. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

542.2 Causes

This condition happens because of complete or near-complete coronary artery occlusion. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

542.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus

- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

542.4 Signs and Symptoms

You may experience crushing chest pain, radiation to the left arm or jaw, diaphoresis, nausea, and shortness of breath. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

542.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

542.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

542.7 Diagnosis

Doctors diagnose this based on ECG (ST-elevation MI shows ST-segment elevation

- Non-ST-elevation MI presents without ST elevation but with elevated cardiac biomarkers) and elevated cardiac troponin. Treatment for STEMI requires emergent reperfusion therapy with primary through the skin coronary intervention within 90 minutes or fibrinolysis within 30 minutes of first medical contact
- Treatment options include dual antiplatelet therapy, anticoagulation, beta-blockers, ACE inhibitors, and statins.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

542.8 Prevention

The outlook has improved with timely reperfusion and long-term secondary prevention.

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

542.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications

- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

542.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

542.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 543

Myocarditis

543.1 Overview

Myocarditis is an inflammation of the myocardium, most commonly caused by viral infection (Coxsackievirus B, adenovirus, parvovirus B19, SARS-CoV-2), but also by autoimmune diseases, drugs (immune checkpoint inhibitors), and toxins. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

543.2 Causes

This condition happens because of direct viral injury or immune-mediated myocardial damage. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

543.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

543.4 Signs and Symptoms

Symptoms can range from mild chest pain and myalgias to fulminant heart failure, cardiogenic shock, and sudden death. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

543.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

543.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

543.7 Diagnosis

- Diagnosis typically involves elevated troponin, ECG changes (diffuse ST elevation), echocardiography showing regional or global wall motion abnormalities, and cardiac MRI with Lake Louise criteria
- Endomyocardial biopsy remains the gold standard for definitive diagnosis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

543.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

543.9 Treatment Options

- Treatment typically involves supportive, with guideline-directed medical therapy for heart failure
- Immunosuppression may be considered in autoimmune or giant cell myocarditis.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

543.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

543.11 Outlook

In most cases, the outlook is favorable, with most patients recovering, but a subset develops dilated cardiomyopathy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 544

Myofascial Pain Syndrome

544.1 Overview

Myofascial pain syndrome (MPS) is a chronic pain syndrome characterized by the presence of trigger points—hyperirritable, taut bands of skeletal muscle that are painful on palpation and produce referred pain. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

544.2 Causes

This condition happens because of dysfunctional motor endplates with excessive acetylcholine release, sustained sarcomere contraction, local reduced blood flow, and release of inflammatory mediators. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

544.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

544.4 Signs and Symptoms

- You may experience regional pain (not widespread like fibromyalgia), with active trigger points causing spontaneous pain and latent trigger points painful only with palpation
- Common sites include the trapezius, levator scapulae, masseter, and gluteal muscles.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

544.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

544.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

544.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

544.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

544.9 Treatment Options

- Treatment options include trigger point injections (dry needling or with local anesthetic), spray and stretch, physical therapy, posture correction, and ergonomic modifications
- Botulinum toxin injections may be used for refractory cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

544.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

544.11 Outlook

The outlook is generally positive with treatment.

Chapter 545

Myopia

545.1 Overview

Myopia is the most common refractive error worldwide, affecting approximately 30% of the global population and approaching 80–90% in some East Asian countries. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

545.2 Causes

This condition happens because of an axial length of the eye that is too long relative to the refractive power of the cornea and lens, causing light to focus in front of the retina. High myopia (sphere -6.00 diopters or greater) increases the risk of retinal detachment, choroidal neovascularization, glaucoma, and cataract. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

545.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

545.4 Signs and Symptoms

You may experience blurred distance vision with preserved near vision. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

545.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment focuses on correction with concave (minus) lenses, contact lenses, or refractive surgery (LASIK, PRK, ICL)
- Atropine eye drops and orthokeratology lenses slow myopia progression in children.

545.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

545.7 Diagnosis

Doctors diagnose this based on refraction testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

545.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

545.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

545.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

545.11 Outlook

Most people recover well with correction. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 546

Myotonic Dystrophy

546.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Myotonic dystrophy (DM) is the most common muscular dystrophy in adults, with an autosomal dominant inheritance pattern. Type 1 (DM1, Steinert disease) results from a CTG trinucleotide repeat expansion in the DMPK gene on chromosome 19
- Type 2 (DM2, proximal myotonic myopathy) results from a CCTG expansion in the CNBP gene.

546.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

546.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

546.4 Signs and Symptoms

Clinical features in DM1 include distal and facial muscle weakness, myotonia (delayed muscle relaxation after contraction), cataracts (posterior subcapsular "Christmas tree" cataracts), cardiac conduction abnormalities, endocrine dysfunction, and thinking and memory impairment. Genetic anticipation (progressively earlier onset and more severe disease in successive generations) occurs. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

546.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

546.6 Complications

The outlook varies from person to person, with progressive disability and reduced life expectancy due to cardiac and respiratory complications.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

546.7 Diagnosis

Doctors diagnose this based on genetic testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

546.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

546.9 Treatment Options

Treatment typically involves symptomatic: mexiletine for myotonia, cardiac pacemaker/defibrillator for conduction defects, and Treatment for endocrine and other systemic manifestations. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

546.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

546.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 547

Ménière's Disease

547.1 Overview

Ménière's disease is a chronic inner ear disorder characterized by episodic feeling of spinning, fluctuating sensorineural hearing loss (typically low-frequency), ringing in the ears, and aural fullness. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

547.2 Causes

This condition happens because of endolymphatic hydrops, although the exact pathogenesis remains debated. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

547.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

547.4 Signs and Symptoms

You may experience feeling of spinning episodes lasting minutes to hours that may be incapacitating, associated with nausea and vomiting, along with fluctuating hearing loss, ringing in the ears, and aural fullness. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

547.5 Progression and Stages

Onset is typically unilateral, with progression to bilateral disease in 10–30% of cases over decades. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

547.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

547.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by audiometry, vestibular testing, and MRI to exclude other causes. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

547.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

547.9 Treatment Options

Treatment options include dietary modification (low sodium, caffeine restriction), betahistine, diuretics, intratympanic gentamicin or corticosteroid for refractory cases, and labyrinthectomy or vestibular nerve section as a last resort. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

547.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

547.11 Outlook

The outlook varies from person to person, with most patients achieving symptom control with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 548

Narcissistic Personality Disorder

548.1 Overview

Narcissistic personality disorder has a prevalence of 0.5–1% in the general population, with a male predominance (60–75%). The condition may involve exaggerated self-enhancement to compensate for underlying fragile self-esteem, deficits in mentalization, and altered frontolimbic connectivity. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

548.2 Causes

Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

548.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

548.4 Signs and Symptoms

You may experience a pervasive pattern of grandiosity (in fantasy or behavior), need for admiration, and lack of empathy, with required criteria including at least five of the following: grandiose sense of self-importance, preoccupation with fantasies of unlimited success, belief of being special and unique, need for excessive admiration, sense of entitlement, interpersonal exploitativeness, lack of empathy, envy of others, and arrogant behaviors or attitudes. Vulnerable (covert) narcissism is a subtype characterized by hypersensitivity to evaluation and chronic feelings of shame.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

548.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

548.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

548.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

548.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

548.9 Treatment Options

Treatment options include psychotherapy (transfer-focused, mentalization-based, schema therapy). Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

548.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

548.11 Outlook

- The outlook varies from person to person
- Grandiosity may diminish with age.

Chapter 549

Narcolepsy Type 1

549.1 Overview

Narcolepsy type 1 (narcolepsy with cataplexy) is a chronic neurological disorder characterized by excessive daytime sleepiness, cataplexy (sudden, transient episodes of bilateral skeletal muscle weakness triggered by strong emotions), sleep paralysis, hypnagogic hallucinations, and fragmented nighttime sleep. The prevalence is approximately 25–50 per 100,000 individuals, with onset peaking in adolescence and young adulthood. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

549.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of autoimmune-mediated destruction of hypocretin (orexin)-producing neurons in the lateral hypothalamus, leading to severely decreased or absent CSF hypocretin-1 levels
- Genetic susceptibility is strongly linked to HLA-DQB1*06:02 (present in >90% of cases).

549.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

549.4 Signs and Symptoms

You may experience excessive daytime sleepiness, cataplexy (brief episodes of bilateral weakness provoked by laughter or surprise), sleep paralysis, and hypnagogic hallucinations. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

549.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

549.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

549.7 Diagnosis

To make the diagnosis, doctors look for polysomnography followed by MSLT demonstrating mean sleep latency $\leq$8 minutes and two or more SOREM naps, plus either cataplexy or CSF hypocretin-1 deficiency. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

549.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

549.9 Treatment Options

Treatment options include scheduled naps, wake-promoting agents (modafinil, armodafinil, pitolisant, solriamfetol), anti-cataplexy agents (sodium oxybate, antidepressants), and sodium oxybate for both EDS and cataplexy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

549.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

549.11 Outlook

outlook is chronic but symptoms can be controlled with appropriate therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 550

Narcolepsy Type 2

550.1 Overview

Narcolepsy type 2 is a chronic neurological disorder characterized by excessive daytime sleepiness and objective findings on MSLT (mean sleep latency $\leq$8 minutes with two or more SOREMs) but without cataplexy and with normal CSF hypocretin-1 levels. It represents 15–25% of narcolepsy cases. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

550.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

550.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

550.4 Signs and Symptoms

- Clinical features are generally less severe than type 1

- Some patients may eventually develop cataplexy and transition to type 1.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

550.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

550.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

550.7 Diagnosis

To make the diagnosis, doctors look for polysomnography followed by MSLT demonstrating the characteristic findings without cataplexy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

550.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

550.9 Treatment Options

Treatment focuses on wake-promoting agents (modafinil, armodafinil, pitolisant, solriamfetol) and behavioral sleep hygiene. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

550.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

550.11 Outlook

outlook is chronic but manageable. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 551

Nasal Fracture

551.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Nasal fractures are the most common facial fracture, accounting for approximately 40% of all facial fractures, due to the prominent position of the nose
- They result from direct trauma (assault, sports injuries, falls).

551.2 Causes

This condition happens because of direct impact causing bony or cartilaginous disruption. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

551.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

551.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

551.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

551.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

551.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supplemented by CT for complex fractures. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

551.8 Prevention

- You may experience epistaxis, nasal pain, swelling, bruising, and deformity

- Nasal septal hematoma may coexist and requires urgent removal of fluid to prevent septal tissue death and saddle-nose deformity.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

551.9 Treatment Options

- Treatment focuses on closed reduction within 7–10 days of injury
- Severe or comminuted fractures may require open reduction and internal fixation.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

551.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

551.11 Outlook

The outlook is good with timely treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 552

Nasal Polyps

552.1 Overview

Nasal polyps are soft, pale, edematous masses of inflamed sinonasal mucosa that most commonly arise from the ethmoid sinuses and middle meatus, associated with type 2 inflammation, chronic rhinosinusitis, asthma, aspirin-exacerbated respiratory disease, and cystic scarring. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

552.2 Causes

This condition happens because of eosinophilic, IgE-mediated mucosal inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

552.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

552.4 Signs and Symptoms

- You may experience nasal obstruction, loss of smell or hyposmia, rhinorrhea, and postnasal drip
- Large or multiple polyps can cause facial deformity.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

552.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

552.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

552.7 Diagnosis

Doctors diagnose this using anterior rhinoscopy or nasal endoscopy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

552.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

552.9 Treatment Options

Treatment options include intranasal corticosteroids as first-line therapy, short courses of oral corticosteroids, surgical polypectomy, and dupilumab (anti-IL-4Rα monoclonal antibody) for refractory disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

552.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

552.11 Outlook

The outlook varies from person to person, with recurrence being common. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 553

Necrotizing Enterocolitis (NEC)

553.1 Overview

Necrotizing enterocolitis (NEC) is a devastating intestinal inflammatory disease of premature infants, with an incidence of 5–10% in very low birth weight infants. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

553.2 Causes

This condition happens because of a dysregulated inflammatory response to intestinal microbes leading to transmural tissue death, most commonly affecting the terminal ileum and proximal colon. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

553.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes

- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

553.4 Signs and Symptoms

You may experience abdominal distension, feeding intolerance, bloody stools, lethargy, and temperature instability. Pneumatosis intestinalis (gas in the bowel wall) on abdominal radiograph is a hallmark sign. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

553.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

- Treatment options include bowel rest, nasogastric decompression, broad-spectrum antibiotics, and supportive care
- Surgical intervention is required for perforation or progressive disease.

553.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation

- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

553.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with radiographic confirmation. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

553.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

553.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

553.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

553.11 Outlook

- Outlook is guarded
- Mortality ranges from 20–50%.

Chapter 554

Necrotizing Fasciitis

554.1 Overview

Necrotizing fasciitis is a rapidly progressive, life-threatening infection of the subcutaneous tissue and fascia, with secondary involvement of skin and muscle, with 500–1500 cases annually in the US, higher incidence in immunocompromised patients, diabetics, intravenous drug users, and after trauma or surgery, with mortality of 20–40%. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

554.2 Causes

This condition happens because of infection spreading along the superficial and deep fascia, causing thrombotic occlusion of perforating blood vessels, leading to ischemic tissue death, with Type I (polymicrobial, 70%) involving anaerobes and facultative organisms, Type II (monomicrobial, 20%) involving *Streptococcus pyogenes* and *Staphylococcus aureus*, and Type III (gram-negative, marine-related) involving *Vibrio vulnificus*. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

554.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

554.4 Signs and Symptoms

You may experience pain out of proportion to examination (earliest sign), rapidly spreading redness of the skin and swelling, induration, skin discoloration, involving bleeding bullae, crepitus, and systemic toxicity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

554.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

554.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

554.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, with surgical exploration being paramount. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

554.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

554.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

554.10 Living with It

Management requires urgent, aggressive surgical removal of dead tissue, empirical broad-spectrum antibiotics, and intensive supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

554.11 Outlook

outlook is guarded, with mortality heavily dependent on prompt surgical intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 555

Neonatal Abstinence Syndrome

555.1 Overview

Neonatal abstinence syndrome (NAS) is a postnatal drug withdrawal syndrome in newborns exposed to opioids or other substances in utero. Incidence has risen dramatically with the opioid epidemic. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

555.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of chronic fetal exposure to opioids downregulating CNS opioid receptors
- Abrupt cessation at birth causes neurotransmitter dysregulation and autonomic dysfunction.

555.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

555.4 Signs and Symptoms

Clinical features develop within 24–72 hours of birth and include CNS irritability (high-pitched cry, tremors, hypertonia, seizures), autonomic dysfunction (sweating, fever, yawning, sneezing, mottling), and GI dysfunction (poor feeding, vomiting, diarrhea).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

555.5 Progression and Stages

Diagnosis relies on maternal history with the Finnegan Neonatal Abstinence Scoring System quantifying severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

555.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

555.7 Diagnosis

- Treatment: non-pharmacologic care (rooming-in, breastfeeding if mother is on MAT, swaddling, low-stimulation environment) is first-line
- Pharmacologic therapy includes oral morphine, methadone, or buprenorphine.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

555.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

555.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

555.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

555.11 Outlook

- The outlook is generally positive with treatment
- Long-term outcomes include increased risk of developmental delays.

Chapter 556

Neonatal Hypoglycemia

556.1 Overview

Neonatal hypoglycemia is defined as plasma glucose <45–50 mg/dL in the first 48 hours of life, though the precise threshold associated with neurodevelopmental impairment remains debated. It occurs in 5–15% of all newborns and up to 50% of infants of diabetic mothers. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

556.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

556.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

556.4 Signs and Symptoms

Clinical features are nonspecific: jitteriness, irritability, lethargy, seizures, apnea, poor feeding, and hypothermia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

556.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

556.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

556.7 Diagnosis

To make the diagnosis, doctors look for timely glucose measurement. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

556.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

556.9 Treatment Options

- Treatment options include early feeding, intravenous dextrose, and monitoring of glucose levels
- Persistent or severe hypoglycemia requires evaluation for hyperinsulinism, fatty acid oxidation disorders, and other metabolic diseases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

556.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

556.11 Outlook

The outlook is generally positive with timely treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 557

Neonatal Jaundice / Hyperbilirubinemia / Kernicterus

557.1 Overview

Neonatal jaundice (hyperbilirubinemia) is the yellow discoloration of the skin and sclerae due to elevated serum bilirubin, occurring in 60–80% of term and nearly all preterm newborns. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

557.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Physiologic jaundice results from increased bilirubin production, immature liver conjugation, and decreased bilirubin excretion. Pathologic jaundice is defined by onset <24 hours of age, rapid rise (>5 mg/dL/day), elevated direct bilirubin, or persistence beyond 2 weeks
- Causes include hemolytic disease, sepsis, Crigler-Najjar syndrome, Gilbert syndrome, biliary atresia, and breast milk jaundice. Unconjugated bilirubin is neurotoxic
- At high levels it crosses the blood-brain barrier and causes bilirubin-induced nervous system dysfunction and kernicterus. Diagnosis includes total and direct serum bilirubin, blood type and Coombs test, G6PD screening, and evaluation for sepsis.

557.3 Risk Factors

- The right treatment depends on gestational age, risk factors, and bilirubin level: phototherapy is first-line
- Exchange transfusion for severe cases or signs of acute bilirubin encephalopathy.
- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

557.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

557.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

557.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

557.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

557.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

557.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

557.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

557.11 Outlook

The outlook is generally positive with timely treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 558

Neonatal Sepsis (Early-Onset and Late-Onset)

558.1 Overview

Neonatal sepsis is a systemic bacterial infection in the first 28 days of life. Early-onset sepsis ($<$72 hours of age) results from vertical transmission from the maternal genital tract, most commonly Group B Streptococcus and Escherichia coli. Late-onset sepsis ($>$72 hours of age) is nosocomial or community-acquired, with Coagulase-negative staphylococci, S. aureus, E. coli, and Klebsiella being common pathogens. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

558.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

558.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

558.4 Signs and Symptoms

Clinical features are nonspecific: temperature instability, respiratory distress, apnea, bradycardia, feeding intolerance, lethargy, irritability, and seizures. Diagnosis includes blood culture (gold standard), complete blood count, C-reactive protein, and lumbar puncture. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

558.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

558.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

558.7 Diagnosis

Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

558.8 Prevention

- Treatment options include empiric broad-spectrum antibiotics and supportive care
- Preventative intrapartum antibiotic prophylaxis for GBS-colonized women has significantly reduced early-onset sepsis incidence.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures

- Travel precautions and pre-travel consultations

558.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

558.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

558.11 Outlook

Your outlook depends on prompt recognition and treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 559

Nephritic Syndrome

559.1 Overview

Nephritic syndrome is a clinical complex resulting from acute glomerular inflammation characterized by leukocyte infiltration, cellular growth, and reactive glomerular capillary wall injury. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

559.2 Causes

This condition happens because of immune complex deposition or antibody binding within the glomerulus, causing a decrease in glomerular filtration rate (GFR) and breakdown of the capillary barrier to red blood cells and protein. Etiologies include poststreptococcal glomerulonephritis (PSGN), IgA nephropathy (Berger disease), lupus nephritis, and anti-GBM disease (Goodpasture syndrome). Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

559.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

559.4 Signs and Symptoms

You may experience the classic triad of hematuria ('cola-colored' or dysmorphic RBCs and RBC casts), high blood pressure (due to fluid retention), and oliguria with mild-to-moderate proteinuria (< 3.5 g/day) and facial/periorbital swelling. Diagnosis is supported by urinalysis showing hematuria and proteinuria, elevated serum creatinine, low serum complement levels (in PSGN and lupus), and definitive kidney biopsy.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

559.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

559.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

559.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

559.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

559.9 Treatment Options

Treatment focuses on fluid and sodium restriction, antihypertensive therapy (diuretics, ACE inhibitors), and targeted immunosuppressants (corticosteroids, cyclophosphamide) for rapidly progressive cases. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

559.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

559.11 Outlook

Your outlook depends on underlying histology.

Chapter 560

Nephrolithiasis

560.1 Overview

Nephrolithiasis (kidney stones) affects approximately 10% of the population, with higher prevalence in males and hot climates. Stone types include calcium oxalate (most common, 70–80%), calcium phosphate, uric acid (associated with hyperuricemia, low urine pH, and metabolic syndrome), struvite (magnesium ammonium phosphate, associated with urease-producing organisms in chronic UTI—"staghorn calculi"), and cystine (rare, hereditary cystinuria). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

560.2 Causes

This condition happens because of supersaturation of stone-forming salts in urine. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

560.3 Risk Factors

Risk factors include low fluid intake, high dietary sodium and animal protein, hypercalciuria, hyperoxaluria, hypocitraturia, and hyperuricosuria. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

560.4 Signs and Symptoms

You may experience kidney colic (severe, colicky flank pain radiating to the groin), hematuria, nausea, and urinary urgency. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

560.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

560.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

560.7 Diagnosis

Doctors diagnose this based on CT abdomen/pelvis without contrast (gold standard for acute evaluation). Treatment for acute kidney colic includes NSAIDs, opioids, antiemetics, and alpha-blockers (medical expulsive therapy). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

560.8 Prevention

- Prevention includes increased fluid intake (>2.5 L/day), dietary modifications, and pharmacotherapy (thiazide diuretics for hypercalciuria, allopurinol for uric acid stones, potassium citrate for hypocitraturia). The outlook is good with preventive measures
- Recurrence rates are high without prophylaxis.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

560.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

560.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

560.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 561

Nephrotic Syndrome

561.1 Overview

Nephrotic syndrome is a clinical complex resulting from severe glomerular podocyte effacement and disruption of the charge and size barrier of the glomerular basement membrane. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

561.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of primary kidney podocytopathies or systemic disease damage to the filtration barrier. Primary etiologies include minimal change disease (most common in children), focal segmental glomerulosclerosis (FSGS, most common in African Americans), and membranous nephropathy (most common in Caucasian adults)
- Secondary etiologies include diabetic nephropathy, amyloidosis, and lupus nephritis.

561.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

561.4 Signs and Symptoms

You may experience the hallmark tetrad: heavy proteinuria ($>$ 3.5 g/24 hours or spot urine protein-to-creatinine ratio $>$ 3.5), hypoalbuminemia ($<$ 3.0 g/dL), generalized peripheral and periorbital swelling (anasarca due to decreased oncotic pressure and secondary sodium retention), and hyperlipidemia with lipiduria ('oval fat bodies' and Maltese cross appearance under polarized light).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

561.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

561.6 Complications

Complications include deep vein blood clot formation/lung blockage by a blood clot (due to urinary loss of antithrombin III) and infection (loss of immunoglobulins).

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

561.7 Diagnosis

Doctors diagnose this using 24-hour urine collection or spot ratio, serum chemistries, and kidney biopsy.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

561.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

561.9 Treatment Options

Treatment options include ACE inhibitors/ARBs (to reduce intraglomerular pressure), loop diuretics, statins, anticoagulation when indicated, and systemic corticosteroids or immunosuppressive agents. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

561.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

561.11 Outlook

Recovery varies by underlying cause.

Chapter 562

Nesidioblastosis and Congenital Hyperinsulinism

562.1 Overview

Nesidioblastosis (persistent hyperinsulinemic hypoglycemia of infancy) is a rare disorder of pancreatic endocrine cell dysplasia characterized by diffuse enlargement, abnormal increase in cells, and neoformation of insulin-producing pancreatic β-cells from ductal epithelium. Driven by inactivating mutations in ABCC8 or KCNJ11 genes encoding the ATP-sensitive potassium channel (K_{ATP}) in β-cells, the condition leads to unregulated, continuous insulin secretion unresponsive to blood glucose levels. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

562.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

562.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

562.4 Signs and Symptoms

Clinical features present in neonates or infants with severe, intractable hypoglycemia, lethargy, seizures, and risk of irreversible neurodevelopmental damage.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

562.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

562.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

562.7 Diagnosis

Doctors confirm the diagnosis through persistent endogenous hyperinsulinemia, elevated C-peptide, and negative sulfonylurea screening during hypoglycemia.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

562.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

562.9 Treatment Options

Treatment focuses on IV glucose, diazoxide (K_{ATP} channel opener), octreotide, and subtotal or near-total (95%) pancreatectomy for medical treatment failure. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

562.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 563

Neural Tube Defects

563.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Incidence is 0.5–1 per 1000 pregnancies
- Risk is reduced by 50–70% with periconceptional folic acid supplementation. Spina bifida encompasses a spectrum from spina bifida occulta (vertebral involvement only), meningocele (meningeal herniation), to meningomyelocele (herniation of meninges and spinal cord tissue causing nervous system deficits). Anencephaly is a fatal NTD with absence of the cerebral hemispheres.

563.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

563.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

563.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

563.5 Progression and Stages

Neural tube defects (NTDs) are congenital malformations resulting from failure of neural tube closure during the fourth week of embryonic development, classified as open or closed. Your outlook depends on the level and severity of the lesion. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

563.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

563.7 Diagnosis

Doctors diagnose this using prenatal ultrasound and maternal serum alpha-fetoprotein screening. Treatment for meningomyelocele includes prenatal or postnatal surgical repair and multidisciplinary care. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

563.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

563.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

563.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

563.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 564

Neuroblastoma

564.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Neuroblastoma is the most common extracranial solid tumor in children and the most common malignancy in infants. It arises from the sympathetic nervous system, most commonly in the adrenal medulla (40%) or paraspinal sympathetic chain. Incidence is 1 in 7000 children
- Median age at diagnosis is 18 months.

564.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

564.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes

- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

564.4 Signs and Symptoms

Your symptoms depend on the site: abdominal mass (adrenal), Horner syndrome (cervical), opsoclonus-myoclonus-loss of coordination (paraneoplastic), proptosis and periorbital ecchymoses (orbital metastases), and bone pain (skeletal metastases). Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

564.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

564.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

564.7 Diagnosis

Doctors diagnose this using biopsy with histology, MYCN amplification status, and imaging. International Neuroblastoma Risk Group staging classifies tumors as low-risk, intermediate-risk, or high-risk. Treatment varies by risk group: low-risk tumors may be observed or surgically removed

- Intermediate-risk tumors receive chemotherapy
- High-risk tumors require intensive multimodal therapy. outlook ranges from excellent (>95% survival in low-risk) to poor (40–50% survival in high-risk).
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

564.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

564.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

564.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications

- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

564.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 565

Neurocysticercosis

565.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

565.2 Causes

This condition happens because of encystment of larval forms in brain parenchyma, ventricles, or subarachnoid space. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

565.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

565.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

565.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Neurocysticercosis is the most common parasitic infection of the central nervous system, caused by the larval stage of *Taenia solium* (pork tapeworm)
- It is endemic in Latin America, Asia, and Africa. Clinical manifestations depend on cyst location and stage and include seizures (the most common presentation worldwide), headache, focal deficits, and hydrocephalus.

565.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

565.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

565.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

565.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

565.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

565.11 Outlook

In most cases, the outlook is favorable with appropriate treatment. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 566

Neurofibromatosis Type 1

566.1 Overview

Neurofibromatosis type 1 (NF1, von Recklinghausen disease) is an autosomal dominant neurocutaneous disorder caused by mutations in the NF1 gene encoding neurofibromin, a Ras-GTPase tumor suppressor, affecting 1 in 3,000 births. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

566.2 Causes

This condition happens because of loss of neurofibromin function, leading to uncontrolled Ras signaling and increased cell growth. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

566.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

566.4 Signs and Symptoms

You may experience café-au-lait spots ($\geq$ 6 spots > 5 mm in children, > 15 mm in adults), axillary and inguinal freckling, Lisch nodules (iris hamartomas), neurofibromas (cutaneous, subcutaneous, plexiform), optic pathway gliomas, osseous dysplasia, and increased risk of cancerous peripheral nerve sheath tumors. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

566.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

566.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

566.7 Diagnosis

Doctors diagnose this based on NIH consensus criteria ($\geq$ 2 of 7 features).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

566.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

566.9 Treatment Options

Treatment focuses on annual ophthalmologic and nervous system monitoring, surgical removal of symptomatic neurofibromas, MEK inhibitor therapy (selumetinib) for symptomatic plexiform neurofibromas, and surveillance for malignancies. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

566.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

566.11 Outlook

- The outlook varies from person to person
- Life expectancy is reduced by 10–15 years, primarily due to malignancy and cardio-

vascular disease.

Chapter 567

New Daily Persistent Headache

567.1 Overview

New daily persistent headache (NDPH) is a primary headache disorder characterized by a sudden-onset headache that becomes continuous within 24 hours and persists for >3 months. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

567.2 Causes

This condition happens because of unclear Disease mechanism, with triggers including a viral or flu-like illness (40–50%), stressful life event, surgery, or travel. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

567.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

567.4 Signs and Symptoms

- You may experience bilateral, pressing/tightening, moderate intensity pain
- The onset is clearly remembered by the patient ("a thunderclap headache that never went away").
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

567.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

567.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

567.7 Diagnosis

To make the diagnosis, doctors look for rigorous exclusion of secondary causes (subarachnoid bleeding, cerebral venous sinus blood clot formation, meningitis, intracranial hypotension). Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

567.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

567.9 Treatment Options

Treatment typically involves challenging, involving a multidisciplinary approach with amitriptyline, gabapentin, topiramate, and physical therapy. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

567.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

567.11 Outlook

- The outlook varies from person to person
- The self-limiting subtype typically resolves within 2 years, while the refractory form

causes significant disability.

Chapter 568

Nociception

568.1 Overview

Ion channels including TRPV1 (heat, capsaicin), TRPM8 (cold, menthol), TRPA1 (irritants), ASIC (acid), P2X (ATP), and voltage-gated sodium channels (NaV1.7, NaV1.8) mediate transduction.

Transmission: The electrical signal propagates along primary afferent neurons to the dorsal horn of the spinal cord. Fast, sharp pain is carried by thinly myelinated A-delta fibers (5–30 m/s). Slow, dull, burning pain is carried by unmyelinated C fibers (0.5–2 m/s). Second-order neurons cross the midline and ascend via the spinothalamic tract to the thalamus. Third-order neurons project to the somatosensory cortex (sensory-discriminative component) and limbic system (affective-motivational component).

Modulation: Descending pathways from the periaqueductal gray (PAG), rostral ventromedial medulla (RVM), and locus coeruleus modulate nociceptive transmission at the dorsal horn via endogenous opioids (endorphins, enkephalins, dynorphins), serotonin, and norepinephrine. This system is the basis for the analgesic effects of opioids, TCAs, and SNRIs.

Perception: The conscious experience of pain, influenced by attention, emotion, past experience, and context. This explains the variability in pain experience across individuals and situations. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

568.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

568.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

568.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

568.5 Progression and Stages

Nociception is the neural process of encoding noxious stimuli and involves four phases: The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

568.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

568.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

568.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

568.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

568.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

568.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can

be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 569

Non-24-Hour Sleep-Wake Rhythm Disorder

569.1 Overview

Non-24-hour sleep-wake rhythm disorder (N24) is a circadian rhythm sleep-wake disorder characterized by a sleep-wake cycle that is desynchronized from the 24-hour day, typically running at a period longer than 24 hours (free-running), with each day sleep onset and wake time drifting progressively later by 30–60 minutes. It occurs in 50–80% of blind individuals (due to lack of photic entrainment of the SCN) and rarely in sighted individuals. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

569.2 Causes

This condition happens because of absence of light perception to entrain the circadian clock. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

569.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

569.4 Signs and Symptoms

You may experience a progressive daily delay in sleep-wake timing cycling through the 24-hour day. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

569.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

569.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

569.7 Diagnosis

- To make the diagnosis, doctors look for at least 14 days of actigraphy or sleep diaries demonstrating a circadian period other than 24 hours. Management in blind individuals includes melatonin (0.5–10 mg at desired bedtime) and tasimelteon (approved for N24 in blind people)
- In sighted individuals, Treatment options include morning light therapy and evening melatonin.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

569.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

569.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

569.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

569.11 Outlook

outlook is chronic but manageable. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 570

Non-Alcoholic Fatty Liver Disease

570.1 Overview

Non-alcoholic fatty liver disease (NAFLD), now increasingly termed metabolic dysfunction-associated steatotic liver disease (MASLD), is the most common chronic liver disease worldwide, affecting approximately 25% of the global population, and is strongly associated with metabolic syndrome, obesity, type 2 diabetes mellitus, and dyslipidemia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

570.2 Causes

This condition happens because of liver steatosis due to insulin resistance and metabolic dysfunction. The spectrum ranges from simple steatosis to non-alcoholic steatohepatitis (NASH/MASH), which involves hepatocyte injury, inflammation, and scarring, and can progress to cirrhosis and hepatocellular carcinoma. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

570.3 Risk Factors

Treatment focuses on weight loss through diet and exercise (7–10% body weight reduction), Treatment for metabolic risk factors, vitamin E or pioglitazone for non-diabetic NASH, and resmetirom for NASH with scarring. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

570.4 Signs and Symptoms

Clinical features are often absent, with elevated transaminases detected incidentally. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

570.5 Progression and Stages

Your outlook depends on the stage of scarring. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

570.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

570.7 Diagnosis

- Doctors diagnose this using imaging (ultrasound, CT, MRI-PDFF) showing liver steatosis and exclusion of other causes
- Liver biopsy remains the gold standard for staging.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

570.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

570.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

570.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

570.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 571

Non-Allergic Rhinitis

571.1 Overview

Non-allergic rhinitis is a chronic nasal condition that encompasses symptoms without an IgE-mediated mechanism, including vasomotor, drug-induced, hormonal, and gustatory forms. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

571.2 Causes

Vasomotor rhinitis results from hypersensitivity of trigeminal nerve endings leading to neurogenic inflammation triggered by environmental factors (temperature changes, strong odors, irritants). Drug-induced rhinitis results from overuse of topical decongestants (rhinitis medicamentosa). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

571.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

571.4 Signs and Symptoms

You may experience clear rhinorrhea and nasal congestion. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

571.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

571.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

571.7 Diagnosis

Doctors diagnose this using exclusion of allergic and infectious causes. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

571.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

571.9 Treatment Options

Treatment options include intranasal corticosteroids, ipratropium bromide spray for rhinorrhea, and avoidance of triggers. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

571.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

571.11 Outlook

The outlook varies from person to person depending on the ability to avoid triggers. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 572

Non-Gonococcal Urethritis (NGU)

572.1 Overview

Non-gonococcal urethritis is urethral inflammation without *Neisseria gonorrhoeae* infection, the most common urethral syndrome in sexually active men, with *Chlamydia trachomatis* (serovars D–K) accounting for 30–50% of cases, *Mycoplasma genitalium* (15–25%), *Trichomonas vaginalis* (5–10%), and other causes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

572.2 Causes

This condition happens because of infection of the urethral epithelium triggering a neutrophilic inflammatory response. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

572.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

572.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

572.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

572.6 Complications

You may experience an incubation of 2–35 days, with dysuria, urethral discharge (mucoid to mucopurulent), and urethral itching, with many infections being asymptomatic, and complications including epididymitis, prostatitis, and reactive arthritis.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

572.7 Diagnosis

Doctors diagnose this using first-void urine NAAT for *C. trachomatis*, *N. gonorrhoeae*, *M. genitalium*, and *T. vaginalis*. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

572.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

572.9 Treatment Options

Treatment options include azithromycin or doxycycline for chlamydia, with moxifloxacin as second-line for *M. genitalium*. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

572.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

572.11 Outlook

Most people recover well with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 573

Non-Hodgkin Lymphoma

573.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

573.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

573.3 Risk Factors

Non-Hodgkin lymphoma encompasses a heterogeneous group of over 60 subtypes of lymphoid malignancies not classified as Hodgkin lymphoma, with the most common subtypes including diffuse large B-cell lymphoma (DLBCL, most common NHL overall, aggressive), follicular lymphoma (most common indolent NHL), and mantle cell lymphoma, with risk factors including immunosuppression (HIV/AIDS—Burkitt lymphoma, EBV-associated lymphomas), autoimmune diseases, and *H. pylori* infection (gastric MALT lymphoma).

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)

- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

573.4 Signs and Symptoms

- Symptoms vary depending on subtype: aggressive lymphomas (DLBCL) present with rapidly growing masses, B symptoms, and organ compromise
- Indolent lymphomas (follicular) present with waxing and waning painless lymphadenopathy.
- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

573.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

573.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

573.7 Diagnosis

Doctors diagnose this using lymph node biopsy with histopathology, immunophenotyping, cytogenetics, and molecular studies. Treatment for DLBCL is R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone), with approximately 60–70% cure rate, while indolent NHL may be observed (watch and wait) or treated with rituximab-based regimens. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies

- Genetic counseling and testing for hereditary cancer syndromes

573.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

573.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

573.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay

physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

573.11 Outlook

Recovery varies widely by subtype. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 574

Non-Opioid Analgesics

574.1 Overview

Acetaminophen (paracetamol) is a centrally acting analgesic with weak anti-inflammatory effects. Its mechanism involves inhibition of cyclooxygenase (COX) in the central nervous system and interaction with the serotonergic system. It is effective for mild to moderate pain, particularly musculoskeletal pain and osteoarthritis. Maximum daily dose is 4,000 mg (3,000 mg in patients with liver impairment, older adults, or chronic alcohol use). Hepatotoxicity occurs at doses exceeding 4,000 mg/day or with acute overdose. NSAIDs include non-selective COX inhibitors (ibuprofen, naproxen, diclofenac, ketorolac, indomethacin) and selective COX-2 inhibitors (celecoxib, etoricoxib). They are effective for nociceptive pain with an inflammatory component. Mechanism is inhibition of COX-1 and COX-2, reducing prostaglandin synthesis. Adverse effects include digestive tract ulceration and bleeding (especially with COX-1 inhibition), kidney impairment (reduced prostaglandin-mediated kidney blood flow), cardiovascular risk (increased in both COX-2 selective and non-selective NSAIDs), and platelet inhibition (non-selective NSAIDs). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

574.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

574.3 Risk Factors

GI protection with proton pump inhibitors is recommended for patients on chronic NSAID therapy or with risk factors. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

574.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

574.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

574.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

574.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

574.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

574.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

574.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

574.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 575

Normal Pressure Hydrocephalus

575.1 Overview

Normal pressure hydrocephalus (NPH) is a potentially reversible cause of dementia and gait impairment, characterized by the classic triad of gait apraxia (magnetic gait—feet appear glued to the floor), thinking and memory decline (subcortical pattern), and urinary incontinence (late symptom). Prevalence increases with age: up to 5–6% of nursing home residents. Ventriculomegaly is present on brain imaging without significant cortical shrinkage or wasting, and lumbar puncture opening pressure is normal ($\leq$20 cm HH_2O). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

575.2 Causes

This condition happens because of impaired CSF absorption. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

575.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

575.4 Signs and Symptoms

You may experience the triad of gait apraxia, thinking and memory decline, and urinary incontinence. Diagnosis is supported by clinical improvement after high-volume lumbar puncture (CSF tap test) and Evans index >0.3. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

575.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

575.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

575.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

575.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

575.9 Treatment Options

Treatment typically involves ventriculoperitoneal shunt placement, which improves gait in 70–80% of appropriately selected patients. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

575.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

575.11 Outlook

- The outlook is generally positive with treatment
- Patient selection is critical for good outcomes.

Chapter 576

Norovirus

576.1 Overview

Norovirus is a single-stranded RNA virus (family Caliciviridae) and the leading cause of acute gastroenteritis worldwide, responsible for 685 million cases and 200,000 deaths annually, with all age groups affected and peaks in winter months ("winter vomiting disease"). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

576.2 Causes

This condition happens because of the virus infecting enterocytes of the small intestine, causing blunting and shrinkage or wasting of intestinal villi, crypt abnormal increase in cells, and increased apoptosis, reducing absorptive capacity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

576.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

576.4 Signs and Symptoms

You may experience an incubation of 12–48 hours, with sudden onset of nausea, vomiting (more prominent in children), watery diarrhea (non-bloody), abdominal cramps, and low-grade fever, with symptoms typically lasting 24–72 hours. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

576.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

576.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

576.7 Diagnosis

Doctors diagnose this using RT-PCR of stool. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

576.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

576.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

576.10 Living with It

Treatment typically involves supportive care with oral rehydration therapy.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

576.11 Outlook

Most people recover well in immunocompetent hosts. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 577

Novel Synthetic Drugs

577.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Synthetic cannabinoids (Spice, K2) are full agonists at CB1 and CB2 receptors with higher potency and longer half-life than THC. Toxicity: severe agitation, psychosis, seizures, hyperthermia, acute kidney injury, heart attack, and stroke. Synthetic cathinones (bath salts
- MDPV, methylone, mephedrone, alpha-PVP) are potent dopamine, norepinephrine, and serotonin reuptake inhibitors and releasers. Toxicity: severe agitation, psychosis, violent behavior, hyperthermia, hyponatremia, serotonin syndrome, rhabdomyolysis, DIC, and cardiovascular collapse. Fentanyl analogs (carfentanil, acetylfentanyl, furanylfentanyl) are ultra-potent opioids, up to 10,000 times more potent than morphine. Toxicity: rapid-onset respiratory depression and death.

577.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

577.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

577.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

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577.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

577.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

577.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

577.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

577.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

577.10 Living with It

- Treatment: aggressive sedation with benzodiazepines, active cooling, and supportive care
- Large doses of naloxone may be required for fentanyl analogs.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

577.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 578

NREM Parasomnias: Confusional Arousals

578.1 Overview

Confusional arousals are episodes of mental confusion, disorientation, and inappropriate behavior (slow speech, fumbling, ineffective actions) upon arousal from NREM sleep, typically N3. Episodes are more common in children. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

578.2 Causes

This condition happens because of incomplete arousal from slow-wave sleep. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

578.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

578.4 Signs and Symptoms

You may experience confusion and disorientation upon awakening without the complex mobility of sleepwalking or the intense autonomic activation of sleep terrors. Severe confusional arousal in adults is linked to obstructive sleep apnea, shift work, and psychiatric illness. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

578.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

578.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

578.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Management addresses underlying sleep-disordered breathing and triggers
- Clonazepam may be used in severe cases.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

578.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

578.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

578.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

578.11 Outlook

The outlook is generally positive. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 579

NREM Parasomnias: Sleep Terrors

579.1 Overview

Sleep terrors (pavor nocturnus) are dramatic arousals from NREM slow-wave sleep, characterized by sudden sitting up, piercing screams, intense autonomic activation (tachycardia, tachypnea, mydriasis, sweating), and behavioral expressions of intense fear. Peak prevalence is in children ages 1.5–7 years (1–6%). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

579.2 Causes

This condition happens because of incomplete arousal from deep NREM sleep. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

579.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

579.4 Signs and Symptoms

You may experience episodes lasting 1–10 minutes with the individual inconsolable, confused, and potentially thrashing violently, followed by return to sleep with no recall of the event. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

579.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

579.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

579.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

579.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

579.9 Treatment Options

Treatment options include reassurance, parental education, safety measures, and clonazepam for severe cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

579.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

579.11 Outlook

- The outlook is generally positive
- Most children outgrow the condition.

Chapter 580

NREM Parasomnias: Sleep-Related Eating Disorder

580.1 Overview

Sleep-related eating disorder (SRED) is a disorder involving recurrent episodes of involuntary eating and drinking during partial arousals from NREM sleep, with impaired awareness and variable amnesia for the events. Prevalence is 1–3% in the general population, higher in women. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

580.2 Causes

This condition happens because of abnormal arousal from NREM sleep and is strongly associated with other NREM parasomnias and hypnotic use (particularly zolpidem). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

580.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

580.4 Signs and Symptoms

You may experience consumption of unusual or dangerous items during the night with subsequent weight gain, dental injury, and potential ingestion of toxic substances. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

580.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

580.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

580.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

580.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

580.9 Treatment Options

Treatment options include discontinuation of triggering hypnotics, treatment of comorbid sleep apnea or restless legs syndrome, topiramate (50–200 mg nightly), and safety measures. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

580.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

580.11 Outlook

The outlook is generally positive with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 581

NREM Parasomnias: Sleepwalking

581.1 Overview

Sleepwalking (somnambulism) is a disorder of arousal from NREM sleep, typically arising from N3 (slow-wave) sleep during the first third of the night. Prevalence is 10–30% in children and 2–4% in adults. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

581.2 Causes

This condition happens because of incomplete arousal from slow-wave sleep, with genetic factors playing a strong role. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

581.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

581.4 Signs and Symptoms

You may experience complex, semi-purposeful behaviors ranging from sitting up to walking and performing routine activities without conscious awareness, with a blank stare and amnesia for the event. Triggers include sleep deprivation, fever, alcohol, stress, and sedative-hypnotic use. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

581.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

581.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

581.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Polysomnography is indicated for atypical, frequent, or potentially injurious episodes.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

581.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

581.9 Treatment Options

Treatment options include safety precautions, avoidance of triggers, scheduled awakening before typical episode time, and pharmacotherapy (clonazepam, diazepam) for frequent or injurious episodes. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

581.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

581.11 Outlook

- The outlook is generally positive
- Most children outgrow the condition.

Chapter 582

Nummular Dermatitis

582.1 Overview

Nummular (discoid) dermatitis is a chronic inflammatory skin condition presenting as coin-shaped, well-demarcated, erythematous plaques with serous crusting, often on the extremities. It is more common in winter and in patients with dry skin. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

582.2 Causes

This condition happens because of skin barrier disruption and environmental factors. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

582.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

582.4 Signs and Symptoms

You may experience intense itching and characteristic coin-shaped lesions. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

582.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

582.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

582.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- KOH preparation distinguishes it from tinea (negative in nummular dermatitis).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

582.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

582.9 Treatment Options

Treatment options include potent topical corticosteroids, moisturizers, and avoidance of skin irritants. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

582.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

582.11 Outlook

outlook is chronic with flares. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 583

Obesity

583.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Obesity is a chronic, relapsing, multifactorial disease characterized by excessive accumulation of adipose tissue with adverse health consequences. The global prevalence has nearly tripled since 1975
- Over 650 million adults are obese.

583.2 Causes

Clinical consequences affect virtually every organ system: type 2 diabetes mellitus, high blood pressure, dyslipidemia, coronary artery disease, stroke, obstructive sleep apnea, non-alcoholic fatty liver disease, cholelithiasis, osteoarthritis, infertility, depression, and increased risk of multiple cancers. Diagnosis includes BMI measurement, waist circumference, and screening for comorbidities. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

583.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

583.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

583.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- It is defined by a body mass index (BMI) of 30 kg/m^2 or greater in adults, classified into class I (30–34.9), class II (35–39.9), and class III (40 or higher). This condition happens because of a chronic energy imbalance—caloric intake exceeding energy expenditure—driven by complex interactions of genetic, environmental, behavioral, psychosocial, endocrine, and iatrogenic factors
- Adipose tissue is an active endocrine organ secreting adipokines (leptin, adiponectin, resistin, TNFα, IL-6) contributing to a chronic low-grade inflammatory state. Treatment typically involves stepwise: lifestyle intervention (dietary modification, physical activity, behavioral therapy), pharmacotherapy (orlistat, GLP-1 receptor agonists such as semaglutide and liraglutide, naltrexone-bupropion, phentermine-topiramate), and bariatric surgery for class III obesity or class II with comorbidities.

583.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

583.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

583.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

583.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

583.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

583.11 Outlook

- The outlook varies from person to person
- Sustained weight loss improves or resolves many comorbidities.

Chapter 584

Obesity Hypoventilation Syndrome

584.1 Overview

Obesity hypoventilation syndrome (OHS, Pickwickian syndrome) is a sleep-related breathing disorder defined by the triad of obesity (BMI $\geq$30 kg/m^2), awake daytime hypercapnia ($PaCO_2$ >45 mmHg) without other causes, and sleep-disordered breathing (90% have OSA). Prevalence among obese individuals is 10–20%, with higher rates in severe obesity. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

584.2 Causes

This condition happens because of mechanical loading of the respiratory system (reduced chest wall compliance, diaphragmatic restriction), blunted central respiratory drive (decreased chemosensitivity to hypercapnia and hypoxia), and leptin resistance leading to impaired ventilatory stimulation. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

584.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

584.4 Signs and Symptoms

You may experience obesity, hypersomnolence, shortness of breath on exertion, morning headache, cor pulmonale (right heart failure), and lung high blood pressure.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

584.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

584.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

584.7 Diagnosis

To make the diagnosis, doctors look for arterial blood gas showing awake hypercapnia, polysomnography, and exclusion of other causes of hypoventilation (neuromuscular disease, COPD, hypothyroidism).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

584.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

584.9 Treatment Options

Treatment focuses on positive airway therapy (CPAP or bilevel PAP), weight loss (medical or bariatric surgery), and supplemental oxygen if needed. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

584.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

584.11 Outlook

- Outlook is guarded
- Untreated OHS is linked to significantly increased cardiovascular morbidity and mortality.

Chapter 585

Obsessive-Compulsive Disorder (OCD)

585.1 Overview

Obsessive-compulsive disorder has a lifetime prevalence of 2–3% worldwide, with equal distribution across sexes, mean age of onset of 19.5 years, and a bimodal distribution (early-onset peaking in childhood/adolescence, late-onset peaking in the third decade). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

585.2 Causes

This condition happens because of hyperactivity in the cortico-striato-thalamo-cortical (CSTC) loop, specifically the orbitofrontal cortex, anterior cingulate cortex, and caudate nucleus, with glutamatergic hyperactivity and reduced serotonin function. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

585.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

585.4 Signs and Symptoms

You may experience obsessions (recurrent, persistent, intrusive, and unwanted thoughts, urges, or images) and/or compulsions (repetitive behaviors or mental acts that the individual feels driven to perform), with common obsession themes including contamination, harm, symmetry and order, and taboo thoughts, and common compulsions including washing, checking, counting, repeating, and ordering.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

585.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

585.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

585.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

585.8 Prevention

Treatment options include pharmacotherapy (SSRIs are first-line, typically at higher doses than for depression, with clomipramine as an effective alternative) and psychotherapy (exposure and response prevention [ERP] is the gold-standard psychotherapy).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

585.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

585.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

585.11 Outlook

The outlook varies from person to person: 30–40% have a chronic course, 40–50% have moderate improvement, 10–20% have refractory disease. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 586

Obstructive Sleep Apnea

586.1 Overview

Obstructive sleep apnea (OSA) is a sleep-related breathing disorder characterized by recurrent episodes of partial or complete pharyngeal collapse during sleep, resulting in apnea (breathing cessation $\geq$10 seconds) or hypopnea (reduced airflow with desaturation or arousal), causing intermittent hypoxia, hypercapnia, intrathoracic pressure swings, and sleep fragmentation. Prevalence is 15–30% of men and 10–20% of women in the general population, with marked increases in obesity. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

586.2 Causes

This condition happens because of anatomic narrowing of the pharyngeal airway (obesity, craniofacial abnormalities, macroglossia, tonsillar enlargement) combined with reduced dilator muscle activity of the genioglossus during sleep and a low arousal threshold. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

586.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

586.4 Signs and Symptoms

You may experience loud snoring, witnessed apneas, gasping or choking awakenings, nocturia, morning headache, excessive daytime sleepiness, fatigue, thinking and memory impairment, and irritability. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

586.5 Progression and Stages

Doctors diagnose this using attended in-laboratory polysomnography (gold standard) or home sleep apnea testing (HSAT) in uncomplicated adults, with severity classified by the apnea-hypopnea index (AHI): mild (AHI 5–15/hr), moderate (AHI 15–30/hr), and severe (AHI >30/hr). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

586.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

586.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

586.8 Prevention

Treatment focuses on positive airway pressure (PAP) therapy (continuous PAP, auto-CPAP, bilevel PAP) as The first-choice treatment, mandibular advancement devices for mild-to-moderate OSA, hypoglossal nerve stimulation therapy, positional therapy, weight loss and lifestyle modification, and upper airway surgery for patients with surgically correctable anatomy or PAP intolerance.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

586.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

586.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

586.11 Outlook

- The outlook is generally positive with treatment
- Untreated OSA is linked to high blood pressure, atrial fibrillation, heart failure, coronary artery disease, stroke, and type 2 diabetes.

Chapter 587

Obstructive Sleep Apnea

587.1 Overview

Obstructive sleep apnea is a disorder characterized by repetitive episodes of partial or complete upper airway obstruction during sleep, resulting in apneas and hypopneas. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

587.2 Causes

This condition happens because of collapse of the pharyngeal airway during sleep. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

587.3 Risk Factors

Risk factors include obesity, male sex, large neck circumference, retrognathia, tonsillar and adenoid enlargement, and craniofacial abnormalities.

- Family history of mental illness
- Trauma or adverse childhood experiences
- Chronic stress
- Substance use
- Social isolation
- Underlying medical conditions
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Medication use

587.4 Signs and Symptoms

- You may experience loud snoring, witnessed apneas, excessive daytime sleepiness, morning headaches, and thinking and memory impairment
- Untreated OSA increases the risk of high blood pressure, atrial fibrillation, stroke, heart failure, and metabolic syndrome.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

587.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

587.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

587.7 Diagnosis

Doctors diagnose this using polysomnography measuring the apnea-hypopnea index (AHI). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

587.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

587.9 Treatment Options

Treatment options include continuous positive airway pressure (CPAP), weight loss, oral appliances, and surgical options (uvulopalatopharyngoplasty, maxillomandibular advancement, hypoglossal nerve stimulation). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

587.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

587.11 Outlook

The outlook is generally positive with effective treatment and adherence. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 588

Occipital Neuralgia

588.1 Overview

Occipital neuralgia is a neuropathic pain disorder characterized by paroxysmal, stabbing, or shooting pain in the distribution of the greater, lesser, or third occipital nerves (posterior scalp, from occiput to vertex). It may be caused by entrapment of the occipital nerves by cervical muscles or by trauma, cervical arthritis, or C1–C2 pathology. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

588.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

588.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

588.4 Signs and Symptoms

You may experience pain accompanied by tenderness over the occipital nerve and paresthesias or dysesthesia in the affected distribution. Diagnosis is supported by temporary relief with local anesthetic block of the occipital nerve. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

588.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

588.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

588.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

588.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

588.9 Treatment Options

- Treatment options include nerve blocks with local anesthetic and corticosteroid (therapeutic and diagnostic), physical therapy, and medications (gabapentin, pregabalin, TCAs)
- For refractory cases, pulsed radiofrequency ablation or occipital nerve stimulation may be considered.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

588.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

588.11 Outlook

The outlook varies from person to person. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 589

Oculocutaneous Albinism

589.1 Overview

Oculocutaneous albinism (OCA) is a group of autosomal recessive disorders characterized by reduced or absent melanin production in the skin, hair, and eyes, with a global prevalence of 1 in 17,000. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

589.2 Causes

This condition happens because of mutations in genes involved in melanin biosynthesis: OCA1 (TYR, tyrosinase deficiency, most severe, absence of pigment), OCA2 (OCA2/P gene, common in Africans, some residual pigment), OCA3 (TYRP1, rufous albinism in Africans), and OCA4 (SLC45A2, common in Japanese). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

589.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

589.4 Signs and Symptoms

You may experience hypopigmentation of skin and hair ranging from white (OCA1) to light yellow or brown (OCA2/3/4), translucent irides, involuntary eye movements, photophobia, reduced visual acuity, foveal underdevelopment, optic nerve misrouting, and strabismus. Patients have markedly increased risk of skin cancer (squamous cell carcinoma, basal cell carcinoma, melanoma) due to lack of UV protection. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

589.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

589.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

589.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with genetic confirmation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

589.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

589.9 Treatment Options

- Treatment focuses on lifelong photoprotection (broad-spectrum sunscreen, protective clothing, UV-protective eyewear), regular skin surveillance for skin cancer, ophthalmologic care for visual impairment (low vision aids, tinted lenses, strabismus surgery), and skin cancer treatment. outlook: normal life expectancy with adequate sun protection
- Cutaneous malignancies are the major cause of morbidity and mortality.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

589.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

589.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 590

Opioid Analgesics

590.1 Overview

Opioids are agonists at mu, kappa, and delta opioid receptors in the central and peripheral nervous systems. They produce analgesia, euphoria, sedation, constipation, respiratory depression, nausea, and miosis. Morphine is the prototypic mu-opioid receptor agonist, effective for moderate to severe pain. It undergoes liver metabolism to morphine-3-glucuronide (M3G, inactive for analgesia but neuroexcitatory) and morphine-6-glucuronide (M6G, active, more potent than morphine). Hydromorphone is a semi-synthetic opioid 5–10 times more potent than morphine, with similar efficacy and safety profile. Fentanyl is a highly potent synthetic opioid (100 times more potent than morphine) with rapid onset and short duration of action, available in IV, transdermal, intranasal, and transmucosal formulations. The transdermal fentanyl patch provides continuous delivery for 72 hours and is not recommended for opioid-naïve patients. Oxycodone is a semi-synthetic opioid available as immediate-release and extended-release formulations, often combined with acetaminophen or naloxone. Methadone is a synthetic opioid with mu-receptor agonism and NMDA receptor antagonism. It has a long and variable half-life (15–60 hours) requiring cautious titration due to risk of accumulation and QT prolongation (ECG monitoring recommended). Tapentadol is a mu-opioid receptor agonist and norepinephrine reuptake inhibitor, effective for both nociceptive and neuropathic pain. Buprenorphine is a partial mu-opioid receptor agonist and kappa-receptor antagonist, with a ceiling effect for respiratory depression. It is available in sublingual, buccal, transdermal, and implantable formulations. It may be used for chronic pain (especially in patients with opioid use disorder) and has a favorable safety profile. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

590.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

590.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

590.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

590.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

590.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

590.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

590.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

590.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

590.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

590.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 591

Opioid Overdose

591.1 Overview

Opioid overdose is a leading cause of poisoning death, driven by prescription opioids, heroin, and illicit fentanyl analogs. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

591.2 Causes

This condition happens because of mu-opioid receptor agonism in the brainstem producing respiratory depression, miosis (pinpoint pupils), and CNS depression leading to coma. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

591.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

591.4 Signs and Symptoms

You may experience the classic triad of coma, miosis, and respiratory depression. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

591.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

591.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

591.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

591.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

591.9 Treatment Options

- Treatment focuses on airway management and naloxone (0.04–2 mg IV, 0.4–2 mg IM, or 2–4 mg IN) as the antidote
- Naloxone has a short half-life (30–90 minutes) that may require repeat doses or continuous infusion. Patients should be observed for at least 4–6 hours after the last naloxone dose.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

591.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

591.11 Outlook

The outlook is generally positive with timely intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 592

Opioid Therapy Risks

592.1 Overview

Tolerance is the reduced analgesic effect with repeated use, requiring dose escalation to maintain pain relief. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

592.2 Causes

It happens because of mu-receptor desensitization, upregulation of adenylyl cyclase, and NMDA receptor activation. Tolerance develops more slowly for analgesia than for respiratory depression, euphoria, and sedation. Physical dependence is a state of adaptation manifested by a withdrawal syndrome when the opioid is discontinued, the dose is reduced, or an opioid antagonist is administered. Withdrawal symptoms are unpleasant but not life-threatening (except in neonates) and include anxiety, irritability, mydriasis, lacrimation, rhinorrhea, piloerection, diaphoresis, myalgias, arthralgias, abdominal cramping, diarrhea, nausea, and vomiting. Withdrawal is managed by gradual dose tapering. Addiction (opioid use disorder, OUD) is a chronic, relapsing neurobiological disease characterized by impaired control over opioid use, compulsive use, continued use despite harm, and craving. The prevalence of OUD among patients prescribed opioids for chronic pain is estimated at 8–12%. Monitoring tools include prescription drug monitoring programs, urine drug testing, and risk assessment instruments (Opioid Risk Tool, Screener and Opioid Assessment for Patients with Pain). Risk mitigation strategies include avoiding high-dose therapy (morphine milligram equivalents >90 mg/day is linked to increased risk of overdose), avoiding concurrent benzodiazepine use, providing naloxone education and prescription, and establishing treatment agreements. Opioid-induced hyperalgesia (OIH) is a paradoxical increase in pain sensitivity caused by opioid exposure, distinct from tolerance. The mechanism involves activation of central glutamatergic systems (NMDA receptors), descending facilitation from the RVM, and increased dynorphin and substance P in the spinal cord. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

592.3 Risk Factors

Risk factors include personal or family history of substance use disorder, younger age, untreated psychiatric conditions, and history of childhood abuse or trauma. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

592.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

592.5 Progression and Stages

It presents as worsening pain despite increasing opioid dose, with diffuse pain that extends beyond the original pain site. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

592.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life

- Emotional and psychological impact

592.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

592.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

592.9 Treatment Options

Treatment options include opioid dose reduction, rotation to a different opioid (methadone or buprenorphine), addition of NMDA receptor antagonists (ketamine, methadone), and non-opioid analgesic optimization. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

592.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

592.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 593

Opioid Use Disorder

593.1 Overview

Opioid use disorder has a lifetime prevalence of approximately 1–2% and has been a major driver of the opioid epidemic in North America. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

593.2 Causes

This condition happens because of mu-opioid receptor activation in the ventral tegmental area and nucleus accumbens, producing dopamine release and intense euphoria, with chronic use leading to tolerance, severe withdrawal (agitation, myalgias, lacrimation, rhinorrhea, piloerection, yawning, diarrhea, nausea/vomiting, dilated pupils, muscle spasms, intense craving), and hijacking of the reward system, with respiratory depression being the acute lethal risk. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

593.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

593.4 Signs and Symptoms

You may experience a problematic pattern of opioid use (heroin, prescription opioids) leading to clinically significant impairment, with criteria analogous to AUD. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

593.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

593.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

593.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

593.8 Prevention

- Overdose prevention includes naloxone distribution. outlook: MAT is linked to reduced mortality
- Relapse rates after detoxification alone exceed 80% without ongoing medication.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

593.9 Treatment Options

Treatment options include medication-assisted treatment (MAT) as standard of care: methadone (full mu-agonist), buprenorphine (partial mu-agonist), and naltrexone (mu-antagonist), along with psychosocial interventions (CBT, contingency management). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

593.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

593.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 594

Oral Candidiasis

594.1 Overview

Oral candidiasis is a fungal infection of the oral mucosa caused by *Candida* species, most commonly *Candida albicans*, with predisposing factors including immunosuppression (HIV/AIDS, chemotherapy), diabetes, denture use, inhaled corticosteroids, xerostomia, and broad-spectrum antibiotics. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

594.2 Causes

This condition happens because of overgrowth of *Candida* organisms in the oral cavity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

594.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

594.4 Signs and Symptoms

- You may experience acute pseudomembranous candidiasis presenting with white, removable plaques on the buccal mucosa, tongue, palate, and oropharynx, revealing an erythematous base when scraped
- Erythematous (atrophic) and chronic hyperplastic forms also exist.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

594.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

594.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

594.7 Diagnosis

Your doctor confirms the diagnosis with wet mount with potassium hydroxide showing pseudohyphae and blastospores. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

594.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

594.9 Treatment Options

Treatment focuses on topical antifungals (nystatin, clotrimazole) for mild disease or systemic azoles (fluconazole) for moderate-to-severe or refractory disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

594.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

594.11 Outlook

Most people recover well with appropriate antifungal therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 595

Oral Leukoplakia

595.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Oral leukoplakia is a white plaque of the oral mucosa that cannot be scraped off and does not correspond to any other diagnosable condition
- It is the most common oral premalignant lesion, with a cancerous transformation rate of approximately 3–5%.

595.2 Causes

This condition happens because of chronic mucosal irritation leading to epithelial dysplasia. Homogeneous leukoplakia has a lower cancerous potential than non-homogeneous (speckled) leukoplakia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

595.3 Risk Factors

Risk factors include tobacco use (smoking and smokeless), alcohol consumption, and betel nut chewing. The right treatment depends on the degree of dysplasia: observation for mild dysplasia, surgical removal for moderate-to-severe dysplasia, and close follow-up for all patients with cessation of risk factors. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

595.4 Signs and Symptoms

You may experience a white plaque on the oral mucosa. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

595.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

595.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

595.7 Diagnosis

To make the diagnosis, doctors look for biopsy to assess for epithelial dysplasia. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

595.8 Prevention

The outlook is good with early detection and appropriate management.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

595.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

595.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

595.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 596

Oral Lichen Planus

596.1 Overview

Oral lichen planus is a chronic inflammatory condition of the oral mucosa mediated by T-cell autoimmune attack on basal keratinocytes, affecting approximately 1–2% of the population, most commonly middle-aged women. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

596.2 Causes

This condition happens because of autoimmune T-cell-mediated damage to basal keratinocytes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

596.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

596.4 Signs and Symptoms

- You may experience the reticular form with characteristic Wickham striae (white, lacy patches) on the buccal mucosa, tongue, and gingiva, usually asymptomatic
- The erosive form causes painful, erythematous ulcerated areas. There is a small (approximately 1%) risk of cancerous transformation.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

596.5 Progression and Stages

Doctors diagnose this using biopsy showing a band-like lymphocytic infiltrate at the epithelial-connective tissue interface with basal cell deterioration. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

596.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

596.7 Diagnosis

- Treatment for symptomatic disease includes topical corticosteroids and calcineurin inhibitors
- Long-term surveillance is recommended.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

596.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

596.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

596.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

596.11 Outlook

In most cases, the outlook is good, but the condition is chronic and requires surveillance for cancerous transformation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 597

Oral Squamous Cell Carcinoma

597.1 Overview

Oral squamous cell carcinoma is a cancerous tumor that accounts for over 90% of oral malignancies, with approximately 370,000 new cases annually worldwide, most commonly arising on the lateral tongue and floor of the mouth. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

597.2 Causes

This condition happens because of cancerous transformation of oral epithelial cells, driven by tobacco use, heavy alcohol consumption, betel nut chewing, and HPV infection. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

597.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal

factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

597.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

597.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for early disease is surgical removal with or without neck dissection
- Advanced disease requires multimodal therapy with surgery, radiation, and chemotherapy. outlook is related to stage at diagnosis, with 5-year survival rates of approximately 80–90% for stage I and less than 30% for stage IV.

597.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

597.7 Diagnosis

Doctors diagnose this using incisional biopsy with TNM staging. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

597.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

597.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

597.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

597.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 598

Organic Acidemias

598.1 Overview

Organic acidemias are a group of autosomal recessive disorders of amino acid, fatty acid, and carbohydrate catabolism causing accumulation of organic acids in blood and urine. Propionic acidemia (PA) is caused by propionyl-CoA carboxylase deficiency, presenting similarly with severe metabolic acidosis, ketosis, hyperammonemia, pancytopenia, and basal ganglia tissue death due to lack of blood supply. Isovaleric acidemia (IVA) is caused by isovaleryl-CoA dehydrogenase deficiency, producing a characteristic "sweaty feet" odor (isovaleric acid) and presenting with vomiting, metabolic acidosis, lethargy, and pancytopenia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

598.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

598.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

598.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

598.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

598.6 Complications

- Methylmalonic acidemia (MMA) is caused by deficiency of methylmalonyl-CoA mutase or cobalamin metabolism, presenting in infancy with recurrent episodes of metabolic acidosis, hyperammonemia, hypoglycemia, lethargy, vomiting, hypotonia, and failure to thrive
- Long-term complications include chronic kidney disease, pancreatitis, and nervous system deficits.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

598.7 Diagnosis

To diagnose all organic acidemias is by urine organic acid analysis, plasma acylcarnitine profile, and genetic testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

598.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

598.9 Treatment Options

Treatment options include protein restriction, carnitine supplementation, and specific therapies (metronidazole for MMA, glycine and carnitine for IVA). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

598.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

598.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 599

Organophosphate and Carbamate Insecticide Poisoning

599.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Organophosphate (OP) and carbamate insecticides inhibit acetylcholinesterase (AChE) at cholinergic synapses, causing accumulation of acetylcholine and excessive muscarinic, nicotinic, and CNS stimulation. OPs bind irreversibly (aging occurs within 24–48 hours)
- Carbamates bind reversibly.

599.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

599.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

599.4 Signs and Symptoms

Clinical features: cholinergic crisis—SLUDGE syndrome (Salivation, Lacrimation, Urination, Defecation, GI upset, Emesis) plus miosis, bronchospasm, bronchorrhea, bradycardia, excessive sweating, muscle fasciculations, weakness, paralysis, seizures, and CNS depression. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

599.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

599.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

599.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with RBC AChE activity for confirmation. Management: immediate decontamination, atropine (1–4 mg IV, doubled every 5–10 minutes) to block muscarinic effects, pralidoxime (2-PAM, 1–2 g IV) to reactivate AChE if given before aging, and benzodiazepines for seizures. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

599.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

599.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

599.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

599.11 Outlook

The outlook is generally positive with early treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 600

Osteoarthritis

600.1 Overview

Osteoarthritis (OA) is the most common joint disease worldwide, characterized by progressive articular cartilage degradation, subchondral bone remodeling, osteophyte formation, and synovial inflammation. It affects over 500 million people globally, with prevalence increasing with age. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

600.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

600.3 Risk Factors

This condition happens because of mechanical and metabolic factors leading to cartilage breakdown, with risk factors including advancing age, obesity, joint injury, female sex, and genetic predisposition.

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Environmental triggers (infections, smoking)
- Hormonal factors
- Certain medications
- Advancing age
- Genetic predisposition
- Repetitive use or overuse (joints)

- Previous injuries
- Obesity
- Smoking and alcohol use
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

600.4 Signs and Symptoms

- You may experience joint pain (worse with activity and relieved by rest), stiffness (morning stiffness <30 minutes, unlike RA), crepitus, reduced range of motion, and joint enlargement (osteophytes, bony enlargement). Weight-bearing joints (knees, hips) and the hands (distal interphalangeal joints—Heberden nodes, proximal interphalangeal joints—Bouchard nodes, first carpometacarpal joints) are most commonly affected. Diagnosis is primarily clinical
- X-rays show joint space narrowing, subchondral sclerosis, osteophytes, and subchondral cysts.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

600.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment options include patient education, weight loss, exercise, physical therapy, analgesics (acetaminophen, NSAIDs), intra-articular corticosteroid or hyaluronic acid injections, and joint replacement surgery (total knee or hip arthroplasty) for advanced disease. The outlook varies from person to person
- Joint replacement provides excellent outcomes for advanced disease.

600.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

600.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

600.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

600.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

600.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

600.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can

be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 601

Osteogenesis Imperfecta

601.1 Overview

Osteogenesis imperfecta (OI, "brittle bone disease") is a group of genetic disorders primarily caused by mutations in the COL1A1 or COL1A2 genes encoding type I collagen, leading to defective bone formation and increased bone fragility. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

601.2 Causes

This condition happens because of defective collagen production leading to brittle bones. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

601.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

601.4 Signs and Symptoms

You may experience recurrent fractures with minimal trauma, blue sclerae, dentinogenesis imperfecta (brittle, discolored teeth), hearing loss (conductive, sensorineural, or mixed), and joint hyperlaxity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

601.5 Progression and Stages

Severity ranges from perinatal lethal (type II) to mild (type I, most common). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

601.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

601.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by genetic testing and bone densitometry. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

601.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

601.9 Treatment Options

Treatment options include bisphosphonates (to reduce fracture frequency), physical therapy, bracing, surgical correction of deformities (intramedullary rodding), and dental management. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

601.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

601.11 Outlook

- Your outlook depends on the type
- Milder forms have a normal life expectancy.

Chapter 602

Osteomyelitis

602.1 Overview

Osteomyelitis is infection of bone, which may be acute or chronic. Hematogenous osteomyelitis (most common in children) typically affects the metaphysis of long bones (femur, tibia, humerus), with *Staphylococcus aureus* as the predominant organism. Contiguous-focus osteomyelitis occurs after direct inoculation (trauma, surgery, open fractures) or spread from adjacent soft tissue infection. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

602.2 Causes

This condition happens because of bacterial infection of bone leading to inflammation, tissue death, and new bone formation. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

602.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

602.4 Signs and Symptoms

- Clinical features in children include localized bone pain, fever, and elevated inflammatory markers
- In adults, it more commonly involves the vertebrae (vertebral osteomyelitis/diskitis). Chronic osteomyelitis is marked by bone tissue death (sequestrum), new bone formation (involucrum), and sinus tracts draining pus.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

602.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

602.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage

- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

602.7 Diagnosis

Diagnosis typically involves blood cultures, inflammatory markers (ESR, CRP), and imaging (MRI is the gold standard for detecting bone swelling and abscesses

- CT for sequestrum
- Bone scan). Management requires prolonged antibiotic therapy (6 weeks or more, IV initially then oral step-down) and surgical removal of dead tissue for chronic or refractory disease.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

602.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

602.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

602.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

602.11 Outlook

The outlook is generally positive with early diagnosis and appropriate treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 603

Osteoporosis

603.1 Overview

It affects approximately 200 million people worldwide, predominantly postmenopausal women and the elderly. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

603.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

603.3 Risk Factors

This condition happens because of an imbalance between bone resorption and bone formation, with risk factors including female sex, advancing age, low body weight, family history, smoking, excessive alcohol, low calcium and vitamin D intake, sedentary lifestyle, and medications (glucocorticoids, aromatase inhibitors, androgen deprivation therapy).

- Family history
- Obesity and sedentary lifestyle
- Poor dietary habits
- Advancing age
- Hormonal imbalances
- Certain medications
- Genetic predisposition

- Repetitive use or overuse (joints)
- Previous injuries
- Obesity
- Smoking and alcohol use
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

603.4 Signs and Symptoms

You may experience fragility fractures commonly at the hip, vertebrae (compression fractures), and distal radius (Colles' fracture).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

603.5 Progression and Stages

Osteoporosis is a systemic skeletal disease characterized by low bone mineral density (BMD) and deterioration of bone microarchitecture, resulting in increased bone fragility and fracture risk. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

603.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

603.7 Diagnosis

- Doctors diagnose this based on dual-energy X-ray absorptiometry (DXA): T-score $\leq$-2.5 at the hip or spine defines osteoporosis
- T-score between -1 and -2.5 is osteopenia. The FRAX tool estimates 10-year fracture risk.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

603.8 Prevention

- Outlook is improved with early detection and treatment
- Fracture risk is significantly reduced with appropriate therapy.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

603.9 Treatment Options

Treatment options include calcium (1000–1200 mg/day) and vitamin D (800–1000 IU/day), weight-bearing exercise, bisphosphonates (alendronate, zoledronic acid, first-line), denosumab (RANKL inhibitor), teriparatide and abaloparatide (parathyroid hormone analogs), and romosozumab (anti-sclerostin antibody). Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

603.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 604

Osteosarcoma

604.1 Overview

Osteosarcoma is the most common primary cancerous bone tumor (excluding myeloma), with a bimodal age distribution (peak in adolescents during the growth spurt and in older adults). It arises from mesenchymal cells producing osteoid and most commonly affects the metaphysis of long bones (distal femur, proximal tibia, proximal humerus). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

604.2 Causes

This condition happens because of cancerous transformation of osteoblast precursors. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

604.3 Risk Factors

Risk factors include prior radiation, Paget's disease, Li-Fraumeni syndrome, hereditary retinoblastoma, and Bloom syndrome.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Advanced age
- Family history and genetic predisposition
- Head trauma or brain injury
- Vascular risk factors (hypertension, diabetes)
- Exposure to environmental toxins
- Chronic stress
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

604.4 Signs and Symptoms

You may experience localized bone pain, swelling, and a palpable mass. X-rays show a destructive lesion with sunburst periosteal reaction and Codman's triangle. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

604.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

604.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

604.7 Diagnosis

Your doctor confirms the diagnosis with biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. Physical examination including palpation of mass and lymph node assessment

Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

604.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

604.9 Treatment Options

Treatment focuses on neoadjuvant chemotherapy (doxorubicin, cisplatin, high-dose methotrexate), surgical removal (limb-salvage surgery when possible), and adjuvant chemotherapy. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

604.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
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- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

604.11 Outlook

Your outlook depends on tumor tissue death after neoadjuvant chemotherapy (>90% tissue death is a favorable prognostic factor). Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 605

Osteosarcoma in Children

605.1 Overview

Osteosarcoma is the most common primary cancerous bone tumor in children and adolescents (see Chapter 26, Diseases of the Musculoskeletal System for comprehensive coverage). Peak incidence is during the adolescent growth spurt (ages 10–16), with a predilection for the metaphyses of long bones, most commonly the distal femur, proximal tibia, and proximal humerus. Most childhood osteosarcomas are high-grade. Treatment follows neoadjuvant chemotherapy, surgical removal, and adjuvant chemotherapy, with survival of 60–70% for localized disease. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

605.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

605.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

605.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

605.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

605.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

605.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

605.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

605.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

605.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

605.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 606

Other Parasomnias: Exploding Head Syndrome

606.1 Overview

Exploding head syndrome is a parasomnia characterized by episodes of loud imagined noises (explosion, gunshot, crashing cymbals) or a sensation of violent explosion inside the head, occurring at the transition from wakefulness to sleep (hypnagogic) or from sleep to wakefulness. It is relatively common (10–15% of young adults). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

606.2 Causes

This condition happens because of a transient delay in the normal inhibition of the auditory system during sleep onset. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

606.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

606.4 Signs and Symptoms

You may experience frightening but painless auditory sensations lasting seconds.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

606.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

606.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

606.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

606.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

606.9 Treatment Options

- Treatment focuses on reassurance
- Clomipramine and calcium channel blockers have been used in severe cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

606.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

606.11 Outlook

outlook is non-cancerous.

Chapter 607

Other Parasomnias: Sleep Enuresis

607.1 Overview

Sleep enuresis (bedwetting) is involuntary voiding of urine during sleep, occurring at least twice per week for at least three months, in children over age 5. Prevalence decreases with age: 15% at age 5, 5% at age 10, and 1–2% by age 15, with a male predominance. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

607.2 Causes

This condition happens because of a mismatch between nocturnal bladder capacity and overnight urine production, impaired arousal from sleep in response to bladder fullness, and in some cases, nocturnal polyuria due to insufficient antidiuretic hormone secretion. Primary enuresis (never dry for an extended period) is more common than secondary enuresis (relapse after ≥6 months of dryness). Diagnosis includes urinalysis to exclude diabetes mellitus, diabetes insipidus, and urinary tract infection. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

607.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

607.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

607.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

607.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

607.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

607.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

607.9 Treatment Options

- Treatment options include behavioral interventions (voiding before bed, limiting evening fluids, reward systems, bedwetting alarms), desmopressin for nocturnal polyuria, and anticholinergics for bladder overactivity
- Combination therapy may benefit refractory cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

607.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

607.11 Outlook

- The outlook is generally positive
- Most children outgrow the condition.

Chapter 608

Other Parasomnias: Sleep-Related Hallucinations

608.1 Overview

Sleep-related hallucinations are vivid, dream-like perceptual experiences occurring at sleep onset (hypnagogic) or upon awakening (hypnopompic), involving visual, auditory, tactile, or kinetic sensations. They are common in the general population (25–40% of young adults) and typically non-cancerous. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

608.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of intrusion of REM sleep imagery into wakefulness
- Pathological sleep-related hallucinations are hallmark features of narcolepsy type 1 and can occur in Parkinson disease.

608.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

608.4 Signs and Symptoms

You may experience multimodal hallucinations with intact reality testing. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

608.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

608.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

608.7 Diagnosis

To make the diagnosis, doctors look for exclusion of narcolepsy and distinction from psychotic disorders. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

608.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

608.9 Treatment Options

Treatment focuses on reassurance for non-cancerous cases and treatment of underlying narcolepsy or neurodegenerative disease when present. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

608.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

608.11 Outlook

outlook is non-cancerous in isolation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 609

Other Primary Headache Disorders

609.1 Overview

Primary cough headache is a primary headache disorder brought on by coughing, straining, or Valsalva maneuver, typically bilateral, sudden-onset, lasting 1 second to 30 minutes, with a prevalence of approximately 1%. Secondary causes (Chiari malformation, intracranial high blood pressure) must be excluded by MRI. Indomethacin is The first-choice treatment. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

609.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

609.3 Risk Factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

609.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

609.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

609.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

609.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

609.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

609.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

609.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

609.11 Outlook

The outlook is generally positive. Primary exertional headache is a primary headache disorder precipitated by any form of exercise, typically bilateral, pulsating, lasting 5 minutes to 48 hours, more common in hot weather, at high altitude, and in patients with a history of migraine. Treatment options include ergonovine, indomethacin, or triptans taken prior to exercise. Primary stabbing headache is a primary headache disorder presenting with brief, sharp, stabbing pains ("ice pick headache") lasting 1–10 seconds, occurring in single stabs or a series, confined to the first division of the trigeminal nerve. Indomethacin is first-line. Hypnic headache ("alarm clock headache") is a rare primary headache disorder in patients over 50 years of age, with attacks strictly sleep-related, awakening the patient at a consistent time (typically 1–3 hours after sleep onset), bilateral, dull, lasting 15–180 minutes, without autonomic features. Caffeine, lithium, or indomethacin are used. In most cases, the outlook is favorable. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 610

Otitis Externa

610.1 Overview

Otitis externa is an infection of the external auditory canal that commonly affects swimmers and individuals with predisposing factors such as water exposure, cotton-swab use, hearing aids, and humid climates. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

610.2 Causes

This condition happens because of inflammation and infection of the canal skin, most commonly caused by *Pseudomonas aeruginosa* and *Staphylococcus aureus*. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

610.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

610.4 Signs and Symptoms

- You may experience ear pain (otalgia), itching, otorrhea, and a sensation of ear fullness
- Examination reveals redness of the skin, swelling, and tenderness of the canal wall with purulent debris. In severe cases, infection may spread to the surrounding skin (cancerous otitis externa), particularly in diabetics and immunocompromised patients.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

610.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

610.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

610.7 Diagnosis

Doctors diagnose this based on clinical examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

610.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

610.9 Treatment Options

- Treatment focuses on thorough canal removal of dead tissue and topical antimicrobial drops (fluoroquinolones or aminoglycosides with corticosteroids)
- Oral antibiotics are reserved for extensive or invasive disease.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

610.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

610.11 Outlook

Most people recover well with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 611

Otosclerosis

611.1 Overview

Otosclerosis is a primary bone remodeling disorder of the otic capsule that causes progressive conductive hearing loss, typically bilateral, beginning in the third to fourth decade of life, with a female predominance and genetic component (autosomal dominant with variable penetrance). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

611.2 Causes

This condition happens because of abnormal spongy bone formation (active otosclerosis) that replaces normal endochondral bone, most commonly fixing the footplate of the stapes in the oval window. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

611.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

611.4 Signs and Symptoms

You may experience progressive conductive hearing loss. Diagnosis is suggested by a positive Schwartze sign (red hue of the promontory on otoscopy) and confirmed by audiometry showing conductive or mixed hearing loss. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

611.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment focuses on stapedectomy or stapedotomy with prosthetic stapes replacement, which is highly effective
- Sodium fluoride may slow disease progression.

611.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

611.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

611.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

611.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

611.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

611.11 Outlook

Most people recover well with surgical intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 612

Ovarian Cancer

612.1 Overview

Risk-reducing bilateral salpingo-oophorectomy (RRSO) is recommended for women at high risk (BRCA1 carriers by age 35–40, BRCA2 carriers by age 40–45). Oral contraceptive use reduces ovarian cancer risk by 50% after five years of use. Screening with CA-125 and transvaginal ultrasound is not recommended for average-risk women but may be considered for high-risk women beginning age 30–35. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

612.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

612.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)

- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

612.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

612.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

612.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

612.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

612.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin

from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

612.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

612.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

612.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 613

Ovarian Cancer

613.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

613.2 Causes

The most common histological type is high-grade serous carcinoma (70%), arising from the fallopian tube epithelium. This condition happens because of cancerous transformation of ovarian or fallopian tube epithelium. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

613.3 Risk Factors

Risk factors include BRCA1/2 mutations (lifetime risk 40–60% and 10–30%, respectively), family history, nulliparity, early menarche, and late menopause, with protective factors including oral contraceptive use and tubal ligation.

- Advancing age
- Family history of cancer

- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

613.4 Signs and Symptoms

You may experience abdominal bloating, pelvic/abdominal pain, urinary urgency, and early satiety. CA-125 is a serum tumor marker used for monitoring treatment response. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

613.5 Progression and Stages

Ovarian cancer is the fifth leading cause of cancer-related death in women, with approximately 313,000 new cases and 207,000 deaths annually worldwide, often diagnosed at an advanced stage due to vague symptoms. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

613.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

613.7 Diagnosis

Doctors diagnose this using imaging, surgical exploration, and histopathology. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)

- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

613.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

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- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

613.9 Treatment Options

Treatment typically involves optimal cytoreductive surgery followed by platinum-based chemotherapy (carboplatin + paclitaxel), with maintenance therapy using PARP inhibitors for BRCA-mutated tumors. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations

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613.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

613.11 Outlook

In most cases, the outlook is poor due to late diagnosis. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 614

Paget's Disease of Bone

614.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Paget's disease of bone is a chronic disorder of bone remodeling characterized by excessive osteoclastic bone resorption followed by disordered, chaotic new bone formation, resulting in enlarged, structurally abnormal, weakened bones. It is the second most common metabolic bone disease after osteoporosis, affecting 1–3% of adults over 55
- The pelvis, spine, skull, and femur are most commonly involved.

614.2 Causes

This condition happens because of abnormal osteoclast activity with increased bone turnover. Most patients are asymptomatic, discovered incidentally by elevated alkaline phosphatase or radiographic findings. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

614.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

614.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

614.5 Progression and Stages

Doctors diagnose this based on elevated serum alkaline phosphatase, characteristic radiographic changes (osteolytic, mixed, or osteosclerotic phases, cotton-wool appearance of the skull), and bone scan showing increased uptake. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

614.6 Complications

You may experience bone pain, skeletal deformity, pathologic fractures, and neurological complications (compression of cranial nerves, spinal stenosis, hearing loss from skull base involvement).

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

614.7 Diagnosis

Management with bisphosphonates (zoledronic acid most effective) inhibits osteoclastic activity and normalizes bone turnover. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

614.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

614.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

614.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

614.11 Outlook

In most cases, the outlook is good with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 615

Paget's Disease of the Nipple

615.1 Overview

Paget's disease of the nipple is a rare presentation of breast cancer (1–3% of cases) involving the nipple-areolar complex. The condition is marked by an eczematous or psoriasiform eruption of the nipple, often with itching, burning, and nipple discharge, with nearly all cases being associated with an underlying breast carcinoma (in situ or invasive). Histologically, Paget cells (large, pale, intraepidermal adenocarcinoma cells) are found within the epidermis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

615.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

615.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

615.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

615.5 Progression and Stages

Your outlook depends on the stage of the underlying carcinoma rather than the Paget's disease itself. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

615.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

615.7 Diagnosis

To make the diagnosis, doctors look for biopsy of the affected nipple. Management typically involves mastectomy with sentinel lymph node biopsy, or breast-conserving surgery if the underlying carcinoma can be adequately excised. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

615.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

615.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
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- Surgical intervention when indicated
- Regular monitoring and follow-up care

615.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

615.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 616

Pain Management in the Elderly

616.1 Overview

Chronic pain affects 50–75% of older adults, most commonly from osteoarthritis, degenerative spine disease, neuropathic conditions, and musculoskeletal disorders. Pain is often under-recognized and under-treated in older adults, especially those with thinking and memory impairment (use of PAINAD scale or Abbey Pain Scale for non-verbal patients). The WHO analgesic ladder guides treatment, but geriatric-specific cautions apply: This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

Acetaminophen (paracetamol): First-line for musculoskeletal pain; maximum 3000 mg/day to avoid hepatotoxicity. **NSAIDs:** Avoid chronic use in older adults due to digestive tract, kidney, and cardiovascular risks; use lowest dose, shortest duration if unavoidable, with PPI for GI protection. **Opioids:** Reserved for moderate-to-severe pain not responding to non-opioid therapies; start low, go slow; monitor for constipation, sedation, respiratory depression, and falls. **Gabapentinoids (gabapentin, pregabalin):** Effective for neuropathic pain; renally dosed; associated with sedation, dizziness, falls.

- **Topical analgesics:** Lidocaine patches, topical NSAIDs (diclofenac), capsaicin—preferred for localized pain with minimal systemic side effects.
- **Adjuvants:** SNRIs (duloxetine) and TCAs (nortriptyline preferred over amitriptyline for lower anticholinergic burden) for neuropathic pain and fibromyalgia.

616.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

616.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

616.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

616.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

616.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

616.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

616.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

616.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

616.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

616.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 617

Pancreatic Adenocarcinoma

617.1 Overview

Pancreatic adenocarcinoma is one of the most lethal malignancies, with a 5-year survival rate of approximately 12%. It is the fourth leading cause of cancer death worldwide. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

617.2 Causes

Approximately 80% of tumors arise in the pancreatic head. This condition happens because of accumulation of genetic mutations (KRAS, CDKN2A, TP53, SMAD4) driving cancerous transformation. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

617.3 Risk Factors

Risk factors include smoking, obesity, chronic pancreatitis, diabetes mellitus (new-onset in an older patient), family history, and BRCA2 mutations.

- Advancing age
- Family history of cancer

- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

617.4 Signs and Symptoms

You may experience painless jaundice (Courvoisier's sign—palpable, non-tender gallbladder), new-onset diabetes, weight loss, epigastric pain radiating to the back, and depression. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

617.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage,

small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

617.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

617.7 Diagnosis

Doctors diagnose this based on CT abdomen with pancreatic protocol, CA 19-9 (tumor marker), and using a thin camera tube ultrasound-guided fine needle removal by suction for tissue diagnosis. Staging uses the TNM system. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging

- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

617.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

617.9 Treatment Options

Treatment focuses on surgical removal (Whipple procedure for head tumors, distal pancreatectomy for body/tail tumors) as the only curative option, possible in only 15–20% of patients

- Adjuvant chemotherapy (FOLFIRINOX or gemcitabine/capecitabine) improves survival. Metastatic disease doctors typically treat it with systemic chemotherapy
- Molecular profiling may identify targets for precision therapy (BRCA mutations, MSI-high tumors). outlook remains poor, with a 5-year survival of approximately 12%.
- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

617.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

617.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 618

Pancreatic Exocrine Insufficiency

618.1 Overview

Pancreatic exocrine insufficiency (PEI) is the failure of the pancreas to produce sufficient digestive enzymes, leading to maldigestion and malabsorption. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

618.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of loss of pancreatic acinar cell function
- Lipase deficiency is the most important, leading to steatorrhea.

618.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

618.4 Signs and Symptoms

You may experience fatty, foul-smelling, floating stools, weight loss, and fat-soluble vitamin deficiencies (A, D, E, K). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

618.5 Progression and Stages

Common causes include chronic pancreatitis, cystic scarring, pancreatic surgery (Whipple procedure, distal pancreatectomy), and advanced pancreatic cancer. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

618.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

618.7 Diagnosis

Doctors diagnose this based on fecal elastase-1 level (<200 μg/g indicates moderate insufficiency, <100 μg/g severe) or 72-hour fecal fat collection. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

618.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

618.9 Treatment Options

Treatment typically involves pancreatic enzyme replacement therapy (PERT) with enteric-coated enzyme capsules taken with meals, with dose titrated based on symptoms and nutritional status. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

618.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

618.11 Outlook

The outlook is good with adequate enzyme replacement. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 619

Pancreatic Neuroendocrine Tumors

619.1 Overview

Pancreatic neuroendocrine tumors (PanNETs) are tumors arising from the islet cells of the pancreas, accounting for 1–2% of pancreatic tumors. They may be functional (hormone-secreting, 40–60%) or non-functional. Functional PanNETs include insulinoma (most common functional type, presenting with hypoglycemia), gastrinoma (Zollinger-Ellison syndrome with recurrent peptic ulcers), glucagonoma (necrolytic migratory redness of the skin, diabetes, weight loss), VIPoma (watery diarrhea, hypokalemia, achlorhydria), and somatostatinoma (diabetes, steatorrhea, gallstones). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

619.2 Causes

This condition happens because of neoplastic transformation of pancreatic islet cells. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

619.3 Risk Factors

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

619.4 Signs and Symptoms

- Your symptoms depend on the hormone secreted
- Non-functional tumors are often discovered incidentally.
- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

619.5 Progression and Stages

In most cases, the outlook is better than pancreatic adenocarcinoma, depending on tumor grade and stage. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs.

Higher stages generally indicate more advanced disease and poorer prognosis.

619.6 Complications

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

619.7 Diagnosis

Diagnosis typically involves hormone levels for functional tumors and imaging including CT, MRI, and somatostatin receptor scintigraphy (Octreoscan) or Ga-68 DOTATATE PET/CT. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade.

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

619.8 Prevention

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

619.9 Treatment Options

- Treatment typically involves surgical removal for localized disease
- Metastatic disease doctors typically treat it with somatostatin analogs (octreotide, lanreotide), targeted therapy (sunitinib, everolimus), PRRT (peptide receptor radionuclide therapy with Lu-177 DOTATATE), and chemotherapy.
- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

619.10 Living with It

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

619.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 620

Pancreatic Pseudocyst

620.1 Overview

Unlike true pancreatic cysts, pseudocysts lack an epithelial lining. Driven by pancreatic ductal disruption, extravasated pancreatic juice digests surrounding tissues until contained by inflammatory adhesions of adjacent organs (stomach, duodenum, transverse mesocolon). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

620.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

620.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

620.4 Signs and Symptoms

You may experience persistent abdominal pain, early satiety, nausea, vomiting, and a palpable epigastric mass following an episode of pancreatitis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

620.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

620.6 Complications

Complications include infection (abscess formation), gastric outlet obstruction, rupture, and pseudoaneurysm erosion into adjacent vessels.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

620.7 Diagnosis

- Doctors confirm the diagnosis through CT or MRI showing a well-circumscribed fluid collection with a mature fibrous wall. Treatment for asymptomatic pseudocysts is observation
- Symptomatic or enlarging cysts (> 6 cm) undergo using a thin camera tube cystogastrostomy, through the skin removal of fluid, or surgical cystojejunostomy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

620.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

620.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

620.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

620.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 621

Panic Disorder / Panic Attacks

621.1 Overview

Panic disorder has a lifetime prevalence of 3–5%, with a mean age of onset in early 20s to 30s. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

621.2 Causes

This condition happens because of a hypersensitive fear network centered on the amygdala, locus coeruleus (noradrenergic dysregulation), and periaqueductal gray, with faulty interoceptive conditioning and catastrophic misinterpretation of somatic sensations. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

621.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

621.4 Signs and Symptoms

You may experience recurrent unexpected panic attacks (discrete periods of intense fear or discomfort that reach a peak within minutes, with at least four of: palpitations, sweating, trembling, shortness of breath, sensations of choking, chest pain, nausea, dizziness, chills or heat sensations, paresthesias, derealization, fear of losing control or dying) plus at least one month of persistent concern about additional attacks.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

621.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

621.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

621.7 Diagnosis

Doctors diagnose this using clinical interview, ruling out cardiac, lung, and endocrine causes.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

621.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

621.9 Treatment Options

Treatment options include pharmacotherapy (SSRIs and SNRIs are first-line) and psychotherapy (CBT with interoceptive exposure is highly effective). Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

621.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

621.11 Outlook

The outlook is good with appropriate treatment. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 622

Parkinson Disease

622.1 Overview

Parkinson disease (PD) is a progressive neurodegenerative disorder affecting 1–2% of adults over 65. Geriatric-specific considerations include Treatment for orthostatic hypotension (common with levodopa and age-related autonomic dysfunction), careful use of dopamine agonists (increased risk of impulse control disorders, hallucinations, and sedation in older adults), and switching to levodopa monotherapy when possible. Deep brain stimulation (DBS) is effective but requires careful patient selection in older adults. Falls are a major concern, and physical therapy (LSVT BIG, balance training) is essential. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

622.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

622.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

622.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

622.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

622.6 Complications

In the geriatric population, PD is linked to higher rates of postural instability, falls, dementia, difficulty swallowing, and medication complications (motor fluctuations, dyskinesias, psychosis).

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

622.7 Diagnosis

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

622.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

622.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

622.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

622.11 Outlook

outlook is progressive. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 623

Parkinson's Disease

623.1 Overview

Parkinson's disease is a chronic, progressive neurodegenerative disorder affecting the dopaminergic neurons of the substantia nigra pars compacta in the midbrain. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

623.2 Causes

This condition happens because of loss of pigmented dopaminergic neurons and the presence of Lewy bodies composed of misfolded alpha-synuclein. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

623.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

623.4 Signs and Symptoms

You may experience cardinal motor manifestations—bradykinesia, resting tremor, rigidity, and postural instability—as well as non-motor symptoms such as loss of smell, constipation, REM sleep behavior disorder, depression, and autonomic dysfunction, which often precede motor manifestations by years. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

623.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Treatment focuses on dopamine replacement with levodopa/carbidopa, dopamine agonists, MAO-B inhibitors, and COMT inhibitors
- Advanced disease may be managed with deep brain stimulation of the subthalamic nucleus or globus pallidus interna.

623.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

623.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by response to dopaminergic therapy and DaT-SPECT imaging. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

623.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

623.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

623.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

623.11 Outlook

outlook is progressive, with disease-modifying therapies remaining under investigation. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 624

Paroxysmal Hemicrania

624.1 Overview

Paroxysmal hemicrania is a trigeminal autonomic cephalalgia characterized by strictly unilateral, severe pain attacks lasting 2–30 minutes, occurring several to many times per day (typically >5/day). Onset is typically in adulthood, with a slight female predominance. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

624.2 Causes

This condition happens because of activation of the trigeminovascular system with hypothalamic dysfunction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

624.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

624.4 Signs and Symptoms

You may experience unilateral pain with ipsilateral autonomic features identical to cluster headache. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

624.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

624.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

624.7 Diagnosis

- To make the diagnosis, doctors look for a therapeutic trial of indomethacin
- The defining feature is absolute response to indomethacin (75–150 mg/day).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

624.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

624.9 Treatment Options

Treatment focuses on indomethacin with digestive tract protection. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

624.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

624.11 Outlook

- The outlook is generally positive with treatment
- Episodic and chronic forms exist.

Chapter 625

Patau Syndrome (Trisomy 13)

625.1 Overview

Patau syndrome is a chromosomal disorder caused by trisomy 13, with an incidence of 1 in 10,000 live births. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

625.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

625.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

625.4 Signs and Symptoms

- You may experience severe intellectual disability, microcephaly, holoprosencephaly, cleft lip and palate, polydactyly, low-set ears, microphthalmia, and congenital heart disease. Median survival is 3–10 days
- Less than 10% survive to age 1 year.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

625.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

625.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

625.7 Diagnosis

Doctors diagnose this using karyotype.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

625.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

625.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

625.10 Living with It

Treatment typically involves supportive and palliative.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 626

Patent Ductus Arteriosus (PDA) in Prematurity

626.1 Overview

Patent ductus arteriosus (PDA) is the persistent patency of the fetal vascular connection between the descending aorta and the lung artery. In preterm infants, ductal closure is delayed, and a hemodynamically significant PDA occurs in up to 60% of extremely preterm infants. Left-to-right shunting through the PDA leads to lung overcirculation, increased left ventricular preload, and systemic steal. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

626.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

626.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

626.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

626.5 Progression and Stages

You may experience a continuous murmur, bounding pulses, wide pulse pressure, respiratory deterioration, and hypotension. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

626.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

626.7 Diagnosis

Doctors diagnose this using echocardiography. Treatment options include fluid restriction, diuretics, pharmacologic closure with indomethacin or ibuprofen, and surgical ligation for failed medical therapy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

626.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

626.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

626.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

626.11 Outlook

The outlook is generally positive with closure. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 627

Pediatric Appendicitis

627.1 Overview

Appendicitis is the most common surgical emergency in children (see Chapter 16, Diseases of the Intestines for comprehensive coverage). In pediatric patients, diagnosis may be more challenging due to atypical presentations, especially in younger children. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

627.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

627.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

627.4 Signs and Symptoms

You may experience periumbilical pain migrating to the right lower quadrant, fever, anorexia, nausea and vomiting, and rebound tenderness. Ultrasound is the first-line imaging modality. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

627.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

627.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

627.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

627.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

627.9 Treatment Options

Treatment typically involves using small incisions and a camera appendectomy with perioperative antibiotics. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

627.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

627.11 Outlook

- Most people recover well with timely treatment
- Young children have higher rates of perforation.

Chapter 628

Pelvic Inflammatory Disease

628.1 Overview

Pelvic inflammatory disease (PID) is a gynecologic infection recognized as the most common preventable cause of infertility and ectopic pregnancy worldwide. It is primarily sexually transmitted, with *Chlamydia trachomatis* and *Neisseria gonorrhoeae* as the predominant pathogens. The CDC-recommended treatment regimen includes ceftriaxone 500 mg IM plus doxycycline 100 mg BID for 14 days, with metronidazole for anaerobic coverage. All sexual partners should be treated. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

628.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

628.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

628.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

628.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

628.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

628.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

628.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

628.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

628.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

628.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 629

Pelvic Inflammatory Disease

629.1 Overview

Pelvic inflammatory disease is an infection of the female upper genital tract, including the endometrium (endometritis), fallopian tubes (salpingitis), ovaries (oophoritis), and pelvic peritoneum. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

629.2 Causes

This condition happens because of ascending infection from the lower genital tract. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

629.3 Risk Factors

Risk factors include multiple sexual partners, prior PID, younger age, and lack of barrier contraception. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

629.4 Signs and Symptoms

You may experience lower abdominal pain (often bilateral), cervical motion tenderness ("chandelier sign"), uterine tenderness, adnexal tenderness, and abnormal cervical/vaginal discharge. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

629.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

629.6 Complications

It is the most common and most serious complication of sexually transmitted infections, with *Chlamydia trachomatis* and *Neisseria gonorrhoeae* being the most common causative organisms. The outlook is good with prompt treatment, though complications include tubo-ovarian abscess, infertility, and ectopic pregnancy.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

629.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (CDC criteria). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

629.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

629.9 Treatment Options

Treatment options include empiric antibiotics covering *N. gonorrhoeae* and *C. trachomatis*: ceftriaxone plus doxycycline plus metronidazole. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

629.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

629.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 630

Pemphigus Vulgaris

630.1 Overview

Pemphigus vulgaris (PV) is a potentially life-threatening autoimmune blistering disease characterized by IgG autoantibodies against desmoglein 3 (and sometimes desmoglein 1), components of desmosomes that maintain keratinocyte adhesion. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

630.2 Causes

This condition happens because of autoantibody-mediated loss of keratinocyte adhesion leading to intraepidermal blisters (suprabasal acantholysis). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

630.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

630.4 Signs and Symptoms

- You may experience mucosal involvement (oral erosions) in 50–70% of cases, often preceding cutaneous lesions
- Skin blisters are flaccid, easily ruptured (positive Nikolsky sign), and heal without scarring.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

630.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

630.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

630.7 Diagnosis

- To make the diagnosis, doctors look for skin biopsy showing suprabasal acantholysis, direct immunofluorescence showing intercellular IgG and C3 deposition ("chicken-wire" pattern), and detection of anti-desmoglein antibodies by ELISA. Management requires immunosuppression: systemic corticosteroids combined with steroid-sparing agents (azathioprine, mycophenolate, rituximab)
- Rituximab (anti-CD20) has become a The first-choice treatment, often achieving drug-free remission.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

630.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

630.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

630.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

630.11 Outlook

The outlook has improved significantly with modern immunosuppressive therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 631

Penile Cancer

631.1 Overview

Penile cancer is rare in developed countries (0.5–1 per 100,000) but accounts for up to 10% of male malignancies in parts of Africa, South America, and Asia where circumcision is less common and HPV prevalence is higher, with squamous cell carcinoma representing >95% of cases. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

631.2 Causes

This condition happens because of cancerous transformation of squamous epithelium. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

631.3 Risk Factors

Risk factors include lack of neonatal circumcision, phimosis, poor hygiene, HPV infection (types 16, 18, 45), smoking, and chronic balanoposthitis, with penile intraepithelial neoplasia (PeIN) being the precursor.

- Advancing age

- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

631.4 Signs and Symptoms

You may experience a painless ulcer, mass, or wart-like lesion on the glans, prepuce, or shaft. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

631.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage,

small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

631.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

631.7 Diagnosis

Doctors diagnose this using biopsy, with staging using the TNM system. Treatment for the primary tumor includes topical therapy for PeIN, laser ablation or wide local removal for superficial tumors, and partial or total penectomy for invasive tumors. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)

- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

631.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

631.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies

- Surveillance: Active monitoring after definitive treatment

631.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

631.11 Outlook

- Most people recover well for organ-confined disease
- 5-year survival drops to 50% with nodal involvement.

Chapter 632

Penile Fracture

632.1 Overview

Penile fracture is a traumatic rupture of the tunica albuginea of the corpus cavernosum, occurring during erectile activity when the penis is forcibly bent, most commonly during vigorous sexual intercourse or masturbation. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

632.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

632.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

632.4 Signs and Symptoms

You may experience a sudden snapping sound (audible crack), immediate detumescence, severe pain, and rapid development of swelling and bruising (the "eggplant deformity"), with urethral injury occurring in 10–20% of cases. This condition happens because of traumatic rupture of the tunica albuginea. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

632.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

632.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

632.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

632.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

632.9 Treatment Options

Treatment typically involves immediate surgical repair: degloving of the penis, evacuation of hematoma, identification and primary closure of the tunical defect. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

632.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

632.11 Outlook

Most people recover well with early surgical repair (erectile function preserved in >90%), while conservative management leads to high rates of erectile dysfunction. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 633

Peptic Ulcer Bleeding

633.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

633.2 Causes

This condition happens because of erosion of the ulcer into underlying blood vessels. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

633.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

633.4 Signs and Symptoms

Symptoms can range from hematemesis (vomiting of blood, either bright red or "coffee-ground") to melena (black, tarry stools) or occult GI bleeding with iron deficiency anemia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

633.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

633.6 Complications

Peptic ulcer bleeding is the most common complication of peptic ulcer disease and the most frequent cause of upper digestive tract bleeding, accounting for approximately 50% of all upper GI bleeds.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

633.7 Diagnosis

Doctors diagnose this based on upper endoscopy, which identifies stigmata of recent bleeding including active bleeding, visible vessel, adherent clot, and flat pigmented spot. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

633.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

633.9 Treatment Options

Treatment focuses on hemodynamic resuscitation (IV fluids, blood transfusion), intravenous PPI infusion (to raise gastric pH and stabilize clots), and urgent endoscopy (within 24 hours) with using a thin camera tube therapy (injection, thermal coagulation, or hemoclips) for high-risk stigmata. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

633.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

633.11 Outlook

- In most cases, the outlook is favorable with intervention
- Re-bleeding occurs in 5–10% of cases and may require repeat endoscopy or angiographic embolization.

Chapter 634

Peptic Ulcer Disease

634.1 Overview

Peptic ulcer disease (PUD) involves mucosal defects extending through the muscularis mucosae in the stomach or duodenum. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

634.2 Causes

This condition happens because of mucosal barrier disruption caused primarily by Helicobacter pylori infection (accounting for 70–90% of duodenal and 60–70% of gastric ulcers) and chronic nonsteroidal anti-inflammatory drug (NSAID) use. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

634.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

634.4 Signs and Symptoms

You may experience epigastric pain: duodenal ulcer pain typically improves with food ingestion and recurs 2–3 hours later, whereas gastric ulcer pain is often precipitated by meals. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

634.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

634.6 Complications

Complications include digestive tract bleeding, ulcer perforation (presenting with sudden severe pain and pneumoperitoneum), and gastric outlet obstruction.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

634.7 Diagnosis

Doctors diagnose this using upper endoscopy with mucosal biopsy for H. pylori testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

634.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

634.9 Treatment Options

Treatment options include proton pump inhibitors (PPIs) and H. pylori eradication therapy (quadruple therapy with PPI, bismuth, metronidazole, tetracycline, or triple therapy). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

634.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

634.11 Outlook

Most people recover well with eradication. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 635

Pericarditis

635.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Pericarditis is an inflammation of the pericardium, the fibrous sac surrounding the heart
- Acute pericarditis is the most common pericardial disease, with viral cause being most frequent (Coxsackievirus, echovirus, adenovirus, SARS-CoV-2). Other causes include bacterial infection, uremia, autoimmune diseases, malignancy, and post-MI (Dressler syndrome).

635.2 Causes

This condition happens because of inflammation of the pericardial layers. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

635.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

635.4 Signs and Symptoms

You may experience sharp, pleuritic chest pain that worsens with inspiration and improves with sitting forward, pericardial friction rub on auscultation, and diffuse ST-segment elevation on ECG. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

635.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

635.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection

- Chronic kidney disease from reduced perfusion

635.7 Diagnosis

Doctors diagnose this based on clinical features, ECG, and echocardiography (may show pericardial effusion). Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

635.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

635.9 Treatment Options

- Treatment options include NSAIDs (ibuprofen, aspirin) and colchicine for 3 months
- Corticosteroids are used for refractory or recurrent cases.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)

- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

635.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

635.11 Outlook

In most cases, the outlook is good, with most cases resolving with treatment. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 636

Perinatal Asphyxia / Hypoxic-Ischemic Encephalopathy (HIE)

636.1 Overview

Hypoxic-ischemic encephalopathy (HIE) is brain injury resulting from perinatal asphyxia—impaired cerebral blood flow and oxygen delivery around the time of birth. Incidence is 1–3 per 1000 live births in developed countries. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

636.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

636.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

636.4 Signs and Symptoms

You may experience low Apgar scores, metabolic acidosis, seizures, hypotonia, abnormal neurological examination, and multi-organ dysfunction. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

636.5 Progression and Stages

This condition happens because of primary energy failure (decreased ATP, cellular tissue death) followed by a secondary phase of inflammation, excitotoxicity, and apoptosis hours to days later. HIE is graded as mild (Sarnat Stage 1), moderate (Stage 2), or severe (Stage 3). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

636.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

636.7 Diagnosis

- Doctors diagnose this based on perinatal history, blood gas analysis, and amplitude-integrated EEG
- MRI within the first week shows characteristic patterns of injury.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

636.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

636.9 Treatment Options

Treatment typically involves therapeutic hypothermia (33.5°C for 72 hours), initiated within 6 hours of birth, which reduces mortality and neurodevelopmental disability. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

636.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

636.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 637

Periodic Fever Syndromes

637.1 Overview

Periodic fever syndromes (also called autoinflammatory diseases) are a group of inherited disorders of the innate immune system characterized by recurrent, self-limited episodes of fever and systemic inflammation in the absence of autoantibodies or antigen-specific T cells. They are distinct from autoimmune diseases and arise from dysregulation of the inflammasome and IL-1β production (the "IL-1opathies").

Familial Mediterranean Fever

Familial Mediterranean fever is the most common hereditary periodic fever syndrome, caused by mutations in the *MEFV* gene encoding pyrin, a component of the pyrin inflammasome, most prevalent in Mediterranean populations (Armenian, Turkish, Arab, Jewish—carrier frequency 1:5 in Armenians). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

637.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

637.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

637.4 Signs and Symptoms

Clinical features begin in childhood (90% before age 20) and consist of recurrent, self-limited attacks (12–72 hours) of high fever, serositis (peritonitis—severe abdominal pain resembling acute abdomen, pleuritis, pericarditis), arthritis (usually monoarticular of the lower extremity), and an erysipelas-like redness of the skin on the lower leg, with the diagnostic hallmark being a dramatic response to colchicine.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

637.5 Progression and Stages

The outlook varies from person to person, with amyloidosis risk reduced with effective anti-IL-1 therapy.

Cryopyrin-Associated Periodic Syndromes

Cryopyrin-associated periodic syndromes are a spectrum of autoinflammatory diseases caused by gain-of-function mutations in the *NLRP3* gene encoding cryopyrin (NALP3), a key component of the NLRP3 inflammasome, encompassing three overlapping phenotypes in order of increasing severity: familial cold autoinflammatory syndrome (FCAS—mildest), Muckle-Wells syndrome (MWS—intermediate), and neonatal-onset multisystem inflammatory disease/chronic infantile nervous system cutaneous articular syndrome (NOMID/CINCA—most severe), with an incidence of approximately 1 per 1,000,000. Treatment typically involves with IL-1 inhibitors: anakinra, canakinumab (long-acting, FDA-approved for CAPS), and rilonacept (IL-1 trap), which are dramatically effective, normalizing symptoms and inflammatory markers within hours and preventing AA amyloidosis and hearing loss progression.

Hyper-IgD Syndrome

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Hyper-IgD syndrome (also known as mevalonate kinase deficiency, MKD) is an autosomal recessive periodic fever syndrome caused by mutations in the *MVK* gene encoding mevalonate kinase, an enzyme in the cholesterol biosynthesis pathway, leading to accumulation of mevalonic acid and activation of the inflammasome, with approximately 300 reported cases, predominantly in Northern European populations (especially Dutch). The outlook is good
- Attacks decrease in frequency and severity with age, and life expectancy is normal.

637.6 Complications

The most serious long-term complication is amyloidosis (AA type) if untreated.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

637.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (Tel Hashomer criteria) or by genetic confirmation of biallelic *MEFV* mutations.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

637.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

637.9 Treatment Options

Treatment typically involves daily colchicine (0.5–2 mg/day), which prevents attacks and amyloidosis in 95% of patients, with IL-1 inhibitors (anakinra, canakinumab) used for colchicine-resistant FMF. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications

Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

637.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

637.11 Outlook

- Most people recover well with colchicine adherence
- AA amyloidosis risk is nearly eliminated.

Chapter 638

Periodic Limb Movement Disorder

638.1 Overview

Periodic limb movement disorder (PLMD) is a sleep-related movement disorder characterized by recurrent, stereotypical, involuntary movements of the lower extremities during sleep, most commonly involving dorsiflexion of the great toe, fanning of the toes, and flexion of the ankle, knee, or hip, occurring at intervals of 5–90 seconds. Prevalence increases with age, affecting 10–30% of adults over 60, but is much lower as a primary diagnosis. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

638.2 Causes

This condition happens because of unknown Disease mechanism, likely related to dopaminergic dysfunction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

638.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

638.4 Signs and Symptoms

You may experience sleep fragmentation from movements associated with microarousals, leading to daytime impairment. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

638.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

638.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

638.7 Diagnosis

To make the diagnosis, doctors look for polysomnography demonstrating a periodic limb movement index (PLMI) >15 per hour in adults (>5 in children) with clinically significant sleep disturbance, in the absence of RLS or other explanatory disorders. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

638.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

638.9 Treatment Options

Treatment typically involves indicated only when PLMS are associated with sleep disturbance and includes alpha-2-delta ligands (gabapentin, pregabalin) preferred over dopamine agonists due to augmentation risk. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

638.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

638.11 Outlook

outlook is chronic but manageable. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 639

Periodic Paralysis

639.1 Overview

Periodic paralysis is a rare autosomal dominant channelopathy characterized by episodic attacks of muscle weakness or paralysis. Hypokalemic periodic paralysis presents with attacks of flaccid paralysis with low serum potassium, triggered by carbohydrate-rich meals, rest after exercise, or stress. Hyperkalemic periodic paralysis features attacks with elevated potassium, triggered by fasting or exercise. Andersen-Tawil syndrome combines periodic paralysis with irregular heartbeats and skeletal developmental abnormalities. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

639.2 Causes

This condition happens because of mutations in ion channel genes (CACNA1S, SCN4A, KCNJ2). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

639.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

639.4 Signs and Symptoms

- You may experience attacks lasting hours to days
- Between attacks, patients may be asymptomatic or have mild fixed weakness.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

639.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

639.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

639.7 Diagnosis

Doctors diagnose this based on genetic testing, potassium levels during attacks, and provocation testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

639.8 Prevention

- Treatment options include potassium supplementation during attacks and acetazolamide or dichlorphenamide prophylaxis for hypokalemic type
- Inhaled beta-agonists, glucose, and avoidance of fasting for hyperkalemic type.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

639.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

639.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

639.11 Outlook

- The outlook varies from person to person
- Attacks are typically reversible with appropriate management.

Chapter 640

Peripheral Arterial Disease

640.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Peripheral arterial disease is atherosclerotic narrowing of the arteries supplying the extremities, predominantly the lower limbs, resulting in reduced blood flow
- It affects approximately 10–15% of adults over 50 and is a marker of systemic atherosclerosis.

640.2 Causes

This condition happens because of atherosclerotic plaque accumulation in peripheral arteries. Most patients are asymptomatic The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Intermittent claudication (cramping leg pain triggered by exertion and relieved by rest within 10 minutes) is the hallmark symptom
- Severe disease presents with critical limb reduced blood flow (rest pain, non-healing ulcers, gangrene).

640.3 Risk Factors

The most important risk factor is cigarette smoking, followed by diabetes mellitus, high blood pressure, hyperlipidemia, and chronic kidney disease. Your outlook depends on disease severity and risk factor modification. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

640.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

640.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

640.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

640.7 Diagnosis

Doctors diagnose this using the ankle-brachial index: a ratio of 0.90 or less is diagnostic. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

640.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

640.9 Treatment Options

- Treatment options include smoking cessation (most important intervention), antiplatelet therapy, statin therapy, supervised exercise programs, and Treatment for comorbidities
- Revascularization is indicated for lifestyle-limiting claudication or critical limb reduced blood flow.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

640.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

640.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 641

Peripheral Artery Disease

641.1 Overview

Peripheral artery disease (PAD) is a chronic atherosclerotic occlusion of systemic arteries supplying the lower extremities. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

641.2 Causes

This condition happens because of endothelial dysfunction and atheroma accumulation reducing tissue perfusion. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

641.3 Risk Factors

Risk factors include cigarette smoking, diabetes mellitus, high blood pressure, hyperlipidemia, and advanced age. Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

641.4 Signs and Symptoms

You may experience intermittent claudication (reproducible lower extremity muscle pain precipitated by exertion and relieved by rest), cold extremities, hair loss, thickened toenails, and resting pain or ischemic ulceration/gangrene (critical limb reduced blood flow). Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

641.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

641.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia

- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

641.7 Diagnosis

Your doctor confirms the diagnosis with an Ankle-Brachial Index (ABI) < 0.90, with duplex ultrasonography or CT angiography defining anatomical stenosis. History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

641.8 Prevention

Treatment options include lifestyle modification (smoking cessation, structured exercise program), antiplatelet therapy (aspirin or clopidogrel), statin therapy, cilostazol for claudication, and endovascular angioplasty/stenting or surgical bypass for severe or refractory disease.

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

641.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

641.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

641.11 Outlook

outlook is strongly correlated with coexisting coronary and cerebrovascular disease. With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 642

Periventricular Leukomalacia (PVL)

642.1 Overview

Periventricular leukomalacia (PVL) is ischemic tissue death of the periventricular white matter in preterm infants, resulting from the vulnerability of oligodendrocyte precursor cells to hypoxic-ischemic injury. It occurs in 3–10% of preterm infants and is a major cause of cerebral palsy, particularly spastic diplegia. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

642.2 Causes

This condition happens because of impaired perfusion in the watershed zone between the anterior, middle, and posterior cerebral arteries. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

642.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

642.4 Signs and Symptoms

You may experience neurodevelopmental impairment, with spastic diplegia as a major consequence. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

642.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

642.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

642.7 Diagnosis

Doctors diagnose this using cranial ultrasound (cystic echolucencies) or MRI. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

642.8 Prevention

Treatment focuses on prevention (avoiding hypotension, hypoxia, and hypocarbia).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

642.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

642.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

642.11 Outlook

- Outlook is guarded
- Long-term outcomes include spastic diplegia, visual impairment, and thinking and memory deficits.

Chapter 643

Persistent Depressive Disorder (Dysthymia)

643.1 Overview

Persistent depressive disorder (dysthymia) is a chronic form of depression with symptoms present most of the day, for more days than not, for at least 2 years (1 year in children and adolescents), with a lifetime prevalence of 3–6%. The condition is marked by depressed mood plus at least two of the following: poor appetite or overeating, insomnia or hypersomnia, low energy or fatigue, low self-esteem, poor concentration, or feelings of hopelessness, with symptoms being less severe than MDD but more persistent. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

643.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

643.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

643.4 Signs and Symptoms

You may experience being “depressed for as long as I can remember,” with double depression referring to MDD superimposed on pre-existing dysthymia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

643.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

643.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

643.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

643.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

643.9 Treatment Options

Treatment options include SSRIs/SNRIs (same as MDD) and psychotherapy (CBT, IPT). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

643.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

643.11 Outlook

outlook is guarded, with high rates of recurrence, and Treatment typically involves often recommended for at least 2 years. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 644

Persistent Pulmonary Hypertension of the Newborn (PPHN)

644.1 Overview

Persistent lung high blood pressure of the newborn (PPHN) is the failure of the lung vascular resistance to decrease normally after birth, leading to right-to-left shunting across the foramen ovale and ductus arteriosus and profound hypoxemia. Incidence is approximately 2 per 1000 live births. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

644.2 Causes

This condition happens because of failure of the normal postnatal drop in lung vascular resistance. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

644.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age

- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

644.4 Signs and Symptoms

You may experience severe hypoxemia out of proportion to the degree of lung disease, with a preductal-to-postductal oxygen saturation gradient. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

644.5 Progression and Stages

Your outlook depends on the underlying cause and severity. Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

644.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection

- Chronic kidney disease from reduced perfusion

644.7 Diagnosis

Doctors diagnose this using echocardiography demonstrating elevated lung artery pressure and right-to-left shunt. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

644.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

644.9 Treatment Options

Treatment options include optimization of lung inflation, sedation, inotropic support, inhaled nitric oxide, and ECMO for refractory cases. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting)

Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

644.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

644.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 645

Persistent Vegetative State

645.1 Overview

A persistent vegetative state is a condition of wakeful unconsciousness in which the patient has sleep-wake cycles and may open their eyes spontaneously but shows no evidence of awareness of self or environment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

645.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of severe diffuse brain injury, cardiac arrest, or massive stroke. When it persists beyond one month, it is termed persistent
- Beyond three months for non-traumatic and twelve months for traumatic etiologies, it is termed permanent.

645.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

645.4 Signs and Symptoms

You may experience spontaneous eye opening with sleep-wake cycles but no purposeful responses. Diagnosis is challenging and may be aided by functional neuroimaging (fMRI) and EEG. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

645.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

645.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

645.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

645.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

645.9 Treatment Options

- Treatment typically involves supportive
- Ethical and legal considerations surrounding withdrawal of life-sustaining treatment are complex. outlook for recovery of consciousness is poor once permanent criteria are met.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

645.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

645.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 646

Pertussis (Whooping Cough)

646.1 Overview

Pertussis is a highly contagious respiratory infection caused by *Bordetella pertussis*, a gram-negative coccobacillus, with 24 million cases and 160,000 deaths annually worldwide, with infants < 6 months having the highest morbidity and mortality, and transmission by respiratory droplets. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

646.2 Causes

This condition happens because of *B. pertussis* adhering to ciliated respiratory epithelial cells and producing pertussis toxin, which ADP-ribosylates G proteins and promotes lymphocytosis, with tracheal cytotoxin damaging ciliated epithelial cells. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

646.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

646.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

646.5 Progression and Stages

Treatment options include azithromycin (which does not alter the clinical course if begun after the paroxysmal stage but eradicates the organism). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

646.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

646.7 Diagnosis

Doctors diagnose this using PCR of nasopharyngeal swab (test of choice). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

646.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

646.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

646.10 Living with It

Most people recover well with supportive care, though pneumonia is the most common cause of death in infants.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

646.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 647

Peyronie's Disease

647.1 Overview

Peyronie's disease is an acquired penile curvature caused by the formation of fibrous plaques in the tunica albuginea of the corpora cavernosa, affecting 3–9% of men. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

647.2 Causes

This condition happens because of repetitive microtrauma during sexual activity in genetically susceptible individuals, leading to aberrant wound healing and excessive collagen deposition. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

647.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

647.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

647.5 Progression and Stages

Treatment for the acute phase includes conservative management with observation, oral therapy, and intralesional collagenase, while chronic stable disease may require traction therapy, vacuum devices, intralesional verapamil or interferon, or surgical correction. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

647.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

647.7 Diagnosis

- Doctors diagnose this using history and physical examination
- Ultrasound or MRI can quantify plaque size and calcium buildup.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

647.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

647.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

647.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

647.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 648

Phantom Limb Pain

648.1 Overview

Phantom limb pain is pain perceived in a body part that has been amputated. It occurs in 60–80% of amputees, typically within the first week after amputation, and persists in 50–80% at 2 years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

648.2 Causes

This condition happens because of peripheral neuroma formation with ectopic discharges, spinal cord reorganization, and cortical reorganization (invasion of the deafferented somatosensory cortex by adjacent cortical inputs). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

648.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

648.4 Signs and Symptoms

You may experience pain perceived in the absent limb, distinct from residual limb pain and non-painful phantom limb sensations. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

648.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

648.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

648.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

648.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

648.9 Treatment Options

Treatment typically involves challenging

- First-line treatments include gabapentin, pregabalin, TCAs, and tramadol
- Non-pharmacological approaches include mirror therapy, graded motor imagery, and virtual reality therapy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

648.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

648.11 Outlook

- The outlook varies from person to person
- Treatment typically involves often challenging.

Chapter 649

Pharyngitis

649.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Pharyngitis is an inflammation of the pharynx, most commonly caused by viral infections (rhinoviruses, coronaviruses, adenovirus, influenza, Epstein-Barr virus)
- Group A *Streptococcus pyogenes* (GAS) is the most important bacterial cause, accounting for 15–30% of pharyngitis cases in children and 5–15% in adults.

649.2 Causes

This condition happens because of infectious inflammation of the pharyngeal mucosa. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

649.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

649.4 Signs and Symptoms

Symptoms of GAS pharyngitis include acute onset of sore throat, fever, tonsillar exudates, tender anterior cervical lymphadenopathy, and palatal tiny red or purple spots, typically without cough. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

649.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

649.6 Complications

- Most people recover well with appropriate treatment
- Complications include peritonsillar abscess and poststreptococcal glomerulonephritis.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

649.7 Diagnosis

Doctors diagnose this based on the Centor/McIsaac score, rapid antigen detection testing, and throat culture. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

649.8 Prevention

Treatment focuses on penicillin or amoxicillin for GAS pharyngitis to prevent acute rheumatic fever.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

649.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

649.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

649.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 650

Phenylketonuria

650.1 Overview

Phenylketonuria (PKU) is an autosomal recessive disorder of phenylalanine metabolism caused by deficiency of phenylalanine hydroxylase (PAH), the enzyme that converts phenylalanine to tyrosine. The incidence is approximately 1 in 10,000–15,000 live births. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

650.2 Causes

This condition happens because of accumulation of phenylalanine and its metabolites in blood and tissues, causing impaired brain development and function. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

650.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

650.4 Signs and Symptoms

You may experience progressive developmental delay, intellectual disability, microcephaly, seizures, behavioral problems, a characteristic musty odor of urine and skin, and hypopigmentation (fair skin, light hair, blue eyes) due to impaired melanin synthesis. Newborns are asymptomatic at birth. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

650.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

650.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

650.7 Diagnosis

Doctors diagnose this using newborn screening (elevated phenylalanine greater than 120 µmol/L) confirmed by elevated blood phenylalanine and normal or low tyrosine. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

650.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

650.9 Treatment Options

Treatment focuses on lifelong dietary restriction of phenylalanine (elimination of high-protein foods and use of phenylalanine-free medical formulas)

- Sapropterin (BH4 supplementation) is effective in a subset of patients
- Pegvaliase is an enzyme substitution therapy for adults with poorly controlled PKU.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

650.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

650.11 Outlook

Most people recover well with early and consistent dietary treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 651

Pheochromocytoma

651.1 Overview

Pheochromocytoma is a catecholamine-secreting tumor arising from chromaffin cells of the adrenal medulla, accounting for 0.1–0.6% of hypertensive patients, with paragangliomas being extra-adrenal catecholamine-secreting tumors. Approximately 10% are bilateral, 10% are extra-adrenal, and 10% are cancerous, with up to 40% being hereditary (MEN2, von Hippel-Lindau, neurofibromatosis type 1, SDH mutations). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

651.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

651.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

651.4 Signs and Symptoms

You may experience the classic triad of episodic headache, sweating, and palpitations with paroxysmal or sustained high blood pressure. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

651.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

651.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

651.7 Diagnosis

Doctors diagnose this using plasma free metanephrines or 24-hour urinary metanephrines and catecholamines (high sensitivity), with imaging including CT or MRI of the abdomen and MIBG scan. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

651.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

651.9 Treatment Options

Treatment typically involves alpha-adrenergic blockade (phenoxybenzamine or doxazosin) for at least 10–14 days preoperatively, followed by beta-blockade (only after adequate alpha-blockade), volume expansion, and using small incisions and a camera adrenalectomy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

651.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

651.11 Outlook

Most people recover well after removal for non-cancerous disease. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 652

Phimosis and Paraphimosis

652.1 Overview

Phimosis is the inability to retract the foreskin over the glans penis, a normal finding in uncircumcised boys before age 5 (physiologic phimosis), while pathologic phimosis results from recurrent balanoposthitis, poor hygiene, or lichen sclerosus et atrophicus, causing scarring and a contracted, fibrous preputial ring. Paraphimosis is a urologic emergency in which the retracted foreskin cannot be returned to its normal position, constricting the glans and leading to vascular compromise, swelling, and pain, typically occurring after catheterization, sexual activity, or inadequate reduction following examination. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

652.2 Causes

This condition happens because of preputial abnormalities. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

652.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

652.4 Signs and Symptoms

- Symptoms of phimosis include difficulty with urination, recurrent infections, and painful erections
- Paraphimosis presents with a constricted glans and pain.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

652.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

652.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

652.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Treatment for paraphimosis involves gentle manual compression of the edematous glans followed by reduction
- Elective circumcision is the The main treatment for pathologic phimosis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

652.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

652.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

652.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

652.11 Outlook

Most people recover well with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 653

Phyllodes Tumor

653.1 Overview

Phyllodes tumors are rare fibroepithelial tumors that can be non-cancerous, borderline, or cancerous (10–15% of all phyllodes tumors), typically presenting as large, rapidly growing, painless breast masses, more common in women 40–50 years of age. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

653.2 Causes

This condition happens because of stromal growth in fibroepithelial tissue. Histologically, they are distinguished from fibroadenomas by increased stromal cellularity, stromal overgrowth, atypia, and mitotic activity. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce

DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

653.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

653.4 Signs and Symptoms

You may experience a rapidly enlarging breast mass. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

653.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

653.6 Complications

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

653.7 Diagnosis

Doctors diagnose this using core needle biopsy and definitive surgical removal with wide margins (>1 cm) is required for all phyllodes tumors, regardless of grade. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

653.8 Prevention

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains

- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

653.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

653.10 Living with It

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

653.11 Outlook

- The outlook is good with complete removal
- Cancerous phyllodes tumors can metastasize hematogenously.

Chapter 654

Physical and Rehabilitation Approaches

654.1 Overview

Physical therapy is essential for chronic pain management. Therapeutic exercise includes range of motion, stretching, strengthening, and aerobic conditioning. Graded activity programs gradually increase activity levels based on time rather than pain, countering fear-avoidance behavior. Manual therapy (joint mobilization, soft tissue mobilization, myofascial release) addresses musculoskeletal dysfunction. Modalities such as transcutaneous electrical nerve stimulation (TENS), ultrasound, heat, cold, and traction provide short-term symptomatic relief. Occupational therapy focuses on activity modification, ergonomics, energy conservation, and adaptive equipment to improve function in daily activities. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

654.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

654.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

654.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

654.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

654.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

654.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

654.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

654.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

654.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

654.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 655

Pilonidal Sinus

655.1 Overview

Pilonidal sinus (pilonidal disease) is a chronic inflammatory condition affecting the skin and subcutaneous tissue of the intergluteal cleft (natal cleft), characterized by hair-containing sinus tracts. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

655.2 Causes

This condition happens because of mechanical occlusion of hair follicles in the sacrococcygeal region, leading to follicle rupture, foreign-body granulomatous reaction to entrapped shed hairs, and secondary bacterial infection (Staphylococcus aureus, anaerobes). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

655.3 Risk Factors

Risk factors include male sex, hirsutism, deep natal cleft, sedentary lifestyle, and obesity. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

655.4 Signs and Symptoms

Symptoms can range from an asymptomatic small pit in the natal cleft to an acute pilonidal abscess (severe localized pain, redness of the skin, fluctuating mass, fever) or chronic purulent sinus removal of fluid. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

655.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

655.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

655.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Treatment for acute abscess requires prompt incision and removal of fluid
- Chronic sinus disease doctors typically treat it with surgical removal or unroofing (lay-open) with primary closure or secondary healing, accompanied by local hair removal.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

655.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

655.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

655.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

655.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 656

Pituitary Adenoma

656.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

656.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of clonal growth of pituitary cells. Functioning adenomas secrete excess hormones: prolactinomas (galactorrhea, amenorrhea, hypogonadism), GH-secreting adenomas (acromegaly or gigantism), ACTH-secreting adenomas (Cushing's disease), and TSH-secreting adenomas (central hyperthyroidism)
- Non-functioning adenomas present with mass effects including bitemporal hemianopia from optic chiasm compression, headaches, and hypopituitarism.

656.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

656.4 Signs and Symptoms

Your symptoms depend on hormone secretion and tumor size.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

656.5 Progression and Stages

Pituitary adenomas are non-cancerous tumors of the anterior pituitary gland, classified as microadenomas (less than 10 mm) or macroadenomas (10 mm or greater). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

656.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

656.7 Diagnosis

- Diagnosis typically involves hormonal evaluation and MRI of the sella. Management varies: prolactinomas respond to dopamine agonists (cabergoline)
- Other functioning and non-functioning adenomas typically require transsphenoidal surgery, with radiation for residual or recurrent disease.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

656.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

656.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

656.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

656.11 Outlook

In most cases, the outlook is favorable. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy.

Chapter 657

Placenta Accreta Spectrum

657.1 Overview

Incidence has risen dramatically with increasing cesarean rates, now 1 in 500–1000 pregnancies. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

657.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

657.3 Risk Factors

The strongest risk factor is placenta previa in the setting of prior cesarean delivery. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

657.4 Signs and Symptoms

You may experience difficulty with placental separation at delivery. Prenatal diagnosis by ultrasound and MRI allows planned management. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

657.5 Progression and Stages

Placenta accreta spectrum (PAS) is abnormal placental adherence to the myometrium due to a deficient decidual layer, classified by depth of invasion: accreta, increta, and percreta. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

657.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

657.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

657.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

657.9 Treatment Options

Treatment typically involves controversial but increasingly favors conservative approaches when possible: planned cesarean hysterectomy with the placenta left in situ, or delayed hysterectomy after uterine artery embolization. Massive transfusion protocols and multidisciplinary teams are essential. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

657.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

657.11 Outlook

Your outlook depends on depth of invasion and preparedness. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 658

Placenta Previa

658.1 Overview

Placenta previa is the implantation of the placenta over or near the internal cervical os, occurring in approximately 1 in 200 pregnancies at term. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

658.2 Causes

This condition happens because of abnormal trophoblast implantation in the lower uterine segment. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

658.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

658.4 Signs and Symptoms

You may experience painless vaginal bleeding in the second or third trimester, typically after 28 weeks gestation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

658.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Doctors diagnose this using transvaginal ultrasound
- Placenta previa is classified as complete, partial, marginal, or low-lying.

658.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

658.7 Diagnosis

Treatment for asymptomatic women is conservative: pelvic rest, avoidance of intercourse and vaginal exams, and elective cesarean delivery at 36–37 weeks. Massive bleeding may require emergency cesarean delivery, transfusion, and potentially hysterectomy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

658.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

658.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

658.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

658.11 Outlook

The outlook is generally positive with planned cesarean delivery. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 659

Placental Abruption

659.1 Overview

Placental abruption (abruptio placentae) is the premature separation of the normally implanted placenta from the uterine wall before delivery, occurring in approximately 1% of pregnancies. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

659.2 Causes

This condition happens because of rupture of decidual spiral arteries, causing retroplacental hematoma formation and further separation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

659.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

659.4 Signs and Symptoms

- You may experience vaginal bleeding (80%, though 20% have concealed abruption), sudden sustained abdominal pain, uterine hypertonicity ("wooden" uterus), fetal distress, and signs of hypovolemic shock that may appear out of proportion to visible blood loss. Diagnosis is primarily clinical
- Ultrasound has limited sensitivity (50%).
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

659.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- The right treatment depends on severity and gestational age: mild abruption at term is managed by delivery
- Severe abruption requires immediate cesarean delivery with massive transfusion protocol.

659.6 Complications

- Your outlook depends on severity
- Complications include DIC, involving bleeding shock, acute kidney injury, and fetal death (10–20%).
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

659.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

659.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

659.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

659.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

659.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 660

Plague (Yersinia pestis)

660.1 Overview

Plague is a zoonotic infection caused by *Yersinia pestis*, a gram-negative, bipolar-staining coccobacillus, with 1000–5000 cases annually worldwide, endemic in Madagascar, Democratic Republic of Congo, and Peru, transmitted from infected rodents to humans by fleas, and is a category A bioterrorism agent. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

660.2 Causes

This condition happens because of *Y. pestis* being injected by flea bite, phagocytosed by macrophages, and resisting intracellular killing via a type III secretion system, with virulence factors including the F1 capsular antigen and Yops. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

660.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

660.4 Signs and Symptoms

You may experience bubonic plague (most common, 80%): sudden fever, chills, headache, and tender, swollen, matted lymph nodes (buboes)

- Septicemic plague: high fever, hypotension, DIC, purpuric rash, and acral tissue death
- Pneumonic plague: fever, productive cough, hemoptysis, and rapidly progressive respiratory failure.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

660.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

660.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

660.7 Diagnosis

Doctors diagnose this using Gram stain and culture from bubo aspirate, blood, or sputum. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

660.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

660.9 Treatment Options

Treatment options include streptomycin or gentamicin, with alternatives including doxycycline, ciprofloxacin, and levofloxacin. outlook: bubonic 5–15% mortality (treated)

- Septicemic 30–50%
- Pneumonic nearly 100% if untreated $>$ 24 hours.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

660.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

660.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 661

Plantar Fasciitis

661.1 Overview

It affects approximately 10% of the population at some point in their life, with peak incidence between ages 40 and 60. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

661.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

661.3 Risk Factors

Risk factors include obesity, flat feet, high arches, prolonged standing, running, and tight Achilles tendon. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

661.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance

or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

661.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Plantar fasciitis is the most common cause of heel pain in adults, resulting from overuse injury and deterioration of the plantar fascia, a thick band of connective tissue originating from the medial tubercle of the calcaneus and inserting into the toes. This condition happens because of repetitive microtrauma and deterioration of the plantar fascia. The classic The condition usually appears as sharp heel pain with the first steps in the morning or after rest ("start-up pain"), which improves with activity but may worsen with prolonged standing
- Pain is maximal at the medial heel.

661.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

661.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Imaging (X-ray, ultrasound, or MRI) is rarely needed but may show calcaneal spur (present in 50% of cases but often asymptomatic) or plantar fascia thickening.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

661.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

661.9 Treatment Options

Treatment options include rest, activity modification, stretching exercises (calf and plantar fascia), ice massage, orthotics, and night splints

- Corticosteroid injection may provide short-term relief but should be limited
- Extracorporeal shockwave therapy is an alternative for refractory cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

661.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

661.11 Outlook

- Most people recover well
- Most cases resolve within 6–18 months with conservative management.

Chapter 662

Pleurisy

662.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Pleurisy is an inflammation of the pleural membranes causing sharp, stabbing chest pain that worsens with breathing, coughing, or sneezing
- Causes include viral infections (most common), bacterial pneumonia with parapneumonic effusion, lung blockage by a blood clot, autoimmune diseases, malignancy, tuberculosis, and uremia.

662.2 Causes

This condition happens because of inflammation of the parietal and visceral pleura. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

662.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

662.4 Signs and Symptoms

- You may experience pleuritic chest pain (worse on inspiration, improved by holding breath) and possibly a pleural friction rub on auscultation
- Pleural effusion may develop.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

662.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

662.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

662.7 Diagnosis

Diagnosis typically involves chest X-ray, thoracic ultrasound, diagnostic thoracentesis with Light's criteria, and CT chest for underlying pathology. Management targets the underlying cause

- Pleuritic pain is managed with NSAIDs
- Large symptomatic effusions may require therapeutic thoracentesis
- Recurrent pleurisy may necessitate pleurodesis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

662.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

662.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

662.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

662.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 663

Pneumoconiosis

663.1 Overview

Pneumoconiosis is a group of interstitial lung diseases caused by inhalation of mineral dusts and fibrogenic particles. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

663.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Asbestosis results from asbestos fiber inhalation, causing basal-predominant interstitial scarring and pleural plaques, with a latency period of 10–20 years
- Silicosis (from silica dust) presents in chronic, accelerated, and acute forms
- Coal workers' pneumoconiosis results from coal dust exposure
- Berylliosis is a granulomatous disease from beryllium exposure. This condition happens because of accumulation of inhaled mineral dusts in the lungs, triggering chronic inflammation and scarring.

663.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

663.4 Signs and Symptoms

You may experience progressive shortness of breath and cough. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

663.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

663.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

663.7 Diagnosis

Doctors diagnose this using history of occupational exposure, HRCT, and lung function tests showing restriction with reduced DLCO. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

663.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

663.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

663.10 Living with It

Treatment focuses on exposure removal and supportive care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

663.11 Outlook

- Your outlook depends on the type and duration of exposure
- These conditions are largely preventable through occupational safety measures.

Chapter 664

Pneumothorax

664.1 Overview

Pneumothorax is the accumulation of air in the pleural space, leading to partial or complete lung collapse. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

664.2 Causes

This condition happens because of rupture of subpleural apical blebs (primary spontaneous pneumothorax, typically in tall, thin young men), underlying lung disease (secondary spontaneous pneumothorax in COPD, cystic scarring), or chest trauma. Tension pneumothorax occurs when a one-way valve mechanism allows air to accumulate continuously under pressure, causing mediastinal shift and hemodynamic collapse. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

664.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

664.4 Signs and Symptoms

You may experience sudden-onset ipsilateral pleuritic chest pain and shortness of breath, with hyperresonance and decreased breath sounds on the affected side. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

664.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

664.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

664.7 Diagnosis

Doctors diagnose this using chest radiograph showing a visceral pleural line without peripheral lung markings

- Tension pneumothorax is a clinical diagnosis requiring immediate needle decompression at the second intercostal space in the midclavicular line prior to imaging. Treatment for small spontaneous pneumothoraces involves observation and oxygen administration
- Larger pneumothoraces require tube thoracostomy (chest tube insertion).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

664.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

664.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

664.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

664.11 Outlook

The outlook is good with prompt treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 665

Poliomyelitis

665.1 Overview

Poliomyelitis is a highly infectious viral disease caused by poliovirus (serotypes 1, 2, 3) of the *Enterovirus* genus, with endemic transmission persisting in Afghanistan and Pakistan. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

665.2 Causes

This condition happens because of fecal-oral transmission, viral replication in the oropharynx and gut, and hematogenous spread to the central nervous system, where it infects and destroys anterior horn cells of the spinal cord and motor nuclei of the brainstem. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

665.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

665.4 Signs and Symptoms

You may experience asymptomatic infection ($> 90\%$), abortive poliomyelitis (febrile illness with sore throat, nausea), non-paralytic aseptic meningitis (1–5%), and paralytic poliomyelitis (0.1–2%) presenting with asymmetric, flaccid paralysis of lower limbs more than upper limbs, areflexia, without sensory loss. Bulbar paralysis affects cranial nerves and respiratory muscles. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

665.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

665.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

665.7 Diagnosis

Doctors diagnose this based on stool viral culture, RT-PCR of stool or CSF, and serology. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

665.8 Prevention

- Treatment typically involves supportive: physical therapy, orthotics, respiratory support for bulbar involvement, and prevention of complications. outlook for paralytic poliomyelitis: 10% die from respiratory paralysis, 30% recover some function, 60% have permanent paralysis. Prevention is through OPV (Sabin) or IPV (Salk) vaccination
- Global eradication is 99.9% complete.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification

- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

665.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

665.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

665.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 666

Polyarteritis Nodosa

666.1 Overview

Polyarteritis nodosa is a systemic necrotizing vasculitis affecting medium-sized arteries, characterized by inflammation and fibrinoid tissue death of the vessel wall, with aneurysm formation and blood clot formation, sparing small vessels (capillaries, venules) and not associated with ANCA, with an incidence of 2–3 per million, a slight male predominance, and affecting all ages. The condition involves immune complex deposition in the arterial wall (especially in HBV-associated PAN), complement activation, and neutrophil recruitment leading to vessel wall tissue death, with approximately 10–20% of cases associated with chronic hepatitis B virus (HBV) infection. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

666.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

666.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

666.4 Signs and Symptoms

You may experience constitutional symptoms (fever, weight loss, muscle pain, joint pain), mononeuritis multiplex (the most characteristic neurological manifestation), livedo reticularis, testicular pain, kidney artery involvement (kidney tissue death due to lack of blood supply, high blood pressure, but not glomerulonephritis—distinguishing PAN from microscopic polyangiitis), mesenteric reduced blood flow, and cutaneous manifestations (nodules, ulcers, gangrene). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

666.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

666.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

666.7 Diagnosis

- To make the diagnosis, doctors look for biopsy of affected tissue showing necrotizing arteritis with fibrinoid tissue death, or angiography showing microaneurysms and irregular stenoses. Treatment for HBV-negative PAN is high-dose corticosteroids plus cyclophosphamide for severe disease
- HBV-associated PAN requires antiviral therapy plus a shorter course of corticosteroids and plasmapheresis.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

666.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

666.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

666.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

666.11 Outlook

The outlook is good with treatment, but severe cases can be fatal. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 667

Polycystic Ovary Syndrome

667.1 Overview

Polycystic ovary syndrome (PCOS) is the most common cause of anovulatory infertility and oligomenorrhea. Key management considerations include endometrial protection from unopposed estrogen: annual ultrasound assessment of endometrial thickness and biopsy if indicated, with progestins or combined oral contraceptives for cycle regulation. Long-term health risks include a 3–7-fold increased risk of endometrial cancer, 50–70% lifetime risk of metabolic syndrome, and increased cardiovascular risk. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

667.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

667.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

667.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

667.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

667.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

667.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

667.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

667.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

667.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 668

Polycystic Ovary Syndrome

668.1 Overview

Polycystic ovary syndrome is the most common endocrine disorder in women of reproductive age, affecting 6–12%. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

668.2 Causes

This condition happens because of insulin resistance (present in 50–70%), hyperinsulinemia, and relative hyperandrogenism. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

668.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

668.4 Signs and Symptoms

You may experience menstrual irregularity, infertility, hirsutism, acne, obesity (60–80%), metabolic syndrome, type 2 diabetes mellitus, and increased cardiovascular risk.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

668.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

668.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

668.7 Diagnosis

To make the diagnosis, doctors look for two of three Rotterdam criteria: oligo-ovulation or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovarian morphology on ultrasound, with exclusion of other causes.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

668.8 Prevention

Treatment typically involves tailored to goals: combined oral contraceptives for menstrual regulation and hyperandrogenism, metformin or inositol for insulin resistance, letrozole for ovulation induction in infertility, and lifestyle modification.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

668.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

668.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

668.11 Outlook

The outlook is good with appropriate management.

Chapter 669

Polycythemia Vera

669.1 Overview

Polycythemia vera is a Philadelphia chromosome-negative myeloproliferative tumor characterized by clonal growth of hematopoietic stem cells, predominantly producing excess red blood cells, and often accompanied by leukocytosis and thrombocytosis, with over 95% of cases harboring the JAK2 V617F mutation (or JAK2 exon 12 mutations). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

669.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

669.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

669.4 Signs and Symptoms

The WHO 2022 diagnostic criteria require hemoglobin >16.5 g/dL in men or >16.0 g/dL in women, or increased red cell mass, plus bone marrow biopsy showing hypercellularity with trilineage growth, or JAK2 mutation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

669.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

669.6 Complications

You may experience erythromelalgia (burning pain and redness of extremities), itching (especially aquagenic—after bathing), headache, plethora, hepatosplenomegaly, and thrombotic or involving bleeding complications.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

669.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

669.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

669.9 Treatment Options

Treatment options include phlebotomy (target hematocrit $<45\%$), low-dose aspirin, and cytoreductive therapy (hydroxyurea for high-risk patients), with ruxolitinib (JAK1/2 inhibitor) used for hydroxyurea-resistant or intolerant disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

669.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

669.11 Outlook

In most cases, the outlook is favorable but carries a risk of transformation to myelofibrosis (10–20%) or AML (5–10%). The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 670

Polyhydramnios and Oligohydramnios

670.1 Overview

Polyhydramnios is an excessive volume of amniotic fluid, defined by an amniotic fluid index (AFI) >24 cm or a deepest vertical pocket >8 cm, occurring in 1–2% of pregnancies. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

670.2 Causes

Causes include idiopathic (50–60%), maternal diabetes (10–15%), fetal anomalies impairing swallowing, and twin-twin transfusion syndrome. Oligohydramnios is deficient amniotic fluid (AFI <5 cm or deepest vertical pocket <2 cm), affecting 1–2% of pregnancies. Causes include fetal anomalies (kidney agenesis, posterior urethral valves, obstructive uropathy), preterm PROM, placental insufficiency, and post-term pregnancy. Severe oligohydramnios in the second trimester causes lung underdevelopment (Potter sequence) and limb compression deformities. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

670.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

670.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

670.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

670.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

670.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

670.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

670.9 Treatment Options

The right treatment depends on cause and gestational age. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

670.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

670.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 671

Polymyalgia Rheumatica

671.1 Overview

Polymyalgia rheumatica is an inflammatory disorder of the elderly, characterized by pain and stiffness of the shoulder and pelvic girdles, affecting individuals over 50 years of age (peak incidence 70–80 years), with a female predominance of 2–3:1. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

671.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

671.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

671.4 Signs and Symptoms

You may experience acute or subacute onset of severe morning stiffness lasting over 30 minutes in the shoulders, neck, and hips, with associated muscle pain, constitutional symptoms (low-grade fever, fatigue, weight loss), and difficulty raising arms or rising from a chair, with peripheral arthritis (knees, wrists) occurring in 25–50%. PMR overlaps significantly with giant cell arteritis (GCA): 15–30% of PMR patients have concurrent GCA, and 40–60% of GCA patients have PMR symptoms. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

671.5 Progression and Stages

The cause is unknown, but genetic associations with HLA-DR4 and environmental triggers have been implicated, with Disease mechanism involving local inflammation of proximal joints and extra-articular synovial structures, infiltration of macrophages and CD4+ T cells, elevated IL-6 levels, and increased acute-phase reactants. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

671.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

671.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by elevated ESR (often >50–100 mm/hr) and CRP, with imaging (ultrasound, MRI) showing synovitis, bursitis, and tenosynovitis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

671.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

671.9 Treatment Options

Treatment typically involves with low-to-moderate dose corticosteroids (prednisone 15–25 mg/day), typically with rapid clinical response within 24–48 hours and slow tapering over 12–18 months. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

671.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

671.11 Outlook

The outlook is good, with most patients achieving remission within 1–3 years. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 672

Polymyositis

672.1 Overview

Polymyositis is an idiopathic inflammatory myopathy characterized by symmetric proximal muscle weakness and endomysial CD8+ T-cell infiltration of muscle fibers. It primarily affects adults, with a slight female predominance. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

672.2 Causes

This condition happens because of autoimmune-mediated muscle fiber destruction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

672.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

672.4 Signs and Symptoms

You may experience progressive symmetric proximal muscle weakness (difficulty rising from chairs, climbing stairs, lifting objects overhead), difficulty swallowing, and rarely respiratory muscle weakness. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

672.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

672.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

672.7 Diagnosis

Doctors diagnose this based on elevated CK (10–50 times normal), EMG showing myopathic changes, MRI showing muscle swelling, and definitive muscle biopsy. Muscle-specific autoantibodies (anti-Jo-1, anti-Mi-2, anti-MDA5) may support diagnosis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

672.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

672.9 Treatment Options

- Treatment options include corticosteroids and steroid-sparing immunosuppressive agents (methotrexate, azathioprine, mycophenolate)
- IVIG may be used for refractory disease.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

672.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

672.11 Outlook

The outlook varies from person to person, with many patients responding to immunosuppression. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 673

Polypharmacy

673.1 Overview

Polypharmacy—commonly defined as the concurrent use of five or more medications—affects 40–70% of older adults. It is linked to increased risk of adverse drug reactions (ADRs), drug-drug interactions, medication non-adherence, falls, thinking and memory impairment, hospitalization, and mortality. Factors driving polypharmacy include multimorbidity, multiple prescribers, lack of medication reconciliation, and prescribing cascades (where ADRs are misinterpreted as new medical conditions and treated with additional medications). Management requires regular medication review, deprescribing (systematic process of discontinuing potentially inappropriate medications), and the use of explicit criteria to guide prescribing decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

673.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

673.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

673.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

673.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

673.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

673.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

673.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

673.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

673.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

673.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 674

Porphyrias

674.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- The porphyrias are a group of inherited or acquired disorders of heme biosynthesis caused by deficiency of specific enzymes in the heme biosynthetic pathway, leading to accumulation of porphyrins or their precursors. Acute intermittent porphyria (AIP) is caused by porphobilinogen deaminase deficiency (autosomal dominant with low penetrance), the most common acute liver porphyria
- Attacks are precipitated by drugs, fasting, infections, and hormonal changes.

674.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

674.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

674.4 Signs and Symptoms

- You may experience severe abdominal pain (90%), neuropsychiatric symptoms, autonomic dysfunction, and peripheral motor neuropathy
- Cutaneous photosensitivity is absent.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

674.5 Progression and Stages

They are classified as liver or erythropoietic. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

674.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

674.7 Diagnosis

Doctors diagnose this using elevated urinary porphobilinogen (PBG) and aminolevulinic acid (ALA) during attacks. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

674.8 Prevention

Treatment options include removing precipitating factors, high-carbohydrate loading, intravenous hemin, and givosiran for prevention.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

674.9 Treatment Options

- Porphyria cutanea tarda (PCT) is caused by uroporphyrinogen decarboxylase deficiency, the most common porphyria, associated with alcohol, hepatitis C, estrogen therapy, and hemochromatosis
- You may experience blistering skin lesions on sun-exposed areas
- Treatment options include phlebotomy or low-dose hydroxychloroquine. Erythropoietic protoporphyria (EPP) is caused by ferrochelatase deficiency, causing photosensitivity in childhood.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

674.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

674.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 675

Portal Hypertension

675.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Portal high blood pressure is defined as a liver venous pressure gradient (HVPG) of 5 mmHg or greater, with clinically significant portal high blood pressure at HVPG of 10 mmHg or greater
- The most common cause is cirrhosis.

675.2 Causes

This condition happens because of increased resistance to portal blood flow. Consequences include esophageal and gastric varices, ascites, portosystemic collaterals, splenomegaly, and hepatorenal syndrome. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

675.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

675.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

675.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

675.6 Complications

Your symptoms depend on the complications present.

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

675.7 Diagnosis

- Your doctor confirms the diagnosis with HVPG measurement via right heart catheterization
- Non-invasive assessment includes platelet count, transient elastography, and endoscopy.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

675.8 Prevention

Treatment options include non-selective beta-blockers (propranolol, nadolol) and using a thin camera tube variceal band ligation for primary and secondary prophylaxis of variceal bleeding.

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

675.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)

- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

675.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

675.11 Outlook

Your outlook depends on the underlying liver disease and response to treatment. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 676

Post-Traumatic Headache

676.1 Overview

Post-traumatic headache (PTH) is a secondary headache disorder and the most common symptom following traumatic brain injury (TBI), affecting 30–90% of patients. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

676.2 Causes

This condition happens because of a neurometabolic cascade after TBI including axonal injury, excitotoxicity, mitochondrial dysfunction, altered cerebral blood flow, and activation of the trigeminovascular system. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

676.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

676.4 Signs and Symptoms

Clinical features most commonly mimic migraine or tension-type headache. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

676.5 Progression and Stages

It is classified as acute ($<$3 months) or persistent ($\geq$3 months). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

676.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

676.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam with history of TBI. Management follows standard primary headache guidelines with

amitriptyline, topiramate, and gabapentin commonly used
- Physical therapy for associated cervical dysfunction is essential.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

676.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

676.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

676.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

676.11 Outlook

- The outlook varies from person to person
- Most improve within 3–6 months but some remain symptomatic for years.

Chapter 677

Post-Traumatic Stress Disorder (PTSD)

677.1 Overview

Post-traumatic stress disorder has a lifetime prevalence of 6–8% in the general population, with higher rates in combat veterans (10–30%), sexual assault survivors (30–50%), and first responders (10–20%), and is twice as common in women. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

677.2 Causes

This condition happens because of an overly responsive amygdala (hypervigilance, exaggerated fear), a hypoactive medial prefrontal cortex (impaired fear extinction), reduced hippocampal volume, dysregulation of the HPA axis, and heightened sympathetic reactivity, with fear conditioning to trauma-related cues failing to extinguish normally. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

677.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

677.4 Signs and Symptoms

Clinical features require exposure to actual or threatened death, serious injury, or sexual violence, followed by four symptom clusters for at least 1 month: intrusion symptoms (recurrent, involuntary, intrusive memories, distressing dreams, dissociative reactions, flashbacks), avoidance (persistent avoidance of trauma-related stimuli), negative alterations in thinking and mood, and marked alterations in arousal and reactivity (irritable behavior, hypervigilance, exaggerated startle response, sleep disturbance).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

677.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

677.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

677.7 Diagnosis

Doctors diagnose this using clinical interview using DSM-5-TR criteria.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

677.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

677.9 Treatment Options

Treatment options include trauma-focused psychotherapy as first-line (TF-CBT, prolonged exposure, CPT, EMDR) and pharmacotherapy (SSRIs and SNRIs are FDA-approved). Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

677.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

677.11 Outlook

- Outlook is chronic in many cases
- About 30–40% recover fully, 30–40% have partial improvement, and 20–30% have persistent symptoms.

Chapter 678

Postherpetic Neuralgia

678.1 Overview

Postherpetic neuralgia (PHN) is neuropathic pain persisting for $\geq$3 months after the resolution of herpes zoster (shingles) rash. It occurs in 10–20% of patients with herpes zoster, increasing to >30% in patients over 60 years of age. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

678.2 Causes

This condition happens because of peripheral and central sensitization from varicella-zoster virus reactivation in the dorsal root ganglia, leading to nerve damage, inflammation, and neuronal hyperexcitability. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

678.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

678.4 Signs and Symptoms

You may experience burning, aching, electric, and lancinating pain with allodynia and hyperalgesia in a dermatomal distribution (most commonly thoracic). Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

678.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

678.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

678.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. First-line pharmacotherapy includes topical lidocaine 5% patches, pregabalin, gabapentin,

and TCAs
- Second-line includes tramadol, capsaicin 8% patches, and opioid analgesics.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

678.8 Prevention

- The outlook varies from person to person
- Prevention is the best treatment, with herpes zoster vaccination (recombinant zoster vaccine) reducing the incidence of PHN by approximately 90%.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

678.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

678.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

678.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 679

Postpartum Cardiomyopathy

679.1 Overview

Peripartum cardiomyopathy (PPCM) is an idiopathic dilated cardiomyopathy presenting with left ventricular systolic dysfunction (LVEF <45%) in the last month of pregnancy through five months postpartum, occurring in 1 in 1000–4000 pregnancies. The Disease mechanism involves oxidative stress, a 16-kDa prolactin cleavage fragment (vasoinhibin) that impairs cardiac microvascular function, and inflammation. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

679.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

679.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

679.4 Signs and Symptoms

You may experience shortness of breath, orthopnea, paroxysmal nocturnal shortness of breath, peripheral swelling, cough, and signs of left heart failure. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

679.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

679.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

679.7 Diagnosis

Doctors diagnose this using echocardiogram showing left ventricular dilation and reduced ejection fraction. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

679.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

679.9 Treatment Options

- Treatment typically involves standard heart failure therapy adjusted for pregnancy and lactation
- Bromocriptine may be used in some centers.
- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease

- Cardiac rehabilitation programs

679.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

679.11 Outlook

Recovery varies: 50–60% recover LVEF to ≥50% within 6 months. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 680

Postpartum Depression

680.1 Overview

Postpartum depression (PPD) is a major depressive episode with onset within 4–6 weeks of childbirth (DSM-5 specifies peripartum onset). The Disease mechanism involves a rapid drop in estrogen and progesterone after delivery, altered hypothalamic-pituitary-adrenal axis function, genetic predisposition, and psychosocial factors. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

680.2 Causes

Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

680.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

680.4 Signs and Symptoms

You may experience persistent sadness, anhedonia, crying spells, fatigue, sleep and appetite disturbance, feelings of worthlessness, difficulty bonding with the infant, and in severe cases, thoughts of harming self or baby.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

680.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

680.6 Complications

It is the most common complication of the puerperium, affecting 10–15% of postpartum women.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

680.7 Diagnosis

Doctors diagnose this using clinical assessment using the Edinburgh Postnatal Depression Scale (EPDS, score $\geq$10).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

680.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

680.9 Treatment Options

- Treatment options include psychotherapy and antidepressant medication (SSRIs—sertraline and fluoxetine are first-line)
- For postpartum psychosis (rare, 1–2 per 1000 deliveries), emergency psychiatric care is required.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

680.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

680.11 Outlook

The outlook is generally positive with treatment. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 681

Postpartum Endometritis

681.1 Overview

Postpartum endometritis is infection of the decidua and myometrium after delivery, most commonly following cesarean delivery (5–30%, versus 1–3% after vaginal delivery). This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

681.2 Causes

This condition happens because of ascending infection from the lower genital tract, most commonly polymicrobial with aerobes and anaerobes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

681.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

681.4 Signs and Symptoms

You may experience fever (typically within 24–72 hours of delivery), tachycardia, uterine tenderness, and foul-smelling lochia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

681.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

681.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

681.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

681.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

681.9 Treatment Options

- Treatment typically involves IV broad-spectrum antibiotics (clindamycin plus gentamicin is the standard regimen)
- If no response within 48 hours, add ampicillin for enterococcal coverage.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

681.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

681.11 Outlook

The outlook is generally positive with antibiotics. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 682

Postpartum Hemorrhage

682.1 Overview

Postpartum bleeding (PPH) is defined as blood loss >500 mL after vaginal delivery or >1000 mL after cesarean delivery. It is the leading cause of maternal mortality worldwide, accounting for 25% of maternal deaths. Causes are categorized by the “four Ts”: Tone (uterine atony—70–80% of cases), Trauma (lacerations, episiotomy, uterine rupture), Tissue (retained placenta, placenta accreta), and Thrombin (coagulopathy). This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

682.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

682.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

682.4 Signs and Symptoms

You may experience excessive bleeding after delivery with signs of hemodynamic compromise. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

682.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

682.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

682.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam. Management follows a systematic protocol: step 1—medical management (uterine massage, oxytocin, methergine, misoprostol, carboprost)
- Step 2—mechanical measures (uterine balloon tamponade, compression sutures)
- Step 3—surgical intervention (surgical opening of the abdomen, uterine artery ligation, hysterectomy). Concurrent Treatment options include aggressive fluid resuscitation, transfusion of blood products, and monitoring for DIC.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

682.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

682.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

682.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

682.11 Outlook

Your outlook depends on prompt recognition and treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 683

Postpartum Thyroiditis

683.1 Overview

Postpartum thyroiditis (PPT) is an autoimmune inflammation of the thyroid gland occurring within the first year after delivery, affecting 5–10% of women. It is caused by a rebound of immune activity after the relative immunosuppression of pregnancy resolves, in susceptible women who are positive for thyroid peroxidase (TPO) antibodies. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

683.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

683.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)

- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

683.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

683.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

- You may experience a transient thyrotoxic phase (4–8 weeks, 30% of cases) followed by a hypothyroid phase (4–12 weeks, 40%), with 25% remaining hypothyroid permanently. The thyrotoxic phase is managed symptomatically with beta-blockers (not antithyroid drugs)
- The hypothyroid phase doctors typically treat it with levothyroxine.

683.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension

- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

683.7 Diagnosis

Doctors diagnose this based on TSH and TPO antibody testing. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

683.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

683.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

683.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

683.11 Outlook

- The outlook is generally positive
- Recurrence in subsequent pregnancies is common.

Chapter 684

Prader-Willi Syndrome

684.1 Overview

Prader-Willi syndrome (PWS) is a neurogenetic disorder caused by loss of function of the paternal copy of genes on chromosome 15q11-q13, the region imprinted on the maternal allele. Mechanisms include paternal deletion (70%), maternal uniparental disomy (25%), and imprinting defects (5%). Incidence is 1 in 10,000–30,000. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

684.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

684.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

684.4 Signs and Symptoms

- Clinical features are biphasic: in infancy, hypotonia, poor feeding, and failure to thrive
- In childhood, hyperphagia (insatiable appetite), obesity, short stature, hypogonadism, small hands and feet, and mild to moderate intellectual disability.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

684.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

684.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

684.7 Diagnosis

Doctors diagnose this using methylation analysis.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

684.8 Prevention

Treatment options include growth hormone therapy, strict dietary supervision to prevent obesity, and behavioral interventions.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

684.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

684.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 685

Preeclampsia

685.1 Overview

Preeclampsia is a multisystem disorder of pregnancy characterized by high blood pressure and proteinuria developing after 20 weeks, affecting 2–8% of pregnancies worldwide. It is a leading cause of maternal death, preterm birth, and intrauterine growth restriction. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

685.2 Causes

This condition happens because of abnormal placental implantation with inadequate spiral artery remodeling, placental reduced blood flow, systemic endothelial dysfunction, and an imbalance of angiogenic factors (elevated soluble Flt-1, reduced PlGF). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

685.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

685.4 Signs and Symptoms

You may experience high blood pressure and proteinuria, with severe features including BP ≥160/110 mmHg, thrombocytopenia, elevated liver enzymes, serum creatinine >1.1 mg/dL, and persistent cerebral symptoms. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

685.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

685.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

685.7 Diagnosis

- To make the diagnosis, doctors look for BP ≥140/90 mmHg on two occasions four hours apart plus proteinuria ≥300 mg/24 hours
- In the absence of proteinuria, new-onset high blood pressure with thrombocytopenia, kidney insufficiency, impaired liver function, lung swelling, or cerebral symptoms may be diagnostic.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

685.8 Prevention

- Treatment for term preeclampsia is delivery
- Preterm preeclampsia is managed expectantly with antihypertensives, magnesium sulfate for seizure prophylaxis, and corticosteroids for fetal lung maturity
- Low-dose aspirin initiated before 16 weeks reduces risk in high-risk women.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

685.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

685.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

685.11 Outlook

In most cases, the outlook is favorable with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 686

Premature Ovarian Insufficiency

686.1 Overview

Premature ovarian insufficiency (POI) is the loss of ovarian function before age 40, characterized by amenorrhea, elevated FSH (>40 IU/L), and low estradiol. It affects 1% of women under 40. Causes are idiopathic (70–90%), genetic, autoimmune, and iatrogenic. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

686.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

686.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

686.4 Signs and Symptoms

You may experience oligomenorrhea or amenorrhea, infertility, vasomotor symptoms, vaginal dryness, and psychological distress. POI has significant long-term health consequences due to prolonged estrogen deficiency: osteoporosis, cardiovascular disease, and thinking and memory decline. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

686.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

686.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

686.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

686.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

686.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

686.10 Living with It

Treatment options include MHT (estrogen plus progestin continued until the average age of menopause), calcium and vitamin D supplementation, and fertility counseling.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

686.11 Outlook

The outlook is generally positive with hormone replacement. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 687

Premature Rupture of Membranes

687.1 Overview

Premature rupture of membranes (PROM) is rupture of the amniotic sac before the onset of labor. PROM at term ($\geq$37 weeks) occurs in 8% of pregnancies and is managed with induction if labor does not ensue spontaneously within 12–24 hours. Preterm PROM (PPROM, <37 weeks) occurs in 3% of pregnancies and is linked to ascending infection, chorioamnionitis, and preterm delivery. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

687.2 Causes

This condition happens because of weakening of the fetal membranes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

687.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

687.4 Signs and Symptoms

You may experience fluid leakage from the vagina. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

687.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

687.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

687.7 Diagnosis

- Doctors diagnose this based on history, sterile speculum examination (pooling of fluid, positive Nitrazine test, ferning pattern on microscopy). Treatment for PPROM remote from term includes antenatal corticosteroids, antibiotics to prolong latency, and close surveillance for infection
- Delivery is indicated for chorioamnionitis, non-reassuring fetal status, or at 34–37 weeks.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

687.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

687.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

687.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

687.11 Outlook

Your outlook depends on gestational age at delivery. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 688

Premenstrual Syndrome / Premenstrual Dysphoric Disorder

688.1 Overview

This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

688.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

688.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

688.4 Signs and Symptoms

Symptoms include emotional (irritability, anxiety, depression, mood swings), physical (breast tenderness, bloating, fatigue, headaches), and behavioral changes that resolve within days of menstruation. Premenstrual dysphoric disorder (PMDD) is a severe form affecting 3–8%, characterized by marked affective lability, irritability, depressed mood, and anxiety, causing significant functional impairment. The Disease mechanism involves abnormal central nervous system response to normal cyclical fluctuations of estrogen and progesterone.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

688.5 Progression and Stages

Premenstrual syndrome (PMS) is a cyclical symptom complex occurring in the luteal phase of the menstrual cycle, affecting 20–30% of reproductive-age women. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

688.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

688.7 Diagnosis

Doctors use DSM-5 criteria requiring prospective daily symptom charting across two cycles.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

688.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

688.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

688.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

688.11 Outlook

The outlook is generally positive with treatment.

Chapter 689

Presbycusis

689.1 Overview

Presbycusis is age-related hearing loss affecting two-thirds of adults over 70. In older adults, untreated hearing loss is linked to thinking and memory decline (increased dementia risk), social isolation, depression, and falls. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

689.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

689.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

689.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance

or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

689.5 Progression and Stages

It is a sensorineural hearing loss, typically bilateral, symmetrical, and high-frequency (above 2000 Hz), resulting from deterioration of cochlear hair cells, stria vascularis, and spiral ganglion neurons. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

689.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

689.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

689.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

689.9 Treatment Options

Treatment options include hearing aids (rehabilitative benefit is well established), cochlear implants for severe-to-profound loss, communication strategies, and treating cerumen impaction (common, easily reversible cause of hearing loss in older adults). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

689.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

689.11 Outlook

The outlook is generally positive with amplification. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 690

Presbyopia

690.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Presbyopia is the age-related loss of accommodation, resulting from hardening of the crystalline lens and decreased ciliary muscle function
- It is universal, typically becoming symptomatic after age 40.

690.2 Causes

This condition happens because of age-related changes in the lens and ciliary muscle. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

690.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

690.4 Signs and Symptoms

You may experience difficulty reading small print, especially in dim light, and the need to hold reading material farther away. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

690.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

690.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

690.7 Diagnosis

Doctors diagnose this based on clinical history and refraction testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

690.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

690.9 Treatment Options

- Treatment options include reading glasses, bifocal or progressive lenses, monovision contact lens arrangements, and multifocal intraocular lenses
- Surgical options include corneal inlays, scleral implants, and laser procedures.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

690.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

690.11 Outlook

Most people recover well with appropriate correction. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 691

Presbyopia and Cataracts

691.1 Overview

Presbyopia is the age-related loss of accommodation resulting from lens hardening, affecting virtually all adults over 45, corrected with reading glasses, bifocals, or progressive lenses. Surgical removal with intraocular lens (IOL) implantation is The main treatment, with phacoemulsification being the standard technique, yielding >95% improvement in visual acuity. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

691.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

691.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

691.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

691.5 Progression and Stages

Cataracts—opacification of the crystalline lens—affect 50% of adults over 75 and are the leading cause of reversible vision loss worldwide, classified by location: nuclear sclerotic (most common, progressive myopic shift), cortical (spoke-like opacities), and posterior subcapsular (associated with corticosteroid use, diabetes, and radiation). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

691.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

691.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

691.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

691.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

691.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

691.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 692

Pressure Ulcers

692.1 Overview

Pressure ulcers (bedsores, decubitus ulcers) are localized injury to skin and underlying tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences (sacrum, heels, greater trochanters, ischial tuberosities). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

692.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

692.3 Risk Factors

Risk factors include immobility, malnutrition, incontinence, sensory loss, advanced age, and comorbidities such as diabetes and peripheral vascular disease. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

692.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

692.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Symptoms can range from non-blanchable redness of the skin (Stage 1) to full-thickness tissue loss with exposed bone/tendon/muscle (Stage 4), with stages defined by the National Pressure Injury Advisory Panel (NPIAP). The outlook varies from person to person
- Healing depends on the stage and underlying patient condition.

692.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

692.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with staging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

692.8 Prevention

Prevention is paramount: risk assessment (Braden Scale), pressure-redistribution surfaces, regular repositioning every 2 hours, nutritional support, and skin care.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

692.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

692.10 Living with It

Prevalence is 5–15% in hospitalized patients and up to 25% in long-term care.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

692.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 693

Preterm Birth

693.1 Overview

Preterm birth is a perinatal condition defined as delivery before 37 weeks of gestation. It affects approximately 10% of all pregnancies worldwide and is the leading cause of neonatal morbidity and mortality. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

693.2 Causes

This condition happens because of multiple pathogenic pathways including infection, inflammation, and uteroplacental reduced blood flow. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

693.3 Risk Factors

Risk factors include multiple gestation, intrauterine infection, cervical insufficiency, prior preterm birth, smoking, and maternal stress. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

693.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

693.5 Progression and Stages

You may experience delivery before 37 weeks, classified as late preterm (34–36 weeks), moderate preterm (32–33 weeks), very preterm (28–31 weeks), and extremely preterm (<28 weeks), with complications inversely related to gestational age. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

693.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

693.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

693.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

693.9 Treatment Options

- Treatment options include antenatal corticosteroids (betamethasone) to accelerate fetal lung maturation, magnesium sulfate for neuroprotection, and transfer to a facility with a level III NICU
- Tocolysis may be used to delay delivery for corticosteroid administration.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

693.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

693.11 Outlook

Your outlook depends on gestational age and birth weight. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 694

Preterm Labor and Preterm Birth

694.1 Overview

Preterm labor is defined as regular uterine contractions causing cervical change between 20 and 36 weeks 6 days of gestation. Preterm birth (delivery before 37 weeks) affects 10% of pregnancies worldwide and is the leading cause of neonatal mortality and morbidity. The Disease mechanism involves multiple pathways: infection/inflammation (40% of early preterm births), uteroplacental reduced blood flow, uterine overdistension, and immunologically mediated rejection. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

694.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

694.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

694.4 Signs and Symptoms

You may experience regular contractions with cervical change before 37 weeks. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

694.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

694.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

694.7 Diagnosis

- To make the diagnosis, doctors look for documented cervical change on exam
- Transvaginal ultrasound measurement of cervical length (<25 mm at 20–24 weeks is high risk) and fetal fibronectin testing help stratify risk. Acute Treatment options include antenatal corticosteroids for fetal lung maturity (24–34 weeks), magnesium sulfate for fetal neuroprotection (24–32 weeks), and tocolysis to delay delivery for 48 hours to allow corticosteroid benefit.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

694.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

694.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

694.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

694.11 Outlook

Your outlook depends on gestational age at delivery. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 695

Priapism

695.1 Overview

Priapism is a persistent, often painful, penile erection lasting longer than 4 hours unrelated to sexual stimulation. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

695.2 Causes

Causes include sickle cell disease (most common cause in children), intracavernosal injections, psychotropic medications, illicit drugs, anticoagulant therapy, and hematologic malignancies. Non-ischemic (high-flow, arterial) priapism results from unregulated arterial inflow, typically from perineal trauma causing arteriocavernosal fistula. This condition happens because of dysregulation of penile blood flow. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

695.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

695.4 Signs and Symptoms

You may experience a rigid, tender penis with soft glans in ischemic type. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

695.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

695.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

695.7 Diagnosis

Diagnosis typically involves history, physical examination, and cavernous blood gas analysis. Treatment for ischemic priapism follows a stepwise approach: corporal removal by suction with irrigation, intracavernosal phenylephrine, and surgical shunting. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

695.8 Prevention

Ischemic (low-flow, veno-occlusive) priapism accounts for >95% of cases and is a urologic emergency requiring prompt intervention to prevent corporal scarring and permanent erectile dysfunction.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

695.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

695.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

695.11 Outlook

Your outlook depends on promptness of intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 696

Primary Amebic Meningoencephalitis (Naegleria fowleri)

696.1 Overview

Primary amebic meningoencephalitis is a rare but nearly always fatal CNS infection caused by *Naegleria fowleri*, a free-living amoeba found in warm freshwater environments, with fewer than 20 cases annually in the United States, primarily in children and young adults with a history of swimming in warm freshwater lakes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

696.2 Causes

This condition happens because of the trophozoite form invading the olfactory neuroepithelium, migrating along the olfactory nerve, and entering the subarachnoid space, producing a involving bleeding, necrotizing meningoencephalitis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

696.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

696.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

696.5 Progression and Stages

You may experience an incubation of 1–9 days, with fulminant onset of severe frontal headache, fever, nausea, vomiting, and meningismus, with rapid progression to altered mental status, seizures, coma, and death within 3–7 days. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

696.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

696.7 Diagnosis

Doctors diagnose this using PCR of CSF. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

696.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

696.9 Treatment Options

Treatment options include miltefosine plus fluconazole, azithromycin, and rifampin, with therapeutic hypothermia and Treatment for increased intracranial pressure. outlook: survival is extremely rare (fewer than 15 survivors documented worldwide). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

696.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

696.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 697

Primary Angle-Closure Glaucoma

697.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Primary angle-closure glaucoma is an eye emergency that occurs when the peripheral iris physically obstructs the trabecular meshwork, blocking aqueous humor removal of fluid and causing a rapid rise in intraocular pressure
- It is more common in hyperopic eyes with shallow anterior chambers.

697.2 Causes

This condition happens because of anatomical predisposition leading to iridotrabecular contact. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

697.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

697.4 Signs and Symptoms

Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

697.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Symptoms of acute angle closure include severe eye pain, headache, nausea, blurred vision, halos around lights, a mid-dilated fixed pupil, and markedly elevated IOP
- Chronic angle closure may be asymptomatic until advanced.

697.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

697.7 Diagnosis

- Doctors diagnose this based on tonometry and gonioscopy. Treatment for acute angle closure includes immediate topical and systemic IOP-lowering agents followed by laser peripheral iridotomy
- Fellow eyes should undergo prophylactic iridotomy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

697.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

697.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

697.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

697.11 Outlook

Most people recover well with prompt treatment. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 698

Primary Biliary Cholangitis

698.1 Overview

Primary biliary cholangitis (PBC, formerly primary biliary cirrhosis) is a chronic, progressive autoimmune liver disease characterized by granulomatous destruction of intrahepatic interlobular bile ducts, occurring predominantly in women aged 40–60. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

698.2 Causes

This condition happens because of autoimmune-mediated destruction of biliary epithelial cells. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

698.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

698.4 Signs and Symptoms

You may experience fatigue, severe generalized itching (often preceding jaundice), hyperpigmentation, and xanthomas. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

698.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Management with ursodeoxycholic acid (UDCA) slows disease progression
- Obeticholic acid is used in UDCA non-responders
- Liver transplantation is definitive for end-stage disease.

698.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

698.7 Diagnosis

Your doctor confirms the diagnosis with elevated serum alkaline phosphatase, positive antimitochondrial antibodies (AMA, present in $> 95\%$ of patients), and liver biopsy showing nonsuppurative destructive cholangitis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

698.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

698.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

698.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

698.11 Outlook

outlook is significantly improved with UDCA treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 699

Primary Open-Angle Glaucoma

699.1 Overview

Primary open-angle glaucoma is the most common form of glaucoma and a leading cause of irreversible blindness worldwide, characterized by progressive optic nerve damage and visual field loss, typically associated with elevated intraocular pressure. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This degenerative condition involves progressive deterioration of affected tissues, leading to gradual loss of function. It is more common with advancing age. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

699.2 Causes

This condition happens because of increased resistance to aqueous humor outflow through the trabecular meshwork despite an open angle between the iris and cornea. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Degenerative processes involve progressive cellular dysfunction and tissue breakdown over time. Contributing factors include oxidative stress, inflammation, accumulated cellular damage, and reduced regenerative capacity. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

699.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

699.4 Signs and Symptoms

You may experience progressive visual field loss, often beginning peripherally. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

699.5 Progression and Stages

The outlook varies from person to person, with treatment slowing but not reversing disease progression. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

699.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

699.7 Diagnosis

Diagnosis typically involves tonometry, gonioscopy, optic disc evaluation (cup-to-disc ratio), automated perimetry, and optical coherence tomography. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

699.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

699.9 Treatment Options

Treatment options include topical hypotensive medications (prostaglandin analogs, beta-blockers, alpha-agonists, carbonic anhydrase inhibitors), laser trabeculoplasty, and trabeculectomy or tube shunt surgery for refractory cases. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

699.10 Living with It

- Follow your treatment plan as prescribed

- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

699.11 Outlook

Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 700

Primary Sclerosing Cholangitis

700.1 Overview

Primary sclerosing cholangitis (PSC) is a chronic cholestatic liver disease characterized by inflammation, fibrotic stricturing, and obliteration of intrahepatic and extrahepatic bile ducts, strongly associated with ulcerative colitis (present in 70–80% of PSC patients). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

700.2 Causes

This condition happens because of immune-mediated bile duct injury leading to concentric onion-skin scarring around bile ducts. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

700.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

700.4 Signs and Symptoms

You may experience fatigue, itching, right upper quadrant pain, recurrent acute cholangitis, and progressive jaundice. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

700.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

700.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

700.7 Diagnosis

Doctors confirm the diagnosis through magnetic resonance cholangiopancreatography (MRCP) or ERCP showing characteristic beaded appearance of bile ducts. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

700.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

700.9 Treatment Options

Treatment typically involves supportive

- Using a thin camera tube balloon dilation or stenting is used for dominant strictures
- Liver transplantation is the only curative option. PSC carries a 10–15% lifetime risk of cholangiocarcinoma.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

700.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

700.11 Outlook

outlook is guarded without liver transplantation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 701

Propriospinal Myoclonus at Sleep Onset

701.1 Overview

Propriospinal myoclonus at sleep onset is a rare sleep-related movement disorder characterized by sudden, brief, involuntary jerks of the axial muscles occurring during the transition from wakefulness to sleep. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

701.2 Causes

This condition happens because of myoclonus originating in the spinal cord and propagating rostrally and caudally via propriospinal pathways. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

701.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

701.4 Signs and Symptoms

You may experience jerks triggered by voluntary attempts to relax or fall asleep, severely delaying sleep onset and resulting in sleep-onset insomnia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

701.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

701.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

701.7 Diagnosis

To make the diagnosis, doctors look for polygraphic EMG recording across multiple spinal segments and polysomnography, with differentiation from hypnic jerks (physiologic sleep starts) and epileptic myoclonus. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

701.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

701.9 Treatment Options

- Treatment typically involves challenging
- Clonazepam, levetiracetam, and other anticonvulsants have been used with variable success.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

701.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

701.11 Outlook

- The outlook varies from person to person
- Some patients experience spontaneous improvement, while others require long-term pharmacotherapy.

Chapter 702

Prostate Cancer

702.1 Overview

Prostate cancer is the second most commonly diagnosed cancer and the fifth leading cause of cancer-related death in men worldwide. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

702.2 Causes

This condition happens because of cancerous transformation of prostatic epithelial cells. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

702.3 Risk Factors

Risk factors include age (rare under 50, risk increases sharply after 65), race (highest incidence in African Americans), family history, and genetic factors (BRCA2 mutations), with most prostate cancers being adenocarcinomas arising in the peripheral zone.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption

- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

702.4 Signs and Symptoms

Screening involves prostate-specific antigen (PSA) testing and digital rectal examination. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

702.5 Progression and Stages

Clinical features are often asymptomatic in early stages, with advanced disease causing urinary symptoms, hematuria, and bone pain from metastases. Most people recover well for localized disease and variable for advanced disease. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

702.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

702.7 Diagnosis

Doctors diagnose this using transrectal or transperineal MRI-guided biopsy, with Gleason score and ISUP grade group guiding risk stratification. Treatment for low-risk disease may be active surveillance

- Intermediate-risk disease doctors typically treat it with radical prostatectomy or radiation therapy
- High-risk and metastatic disease require multimodal therapy including androgen deprivation therapy (ADT), radiation, chemotherapy (docetaxel), and novel hormonal agents (abiraterone, enzalutamide).
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies

- Genetic counseling and testing for hereditary cancer syndromes

702.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

702.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

702.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay

physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

702.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 703

Psoriasis

703.1 Overview

Psoriasis is a chronic, immune-mediated inflammatory skin disease characterized by well-demarcated, erythematous plaques with silvery-white scales, affecting 2–3% of the global population. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

703.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of dysregulated Th1/Th17 immune responses leading to keratinocyte hyperproliferation (epidermal turnover time reduced from 28 days to 3–5 days). Plaque psoriasis (psoriasis vulgaris) is the most common form, presenting with symmetrically distributed plaques on extensor surfaces (elbows, knees), scalp, and lower back. Other forms include guttate (small, drop-shaped lesions, often post-streptococcal), inverse (flexural areas), pustular, and erythrodermic psoriasis. Nail changes (pitting, onycholysis, oil-drop discoloration) are common
- Up to 30% of patients develop psoriatic arthritis.

703.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

703.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

703.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Management follows a stepwise approach based on severity: topical therapies (corticosteroids, vitamin D analogues [calcipotriene], topical calcineurin inhibitors) for mild disease
- Phototherapy (narrowband UVB) for moderate disease
- And systemic therapy for moderate-to-severe disease: conventional agents (methotrexate, cyclosporine, acitretin), and biologics targeting specific cytokines (TNF inhibitors, IL-17 inhibitors, IL-23 inhibitors, IL-12/23 inhibitor [ustekinumab]).

703.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

703.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

703.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

703.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

703.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

703.11 Outlook

outlook is chronic but can be well-controlled with appropriate therapy. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can

be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 704

Psoriatic Arthritis

704.1 Overview

Psoriatic arthritis (PsA) is discussed in Chapter ?? (Skin Diseases). Five clinical patterns include oligoarthritis, polyarthritis, arthritis mutilans, spondylitis, and dactylitis. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

704.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

704.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

704.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

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- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

704.5 Progression and Stages

It is an inflammatory arthropathy associated with psoriasis, classified as a seronegative spondyloarthropathy. Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

704.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

704.7 Diagnosis

Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

704.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

704.9 Treatment Options

Treatment options include NSAIDs, DMARDs (methotrexate), and biologics (TNF inhibitors, IL-17 inhibitors, IL-23 inhibitors, JAK inhibitors). Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

704.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

704.11 Outlook

Autoimmune diseases are generally chronic conditions requiring long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 705

Psoriatic Arthritis

705.1 Overview

Psoriatic arthritis (PsA) is an inflammatory arthropathy associated with psoriasis, affecting 30% of patients with cutaneous psoriasis. It is a seronegative spondyloarthropathy (negative for rheumatoid factor and anti-CCP antibodies). This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

705.2 Causes

This condition happens because of immune-mediated joint inflammation. Five clinical patterns exist: oligoarthritis (most common), polyarthritis, arthritis mutilans (severe, destructive), spondylitis, and dactylitis ("sausage digits"). Extra-articular manifestations include enthesitis (Achilles, plantar fascia), nail changes, eye inflammation, and cardiovascular disease. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

705.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

705.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

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- Raynaud phenomenon (cold-induced color changes in fingers)
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705.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

705.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy

- Lymphoma risk in some autoimmune conditions

705.7 Diagnosis

Doctors diagnose this based on the CASPAR criteria. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

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- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

705.8 Prevention

outlook is improved with early diagnosis and treatment to prevent irreversible joint damage.

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

705.9 Treatment Options

Treatment options include NSAIDs, DMARDs (methotrexate, sulfasalazine), and biologics (TNF inhibitors, IL-17 inhibitors, IL-23 inhibitors, JAK inhibitors, PDE4 inhibitors). Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways

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705.10 Living with It

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- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

705.11 Outlook

Autoimmune diseases are generally chronic conditions requiring long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 706

Psychological Approaches

706.1 Overview

thinking and memory behavioral therapy (CBT) is the most evidence-based psychological treatment for chronic pain. It aims to identify and modify maladaptive thoughts (catastrophizing, fear-avoidance beliefs) and behaviors (activity avoidance, pain behaviors) through thinking and memory restructuring, behavioral activation, graded exposure, and relaxation techniques. Biofeedback uses physiological monitoring (EMG, heart rate variability, skin conductance, breathing) to teach patients voluntary control of physiological processes that contribute to pain and stress. Mindfulness-based stress reduction (MBSR) and acceptance and commitment therapy (ACT) cultivate present-moment awareness, acceptance of pain, and value-driven behavior. These approaches reduce pain-related distress and improve function and are recommended as part of multimodal chronic pain treatment. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

706.2 Causes

Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

706.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

706.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

706.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

706.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

706.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

706.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

706.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

706.10 Living with It

The goal is improved function and quality of life, not necessarily pain elimination.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

706.11 Outlook

With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 707

Psychophysiologic Insomnia

707.1 Overview

Psychophysiologic insomnia is a subtype of chronic insomnia characterized by a learned, conditioned arousal to sleep-related stimuli, such that the bed and bedroom environment become associated with frustration, worry, and heightened alertness rather than sleep. Patients often sleep better away from home (paradoxical improvement in novel environments). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

707.2 Causes

This condition happens because of physiologic hyperarousal with elevated metabolic rate, heart rate, and cortisol at sleep onset, combined with thinking and memory patterns including excessive preoccupation with sleep, catastrophic thinking about sleep loss, and performance anxiety. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

707.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

707.4 Signs and Symptoms

You may experience difficulty sleeping in one's own bed but improved sleep in novel environments.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

707.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

707.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

707.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

707.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

707.9 Treatment Options

Treatment focuses on stimulus control therapy (reassociating the bed with sleep), sleep restriction therapy (consolidating sleep window to improve sleep efficiency), and thinking and memory restructuring to reduce maladaptive beliefs about sleep. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

707.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

707.11 Outlook

The outlook is generally positive with appropriate behavioral therapy. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 708

Pulmonary Embolism

708.1 Overview

lung blockage by a blood clot is a potentially life-threatening condition caused by obstruction of the lung arteries, most commonly by deep vein blood clot formation of the lower extremities. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

708.2 Causes

This condition happens because of thrombotic occlusion of lung arteries. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

708.3 Risk Factors

Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptives, obesity, and inherited thrombophilia.

- Advanced age
- Cigarette smoking
- Hypertension and diabetes
- Elevated cholesterol levels
- Family history of heart disease
- Obesity and physical inactivity

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

708.4 Signs and Symptoms

Symptoms can range from shortness of breath and pleuritic chest pain to hemoptysis, syncope, and hemodynamic instability (massive PE). Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

708.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

- The right treatment depends on severity: anticoagulation for hemodynamically stable disease
- Systemic thrombolysis, catheter-directed therapy, or surgical embolectomy for massive disease with hemodynamic instability
- Inferior vena cava filters are placed when anticoagulation is contraindicated.

708.6 Complications

- Myocardial infarction (heart attack)

- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

708.7 Diagnosis

Diagnosis typically involves clinical probability assessment (Wells score), D-dimer, and CT lung angiography. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

708.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

708.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

708.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

708.11 Outlook

Most people recover well with prompt diagnosis and appropriate treatment. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 709

Pulmonary Hypertension

709.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Lung high blood pressure is defined as a mean lung artery pressure of 20 mmHg or greater at rest, measured by right heart catheterization
- The 2022 classification groups lung high blood pressure into five categories: Group 1 (lung arterial high blood pressure), Group 2 (left heart disease), Group 3 (lung disease or hypoxia), Group 4 (chronic thromboembolic disease), and Group 5 (unclear or multifactorial).

709.2 Causes

This condition happens because of vascular remodeling with intimal scarring, medial enlargement, and plexiform lesions. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

709.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking

- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

709.4 Signs and Symptoms

You may experience progressive shortness of breath, fatigue, syncope, and right heart failure. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

709.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

709.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

709.7 Diagnosis

- Diagnosis typically involves echocardiography, right heart catheterization, and evaluation of the underlying cause. Treatment for lung arterial high blood pressure includes phosphodiesterase-5 inhibitors, endothelin receptor antagonists, prostacyclin analogs, and soluble guanylate cyclase stimulators
- Group 4 disease doctors typically treat it with lung endarterectomy or balloon lung angioplasty.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

709.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

709.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease

- Cardiac rehabilitation programs

709.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

709.11 Outlook

Your outlook depends on the underlying group and response to therapy. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 710

Pyelonephritis

710.1 Overview

Pyelonephritis is an infection of the kidney parenchyma and kidney pelvis, representing an upper urinary tract infection. Acute pyelonephritis typically results from ascending infection from the lower urinary tract, most commonly caused by *E. coli*. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

710.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

710.3 Risk Factors

Risk factors include vesicoureteral reflux, urinary obstruction, diabetes, and immunosuppression. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

710.4 Signs and Symptoms

- You may experience high fever, chills, flank pain, nausea, and vomiting

- CVA tenderness is present on examination.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

710.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

710.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

710.7 Diagnosis

Doctors diagnose this based on urinalysis (pyuria, bacteriuria, hematuria) and urine culture

- Blood cultures are indicated for severe or complicated cases. Treatment for mild cases is outpatient oral fluoroquinolones or initial IV ceftriaxone followed by oral step-down therapy
- Severe cases require hospitalization and IV antibiotics.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

710.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

710.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

710.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

710.11 Outlook

- The outlook is generally positive with appropriate antibiotic therapy
- Chronic pyelonephritis results from recurrent infections and may lead to kidney scarring and CKD.

Chapter 711

Rabies

711.1 Overview

Rabies is a fatal viral encephalitis caused by rabies virus, a single-stranded RNA virus of the genus *Lyssavirus*, with 60,000 deaths annually worldwide, primarily in Asia and Africa, with dogs being the main reservoir and source of transmission (99% of human cases), and rabies being nearly 100% fatal after symptom onset. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

711.2 Causes

This condition happens because of the virus being inoculated through saliva of an infected animal, binding to nicotinic acetylcholine receptors at the neuromuscular junction, traveling by retrograde axonal transport to the CNS, and disseminating throughout the central nervous system. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

711.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational

exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

711.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

711.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- You may experience an incubation of 20–90 days, followed by a prodrome (fever, malaise, tingling or numbness at the bite site), then acute nervous system phase: furious rabies (80%)—hydrophobia, aerophobia, agitated behavior, autonomic instability, hypersalivation, and seizures
- Or paralytic rabies (20%)—ascending flaccid paralysis.

711.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

711.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by RT-PCR of saliva, CSF, or skin biopsy. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

711.8 Prevention

- Treatment typically involves supportive once symptoms develop
- Prevention through pre-exposure prophylaxis and post-exposure prophylaxis (immediate wound cleansing, HRIG, and rabies vaccine) is nearly 100% effective if administered before symptom onset.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings

- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

711.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

711.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

711.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 712

Radiation Sickness (Acute Radiation Syndrome)

712.1 Overview

Acute radiation syndrome (ARS) results from high-dose whole-body or significant partial-body ionizing radiation exposure (>1 Gy). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

712.2 Causes

This condition happens because of radiation causing DNA damage (double-strand breaks), cell death in rapidly dividing tissues (bone marrow, GI tract, skin), and free-radical generation. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

712.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

712.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

712.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Clinical phases: prodromal phase (0–48 hours): nausea, vomiting, diarrhea, anorexia
- Latent phase: transient symptom improvement
- Manifest illness phase depends on dose: hematopoietic syndrome (1–6 Gy): pancytopenia, infection, bleeding, anemia
- Digestive tract syndrome (6–10 Gy): severe diarrhea, fluid loss, mucosal tissue death, sepsis
- Cerebrovascular/cardiovascular syndrome (>10–20 Gy): hypotension, cerebral swelling, convulsions, coma, death.

712.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

712.7 Diagnosis

Doctors diagnose this based on history of radiation exposure, absolute lymphocyte count decline, and cytogenetic analysis.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

712.8 Prevention

Treatment: supportive care (reverse isolation, antimicrobial prophylaxis, blood product support), cytokines (G-CSF, GM-CSF) for neutropenia, and treatment of internal contamination.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

712.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

712.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

712.11 Outlook

Unfortunately, the outlook is often poor for doses >6 Gy.

Chapter 713

Reactive Arthritis

713.1 Overview

Reactive arthritis is an aseptic inflammatory arthritis occurring 1–4 weeks after a genitourinary (usually *Chlamydia trachomatis*) or digestive tract (usually *Salmonella*, *Shigella*, *Yersinia*, *Campylobacter*) infection. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

713.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of an immune response to the initial infection triggering joint inflammation. The classic triad is arthritis, urethritis, and conjunctivitis ("can't see, can't pee, can't climb a tree"). Arthritis is typically asymmetric and oligoarticular, affecting large joints of the lower extremities
- Enthesitis, dactylitis, and oral ulcers are common. Most cases resolve within 3–12 months.

713.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

713.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

713.5 Progression and Stages

It is strongly associated with HLA-B27 and classified as a seronegative spondyloarthropathy. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

713.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

713.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with history of preceding infection. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

713.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

713.9 Treatment Options

- Treatment options include NSAIDs, intra-articular corticosteroids, and for chronic or severe disease, DMARDs (sulfasalazine) or TNF inhibitors
- Antibiotics are not helpful once arthritis has developed.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

713.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

713.11 Outlook

- In most cases, the outlook is good
- Most cases resolve within 3–12 months.

Chapter 714

Recurrent Pregnancy Loss

714.1 Overview

Recurrent pregnancy loss (RPL) is traditionally defined as two or more consecutive pregnancy losses before 20 weeks, affecting 1–2% of couples. Evaluation is recommended after two losses and includes parental karyotyping, uterine evaluation, antiphospholipid antibody testing, and thyroid function tests. The cause remains unexplained in 50%. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

714.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

714.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

714.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

714.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

714.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

714.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

714.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

714.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

714.10 Living with It

- Management addresses identified causes: surgical correction of uterine septum, anticoagulation for antiphospholipid syndrome, and thyroid hormone replacement for hypothyroidism
- For unexplained RPL, progesterone supplementation and supportive care are associated with improved outcomes.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

714.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 715

Relapsing Polychondritis

715.1 Overview

Relapsing polychondritis is a rare, immune-mediated disease characterized by recurrent, progressive inflammation and destruction of cartilaginous structures, particularly in the ears, nose, larynx, tracheobronchial tree, and joints, with an incidence of approximately 0.5–3.5 per million, equal sex distribution, and typically presenting in the fifth to sixth decade. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

715.2 Causes

This condition happens because of an autoimmune attack on type II, IX, and XI collagen and other cartilage matrix components, mediated by autoantibodies, T-cell infiltration, and complement activation, with proteoglycan degradation by matrix metalloproteinases contributing to cartilage destruction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

715.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

715.4 Signs and Symptoms

You may experience auricular chondritis (the most common presenting feature: painful, swollen, erythematous ears sparing the lobule), nasal chondritis (saddle nose deformity), laryngotracheal involvement (subglottic stenosis, hoarseness, stridor, airway collapse—the most life-threatening manifestation), costochondritis, and non-erosive seronegative arthritis, with ocular involvement (episcleritis, scleritis, uveitis) occurring in up to 60%, audiovestibular involvement (conductive and sensorineural hearing loss, feeling of spinning) being common, and association with other autoimmune diseases (SLE, vasculitis, myelodysplastic syndrome) in 30%. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

715.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

715.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

715.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, based on McAdam's criteria or Damiani's criteria, with laboratory findings being non-specific. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

715.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

715.9 Treatment Options

Treatment options include corticosteroids (rapid symptomatic relief), dapsone (mild cases), and immunosuppressants (methotrexate, mycophenolate, cyclophosphamide) for severe disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

715.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

715.11 Outlook

The outlook varies from person to person, with respiratory involvement being the leading cause of death. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 716

REM Parasomnias: Nightmare Disorder

716.1 Overview

Nightmare disorder is a parasomnia involving recurrent, vividly realistic, dysphoric dreams that awaken the individual from REM sleep, typically during the second half of the night, with rapid return to full alertness and clear recall of the frightening content. Prevalence is 2–8% in adults and 10–50% in children. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

716.2 Causes

This condition happens because of abnormal processing of emotional memories during REM sleep and is linked to PTSD, anxiety, borderline personality disorder, and certain medications (beta-blockers, SSRIs). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

716.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

716.4 Signs and Symptoms

You may experience recurrent nightmares causing sleep avoidance and daytime fatigue. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

716.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

716.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

716.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

716.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

716.9 Treatment Options

Treatment options include imagery rehearsal therapy (IRT), prazosin (an alpha-1 adrenergic antagonist) for PTSD-related nightmares, and treatment of comorbid psychiatric disorders. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

716.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

716.11 Outlook

The outlook is generally positive with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 717

REM Parasomnias: Recurrent Isolated Sleep Paralysis

717.1 Overview

Recurrent isolated sleep paralysis is a parasomnia consisting of episodes of inability to perform voluntary movements at sleep onset (hypnagogic) or upon awakening (hypnopompic), with preserved consciousness and awareness of the environment. Prevalence is 8–50% depending on population, higher in those with anxiety, panic disorder, and sleep deprivation. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

717.2 Causes

This condition happens because of a dissociation between REM sleep atonia and conscious awareness. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

717.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

717.4 Signs and Symptoms

You may experience episodes lasting seconds to minutes, often accompanied by terrifying hallucinations (intruder, pressure on the chest), and the ability to terminate episodes by external stimulation or strong voluntary effort to move a small muscle group. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

717.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

717.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

717.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

717.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

717.9 Treatment Options

Treatment options include reassurance and sleep hygiene. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

717.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

717.11 Outlook

outlook is non-cancerous. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 718

REM Parasomnias: REM Sleep Behavior Disorder

718.1 Overview

REM sleep behavior disorder (RBD) is a parasomnia characterized by loss of the normal muscle atonia of REM sleep, allowing patients to act out their dreams, often with violent or injurious behaviors. Prevalence is 1–2% in the general population, with a strong male predominance (>80% of cases), typically occurring after age 50. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

718.2 Causes

This condition happens because of dysfunction of the brainstem REM-atonia circuits (sublaterodorsal nucleus in the pons). Critically, idiopathic RBD is a highly specific prodromal marker for synucleinopathies: 80–90% of patients will eventually develop Parkinson disease, dementia with Lewy bodies, or multiple system shrinkage or wasting, with a mean latency of 10–15 years. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

718.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

718.4 Signs and Symptoms

You may experience dream-enactment behaviors (punching, kicking, jumping out of bed) corresponding to aggressive dream content. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

718.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

718.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

718.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with polysomnography demonstrating elevated chin EMG tone during REM. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

718.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

718.9 Treatment Options

Treatment options include safety measures, clonazepam (first-line, 0.5–2 mg nightly), and melatonin (3–15 mg) as an effective alternative. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

718.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

718.11 Outlook

- Outlook is chronic
- The condition is manageable but carries a high risk of eventual neurodegenerative disease.

Chapter 719

Renal Osteodystrophy

719.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

719.2 Causes

This condition happens because of phosphate retention, decreased kidney production of active vitamin D (calcitriol), hypocalcemia, and secondary hyperparathyroidism. Bone histology subtypes include osteitis fibrosa cystica (high-turnover, from severe hyperparathyroidism), adynamic bone disease (low-turnover, from oversuppression of PTH with calcitriol and calcium-based binders, common in diabetics and the elderly), osteomalacia (defective mineralization, from vitamin D deficiency or aluminum toxicity), and mixed uremic osteodystrophy. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

719.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

719.4 Signs and Symptoms

You may experience bone pain (most commonly in lower back, hips, and legs), proximal muscle weakness, fractures, skeletal deformities, growth retardation in children, and extraosseous calcium buildup (vascular calcium buildup, calciphylaxis). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

719.5 Progression and Stages

kidney osteodystrophy is a spectrum of bone disorders that develop in patients with chronic kidney disease (CKD), resulting from disturbances in mineral metabolism, affecting virtually all patients with CKD stage 5. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

719.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

719.7 Diagnosis

Doctors diagnose this based on elevated PTH, hyperphosphatemia, hypocalcemia or hypercalcemia, low vitamin D levels, and bone biopsy (gold standard but rarely performed). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

719.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

719.9 Treatment Options

Treatment focuses on phosphate restriction (dietary), phosphate binders (calcium-based, sevelamer, lanthanum carbonate, iron-based binders), active vitamin D analogs (calcitriol, paricalcitol, doxercalciferol), calcimimetics (cinacalcet, etelcalcetide), and optimization of dialysis. Parathyroidectomy for refractory hyperparathyroidism. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

719.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

719.11 Outlook

- Outlook is improved by early recognition and treatment
- Kidney osteodystrophy improves after successful kidney transplantation.

Chapter 720

Renal Tubular Acidosis

720.1 Overview

kidney tubular acidosis (RTA) is a group of disorders in which the kidneys fail to appropriately acidify the urine despite normal or near-normal GFR. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

720.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Type 2 (proximal) RTA involves impaired bicarbonate reabsorption in the proximal tubule, causing metabolic acidosis with appropriately acidic urine and bicarbonaturia
- It is often part of Fanconi syndrome. Type 4 RTA is caused by aldosterone deficiency or resistance, leading to hyperkalemic metabolic acidosis. This condition happens because of the specific tubular transport defect.

720.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

720.4 Signs and Symptoms

Symptoms vary depending on type and include metabolic acidosis, growth retardation, and nephrocalcinosis. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

720.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

720.6 Complications

- Type 1 (distal) RTA results from impaired hydrogen ion secretion in the collecting duct, causing hyperchloremic metabolic acidosis with inappropriately alkaline urine (pH >5.5)
- Complications include nephrocalcinosis and rickets.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

720.7 Diagnosis

Doctors diagnose this based on serum electrolytes, urine pH, and acid-base testing. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

720.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

720.9 Treatment Options

Treatment options include oral bicarbonate supplementation (Types 1 and 2) and Treatment for hyperkalemia and aldosterone deficiency (Type 4). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

720.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

720.11 Outlook

The outlook is generally positive with appropriate supplementation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 721

Respiratory Distress Syndrome (RDS)

721.1 Overview

Respiratory distress syndrome (RDS), also known as hyaline membrane disease, is a lung disorder of prematurity caused by surfactant deficiency in preterm infants. Incidence is inversely proportional to gestational age: approximately 50% of infants born at 28–30 weeks and nearly all infants born before 26 weeks develop RDS. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

721.2 Causes

This condition happens because of deficiency of surfactant, a phospholipoprotein complex produced by type II pneumocytes that reduces alveolar surface tension. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

721.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

721.4 Signs and Symptoms

You may experience tachypnea, grunting, nasal flaring, intercostal retractions, and cyanosis within minutes to hours after birth.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

721.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

721.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

721.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with chest radiograph showing a characteristic ground-glass appearance with air bronchograms. Treatment consists of intratracheal surfactant replacement therapy (beractant, poractant alfa) and respiratory support (nasal continuous positive airway pressure or mechanical ventilation).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

721.8 Prevention

Prevention includes antenatal corticosteroids.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

721.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

721.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

721.11 Outlook

The outlook has dramatically improved with surfactant replacement. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 722

Restless Legs Syndrome

722.1 Overview

Restless legs syndrome (RLS, Willis-Ekbom disease) is a sensorimotor disorder characterized by an irresistible urge to move the legs, usually accompanied by uncomfortable sensations (creeping, crawling, pulling, aching, or tingling), that begins or worsens during periods of rest or inactivity, is partially or completely relieved by movement, occurs predominantly in the evening or night, and is not better explained by another condition. Prevalence is 5–10% in adults of European descent, 2–4% in Asian populations, and increases with age, affecting women 2:1 over men. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

722.2 Causes

This condition happens because of brain iron deficiency (reduced iron content in substantia nigra and striatal dopamine receptors) and dopaminergic dysfunction, with genetic associations (BTBD9, MEIS1, MAP2K5). Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

722.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

722.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

722.5 Progression and Stages

Secondary RLS is caused by iron deficiency anemia, pregnancy, end-stage kidney disease, peripheral neuropathy, and medications (SSRIs, antihistamines, antipsychotics). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

722.6 Complications

- Outlook is chronic but manageable
- Augmentation is the most significant complication of dopaminergic therapy.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

722.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam using the five essential IRLSSG diagnostic criteria

- Serum ferritin levels should be checked (goal >75 μg/L).
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

722.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

722.9 Treatment Options

Treatment options include non-pharmacologic measures (moderate exercise, avoidance of alcohol and caffeine in the evening), correction of iron deficiency, first-line pharmacotherapy with alpha-2-delta calcium channel ligands (gabapentin, pregabalin—preferred due to lower augmentation risk), and dopamine agonists (pramipexole, ropinirole, rotigotine) for refractory cases. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

722.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 723

Restrictive Cardiomyopathy

723.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Restrictive cardiomyopathy is a condition characterized by impaired ventricular filling due to increased myocardial stiffness with preserved or near-preserved systolic function
- Causes include amyloidosis (AL and ATTR types), sarcoidosis, hemochromatosis, endomyocardial scarring, and infiltrative diseases.

723.2 Causes

This condition happens because of myocardial infiltration or scarring restricting ventricular filling. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

723.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

723.4 Signs and Symptoms

You may experience predominant right-sided heart failure (swelling, ascites, hepatomegaly, jugular venous distension). Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

723.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

723.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

723.7 Diagnosis

Doctors diagnose this based on echocardiography showing small to normal-sized ventricles with biatrial enlargement and restrictive filling pattern

- Cardiac MRI demonstrates late gadolinium enhancement
- Endomyocardial biopsy confirms diagnosis. Management targets the underlying cause (tafamidis for ATTR amyloidosis, chemotherapy for AL amyloidosis) with diuretics for symptom management.
- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

723.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

723.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease

- Cardiac rehabilitation programs

723.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

723.11 Outlook

Recovery varies by cause. With proper management and lifestyle modifications, many patients maintain good quality of life. Long-term outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 724

Retinal Detachment

724.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Retinal detachment is a separation of the neurosensory retina from the underlying retinal pigment epithelium, constituting an eye emergency
- Rhegmatogenous retinal detachment, the most common type, results from a retinal break allowing vitreous fluid to enter the subretinal space.

724.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Tractional retinal detachment occurs in proliferative diabetic retinopathy
- Exudative retinal detachment results from fluid accumulation without a break.

724.3 Risk Factors

Risk factors include high myopia, prior cataract surgery, trauma, and lattice deterioration. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

724.4 Signs and Symptoms

You may experience sudden onset of floaters, flashing lights, and a curtain-like visual field defect. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

724.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

724.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

724.7 Diagnosis

Doctors diagnose this based on dilated fundus examination. Treatment for rhegmatogenous retinal detachment requires surgical repair with pneumatic retinopexy, scleral buckling, or vitrectomy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

724.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

724.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

724.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

724.11 Outlook

Your outlook depends on the type, location, and timeliness of repair. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 725

Retinitis Pigmentosa

725.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

725.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

725.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

725.4 Signs and Symptoms

- You may experience night blindness and progressive peripheral visual field constriction (tunnel vision), eventually affecting central vision
- Fundoscopy reveals bone-spicule pigmentation, waxy pallor of the optic disc, and attenuated retinal vessels.
- Pain or discomfort in the affected area

- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

725.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Retinitis pigmentosa is a group of inherited retinal dystrophies characterized by progressive deterioration of photoreceptors, primarily rod cells
- Most cases are inherited (autosomal dominant, autosomal recessive, or X-linked), with over 80 genes identified. This condition happens because of genetic mutations causing photoreceptor deterioration. Treatment typically involves supportive
- Vitamin A supplementation may slow progression in some forms
- Retinal prostheses and gene therapy (voretigene neparvovec for RPE65-mutated RP) offer emerging therapeutic avenues.

725.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

725.7 Diagnosis

Doctors diagnose this based on fundus examination and electroretinography showing diminished or absent responses. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

725.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

725.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

725.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

725.11 Outlook

outlook is progressive with no cure currently available. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 726

Retinoblastoma

726.1 Overview

This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

- Retinoblastoma is the most common intraocular malignancy in children, with an incidence of 1 in 15,000–20,000 live births. It arises from primitive retinal cells and is caused by biallelic inactivation of the RB1 tumor suppressor gene. Retinoblastoma occurs in heritable (40%) and non-heritable (60%) forms
- Heritable retinoblastoma involves a germline RB1 mutation with younger age at presentation, bilateral involvement, and increased risk of second malignancies.

726.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

726.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

726.4 Signs and Symptoms

You may experience leukocoria (white pupillary reflex, the most common presenting sign), strabismus, decreased vision, and orbital inflammation. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

726.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

726.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

726.7 Diagnosis

Doctors diagnose this using ophthalmologic examination under anesthesia. Treatment depends on the extent of disease: intra-arterial chemotherapy, laser photocoagulation, freezing treatment, brachytherapy, systemic chemotherapy, or enucleation. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
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- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

726.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

726.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

726.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and

depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

726.11 Outlook

- Most people recover well
- Survival exceeds 95% in developed countries.

Chapter 727

Retinopathy of Prematurity

727.1 Overview

Retinopathy of prematurity is a vasoproliferative retinal disease affecting premature infants, particularly those born before 32 weeks gestation with low birth weight. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

727.2 Causes

This condition happens because of abnormal vascularization of the developing retina, which can lead to tractional retinal detachment and blindness. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

727.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

727.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

727.5 Progression and Stages

The disease is classified by zone, stage, and extent of vascular involvement, with plus disease indicating vascular activity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

727.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

727.7 Diagnosis

Doctors diagnose this based on screening examinations in at-risk premature infants. Treatment for severe ROP involves anti-VEGF intravitreal injections or laser photocoagulation of the avascular retina. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

727.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

727.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

727.10 Living with It

- The outlook is good with timely screening and treatment
- Long-term follow-up is necessary due to risks of late sequelae including myopia, strabismus, and retinal detachment.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

727.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 728

Retinopathy of Prematurity (ROP)

728.1 Overview

Retinopathy of prematurity (ROP) is a vasoproliferative disorder of the developing retina in preterm infants (see Chapter 23, Diseases of the Eye for comprehensive coverage). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

728.2 Causes

Screening by indirect ophthalmoscopy is performed for at-risk preterm infants. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

728.3 Risk Factors

Risk factors include low gestational age (<30 weeks), low birth weight (<1500 g), and supplemental oxygen therapy. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

728.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

728.5 Progression and Stages

This condition happens because of disruption of normal retinal vascularization, with an initial phase of vessel loss followed by pathologic neovascularization. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

728.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

728.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

728.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

728.9 Treatment Options

Treatment options include laser photocoagulation and intravitreal bevacizumab for severe disease (Type 1 ROP). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

728.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

728.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 729

Rh Incompatibility / Hemolytic Disease of the Newborn

729.1 Overview

Rh incompatibility occurs when an Rh-negative mother carries an Rh-positive fetus, leading to maternal sensitization and production of anti-Rh(D) IgG antibodies that cross the placenta and cause hemolytic disease of the fetus and newborn (HDFN). Sensitization occurs most commonly at delivery but also with miscarriage, ectopic pregnancy, invasive prenatal procedures, and trauma. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

729.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

729.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

729.4 Signs and Symptoms

Symptoms can range from mild hemolytic anemia to severe hydrops fetalis. Diagnosis includes positive direct antiglobulin test on cord blood, elevated bilirubin, and anemia. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

729.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

729.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

729.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

729.8 Prevention

- Most people recover well with prophylaxis
- Prevention has reduced HDFN incidence by 90%.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

729.9 Treatment Options

- Treatment typically involves prophylactic: Rh-immune globulin (RhoGAM, 300 μg) is administered at 28 weeks and within 72 hours of delivery of an Rh-positive infant
- Affected fetuses with severe anemia are treated with intrauterine transfusion.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

729.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

729.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 730

Rhabdomyolysis

730.1 Overview

Rhabdomyolysis is a syndrome of acute skeletal muscle tissue death with release of intracellular muscle contents (myoglobin, CK, potassium, phosphate, uric acid) into the circulation. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

730.2 Causes

Causes include crush injuries, prolonged immobilization, extreme exertion, seizures, cancerous hyperthermia, medications (statins, fibrates, antipsychotics), electrolyte imbalances, infections, and inflammatory myopathies. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

730.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

730.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
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- Symptoms may develop gradually or appear suddenly

730.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Your outlook depends on severity and promptness of treatment
- Most patients recover with adequate hydration.

730.6 Complications

- You may experience muscle pain, muscle weakness, dark (cola-colored) urine, and swelling of affected muscles
- Complications include acute kidney injury (myoglobin nephrotoxicity), hyperkalemia (irregular heartbeats), disseminated intravascular coagulation, and compartment syndrome.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

730.7 Diagnosis

Doctors diagnose this based on elevated CK (typically >5 times the upper limit of normal, often >10,000 IU/L), elevated myoglobin, and elevated transaminases. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

730.8 Prevention

- Treatment focuses on aggressive IV fluid resuscitation (target urine output 200–300 mL/h), urinary alkalinization (to prevent myoglobin precipitation), correction of electrolyte abnormalities, and monitoring for complications
- Kidney replacement therapy may be necessary for severe AKI.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

730.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

730.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

730.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 731

Rhabdomyosarcoma

731.1 Overview

Rhabdomyosarcoma (RMS) is the most common soft tissue sarcoma in children, with an incidence of 4.5 per million children under age 20. It arises from primitive mesenchymal cells with skeletal muscle differentiation. Two major histologic subtypes: embryonal RMS (60%, more favorable outlook) and alveolar RMS (20%, more aggressive). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

731.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

731.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

731.4 Signs and Symptoms

Your symptoms depend on the site: a rapidly growing mass, proptosis (orbital), nasal obstruction (parameningeal), or scrotal or vaginal mass. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
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731.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer

prognosis.

731.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

731.7 Diagnosis

Doctors diagnose this using biopsy with immunohistochemistry. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)

- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

731.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

731.9 Treatment Options

Treatment typically involves multimodal: chemotherapy, surgical removal, and radiation therapy. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Neurological treatment focuses on symptom management, disease modification when possible, and rehabilitation to maintain function. A multidisciplinary team approach is often beneficial. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy

- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

731.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

731.11 Outlook

outlook is 70–80% survival for low-risk and intermediate-risk, but <50% for high-risk disease. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 732

Rheumatic Heart Disease

732.1 Overview

Rheumatic heart disease (RHD) is the chronic valvular damage resulting from acute rheumatic fever, an autoimmune response to group A streptococcal pharyngitis. This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

732.2 Causes

Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process.

- This condition happens because of molecular mimicry between streptococcal M protein and human cardiac tissue, leading to pancarditis and chronic fibrotic scarring of heart valves
- The mitral valve is affected in over 90% of cases, leading to mitral stenosis or regurgitation.

732.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

732.4 Signs and Symptoms

You may experience exertional shortness of breath, fatigue, palpitations, and signs of heart failure or stroke secondary to atrial fibrillation. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

732.5 Progression and Stages

Your outlook depends on valve damage severity. Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

732.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

732.7 Diagnosis

Doctors diagnose this based on echocardiography showing valvular thickening, commissural fusion, and regurgitation/stenosis. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

732.8 Prevention

Treatment options include secondary antibiotic prophylaxis (IM benzathine penicillin G every 3–4 weeks) to prevent recurrent streptococcal infections, medical Treatment for heart failure, and surgical valve repair or replacement.

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

732.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation

- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

732.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

732.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 733

Rheumatoid Arthritis

733.1 Overview

Rheumatoid arthritis (RA) is a chronic, systemic autoimmune disease characterized by symmetric inflammatory synovitis, predominantly affecting the small joints of the hands (metacarpophalangeal and proximal interphalangeal joints) and feet. It affects approximately 0.5–1% of the population, with a 3:1 female predominance. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

733.2 Causes

This condition happens because of autoantibody production (rheumatoid factor, anti-citrullinated peptide antibodies [anti-CCP]), immune complex deposition, and synovial inflammation leading to pannus formation and progressive joint destruction. Extra-articular manifestations include rheumatoid nodules, interstitial lung disease, vasculitis, scleritis, pericarditis, and accelerated atherosclerosis. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

733.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency

Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

733.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

733.5 Progression and Stages

The 2010 ACR/EULAR classification criteria aid early diagnosis. Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

733.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation

- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

733.7 Diagnosis

- Management follows a treat-to-target strategy with early initiation of DMARDs: methotrexate (first-line anchor drug), leflunomide, sulfasalazine, and hydroxychloroquine
- Biologics (TNF inhibitors, IL-6 inhibitors [tocilizumab, sarilumab], T-cell co-stimulation blocker [abatacept], anti-CD20 [rituximab]) and targeted synthetic DMARDs (JAK inhibitors [tofacitinib, baricitinib]) are used for inadequate response.
- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

733.8 Prevention

- To make the diagnosis, doctors look for clinical assessment, serological testing (RF, anti-CCP), and imaging (X-rays showing periarticular osteopenia, joint space narrowing, and erosions
- Ultrasound or MRI for early detection of synovitis).
- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

733.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

733.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

733.11 Outlook

The outlook varies from person to person but has improved significantly with early aggressive treatment. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 734

Rheumatoid Arthritis

734.1 Overview

Rheumatoid arthritis is discussed in detail in Chapter ??. It is a systemic autoimmune disease primarily affecting joints but with significant extrarticular manifestations involving the immune system. Autoantibodies (rheumatoid factor, anti-CCP) are central to diagnosis and outlook. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

734.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

734.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)

- Vitamin D deficiency

734.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

734.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

734.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

734.7 Diagnosis

Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

734.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

734.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

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- Corticosteroids for rapid control of inflammation
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- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

734.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

734.11 Outlook

Autoimmune diseases are generally chronic conditions requiring long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 735

Rickets and Osteomalacia

735.1 Overview

Rickets (in children) and osteomalacia (in adults) are disorders of impaired bone mineralization due to vitamin D deficiency or abnormalities in vitamin D metabolism. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

735.2 Causes

This condition happens because of inadequate vitamin D (produced in the skin upon UV exposure and obtained from diet), which is essential for intestinal calcium absorption, leading to soft, weak bones. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

735.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

735.4 Signs and Symptoms

Clinical features in children (rickets) include growth retardation, skeletal deformities (bowed legs, knock knees, rachitic rosary, Harrison's groove, frontal bossing), dental abnormalities, and hypocalcemic seizures. Clinical features in adults (osteomalacia) include diffuse bone pain, muscle weakness, increased fracture risk, and pathologic fractures. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

735.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

735.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

735.7 Diagnosis

Doctors diagnose this based on serum 25-hydroxyvitamin D level (<20 ng/mL indicates deficiency), elevated alkaline phosphatase, low calcium and phosphate, and radiographic findings (widened growth plates, Looser's zones in adults). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

735.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

735.9 Treatment Options

Treatment typically involves vitamin D supplementation (ergocalciferol or cholecalciferol) and calcium. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

735.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

735.11 Outlook

Most people recover well with adequate vitamin D replacement. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 736

Rocky Mountain Spotted Fever

736.1 Overview

Rocky Mountain spotted fever is a tick-borne illness caused by *Rickettsia rickettsii*, an obligate intracellular gram-negative coccobacillus, and is the most severe tick-borne rickettsial disease in the Americas, with 2000–4000 cases annually in the United States. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

736.2 Causes

This condition happens because of *R. rickettsii* infecting vascular endothelial cells, causing widespread endothelial damage through direct cytotoxicity and immune-mediated injury, leading to increased vascular permeability, swelling, coagulation activation, and microvascular blood clot formation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

736.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

736.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

736.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

736.6 Complications

You may experience an incubation of 2–14 days, with the classic triad of fever, severe headache, and rash appearing on days 2–5, beginning as blanching erythematous macules on the wrists, ankles, and forearms, spreading centripetally to the trunk and palms/soles, with severe complications including encephalitis, acute kidney injury, ARDS, and gangrene.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

736.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam in endemic areas—treatment should not be delayed for laboratory confirmation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

736.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

736.9 Treatment Options

- Treatment options include doxycycline. outlook: mortality 20–30% if untreated
- 3–5% with appropriate treatment.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

736.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

736.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 737

Rosacea

737.1 Overview

Rosacea is a chronic inflammatory facial dermatosis primarily affecting middle-aged adults with fair skin. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

737.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of innate immune dysregulation, neurovascular dysfunction, and *Demodex folliculorum* mite overgrowth
- Triggers include sunlight, alcohol, spicy foods, hot beverages, and emotional stress.

737.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

737.4 Signs and Symptoms

Clinical features present with four subtypes: erythematotelangiectatic (persistent facial redness of the skin, telangiectasia, flushing), papulopustular (redness of the skin with papules and pustules, resembling acne), phymatous (skin thickening, particularly rhinophyma), and ocular (blepharitis, conjunctivitis, keratitis). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

737.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

737.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

737.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

737.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

737.9 Treatment Options

- Treatment options include sun protection, topical metronidazole or azelaic acid, topical ivermectin, oral doxycycline (low-dose, anti-inflammatory), and laser therapy for telangiectasia
- Severe rhinophyma may require surgical treatment.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

737.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

737.11 Outlook

- Outlook is chronic with flares
- Treatment provides control but not cure.

Chapter 738

Rosai-Dorfman Disease

738.1 Overview

Rosai-Dorfman disease (sinus histiocytosis with massive lymphadenopathy) is a rare, non-Langerhans cell histiocytic disorder characterized by massive, painless, non-tender lymphadenopathy. Driven by polyclonal growth of S100-positive histiocytes within lymph node sinuses, a pathognomonic histological hallmark is emperipolesis—the engulfment of intact lymphocytes and erythrocytes within the cytoplasm of histiocytes without cellular degradation. Affecting children and young adults, clinical features present with bilateral massive cervical lymphadenopathy ('bull-neck' appearance), fever, leukocytosis, elevated ESR, and polyclonal hypergammaglobulinemia. Extranodal involvement occurs in 40% of cases (skin, nasal cavity, orbits, bone, central nervous system). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

738.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

738.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

738.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

738.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

738.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

738.7 Diagnosis

Doctors confirm the diagnosis through tissue biopsy showing large S100+, CD68+, CD1a- histiocytes exhibiting emperipolesis. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

738.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

738.9 Treatment Options

- Treatment typically involves observation for asymptomatic self-limiting cases
- Surgical debulking, systemic corticosteroids, or targeted therapy (MEK/BRAF inhibitors) are reserved for compressive extranodal lesions.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

738.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

738.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 739

Rotavirus

739.1 Overview

Transmission is fecal-oral. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

739.2 Causes

This condition happens because of the virus infecting enterocytes of small intestinal villi, causing cell death, villous shrinkage or wasting, and malabsorptive diarrhea. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

739.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

739.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

739.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

739.6 Complications

- Clinical features: incubation period of 1–3 days, acute onset of fever, vomiting followed by profuse watery diarrhea lasting 3–8 days
- Severe dehydration is the major complication.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

739.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Stool antigen detection or PCR is used for confirmation. Treatment: aggressive oral rehydration therapy and zinc supplementation
- No antivirals are available.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

739.8 Prevention

Rotavirus is the leading cause of severe acute gastroenteritis in infants and young children worldwide before vaccine introduction. Prevention: live attenuated oral rotavirus vaccines (Rotarix, RotaTeq) have reduced rotavirus hospitalizations by 80–90%.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

739.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

739.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

739.11 Outlook

Most people recover well with appropriate rehydration. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 740

Rubella

740.1 Overview

Rubella (German measles) is a mild viral illness caused by rubella virus, most significant for its teratogenic effects, with less than 1000 cases reported annually in the US since 2004, and transmission by respiratory droplets and transplacentally. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

740.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of the virus replicating in the respiratory epithelium and spreading hematogenously
- In congenital rubella, the virus infects the placenta and fetus during the first trimester, causing vasculitis and direct cellular damage.

740.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

740.4 Signs and Symptoms

You may experience an incubation of 14–21 days. Postnatal rubella: low-grade fever, headache, coryza, and post-auricular/occipital lymphadenopathy, with a pink, maculopapular rash beginning on the face and spreading downward. Congenital rubella syndrome (CRS) includes the classic triad: sensorineural hearing loss (most common, 60–80%), congenital heart disease, and cataracts, occurring with infection in the first 16 weeks of pregnancy. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

740.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

740.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

740.7 Diagnosis

Doctors diagnose this using RT-PCR or IgM serology. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

740.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification

- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

740.9 Treatment Options

Treatment typically involves supportive. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

740.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

740.11 Outlook

- Most people recover well for postnatal rubella
- Congenital rubella syndrome causes permanent disability.

Chapter 741

Salicylate (Aspirin) Overdose

741.1 Overview

Salicylate overdose is a pharmaceutical overdose that can be acute or chronic (chronic toxicity more dangerous with higher mortality). This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

741.2 Causes

This condition happens because of salicylates uncoupling oxidative phosphorylation, stimulating the respiratory center, inhibiting Kreb's cycle enzymes, and increasing metabolic acid production. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

741.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

741.4 Signs and Symptoms

You may experience ringing in the ears (earliest symptom), hyperventilation, nausea, vomiting, hyperthermia, diaphoresis, and tachycardia

- As toxicity progresses, mixed acid-base disorders emerge (respiratory alkalosis followed by concurrent metabolic acidosis)
- Severe toxicity includes altered mental status, seizures, coma, and lung swelling.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

741.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

741.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

741.7 Diagnosis

Doctors diagnose this using serum salicylate level. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

741.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

741.9 Treatment Options

Treatment options include activated charcoal, IV fluids with dextrose, bicarbonate therapy (goal urine pH 7.5–8.0) to enhance kidney elimination, and hemodialysis for severe toxicity. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

741.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

741.11 Outlook

The outlook is generally positive with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 742

Sarcoidosis

742.1 Overview

Sarcoidosis is a multisystem granulomatous disease of unknown cause, characterized by non-caseating granulomas in affected tissues. It is discussed in detail in Chapter ?? (Section 743). Briefly, it most commonly affects the lungs and intrathoracic lymph nodes (>90%), but can involve any organ, including the skin (lupus pernio, redness of the skin nodosum), eyes (uveitis, dry eyes), liver, spleen, heart (arrhythmias, cardiomyopathy, heart failure), central nervous system (neurosarcoidosis—cranial nerve palsies, meningitis, hypothalamic-pituitary involvement), exocrine glands, and bones. The To make the diagnosis, doctors look for compatible clinical/radiographic findings, histologic demonstration of non-caseating granulomas, and exclusion of other causes. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

742.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

742.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease

- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

742.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

742.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

742.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

742.7 Diagnosis

Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

742.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

742.9 Treatment Options

Treatment typically involves indicated for organ-threatening or symptomatic disease: corticosteroids are first-line

- Hydroxychloroquine is used for cutaneous sarcoidosis
- Methotrexate, leflunomide, mycophenolate, azathioprine, and TNF inhibitors are used for refractory or severe disease.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

742.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

742.11 Outlook

The outlook is generally positive: two-thirds of patients experience spontaneous remission within a decade. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 743

Sarcoidosis

743.1 Overview

Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions.

- Sarcoidosis is a multisystem granulomatous disorder of unknown cause characterized by non-caseating granulomas, most commonly affecting the lungs and intrathoracic lymph nodes
- It disproportionately affects young adults, particularly African Americans and Scandinavians.

743.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

- This condition happens because of an aberrant immune response leading to granuloma formation. lung involvement (present in 90% of cases) manifests as cough, shortness of breath, and chest pain
- Extrapulmonary disease involves the skin (redness of the skin nodosum, lupus pernio), eyes (uveitis), joints, heart, and nervous system.

743.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)

- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

743.4 Signs and Symptoms

Your symptoms depend on the organs involved. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

743.5 Progression and Stages

Treatment typically involves indicated for symptomatic or progressive disease: corticosteroids are first-line, with methotrexate, azathioprine, or infliximab as steroid-sparing agents. Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

743.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy

- Lymphoma risk in some autoimmune conditions

743.7 Diagnosis

Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

743.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

743.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)

- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

743.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

743.11 Outlook

The outlook is generally positive, with many cases resolving spontaneously. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 744

Sarcoidosis

744.1 Overview

Sarcoidosis is discussed in detail in Chapter **??**. It is a multisystem granulomatous disease of unknown cause affecting the lungs, lymph nodes, skin, eyes, and other organs. Non-caseating granulomas on biopsy are the histological hallmark. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

744.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

744.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)

- Vitamin D deficiency

744.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

744.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

744.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

744.7 Diagnosis

Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

744.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

744.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

744.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

744.11 Outlook

Autoimmune diseases are generally chronic conditions requiring long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 745

Sarcopenia

745.1 Overview

Sarcopenia is a geriatric syndrome defined as the progressive, generalized loss of skeletal muscle mass, strength, and function with aging, with prevalence of 5–13% in adults aged 60–70 years and up to 50% in those over 80. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

745.2 Causes

This condition happens because of neural denervation of motor units, reduced anabolic signaling (growth hormone, IGF-1, testosterone), increased catabolic cytokines (IL-6, TNF-alpha), mitochondrial dysfunction, and inadequate protein intake. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

745.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

745.4 Signs and Symptoms

You may experience low muscle mass, weakness, and functional decline. The European Working Group on Sarcopenia in Older People (EWGSOP2) defines sarcopenia by low muscle strength (grip strength <27 kg men, <16 kg women) plus low muscle quantity/quality and confirmed by low physical performance (gait speed $\leq$0.8 m/s). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

745.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

745.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

745.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

745.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

745.9 Treatment Options

- Treatment typically involves resistance exercise (2–3 times weekly, progressive overload) plus adequate protein intake ($\geq$1.2 g/kg/day) and vitamin D (800 IU/day)
- Testosterone replacement may be considered in hypogonadal men.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

745.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

745.11 Outlook

- The outlook varies from person to person
- Early intervention can improve outcomes.

Chapter 746

Sarcopenic Obesity

746.1 Overview

Sarcopenic obesity is a distinct clinical entity characterized by the combination of excess adiposity and low muscle mass and function in the same individual. It is increasingly recognized with aging, but also occurs in the setting of malignancy, chronic disease, and catabolic states. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

746.2 Causes

This condition happens because of a complex interplay of age-related hormonal decline, chronic inflammation, insulin resistance, physical inactivity, and inadequate protein intake. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

746.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

746.4 Signs and Symptoms

You may experience the combination of obesity with low muscle strength and functional decline.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

746.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

746.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

746.7 Diagnosis

To make the diagnosis, doctors look for measurement of body composition (DXA, bioelectrical impedance analysis, CT, or MRI) along with assessment of muscle strength and physical performance.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

746.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

746.9 Treatment Options

Treatment options include resistance exercise training, adequate protein intake, vitamin D supplementation, and intentional weight loss through caloric restriction with emphasis on preserving lean mass. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

746.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

746.11 Outlook

- Outlook is guarded
- Sarcopenic obesity carries additive risks for falls, fractures, physical disability, cardiovascular disease, and mortality.

Chapter 747

Scabies

747.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Scabies is a parasitic infestation of the skin caused by the mite *Sarcoptes scabiei var. hominis*. Transmission occurs through prolonged skin-to-skin contact
- Fomite transmission is less common but possible.

747.2 Causes

This condition happens because of the female mite burrowing into the stratum corneum, laying eggs and triggering a delayed type IV hypersensitivity reaction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

747.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

747.4 Signs and Symptoms

- You may experience intensely pruritic (especially nocturnal), papular, and vesicular eruptions in characteristic locations: interdigital web spaces, wrists, elbows, axillae, periumbilical area, waistband, buttocks, and genitalia in males
- Burrows (thin, linear, grayish-white tracks) are pathognomonic. Crusted (Norwegian) scabies is a highly contagious variant in immunocompromised patients with widespread, thick, hyperkeratotic crusts.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

747.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

747.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

747.7 Diagnosis

Doctors diagnose this based on clinical examination

- Dermoscopy may reveal the "delta wing jet" appearance
- Skin scraping confirms.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

747.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

747.9 Treatment Options

- Treatment typically involves with topical permethrin 5% cream (applied to the entire body from neck down, left on for 8–14 hours, repeated in 1 week) or oral ivermectin (200 μg/kg, repeated in 1–2 weeks)
- Close contacts should be treated simultaneously.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

747.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

747.11 Outlook

- Most people recover well with treatment
- Post-scabetic itching may persist for weeks.

Chapter 748

Scarlet Fever

748.1 Overview

Scarlet fever is an acute systemic infection caused by exotoxin-producing strains of Group A β-hemolytic Streptococcus (GAS, Streptococcus pyogenes), predominantly affecting children aged 5–15 years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

748.2 Causes

This condition happens because of pyrogenic exotoxins (SPE-A, B, C) inducing delayed-type cutaneous hypersensitivity in individuals lacking protective antitoxin antibodies. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

748.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

748.4 Signs and Symptoms

You may experience fever, pharyngitis, strawberry tongue (initially white coat with red papillae, progressing to a beefy-red raw appearance), circumoral pallor, and a diffuse erythematous maculopapular rash with a 'sandpaper' texture that blanches on pressure and accentuates in skin folds (Pastia lines), followed by desquamation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

748.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

748.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

748.7 Diagnosis

Your doctor confirms the diagnosis with rapid antigen detection test or throat culture. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

748.8 Prevention

Management requires 10 days of oral penicillin V or amoxicillin (or macrolides in penicillin-allergic patients) to eradicate GAS and prevent acute rheumatic fever and poststreptococcal glomerulonephritis.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

748.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

748.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

748.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 749

Schistosomiasis

749.1 Overview

Schistosomiasis (bilharziasis) is a trematode infection caused by blood flukes of the genus *Schistosoma*: *S. haematobium* (urinary), *S. mansoni* (intestinal), *S. japonicum*, *S. intercalatum*, and *S. mekongi* (intestinal), with 200 million infected and 800 million at risk, primarily in sub-Saharan Africa, with transmission through contact with fresh water containing infected snails. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

749.2 Causes

This condition happens because of cercariae penetrating human skin, transforming into schistosomula, migrating to the portal venous system, maturing into adult worms, and producing eggs that become trapped in tissues, causing granulomatous inflammation, scarring, organomegaly, and obstruction. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

749.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

749.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

749.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

749.6 Complications

You may experience acute schistosomiasis (Katayama fever): fever, cough, urticaria, muscle pain, and hepatosplenomegaly 2–6 weeks after exposure

- Chronic *S. haematobium*: hematuria, dysuria, hydronephrosis, and bladder scarring, with squamous cell carcinoma as a long-term complication
- Chronic *S. mansoni* and *S. japonicum*: hepatosplenomegaly, periportal scarring (Symmers' pipe-stem scarring), portal high blood pressure, and esophageal varices.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

749.7 Diagnosis

Doctors diagnose this using microscopic identification of eggs in stool or urine. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

749.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

749.9 Treatment Options

Treatment options include praziquantel. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

749.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

749.11 Outlook

The outlook is good with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 750

Schizoaffective Disorder

750.1 Overview

Schizoaffective disorder has a lifetime prevalence of 0.3%. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

750.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

750.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

750.4 Signs and Symptoms

Clinical features require that delusions or hallucinations occur without prominent mood symptoms for at least 2 weeks. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

750.5 Progression and Stages

The condition is marked by an uninterrupted period of illness with concurrent features of schizophrenia (active-phase symptoms of delusions, hallucinations, disorganized speech, negative symptoms) and a major mood episode (depressive or manic) that occurs for the majority of the total duration of the illness. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

750.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

750.7 Diagnosis

Doctors diagnose this using clinical interview. Management combines antipsychotics and mood stabilizers/antidepressants guided by the predominant mood polarity. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

750.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

750.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

750.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

750.11 Outlook

outlook is intermediate between schizophrenia and mood disorders. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 751

Schizophrenia

751.1 Overview

Schizophrenia has a lifetime prevalence of 0.3–0.7%, with a male-to-female ratio of 1.4:1, with men typically presenting earlier (peak onset 20–28 years) than women (peak onset 25–35 years). This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

751.2 Causes

This condition happens because of dopaminergic hyperactivity in the mesolimbic pathway (positive symptoms), dopaminergic hypoactivity in the mesocortical pathway (negative symptoms and thinking and memory deficits), glutamatergic NMDA receptor hypofunction, and widespread gray matter loss, with genetic factors accounting for 80% of risk. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

751.3 Risk Factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

751.4 Signs and Symptoms

You may experience two or more of the following for a significant portion of time during a 1-month period (at least one must be delusions, hallucinations, or disorganized speech): delusions (fixed false beliefs, often bizarre), hallucinations (auditory most common), disorganized speech (loose associations, tangentiality), grossly disorganized or catatonic behavior, and negative symptoms (blunted affect, avolition, alogia, anhedonia, asociality), with continuous signs persisting for at least 6 months.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

751.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

751.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

751.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam—no biomarkers are available.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

751.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

751.9 Treatment Options

Treatment options include antipsychotics (both first-generation and second-generation, with clozapine being the only FDA-approved medication for treatment-resistant schizophrenia), thinking and memory-behavioral therapy for psychosis (CBTp), psychosocial interventions, and thinking and memory remediation therapy. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

751.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

751.11 Outlook

The outlook varies from person to person: 20–30% have a good outcome, with a chronic course and functional decline in many, and suicide risk of 5–10%. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 752

Schwannoma

752.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Schwannomas are non-cancerous nerve sheath tumors arising from Schwann cells
- Vestibular schwannomas (acoustic neuromas) are the most common intracranial schwannomas.

752.2 Causes

This condition happens because of neoplastic growth of Schwann cells. Bilateral vestibular schwannomas are pathognomonic of neurofibromatosis type 2. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

752.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

752.4 Signs and Symptoms

Symptoms of vestibular schwannoma include unilateral sensorineural hearing loss, ringing in the ears, and imbalance. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

752.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

752.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

752.7 Diagnosis

- Doctors diagnose this based on MRI showing a cerebellopontine angle mass with an ice-cream cone appearance in the internal auditory canal. Management options include observation with serial imaging, stereotactic radiosurgery (Gamma Knife), and microsurgical removal
- Facial nerve preservation is a key surgical goal.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

752.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

752.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

752.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

752.11 Outlook

Most people recover well for non-cancerous schwannomas. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 753

Scleritis

753.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

753.2 Causes

This condition happens because of immune-mediated inflammation of the sclera. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

753.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

753.4 Signs and Symptoms

You may experience deep, boring eye pain, redness, and decreased vision

- Anterior scleritis may be diffuse, nodular, or necrotizing
- Posterior scleritis can mimic orbital tumors.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

753.5 Progression and Stages

Your outlook depends on the severity and underlying systemic disease. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

753.6 Complications

- Scleritis is a severe inflammation of the sclera that is potentially sight-threatening due to complications such as glaucoma, cataract, and posterior segment inflammation
- Over 50% of cases are associated with systemic autoimmune disease, particularly rheumatoid arthritis, granulomatosis with polyangiitis, and relapsing polychondritis.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

753.7 Diagnosis

Doctors diagnose this based on clinical examination and imaging when indicated. Management requires systemic corticosteroids and immunosuppressive agents. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

753.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

753.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

753.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

753.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 754

Scoliosis

754.1 Overview

Scoliosis is abnormal lateral curvature of the spine (Cobb angle >10°). Adolescent idiopathic scoliosis (AIS) is the most common form (80% of cases), typically presenting in girls during the adolescent growth spurt, with a right thoracic curve. Neuromuscular scoliosis (from cerebral palsy, muscular dystrophy), congenital scoliosis (vertebral anomalies), and syndromic scoliosis (Marfan, Down syndrome) are other forms. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

754.2 Causes

This condition happens because of abnormal spinal development or neuromuscular imbalance. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

754.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

754.4 Signs and Symptoms

You may experience asymmetrical trunk, shoulder height discrepancy, and rib hump on forward bending. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

754.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- The right treatment depends on severity: mild curves are monitored
- Moderate curves (25–40°) are treated with bracing
- Severe curves (>40–50°) require surgical correction with posterior spinal instrumentation and fusion.

754.6 Complications

- In most cases, the outlook is favorable with appropriate management
- Complications include chronic back pain and cardiopulmonary compromise in severe curves.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

754.7 Diagnosis

Doctors diagnose this based on physical examination and X-ray with Cobb angle measurement. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

754.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

754.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

754.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

754.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 755

Scorpion Stings

755.1 Overview

Centruroides sculpturatus (bark scorpion) is the only medically significant scorpion in North America. Venom is a sodium channel toxin causing prolonged neuronal depolarization and excessive autonomic and somatic motor activity. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

755.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

755.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

755.4 Signs and Symptoms

Clinical features: immediate local pain, paresthesias, involuntary muscle spasms, cranial nerve dysfunction (involuntary eye movements, opsoclonus, hypersalivation, tongue fasciculations, difficulty swallowing), respiratory distress, and seizures. Children are at highest risk for severe toxicity. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

755.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

755.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

755.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Anascorp (Centruroides immune F(ab')2 antivenom) is highly effective, with resolution of symptoms within 1–4 hours. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

755.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

755.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

755.10 Living with It

Management: supportive care, benzodiazepines for agitation and muscle spasms.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

755.11 Outlook

The outlook is generally positive with antivenom administration. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 756

Seborrheic Dermatitis

756.1 Overview

Seborrheic dermatitis is a chronic inflammatory skin condition affecting areas rich in sebaceous glands (scalp, face, chest, back). It is linked to *Malassezia* species (lipophilic yeast) and may be triggered by stress, cold weather, and neurological conditions (Parkinson's disease). In infants, it manifests as "cradle cap" (thick, yellowish scales on the scalp). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

756.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

756.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

756.4 Signs and Symptoms

Clinical features in adults include greasy, yellowish scales on an erythematous base, affecting the scalp (dandruff in mild form), eyebrows, nasolabial folds, and ears. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

756.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

756.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

756.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

756.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

756.9 Treatment Options

Treatment options include topical antifungal agents (ketoconazole shampoo/cream), low-potency topical corticosteroids, and calcineurin inhibitors. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

756.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

756.11 Outlook

outlook is chronic with periodic flares. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 757

Selenium Deficiency

757.1 Overview

Selenium deficiency is a mineral deficiency disorder occurring in regions with low soil selenium content (parts of China, Europe, New Zealand). This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

757.2 Causes

This condition happens because of deficiency of selenium, an essential trace element functioning as a component of selenoproteins, including glutathione peroxidases (antioxidant defense), iodothyronine deiodinases (thyroid hormone metabolism), and selenoprotein P (selenium transport). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

757.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

757.4 Signs and Symptoms

- You may experience Keshan disease (an endemic dilated cardiomyopathy affecting children and women of childbearing age, presenting with acute or chronic heart failure, cardiomegaly, arrhythmias) and Kashin-Beck disease (an osteoarthropathy presenting with joint pain, stiffness, and deformities of the fingers, knees, and ankles)
- Selenium deficiency also impairs immune function, exacerbates viral virulence, and is linked to infertility and myopathy.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

757.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

757.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

757.7 Diagnosis

Doctors diagnose this using low serum selenium (less than 50–70 µg/L) and decreased glutathione peroxidase activity.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

757.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

757.9 Treatment Options

Treatment focuses on sodium selenite or selenomethionine 100–200 µg daily. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

757.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

757.11 Outlook

The outlook varies from person to person.

Chapter 758

Sepsis and Septic Shock

758.1 Overview

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection, with septic shock being a subset with profound circulatory, cellular, and metabolic abnormalities associated with a higher risk of mortality, with an estimated 48.9 million cases and 11 million deaths annually globally. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

758.2 Causes

This condition happens because of infection triggering pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAMPs) that bind toll-like receptors (TLRs), leading to release of pro-inflammatory cytokines (TNF-α, IL-1, IL-6), neutrophil activation, complement cascade, endothelial injury, and microvascular blood clot formation, with mitochondrial dysfunction and immune exhaustion contributing to late mortality. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

758.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to

endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

758.4 Signs and Symptoms

You may experience fever or hypothermia, tachycardia, tachypnea, altered mental status, hypotension, oliguria, mottled skin, and evidence of end-organ dysfunction (elevated lactate, acute kidney injury, liver dysfunction, thrombocytopenia). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

758.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

758.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

758.7 Diagnosis

Doctors diagnose this using clinical criteria per Sepsis-3 guidelines (suspected infection plus SOFA score increase of ≥ 2 points), with blood cultures, lactate, complete blood count, coagulation profile, kidney and liver function, procalcitonin, and imaging to identify the source. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

758.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

758.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

758.10 Living with It

Treatment options include early recognition, source control, empirical broad-spectrum antibiotics (within 1 hour of recognition), intravenous fluids (30 mL/kg crystalloids for hypotension), vasopressors (norepinephrine first-line), and supportive care.

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

758.11 Outlook

Your outlook depends on the underlying infection, timeliness of treatment, and patient factors. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 759

Septic Abortion

759.1 Overview

Septic abortion is a miscarriage or induced abortion complicated by infection of the uterine cavity and its contents, typically following unsafe abortion procedures. It remains a significant cause of maternal mortality in regions with restricted abortion access. Causative organisms include a mix of aerobic and anaerobic bacteria from the lower genital tract. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

759.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

759.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices

- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

759.4 Signs and Symptoms

You may experience fever, chills, foul-smelling vaginal discharge, lower abdominal pain, uterine tenderness, and signs of sepsis. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

759.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

759.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts

- Multi-organ failure in severe cases

759.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with blood cultures and pelvic imaging. Treatment requires IV broad-spectrum antibiotics and prompt surgical evacuation of retained products of conception. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

759.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

759.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

759.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

759.11 Outlook

The outlook is generally positive with prompt treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 760

Septic Arthritis

760.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- It most commonly affects large joints (knee, hip) and is usually monoarticular. Routes of infection include hematogenous spread (most common), direct inoculation (trauma, surgery, injection), and contiguous spread from adjacent osteomyelitis. *S. aureus* is the most common organism in adults
- *Neisseria gonorrhoeae* in young, sexually active adults
- And *Kingella kingae* in children under 4.

760.2 Causes

This condition happens because of bacterial infection of the joint space. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

760.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

760.4 Signs and Symptoms

You may experience acute joint pain, swelling, warmth, redness of the skin, and reduced range of motion, with systemic signs of infection (fever, chills). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

760.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

760.6 Complications

- Sepsis and septic shock

- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

760.7 Diagnosis

To make the diagnosis, doctors look for joint removal by suction showing turbid fluid with elevated WBC count (>50,000 cells/μL with >90% neutrophils), positive Gram stain and culture. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

760.8 Prevention

Septic arthritis is infection of a joint, a medical emergency requiring prompt treatment to prevent irreversible articular cartilage destruction. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules

- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

760.9 Treatment Options

Treatment focuses on urgent joint removal of fluid (repeated removal by suction, using a camera inside a joint lavage, or open removal of fluid) and IV antibiotics (empiric: vancomycin + ceftriaxone, adjusted based on culture and sensitivity). Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

760.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

760.11 Outlook

- The outlook is generally positive with prompt treatment
- Delayed treatment leads to irreversible joint damage.

Chapter 761

Severe Combined Immunodeficiency

761.1 Overview

Severe combined immunodeficiency is a group of rare, life-threatening genetic disorders characterized by severe impairment of both cellular (T-cell) and humoral (B-cell) immunity, with or without natural killer (NK) cell deficiency. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

761.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of over 20 genetic causes, with X-linked SCID (IL2RG mutations, most common) and adenosine deaminase (ADA) deficiency being the best known
- Without treatment, affected infants typically die within the first year of life from severe, recurrent, opportunistic infections.

761.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

761.4 Signs and Symptoms

You may experience recurrent severe infections, failure to thrive, and persistent diarrhea.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

761.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

761.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

761.7 Diagnosis

Newborn screening can detect SCID by measuring T-cell receptor removal circles (TRECs) on dried blood spots, with diagnosis confirmed by lymphocyte subset analysis (flow cytometry), immunoglobulin levels, and genetic testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

761.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

761.9 Treatment Options

Treatment options include hematopoietic stem cell transplantation (curative, ideally before 3.5 months of age), enzyme replacement therapy (for ADA deficiency), and gene therapy (for X-linked SCID and ADA-SCID). Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

761.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

761.11 Outlook

Most people recover well with early transplantation.

Chapter 762

Shift Work Sleep Disorder

762.1 Overview

Shift work sleep disorder (SWSD) is a circadian rhythm sleep-wake disorder affecting individuals whose work schedules overlap with the typical sleep period (night shifts, early morning shifts, rotating shifts), causing circadian misalignment with resultant insomnia, excessive sleepiness, or both. Approximately 20–30% of shift workers meet criteria. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

762.2 Causes

This condition happens because of a misalignment between the endogenous circadian timing system and the work-rest schedule. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

762.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

762.4 Signs and Symptoms

You may experience difficulty initiating or maintaining sleep during the available sleep period and excessive sleepiness during work hours. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

762.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

762.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

762.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with sleep diaries. Management strategies include strategic napping before or during shifts, bright light exposure during work, controlled light avoidance during the commute home, fixed sleep scheduling in a dark/cool room, melatonin at the desired bedtime, and wake-promoting agents (modafinil, armodafinil, or caffeine) for night-shift alertness. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

762.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

762.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

762.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

762.11 Outlook

- The outlook is generally positive with appropriate management
- Chronic misalignment increases risks of cardiovascular disease, metabolic syndrome, and depression.

Chapter 763

Short Bowel Syndrome

763.1 Overview

Short bowel syndrome (SBS) is a malabsorptive condition resulting from surgical removal of a significant portion of the small intestine, leading to insufficient nutrient and fluid absorption. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

763.2 Causes

The most common cause is Crohn's disease, followed by mesenteric reduced blood flow, trauma, and malignancy. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

763.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

763.4 Signs and Symptoms

You may experience profuse diarrhea, steatorrhea, weight loss, dehydration, electrolyte imbalances, and nutritional deficiencies.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

763.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- This condition happens because of reduced intestinal absorptive surface area
- The remaining functional bowel length and the presence or absence of the ileum and colon determine severity.

763.6 Complications

- Your outlook depends on the extent of removal and degree of adaptation
- Intestinal transplantation is reserved for patients with life-threatening complications of parenteral nutrition.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

763.7 Diagnosis

- Doctors diagnose this based on surgical history and clinical presentation
- Adaptation of the remaining bowel occurs over months to years.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

763.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

763.9 Treatment Options

- Treatment options include parenteral nutrition and fluids, gradual enteral feeding to promote adaptation, antidiarrheal agents, and H2 blockers or PPIs to reduce gastric acid
- Teduglutide (a GLP-2 analog) promotes intestinal adaptation and may reduce parenteral nutrition dependence.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

763.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 764

SIADH

764.1 Overview

Syndrome of inappropriate antidiuretic hormone secretion (SIADH) is the most common cause of euvolemic hyponatremia, characterized by excessive ADH secretion despite normal or low plasma osmolality. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

764.2 Causes

Causes include malignancies (small cell lung cancer most common, producing ectopic ADH), lung disorders (pneumonia, tuberculosis, aspergillosis), CNS disorders (meningitis, encephalitis, stroke, traumatic brain injury), and medications (SSRIs, TCAs, carbamazepine, vincristine, cyclophosphamide). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

764.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

764.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

764.5 Progression and Stages

Your symptoms depend on the severity and rate of development of hyponatremia: mild (lethargy, confusion, headache), moderate (nausea, vomiting, gait instability), severe (seizures, coma, respiratory arrest). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

764.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

764.7 Diagnosis

To make the diagnosis, doctors look for hyponatremia (<135 mEq/L), low plasma osmolality (<275 mOsm/kg), inappropriately concentrated urine (urine osmolality >100 mOsm/kg), elevated urine sodium (>40 mEq/L), and exclusion of hypovolemia, hypervolemia, adrenal insufficiency, and hypothyroidism. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

764.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

764.9 Treatment Options

- Treatment options include fluid restriction (first-line), pharmacotherapy (demeclocycline, vaptans/tolvaptan for refractory cases), and for severe symptomatic hyponatremia, cautious 3% hypertonic saline with frequent sodium monitoring to avoid osmotic demyelination syndrome. outlook relates to the underlying cause
- SIADH itself resolves when the causative condition is treated.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

764.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

764.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 765

Sick Sinus Syndrome

765.1 Overview

Sick sinus syndrome (SSS) is a group of irregular heartbeats caused by dysfunction of the sinoatrial node, the heart's primary pacemaker, with an incidence increasing with age, affecting 1 in 600 patients over 65 years. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

765.2 Causes

This condition happens because of age-related degenerative scarring of the SA node, reduced blood flow, infiltrative diseases (amyloidosis, sarcoidosis, hemochromatosis), cardiomyopathy, or surgical trauma. Iatrogenic causes include beta-blockers, calcium channel blockers, digoxin, and antiarrhythmic drugs. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

765.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

765.4 Signs and Symptoms

You may experience sinus bradycardia (often profound, <40 bpm), sinus arrest (pauses >3 seconds causing syncope), sinoatrial exit block, and the tachycardia-bradycardia syndrome (alternating paroxysmal atrial tachyarrhythmias, usually atrial fibrillation, with sinus bradycardia). Symptoms include fatigue, lightheadedness, presyncope, syncope (Stokes-Adams attacks), shortness of breath on exertion, and palpitations.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

765.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

765.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

765.7 Diagnosis

Doctors diagnose this based on 12-lead ECG showing bradyarrhythmia, 24-hour Holter monitoring (most useful for correlating symptoms with rhythm), event monitor, and for equivocal cases, electrophysiology study.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

765.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

765.9 Treatment Options

Treatment focuses on permanent pacemaker implantation (dual-chamber or atrial pacing) for symptomatic SSS. Pacemaker therapy resolves symptoms and prevents syncope but does not reduce mortality. Rate-slowing medications for tachycardia-bradycardia syndrome may be used after pacemaker placement. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

765.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

765.11 Outlook

- The outlook is good with appropriate pacing
- SSS itself is not life-threatening but can cause significant morbidity from syncope and falls.

Chapter 766

Sickle Cell Disease

766.1 Overview

Sickle cell disease is a group of inherited hemoglobinopathies predominantly affecting individuals of African, Mediterranean, Middle Eastern, and Indian descent. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

766.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of mutations in the β-globin gene, leading to production of abnormal hemoglobin S (HbS), with the most common genotype being HbSS (sickle cell anemia, caused by homozygosity for the Glu6Val mutation)
- Deoxygenated HbS polymerizes, causing red blood cells to assume a sickle shape, leading to vaso-occlusion, hemolysis, and end-organ damage.

766.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

766.4 Signs and Symptoms

You may experience vaso-occlusive crises (acute pain crises, the hallmark of SCD), acute chest syndrome (chest pain, fever, lung infiltrates—a leading cause of hospitalization and death), stroke (both ischemic and involving bleeding, occurring in up to 20% of children), avascular tissue death of the femoral head, priapism, splenic sequestration and functional asplenia, delayed growth, and chronic kidney disease.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

766.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

766.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

766.7 Diagnosis

Doctors diagnose this using hemoglobin electrophoresis and HPLC showing elevated HbS.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

766.8 Prevention

Treatment options include hydroxyurea (increases fetal hemoglobin [HbF], reduces crisis frequency), chronic transfusion therapy, pain Treatment for crises, and prophylactic penicillin (for children to prevent *S. pneumoniae* sepsis), with novel therapies including voxelotor (HbS polymerization inhibitor), crizanlizumab (anti-P-selectin antibody), and L-glutamine (oxidative stress reducer).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

766.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

766.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

766.11 Outlook

The outlook has improved significantly with modern therapies, with allogeneic stem cell transplantation offering cure but limited by donor availability, and gene therapy (lovotibeglogene autotemcel, exagamglogene autotemcel) showing remarkable efficacy in clinical trials.

Chapter 767

Sideroblastic Anemia

767.1 Overview

Sideroblastic anemia is a disorder characterized by defective iron utilization in erythropoiesis, leading to iron accumulation in mitochondria of erythroid precursors and forming ring sideroblasts in the bone marrow. The condition may be hereditary (X-linked sideroblastic anemia, ALAS2 mutation) or acquired (myelodysplastic syndrome, alcohol, lead poisoning, isoniazid, pyridoxine deficiency). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

767.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

767.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

767.4 Signs and Symptoms

You may experience symptoms of anemia. Diagnosis shows a dimorphic picture (microcytic and macrocytic cells on peripheral smear), basophilic stippling, and a characteristic ring sideroblast pattern on Prussian blue staining of bone marrow. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

767.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

767.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

767.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

767.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

767.9 Treatment Options

Treatment options include pyridoxine (vitamin B6, especially for hereditary forms), blood transfusions, and iron chelation therapy for iron overload, with allogeneic stem cell transplantation potentially curative. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

767.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

767.11 Outlook

Your outlook depends on the underlying cause and response to treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 768

Silicosis

768.1 Overview

Silicosis is an occupational pneumoconiosis caused by chronic inhalation of crystalline silicon dioxide (silica) dust in miners, sandblasters, stonecutters, and foundry workers. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

768.2 Causes

This condition happens because of silica particle ingestion by alveolar macrophages, triggering cell death, release of fibrogenic cytokines, and formation of dense hyalinized silicotic nodules. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

768.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

768.4 Signs and Symptoms

You may experience exertional shortness of breath, chronic productive cough, and increased susceptibility to lung tuberculosis (silicotuberculosis) due to impaired macrophage function. Diagnosis is supported by chest imaging showing upper lobe eggshell calcium buildup of hilar lymph nodes and nodular opacities. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

768.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

768.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

768.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

768.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

768.9 Treatment Options

- Treatment typically involves preventive and supportive
- Oxygen therapy and prompt treatment of mycobacterial infections are essential.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

768.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

768.11 Outlook

Your outlook depends on cumulative silica exposure. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 769

Sjögren's Syndrome

769.1 Overview

Sjögren's syndrome is a chronic autoimmune disease characterized by lymphocytic infiltration of exocrine glands (salivary and lacrimal), resulting in dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia), predominantly affecting women (9:1), and may be primary (isolated) or secondary (associated with another autoimmune disease, most commonly RA). This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

769.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

769.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

769.4 Signs and Symptoms

You may experience Sicca symptoms (dry eyes, dry mouth), parotid gland enlargement, dental caries, difficulty swallowing, and extraglandular manifestations (arthritis, interstitial lung disease, kidney tubular acidosis, peripheral neuropathy, lymphoma).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

769.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

769.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

769.7 Diagnosis

To make the diagnosis, doctors look for demonstration of glandular inflammation: salivary gland biopsy showing focal lymphocytic sialadenitis (focus score $\geq$1), positive anti-SSA/Ro and anti-SSB/La antibodies, abnormal ocular staining (Schirmer's test, Rose Bengal), and unstimulated salivary flow.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

769.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

769.9 Treatment Options

Treatment typically involves symptomatic: artificial tears, pilocarpine or cevimeline (muscarinic agonists to stimulate salivary and lacrimal secretion), and immunosuppression for severe extraglandular manifestations. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

769.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

769.11 Outlook

In most cases, the outlook is favorable, though patients have a 5–10% lifetime risk of developing B-cell lymphoma (MALT lymphoma of the parotid gland most common).

Chapter 770

Sleep Architecture Across the Lifespan

770.1 Overview

Newborns spend approximately 50% of sleep time in active (REM-like) sleep with an ultradian cycle of 50–60 minutes, and total sleep duration averages 16–18 hours daily. By age 5, REM proportion declines to adult levels (20–25%), and N3 sleep reaches its peak in childhood. Young adults exhibit well-defined sleep cycles with 20–25% REM, 45–55% N2, and 15–25% N3. The prevalence of sleep disorders such as insomnia, sleep-disordered breathing, and REM sleep behavior disorder increases with age. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

770.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

770.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

770.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

770.5 Progression and Stages

Sleep architecture is the pattern of sleep stages that changes dramatically across the lifespan. During adolescence, a physiologic phase delay emerges (delayed melatonin onset), coupled with reduced slow-wave sleep. Aging is linked to progressive decline in slow-wave sleep (especially N3), increased sleep fragmentation, increased N1 and wake after sleep onset (WASO), decreased total sleep time, phase advancement of circadian rhythms, and reduced amplitude of the melatonin rhythm. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

770.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

770.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

770.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

770.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

770.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

770.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 771

Sleep Stages (NREM N1–N3, REM)

771.1 Overview

Sleep is a reversible behavioral state divided into non-rapid eye movement (NREM) and rapid eye movement (REM) sleep, alternating in approximately 90-minute cycles throughout the night. REM sleep is marked by rapid, conjugate eye movements, skeletal muscle atonia (except diaphragm and extraocular muscles), desynchronized low-voltage mixed-frequency EEG, and vivid dreaming. Autonomic instability during REM produces fluctuating heart rate, blood pressure, and respiration. The transition from wakefulness to sleep involves activation of the ventrolateral preoptic nucleus (VLPO) in the hypothalamus, which inhibits arousal-promoting brainstem nuclei (locus coeruleus, raphe nuclei, tuberomammillary nucleus). The flip-flop switch model describes the mutual inhibition between the VLPO and the monoaminergic arousal systems, ensuring rapid and stable state transitions. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

771.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

771.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

771.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

771.5 Progression and Stages

N1 (Stage 1) is the lightest stage, representing the transition from wakefulness, characterized by theta rhythm (4–7 Hz) on EEG, slow rolling eye movements, and easy arousal. N2 (Stage 2) is the predominant sleep stage, occupying approximately 45–55% of total sleep time in adults, characterized by sleep spindles (12–14 Hz bursts) and K-complexes on EEG with decreased heart rate and core body temperature. N3 (Stage 3), or slow-wave sleep, is marked by high-amplitude, low-frequency delta waves (0.5–4 Hz), a high arousal threshold, and the greatest anabolic and restorative functions, including growth hormone secretion, immune cytokine release, and synaptic downscaling. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

771.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

771.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists

Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

771.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

771.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

771.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

771.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 772

Sleep-Related Bruxism

772.1 Overview

Sleep-related bruxism is a sleep-related movement disorder characterized by repetitive jaw-muscle activity during sleep, manifesting as tooth grinding or clenching, often associated with rhythmic masticatory muscle activity (RMMA). Prevalence is 8–12% in adults and 15–30% in children, decreasing with age. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

772.2 Causes

This condition happens because of sleep-related activation of the brainstem masticatory central pattern generator, triggered by spontaneous microarousals and possibly modulated by autonomic activation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

772.3 Risk Factors

Risk factors include anxiety, stress, sleep-disordered breathing, alcohol, smoking, and certain medications (SSRIs, amphetamines).

- Family history of mental illness
- Trauma or adverse childhood experiences
- Chronic stress
- Substance use
- Social isolation
- Underlying medical conditions
- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Medication use

772.4 Signs and Symptoms

You may experience audible tooth grinding, jaw pain or fatigue, temporal headaches, tooth wear, tooth sensitivity, and temporomandibular joint pain. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

772.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

772.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

772.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Polysomnography with masseter EMG is the gold standard.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

772.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

772.9 Treatment Options

Treatment options include occlusal splints for tooth protection, stress reduction, treatment of comorbid OSA, botulinum toxin injections for severe cases, and avoidance of exacerbating substances. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

772.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

772.11 Outlook

- In most cases, the outlook is manageable
- Severe untreated bruxism can result in tooth fracture and TMJ disorders.

Chapter 773

Sleep-Related Rhythmic Movement Disorder

773.1 Overview

Sleep-related rhythmic movement disorder (RMD) is a sleep-related movement disorder involving stereotypical, repetitive movements of large muscle groups occurring primarily during the transition from wakefulness to sleep or during NREM sleep. The most common forms are head banging, head rolling, and body rocking. Prevalence is up to 60% in infants (physiologic), declining to less than 1% in adults. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

773.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of unknown Disease mechanism
- In adults, it is linked to neurodevelopmental disorders, anxiety, and OSA.

773.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

773.4 Signs and Symptoms

You may experience rhythmic movements at a frequency of 0.5–2 Hz lasting less than 15 minutes. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

773.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

773.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

773.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Polysomnography with synchronized video recording is indicated when injury is suspected or to rule out seizure.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

773.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

773.9 Treatment Options

- Treatment typically involves primarily reassurance and safety measures
- Clonazepam, hypnotics, or behavioral therapy may be used in persistent or injurious adult cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

773.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

773.11 Outlook

- Outlook is non-cancerous
- Most children outgrow the condition.

Chapter 774

Small Bowel Obstruction

774.1 Overview

Small bowel obstruction (SBO) is a partial or complete blockage of the small intestine. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

774.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- The most common cause is adhesions from prior abdominal surgery (60% of cases)
- Other causes include hernias (inguinal, incisional), malignancy, volvulus, intussusception (in children), inflammatory bowel disease strictures, and gallstone ileus. This condition happens because of mechanical blockage of intestinal contents.

774.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

774.4 Signs and Symptoms

You may experience crampy abdominal pain (periumbilical), nausea, vomiting (bilious early, feculent late), abdominal distension, and obstipation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

774.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

774.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

774.7 Diagnosis

- Doctors diagnose this based on abdominal X-ray showing air-fluid levels and dilated proximal bowel loops with collapsed distal bowel
- CT abdomen/pelvis is more sensitive and identifies the transition point and cause.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

774.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

774.9 Treatment Options

- Treatment options include nasogastric decompression, IV fluid resuscitation, electrolyte correction, and close monitoring
- Complete obstruction or signs of strangulation (fever, tachycardia, peritoneal signs, metabolic acidosis) require emergent surgical exploration.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

774.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

774.11 Outlook

Your outlook depends on cause and presence of strangulation. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 775

Snake Envenomation

775.1 Overview

Viperidae venom contains hemotoxins causing local tissue destruction, coagulopathy, and thrombocytopenia. Elapidae venom contains neurotoxins causing neuromuscular blockade with descending paralysis, ptosis, bulbar palsy, respiratory failure, and sometimes minimal local symptoms. Sea snakes cause myotoxicity with generalized muscle pain, myoglobinuria, and acute kidney injury. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

775.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

775.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

775.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

775.5 Progression and Stages

Clinically significant snake envenomation is classified by family: Viperidae (vipers, pit vipers—rattlesnakes, copperheads, cottonmouths) and Elapidae (cobras, kraits, mambas, coral snakes, sea snakes). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

775.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

775.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with lab evaluation (CBC, PT/PTT, INR, fibrinogen, D-dimer, creatinine, CK). Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

775.8 Prevention

Management: supportive care, wound management, tetanus prophylaxis, and antivenom (Crotalidae polyvalent immune Fab for North American pit vipers).

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

775.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

775.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

775.11 Outlook

The outlook is generally positive with timely antivenom administration. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 776

Social Anxiety Disorder (Social Phobia)

776.1 Overview

Social anxiety disorder has a lifetime prevalence of 7–13%, making it one of the most common anxiety disorders, typically beginning in adolescence with peak onset at age 13. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

776.2 Causes

This condition happens because of amygdala hyperreactivity, insula hyperactivity, and altered default-mode network connectivity. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

776.3 Risk Factors

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

776.4 Signs and Symptoms

You may experience marked fear or anxiety about one or more social situations in which the individual is exposed to possible scrutiny by others, with fear of being negatively evaluated, embarrassment, rejection, or offending others, with the fear being out of proportion to the actual threat and persistent (at least 6 months). Physical symptoms (blushing, trembling, sweating, palpitations) are common.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

776.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

776.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

776.7 Diagnosis

Doctors diagnose this using clinical interview.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

776.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

776.9 Treatment Options

Treatment options include SSRIs and SNRIs as first-line pharmacotherapy, and CBT (thinking and memory restructuring and exposure therapy) as first-line psychotherapy. Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

776.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

776.11 Outlook

The outlook is good with treatment. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 777

Somatic Symptom Disorder

777.1 Overview

Somatic symptom disorder has an estimated prevalence of 5–7% in the general population and is more common in women. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

777.2 Causes

This condition happens because of hypervigilance to bodily sensations, amplification of normal physiological signals, maladaptive health-related cognitions, and altered brain connectivity in insula, anterior cingulate, and prefrontal cortex. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

777.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

777.4 Signs and Symptoms

You may experience one or more somatic symptoms that are distressing or result in significant disruption of daily life, with excessive thoughts, feelings, or behaviors related to the somatic symptoms manifested by disproportionate and persistent thoughts about the seriousness of symptoms, persistently high level of anxiety about health, or excessive time and energy devoted to these symptoms. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

777.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

777.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

777.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

777.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

777.9 Treatment Options

Treatment options include CBT (thinking and memory restructuring, attention retraining, behavioral activation) and pharmacotherapy focusing on comorbid depression and anxiety (SSRIs). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

777.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

777.11 Outlook

outlook is chronic but function can improve with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 778

Specific Phobias

778.1 Overview

Specific phobias have a lifetime prevalence of 10–13%, making them the most common anxiety disorder in the general population. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

778.2 Causes

This condition happens because of conditioned fear responses. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

778.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

778.4 Signs and Symptoms

You may experience marked, persistent (at least 6 months) fear or anxiety about a specific object or situation (eg., flying, heights, animals, receiving an injection, blood, enclosed spaces), with the phobic stimulus almost invariably provoking an immediate fear response out of proportion to the actual danger. Common subtypes include animal type, natural environment type, blood-injection-injury type (unique vasovagal response with fainting in up to 75%), situational type, and other type.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

778.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

778.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

778.7 Diagnosis

Doctors diagnose this using clinical interview. Treatment for choice is CBT with exposure therapy (systematic desensitization).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

778.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

778.9 Treatment Options

Psychotherapy (cognitive behavioral therapy, interpersonal therapy, psychodynamic therapy) Pharmacotherapy (antidepressants, antipsychotics, mood stabilizers, anxiolytics) Combined treatment approaches for optimal outcomes Hospitalization for acute crisis management Support groups and peer support programs Lifestyle interventions (exercise, sleep hygiene, stress management) Mindfulness and meditation practices Family therapy and psychoeducation Vocational and social rehabilitation Long-term maintenance therapy for chronic conditions

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

778.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

778.11 Outlook

Most people recover well with treatment. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 779

Spider Bites

779.1 Overview

Clinically significant spider bites in North America include black widow (Latrodectus) and brown recluse (Loxosceles). Black widow venom contains alpha-latrotoxin, causing massive release of acetylcholine and norepinephrine. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

779.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

779.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

779.4 Signs and Symptoms

Clinical features: local pinprick sensation, then severe muscle pain and cramping, rigidity, diaphoresis, nausea, vomiting, high blood pressure, tachycardia, and rarely respiratory compromise. Brown recluse venom contains sphingomyelinase D, causing dermonecrotic injury, hemolysis, and coagulopathy. Clinical features: initially painless bite, then over 6–12 hours redness of the skin, blister, and tissue death forming an eschar over 1–2 weeks. Systemic loxoscelism includes fever, chills, nausea, arthralgias, hemolytic anemia, thrombocytopenia, DIC, acute kidney injury. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

779.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

779.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

779.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

779.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

779.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

779.10 Living with It

Treatment: supportive care, benzodiazepines for muscle spasms, and latrodectus antivenom for severe cases. Treatment: wound care, supportive care for hemolysis and kidney failure.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

779.11 Outlook

In most cases, the outlook is favorable. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 780

Splenic Abscess

780.1 Overview

Splenic abscess is a rare, life-threatening localized suppurative infection of the splenic parenchyma, occurring predominantly in immunocompromised individuals or patients with infective endocarditis. Disease mechanism involves hematogenous seeding from distant septic foci (endocarditis, urinary tract infections, osteomyelitis), direct extension from contiguous abdominal infections, or secondary infection of ischemic splenic infarcts. Common pathogens include Staphylococcus aureus, Streptococcus species, Enterobacteriaceae, and anaerobes. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

780.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

780.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

780.4 Signs and Symptoms

Symptoms often appear as the classic triad of fever, left upper quadrant abdominal pain, and leukocytosis, frequently accompanied by left pleural effusion or referred left shoulder pain (Kehr sign). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

780.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

780.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage

- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

780.7 Diagnosis

Doctors confirm the diagnosis through abdominal CT or ultrasound showing solitary or multiple hypodense lesions. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

780.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)

- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

780.9 Treatment Options

Treatment focuses on parenteral broad-spectrum antimicrobial therapy combined with through the skin image-guided catheter removal of fluid or splenectomy for multiloculated or refractory abscesses. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

780.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

780.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 781

Splenic Artery Aneurysm

781.1 Overview

Splenic artery aneurysm (SAA) is the most common visceral artery aneurysm, accounting for 60–80% of all splanchnic vascular aneurysms, with a marked female-to-male predominance (4:1). This cardiovascular condition is a leading cause of morbidity and mortality globally. Risk increases with age, and the condition often requires lifelong management. Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

781.2 Causes

Hormonal and hemodynamic changes during pregnancy markedly increase wall stress and rupture risk. Most SAAs are asymptomatic incidental findings on abdominal imaging, but symptomatic cases present with left upper quadrant or epigastric pain. Rupture presents as acute intra-abdominal bleeding ('double-rupture' phenomenon into the lesser sac followed by free peritoneal rupture), precipitating catastrophic involving bleeding shock. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

781.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)
- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

781.4 Signs and Symptoms

Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

781.5 Progression and Stages

This condition happens because of arterial wall media deterioration driven by multiparity, portal high blood pressure, arterial dysplasia, or pancreatitis-induced pseudaneurysms. Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

781.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia

- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

781.7 Diagnosis

Your doctor confirms the diagnosis with CT angiography. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

781.8 Prevention

- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

781.9 Treatment Options

Treatment options include endovascular coil embolization, covered stenting, or surgical ligation for symptomatic aneurysms, aneurysms > 2 cm, or any SAA in women of childbearing age or pregnant patients. Management includes lifestyle modifications, pharmacotherapy targeting the underlying mechanisms, and interventional procedures when indicated. Long-term adherence to treatment is essential for preventing disease progression. Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy (pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

781.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

781.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 782

Splenic Cyst

782.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Splenic cysts are fluid-filled parenchymal lesions categorized as primary (true cysts with an epithelial or mesothelial lining, accounting for 20%) or secondary (false pseudocysts lacking cellular lining, accounting for 80%). Primary cysts include congenital epidermoid cysts, mesothelial cysts, and parasitic hydatid cysts (Echinococcus granulosus). Secondary pseudocysts result from prior blunt abdominal trauma, splenic tissue death due to lack of blood supply, or localized hematoma liquefaction. Most small cysts (< 5 cm) are asymptomatic
- Larger lesions cause left upper quadrant fullness, early satiety, dull flank pain, or palpable splenomegaly.

782.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

782.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

782.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

782.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

782.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

782.7 Diagnosis

- Doctors confirm the diagnosis through ultrasound or CT imaging
- Hydatid serology is mandatory in endemic regions. Treatment for small asymptomatic cysts is conservative surveillance
- Large symptomatic or high-risk cysts (> 5 cm) undergo using small incisions and a camera unroofing, partial splenectomy, or cystectomy to preserve splenic immune function while avoiding rupture.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

782.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

782.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

782.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

782.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 783

Splenic Cysts

783.1 Overview

Most are asymptomatic and discovered incidentally on imaging. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

783.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

783.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

783.4 Signs and Symptoms

Symptoms of large cysts include abdominal pain, early satiety, or a palpable mass. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight

changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

783.5 Progression and Stages

Splenic cysts are fluid-filled lesions in the spleen, classified as true cysts (with epithelial lining, including congenital, epidermoid, and dermoid cysts) or pseudocysts (without epithelial lining, usually resulting from prior trauma, tissue death due to lack of blood supply, or infection). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

783.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

783.7 Diagnosis

Doctors diagnose this based on imaging. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

783.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

783.9 Treatment Options

Treatment typically involves observation for small, asymptomatic cysts

- Surgical removal (cystectomy or partial splenectomy) or splenectomy for large or symptomatic cysts
- Through the skin removal by suction and sclerotherapy is an alternative for selected cases.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

783.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

783.11 Outlook

Most people recover well. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 784

Splenic Infarction

784.1 Overview

Splenic tissue death due to lack of blood supply is ischemic tissue death of splenic parenchyma resulting from occlusion of the splenic artery or its branches. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

784.2 Causes

The most common cause is embolization from the heart (atrial fibrillation, endocarditis, mural blood clot), followed by hematologic disorders (myeloproliferative disorders, sickle cell disease, thalassemia) and vasculitis. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

784.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

784.4 Signs and Symptoms

You may experience acute left upper quadrant pain, sometimes radiating to the left shoulder (Kehr's sign). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

784.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

784.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

784.7 Diagnosis

Doctors diagnose this based on CT scan showing wedge-shaped, hypodense, non-enhancing lesions. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

784.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

784.9 Treatment Options

Treatment typically involves conservative (analgesics, observation) with treatment of the underlying cause. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

784.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

784.11 Outlook

- In most cases, the outlook is favorable
- Rarely, complicated tissue death due to lack of blood supply (abscess, rupture) may require splenectomy.

Chapter 785

Splenic Lymphoma

785.1 Overview

Splenic lymphoma is involvement of the spleen by systemic lymphoma or, less commonly, primary splenic origin. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

785.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

- This condition happens because of lymphoproliferation
- Hepatitis C virus infection is linked to SMZL in some cases. Hairy cell leukemia is another important splenic lymphoproliferative disorder characterized by the BRAF V600E mutation.

785.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age

- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

785.4 Signs and Symptoms

You may experience massive splenomegaly, lymphocytosis (villous lymphocytes on blood smear in SMZL), and pancytopenia (in hairy cell leukemia). Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

785.5 Progression and Stages

Splenic marginal zone lymphoma (SMZL) is the most common primary splenic lymphoma, a low-grade B-cell lymphoma. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

785.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

785.7 Diagnosis

- Diagnosis typically involves bone marrow biopsy, flow cytometry, and CT imaging. Management varies by histology: splenectomy and/or rituximab for SMZL
- Cladribine or moxetumomab for hairy cell leukemia.
- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

785.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs

Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

785.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

785.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian

- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

785.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 786

Splenic Rupture

786.1 Overview

Splenic rupture is a life-threatening emergency, most commonly caused by blunt abdominal trauma (motor vehicle accidents, falls, sports injuries). The spleen is the most commonly injured abdominal organ in trauma. The condition may also occur spontaneously in the setting of massive splenomegaly (infectious mononucleosis, hematologic malignancies, myelofibrosis). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

786.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

786.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

786.4 Signs and Symptoms

You may experience left upper quadrant pain, peritoneal signs, hemodynamic instability (tachycardia, hypotension), and Kehr's sign (left shoulder pain). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

786.5 Progression and Stages

Management follows Advanced Trauma Life Support (ATLS) principles: hemodynamic resuscitation, splenic grading, and decision between non-operative management (for hemodynamically stable patients with lower-grade injuries) and emergent splenectomy (for hemodynamic instability or high-grade injuries). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

786.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

786.7 Diagnosis

Doctors diagnose this based on focused assessment with sonography for trauma (FAST) and CT scan, which also grades the injury. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

786.8 Prevention

- Your outlook depends on injury severity and timely intervention
- Post-splenectomy vaccination (pneumococcal, meningococcal, *Haemophilus influenzae* type b) and prophylactic antibiotics are essential to prevent OPSI.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

786.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

786.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

786.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 787

Splenomegaly

787.1 Overview

Splenomegaly is enlargement of the spleen beyond its normal size (normally approximately 12 cm in length). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

787.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

787.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

787.4 Signs and Symptoms

- Your symptoms depend on the degree of enlargement
- Mild splenomegaly may be asymptomatic, while massive splenomegaly causes early satiety, left upper quadrant pain, and discomfort, with a firm, smooth spleen on

palpation.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

787.5 Progression and Stages

It has numerous causes classified as congestive (portal high blood pressure, heart failure, liver vein blood clot formation), infectious (malaria, infectious mononucleosis, infective endocarditis, visceral leishmaniasis, tuberculosis), hematologic (hemolytic anemias, myeloproliferative disorders, lymphomas, leukemias, myelofibrosis), inflammatory/autoimmune (sarcoidosis, Felty syndrome, SLE, rheumatoid arthritis), and infiltrative (Gaucher disease, Niemann-Pick disease, amyloidosis). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

787.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

787.7 Diagnosis

Diagnosis typically involves identifying the underlying cause through blood tests, imaging (ultrasound, CT), and bone marrow biopsy when indicated. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

787.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

787.9 Treatment Options

Treatment typically involves directed at the underlying condition. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

787.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

787.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 788

Spontaneous Abortion (Miscarriage)

788.1 Overview

Spontaneous abortion, or miscarriage, is the loss of a pregnancy before 20 weeks, occurring in 10–20% of clinically recognized pregnancies. Approximately 80% occur in the first trimester. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

788.2 Causes

The most common cause (50–60%) is chromosomal abnormalities, most often autosomal trisomies. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

788.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

788.4 Signs and Symptoms

You may experience vaginal bleeding with or without cramping, passage of tissue, and cervical dilation. Ultrasound evaluation determines type: threatened, inevitable, incomplete, complete, missed, or anembryonic pregnancy. Management options include expectant management, medical management (misoprostol), or surgical evacuation (dilation and curettage) for hemodynamic instability, infection, or incomplete evacuation. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

788.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

788.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

788.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

788.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

788.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

788.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

788.11 Outlook

- In most cases, the outlook is favorable
- Most early miscarriages are isolated events.

Chapter 789

Squamous Cell Carcinoma

789.1 Overview

Cutaneous squamous cell carcinoma (cSCC) is the second most common skin cancer, arising from keratinocytes in the epidermis. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

789.2 Causes

This condition happens because of UV-induced DNA damage and immunosuppression leading to cancerous transformation of keratinocytes. cSCC is more aggressive than BCC, with metastatic potential (2–5% for primary lesions, higher for recurrent or immunosuppressed patients). Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

789.3 Risk Factors

Risk factors include cumulative UV exposure, fair skin, immunosuppression (organ transplant recipients have 65-fold increased risk), chronic wounds, and actinic keratoses (precursor lesions).

- Advancing age

- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

789.4 Signs and Symptoms

Clinical features vary from an erythematous, scaly plaque to a firm, ulcerated nodule. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

789.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment options include surgical removal, Mohs surgery for high-risk lesions, and

radiation therapy for inoperable cases
- Cemiplimab (anti-PD-1) is approved for advanced cSCC.

789.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

789.7 Diagnosis

Doctors diagnose this based on biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies

- Genetic counseling and testing for hereditary cancer syndromes

789.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

789.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

789.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay

physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Stay physically active within your energy limits
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- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

789.11 Outlook

- In most cases, the outlook is favorable but worse than BCC
- Metastatic risk is higher in immunocompromised patients.

Chapter 790

SSRI Overdose and Serotonin Syndrome

790.1 Overview

Selective serotonin reuptake inhibitor (SSRI) overdose is generally less toxic than TCA overdose. SSRIs cause mild CNS depression, nausea, vomiting, and tachycardia. Citalopram and escitalopram are associated with QT prolongation (dose-dependent). Serotonin syndrome is a potentially life-threatening condition caused by excessive serotonergic activity, most commonly from SSRI in combination with MAOIs, linezolid, methylene blue, dextromethorphan, tramadol, meperidine, or MDMA. The triad is altered mental status, autonomic hyperactivity, and neuromuscular abnormalities. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

790.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

790.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

790.4 Signs and Symptoms

You may experience clonus, hyperreflexia, tremor, myoclonus, hyperthermia, diaphoresis, tachycardia, high blood pressure, and agitation.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

790.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

790.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

790.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam using the Hunter Serotonin Toxicity Criteria.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

790.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

790.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

790.10 Living with It

Treatment options include discontinuation of serotonergic agents, supportive care (cooling, IV fluids, benzodiazepines), and cyproheptadine for moderate to severe cases.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

790.11 Outlook

The outlook is generally positive with early recognition and treatment.

Chapter 791

Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis

791.1 Overview

Stevens-Johnson syndrome and toxic epidermal necrolysis are severe, life-threatening mucocutaneous reactions characterized by extensive keratinocyte apoptosis and epidermal detachment, with SJS affecting <10% of body surface area, SJS/TEN overlap affecting 10–30%, and TEN affecting >30% BSA. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

791.2 Causes

The most common cause is drug hypersensitivity (anticonvulsants [carbamazepine, lamotrigine, phenytoin], allopurinol, sulfonamides, NSAIDs, antibiotics). Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

791.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history

- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

791.4 Signs and Symptoms

You may experience prodromal fever and malaise, followed by painful, dusky red to purpuric macules that coalesce and progress to epidermal detachment with flaccid blisters and erosion, with mucosal involvement (oral, ocular, genital) present in over 90%.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

791.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

791.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

791.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Management requires burn unit or ICU care: fluid resuscitation, wound care, nutritional support, ophthalmological consultation, pain management, and withdrawal of the offending drug, with cyclosporine and IVIG having shown some benefit.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

791.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

791.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

791.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

791.11 Outlook

outlook is guarded, with mortality of 10–30% for SJS and up to 50% for TEN.

Chapter 792

Stillbirth (Intrauterine Fetal Demise)

792.1 Overview

Stillbirth is defined as fetal death at ≥20 weeks (or ≥28 weeks by WHO definition), occurring in 1 in 160 deliveries in the United States. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

792.2 Causes

Causes include placental abruption, umbilical cord accidents, intrauterine infection, placental insufficiency, congenital anomalies, and maternal conditions. A cause is identified in 60–80% of cases with thorough evaluation including fetal autopsy, placental pathology, and karyotype. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

792.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

792.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

792.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

792.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

792.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

792.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

792.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

792.10 Living with It

- Management after diagnosis is induction of labor
- Psychological support and counseling are critical components of care.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

792.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 793

Stimulant Overdose (Cocaine, Methamphetamine)

793.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Cocaine is a dopamine, norepinephrine, and serotonin reuptake inhibitor
- Methamphetamine promotes release of these monoamines. Both produce a sympathetic storm: high blood pressure, tachycardia, mydriasis, hyperthermia, psychomotor agitation, psychosis, coronary vasospasm, heart attack, aortic dissection, stroke, rhabdomyolysis, acute kidney injury, and seizures. Cocaine also has sodium channel blockade causing QRS widening. Hyperthermia is the strongest predictor of mortality. Treatment: benzodiazepines (high-dose IV) for agitation and sympathetic hyperactivity, active cooling for hyperthermia, calcium channel blockers or phentolamine for refractory high blood pressure, and nitroglycerin for cocaine-induced coronary vasospasm. Beta-blockers should be avoided in acute cocaine toxicity due to unopposed alpha-adrenergic stimulation.

793.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

793.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

793.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

793.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

793.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

793.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

793.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

793.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

793.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

793.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 794

Stimulant Use Disorder (Cocaine, Amphetamines)

794.1 Overview

Stimulant use disorder affects approximately 0.5–1% of the population for cocaine and 0.2–0.4% for amphetamine-type stimulants. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

794.2 Causes

This condition happens because of cocaine blocking dopamine, norepinephrine, and serotonin transporters, and amphetamines additionally promoting neurotransmitter release from presynaptic vesicles, with chronic use leading to dopamine depletion, neurotoxicity, and extensive structural brain changes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

794.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

794.4 Signs and Symptoms

You may experience a problematic pattern of stimulant use with criteria analogous to other substance use disorders. Withdrawal is marked by dysphoria, anhedonia, fatigue, hypersomnia, hyperphagia, and intense cravings. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

794.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

794.6 Complications

Medical complications include acute coronary syndrome, heart attack, involving bleeding stroke, rhabdomyolysis, seizures, high blood pressure, and for methamphetamine, severe dental decay (meth mouth), lung high blood pressure, and psychosis.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

794.7 Diagnosis

Doctors diagnose this using clinical interview. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

794.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

794.9 Treatment Options

- Treatment options include behavioral interventions as first-line (CBT, contingency management), with no FDA-approved medications for stimulant use disorder. outlook: chronic relapsing course
- Methamphetamine use disorder has particularly high relapse rates.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

794.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

794.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 795

Strabismus

795.1 Overview

Strabismus is a misalignment of the visual axes of the two eyes that may be horizontal (esotropia, exotropia) or vertical (hypertropia, hypotropia), and may be constant or intermittent, comitant or non-comitant. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

795.2 Causes

This condition happens because of neuromuscular imbalance of the extraocular muscles The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Onset in childhood may be accommodative, paralytic, or idiopathic
- In adults, acquired strabismus may result from cranial nerve palsies (III, IV, VI), thyroid eye disease, myasthenia gravis, or orbital pathology.

795.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

795.4 Signs and Symptoms

You may experience visible eye misalignment, which can cause amblyopia in children and diplopia in adults. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

795.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

795.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

795.7 Diagnosis

Doctors diagnose this based on cover testing and ocular motility examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

795.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

795.9 Treatment Options

Treatment options include corrective lenses, prisms, occlusion therapy, botulinum toxin injection, and surgical realignment of the extraocular muscles. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

795.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

795.11 Outlook

In most cases, the outlook is good with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 796

Subdural Hematoma

796.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- A subdural hematoma is a traumatic intracranial bleeding accumulating in the potential space between the dura mater and arachnoid mater, resulting from tearing of bridging veins. Acute subdural hematomas follow high-impact trauma
- Chronic subdural hematomas occur predominantly in elderly patients and alcoholics, where cerebral shrinkage or wasting stretches bridging veins, making them susceptible to rupture from minor head trauma.

796.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

796.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

796.4 Signs and Symptoms

You may experience headache, confusion, fluctuating consciousness, and weakness, numbness, or problems with movement, developing over hours in acute cases or weeks/months in chronic cases. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

796.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

796.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

796.7 Diagnosis

- Doctors confirm the diagnosis through non-contrast head CT: acute SDH appears crescent-shaped hyperdense
- Chronic SDH appears hypodense
- SDH crosses suture lines but does not cross dural reflections (falx/tentorium). Treatment for large or symptomatic hematomas involves surgical evacuation (burr holes or surgical opening of the skull)
- Small asymptomatic SDHs are managed conservatively.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

796.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

796.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

796.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

796.11 Outlook

Your outlook depends on underlying brain injury. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 797

Sudden Sensorineural Hearing Loss

797.1 Overview

Sudden sensorineural hearing loss is an ear emergency defined as a loss of at least 30 dB over three contiguous audiometric frequencies within 72 hours. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

797.2 Causes

This condition happens because of idiopathic causes, although viral cochleitis, vascular occlusion, autoimmune inner ear disease, or perilymphatic fistula may be implicated. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

797.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

797.4 Signs and Symptoms

You may experience abrupt unilateral hearing loss, often accompanied by ringing in the ears, feeling of spinning, and aural fullness. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

797.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

797.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

797.7 Diagnosis

Doctors diagnose this based on audiometry. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

797.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

797.9 Treatment Options

- Treatment focuses on high-dose oral or intratympanic corticosteroids, with prompt treatment within 2 weeks being crucial
- Hyperbaric oxygen therapy may provide additional benefit.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

797.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

797.11 Outlook

The outlook varies from person to person, with some cases recovering spontaneously and others sustaining permanent hearing loss. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 798

SUNCT and SUNA

798.1 Overview

Short-lasting Unilateral Neuralgiform headache attacks with Conjunctival Injection and Tearing (SUNCT) and Short-lasting Unilateral Neuralgiform headache attacks with cranial autonomic symptoms (SUNA) are trigeminal autonomic cephalalgias characterized by brief, stabbing, burning, or electric shock-like pain in the orbital, supraorbital, or temporal region. They are more common in men (1.5–2:1), with onset typically after age 50. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

798.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

798.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

798.4 Signs and Symptoms

You may experience attacks lasting 1–600 seconds (typically 5–240 seconds) occurring 1–200 times per day. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

798.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

798.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

798.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

798.8 Prevention

- Treatment options include lamotrigine as first-line for prevention, with gabapentin, topiramate, and carbamazepine as alternatives
- Surgical options are reserved for refractory cases.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

798.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

798.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

798.11 Outlook

- The outlook varies from person to person
- The condition is often refractory to treatment.

Chapter 799

Syphilis (Treponema pallidum)

799.1 Overview

Syphilis is a sexually transmitted infection caused by the spirochete *Treponema pallidum* subspecies *pallidum*, with 8 million new cases globally per year, with incidence rising in high-income countries, particularly among men who have sex with men (MSM). This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

799.2 Causes

This condition happens because of *T. pallidum* penetrating mucous membranes or abraded skin through sexual contact or transplacentally, with the organism rapidly disseminating hematogenously and lymphatically. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

799.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

799.4 Signs and Symptoms

Congenital syphilis includes stillbirth, rhinitis ("snuffles"), rash, osteochondritis, and Hutchinson's triad. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

799.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Clinical features occur in four stages: Primary (3–90 days): painless chancre (ulcer with clean base, firm, indurated margins) at the site of inoculation, resolving spontaneously in 3–6 weeks
- Secondary (2–8 weeks after primary): disseminated rash (non-pruritic, maculopapular, involving palms and soles), condylomata lata, generalized lymphadenopathy,

fever, malaise, and mucocutaneous lesions
- Latent: asymptomatic
- Tertiary (10–30 years): gummatous disease, cardiovascular syphilis (aortitis, aortic regurgitation), and neurosyphilis (general paresis, tabes dorsalis).

799.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

799.7 Diagnosis

Doctors diagnose this using dark-field microscopy and serology (non-treponemal tests for screening, treponemal tests for confirmation). Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

799.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use

Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

799.9 Treatment Options

Treatment typically involves benzathine penicillin G (IM) for primary, secondary, and early latent syphilis, with neurosyphilis requiring aqueous crystalline penicillin G IV. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

799.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended

- Gradually resume normal activities as symptoms improve

799.11 Outlook

Most people recover well with appropriate treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 800

Systemic Lupus Erythematosus

800.1 Overview

Systemic lupus erythematosus is a chronic, multisystem autoimmune disease characterized by loss of tolerance to self-antigens and production of a wide array of autoantibodies, leading to immune complex deposition and inflammation in multiple organs, predominantly affecting women of childbearing age (9:1 female predominance) and being more common in African Americans, Hispanics, and Asians. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

800.2 Causes

This condition happens because of autoantibody production and immune complex deposition. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

800.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease

- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

800.4 Signs and Symptoms

Clinical features are protean and include malar rash, photosensitivity, oral ulcers, arthritis, serositis (pleuritis, pericarditis), nephritis (lupus nephritis—the most serious manifestation), cytopenias, seizures, and psychosis. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

800.5 Progression and Stages

The 2019 EULAR/ACR classification criteria require an ANA titer $\geq$1:80 plus weighted criteria across clinical and immunological domains. Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

800.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation

- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

800.7 Diagnosis

Diagnosis typically involves antinuclear antibody (ANA, sensitive but not specific), anti-dsDNA (specific, correlates with nephritis), anti-Smith (highly specific), complement levels (C3, C4 decreased during flares), and kidney biopsy for lupus nephritis classification (ISN/RPS). Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

800.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

800.9 Treatment Options

Treatment options include hydroxychloroquine (cornerstone for all patients, reduces flares and mortality), corticosteroids, and immunosuppressive agents (mycophenolate, azathioprine, cyclophosphamide for severe organ involvement), with newer targeted therapies including belimumab (anti-BLys/BAFF monoclonal antibody), voclosporin, and anifrolumab (anti-type I interferon receptor). Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

800.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

800.11 Outlook

The outlook varies from person to person, with improved outcomes due to early diagnosis and better management. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 801

Systemic Lupus Erythematosus (Skin Manifestations)

801.1 Overview

Cutaneous lupus erythematosus (CLE) encompasses a spectrum of skin involvement in systemic lupus erythematosus (SLE) or as a primary skin-limited condition. Acute cutaneous lupus presents as a malar (butterfly) rash across the cheeks and bridge of the nose, sparing the nasolabial folds, and is linked to active systemic disease. Subacute cutaneous lupus presents as annular polycyclic or papulosquamous plaques on sun-exposed areas, strongly associated with anti-Ro/SSA antibodies. Chronic cutaneous lupus (discoid lupus) presents as well-demarcated, scaly, erythematous plaques with follicular plugging and atrophic scarring, predominantly on the scalp, face, and ears. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

801.2 Causes

This condition happens because of autoimmune-mediated skin inflammation triggered by UV exposure. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

801.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

801.4 Signs and Symptoms

Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

801.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

801.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

801.7 Diagnosis

Doctors diagnose this based on clinical appearance, skin biopsy, and serological testing (ANA, anti-Ro/SSA, anti-dsDNA). Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

801.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

801.9 Treatment Options

Treatment options include sun protection, topical corticosteroids, antimalarials (hydroxychloroquine, first-line), and for refractory disease, immunosuppressants (methotrexate, mycophenolate, belimumab). Treatment aims to control inflammation, manage symptoms, and prevent disease flares. A step-up approach is often used, starting with symptomatic treatments and progressing to immunosuppressive therapies as needed. Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

801.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

801.11 Outlook

Your outlook depends on the subtype and systemic involvement. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 802

Systemic Sclerosis (Scleroderma)

802.1 Overview

Systemic sclerosis is a chronic autoimmune connective tissue disease characterized by widespread vasculopathy, immune dysregulation, and progressive scarring of the skin and internal organs, with an incidence of approximately 1–2 per 100,000, a female-to-male ratio of 4–6:1, and peaking in the fifth decade. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

802.2 Causes

This condition happens because of excessive collagen deposition by activated fibroblasts, driven by endothelial injury, autoimmune inflammation, and transforming growth factor-beta (TGF-β) signaling. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

802.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease

- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

802.4 Signs and Symptoms

- You may experience Raynaud's phenomenon (the most frequent initial manifestation, present in over 95%), and three clinical variants exist: limited cutaneous SSc (lcSSc, skin thickening distal to elbows/knees, formerly CREST syndrome: Calcinosis cutis, Raynaud's phenomenon, Esophageal dysmotility, Sclerodactyly, Telangiectasias), diffuse cutaneous SSc (dcSSc, skin thickening proximal to elbows/knees and trunk, more rapid and severe internal organ involvement), and systemic sclerosis sine scleroderma (Raynaud's phenomenon and internal organ involvement without clinically apparent skin scarring). lung involvement is the leading cause of death: lung scarring (more common in dcSSc) presents with shortness of breath, cough, and restrictive lung disease
- Lung high blood pressure (more common in lcSSc) presents with exertional shortness of breath, right heart failure, and carries a poor outlook. Other features include gastroesophageal reflux, gastric antral vascular ectasia (watermelon stomach), telangiectasias, and myositis.
- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

802.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

802.6 Complications

Scleroderma kidney crisis (SRC) is a life-threatening complication of dcSSc (incidence 5–10%), characterized by cancerous high blood pressure, rapidly progressive kidney failure, microangiopathic hemolytic anemia, and thrombocytopenia.

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

802.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by autoantibodies (anti-centromere in lcSSc, anti-Scl-70/anti-topoisomerase I in dcSSc, anti-RNA polymerase III associated with SRC risk), nailfold capillaroscopy, and organ-specific testing (PFTs, echocardiogram, high-resolution CT chest). Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

802.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis

- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

802.9 Treatment Options

Treatment typically involves organ-specific and multidisciplinary: calcium channel blockers, prostacyclin analogs, and endothelin receptor antagonists for Raynaud's phenomenon and lung high blood pressure

- Proton pump inhibitors and prokinetics for GI involvement
- ACE inhibitors for scleroderma kidney crisis (first-line, which has transformed outcomes—SRC mortality has decreased from 80% to 15–20%)
- Mycophenolate, cyclophosphamide, and nintedanib for interstitial lung disease
- And methotrexate for skin and joint involvement.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

802.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

802.11 Outlook

The outlook varies from person to person, with the diffuse cutaneous variant having a worse outlook, and lung involvement being the leading cause of death. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 803

Takayasu Arteritis

803.1 Overview

Takayasu arteritis is a chronic, granulomatous large-vessel vasculitis affecting the aorta and its major branches, predominantly in young women, with an incidence of 1–2 per million, a 9:1 female predominance, typically presenting between 10–40 years of age, and being more common in Asia, India, and Latin America. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

803.2 Causes

This condition happens because of Th1-mediated granulomatous inflammation, intimal thickening, scarring, and stenosis of large elastic arteries, with skip lesions being characteristic. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

803.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

803.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

803.5 Progression and Stages

Clinical features occur in two phases: an early systemic phase (non-specific constitutional symptoms: fever, malaise, weight loss, night sweats, arthralgias) and a late occlusive phase (claudication of extremities, absence or asymmetry of pulses—pulseless disease, blood pressure differences between arms, vascular bruits, syncope, stroke, visual disturbances, renovascular high blood pressure, and aortic regurgitation). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

803.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

803.7 Diagnosis

Doctors diagnose this based on the 1990 ACR criteria, with imaging (conventional angiography, CT angiography, or MR angiography) showing stenosis, occlusion, or aneurysm of the aorta and its branches, and PET-CT showing vessel wall FDG avidity in active disease. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

803.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

803.9 Treatment Options

Treatment options include high-dose corticosteroids as first-line therapy, with methotrexate, azathioprine, mycophenolate, TNF inhibitors, and tocilizumab used as steroid-sparing agents or for refractory disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

803.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

803.11 Outlook

- The outlook varies from person to person
- Disease is chronic with relapses requiring long-term immunosuppression.

Chapter 804

Tapeworm Infections (Taeniasis / Cysticercosis)

804.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

804.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of taeniasis: ingestion of raw or undercooked meat containing cysticerci, with the scolex attaching to the intestinal mucosa
- Cysticercosis: ingestion of *T. solium* eggs, with oncospheres penetrating the intestinal wall, disseminating hematogenously, and encysting in tissues.

804.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)

- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

804.4 Signs and Symptoms

- You may experience taeniasis: usually asymptomatic or mild abdominal pain and passage of proglottids
- Cysticercosis: neurocysticercosis—seizures (most common presentation), headache, focal nervous system deficits, increased intracranial pressure, and hydrocephalus.
- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

804.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Taeniasis is intestinal infection with adult tapeworms (*Taenia saginata*—beef tapeworm
- *Taenia solium*—pork tapeworm
- *Diphyllobothrium latum*—fish tapeworm), and cysticercosis is tissue infection with the larval stage of *T. solium*, a leading cause of acquired epilepsy in endemic areas, with taeniasis affecting 50 million people and neurocysticercosis affecting 2.5 million.

804.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

804.7 Diagnosis

- To diagnose taeniasis is by macroscopic or microscopic identification of proglottids in stool
- To diagnose cysticercosis is by neuroimaging showing ring-enhancing cysts.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

804.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures

- Travel precautions and pre-travel consultations

804.9 Treatment Options

Treatment options include praziquantel or niclosamide for taeniasis, and albendazole plus corticosteroids for neurocysticercosis. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

804.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

804.11 Outlook

- Most people recover well for taeniasis
- Neurocysticercosis Your outlook depends on cyst burden and location.

Chapter 805

Temporal Lobe Epilepsy

805.1 Overview

Temporal lobe epilepsy is the most common form of focal epilepsy in adults, often arising from mesial temporal sclerosis (hippocampal sclerosis). This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

805.2 Causes

This condition happens because of hippocampal sclerosis, low-grade gliomas, focal cortical dysplasia, or dual pathology. Seizures typically begin with an aura (déjà vu, epigastric rising, fear, olfactory or gustatory hallucinations), followed by impaired awareness, automatisms (lip smacking, hand fumbling), and postictal confusion. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

805.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

805.4 Signs and Symptoms

You may experience focal seizures with impaired awareness. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

805.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

805.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

805.7 Diagnosis

Doctors diagnose this based on EEG showing temporal interictal epileptiform discharges and MRI revealing hippocampal shrinkage or wasting and T2/FLAIR signal changes in mesial temporal sclerosis. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

805.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

805.9 Treatment Options

- Treatment focuses on antiseizure medications
- Drug resistance is common, and temporal lobe epilepsy is the most surgically amenable epilepsy, with anterior temporal lobectomy achieving seizure freedom in 60–80% of selected patients.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

805.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

805.11 Outlook

The outlook is generally positive with surgical treatment for drug-resistant cases. Neurological conditions vary widely in their course and outcomes. Some are progressive, while others follow a relapsing-remitting pattern. Early intervention and rehabilitation can improve outcomes. Neurological conditions vary significantly in their course, from acute and self-limited to chronic and progressive. Early diagnosis and intervention can slow progression and improve quality of life. Rehabilitation and supportive therapies play crucial roles in maintaining function. Research continues to develop disease-modifying treatments for previously untreatable conditions. Multidisciplinary care addressing medical, functional, and psychosocial needs optimizes outcomes.

Chapter 806

Temporomandibular Joint Disorders

806.1 Overview

Temporomandibular joint disorders (TMD/TMJ) encompass a group of conditions affecting the temporomandibular joint, masticatory muscles, and associated structures. Prevalence is 5–12%, with a female predominance (4:1) and peak incidence in reproductive years. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

806.2 Causes

This condition happens because of multifactorial causes including occlusal factors, parafunctional habits, trauma, psychosocial stress, and genetic predisposition. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

806.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

806.4 Signs and Symptoms

You may experience jaw pain, clicking or crepitus, limited mouth opening, locking, headache, and ear pain. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

806.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

806.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

806.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam with palpation of TMJ and muscles and range of motion assessment
- Imaging is reserved for atypical presentations. Conservative Treatment typically involves first-line: patient education, self-management, physical therapy, occlusal splints, and NSAIDs
- For persistent pain, TCAs, muscle relaxants, trigger point injections, or botulinum toxin may be used.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

806.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

806.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

806.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

806.11 Outlook

The outlook is generally positive with conservative management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 807

Tension-Type Headache

807.1 Overview

Tension-type headache (TTH) is a primary headache disorder and the most prevalent headache disorder, with a lifetime prevalence of 30–78%. It is more common in women and peaks in the 30–40 age range. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

807.2 Causes

This condition happens because of peripheral myofascial pain mechanisms in episodic TTH and central sensitization in chronic TTH, with muscle tenderness (pericranial, cervical, shoulder girdle) as a prominent feature. Neurological disorders involve dysfunction of neurons, glial cells, or neural circuits. The underlying mechanisms may include protein aggregation, neurotransmitter imbalances, demyelination, or structural abnormalities. Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

807.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

807.4 Signs and Symptoms

You may experience bilateral, pressing/tightening, non-pulsating headache of mild to moderate intensity, lasting 30 minutes to 7 days, not aggravated by routine physical activity, and not associated with nausea or vomiting. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

807.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

807.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

807.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam based on ICHD-3 criteria. Acute Treatment options include simple analgesics (ibuprofen,

naproxen, acetaminophen, aspirin). Preventive therapy for chronic TTH includes amitriptyline, mirtazapine, or venlafaxine
- Non-pharmacological approaches include physical therapy, relaxation training, biofeedback, and thinking and memory behavioral therapy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

807.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

807.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

807.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

807.11 Outlook

- The outlook is generally positive

- Episodic TTH typically resolves with treatment.

Chapter 808

Testicular Cancer

808.1 Overview

Testicular cancer is the most common solid tumor in young men (peak age 15–40) and is highly curable. The vast majority (95%) are germ cell tumors (GCTs): seminoma (40%) and non-seminomatous germ cell tumors (NSGCT, 60%)—embryonal carcinoma, yolk sac tumor, choriocarcinoma, teratoma, and mixed germ cell tumors. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

808.2 Causes

This condition happens because of cancerous transformation of germ cells. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

808.3 Risk Factors

Risk factors include cryptorchidism, prior testicular cancer, Klinefelter syndrome, and infertility.

- Advancing age
- Family history of cancer

- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

808.4 Signs and Symptoms

Clinical features typically include a painless testicular mass, though some present with scrotal pain, hydrocele, or symptoms of metastatic disease. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

808.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

- Treatment for stage I seminoma involves surveillance or adjuvant carboplatin

- Advanced disease doctors typically treat it with cisplatin-based chemotherapy (BEP—bleomycin, etoposide, cisplatin) with surgery for residual masses.

808.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

808.7 Diagnosis

Diagnosis typically involves scrotal ultrasound (hypoechoic intratesticular mass), serum tumor markers (AFP, β-hCG, LDH), and CT abdomen/pelvis/chest for staging. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)

- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

808.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

808.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

808.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

808.11 Outlook

Most people recover well, with cure rates exceeding 95% for seminoma and 80–90% for NSGCT. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 809

Tetanus (Clostridium tetani)

809.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

809.2 Causes

This condition happens because of spores germinating in anaerobic wound conditions, producing tetanospasmin (a zinc-dependent metalloproteinase) that binds to presynaptic nerve terminals, is transported retrogradely to the CNS, and cleaves synaptobrevin-2 (VAMP), preventing release of inhibitory neurotransmitters (GABA, glycine) from Renshaw cells, producing unopposed motor neuron activity. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

809.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

809.4 Signs and Symptoms

You may experience an incubation period of 3–21 days, with trismus (lockjaw) being the earliest sign in 75% of cases, followed by risus sardonicus, opisthotonos, generalized muscle rigidity, and painful spasms triggered by minimal stimuli, with autonomic instability occurring in severe cases. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

809.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

809.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

809.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

809.8 Prevention

Tetanus is a toxin-mediated disease caused by *Clostridium tetani*, a spore-forming, obligate anaerobic bacillus found in soil and animal feces, with approximately 200,000 cases annually worldwide, primarily in low-income countries with inadequate vaccination, and neonatal tetanus accounting for 50% of tetanus deaths. Treatment options include human tetanus immune globulin (HTIG) to neutralize unbound toxin, wound removal of dead tissue, metronidazole, and supportive care (benzodiazepines for spasms, magnesium sulfate for autonomic instability). outlook: mortality remains 10–60% in severe cases, but prevention is effective through tetanus toxoid vaccination. Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

809.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

809.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

809.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 810

Thalassemia

810.1 Overview

Thalassemia is a group of inherited hemoglobin disorders characterized by reduced or absent synthesis of α- or β-globin chains, leading to ineffective erythropoiesis and hemolytic anemia. β-Thalassemia major (Cooley's anemia, homozygous β^0 mutations) presents in the first year of life with severe anemia, failure to thrive, hepatosplenomegaly, and skeletal deformities ("hair-on-end" appearance on skull X-ray from marrow expansion). β-Thalassemia intermedia has milder anemia and may not require regular transfusions. α-Thalassemia major (Hb Bart's hydrops fetalis, deletion of all four α-globin genes) is uniformly fatal in utero. α-Thalassemia trait (deletion of two genes) causes mild microcytic anemia. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

810.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

810.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age

- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

810.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

810.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

810.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

810.7 Diagnosis

Doctors diagnose this using CBC, hemoglobin electrophoresis, and genetic testing.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

810.8 Prevention

Treatment for β-thalassemia major requires lifelong transfusion therapy and iron chelation (deferasirox, deferoxamine, deferiprone) to prevent transfusional iron overload, which causes cardiomyopathy, liver cirrhosis, and endocrinopathies.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

810.9 Treatment Options

Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

810.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

810.11 Outlook

The outlook has improved with stem cell transplantation (curative for β-thalassemia major) and gene therapy (betibeglogene autotemcel).

Chapter 811

Theophylline Overdose

811.1 Overview

Theophylline overdose is a pharmaceutical overdose causing severe neurological and cardiovascular toxicity. Theophylline is a phosphodiesterase and adenosine receptor antagonist, leading to increased catecholamine release and beta-adrenergic stimulation. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

811.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

811.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

811.4 Signs and Symptoms

- Clinical features: acute toxicity presents with nausea, vomiting, hypokalemia, hypophosphatemia, hyperglycemia, metabolic acidosis, tremor, agitation, and seizures (often refractory)
- Cardiovascular effects include sinus tachycardia, atrial fibrillation, ventricular tachycardia, and hypotension.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

811.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

811.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

811.7 Diagnosis

Doctors diagnose this using serum theophylline level. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

811.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

811.9 Treatment Options

Treatment options include multidose activated charcoal (MDAC, 0.5 g/kg every 2–4 hours), beta-blockers for hemodynamically significant tachycardia, and hemodialysis for severe toxicity. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

811.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

811.11 Outlook

The outlook is generally positive with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 812

Theories of Aging

812.1 Overview

Numerous theories attempt to explain the molecular and cellular basis of aging. The **telomere shortening theory** posits that progressive shortening of telomeres—the protective caps at chromosome ends—with each cell division eventually triggers replicative senescence or apoptosis. Telomerase, the enzyme that elongates telomeres, is silenced in most somatic cells. The **free radical theory** (Harman, 1956) proposes that reactive oxygen species (ROS) generated by mitochondrial metabolism cause cumulative oxidative damage to DNA, proteins, and lipids, driving cellular aging. The **mitochondrial theory** extends this concept, focusing on the accumulation of mitochondrial DNA mutations and impaired oxidative phosphorylation with age. The **epigenetic theory** emphasizes age-related changes in DNA methylation, histone modification, and chromatin remodeling that alter gene expression patterns without changing the DNA sequence. Epigenetic clocks (eg., Horvath clock) use DNA methylation patterns to estimate biological age. **Inflammaging** describes the chronic, low-grade, sterile inflammatory state that characterizes aging, driven by activation of the innate immune system, accumulation of damage-associated molecular patterns (DAMPs), and dysregulation of cytokine networks (elevated IL-6, TNF-alpha, CRP). This promotes many age-related diseases including atherosclerosis, diabetes, dementia, and frailty. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

812.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

812.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

812.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

812.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

812.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

812.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

812.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

812.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

812.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

812.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 813

Thrombotic Thrombocytopenic Purpura

813.1 Overview

Thrombotic thrombocytopenic purple spots on the skin (TTP) is a rare, life-threatening thrombotic microangiopathy characterized by microvascular blood clot formation, thrombocytopenia, and microangiopathic hemolytic anemia, with an incidence of 3–4 per million. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

813.2 Causes

This condition happens because of severe deficiency of ADAMTS13, a von Willebrand factor (VWF) cleaving protease: immune-mediated (acquired TTP, IgG autoantibodies, adulthood) or congenital (Upshaw-Schulman syndrome, ADAMTS13 gene mutations, childhood). ADAMTS13 deficiency leads to accumulation of ultralarge VWF multimers, causing spontaneous platelet adhesion and microthrombi formation in arterioles and capillaries. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

813.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

813.4 Signs and Symptoms

- You may experience the classic pentad: thrombocytopenia (tiny red or purple spots, purple spots on the skin, bleeding), microangiopathic hemolytic anemia (fever, pallor, jaundice), nervous system involvement (headache, confusion, seizures, focal deficits, coma), kidney involvement (elevated creatinine, hematuria), and fever. Not all five features are present simultaneously
- Thrombocytopenia and MAHA are sufficient to suspect TTP.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

813.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

813.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

813.7 Diagnosis

Doctors diagnose this based on peripheral blood smear showing schistocytes (helmet cells, fragmented RBCs), elevated LDH, low haptoglobin, negative direct Coombs test, and ADAMTS13 activity <10%. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function

- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

813.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

813.9 Treatment Options

Treatment typically involves a medical emergency: urgent plasma exchange (PLEX, 1–1.5 plasma volume daily) is the cornerstone of therapy, combined with high-dose corticosteroids (prednisone 1 mg/kg). For refractory/relapsed immune TTP: rituximab, caplacizumab (anti-VWF nanobody, blocks platelet adhesion), or immunosuppression (vincristine, cyclophosphamide). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

813.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

813.11 Outlook

The outlook has improved dramatically: mortality declined from >90% to 10–20% with prompt PLEX. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 814

Thyroid Nodules and Cancer

814.1 Overview

Thyroid nodules are extremely common, palpable in 4–7% of adults and incidentally discovered on imaging in up to 50%, with the vast majority (90–95%) being non-cancerous. Thyroid cancer accounts for approximately 58,000 new cases annually in the US, with papillary thyroid carcinoma being the most common histological type (80%, with excellent outlook and 20-year survival >95%), followed by follicular thyroid carcinoma (10–15%), medullary thyroid carcinoma (5%, arising from parafollicular C cells and producing calcitonin, associated with MEN2), and anaplastic thyroid carcinoma (<1%, highly aggressive and usually fatal). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

814.2 Causes

This condition happens because of neoplastic transformation of thyroid follicular or parafollicular cells. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

814.3 Risk Factors

The American Thyroid Association (ATA) guidelines guide management based on ultrasound characteristics (TI-RADS classification), size, and risk factors, with fine-needle removal by suction (FNA) biopsy being the standard for evaluating suspicious nodules (Bethesda classification for cytology).

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption
- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Family history
- Poor dietary habits
- Hormonal imbalances
- Certain medications
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

814.4 Signs and Symptoms

You may experience a palpable thyroid nodule, sometimes with compressive symptoms. Treatment for differentiated thyroid cancer includes total thyroidectomy (or lobectomy for low-risk papillary cancer <4 cm), radioactive iodine ablation for intermediate/high-risk disease, and TSH suppression with levothyroxine. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats

- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

814.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

814.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

814.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

814.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

814.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

814.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
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- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

814.11 Outlook

Most people recover well for most differentiated thyroid cancers. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 815

Thyroiditis

815.1 Overview

Thyroiditis encompasses inflammatory conditions of the thyroid gland, with Hashimoto's thyroiditis being the most common cause of hypothyroidism in iodine-sufficient regions. Subacute (de Quervain's) thyroiditis is a painful, granulomatous thyroiditis thought to be viral in cause, presenting with thyroid pain, tenderness, fever, and thyrotoxicosis followed by hypothyroidism and eventual recovery. Silent (painless) thyroiditis and postpartum thyroiditis are autoimmune conditions presenting with transient thyrotoxicosis followed by hypothyroidism. Acute suppurative thyroiditis is a rare bacterial infection. Drug-induced thyroiditis may be caused by amiodarone, lithium, interferon-alpha, and immune checkpoint inhibitors. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

815.2 Causes

This condition happens because of inflammation of the thyroid gland. In autoimmune conditions, a combination of genetic susceptibility and environmental triggers initiates an inappropriate immune response against self-antigens. This leads to chronic inflammation and tissue damage in affected organs. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

815.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes
- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

815.4 Signs and Symptoms

Symptoms vary depending on type. Fatigue and general malaise Joint pain, swelling, and stiffness Skin rashes and lesions Low-grade fever Muscle weakness and pain Organ-specific symptoms based on affected system Weight changes (loss or gain) Dry eyes and dry mouth (Sjogren syndrome) Raynaud phenomenon (cold-induced color changes in fingers) Fluctuating symptoms with periods of flare and remission

- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

815.5 Progression and Stages

Management varies by type: NSAIDs or corticosteroids for subacute thyroiditis, levothyroxine for hypothyroid phases, beta-blockers for symptomatic thyrotoxicosis, and observation for most self-limited forms. Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

815.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy
- Lymphoma risk in some autoimmune conditions

815.7 Diagnosis

Doctors diagnose this based on clinical presentation, thyroid function tests, and imaging. Diagnosis is based on a combination of clinical features, laboratory findings (autoantibodies, inflammatory markers), and imaging studies. Classification criteria help standardize the diagnostic process. Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

815.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

815.9 Treatment Options

Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief Corticosteroids for rapid control of inflammation Disease-modifying antirheumatic drugs (DMARDs) Biologic therapies targeting specific immune pathways Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide) Antimalarial drugs (hydroxychloroquine) for mild disease Physical and occupational therapy to maintain function Pain management for chronic pain Lifestyle modifications including diet and stress reduction Regular monitoring for disease activity and treatment side effects

- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain
- Lifestyle modifications including diet and stress reduction

815.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

815.11 Outlook

In most cases, the outlook is excellent. Autoimmune conditions are typically chronic and require long-term management. With appropriate treatment, many patients achieve good symptom control and maintain quality of life. Autoimmune diseases are generally chronic conditions requiring long-term management. Disease activity can fluctuate, requiring adjustments in treatment over time. Early diagnosis and treatment improve outcomes and prevent irreversible organ damage. Research continues to develop more targeted therapies with fewer side effects.

Chapter 816

Tinea Infections

816.1 Overview

Tinea infections (dermatophytosis) are superficial fungal infections caused by dermatophytes (Trichophyton, Microsporum, and Epidermophyton species) that infect keratinized tissue. Named by anatomic location: tinea pedis (athlete's foot, the most common, affecting the feet, particularly between toes), tinea corporis (ringworm of the body, annular, erythematous, scaly plaques with central clearing), tinea cruris (jock itch, affecting the groin and inner thighs), tinea capitis (scalp infection, most common in children, may cause alopecia and kerion formation), tinea unguium (onychomycosis, nail thickening, discoloration, and crumbling), and tinea versicolor (caused by *Malassezia* species, actually a pityriasis, presenting with hypo- or hyperpigmented macules on the trunk). This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

816.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

816.3 Risk Factors

Risk factors include warm, moist environments, tight clothing, shared bathing facilities, immunosuppression, and diabetes.

- Close contact with infected individuals
- Compromised immune system

- Travel to endemic areas
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

816.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

816.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- The right treatment depends on location and severity: topical antifungals (terbinafine, clotrimazole, miconazole) for localized tinea corporis, cruris, and pedis
- Oral antifungals (terbinafine, itraconazole, fluconazole) for tinea capitis, extensive tinea corporis, tinea unguium, and refractory cases.

816.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

816.7 Diagnosis

Doctors diagnose this based on KOH preparation showing hyphae and spores, fungal culture, dermoscopy, or Wood's lamp examination. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

816.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules

- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

816.9 Treatment Options

Tinea capitis requires systemic therapy due to follicular involvement. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

816.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

816.11 Outlook

Most people recover well with appropriate antifungal therapy. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications.

Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 817

Tinnitus

817.1 Overview

ringing in the ears is the perception of sound in the absence of an external auditory stimulus, affecting approximately 10–15% of adults, with higher prevalence in the elderly. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

817.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of maladaptive neural plasticity in the central auditory pathways following peripheral auditory deafferentation
- Causes include noise-induced hearing loss (the most common cause), age-related hearing loss, ototoxic medications, Ménière's disease, vestibular schwannoma, otosclerosis, impacted cerumen, and temporomandibular joint disorders.

817.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

817.4 Signs and Symptoms

- You may experience ringing, buzzing, hissing, roaring, or clicking in one or both ears
- Ringing in the ears may be associated with depression, anxiety, and insomnia.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

817.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

817.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

817.7 Diagnosis

- Doctors diagnose this based on thorough history, otoscopic examination, and audiometry to assess for hearing loss
- MRI with gadolinium is indicated for asymmetric ringing in the ears or focal neurological findings.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

817.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

817.9 Treatment Options

Treatment options include patient education and reassurance, treatment of reversible causes, hearing aids for those with hearing loss, thinking and memory-behavioral therapy, sound therapy, and ringing in the ears retraining therapy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

817.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

817.11 Outlook

The outlook varies from person to person, with many patients achieving habituation through appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 818

Tonsillar Hypertrophy

818.1 Overview

Tonsillar enlargement is an enlargement of the palatine and adenoid tonsils that is common in children and is the most common cause of obstructive sleep apnea in the pediatric population. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

818.2 Causes

This condition happens because of lymphoid abnormal increase in cells, often in response to recurrent infections. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

818.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

818.4 Signs and Symptoms

- You may experience mouth breathing, snoring, restless sleep, recurrent otitis media, and difficulty swallowing
- Severe enlargement may cause failure to thrive and cor pulmonale.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

818.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

818.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

818.7 Diagnosis

Doctors diagnose this based on clinical examination and history. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

818.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

818.9 Treatment Options

Treatment focuses on tonsillectomy with or without adenoidectomy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

818.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

818.11 Outlook

Most people recover well following surgical intervention. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 819

Tonsillitis

819.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

819.2 Causes

This condition happens because of infectious inflammation of the tonsillar tissue. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

819.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

819.4 Signs and Symptoms

- You may experience sore throat, odynophagia, fever, halitosis, and cervical lymphadenopathy
- Tonsillar enlargement may cause obstructive sleep-disordered breathing.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

819.5 Progression and Stages

Tonsillitis is an inflammation of the palatine tonsils, which are lymphoid tissues forming part of Waldeyer's ring, and may be acute (viral or bacterial, with GAS being the most common bacterial cause) or chronic (persistent low-grade infection with recurrent exacerbations). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

819.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

819.7 Diagnosis

- Doctors diagnose this based on clinical examination. Treatment for recurrent tonsillitis follows the Paradise criteria for tonsillectomy
- Peritonsillar abscess (quinsy) requires needle removal by suction or incision and removal of fluid with antibiotic therapy.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

819.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

819.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

819.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

819.11 Outlook

Most people recover well with appropriate treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 820

Tourette Syndrome

820.1 Overview

Tourette syndrome is a neurodevelopmental disorder characterized by multiple motor tics and one or more vocal tics persisting for more than one year, with onset before age 18, affecting 0.3–0.8% of children. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

820.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of dysfunction of cortico-striato-thalamo-cortical circuits, involving dopaminergic and GABAergic dysregulation, with genetic contributions (inherited as autosomal dominant with variable penetrance
- SLITRK1, HDC, CNTN6 genes implicated).

820.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

820.4 Signs and Symptoms

You may experience motor tics (eye blinking, facial grimacing, head jerking, shoulder shrugging, complex tics such as hopping, touching, or echopraxia) and vocal tics (throat clearing, grunting, sniffing, coprolalia, palilalia, echolalia). Comorbidities include ADHD (50–60%), OCD (30–60%), anxiety, and autism spectrum disorder.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

820.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- Tics wax and wane in severity, are preceded by premonitory urges, and can be temporarily suppressed. The outlook is generally positive
- Most patients experience reduction in tic severity by early adulthood.

820.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

820.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, based on DSM-5-TR criteria.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

820.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

820.9 Treatment Options

Treatment options include psychoeducation, habit reversal training and comprehensive behavioral intervention for tics (CBIT), pharmacotherapy for moderate to severe tics (alpha-2 agonists: clonidine, guanfacine as first-line), and antipsychotics (aripiprazole, risperidone, haloperidol) for refractory tics. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

820.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 821

Toxicokinetics and Toxicodynamics

821.1 Overview

Toxicokinetics describes the absorption, distribution, metabolism, and elimination of toxins. Absorption occurs via ingestion (most common), inhalation, dermal contact, or injection. Distribution depends on protein binding, lipid solubility, and volume of distribution (V_d). Elimination is via kidney, liver (biliary), or lung routes. Toxicodynamics describes the molecular mechanisms by which toxins produce their effects—receptor binding (opioids, benzodiazepines), enzyme inhibition (organophosphates, cyanide), ion channel blockade (TCA, calcium channel blockers), or mitochondrial toxicity (acetaminophen NAPQI, cyanide). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

821.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

821.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

821.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

821.5 Progression and Stages

Metabolism primarily occurs in the liver via Phase I (cytochrome P450 oxidation, reduction, hydrolysis) and Phase II (glucuronidation, sulfation, acetylation, glutathione conjugation) reactions. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

821.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

821.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

821.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

821.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

821.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

821.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 822

Toxoplasmosis

822.1 Overview

Toxoplasmosis is a protozoan infection caused by *Toxoplasma gondii*, an obligate intracellular parasite, with approximately 30% of the global population seropositive, with cats being the definitive host, and humans acquiring infection through ingestion of oocysts or tissue cysts, or transplacentally. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

822.2 Causes

This condition happens because of oocysts or tissue cysts releasing bradyzoites/tachyzoites in the GI tract, with tachyzoites disseminating hematogenously and infecting all nucleated cells, with the host cellular immune response driving tachyzoites into dormant bradyzoite forms within tissue cysts. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

822.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

822.4 Signs and Symptoms

- Clinical features in immunocompetent hosts are usually asymptomatic or self-limited mononucleosis-like illness
- Immunocompromised hosts (HIV/AIDS) develop toxoplasmic encephalitis—most common cerebral mass lesion in AIDS—presenting with headache, confusion, seizures, and ring-enhancing lesions on MRI
- Congenital toxoplasmosis presents with the classic triad of chorioretinitis, hydrocephalus, and intracranial calcifications.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

822.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

822.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

822.7 Diagnosis

Doctors diagnose this using serology, PCR, and MRI for CNS disease. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

822.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

822.9 Treatment Options

Treatment options include pyrimethamine plus sulfadiazine plus folinic acid for active disease. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

822.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

822.11 Outlook

- Most people recover well for immunocompetent hosts
- Congenital infection has significant morbidity.

Chapter 823

Transient Ischemic Attack

823.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- A transient ischemic attack (TIA) is a temporary episode of neurological dysfunction caused by focal brain, spinal cord, or retinal reduced blood flow without acute tissue death due to lack of blood supply
- TIAs are critical warning signs, with the risk of subsequent stroke being highest in the first 48 hours (up to 10%).

823.2 Causes

This condition happens because of temporary focal reduced blood flow. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

823.3 Risk Factors

The outlook is good with prompt intervention and risk factor modification. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

823.4 Signs and Symptoms

Clinical features mirror those of ischemic stroke but resolve completely, typically within minutes to hours. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

823.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

823.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

823.7 Diagnosis

- Diagnosis typically involves brain imaging, vascular imaging (carotid ultrasound, CT or MR angiography), cardiac evaluation, and laboratory studies
- The ABCD2 score helps stratify short-term stroke risk.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

823.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

823.9 Treatment Options

Treatment options include antiplatelet therapy (aspirin, clopidogrel, or combination), statins, antihypertensives, and carotid endarterectomy or stenting for significant ipsilateral carotid stenosis. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

823.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

823.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 824

Trichomoniasis

824.1 Overview

Trichomoniasis is a sexually transmitted infection caused by the protozoan *Trichomonas vaginalis*, affecting approximately 170 million people worldwide. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

824.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

824.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

824.4 Signs and Symptoms

- Clinical features in women include a diffuse, yellow-green, frothy, purulent vaginal discharge with a pungent odor, vulvar itching, dysuria, and dyspareunia
- Most men are asymptomatic. Trichomoniasis is linked to adverse pregnancy outcomes

and increased HIV acquisition.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

824.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

824.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

824.7 Diagnosis

Doctors diagnose this using saline wet mount showing motile trichomonads (50–70% sensitivity), culture, or NAAT. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

824.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

824.9 Treatment Options

Treatment typically involves metronidazole 2 g single dose or tinidazole 2 g single dose, with partner treatment essential. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

824.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

824.11 Outlook

The outlook is generally positive with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 825

Tricuspid Regurgitation

825.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Tricuspid regurgitation is backflow of blood through the tricuspid valve during systole
- Functional tricuspid regurgitation (secondary to right ventricular dilation from lung high blood pressure, left-sided heart disease, or atrial fibrillation) is the most common type. Primary tricuspid regurgitation results from endocarditis, rheumatic disease, carcinoid syndrome, or pacemaker lead infection.

825.2 Causes

This condition happens because of structural or functional abnormalities of the tricuspid valve apparatus. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

825.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

825.4 Signs and Symptoms

Symptoms of severe tricuspid regurgitation include right heart failure with peripheral swelling, ascites, hepatomegaly, and pulsatile jugular venous distension. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

825.5 Progression and Stages

Your outlook depends on the underlying cause and severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

825.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

825.7 Diagnosis

- Doctors diagnose this using echocardiography with Doppler. Management targets the underlying cause
- Surgical valve repair or replacement is considered for severe symptomatic disease.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

825.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

825.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

825.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

825.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 826

Tricyclic Antidepressant (TCA) Overdose

826.1 Overview

TCA overdose (amitriptyline, nortriptyline, imipramine, desipramine, doxepin) is a highly lethal pharmaceutical overdose due to cardiovascular and neurological toxicity. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

826.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

826.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

826.4 Signs and Symptoms

You may experience altered mental status (often rapid onset coma), anticholinergic toxidrome, seizures, and irregular heartbeats, with the most specific finding being QRS widening >100 ms. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

826.5 Progression and Stages

This condition happens because of sodium channel blockade (class IA antiarrhythmic effect) slowing Phase 0 depolarization and prolonging QRS duration, alpha-adrenergic blockade causing hypotension, anticholinergic effects, and $GABA_A$ antagonism lowering seizure threshold. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

826.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

826.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam and by ECG. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

826.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

826.9 Treatment Options

Treatment options include airway protection, sodium bicarbonate (1–2 mEq/kg IV bolus) for QRS >100 ms or hemodynamically significant arrhythmias, benzodiazepines for seizures, and vasopressors (norepinephrine preferred) for refractory hypotension. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

826.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

826.11 Outlook

The outlook is generally positive with early aggressive treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 827

Trigeminal Autonomic Cephalalgias

827.1 Overview

The trigeminal autonomic cephalalgias (TACs) are a group of primary headache disorders characterized by unilateral trigeminal distribution of pain with ipsilateral cranial autonomic features. They include cluster headache, paroxysmal hemicrania, SUNCT/SUNA, and hemicrania continua. The unifying Disease mechanism involves activation of the trigeminovascular system and cranial parasympathetic outflow from the superior salivatory nucleus, with dysfunction of the hypothalamic network. Differential diagnosis among TACs is based on attack duration, frequency, periodicity, and response to indomethacin. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

827.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

827.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

827.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

827.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

827.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

827.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

827.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

827.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

827.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

827.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 828

Trigeminal Neuralgia

828.1 Overview

Trigeminal neuralgia (TN, tic douloureux) is a neuropathic pain disorder characterized by brief, electric shock-like, stabbing, or shooting paroxysmal pain in one or more divisions of the trigeminal nerve (typically V2 and V3). Prevalence is 0.01–0.03%, with a female predominance (2:1) and peak onset after age 50. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

828.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of neurovascular compression of the trigeminal nerve root entry zone by a blood vessel (most commonly the superior cerebellar artery), leading to focal demyelination and ephaptic transmission
- Secondary TN may be caused by multiple sclerosis, tumors, or arteriovenous malformations.

828.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

828.4 Signs and Symptoms

You may experience attacks lasting a fraction of a second to 2 minutes triggered by trivial stimuli (light touch, chewing, talking, toothbrushing), with no pain between attacks. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

828.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

828.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

828.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam with MRI identifying neurovascular compression. Neurological diagnosis combines clinical history and examination findings with neuroimaging (MRI, CT), electrophysiological studies (EEG, EMG, nerve conduction), and laboratory testing of blood and cerebrospinal fluid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures

- Specialized testing depending on the affected system
- Consultation with relevant specialists

828.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

828.9 Treatment Options

- Treatment options include carbamazepine (200–1200 mg/day) as first-line, with a highly specific response that is itself diagnostic, or oxcarbazepine as an alternative
- For refractory cases, surgical options include microvascular decompression (80–90% long-term pain-free), stereotactic radiosurgery, or through the skin rhizotomy.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

828.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

828.11 Outlook

- The outlook is generally positive with treatment
- Microvascular decompression addresses the underlying cause.

Chapter 829

Trigger Finger

829.1 Overview

Trigger finger (stenosing tenosynovitis) is a common tenosynovial disorder characterized by painful catching, snapping, or locking of the digit during flexion and extension. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

829.2 Causes

This condition happens because of a mismatch between the diameter of the flexor digitorum tendon and the narrowed first annular (A1) pulley at the metacarpal head, driven by repetitive microtrauma, tendon nodule formation, and fibrocartilaginous metaplasia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

829.3 Risk Factors

Risk factors include diabetes mellitus, rheumatoid arthritis, thyroid dysfunction, and repetitive hand gripping. Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

829.4 Signs and Symptoms

You may experience localized pain and a palpable tender nodule over the palmar MTP joint, with mechanical clicking or locking of the digit that may require passive manual extension to unlock. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

829.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

829.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

829.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

829.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

829.9 Treatment Options

Treatment options include activity modification, night splinting, local corticosteroid injection into the tendon sheath (highly effective), and surgical A1 pulley release for refractory cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

829.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

829.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 830

Triple-Negative Breast Cancer (TNBC)

830.1 Overview

Triple-negative breast cancer (TNBC) is a highly aggressive molecular subtype of breast cancer defined by the complete lack of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2) expression, accounting for 10–20% of all breast malignancies. TNBC occurs with higher frequency in younger women, women of African descent, and individuals harboring germline BRCA1 mutations. Lacking targeted hormonal or HER2 pathways, TNBC exhibits rapid cellular growth (high Ki-67 index), high histological grade, and early hematogenous dissemination, with a peak incidence of visceral (lung, liver) and central nervous system metastases within 3 years of diagnosis. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

830.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

830.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

830.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
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- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

830.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs

Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

830.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

830.7 Diagnosis

Your doctor confirms the diagnosis with immunohistochemistry ($< 1\%$ ER/PR staining) and HER2 FISH testing. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)

- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

830.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

830.9 Treatment Options

Treatment focuses on neoadjuvant combination chemotherapy (anthracyclines, taxanes, platinum agents) combined with immune checkpoint inhibitors (pembrolizumab) for PD-L1-positive tumors, followed by definitive surgery and post-neoadjuvant capecitabine or PARP inhibitors (olaparib) for BRCA carriers. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells

- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

830.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

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830.11 Outlook

outlook is worse than receptor-positive subtypes, with a 5-year survival rate of approximately 77%. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 831

Tuberculosis

831.1 Overview

Tuberculosis is a chronic infectious disease caused by *Mycobacterium tuberculosis*, transmitted via airborne droplets, and remains one of the leading infectious causes of death worldwide (approximately 1.3 million deaths annually). This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

831.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of infection with *M. tuberculosis*, which may remain latent or progress to active disease. Primary tuberculosis may be asymptomatic or present with lower lobe infiltrates and hilar lymphadenopathy
- Reactivation tuberculosis typically affects the upper lobes with cavitary disease, hemoptysis, night sweats, weight loss, and fever.

831.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals

- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

831.4 Signs and Symptoms

Symptoms of active disease include cough, hemoptysis, fever, night sweats, and weight loss. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

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- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

831.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

831.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage

- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

831.7 Diagnosis

- To diagnose active disease requires sputum smear and culture for acid-fast bacilli, nucleic acid amplification testing, and chest imaging
- Latent infection doctors detect it using tuberculin skin testing or interferon-gamma release assays. Treatment for drug-susceptible disease involves a 6-month regimen of isoniazid, rifampicin, pyrazinamide, and ethambutol for 2 months, followed by isoniazid and rifampicin for 4 months
- Multidrug-resistant disease requires prolonged complex regimens.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

831.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

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- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

831.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

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- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

831.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

831.11 Outlook

Most people recover well for drug-susceptible disease with appropriate treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 832

Tuberculosis (Extrapulmonary)

832.1 Overview

This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

832.2 Causes

This condition happens because of primary infection typically occurring in the lungs, with local containment in granulomas, and reactivation or progressive primary disease leading to dissemination to extrapulmonary sites. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

832.3 Risk Factors

Extrapulmonary tuberculosis accounts for 15–20% of all tuberculosis cases, occurring when *Mycobacterium tuberculosis* disseminates from the primary lung focus via lymphatic or hematogenous routes, with risk factors including HIV co-infection, immunosuppression, diabetes, and young age.

- Close contact with infected individuals
- Compromised immune system
- Travel to endemic areas
- Poor sanitation and hygiene practices

- Lack of vaccination
- Exposure to contaminated food or water
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

832.4 Signs and Symptoms

Your symptoms depend on the site: pleuritic involvement (tuberculous pleural effusion, most common EPTB), lymphatic (scrofula—painless, matted, cervical lymphadenopathy), skeletal (Pott's disease, tuberculous arthritis), central nervous system (tuberculous meningitis), genitourinary (sterile pyuria, flank pain), and miliary TB (diffuse hematogenous dissemination with small granulomas throughout the body). Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

832.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

832.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

832.7 Diagnosis

Doctors diagnose this using acid-fast stain and culture of specimens, PCR/Xpert MTB/RIF, and imaging. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

832.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification

- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

832.9 Treatment Options

Treatment options include standard 6-month regimen (2 months rifampin, isoniazid, pyrazinamide, ethambutol, then 4 months rifampin, isoniazid), with longer therapy for CNS TB and corticosteroids. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
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- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

832.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

832.11 Outlook

Most people recover well with appropriate treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications.

Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 833

Turner Syndrome (45,X)

833.1 Overview

Turner syndrome is a chromosomal disorder resulting from complete or partial monosomy X (45,X, 45,X/46,XX mosaicism, or structural X abnormalities), with an incidence of 1 in 2500 live-born females. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

833.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

833.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

833.4 Signs and Symptoms

You may experience short stature, webbed neck, low posterior hairline, shield chest, cubitus valgus, lymphedema of hands and feet at birth, and gonadal dysgenesis (streak gonads, primary amenorrhea, infertility). Associated conditions include coarctation of the aorta, bicuspid aortic valve, horseshoe kidney, sensorineural hearing loss, hypothyroidism, and celiac disease. Intelligence is typically normal.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

833.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

833.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

833.7 Diagnosis

Doctors diagnose this using karyotype.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

833.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

833.9 Treatment Options

Treatment options include recombinant growth hormone therapy for short stature, estrogen replacement for pubertal induction, and ongoing surveillance for associated comorbidities. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

833.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 834

Tympanic Membrane Perforation

834.1 Overview

Tympanic membrane perforation is a tear in the eardrum that may result from acute otitis media, trauma (penetrating injury, barotrauma, blast injury), or prior tympanostomy tubes. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Mental health conditions affect thoughts, emotions, behaviors, and overall well-being. These conditions are among the most common health concerns worldwide, affecting people across all age groups. Mental health disorders result from complex interactions of biological, psychological, and social factors. Effective treatments are available, and recovery is possible with appropriate support. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

834.2 Causes

This condition happens because of mechanical or infectious disruption of the tympanic membrane. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. Neurotransmitter imbalances (serotonin, dopamine, norepinephrine) play key roles in many mental health conditions. Genetic factors contribute to vulnerability, with heritability estimates varying by condition. Early life experiences, trauma, and chronic stress can alter brain development and function. Environmental factors including social support, socioeconomic status, and life events influence mental health. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

834.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)

- Environmental exposures
- Underlying medical conditions
- Medication use

834.4 Signs and Symptoms

You may experience hearing loss, brief otalgia, ringing in the ears, and otorrhea.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

834.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

834.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

834.7 Diagnosis

Doctors diagnose this based on otoscopic examination.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

834.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

834.9 Treatment Options

- Treatment focuses on observation for small perforations, which often heal spontaneously
- Larger or persistent perforations may require surgical repair (myringoplasty or tympanoplasty).
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

834.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

834.11 Outlook

The outlook is good, with healing enhanced by keeping the ear dry and avoiding nose blowing. With appropriate treatment, most people with mental health conditions experience significant improvement. Early intervention is associated with better long-term outcomes. Mental health conditions are chronic for some individuals, requiring ongoing management. Reducing stigma and improving access to care remain important public health priorities.

Chapter 835

Type 1 Diabetes Mellitus

835.1 Overview

Type 1 diabetes mellitus (T1DM) is an autoimmune disease characterized by destruction of pancreatic beta cells in the islets of Langerhans, leading to absolute insulin deficiency. It accounts for 5–10% of diabetes cases. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

835.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of autoreactive T cells (with CD8+ cytotoxic T cells being primary effectors) that destroy beta cells, with autoantibodies (anti-GAD65, anti-IA-2, anti-insulin, anti-ZnT8) serving as biomarkers. Genetic susceptibility is linked to HLA-DR3/DR4 and HLA-DQ2/DQ8 haplotypes
- Environmental triggers (viral infections, early cow's milk exposure) may initiate autoimmunity in predisposed individuals.

835.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

835.4 Signs and Symptoms

You may experience the classic triad of polyuria, polydipsia, and weight loss, with diabetic ketoacidosis (DKA) as the initial presentation in 30–40% of children.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

835.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

835.6 Complications

- Your outlook depends on glycemic control
- Microvascular and macrovascular complications determine long-term outcomes.
- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

835.7 Diagnosis

To make the diagnosis, doctors look for fasting plasma glucose ≥126 mg/dL, random glucose ≥200 mg/dL with symptoms, HbA1c ≥6.5%, or 2-hour OGTT glucose ≥200 mg/dL.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

835.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

835.9 Treatment Options

- Treatment typically involves lifelong insulin therapy via multiple daily injections (basal-bolus regimen) or continuous subcutaneous insulin pump
- Continuous glucose monitoring (CGM) and insulin pump systems (closed-loop "artificial pancreas" systems) improve glycemic control.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

835.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

835.11 Outlook

Islet transplantation and Teplizumab (anti-CD3 antibody) are emerging therapies.

Chapter 836

Type 2 Diabetes Mellitus

836.1 Overview

Type 2 diabetes mellitus (T2DM) is the most common form of diabetes, accounting for 90–95% of cases. It is marked by insulin resistance in peripheral tissues (muscle, liver, adipose tissue) combined with progressive beta-cell dysfunction and relative insulin deficiency. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

836.2 Causes

The Disease mechanism involves the "ominous octet"—eight mechanisms contributing to hyperglycemia: decreased insulin secretion, increased glucagon secretion, increased liver glucose production, decreased muscle glucose uptake, increased lipolysis, decreased incretin effect, increased kidney glucose reabsorption, and neurotransmitter dysfunction. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

836.3 Risk Factors

- Risk factors include obesity (particularly visceral adiposity), physical inactivity, family history, age >45, ethnicity, history of gestational diabetes, and polycystic ovary syndrome. Your outlook depends on glycemic control and Treatment for cardiovascular risk factors
- Cardiovascular benefit is demonstrated with SGLT2 inhibitors and GLP-1 receptor

agonists.
- Family history
- Obesity and sedentary lifestyle
- Poor dietary habits
- Advancing age
- Hormonal imbalances
- Certain medications
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

836.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

836.5 Progression and Stages

Symptoms of advanced disease include polyuria, polydipsia, and weight loss. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

836.6 Complications

Complications include microvascular disease (retinopathy, nephropathy, neuropathy) and macrovascular disease (coronary artery disease, stroke, peripheral arterial disease).

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

836.7 Diagnosis

Doctors diagnose this based on fasting plasma glucose, HbA1c, or OGTT.

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

836.8 Prevention

Many patients are asymptomatic and diagnosed on routine screening. Management follows a stepwise approach: lifestyle modification (diet, exercise, weight loss of 5–10%), metformin as first-line pharmacotherapy, followed by SGLT2 inhibitors, GLP-1 receptor agonists, DPP-4 inhibitors, thiazolidinediones, sulfonylureas, and insulin as needed.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

836.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

836.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 837

Types of Pain

837.1 Overview

It is typically well-localized and proportional to the degree of tissue injury. Subtypes include somatic pain (sharp, aching, from skin, muscle, bone) and visceral pain (cramping, gnawing, poorly localized, from internal organs). Responds to NSAIDs and opioids.

Neuropathic Pain: Caused by a lesion or disease of the somatosensory nervous system. Characterized by burning, shooting, electric shock-like pain, often with allodynia (pain from normally non-painful stimuli) and hyperalgesia (increased pain from normally painful stimuli). Examples include diabetic neuropathy, postherpetic neuralgia, trigeminal neuralgia. Responds to gabapentinoids, TCAs, SNRIs.

Nociplastic Pain (Central Sensitization): Pain arising from altered nociception despite no clear evidence of actual or threatened tissue damage or disease of the somatosensory system. Defined by the IASP in 2017. Features include widespread pain, fatigue, sleep disturbance, mood disturbance, and hyperalgesia/allodynia. Central sensitization involves increased excitability of spinal and supraspinal neurons. Examples include fibromyalgia, chronic tension-type headache, and some chronic back pain. Responds to multimodal therapy including TCAs, SNRIs, exercise, and CBT. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

837.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

837.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

837.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

837.5 Progression and Stages

Pain is classified into three mechanistic categories: The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

837.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

837.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures

Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

837.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

837.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

837.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

837.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 838

Typhoid Fever (Salmonella typhi)

838.1 Overview

Typhoid fever is a systemic febrile illness caused by *Salmonella enterica* serotype Typhi, with 11–21 million cases and 200,000 deaths annually worldwide, endemic in South Asia, Southeast Asia, sub-Saharan Africa, and parts of Latin America, with transmission being fecal-oral through contaminated food or water. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

838.2 Causes

This condition happens because of *S.* Typhi invading intestinal M cells, disseminating to mesenteric lymph nodes, and multiplying within macrophages of the reticuloendothelial system before being released into the bloodstream. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

838.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational

exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

838.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

838.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

838.6 Complications

You may experience an incubation of 7–14 days, with gradual onset of fever (stepwise ladder pattern), headache, malaise, dry cough, abdominal pain, relative bradycardia, rose spots, hepatosplenomegaly, and “pea soup” diarrhea or constipation, with intestinal bleeding or perforation being the classic complication.

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

838.7 Diagnosis

Doctors diagnose this using blood culture, bone marrow culture (gold standard), stool culture, and serology. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

838.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

838.9 Treatment Options

Treatment options include azithromycin, ceftriaxone, or fluoroquinolones for uncomplicated disease, with MDR and XDR strains increasing. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

838.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

838.11 Outlook

- The outlook is good with appropriate antibiotics
- Mortality is 10–30% without treatment.

Chapter 839

Ulcerative Colitis

839.1 Overview

Ulcerative colitis (UC) is a chronic inflammatory bowel disease limited to the colon and rectum, characterized by continuous mucosal and submucosal inflammation beginning at the rectum and extending proximally in a continuous fashion. This autoimmune condition occurs when the immune system mistakenly attacks the body's own tissues. It follows a chronic course with periods of flare and remission. Autoimmune diseases result from an abnormal immune response where the body mistakenly attacks its own healthy tissues. These conditions are typically chronic, with alternating periods of flare and remission. They can affect a single organ or be systemic, involving multiple organ systems. Women are affected more frequently than men for most autoimmune conditions. The exact prevalence varies by condition, but collectively autoimmune diseases affect a significant portion of the population.

839.2 Causes

A combination of genetic susceptibility and environmental triggers initiates the autoimmune process. Specific HLA genotypes are strongly associated with many autoimmune conditions. Environmental triggers may include infections, smoking, medications, and hormonal changes. Loss of self-tolerance leads to activation of autoreactive T cells and B cells producing autoantibodies. Molecular mimicry between pathogen antigens and self-antigens may trigger cross-reactive immune responses.

- This condition happens because of a dysregulated immune response to gut microbiota in genetically susceptible individuals
- Unlike Crohn's disease, it does not involve the small intestine (except for backwash ileitis in pancolitis) and does not cause transmural disease, strictures, or fistulas.

839.3 Risk Factors

Family history of autoimmune disease Female sex (higher prevalence) Specific HLA genotypes Epstein-Barr virus infection Cigarette smoking Certain medications (drug-induced autoimmunity) Hormonal factors (pregnancy, postpartum) Vitamin D deficiency Stress and psychological factors Geographic and ethnic background

- Family history of autoimmune disease
- Female sex (higher prevalence)
- Specific HLA genotypes

- Epstein-Barr virus infection
- Cigarette smoking
- Certain medications (drug-induced autoimmunity)
- Hormonal factors (pregnancy, postpartum)
- Vitamin D deficiency

839.4 Signs and Symptoms

- You may experience bloody diarrhea, urgency, tenesmus, and abdominal cramping
- Extraintestinal manifestations similar to Crohn's disease include arthritis, redness of the skin nodosum, and primary sclerosing cholangitis (occurring in 2–5% of UC patients).
- Fatigue and general malaise
- Joint pain, swelling, and stiffness
- Skin rashes and lesions
- Low-grade fever
- Muscle weakness and pain
- Organ-specific symptoms based on affected system
- Weight changes (loss or gain)
- Dry eyes and dry mouth (Sjogren syndrome)
- Raynaud phenomenon (cold-induced color changes in fingers)
- Fluctuating symptoms with periods of flare and remission

839.5 Progression and Stages

Autoimmune diseases typically follow a relapsing-remitting course with unpredictable flares. Early disease may present with nonspecific symptoms before organ-specific manifestations develop. Chronic inflammation over time can lead to irreversible tissue damage and organ dysfunction. With appropriate treatment, many patients achieve remission or low disease activity states.

- Doctors diagnose this based on colonoscopy with biopsies showing continuous inflammation, crypt abscesses, and mucosal ulceration
- Severity is classified as mild, moderate, or severe based on stool frequency, rectal bleeding, hemoglobin, and CRP.

839.6 Complications

- Irreversible organ damage from chronic inflammation
- Joint deformities and erosions in inflammatory arthritis
- Renal failure in lupus nephritis
- Pulmonary fibrosis or hypertension
- Cardiovascular complications from systemic inflammation
- Osteoporosis from chronic corticosteroid use
- Increased risk of infections from immunosuppressive therapy

- Lymphoma risk in some autoimmune conditions

839.7 Diagnosis

Clinical history and physical examination Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.) Inflammatory markers (ESR, CRP) Complete blood count and comprehensive metabolic panel Imaging studies (X-ray, MRI, ultrasound) of affected areas Biopsy of affected tissue for histological examination Classification criteria for specific autoimmune diseases Rheumatology consultation for suspected autoimmune disease Monitoring for organ involvement (renal, pulmonary, cardiac) Genetic testing for HLA associations when indicated

- Clinical history and physical examination
- Autoantibody testing (ANA, RF, anti-CCP, anti-dsDNA, etc.)
- Inflammatory markers (ESR, CRP)
- Complete blood count and comprehensive metabolic panel
- Imaging studies (X-ray, MRI, ultrasound) of affected areas
- Biopsy of affected tissue for histological examination
- Classification criteria for specific autoimmune diseases
- Rheumatology consultation for suspected autoimmune disease

839.8 Prevention

- No known prevention for most autoimmune diseases
- Avoid known triggers when possible
- Maintain a healthy lifestyle to support immune function
- Early recognition of symptoms for prompt diagnosis
- Regular medical check-ups for monitoring
- Adequate vitamin D levels through sun exposure or supplementation

839.9 Treatment Options

- Treatment options include 5-aminosalicylates (mesalamine) for mild-moderate disease, corticosteroids for flares, immunomodulators, and biologics (anti-TNF, vedolizumab, ustekinumab, tofacitinib [JAK inhibitor])
- Acute severe UC is a medical emergency treated with IV corticosteroids, with colectomy for refractory disease.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom relief
- Corticosteroids for rapid control of inflammation
- Disease-modifying antirheumatic drugs (DMARDs)
- Biologic therapies targeting specific immune pathways
- Immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide)
- Antimalarial drugs (hydroxychloroquine) for mild disease
- Physical and occupational therapy to maintain function
- Pain management for chronic pain

- Lifestyle modifications including diet and stress reduction

839.10 Living with It

- Take medications consistently as prescribed
- Identify and avoid personal flare triggers
- Maintain a balanced diet and regular gentle exercise
- Practice stress management techniques
- Join support groups for emotional support
- Work with a rheumatologist for ongoing disease management
- Monitor for medication side effects
- Plan activities around energy levels during flares

839.11 Outlook

- The outlook varies from person to person
- Total proctocolectomy is curative.

Chapter 840

Undifferentiated Connective Tissue Disease

840.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

840.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

840.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

840.4 Signs and Symptoms

Clinical features are mild and often limited to a few domains: the most common are arthralgias or non-erosive arthritis (50–80%), Raynaud's phenomenon (40–60%), rash (malar-like or photosensitive), oral ulcers, sicca symptoms, serositis, and constitutional symptoms. Laboratory findings typically include a positive ANA (low-to-moderate titer, often speckled or homogeneous pattern). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

840.5 Progression and Stages

Undifferentiated connective tissue disease is a diagnosis applied to patients who have clinical and laboratory features suggestive of a systemic autoimmune disease but do not meet classification criteria for a defined connective tissue disease, estimated to account for 10–50% of patients presenting to rheumatology clinics with early connective tissue disease symptoms, with a female-to-male ratio of approximately 5:1 and onset typically in the third to fifth decade. Diagnosis is one of exclusion: the patient must have symptoms compatible with a connective tissue disease for at least 1 year without fulfilling any classification criteria. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

840.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

840.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

840.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

840.9 Treatment Options

Treatment typically involves symptomatic: NSAIDs for arthralgias, hydroxychloroquine for constitutional and cutaneous symptoms, calcium channel blockers for Raynaud's phenomenon, and low-dose corticosteroids for severe symptoms. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

840.10 Living with It

The hallmark of UCTD is that symptoms remain stable and mild over long-term follow-up, with only 10–30% of patients evolving into a defined connective tissue disease (most commonly SLE or RA) after 3–5 years.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

840.11 Outlook

Most people recover well, with most patients maintaining a stable, mild disease course over decades. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 841

Urea Cycle Disorders

841.1 Overview

Urea cycle disorders (UCDs) are a group of inborn errors of metabolism caused by deficiency of one of the six enzymes or two transporters involved in liver conversion of ammonia to urea. X-linked ornithine transcarbamylase (OTC) deficiency is the most common (incidence 1 in 14,000). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

841.2 Causes

This condition happens because of impaired liver conversion of ammonia to urea, leading to hyperammonemia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

841.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

841.4 Signs and Symptoms

- You may experience severe enzyme deficiencies presenting in neonates after 24–72 hours with poor feeding, vomiting, lethargy, tachypnea with respiratory alkalosis, hypothermia, seizures, and rapidly progressive coma
- Late-onset forms present with recurrent hyperammonemia triggered by protein loads or catabolic stress, manifesting as developmental delay, loss of coordination, behavioral changes, and headaches.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

841.5 Progression and Stages

Your outlook depends on severity and prompt treatment. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

841.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

841.7 Diagnosis

Doctors diagnose this using plasma ammonia (elevated, often greater than 200 µmol/L), plasma amino acids, urinary orotic acid, and genetic testing. Acute Treatment options include discontinuing protein intake, high-calorie dextrose and lipids to promote anabolism, nitrogen scavengers (IV sodium benzoate and sodium phenylacetate), arginine supplementation, and hemodialysis for severe hyperammonemia. Chronic Treatment options include protein restriction, essential amino acid supplementation, oral nitrogen scavengers, and consideration of orthotopic liver transplantation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination

- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

841.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

841.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

841.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

841.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 842

Urinary Incontinence

842.1 Overview

Urinary incontinence (UI) is a geriatric syndrome affecting 30–60% of older adults, increasing with age and more common in women. Types include stress (leakage with coughing, sneezing, laughing—urethral hypermobility or intrinsic sphincter deficiency), urge (overactive bladder—leakage with sudden strong urge to void, due to detrusor overactivity), overflow (incomplete bladder emptying with continuous dribbling—often from outlet obstruction or detrusor underactivity), functional (inability to reach toilet due to mobility impairment, thinking and memory deficit, or environmental barriers), and mixed. In the geriatric population, transient causes must be excluded (DIAPPERS mnemonic: Delirium, Infection, Atrophic urethritis/vaginitis, Pharmaceuticals, Psychological, Excess urine output, Restricted mobility, Stool impaction). Diagnosis includes history, bladder diary, post-void residual by ultrasound, urinalysis, and cough stress test. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

842.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

842.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

842.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

842.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

842.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

842.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

842.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

842.9 Treatment Options

Treatment options include behavioral therapy (pelvic floor exercises, bladder training, scheduled voiding, fluid management), medications (antimuscarinics or beta-3 agonists for overactive bladder), absorbent products, pessaries, and surgical options for refractory cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

842.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

842.11 Outlook

The outlook varies from person to person. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 843

Urinary Incontinence

843.1 Overview

Urinary incontinence is the involuntary loss of urine, affecting approximately 25–45% of women and 10–25% of men, with prevalence increasing with age. Types include stress urinary incontinence (SUI, leakage with coughing, sneezing, laughing, or exercise, caused by urethral hypermobility or intrinsic sphincter deficiency), urge urinary incontinence (UUI, involuntary loss of urine preceded by a sudden, strong urge, associated with detrusor overactivity), mixed urinary incontinence (combination of SUI and UUI), and overflow incontinence (continuous dribbling from an overdistended bladder due to bladder outlet obstruction or detrusor underactivity). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

843.2 Causes

This condition happens because of dysfunction of the pelvic floor, urethral sphincter, or detrusor muscle. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

843.3 Risk Factors

- Risk factors for SUI include vaginal delivery, menopause, obesity, and chronic cough
- UUI may be idiopathic or secondary to neurogenic bladder, bladder stones, infection, or malignancy.
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

843.4 Signs and Symptoms

Your symptoms depend on the type of incontinence. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

843.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

843.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

843.7 Diagnosis

- Doctors diagnose this based on history, bladder diary, physical examination (cough stress test, pelvic organ prolapse assessment), urinalysis, and post-void residual volume measurement
- Urodynamic studies are indicated for refractory cases.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

843.8 Prevention

- Treatment for SUI includes pelvic floor muscle training (Kegel exercises, first-line), lifestyle modification, vaginal pessaries, and surgical options (mid-urethral sling procedures, colposuspension)
- Treatment for UUI includes bladder retraining, timed voiding, antimuscarinics, beta-3 agonists (mirabegron, vibegron), and onabotulinumtoxinA for refractory cases
- Overflow incontinence is managed by treating the underlying obstruction or optimizing bladder emptying.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

843.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

843.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

843.11 Outlook

In most cases, the outlook is favorable with appropriate management tailored to the type of incontinence. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 844

Urosepsis

844.1 Overview

Urosepsis is sepsis originating from the urinary tract, accounting for approximately 25% of all sepsis cases. It is more common in the elderly, immunocompromised, and those with urological abnormalities. Common pathogens include *E. coli*, *Klebsiella*, *Pseudomonas*, and *Enterococcus*. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

844.2 Causes

This condition happens because of a dysregulated host response to urinary tract infection leading to life-threatening organ dysfunction. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

844.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

844.4 Signs and Symptoms

You may experience fever, hypotension, altered mental status, and signs of organ dysfunction. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

844.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

- Outlook is guarded
- Mortality ranges from 20–40% depending on severity.

844.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

844.7 Diagnosis

Doctors diagnose this based on clinical criteria for sepsis with urinary source identified. Management follows the Surviving Sepsis Campaign guidelines: early recognition, broad-spectrum IV antibiotics within 1 hour, aggressive fluid resuscitation, vasopressors for septic shock, source control (eg., decompression of obstructed urinary tract), and organ support. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

844.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

844.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

844.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

844.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 845

Urticaria

845.1 Overview

Urticaria is a common skin condition characterized by transient, itchy, erythematous wheals (hives) with central clearing, affecting 15–25% of the population at some point in life. Acute urticaria (duration <6 weeks) is usually caused by IgE-mediated reactions to foods (shellfish, nuts, eggs), medications (antibiotics, NSAIDs), infections (viral URIs), insect stings, or physical stimuli. Chronic urticaria (duration >6 weeks) is idiopathic in 50% and autoimmune in 30–50% (IgG autoantibodies against FcεRI or IgE). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

845.2 Causes

This condition happens because of mast cell degranulation releasing histamine and other vasoactive mediators, causing dermal swelling and vasodilation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

845.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

845.4 Signs and Symptoms

You may experience well-circumscribed, raised, erythematous wheals with intense itching, ranging from a few millimeters to several centimeters. Individual wheals resolve within 24 hours without scarring. Angioedema accompanies 40% of cases. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

845.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

845.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

845.7 Diagnosis

- Doctors diagnose this using history and physical examination
- Chronic urticaria requires CBC, ESR, TSH, and autoantibody testing.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

845.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

845.9 Treatment Options

Treatment options include first-generation and second-generation H1 antihistamines (cetirizine, fexofenadine, loratadine) as first-line therapy

- H2 antihistamines (ranitidine, famotidine) may be added. For refractory chronic urticaria: omalizumab (anti-IgE monoclonal antibody, highly effective), leukotriene receptor antagonists, cyclosporine, or hydroxychloroquine. Short courses of oral corticosteroids for severe exacerbations. outlook: acute urticaria resolves with trigger avoidance
- Chronic urticaria resolves within 1 year in 50% and within 5 years in 80%.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

845.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

845.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 846

Uterine Fibroids

846.1 Overview

Uterine fibroids (leiomyomas) are the most common non-cancerous tumors of the female reproductive tract, affecting up to 80% of women by age 50, with higher prevalence and symptomatic disease in African American women. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

846.2 Causes

This condition happens because of monoclonal smooth muscle tumors of the myometrium, driven by estrogen and progesterone. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

846.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

846.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

846.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

846.6 Complications

Your symptoms depend on location (submucosal, intramural, subserosal, pedunculated) and may include abnormal uterine bleeding (menorrhagia leading to iron deficiency anemia), pelvic pain and pressure, urinary frequency, constipation, infertility, and pregnancy complications.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

846.7 Diagnosis

Doctors diagnose this using pelvic ultrasound (usually sufficient) or MRI. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

846.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

846.9 Treatment Options

The right treatment depends on symptoms and reproductive goals: medical management (combined oral contraceptives, progestins, GnRH agonists, tranexamic acid), uterine artery embolization, myomectomy, and hysterectomy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

846.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

846.11 Outlook

Most people recover well, with fibroids being non-cancerous. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 847

Uterine Rupture

847.1 Overview

Uterine rupture is a complete disruption of all layers of the uterine wall, occurring in 0.5–1% of women with a prior cesarean delivery attempting vaginal birth after cesarean (VBAC). It is a catastrophic obstetric emergency. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

847.2 Causes

This condition happens because of disruption of a uterine scar or traumatic delivery. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

847.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

847.4 Signs and Symptoms

You may experience acute onset fetal distress (fetal bradycardia most common sign—70%), cessation of uterine contractions, loss of station of the presenting part, severe abdominal pain, and maternal hypovolemic shock. Diagnosis is often made at surgical opening of the abdomen. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

847.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

847.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

847.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

847.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

847.9 Treatment Options

- Treatment typically involves immediate emergency surgical opening of the abdomen for delivery of the fetus and repair of the uterus
- Hysterectomy may be required for extensive rupture.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

847.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

847.11 Outlook

- Your outlook depends on prompt intervention
- Fetal mortality is 6–10% with timely treatment.

Chapter 848

Uveitis

848.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

848.2 Causes

This condition happens because of infectious or autoimmune causes, including HLA-B27-associated conditions (ankylosing spondylitis, reactive arthritis), Behçet's disease, sarcoidosis, and infections (herpes, syphilis, tuberculosis). The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

848.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

848.4 Signs and Symptoms

Symptoms of anterior uveitis include eye pain, redness, photophobia, and hypopyon

- Intermediate uveitis features vitreous cells and floaters
- Posterior uveitis presents with floaters, visual field defects, or decreased vision.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

848.5 Progression and Stages

Uveitis is an intraocular inflammation involving the uveal tract (iris, ciliary body, choroid), classified by anatomical location as anterior uveitis (iritis or iridocyclitis), intermediate uveitis (pars planitis), posterior uveitis (choroiditis), or panuveitis. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

848.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

848.7 Diagnosis

- Doctors diagnose this based on slit-lamp and dilated fundus examination. Management uses topical, periocular, or systemic corticosteroids and steroid-sparing immunosuppressive agents for chronic disease
- Biologics (anti-TNF agents) are used for refractory cases.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

848.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection

- Follow recommended screening guidelines

848.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

848.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

848.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 849

Vaginal Cancer

Clear cell adenocarcinoma of the vagina is a rare variant associated with in utero diethylstilbestrol (DES) exposure, affecting women born between 1940 and 1970.

849.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

849.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

849.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

849.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

849.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

849.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

849.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

849.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin

from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

849.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

849.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team

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- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

849.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 850

Vaginal Cancer

850.1 Overview

Vaginal cancer is the rarest gynecologic malignancy, with squamous cell carcinoma (70–80%) being the most common type, often related to HPV infection and preceded by vaginal intraepithelial neoplasia (VAIN). This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

850.2 Causes

This condition happens because of cancerous transformation of vaginal epithelium. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

850.3 Risk Factors

Risk factors include prior cervical cancer/VIN, HPV infection, smoking, and immunosuppression.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption

- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

850.4 Signs and Symptoms

You may experience vaginal bleeding, discharge, dyspareunia, and pelvic pain. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

850.5 Progression and Stages

The right treatment depends on stage and location: radiation therapy (most common treatment) or local removal for small, superficial lesions. Your outlook depends on stage at diagnosis. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage,

small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

850.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

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- Malignant effusions (pleural, peritoneal, pericardial)
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- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

850.7 Diagnosis

Doctors diagnose this using pelvic examination with biopsy. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
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- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

850.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

850.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
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- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

850.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

850.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 851

Variant CJD (Mad Cow Disease)

851.1 Overview

Variant Creutzfeldt-Jakob disease is a prion disease caused by transmission of bovine spongiform encephalopathy (BSE, "mad cow disease") to humans through ingestion of BSE-contaminated beef products, with 232 cases reported worldwide, mean age at onset 28 years (much younger than sCJD), and a longer duration (14 months). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

851.2 Causes

This condition happens because of the same mechanism as sCJD (PrP^{Sc} replication and neurotoxicity) but with a distinct prion strain and unique glycoform banding pattern, with PrP^{Sc} also found in lymphoid tissue, unlike sCJD. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

851.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

851.4 Signs and Symptoms

You may experience early psychiatric symptoms (depression, anxiety, withdrawal, hallucinations) predominating for several months, followed by cerebellar loss of coordination, thinking and memory decline, myoclonus, choreoathetosis, and akinetic mutism. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

851.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

851.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

851.7 Diagnosis

Doctors diagnose this using MRI showing the pulvinar sign (bilateral high signal in the posterior thalamus) and tonsil biopsy detecting PrP^{Sc}. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

851.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

851.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

851.10 Living with It

Treatment typically involves palliative.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

851.11 Outlook

outlook is uniformly fatal. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 852

Varicella-Zoster (Systemic Aspects)

852.1 Overview

Varicella-zoster virus is a DNA herpesvirus that causes two distinct clinical syndromes: primary infection (varicella/chickenpox) and reactivation (herpes zoster/shingles), with varicella affecting nearly all individuals in unvaccinated populations and herpes zoster affecting 30% of individuals over a lifetime. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

852.2 Causes

This condition happens because of VZV entering through the respiratory mucosa, replicating in the tonsils and regional lymph nodes, and causing a primary viremia that seeds the reticuloendothelial system, with a secondary viremia disseminating the virus to the skin, after which VZV establishes latency in dorsal root ganglia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

852.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

852.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

852.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

852.6 Complications

Symptoms of varicella include fever, malaise, and a vesicular rash (centripetal—trunk, face, proximal extremities), with systemic complications including varicella pneumonia, encephalitis, hepatitis, myocarditis, and glomerulonephritis.

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

852.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, confirmed by PCR of vesicular fluid. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

852.8 Prevention

- Most people recover well for immunocompetent children
- Prevention through varicella vaccine and shingles vaccine is highly effective.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

852.9 Treatment Options

Treatment options include acyclovir (initiated within 24 hours of rash) and varicella immune globulin for exposed high-risk individuals. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

852.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

852.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 853

Vasa Previa

853.1 Overview

Vasa previa is a rare involving bleeding disorder of pregnancy (1 in 2500 pregnancies) in which fetal blood vessels from the umbilical cord traverse the fetal membranes overlying the internal cervical os, unsupported by the placenta or umbilical cord. Fetal mortality without prenatal diagnosis is 50–60%. This pregnancy-related condition requires careful monitoring to ensure the safety of both mother and baby. Early detection and management are essential. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

853.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

853.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

853.4 Signs and Symptoms

You may experience painless vaginal bleeding at amniotomy or with spontaneous rupture of membranes, accompanied by acute fetal distress due to fetal exsanguination. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

853.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

853.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

853.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

853.8 Prevention

Prenatal diagnosis by color Doppler ultrasound allows planned cesarean delivery at 34–35 weeks to prevent catastrophic bleeding.

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

853.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

853.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

853.11 Outlook

Most people recover well with prenatal diagnosis. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 854

Vascular Dementia

854.1 Overview

Vascular dementia (VaD) is a progressive neurodegenerative disorder and the second most common dementia, resulting from cerebrovascular disease—large vessel infarcts (multi-infarct dementia), small vessel disease (subcortical ischemic vascular dementia), strategic single infarcts, cerebral amyloid angiopathy, or CADASIL. This neurological disorder affects the central or peripheral nervous system, leading to progressive or episodic symptoms. It significantly impacts quality of life and often requires multidisciplinary care. This condition is among the most common mental health disorders worldwide. It affects people across all age groups, socioeconomic backgrounds, and geographic regions. Neurological disorders affect the central or peripheral nervous system, leading to a wide range of symptoms affecting movement, sensation, cognition, and autonomic function. The nervous system's complexity means that even small disruptions can cause significant functional impairment. Many neurological conditions are chronic and progressive, requiring long-term management and rehabilitation. Advances in neuroimaging and molecular diagnostics have improved understanding and treatment of these conditions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

854.2 Causes

Neurological dysfunction can result from structural abnormalities, neurodegenerative processes, demyelination, vascular injury, or neurotransmitter imbalances. Protein aggregation and accumulation is a common mechanism in many neurodegenerative disorders. Excitotoxicity, oxidative stress, and neuroinflammation contribute to neuronal injury. Genetic mutations account for some neurological conditions, while others are sporadic or environmentally triggered. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

854.3 Risk Factors

In the geriatric patient, high blood pressure, diabetes, atrial fibrillation, and smoking are the main modifiable risk factors. Treatment focuses on vascular risk factor control (strict blood pressure control, antiplatelet therapy, statins, glycemic control) and symptomatic treatment of thinking and memory symptoms.

- Advanced age
- Family history and genetic predisposition
- Head trauma or brain injury
- Vascular risk factors (hypertension, diabetes)
- Exposure to environmental toxins
- Chronic stress
- Family history of mental illness
- Trauma or adverse childhood experiences
- Substance use
- Social isolation
- Underlying medical conditions
- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Medication use

854.4 Signs and Symptoms

Vascular dementia frequently coexists with Alzheimer disease (mixed dementia). Brain MRI shows white matter hyperintensities, lacunar infarcts, and shrinkage or wasting. Headache and facial pain Weakness or paralysis of muscles Numbness, tingling, or abnormal sensations Vision changes including blurred or double vision Difficulty with balance and coordination Cognitive changes (memory loss, confusion, difficulty concentrating) Speech and language difficulties Seizures or involuntary movements Dizziness and vertigo Fatigue and sleep disturbances

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

854.5 Progression and Stages

You may experience stepwise deterioration, patchy thinking and memory deficits (executive function disproportionately affected), focal neurological signs, and often a history of stroke or TIA. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

854.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

854.7 Diagnosis

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

854.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

854.9 Treatment Options

Pharmacologic therapy targeting specific neurotransmitter systems or disease mechanisms Physical, occupational, and speech therapy for functional maintenance Surgical interventions when appropriate (deep brain stimulation, tumor resection) Cognitive rehabilitation and memory support Pain management for neuropathic pain Assistive devices and mobility aids Lifestyle modifications including exercise, nutrition, and sleep hygiene Support services and caregiver education Regular neurologic monitoring and treatment adjustment Clinical trials for emerging therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

854.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

854.11 Outlook

- Outlook is progressive
- Geriatric patients require careful blood pressure management to avoid orthostatic hypotension and falls.

Chapter 855

Ventricular Tachycardia

855.1 Overview

Cardiovascular disease encompasses conditions affecting the heart and blood vessels, representing a leading cause of morbidity and mortality worldwide. These conditions range from acute events like heart attack and stroke to chronic conditions requiring lifelong management. Atherosclerosis, the buildup of plaque in arterial walls, underlies many cardiovascular conditions. Risk increases with age, and the global burden continues to rise with aging populations and lifestyle changes.

- Ventricular tachycardia is a potentially life-threatening arrhythmia originating from the ventricles at a rate of 100–250 beats per minute
- It may be monomorphic or polymorphic.

855.2 Causes

The most common cause is structural heart disease, particularly prior heart attack with scar-related reentry. This condition happens because of abnormal electrical activity in ventricular myocardium. Hemodynamically unstable ventricular tachycardia requires immediate synchronized cardioversion. Cardiovascular disease develops through a complex interplay of genetic, environmental, and lifestyle factors. Atherosclerosis, characterized by plaque buildup in arterial walls, is a common underlying mechanism. Atherosclerosis begins with endothelial injury and dysfunction, leading to lipid accumulation and inflammatory cell infiltration in arterial walls. Plaque formation narrows vessels and reduces blood flow; plaque rupture can trigger acute thrombosis and vessel occlusion. Hemodynamic factors such as hypertension increase mechanical stress on vessel walls. Metabolic abnormalities including diabetes and dyslipidemia accelerate the atherosclerotic process. Genetic factors influence lipid metabolism, blood pressure regulation, and vascular health.

855.3 Risk Factors

Advanced age Cigarette smoking Hypertension (high blood pressure) Diabetes mellitus Elevated LDL cholesterol and low HDL cholesterol Family history of premature cardiovascular disease Obesity (especially abdominal obesity) Physical inactivity Unhealthy diet (high in saturated fat, salt, processed foods) Chronic kidney disease

- Advanced age
- Cigarette smoking
- Hypertension (high blood pressure)

- Diabetes mellitus
- Elevated LDL cholesterol and low HDL cholesterol
- Family history of premature cardiovascular disease
- Obesity (especially abdominal obesity)
- Physical inactivity
- Unhealthy diet (high in saturated fat, salt, processed foods)

855.4 Signs and Symptoms

You may experience palpitations, dizziness, syncope, and cardiac arrest. Chest pain, pressure, or discomfort (angina) Shortness of breath with exertion or at rest Palpitations or irregular heartbeat Fatigue and reduced exercise tolerance Swelling in the legs, ankles, or feet (edema) Dizziness or lightheadedness Pain radiating to arm, jaw, back, or neck Nausea and indigestion Cold sweats Sudden weakness or numbness (stroke symptoms)

- Chest pain, pressure, or discomfort (angina)
- Shortness of breath with exertion or at rest
- Palpitations or irregular heartbeat
- Fatigue and reduced exercise tolerance
- Swelling in the legs, ankles, or feet (edema)
- Dizziness or lightheadedness
- Pain radiating to arm, jaw, back, or neck
- Nausea and indigestion
- Cold sweats

855.5 Progression and Stages

Cardiovascular disease develops gradually over years, often beginning with asymptomatic risk factors. Early stages involve endothelial dysfunction and fatty streak formation in arterial walls. Progressive plaque buildup leads to hemodynamically significant stenosis causing symptoms with exertion. Advanced disease involves unstable plaques prone to rupture, causing acute coronary syndromes. End-stage disease results in chronic heart failure or critical limb ischemia.

855.6 Complications

- Myocardial infarction (heart attack)
- Heart failure with reduced or preserved ejection fraction
- Stroke from embolic or thrombotic events
- Arrhythmias including atrial fibrillation and ventricular tachycardia
- Peripheral artery disease causing limb ischemia
- Aortic aneurysm and dissection
- Chronic kidney disease from reduced perfusion

855.7 Diagnosis

Doctors diagnose this using ECG. Cardiovascular diagnosis relies on a combination of clinical history, physical examination, electrocardiography, imaging (echocardiography, CT angiography), and laboratory markers (troponin, BNP). History and physical examination including blood pressure measurement Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia Echocardiogram to evaluate cardiac structure and function Stress testing (exercise or pharmacologic) for ischemic evaluation Cardiac biomarkers (troponin, CK-MB, BNP) Lipid profile and glucose testing Coronary angiography or CT angiography for coronary anatomy Holter monitoring for arrhythmia detection Cardiac MRI for detailed structural assessment Ankle-brachial index for peripheral artery disease

- History and physical examination including blood pressure measurement
- Electrocardiogram (ECG/EKG) to assess heart rhythm and ischemia
- Echocardiogram to evaluate cardiac structure and function
- Stress testing (exercise or pharmacologic) for ischemic evaluation
- Cardiac biomarkers (troponin, CK-MB, BNP)
- Lipid profile and glucose testing
- Coronary angiography or CT angiography for coronary anatomy
- Holter monitoring for arrhythmia detection

855.8 Prevention

- Treatment for stable monomorphic ventricular tachycardia involves intravenous amiodarone or procainamide followed by catheter ablation
- Implantable cardioverter-defibrillators are indicated for secondary prevention (survivors of cardiac arrest) and primary prevention in high-risk patients
- Torsades de pointes doctors typically treat it with intravenous magnesium. Your outlook depends on the underlying heart disease and effectiveness of prevention.
- Maintain a heart-healthy diet low in saturated fat and sodium
- Engage in regular aerobic exercise (150 minutes per week)
- Avoid tobacco products and limit alcohol
- Control blood pressure, cholesterol, and blood glucose
- Maintain a healthy body weight
- Manage stress through relaxation techniques
- Take prescribed preventive medications (statins, aspirin)

855.9 Treatment Options

Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics) Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors) Antiplatelet therapy (aspirin, clopidogrel) Anticoagulation for atrial fibrillation and thromboembolic risk Nitrates and antianginal medications Revascularization: percutaneous coronary intervention (stenting) Coronary artery bypass grafting for severe disease Cardiac rehabilitation programs Device therapy

(pacemakers, ICDs) for arrhythmias

- Lifestyle modifications: heart-healthy diet, regular exercise, smoking cessation
- Antihypertensive medications (ACE inhibitors, ARBs, beta-blockers, diuretics)
- Lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors)
- Antiplatelet therapy (aspirin, clopidogrel)
- Anticoagulation for atrial fibrillation and thromboembolic risk
- Nitrates and antianginal medications
- Revascularization: percutaneous coronary intervention (stenting)
- Coronary artery bypass grafting for severe disease
- Cardiac rehabilitation programs

855.10 Living with It

- Take all medications exactly as prescribed
- Monitor your blood pressure and weight regularly
- Follow a heart-healthy diet and stay physically active
- Attend cardiac rehabilitation if recommended
- Recognize warning signs of heart attack and stroke
- Keep all follow-up appointments with your cardiologist
- Manage stress through relaxation and mindfulness
- Carry a list of your medications and dosages

855.11 Outlook

With proper management and lifestyle modifications, many patients maintain good quality of life. Outcomes depend on the extent of disease, adherence to treatment, and control of risk factors. Early intervention for acute events significantly improves survival. Regular monitoring and medication adherence are essential for preventing disease progression. Advances in interventional cardiology continue to improve outcomes for complex cases.

Chapter 856

Vertigo

856.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Feeling of spinning is the sensation of spinning or movement of the self or surroundings when there is no actual movement
- It is a symptom, not a disease, and may originate from peripheral (inner ear) or central (brainstem, cerebellar) causes. Peripheral feeling of spinning is more common and results from dysfunction of the vestibular apparatus or nerve
- Causes include non-cancerous paroxysmal positional feeling of spinning, vestibular neuritis, labyrinthitis, and Ménière's disease.

856.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- Central feeling of spinning results from brainstem or cerebellar pathology
- Causes include posterior circulation stroke, vestibular migraine, multiple sclerosis, and cerebellar tumors.

856.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

856.4 Signs and Symptoms

- Symptoms of peripheral feeling of spinning include severe rotational feeling of spinning with involuntary eye movements (beating away from the affected side), nausea, and vomiting
- Central feeling of spinning tends to be less severe but may be continuous with diplopia, slurred speech, and limb loss of coordination.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

856.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

856.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

856.7 Diagnosis

Diagnosis typically involves detailed history, neurological examination, Dix-Hallpike test for BPPV, and neuroimaging when central causes are suspected. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

856.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

856.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

856.10 Living with It

- The right treatment depends on the underlying cause: canalith repositioning maneuvers for BPPV, vestibular rehabilitation for vestibular neuritis, and stroke protocols for central feeling of spinning
- Vestibular suppressants provide symptomatic relief but should be used short-term.
- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

856.11 Outlook

Your outlook depends on the underlying cause. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 857

Vestibular Neuritis

857.1 Overview

Vestibular neuritis is an acute, isolated episode of severe feeling of spinning caused by inflammation of the vestibular nerve without hearing loss. The condition is thought to be viral in cause, possibly reactivation of herpes simplex virus. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

857.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

857.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

857.4 Signs and Symptoms

You may experience sudden-onset, continuous feeling of spinning lasting days to weeks with severe imbalance and involuntary eye movements beating away from the affected side.

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

857.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

857.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

857.7 Diagnosis

Doctors diagnose this based on clinical examination with the head impulse test showing reduced vestibulo-ocular reflex on the affected side. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

857.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

857.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

857.10 Living with It

Treatment focuses on short-term vestibular suppressants (benzodiazepines, antihistamines) followed by early vestibular rehabilitation.

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

857.11 Outlook

The outlook is generally positive, with most patients recovering fully over weeks. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 858

Viral Encephalitis

858.1 Overview

Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

- Viral encephalitis is an inflammation of the brain parenchyma caused by viral infection
- Herpes simplex virus type 1 (HSV-1) is the most common cause of sporadic fatal encephalitis, typically affecting the temporal and frontal lobes, while other causes include enteroviruses, arboviruses (West Nile, Japanese encephalitis), and varicella-zoster virus.

858.2 Causes

This condition happens because of direct viral invasion of brain parenchyma. Following exposure, the pathogen invades host tissues and replicates, triggering an immune response that contributes to the symptoms. The severity of infection depends on pathogen virulence, host immune status, and timely treatment. Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

858.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

858.4 Signs and Symptoms

You may experience fever, headache, altered consciousness, behavioral changes, seizures, and weakness, numbness, or problems with movement. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

858.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

858.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)

- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

858.7 Diagnosis

Diagnosis relies on MRI (temporal lobe involvement in HSV), CSF analysis (lymphocytic pleocytosis, PCR for viral DNA), and EEG. Laboratory diagnosis includes pathogen identification through culture, PCR testing, or antigen detection. Serological tests can confirm exposure or immune response to the pathogen. Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

858.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)

- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

858.9 Treatment Options

Treatment focuses on intravenous acyclovir for HSV encephalitis, which should be initiated empirically. Antimicrobial therapy is tailored to the specific pathogen and its susceptibility patterns. Supportive care includes hydration, rest, and management of symptoms such as fever and pain. Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

858.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

858.11 Outlook

outlook is guarded, with significant neurological sequelae in survivors despite treatment. Most patients recover fully with appropriate treatment, though some infections may lead to long-term complications. Outcomes depend on the pathogen, host immune status, and timeliness of treatment. Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 859

Viral Exanthems

859.1 Overview

Viral exanthems are rashes associated with systemic viral infections, most common in childhood. Measles, rubella, and mumps are covered in Chapter 20, Infectious Diseases. Roseola infantum (sixth disease) is caused by human herpesvirus 6 (HHV-6) and less commonly HHV-7, with peak incidence at 6–24 months, presenting with high fever for 3–5 days followed by defervescence and a rose-colored maculopapular rash. redness of the skin infectiosum (fifth disease) is caused by parvovirus B19, with peak incidence at 5–14 years, presenting with a "slapped cheek" facial redness of the skin followed by a lacy reticular rash. Hand-foot-and-mouth disease is caused by enteroviruses, most commonly Coxsackievirus A16 and Enterovirus 71, with peak incidence in children under 5 years, presenting with fever and vesicular rash on hands, feet, and oral mucosa. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

859.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

859.3 Risk Factors

Close contact with infected individuals Compromised immune system (HIV, immunosuppressive therapy, malnutrition) Extremes of age (very young or elderly) Travel to endemic regions Poor sanitation and hygiene practices Lack of vaccination Exposure to contaminated food or water Healthcare exposure (nosocomial infections) Occupational exposure (healthcare workers, laboratory personnel) Crowded living conditions

- Close contact with infected individuals
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

859.4 Signs and Symptoms

Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

859.5 Progression and Stages

The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

859.6 Complications

- Sepsis and septic shock

- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

859.7 Diagnosis

Detailed history including exposure, travel, and vaccination status Physical examination focusing on the affected system Complete blood count with differential Microbiological culture of blood, sputum, urine, or wound PCR and nucleic acid amplification tests for rapid pathogen detection Antigen detection tests Serological testing for antibodies (IgM, IgG) Imaging studies (chest X-ray, CT, ultrasound) Gram stain and microscopy of clinical specimens Antimicrobial susceptibility testing to guide therapy

- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

859.8 Prevention

Hand hygiene with soap and water or alcohol-based sanitizer Vaccination according to recommended schedules Safe food handling and water purification Use of personal protective equipment in healthcare settings Safe sex practices including condom use Mosquito avoidance (nets, repellents, insecticides) Respiratory etiquette (covering coughs and sneezes) Isolation of infected individuals when indicated Prophylactic antibiotics or antivirals for high-risk exposures Travel precautions and pre-travel consultations

- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures

- Travel precautions and pre-travel consultations

859.9 Treatment Options

Antibiotics for bacterial infections (targeted or empiric) Antiviral medications for viral infections Antifungal therapy for fungal infections Antiparasitic medications for parasitic infections Supportive care: fluids, rest, nutrition, fever control Hospitalization for severe cases requiring intravenous therapy Oxygen therapy and respiratory support if needed Surgical drainage of abscesses when necessary Pain management and symptom relief Treatment of complications and underlying conditions

- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

859.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

859.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 860

Vitamin A Deficiency

860.1 Overview

Vitamin A deficiency (VAD) is a vitamin deficiency disorder that is the leading cause of preventable blindness in children worldwide and a major contributor to morbidity from infections. It is most prevalent in sub-Saharan Africa and South Asia, where diets are low in preformed vitamin A (retinol, retinyl esters) and provitamin A carotenoids. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

860.2 Causes

This condition happens because of inadequate dietary intake of vitamin A, which is essential for retinal photoreceptor function (11-*cis*-retinal in rhodopsin), epithelial integrity, and immune competence. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

860.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

860.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

860.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- You may experience night blindness (nyctalopia) as the earliest symptom, followed by xerophthalmia including conjunctival xerosis, Bitot spots, corneal xerosis, and ultimately keratomalacia causing irreversible blindness
- VAD also increases the severity of measles, diarrheal diseases, and respiratory infections.

860.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

860.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam (ocular signs) and confirmed by serum retinol less than 0.70 µmol/L.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

860.8 Prevention

- Treatment focuses on high-dose vitamin A supplementation (200,000 IU orally for children over 12 months, 100,000 IU for 6–12 months, 50,000 IU for infants under 6 months) given on diagnosis, repeated the next day, and at least 2 weeks later
- Prevention includes dietary diversification, food fortification, and periodic high-dose supplementation.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

860.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

860.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 861

Vitamin B1 (Thiamine) Deficiency — Beriberi and Wernicke-Korsakoff Syndrome

861.1 Overview

Thiamine (vitamin B1) deficiency is a vitamin deficiency disorder affecting populations consuming polished white rice as a dietary staple, individuals with chronic alcoholism, hyperemesis gravidarum, after bariatric surgery, and in prolonged parenteral nutrition without thiamine supplementation. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

861.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of thiamine deficiency
- Thiamine functions as a cofactor for several enzymes involved in carbohydrate metabolism, including transketolase, pyruvate dehydrogenase, and α-ketoglutarate dehydrogenase.

861.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

861.4 Signs and Symptoms

You may experience two major syndromes: dry beriberi (symmetric peripheral neuropathy with stocking-glove sensory loss, muscle weakness, absent deep tendon reflexes, paresthesias), wet beriberi (high-output heart failure with dilated cardiomyopathy, peripheral vasodilation, swelling, tachycardia, warm extremities, bounding pulses, lactic acidosis), and Wernicke encephalopathy (acute neuropsychiatric emergency with confusion, loss of coordination, and ophthalmoplegia) which may progress to Korsakoff syndrome (irreversible amnestic disorder with anterograde and retrograde amnesia and confabulation).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

861.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

861.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

861.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Whole blood thiamine or erythrocyte transketolase activity may be measured but treatment should not be delayed.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

861.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

861.9 Treatment Options

- Treatment focuses on thiamine 100–500 mg IV/IM daily for 5–7 days followed by oral maintenance
- Wernicke encephalopathy requires immediate high-dose parenteral thiamine before glucose administration.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

861.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

861.11 Outlook

- The outlook is generally positive for beriberi with treatment
- Wernicke encephalopathy is reversible with prompt treatment, but Korsakoff syndrome is irreversible.

Chapter 862

Vitamin B12 (Cobalamin) Deficiency

862.1 Overview

Vitamin B12 (cobalamin) deficiency is a vitamin deficiency disorder common in the elderly due to atrophic gastritis and declining intrinsic factor production (food-cobalamin malabsorption). Pernicious anemia is an autoimmune disorder with antibodies against gastric parietal cells and/or intrinsic factor, leading to profound B12 deficiency. Other causes include gastric or ileal removal, Crohn disease, bacterial overgrowth, fish tapeworm (Diphyllobothrium latum) infection, strict vegetarian/vegan diet, and metformin use. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

862.2 Causes

This condition happens because of deficiency of vitamin B12, which is essential for DNA synthesis, myelin formation, and homocysteine metabolism. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

862.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

862.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

862.5 Progression and Stages

You may experience megaloblastic anemia and the hallmark distinguishing feature from folate deficiency: nervous system involvement (subacute combined deterioration of the spinal cord manifesting as symmetric paresthesias, loss of vibration and proprioception, loss of coordination, spastic paraparesis, and a positive Romberg sign), peripheral neuropathy, visual disturbances (optic shrinkage or wasting), memory loss, and psychiatric symptoms. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

862.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

862.7 Diagnosis

- Doctors diagnose this using low serum B12 (less than 200 pg/mL) with elevated serum methylmalonic acid (MMA) and homocysteine
- Elevated MMA is the most sensitive indicator.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

862.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

862.9 Treatment Options

- Treatment focuses on oral B12 1000–2000 µg daily for deficiency without nervous system involvement
- For pernicious anemia, nervous system symptoms, or severe deficiency, parenteral B12 (hydroxocobalamin or cyanocobalamin) 1000 µg IM daily for 1 week, weekly for 4 weeks, then monthly for life.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

862.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

862.11 Outlook

- Most people recover well with treatment
- Hematologic response is rapid (reticulocytosis within 3–5 days, hemoglobin normalization in 6–8 weeks).

Chapter 863

Vitamin B12 Deficiency Anemia

863.1 Overview

Vitamin B12 deficiency anemia is a megaloblastic anemia that can cause irreversible neurological damage. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

863.2 Causes

This condition happens because of vitamin B12 (cobalamin) deficiency, with causes including pernicious anemia (autoimmune destruction of gastric parietal cells producing intrinsic factor), malabsorption (gastrectomy, ileal removal, Crohn's disease, bacterial overgrowth, celiac disease), dietary deficiency (strict vegans), and medications (metformin, proton pump inhibitors). Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

863.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

863.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

863.5 Progression and Stages

You may experience fatigue, weakness, jaundice (mild, from ineffective erythropoiesis), glossitis (beefy red tongue), and neurological manifestations (subacute combined deterioration of the spinal cord—demyelination of dorsal columns and lateral corticospinal tracts, peripheral neuropathy, thinking and memory impairment, depression). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

863.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

863.7 Diagnosis

Doctors diagnose this using macrocytic anemia (MCV >100 fL), hypersegmented neutrophils on peripheral smear, low serum B12, elevated methylmalonic acid and homocysteine, and elevated LDH.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

863.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

863.9 Treatment Options

Treatment focuses on intramuscular cyanocobalamin or hydroxocobalamin injections, initially daily for one week, then monthly lifelong for pernicious anemia. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

863.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

863.11 Outlook

The outlook is good with treatment, though neurological damage may be partially irreversible if prolonged.

Chapter 864

Vitamin B2 (Riboflavin) Deficiency — Ariboflavinosis

864.1 Overview

Riboflavin (vitamin B2) deficiency (ariboflavinosis) is a vitamin deficiency disorder that is rare in developed countries but occurs in alcoholism, malabsorption syndromes, and populations with limited dairy and meat intake. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

864.2 Causes

This condition happens because of deficiency of riboflavin, a precursor for flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD), cofactors for numerous redox enzymes. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

864.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

864.4 Signs and Symptoms

You may experience angular stomatitis (cheilosis), glossitis (smooth, magenta-colored tongue), seborrheic dermatitis (nasolabial folds, scrotum, vulva), photophobia, corneal vascularization, and normocytic anemia.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

864.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

864.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

864.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam and confirmed by low erythrocyte glutathione reductase activity (EGRAC).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

864.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

864.9 Treatment Options

Treatment focuses on riboflavin 5–10 mg orally daily. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

864.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

864.11 Outlook

- Most people recover well
- Mucocutaneous lesions resolve promptly with treatment.

Chapter 865

Vitamin B3 (Niacin) Deficiency — Pellagra

865.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Niacin (vitamin B3) deficiency (pellagra) is a vitamin deficiency disorder classically associated with diets based on maize (corn), which contains niacin in a bound, non-bioavailable form unless treated with alkali (nixtamalization)
- It also occurs in alcoholism, carcinoid syndrome, and with isoniazid therapy.

865.2 Causes

This condition happens because of deficiency of niacin, a precursor for nicotinamide adenine dinucleotide (NAD) and NADP, essential for redox reactions and cellular energy metabolism. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

865.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

865.4 Signs and Symptoms

You may experience the classic triad of dermatitis, dementia, and diarrhea (the "three D's")

- Dermatitis is photosensitive, symmetric, with well-demarcated, erythematous, hyperpigmented, scaling plaques in sun-exposed areas (Casal necklace)
- Digestive tract involvement includes diarrhea, glossitis, and stomatitis
- Neuropsychiatric manifestations include depression, apathy, confusion, memory loss, and psychosis.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

865.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

865.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

865.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

865.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

865.9 Treatment Options

Treatment focuses on nicotinamide (preferred over niacin to avoid flushing) 100–500 mg orally daily. Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

865.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

865.11 Outlook

Most people recover well with treatment

- Dermatitis and digestive tract symptoms improve within days
- Nervous system symptoms resolve more slowly.

Chapter 866

Vitamin B6 (Pyridoxine) Deficiency

866.1 Overview

Pyridoxine (vitamin B6) deficiency is a vitamin deficiency disorder occurring in alcoholism, chronic kidney failure, autoimmune disorders (rheumatoid arthritis), and with medications such as isoniazid, cycloserine, hydralazine, and oral contraceptives. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

866.2 Causes

This condition happens because of deficiency of pyridoxine, which functions as a cofactor (pyridoxal phosphate, PLP) for over 100 enzymes involved in amino acid metabolism, neurotransmitter synthesis, and heme biosynthesis. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

866.3 Risk Factors

- You may experience microcytic hypochromic anemia (sideroblastic anemia with ringed sideroblasts in bone marrow), seborrheic dermatitis, glossitis, cheilosis, peripheral neuropathy, depression, confusion, and in infants, irritability and seizures
- Homocysteine elevation due to B6 deficiency is a cardiovascular risk factor.
- Family history
- Obesity and sedentary lifestyle

- Poor dietary habits
- Advancing age
- Hormonal imbalances
- Certain medications
- Genetic predisposition and family history
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

866.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

866.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

866.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

866.7 Diagnosis

Doctors diagnose this using low plasma PLP levels.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

866.8 Prevention

- Treatment focuses on pyridoxine 50–100 mg orally daily
- Isoniazid-induced deficiency requires 50–100 mg of pyridoxine daily as prophylaxis.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

866.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

866.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

866.11 Outlook

Most people recover well with treatment.

Chapter 867

Vitamin B7 (Biotin) Deficiency

867.1 Overview

Biotin (vitamin B7) deficiency is a rare vitamin deficiency disorder because biotin is widely distributed in foods and synthesized by gut microbiota. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

867.2 Causes

Causes include high consumption of raw egg whites (avidin binds biotin, preventing absorption), parenteral nutrition without biotin, prolonged antibiotic use, and biotinidase deficiency (an inborn error). This condition happens because of deficiency of biotin, a cofactor for carboxylase enzymes involved in gluconeogenesis, fatty acid synthesis, and amino acid catabolism. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

867.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

867.4 Signs and Symptoms

- You may experience periorificial dermatitis, alopecia, brittle nails, glossitis, conjunctivitis, nervous system symptoms (depression, lethargy, hallucinations, paresthesias), and metabolic acidosis
- In biotinidase deficiency, The condition usually appears as in infancy with seizures, hypotonia, loss of coordination, hearing loss, and developmental delay.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

867.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

867.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

867.7 Diagnosis

Doctors diagnose this using low serum biotin or elevated urinary 3-hydroxyisovaleric acid.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

867.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

867.9 Treatment Options

Treatment focuses on biotin 5–10 mg orally daily (or 10–40 mg daily in biotinidase deficiency). Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

867.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

867.11 Outlook

Most people recover well with treatment.

Chapter 868

Vitamin B9 (Folate) Deficiency

868.1 Overview

Folate (vitamin B9) deficiency is one of the most common vitamin deficiencies worldwide. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

868.2 Causes

Causes include inadequate dietary intake (especially in the elderly, alcoholics, and those with poor vegetable intake), malabsorption (celiac disease, tropical sprue, inflammatory bowel disease), increased requirements (pregnancy, lactation, hemolytic anemias, malignancy), and medications (methotrexate, trimethoprim, sulfasalazine, phenytoin). This condition happens because of deficiency of folate, a water-soluble B vitamin essential for one-carbon transfer reactions, DNA synthesis, purine and pyrimidine synthesis, and homocysteine remethylation. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

868.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

868.4 Signs and Symptoms

- Clinical features overlap with vitamin B12 deficiency and include megaloblastic anemia (macrocytic RBCs with oval macrocytes, hypersegmented neutrophils, pancytopenia), glossitis (smooth, beefy red tongue), diarrhea, and infertility
- Unlike B12 deficiency, folate deficiency does NOT cause nervous system manifestations.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

868.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

868.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

868.7 Diagnosis

Your doctor confirms the diagnosis with serum folate less than 2 ng/mL and elevated homocysteine with normal methylmalonic acid (key to distinguish from B12 deficiency).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

868.8 Prevention

- Most people recover well with treatment
- Before treating macrocytic anemia, vitamin B12 deficiency must be ruled out or treated concurrently to prevent precipitating subacute combined deterioration.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

868.9 Treatment Options

- Treatment focuses on folic acid 1–5 mg orally daily
- Periconceptional folic acid supplementation (400 µg/day) prevents neural tube defects.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

868.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 869

Vitamin C Deficiency — Scurvy

869.1 Overview

Ascorbic acid (vitamin C) deficiency (scurvy) is a vitamin deficiency disorder that is rare in developed countries but occurs in individuals with severely restricted diets (alcoholics, elderly, psychiatric patients, food faddism), malabsorption, and in infants fed only cow's milk without supplementation. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

869.2 Causes

This condition happens because of deficiency of ascorbic acid, which is essential for collagen synthesis (hydroxylation of proline and lysine in procollagen), carnitine synthesis, neurotransmitter synthesis, and as an antioxidant. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

869.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

869.4 Signs and Symptoms

- Clinical features reflect impaired collagen synthesis: perifollicular hemorrhages, tiny red or purple spots, ecchymoses, gingival abnormal increase in cells with bleeding, impaired wound healing, hyperkeratosis, coiled hairs (corkscrew hairs), and leg swelling
- Musculoskeletal manifestations include arthralgias, myalgias, and subperiosteal hemorrhages (particularly painful in infants presenting with pseudoparalysis)
- Systemic symptoms include fatigue, irritability, depression, and anemia.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

869.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

869.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

869.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Serum ascorbic acid less than 0.2 mg/dL confirms deficiency.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

869.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

869.9 Treatment Options

- Treatment focuses on vitamin C 100–200 mg orally three times daily for 1 week, then 100 mg daily
- Infantile scurvy responds to 50 mg daily.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

869.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

869.11 Outlook

- Most people recover well
- Clinical improvement occurs within days.

Chapter 870

Vitamin D Deficiency — Rickets and Osteomalacia

870.1 Overview

Vitamin D deficiency is a vitamin deficiency disorder resulting from inadequate sunlight exposure, low dietary intake (or lack of fortified foods), malabsorption (celiac disease, cystic scarring, short bowel syndrome), kidney disease (impaired 1α-hydroxylation), obesity (sequestration in adipose tissue), and medications (antiepileptics, rifampin, glucocorticoids). This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

870.2 Causes

This condition happens because of deficiency of vitamin D, a fat-soluble secosteroid essential for intestinal calcium absorption, bone mineralization, and calcium homeostasis. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

870.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

870.4 Signs and Symptoms

Clinical features in children include rickets (defective mineralization of growing bone) with craniotabes, frontal bossing, delayed fontanelle closure, rachitic rosary, Harrison groove, bowing of the legs (genu varum), widened wrists and ankles, myopathy, and hypocalcemic tetany. In adults, osteomalacia presents with bone pain, proximal muscle weakness (waddling gait), and fragility fractures.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

870.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

870.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

870.7 Diagnosis

Doctors diagnose this using serum 25-hydroxyvitamin D less than 20 ng/mL (50 nmol/L), elevated alkaline phosphatase, low-normal calcium, and low phosphate

- Radiographs of rickets show cupping, fraying, and widening of the metaphysis
- In osteomalacia, Looser zones (pseudofractures) are seen.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

870.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

870.9 Treatment Options

- Treatment focuses on vitamin D 2000–5000 IU daily for 12 weeks or 50,000 IU weekly for 8 weeks, followed by maintenance 800–2000 IU daily
- Correction of calcium and phosphate is essential.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

870.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

870.11 Outlook

Most people recover well with treatment.

Chapter 871

Vitamin E Deficiency

871.1 Overview

Vitamin E (α-tocopherol) deficiency is a rare vitamin deficiency disorder occurring primarily in fat malabsorption syndromes (cystic scarring, cholestatic liver disease, abetalipoproteinemia, short bowel syndrome) and in genetic disorders of α-tocopherol transfer protein (loss of coordination with isolated vitamin E deficiency, AVED). This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

871.2 Causes

This condition happens because of deficiency of vitamin E, a fat-soluble antioxidant that protects cell membranes from lipid peroxidation. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

871.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

871.4 Signs and Symptoms

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

871.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

- You may experience progressive nervous system dysfunction: spinocerebellar deterioration with loss of coordination, areflexia, loss of vibration and proprioception, ophthalmoplegia, and pigmentary retinopathy
- Hemolytic anemia may occur in premature infants.

871.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

871.7 Diagnosis

Doctors diagnose this using low plasma α-tocopherol corrected for serum lipids.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

871.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

871.9 Treatment Options

Treatment focuses on α-tocopherol 800–1200 IU daily orally (or higher doses in severe malabsorption). Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

871.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

871.11 Outlook

- The outlook varies from person to person
- Nervous system deficits may stabilize but are not fully reversible.

Chapter 872

Vitamin K Deficiency

872.1 Overview

Vitamin K deficiency is a vitamin deficiency disorder leading to impaired liver synthesis of functional clotting factors. This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

872.2 Causes

Causes include inadequate dietary intake (rare in healthy adults), fat malabsorption, liver disease, antibiotic therapy (depleting gut flora), and parenteral nutrition. This condition happens because of deficiency of vitamin K, a fat-soluble vitamin required for the γ-carboxylation of glutamic acid residues on coagulation factors II (prothrombin), VII, IX, and X, as well as proteins C and S and bone matrix proteins. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

872.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions

- Medication use

872.4 Signs and Symptoms

- You may experience vitamin K deficiency bleeding (VKDB) in neonates presenting with bruising, digestive tract bleeding, umbilical bleeding, and life-threatening intracranial bleeding
- In adults, deficiency presents with easy bruising, epistaxis, hematuria, and prolonged prothrombin time (PT) with normal PTT.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

872.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

872.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

872.7 Diagnosis

Doctors diagnose this using prolonged PT (factor VII has the shortest half-life) that corrects with normal plasma, low factor II, VII, IX, X activity, and normal platelet count and fibrinogen.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

872.8 Prevention

Treatment focuses on vitamin K1 (phylloquinone) 1–10 mg orally or 1–2 mg subcutaneously/IM for adults

- VKDB prophylaxis includes 1 mg IM at birth
- Active bleeding requires fresh frozen plasma (FFP) or prothrombin complex concentrates (PCC) with parenteral vitamin K.
- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

872.9 Treatment Options

Dietary modifications and nutritional counseling Pharmacotherapy to correct metabolic abnormalities Hormone replacement therapy when indicated Regular monitoring of metabolic parameters Weight management programs Exercise prescription Management of associated conditions Patient education for self-management

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

872.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

872.11 Outlook

Most people recover well with treatment.

Chapter 873

Vitiligo

873.1 Overview

Vitiligo is an acquired autoimmune disorder characterized by selective destruction of melanocytes, resulting in well-demarcated, milky-white macules and patches, affecting 0.5–2% of the global population. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

873.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of autoimmune-mediated melanocyte destruction. Loss of pigment may occur in any area but commonly affects the face, hands, elbows, knees, and genitalia. Vitiligo is linked to other autoimmune diseases (thyroid disease, type 1 diabetes, Addison's disease, pernicious anemia)
- The Koebner phenomenon (new lesions at sites of trauma) may occur.

873.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

873.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

873.5 Progression and Stages

It may be segmental (unilateral, dermatomal, often with early onset and stable course) or non-segmental (generalized, bilateral, with variable progression). The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

873.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

873.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam, supported by Wood's lamp examination. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

873.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

873.9 Treatment Options

- Treatment options include topical corticosteroids, topical calcineurin inhibitors (tacrolimus, especially for facial lesions), narrowband UVB phototherapy, and for refractory localized disease, surgical melanocyte transplantation
- Ruxolitinib cream (JAK1/2 inhibitor) has been approved for non-segmental vitiligo. Sun protection is essential.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

873.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

873.11 Outlook

- The outlook varies from person to person
- Repigmentation is achievable but recurrence may occur.

Chapter 874

Vocal Cord Nodules

874.1 Overview

Vocal cord nodules are bilateral, symmetric, callus-like thickenings of the true vocal folds at the junction of the anterior and middle thirds, the point of maximal vibratory impact, and are the most common cause of chronic hoarseness in adults. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

874.2 Causes

This condition happens because of chronic vocal abuse leading to repetitive trauma of the vocal folds. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

874.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

874.4 Signs and Symptoms

You may experience hoarseness, breathy voice, vocal fatigue, and decreased vocal range. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

874.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

874.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

874.7 Diagnosis

Doctors diagnose this using laryngoscopy or videostroboscopy showing smooth, bilateral masses. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

874.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

874.9 Treatment Options

Treatment focuses on voice therapy as conservative management, with surgical removal reserved for refractory cases. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

874.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

874.11 Outlook

Most people recover well with appropriate voice therapy. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 875

Vocal Cord Paralysis

875.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Vocal cord paralysis is a condition resulting from disruption of the recurrent laryngeal nerve, superior laryngeal nerve, or vagus nerve, with the most common cause being iatrogenic (thyroid surgery, thoracic and neck surgery)
- Other causes include tumors, stroke, viral infections, and idiopathic cause.

875.2 Causes

This condition happens because of nerve injury leading to loss of vocal fold movement. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

875.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

875.4 Signs and Symptoms

- Symptoms of unilateral paralysis include hoarseness, removal by suction, and a breathy voice
- Bilateral paralysis presents with stridor and respiratory distress.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

875.5 Progression and Stages

Your outlook depends on the underlying cause and severity. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

875.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

875.7 Diagnosis

Doctors diagnose this using flexible laryngoscopy showing an immobile vocal cord. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

875.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

875.9 Treatment Options

Treatment options include voice therapy, injection laryngoplasty, medialization thyroplasty for unilateral paralysis, and tracheostomy or posterior cordotomy for bilateral paralysis with airway compromise. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

875.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

875.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 876

Von Willebrand Disease

876.1 Overview

Von Willebrand disease is the most common inherited bleeding disorder, affecting up to 1% of the population. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

876.2 Causes

This condition happens because of quantitative or qualitative defects in von Willebrand factor (VWF), a large multimeric glycoprotein essential for primary hemostasis (platelet adhesion to injured vessel walls) and carrier/stabilizer of factor VIII, with type 1 (partial quantitative deficiency, 70–80%) being the most common and mildest, type 2 (qualitative deficiency) having various functional abnormalities, and type 3 (complete deficiency) being the most severe. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

876.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

876.4 Signs and Symptoms

You may experience mucocutaneous bleeding: epistaxis, menorrhagia, easy bruising, prolonged bleeding after dental procedures or surgery, and GI bleeding. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

876.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

876.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

876.7 Diagnosis

Doctors diagnose this using decreased VWF antigen, decreased VWF activity (ristocetin cofactor assay), and decreased or normal factor VIII. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

876.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

876.9 Treatment Options

Treatment options include desmopressin (DDAVP, for type 1 and some type 2A patients), VWF-containing factor VIII concentrates (for type 3 or DDAVP non-responders), and antifibrinolytics (tranexamic acid, for mucosal bleeding). Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

876.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

876.11 Outlook

In most cases, the outlook is good with appropriate management. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 877

Vulvar Cancer

Early recognition and biopsy of suspicious vulvar lesions (ulceration, pigmentation changes, persistent itching or pain) are critical for diagnosis at an early stage.

877.1 Overview

This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

877.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

877.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

877.4 Signs and Symptoms

Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

877.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

877.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

877.7 Diagnosis

Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

877.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin

from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

877.9 Treatment Options

Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

877.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team

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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

877.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 878

Vulvar Cancer

878.1 Overview

Vulvar cancer is a rare malignancy accounting for approximately 5% of gynecologic cancers, typically affecting postmenopausal women. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

878.2 Causes

This condition happens because of cancerous transformation of vulvar epithelium. Cancer develops through a multistep process involving genetic mutations that drive uncontrolled cell growth and proliferation. These mutations may be inherited or acquired through exposure to carcinogens, radiation, or random errors during cell division. Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

878.3 Risk Factors

Squamous cell carcinoma (80–90%) is the most common histological type, with risk factors including HPV infection (particularly in younger women), lichen sclerosus, vulvar intraepithelial neoplasia (VIN), smoking, and immunosuppression.

- Advancing age
- Family history of cancer
- Tobacco use and alcohol consumption

- Exposure to carcinogens (radiation, chemicals)
- Obesity and sedentary lifestyle
- Chronic infections (HPV, hepatitis B/C)
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption
- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

878.4 Signs and Symptoms

You may experience a vulvar lump, itching, pain, bleeding, and ulceration. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

878.5 Progression and Stages

outlook is related to stage and lymph node status. Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

878.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
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- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

878.7 Diagnosis

To make the diagnosis, doctors look for biopsy of any suspicious lesion. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

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- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)

- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

878.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

878.9 Treatment Options

Treatment typically involves wide local removal with adequate margins, with inguinal lymph node dissection for invasive tumors >1 mm. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support

- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

878.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
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- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

878.11 Outlook

Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 879

Vulvovaginal Candidiasis

879.1 Overview

Vulvovaginal candidiasis (VVC) is a fungal infection caused by *Candida* species, with *Candida albicans* responsible for 80–90% of cases. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

879.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

879.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

879.4 Signs and Symptoms

You may experience intense vulvar itching, burning, vaginal soreness, dyspareunia, thick white "cottage cheese" discharge, vulvar redness of the skin, fissures, and swelling. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

879.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

879.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

879.7 Diagnosis

Doctors diagnose this using saline and 10% KOH wet mount showing pseudohyphae and budding yeast, or by culture. Treatment of uncomplicated VVC: single oral dose of fluconazole 150 mg or topical azoles. Complicated VVC requires extended therapy. Recurrent VVC ($\geq$4 episodes/year) requires maintenance therapy with fluconazole 150 mg weekly for 6 months. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

879.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise

- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

879.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

879.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

879.11 Outlook

The outlook is generally positive with treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 880

Warts

880.1 Overview

Warts (verrucae) are non-cancerous epithelial proliferations caused by human papillomavirus (HPV) infection, with over 100 HPV types identified. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

880.2 Causes

This condition happens because of HPV infection of keratinocytes causing hyperkeratosis. Common warts (verruca vulgaris) present as rough, hyperkeratotic papules, typically on the hands and fingers. Plantar warts (verruca plantaris) occur on the soles of the feet, growing inward due to pressure, causing pain with walking. Flat warts (verruca plana) are smooth, flat-topped papules on the face and dorsum of hands. Genital warts (condylomata acuminata) are caused by HPV types 6 and 11. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

880.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

880.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

880.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

880.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

880.7 Diagnosis

Doctors usually diagnose this based on your symptoms and a physical exam. Management options include salicylic acid (first-line for common and plantar warts), freezing treatment, cantharidin, imiquimod (for genital warts), and surgical destruction for refractory lesions. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

880.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

880.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

880.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

880.11 Outlook

- Most people recover well
- Many warts resolve spontaneously within 1–2 years due to cell-mediated immunity.

Chapter 881

West Nile Virus

881.1 Overview

West Nile virus is a mosquito-borne flavivirus, first isolated in Uganda (1937) and introduced to the United States in 1999, transmitted by *Culex* species mosquitoes with birds as the primary amplifying host, with 2000–5000 cases reported annually in the US. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

881.2 Causes

This condition happens because of the virus being inoculated by mosquito bite, replicating in dendritic cells, and spreading to lymph nodes and blood, crossing the blood-brain barrier and infecting neurons, causing apoptosis and perivascular inflammation. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

881.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

881.4 Signs and Symptoms

You may experience an incubation of 2–14 days, with 80% being asymptomatic, 20% developing West Nile fever (acute fever, headache, malaise, muscle pain, and maculopapular rash), and less than 1% developing neuroinvasive disease (meningitis, encephalitis, and acute flaccid paralysis). Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

881.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

881.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

881.7 Diagnosis

Doctors diagnose this using WNV-specific IgM in CSF or serum. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

881.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

881.9 Treatment Options

Treatment typically involves supportive. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

881.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

881.11 Outlook

- Most people recover well for non-neuroinvasive disease
- Neuroinvasive disease has significant morbidity and mortality.

Chapter 882

WHO Analgesic Ladder

882.1 Overview

The WHO analgesic ladder, originally developed for cancer pain, is a stepwise framework for analgesic selection: This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- **Step 1:** Non-opioid analgesics (acetaminophen, NSAIDs), with or without adjuvant analgesics, for mild pain (pain score 1–3/10).
- **Step 2:** Weak opioids (codeine, tramadol) or low-dose strong opioids plus non-opioids and adjuvants for moderate pain (pain score 4–6/10).
- **Step 3:** Strong opioids (morphine, oxycodone, hydromorphone, fentanyl, methadone) plus non-opioids and adjuvants for severe pain (pain score 7–10/10).

882.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

882.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

882.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
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- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

882.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

882.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

882.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

882.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors

- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

882.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

882.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

882.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 883

Williams Syndrome

883.1 Overview

Williams syndrome (Williams-Beuren syndrome) is a neurodevelopmental disorder caused by a microdeletion on chromosome 7q11.23, encompassing the ELN (elastin) gene. Incidence is 1 in 7500–10,000. This genetic disorder results from specific gene mutations or chromosomal abnormalities. It may be present at birth or manifest later in life depending on the underlying mechanism. Genetic disorders result from abnormalities in an individual's DNA, ranging from single-gene mutations to chromosomal abnormalities. These conditions may be inherited from parents or arise from new mutations. Pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. Genetic counseling helps families understand risks and make informed decisions. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

883.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

883.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

883.4 Signs and Symptoms

You may experience distinctive facial appearance (elfin facies), supravalvular aortic stenosis, peripheral lung stenosis, hypercalcemia in infancy, hyperacusis, and a characteristic thinking and memory profile (strong verbal skills, friendly extroverted personality, but poor visuospatial skills).

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

883.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

883.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

883.7 Diagnosis

Doctors diagnose this using FISH or chromosomal microarray for the 7q11.23 deletion.

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

883.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

883.9 Treatment Options

Treatment options include cardiac monitoring, dietary calcium restriction during infancy, and educational and behavioral support. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

883.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

Chapter 884

Wilms Tumor (Nephroblastoma)

884.1 Overview

Wilms tumor is the most common kidney malignancy in children, with an incidence of 1 in 10,000 children and median age at To diagnose 3.5 years. It is linked to genetic syndromes such as WAGR complex, Denys-Drash syndrome, and Beckwith-Wiedemann syndrome. This condition accounts for a significant proportion of cancer-related morbidity and mortality worldwide. Early detection through routine screening and advances in treatment have improved survival rates in recent decades. This type of cancer arises from uncontrolled cell growth in affected tissues, with potential to invade nearby structures and spread to distant sites through the bloodstream or lymphatic system. The incidence varies by geographic region, age, and genetic background. Screening programs have improved early detection rates in many populations. Histological classification and molecular subtyping guide treatment decisions and provide prognostic information. Multidisciplinary care involving surgeons, medical oncologists, radiation oncologists, and pathologists is standard for optimal management.

884.2 Causes

Cancer develops through accumulation of genetic and epigenetic alterations that drive uncontrolled cell proliferation, evade growth suppression, resist cell death, and enable replicative immortality. Key molecular pathways involved include tumor suppressor genes (p53, RB, APC), oncogenes (KRAS, MYC, EGFR), and DNA repair mechanisms. Environmental carcinogens such as tobacco smoke, ultraviolet radiation, certain chemicals, and ionizing radiation can induce DNA damage. Chronic inflammation and persistent infections (HPV, hepatitis B/C, H. pylori) contribute to cancer development in some sites.

884.3 Risk Factors

Advancing age Tobacco use (active and secondhand smoke) Excessive alcohol consumption Family history of cancer and known genetic syndromes Obesity and physical inactivity Unhealthy diet (processed meats, low fiber) Exposure to carcinogens (asbestos, benzene, radiation) Chronic infections (HPV, hepatitis B/C, HIV) Immunosuppression Hormonal factors (early menarche, late menopause)

- Advancing age
- Tobacco use (active and secondhand smoke)
- Excessive alcohol consumption

- Family history of cancer and known genetic syndromes
- Obesity and physical inactivity
- Unhealthy diet (processed meats, low fiber)
- Exposure to carcinogens (asbestos, benzene, radiation)
- Chronic infections (HPV, hepatitis B/C, HIV)
- Immunosuppression
- Hormonal factors (early menarche, late menopause)

884.4 Signs and Symptoms

You may experience a palpable abdominal mass, abdominal pain, hematuria (15–20%), and high blood pressure. Persistent lump or mass in the affected area Unexplained weight loss and loss of appetite Chronic fatigue and weakness Persistent pain that worsens over time Changes in bowel or bladder habits Unexplained fever or night sweats Unusual bleeding or discharge Persistent cough or hoarseness Changes in skin appearance (new mole, changing wart) Difficulty swallowing or persistent indigestion

- Persistent lump or mass in the affected area
- Unexplained weight loss and loss of appetite
- Chronic fatigue and weakness
- Persistent pain that worsens over time
- Changes in bowel or bladder habits
- Unexplained fever or night sweats
- Unusual bleeding or discharge
- Persistent cough or hoarseness
- Changes in skin appearance (new mole, changing wart)
- Difficulty swallowing or persistent indigestion

884.5 Progression and Stages

Cancer staging follows the TNM system: Tumor (size and extent), Node (lymph node involvement), and Metastasis (spread to distant sites). Stage 0: Carcinoma in situ (abnormal cells confined to the original location) Stage I: Early-stage, small tumor confined to the organ of origin Stage II: Locally advanced tumor with possible regional lymph node involvement Stage III: More extensive local disease with significant lymph node involvement Stage IV: Advanced disease with distant metastases to other organs Higher stages generally indicate more advanced disease and poorer prognosis. Stage 0: Carcinoma in situ (abnormal cells confined to the original location). Stage I: Early-stage, small tumor confined to the organ of origin. Stage II: Locally advanced tumor with possible regional lymph node involvement. Stage III: More extensive local disease with significant lymph node involvement. Stage IV: Advanced disease with distant metastases to other organs. Higher stages generally indicate more advanced disease and poorer prognosis.

884.6 Complications

Metastatic spread to vital organs (brain, liver, lungs, bones) Malignant effusions (pleural, peritoneal, pericardial) Superior vena cava syndrome (chest tumors) Spinal cord compression (spinal metastases) Pathological fractures (bone metastases) Tumor lysis syndrome (rapid cell death during treatment) Paraneoplastic syndromes (hormonal, neurologic, hematologic) Treatment-related complications (chemotherapy toxicity, radiation damage) Infections due to immunosuppression Venous thromboembolism

- Metastatic spread to vital organs (brain, liver, lungs, bones)
- Malignant effusions (pleural, peritoneal, pericardial)
- Superior vena cava syndrome (chest tumors)
- Spinal cord compression (spinal metastases)
- Pathological fractures (bone metastases)
- Tumor lysis syndrome (rapid cell death during treatment)
- Paraneoplastic syndromes (hormonal, neurologic, hematologic)
- Treatment-related complications (chemotherapy toxicity, radiation damage)
- Infections due to immunosuppression
- Venous thromboembolism

884.7 Diagnosis

Doctors diagnose this using abdominal ultrasound and CT scan, which shows an intrarenal mass with heterogeneous density. Diagnosis typically involves imaging studies (CT, MRI, PET scans) to identify the location and extent of disease. Biopsy with histopathological examination remains the gold standard for confirming the diagnosis and determining the specific type and grade. Physical examination including palpation of mass and lymph node assessment Complete blood count and comprehensive metabolic panel Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP) Imaging: Ultrasound, CT scan, MRI, PET-CT for staging Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology Immunohistochemistry to determine tumor subtype and receptor status Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status) Sentinel lymph node biopsy for surgical staging Bone marrow biopsy for hematologic malignancies Genetic counseling and testing for hereditary cancer syndromes

- Physical examination including palpation of mass and lymph node assessment
- Complete blood count and comprehensive metabolic panel
- Tumor markers specific to cancer type (e.g., PSA, CA-125, CEA, AFP)
- Imaging: Ultrasound, CT scan, MRI, PET-CT for staging
- Biopsy: Core needle, excisional, or endoscopic biopsy for histopathology
- Immunohistochemistry to determine tumor subtype and receptor status
- Molecular and genetic testing (EGFR, KRAS, BRCA, MSI status)
- Sentinel lymph node biopsy for surgical staging
- Bone marrow biopsy for hematologic malignancies
- Genetic counseling and testing for hereditary cancer syndromes

884.8 Prevention

Avoid tobacco use and limit alcohol consumption Maintain a healthy weight through diet and exercise Eat a balanced diet rich in fruits, vegetables, and whole grains Protect skin from excessive sun exposure and avoid tanning beds Get vaccinated against cancer-causing infections (HPV, hepatitis B) Participate in recommended cancer screening programs Know your family history and consider genetic testing if indicated Avoid exposure to known carcinogens in the workplace and environment

- Avoid tobacco use and limit alcohol consumption
- Maintain a healthy weight through diet and exercise
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Protect skin from excessive sun exposure and avoid tanning beds
- Get vaccinated against cancer-causing infections (HPV, hepatitis B)
- Participate in recommended cancer screening programs
- Know your family history and consider genetic testing if indicated
- Avoid exposure to known carcinogens in the workplace and environment

884.9 Treatment Options

Treatment usually involves a multidisciplinary approach: neoadjuvant chemotherapy followed by nephrectomy or primary nephrectomy followed by chemotherapy. Treatment planning is multidisciplinary and depends on the cancer type, stage, and patient factors. Management may include surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy), or a combination of these modalities. Surgery: Wide local excision with clear margins, lymph node dissection Radiation therapy: External beam or brachytherapy for localized disease Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells Targeted therapy: Drugs targeting specific molecular alterations Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines Hormone therapy: For hormone-sensitive cancers (breast, prostate) Combination approaches: Concurrent chemoradiation, sequential therapy Palliative care: Symptom management, pain control, quality of life support Clinical trials: Access to experimental therapies Surveillance: Active monitoring after definitive treatment

- Surgery: Wide local excision with clear margins, lymph node dissection
- Radiation therapy: External beam or brachytherapy for localized disease
- Chemotherapy: Systemic cytotoxic drugs targeting rapidly dividing cells
- Targeted therapy: Drugs targeting specific molecular alterations
- Immunotherapy: Checkpoint inhibitors, CAR-T cells, cancer vaccines
- Hormone therapy: For hormone-sensitive cancers (breast, prostate)
- Combination approaches: Concurrent chemoradiation, sequential therapy
- Palliative care: Symptom management, pain control, quality of life support
- Clinical trials: Access to experimental therapies
- Surveillance: Active monitoring after definitive treatment

884.10 Living with It

Regular follow-up appointments with your oncology team Manage treatment side effects with supportive medications Maintain adequate nutrition with help from a dietitian Stay physically active within your energy limits Join cancer support groups for emotional support and shared experiences Seek counseling or mental health support for anxiety and depression Communicate openly with your healthcare team about symptoms Plan for work and financial considerations during treatment Consider complementary therapies (acupuncture, massage, meditation) Build a strong support network of family and friends

- Regular follow-up appointments with your oncology team
- Manage treatment side effects with supportive medications
- Maintain adequate nutrition with help from a dietitian
- Stay physically active within your energy limits
- Join cancer support groups for emotional support and shared experiences
- Seek counseling or mental health support for anxiety and depression
- Communicate openly with your healthcare team about symptoms
- Plan for work and financial considerations during treatment
- Consider complementary therapies (acupuncture, massage, meditation)
- Build a strong support network of family and friends

884.11 Outlook

Most people recover well, with 5-year survival >90% for localized disease. Outcomes have improved significantly with advances in early detection and treatment. Prognosis depends on the cancer type, stage at diagnosis, molecular characteristics, and response to therapy. Prognosis depends on cancer type, stage at diagnosis, molecular features, and response to treatment. Early-stage cancers have significantly better outcomes, with many achieving long-term remission or cure. Survival rates have improved substantially with advances in screening, targeted therapy, and immunotherapy. Regular surveillance after treatment is essential for detecting recurrence early. Palliative care continues to advance, improving quality of life even for advanced disease.

Chapter 885

Wilson's Disease

885.1 Overview

Wilson's disease is an autosomal recessive disorder of copper metabolism caused by mutations in the ATP7B gene, with a prevalence of approximately 1 in 30,000. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

885.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

- This condition happens because of impaired biliary copper excretion leading to accumulation of copper in the liver, brain, cornea, and other organs. liver manifestations range from asymptomatic transaminase elevation to acute hepatitis, chronic hepatitis, or cirrhosis
- Neuropsychiatric features include tremor, slurred speech, dystonia, parkinsonism, personality changes, and depression. Kayser-Fleischer rings (copper deposition in Descemet's membrane of the cornea) are a hallmark finding.

885.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

885.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

885.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

885.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

885.7 Diagnosis

Diagnosis typically involves low serum ceruloplasmin (less than 20 mg/dL), elevated 24-hour urinary copper excretion, and liver copper quantification on liver biopsy. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

885.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

885.9 Treatment Options

- Treatment options include copper chelation (penicillamine, trientine), zinc therapy to block copper absorption, and liver transplantation in fulminant or decompensated disease
- Screening of first-degree relatives is essential.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

885.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

885.11 Outlook

Most people recover well with early diagnosis and lifelong treatment. The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 886

Wound Infections and Surgical Site Infections

886.1 Overview

Wound infections and surgical site infections (SSIs) are infections occurring within 30 days of surgery (or 1 year for implant procedures) at the incision site or in deep tissues, accounting for approximately 20% of healthcare-associated infections. This infection is transmitted through various routes depending on the specific pathogen. It remains a significant public health concern, particularly in regions with limited healthcare access. This condition results from acute injury or exposure to harmful agents. Prompt medical intervention is critical for optimal outcomes. Infectious diseases are caused by pathogenic microorganisms including bacteria, viruses, fungi, and parasites that invade the body and multiply, leading to tissue damage and immune response. Transmission occurs through various routes: respiratory droplets, direct contact, contaminated food or water, vectors, or blood and body fluids. The incubation period varies from hours to years depending on the pathogen and host factors. Antimicrobial resistance is an emerging concern that complicates treatment and increases morbidity.

886.2 Causes

Infection begins with pathogen entry through a susceptible portal (respiratory tract, gastrointestinal tract, skin, mucous membranes). The pathogen must overcome host defenses including physical barriers, innate immunity, and adaptive immune responses. Microbial virulence factors such as toxins, adhesins, and capsules enable the pathogen to establish infection and evade immune clearance. Host factors including age, nutritional status, immune competence, and comorbidities influence susceptibility and disease severity.

- This condition happens because of bacterial inoculation during or after surgery
- Most SSIs are caused by *S. aureus* (including MRSA) and coagulase-negative staphylococci.

886.3 Risk Factors

Risk factors include patient factors (diabetes, obesity, immunosuppression, smoking, malnutrition) and procedural factors (contamination level, operative duration, implant placement).

- Close contact with infected individuals
- Compromised immune system
- Travel to endemic areas
- Poor sanitation and hygiene practices
- Lack of vaccination
- Exposure to contaminated food or water
- Compromised immune system (HIV, immunosuppressive therapy, malnutrition)
- Extremes of age (very young or elderly)
- Travel to endemic regions
- Healthcare exposure (nosocomial infections)
- Occupational exposure (healthcare workers, laboratory personnel)
- Crowded living conditions

886.4 Signs and Symptoms

You may experience redness of the skin, warmth, swelling, pain, and purulent removal of fluid from the wound. Fever and chills Fatigue and malaise Localized pain and inflammation at infection site Swelling, redness, and warmth (local infection) Cough and respiratory symptoms (respiratory infections) Nausea, vomiting, and diarrhea (gastrointestinal infections) Headache and body aches Skin rash or lesions Enlarged lymph nodes Systemic symptoms in severe cases (hypotension, organ dysfunction)

- Fever and chills
- Fatigue and malaise
- Localized pain and inflammation at infection site
- Swelling, redness, and warmth (local infection)
- Cough and respiratory symptoms (respiratory infections)
- Nausea, vomiting, and diarrhea (gastrointestinal infections)
- Headache and body aches
- Skin rash or lesions
- Enlarged lymph nodes
- Systemic symptoms in severe cases (hypotension, organ dysfunction)

886.5 Progression and Stages

SSIs are classified as superficial incisional (involving skin and subcutaneous tissue), deep incisional (involving fascia and muscle), or organ/space (involving any part opened or manipulated during surgery). The incubation period is the time between pathogen exposure and onset of symptoms, during which the pathogen replicates within the host. The prodromal phase involves early, nonspecific symptoms such as fever, fatigue, and malaise before characteristic symptoms appear. During the acute phase, hallmark signs and symptoms of the specific infection develop fully. The convalescent phase involves gradual resolution of symptoms as the immune system clears the infection. Some infections may progress to chronic or latent stages if the pathogen is not fully eliminated.

886.6 Complications

- Sepsis and septic shock
- Organ-specific complications (pneumonia, meningitis, endocarditis)
- Abscess formation requiring drainage
- Chronic infection (HIV, hepatitis B/C, tuberculosis)
- Reactive arthritis or post-infectious syndromes
- Antimicrobial resistance complicating treatment
- Disseminated infection in immunocompromised hosts
- Multi-organ failure in severe cases

886.7 Diagnosis

- Doctors usually diagnose this based on your symptoms and a physical exam
- Wound cultures guide antibiotic therapy.
- Detailed history including exposure, travel, and vaccination status
- Physical examination focusing on the affected system
- Complete blood count with differential
- Microbiological culture of blood, sputum, urine, or wound
- PCR and nucleic acid amplification tests for rapid pathogen detection
- Antigen detection tests
- Serological testing for antibodies (IgM, IgG)
- Imaging studies (chest X-ray, CT, ultrasound)
- Gram stain and microscopy of clinical specimens
- Antimicrobial susceptibility testing to guide therapy

886.8 Prevention

- In most cases, the outlook is favorable with appropriate management
- Prevention strategies include appropriate antibiotic prophylaxis, skin preparation with chlorhexidine-alcohol, and aseptic technique.
- Hand hygiene with soap and water or alcohol-based sanitizer
- Vaccination according to recommended schedules
- Safe food handling and water purification
- Use of personal protective equipment in healthcare settings
- Safe sex practices including condom use
- Mosquito avoidance (nets, repellents, insecticides)
- Respiratory etiquette (covering coughs and sneezes)
- Isolation of infected individuals when indicated
- Prophylactic antibiotics or antivirals for high-risk exposures
- Travel precautions and pre-travel consultations

886.9 Treatment Options

- Treatment focuses on wound opening and removal of fluid for superficial SSIs, wound packing, and targeted antibiotic therapy
- Deep SSIs and organ/space infections may require surgical removal of dead tissue or reoperation.
- Antibiotics for bacterial infections (targeted or empiric)
- Antiviral medications for viral infections
- Antifungal therapy for fungal infections
- Antiparasitic medications for parasitic infections
- Supportive care: fluids, rest, nutrition, fever control
- Hospitalization for severe cases requiring intravenous therapy
- Oxygen therapy and respiratory support if needed
- Surgical drainage of abscesses when necessary
- Pain management and symptom relief
- Treatment of complications and underlying conditions

886.10 Living with It

- Complete the full course of prescribed medications
- Get adequate rest and maintain hydration during recovery
- Monitor for worsening symptoms that require medical attention
- Practice good hygiene to prevent spread to others
- Follow up with your healthcare provider as recommended
- Gradually resume normal activities as symptoms improve

886.11 Outlook

Most infectious diseases resolve completely with appropriate treatment, especially when diagnosed early. Outcomes depend on the pathogen, host immune status, and timeliness of intervention. Some infections can become chronic (HIV, hepatitis B/C, herpes) requiring long-term management. Antimicrobial resistance may complicate treatment and worsen outcomes. Public health measures and vaccination programs continue to reduce the global burden of infectious diseases.

Chapter 887

Yellow Fever

887.1 Overview

Yellow fever is a mosquito-borne involving bleeding fever caused by yellow fever virus (YFV), a flavivirus transmitted by *Aedes* and *Haemogogus* mosquitoes, with 200,000 cases and 30,000 deaths annually in tropical Africa and South America. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

887.2 Causes

This condition happens because of the virus infecting dendritic cells, spreading to lymph nodes and liver, with hepatitis resulting from direct viral cytopathology of hepatocytes (midzonal tissue death, Councilman bodies), and the pathogenesis of bleeding including liver coagulopathy and thrombocytopenia. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

887.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

887.4 Signs and Symptoms

Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

887.5 Progression and Stages

You may experience an incubation of 3–6 days, with three phases: acute (fever, chills, headache, muscle pain, conjunctival injection, Faget's sign), remission (hours to 1 day), and intoxication phase (15–25% progress): recurrence of fever, jaundice, liver failure, kidney failure, involving bleeding manifestations (hematemesis—"black vomit"), and shock. Treatment typically involves supportive. outlook: mortality in the intoxication phase is 20–50%. The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

887.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

887.7 Diagnosis

Doctors diagnose this using RT-PCR or antigen detection and serology. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system

- Consultation with relevant specialists

887.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

887.9 Treatment Options

Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

887.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

887.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 888

Zenker Diverticulum

888.1 Overview

Zenker diverticulum (pharyngoesophageal diverticulum) is a false mucosal herniation through Killian triangle—a anatomically vulnerable focal weakness in the posterior hypopharyngeal wall bounded by the inferior pharyngeal constrictor and the cricopharyngeus muscle. Occurring predominantly in elderly males (> 70 years), the underlying Disease mechanism involves cricopharyngeal motor dysfunction, incoordination, or hypertonicity during swallowing, generating elevated intrabuminal hypopharyngeal pressure that forces mucosa and submucosa outward. As swallowed food and saliva accumulate within the pouch, the diverticulum expands inferiorly, compressing the esophagus. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

888.2 Causes

The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

888.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

888.4 Signs and Symptoms

Symptoms often appear as progressive difficulty swallowing, regurgitation of undigested food hours after meals, halitosis (foul breath due to food decomposition), chronic cough, nocturnal removal by suction pneumonia, and a palpable left neck mass that gargles upon compression (Boyce sign). Barium swallow esophagography is diagnostic, demonstrating a pouch originating from the posterior hypopharynx. Symptoms vary depending on the severity and extent of the condition Pain or discomfort in the affected area Functional impairment affecting daily activities Changes in appearance or function of affected tissues Systemic symptoms may include fatigue, fever, or weight changes Symptoms may develop gradually or appear suddenly

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

888.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

888.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

888.7 Diagnosis

Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

888.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

888.9 Treatment Options

Treatment focuses on surgical diverticulectomy with cricopharyngeal myotomy or using a thin camera tube stapled diverticulotomy. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

888.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

888.11 Outlook

The outlook varies depending on the specific condition, severity at diagnosis, and response to treatment. Early intervention generally leads to better outcomes. Many conditions can be effectively managed with appropriate treatment. Regular follow-up care is important for monitoring and treatment adjustment.

Chapter 889

Zika Virus

889.1 Overview

Zika virus is a mosquito-borne flavivirus transmitted primarily by *Aedes aegypti* and *A. albopictus*, with other routes including sexual, vertical, blood transfusion, and laboratory exposure, with major outbreaks in Micronesia (2007), French Polynesia (2013–2014), and South America (2015–2016). This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Prompt diagnosis and appropriate management are essential for optimal outcomes. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

889.2 Causes

This condition happens because of the virus replicating in dendritic cells and spreading hematogenously, with tropism for neural progenitor cells, inducing apoptosis and impairing neurogenesis—the basis for congenital Zika syndrome. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors that disrupt normal physiological processes. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition. Understanding the specific cause helps guide targeted treatment approaches.

889.3 Risk Factors

Genetic predisposition and family history Advancing age Lifestyle factors (diet, exercise, smoking, alcohol) Environmental exposures Underlying medical conditions Medication use Stress and psychological factors Socioeconomic factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

889.4 Signs and Symptoms

- You may experience an incubation of 3–14 days, with 80% being asymptomatic
- Symptomatic disease presents with self-limited febrile illness with maculopapular rash (pruritic, descending), conjunctivitis (non-purulent), joint pain, and headache. Congenital Zika syndrome includes microcephaly, intracranial calcifications, ventriculomegaly, and arthrogryposis. Guillain-Barré syndrome is linked to Zika infection.
- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

889.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

889.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

889.7 Diagnosis

Doctors diagnose this using RT-PCR of serum or urine within 14 days of symptom onset, with serology for confirmation. Comprehensive medical history and physical examination Laboratory tests to assess organ function and rule out other conditions Imaging studies to visualize affected structures Specialized testing depending on the affected system Consultation with relevant specialists Monitoring of disease progression over time

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

889.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

889.9 Treatment Options

Treatment typically involves supportive. Medications to manage symptoms and address underlying mechanisms Lifestyle modifications and self-care measures Physical therapy and rehabilitation Surgical intervention when indicated Regular monitoring and follow-up care Patient education and support services

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

889.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

889.11 Outlook

- Most people recover well for acute illness in adults
- Congenital infection has significant morbidity.

Chapter 890

Zinc Deficiency

890.1 Overview

Zinc deficiency is a mineral deficiency disorder common in low-resource settings due to low dietary intake of animal-source foods and high phytate content (inhibits zinc absorption). At-risk groups include pregnant and lactating women, children (especially with diarrhea), the elderly, vegetarians, and those with malabsorption (acrodermatitis enteropathica—autosomal recessive zinc transporter defect). This metabolic disorder involves dysregulation of normal bodily processes, often requiring ongoing monitoring and management. It is increasingly common due to lifestyle and dietary factors. Metabolic disorders involve disruption of normal biochemical processes essential for health. These conditions may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors. Many metabolic disorders are chronic and require ongoing management. Lifestyle modifications are often the cornerstone of treatment. This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

890.2 Causes

This condition happens because of deficiency of zinc, an essential trace element that serves as a cofactor for over 300 enzymes involved in growth, immune function, protein synthesis, wound healing, and DNA synthesis. Metabolic disorders arise from dysregulation of normal biochemical pathways. This may result from enzyme deficiencies, hormonal imbalances, or lifestyle factors that overwhelm normal homeostatic mechanisms. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

890.3 Risk Factors

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures

- Underlying medical conditions
- Medication use

890.4 Signs and Symptoms

You may experience growth retardation, delayed sexual maturation, hypogonadism, impaired immune function (recurrent infections, especially diarrheal diseases), alopecia, dermatitis (periorificial and acral, vesiculobullous or eczematous—characteristic of acrodermatitis enteropathica), poor wound healing, night blindness, anorexia, dysgeusia, and neuropsychiatric changes.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

890.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

890.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

890.7 Diagnosis

Doctors diagnose this using low plasma zinc (less than 60–70 µg/dL).

- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

890.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

890.9 Treatment Options

- Treatment focuses on zinc sulfate or acetate 1–3 mg/kg/day (adults: 25–50 mg elemental zinc daily)
- For acrodermatitis enteropathica, lifelong supplementation is required.
- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

890.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

890.11 Outlook

Most people recover well with supplementation.

Chapter 891

Zollinger-Ellison Syndrome

891.1 Overview

This condition affects a significant number of people worldwide and can range from mild to severe in presentation. Early recognition and appropriate management are essential for optimal outcomes. A thorough understanding of the underlying mechanisms guides treatment decisions. Patient education and self-management play important roles in long-term care.

- Zollinger-Ellison syndrome (ZES) is a rare disorder caused by a gastrin-secreting neuroendocrine tumor (gastrinoma), located most commonly in the duodenum or pancreas (gastrinoma triangle)
- Approximately 25–30% of cases occur as part of Multiple Endocrine Neoplasia Type 1 (MEN1).

891.2 Causes

This condition happens because of autonomous hypergastrinemia, causing massive gastric acid hypersecretion and severe recurrent peptic ulceration. Genetic disorders result from mutations in specific genes that encode proteins essential for normal cellular function. The pattern of inheritance depends on whether the mutation is dominant, recessive, or X-linked. The underlying mechanisms involve complex interactions between genetic predisposition and environmental factors. Disruption of normal physiological processes leads to the characteristic features of the condition. Multiple contributing factors may act synergistically to trigger or worsen the condition.

891.3 Risk Factors

Family history of the genetic condition Consanguineous parents (increased risk of recessive disorders) Advanced parental age (new mutations) Carrier status in parents Ethnic background (specific founder mutations) Previous child with a genetic disorder

- Genetic predisposition and family history
- Advancing age
- Lifestyle factors (diet, exercise, smoking, alcohol)
- Environmental exposures
- Underlying medical conditions
- Medication use

891.4 Signs and Symptoms

You may experience refractory or multiple peptic ulcers (often extending to the jejunum), chronic watery diarrhea, and esophagitis.

- Pain or discomfort in the affected area
- Functional impairment affecting daily activities
- Changes in appearance or function of affected tissues
- Systemic symptoms may include fatigue, fever, or weight changes
- Symptoms may develop gradually or appear suddenly

891.5 Progression and Stages

The condition may progress through different stages of severity over time. Early stages often involve mild or subtle symptoms that may be overlooked. Moderate stages involve more noticeable symptoms affecting daily function. Advanced stages may involve significant impairment and complications.

891.6 Complications

- Disease progression leading to worsening symptoms
- Secondary infections or injuries
- Side effects of treatment
- Reduced quality of life
- Emotional and psychological impact

891.7 Diagnosis

- Doctors confirm the diagnosis through elevated fasting serum gastrin ($>$ 1000 pg/mL) and confirmed by secretin stimulation test (showing a rise in gastrin $>$ 120 pg/mL)
- Tumor localization is achieved via somatostatin receptor scintigraphy (OctreoScan) or Ga-68 DOTATATE PET-CT.
- Comprehensive medical history and physical examination
- Laboratory tests to assess organ function
- Imaging studies to visualize affected structures
- Specialized testing depending on the affected system
- Consultation with relevant specialists

891.8 Prevention

- Maintain a healthy lifestyle with balanced diet and exercise
- Avoid known risk factors
- Attend regular medical check-ups for early detection
- Follow recommended screening guidelines

891.9 Treatment Options

Treatment focuses on high-dose proton pump inhibitors and surgical removal of localized tumors. Symptomatic management and supportive care Enzyme replacement therapy for certain conditions Gene therapy (emerging for select conditions) Dietary modifications (e.g., phenylketonuria) Regular monitoring for associated complications Physical and occupational therapy Genetic counseling for family planning Participation in clinical trials for new therapies

- Medications to manage symptoms and address underlying mechanisms
- Lifestyle modifications and self-care measures
- Physical therapy and rehabilitation
- Surgical intervention when indicated
- Regular monitoring and follow-up care

891.10 Living with It

- Follow your treatment plan as prescribed
- Maintain regular follow-up with your healthcare provider
- Make necessary lifestyle adjustments
- Seek support from family, friends, or support groups
- Stay informed about your condition

891.11 Outlook

Your outlook depends on tumor resectability and metastatic status.

Organ–Disease Dictionary

This appendix lists every disease in this book organized by the primary organ or organ system it affects.

Blood

- Acute Lymphoblastic Leukemia
- Acute Myeloid Leukemia
- Aplastic Anemia
- Chronic Lymphocytic Leukemia
- Deep Vein Thrombosis
- Disseminated Intravascular Coagulation
- Essential Thrombocythemia
- Folate Deficiency Anemia
- Hemolytic Anemias
- Hemophilia A
- Hemophilia B
- Hereditary Spherocytosis
- Hodgkin Lymphoma
- Immune Thrombocytopenic Purpura
- Iron Deficiency Anemia
- Methemoglobinemia
- Multiple Myeloma
- Myelodysplastic Syndromes
- Myelofibrosis
- Non-Hodgkin Lymphoma
- Polycythemia Vera
- Sickle Cell Disease
- Sideroblastic Anemia
- Thalassemia
- Thrombotic Thrombocytopenic Purpura
- Vitamin B12 Deficiency Anemia
- Von Willebrand Disease

Bones

- Achondroplasia
- Adhesive Capsulitis
- Ankylosing Spondylitis
- Bursitis
- Calcium Pyrophosphate Deposition Disease

- Dupuytren Contracture
- Ewing Sarcoma
- Fibrous Dysplasia
- Fractures
- Gout
- Hallux Valgus
- Kyphosis
- Morton Neuroma
- Multiple Myeloma (Bone Manifestations)
- Osteoarthritis
- Osteogenesis Imperfecta
- Osteomyelitis
- Osteoporosis
- Osteosarcoma
- Paget's Disease of Bone
- Plantar Fasciitis
- Psoriatic Arthritis
- Reactive Arthritis
- Rheumatoid Arthritis
- Rickets and Osteomalacia
- Scoliosis
- Septic Arthritis
- Trigger Finger

Brain

- Alzheimer's Disease
- Amyotrophic Lateral Sclerosis
- Bacterial Meningitis
- Bell's Palsy
- Brain Abscess
- Carpal Tunnel Syndrome
- Central Pontine Myelinolysis
- Cerebral Aneurysm
- Cerebral Venous Thrombosis
- Charcot-Marie-Tooth Disease
- Chronic Traumatic Encephalopathy
- Coma
- Concussion
- Dystonia
- Epidural Hematoma
- Epilepsy
- Essential Tremor
- Febrile Seizures
- Friedreich's Ataxia

- Frontotemporal Dementia
- Glioblastoma
- Guillain-Barré Syndrome
- Hemorrhagic Stroke
- Huntington's Disease
- Ischemic Stroke
- Medulloblastoma
- Meningioma
- Motor Neuron Disease
- Multiple Sclerosis
- Neurocysticercosis
- Neurofibromatosis Type 1
- Parkinson's Disease
- Persistent Vegetative State
- Pituitary Adenoma
- Schwannoma
- Subdural Hematoma
- Temporal Lobe Epilepsy
- Transient Ischemic Attack
- Vertigo
- Viral Encephalitis

Breast

- Breast Cancer
- Ductal Carcinoma In Situ (DCIS)
- Fat Necrosis of the Breast
- Fibroadenoma
- Fibrocystic Changes
- Galactocele
- Gynecomastia
- Idiopathic Granulomatous Mastitis
- Inflammatory Breast Cancer
- Intraductal Papilloma
- Invasive Lobular Carcinoma (ILC)
- Lobular Carcinoma In Situ (LCIS)
- Male Breast Cancer
- Mammary Duct Ectasia
- Mastitis and Breast Abscess
- Paget's Disease of the Nipple
- Phyllodes Tumor
- Triple-Negative Breast Cancer (TNBC)

Connective Tissue

- Adult-Onset Still's Disease
- Amyloidosis
- Anti-GBM Disease
- Behçet's Disease
- Cogan's Syndrome
- Cryoglobulinemic Vasculitis
- Ehlers-Danlos Syndrome
- Eosinophilic Fasciitis (Shulman Syndrome)
- Eosinophilic Granulomatosis with Polyangiitis
- Giant Cell Arteritis
- Granulomatosis with Polyangiitis
- Hemophagocytic Lymphohistiocytosis and Macrophage Activation Syndrome
- IgA Vasculitis
- IgG4-Related Disease
- Kawasaki Disease
- Marfan Syndrome
- Microscopic Polyangiitis
- Mixed Connective Tissue Disease
- Periodic Fever Syndromes
- Polyarteritis Nodosa
- Polymyalgia Rheumatica
- Relapsing Polychondritis
- Sarcoidosis
- Systemic Sclerosis (Scleroderma)
- Takayasu Arteritis
- Undifferentiated Connective Tissue Disease

Ears

- Acoustic Neuroma (Vestibular Schwannoma)
- Acute Otitis Media
- Barotrauma of the Ear
- Benign Paroxysmal Positional Vertigo
- Cerumen Impaction
- Chondrodermatitis Nodularis Helicis
- Chronic Otitis Media
- Labyrinthitis
- Ménière's Disease
- Otitis Externa
- Otosclerosis
- Sudden Sensorineural Hearing Loss
- Tinnitus
- Tympanic Membrane Perforation

- Vestibular Neuritis

Endocrine

- Acromegaly
- Addison Disease
- Addison's Disease
- Adrenal Incidentaloma
- Cushing Syndrome
- Cushing's Syndrome
- Diabetes Insipidus
- Graves Disease
- Hashimoto Thyroiditis
- Hyperparathyroidism
- Hyperprolactinemia
- Hyperthyroidism
- Hypoparathyroidism
- Hypopituitarism
- Hypothyroidism
- Multiple Endocrine Neoplasia
- Pheochromocytoma
- SIADH
- Thyroid Nodules and Cancer
- Thyroiditis

Eyes

- Age-Related Cataract
- Age-Related Macular Degeneration
- Amblyopia
- Astigmatism
- Congenital Cataract
- Conjunctivitis
- Diabetic Retinopathy
- Dry Eye Disease
- Fuchs' Endothelial Dystrophy
- Glaucoma in Children
- Hyperopia
- Keratitis
- Keratoconus
- Myopia
- Presbyopia
- Primary Angle-Closure Glaucoma
- Primary Open-Angle Glaucoma
- Retinal Detachment

- Retinitis Pigmentosa
- Retinopathy of Prematurity
- Scleritis
- Strabismus
- Uveitis

Geriatric

- Advance Care Planning
- Age-Related Bone Loss
- Age-Related Macular Degeneration
- Alzheimer Disease
- Aortic Stenosis
- Atrial Fibrillation in the Elderly
- Beers Criteria / STOPP/START Criteria
- Cellular Senescence
- Delirium
- Dementia with Lewy Bodies
- Drug Interactions in the Elderly
- Ethical Issues
- Failure to Thrive
- Falls in the Elderly
- Fear of Falling
- Fecal Incontinence
- Frailty
- Frailty as a Syndrome
- Frontotemporal Dementia
- Gait Disorders
- Heart Failure with Preserved Ejection Fraction
- Hip Fracture
- Hospice Care
- Hypertension in the Elderly
- Malnutrition in the Elderly
- Medication Non-Adherence
- Mild Cognitive Impairment
- Normal Pressure Hydrocephalus
- Pain Management in the Elderly
- Parkinson Disease
- Polypharmacy
- Presbycusis
- Presbyopia and Cataracts
- Pressure Ulcers
- Sarcopenia
- Theories of Aging
- Urinary Incontinence

- Vascular Dementia

Heart

- Acute Decompensated Heart Failure
- Angina Pectoris
- Aortic Dissection
- Aortic Regurgitation
- Aortic Stenosis
- Atrial Fibrillation
- Brugada Syndrome
- Cardiac Tamponade
- Coarctation of the Aorta
- Constrictive Pericarditis
- Dilated Cardiomyopathy
- Heart Failure with Preserved Ejection Fraction
- Heart Failure with Reduced Ejection Fraction
- Hypertension
- Hypertrophic Cardiomyopathy
- Infective Endocarditis
- Ischemic Cardiomyopathy
- Long QT Syndrome
- Mitral Regurgitation
- Mitral Stenosis
- Mitral Valve Prolapse
- Myocardial Infarction
- Myocarditis
- Pericarditis
- Peripheral Arterial Disease
- Peripheral Artery Disease
- Restrictive Cardiomyopathy
- Rheumatic Heart Disease
- Sick Sinus Syndrome
- Tricuspid Regurgitation
- Ventricular Tachycardia

Infectious

- African Trypanosomiasis (Sleeping Sickness)
- Amebiasis (Entamoeba histolytica)
- Anthrax
- Arboviral Encephalitis
- Aspergillosis (Invasive)
- Blastomycosis
- Botulism (Clostridium botulinum)

- Brucellosis
- Candida auris
- Candidiasis (Invasive / Systemic)
- Chagas Disease (American Trypanosomiasis)
- Chancroid
- Chickenpox
- Chikungunya
- Cholera (Vibrio cholerae)
- Coccidioidomycosis (Valley Fever)
- Creutzfeldt-Jakob Disease (CJD)
- Cryptococcosis
- Cyclospora cayetanensis
- Cytomegalovirus (CMV)
- Dengue Fever
- Diphtheria (Corynebacterium diphtheriae)
- Eastern Equine Encephalitis
- Ebola Virus Disease
- Ehrlichiosis
- Epstein-Barr Virus / Infectious Mononucleosis (Systemic Aspects)
- Filariasis (Lymphatic Filariasis)
- Gas Gangrene (Clostridial Myonecrosis)
- Giardiasis
- Hand, Foot, and Mouth Disease
- Hantavirus Pulmonary Syndrome
- Histoplasmosis
- Hookworm / Soil-Transmitted Helminths
- Human Metapneumovirus (hMPV)
- Kuru
- Leishmaniasis
- Leprosy (Hansen's Disease)
- Leptospirosis
- Lyme Disease (Borrelia burgdorferi)
- Lymphogranuloma Venereum (LGV)
- Malaria (Plasmodium spp.)
- Marburg Virus
- Measles (Rubeola)
- Mpox
- Mucormycosis
- Mumps
- Necrotizing Fasciitis
- Non-Gonococcal Urethritis (NGU)
- Norovirus
- Pertussis (Whooping Cough)
- Plague (Yersinia pestis)
- Poliomyelitis
- Primary Amebic Meningoencephalitis (Naegleria fowleri)

- Rabies
- Rocky Mountain Spotted Fever
- Rubella
- Scarlet Fever
- Schistosomiasis
- Sepsis and Septic Shock
- Syphilis (Treponema pallidum)
- Tapeworm Infections (Taeniasis / Cysticercosis)
- Tetanus (Clostridium tetani)
- Toxoplasmosis
- Tuberculosis (Extrapulmonary)
- Typhoid Fever (Salmonella typhi)
- Variant CJD (Mad Cow Disease)
- Varicella-Zoster (Systemic Aspects)
- West Nile Virus
- Yellow Fever
- Zika Virus

Intestines

- Anal Fissure
- Appendicitis
- Celiac Disease
- Colorectal Adenoma
- Colorectal Cancer
- Crohn's Disease
- Diverticulitis
- Diverticulosis
- Familial Adenomatous Polyposis
- Indeterminate Colitis
- Irritable Bowel Syndrome
- Large Bowel Obstruction
- Lynch Syndrome
- Short Bowel Syndrome
- Small Bowel Obstruction
- Ulcerative Colitis

Kidneys

- Acute Interstitial Nephritis
- Acute Kidney Injury
- Autosomal Dominant Polycystic Kidney Disease
- Bladder Cancer
- Chronic Kidney Disease
- Chronic Tubulointerstitial Nephritis

- Cystitis
- Diabetic Nephropathy
- End-Stage Renal Disease
- Fanconi Syndrome
- Focal Segmental Glomerulosclerosis
- IgA Nephropathy
- Kidney Cancer (Renal Cell Carcinoma)
- Membranous Nephropathy
- Minimal Change Disease
- Nephritic Syndrome
- Nephrolithiasis
- Nephrotic Syndrome
- Pyelonephritis
- Renal Osteodystrophy
- Renal Tubular Acidosis
- Urinary Incontinence
- Urosepsis

Liver

- Alcoholic Liver Disease
- Ascites
- Autoimmune Hepatitis
- Budd-Chiari Syndrome
- Cirrhosis
- Hepatic Encephalopathy
- Hepatitis A
- Hepatitis B
- Hepatitis C
- Hepatitis D
- Hepatitis E
- Hepatocellular Carcinoma
- Hereditary Hemochromatosis
- Non-Alcoholic Fatty Liver Disease
- Portal Hypertension
- Primary Biliary Cholangitis
- Primary Sclerosing Cholangitis
- Wilson's Disease

Lungs

- Acute Bronchitis
- Asbestosis
- Asthma
- Asthma-COPD Overlap

- Atelectasis
- Bronchiectasis
- Chronic Obstructive Pulmonary Disease
- Community-Acquired Pneumonia
- COVID-19
- Cystic Fibrosis
- Empyema
- Idiopathic Pulmonary Fibrosis
- Influenza
- Lung Abscess
- Lung Cancer
- Mesothelioma
- Pleurisy
- Pneumoconiosis
- Pneumothorax
- Pulmonary Embolism
- Pulmonary Hypertension
- Sarcoidosis
- Silicosis
- Tuberculosis

Lymph

- Anaphylaxis
- Antiphospholipid Syndrome
- Behçet's Disease
- Chronic Fatigue Syndrome
- Common Variable Immunodeficiency
- DiGeorge Syndrome
- Drug Hypersensitivity
- Hereditary Angioedema
- HIV/AIDS
- Infectious Mononucleosis
- Kikuchi-Fujimoto Disease
- Rheumatoid Arthritis
- Rosai-Dorfman Disease
- Sarcoidosis
- Severe Combined Immunodeficiency
- Sjögren's Syndrome
- Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis
- Systemic Lupus Erythematosus

Muscles

- Becker Muscular Dystrophy

- Compartment Syndrome
- Dermatomyositis
- Duchenne Muscular Dystrophy
- Facioscapulohumeral Muscular Dystrophy
- Fibromyalgia
- Inclusion Body Myositis
- Lambert-Eaton Myasthenic Syndrome (LEMS)
- Malignant Hyperthermia
- McArdle Disease
- Myasthenia Gravis
- Myotonic Dystrophy
- Periodic Paralysis
- Polymyositis
- Rhabdomyolysis

Nose

- Acute Bacterial Sinusitis
- Allergic Fungal Rhinosinusitis (AFRS)
- Allergic Rhinitis
- Choanal Atresia
- Chronic Rhinosinusitis
- Deviated Nasal Septum
- Epistaxis
- Esthesioneuroblastoma
- Fungal Sinusitis
- Inverted Papilloma
- Juvenile Nasopharyngeal Angiofibroma (JNA)
- Nasal Fracture
- Nasal Polyps
- Non-Allergic Rhinitis

Nutrition

- Alkaptonuria — Ochronosis
- Alpha-1 Antitrypsin Deficiency
- Calcium and Phosphorus Disorders
- Copper Deficiency and Excess
- Dysbetalipoproteinemia
- Familial Combined Hyperlipidemia
- Familial Hypercholesterolemia
- Fluoride Excess — Fluorosis
- Galactosemia
- Glycogen Storage Diseases
- Hereditary Hemochromatosis

- Homocystinuria
- Iodine Deficiency Disorders
- Iron Deficiency
- Kwashiorkor
- Lipoprotein(a) Disorders
- Lysosomal Storage Diseases
- Maple Syrup Urine Disease
- Marasmic-Kwashiorkor
- Marasmus
- Metabolic Syndrome
- Mucopolysaccharidoses
- Obesity
- Organic Acidemias
- Phenylketonuria
- Porphyrias
- Sarcopenic Obesity
- Selenium Deficiency
- Urea Cycle Disorders
- Vitamin A Deficiency
- Vitamin B1 (Thiamine) Deficiency — Beriberi and Wernicke-Korsakoff Syndrome
- Vitamin B12 (Cobalamin) Deficiency
- Vitamin B2 (Riboflavin) Deficiency — Ariboflavinosis
- Vitamin B3 (Niacin) Deficiency — Pellagra
- Vitamin B6 (Pyridoxine) Deficiency
- Vitamin B7 (Biotin) Deficiency
- Vitamin B9 (Folate) Deficiency
- Vitamin C Deficiency — Scurvy
- Vitamin D Deficiency — Rickets and Osteomalacia
- Vitamin E Deficiency
- Vitamin K Deficiency
- Zinc Deficiency

Obstetric

- Abnormal Uterine Bleeding
- Amenorrhea (Primary, Secondary)
- Bacterial Vaginosis
- Bartholin Cyst
- Cervical Cancer
- Cervicitis (Chlamydia, Gonorrhea)
- Chronic Hypertension in Pregnancy
- Dysmenorrhea
- Eclampsia
- Ectopic Pregnancy
- Endometrial / Uterine Cancer

- Endometriosis
- Genital Herpes (HSV)
- Genital Warts (HPV)
- Gestational Diabetes Mellitus (Obstetric Focus)
- Gestational Hypertension
- Gestational Trophoblastic Disease
- HELLP Syndrome
- Hyperemesis Gravidarum
- Intrauterine Growth Restriction
- Mastitis (Puerperal)
- Menopause / Perimenopause
- Multiple Gestation Complications
- Ovarian Cancer
- Pelvic Inflammatory Disease
- Placenta Accreta Spectrum
- Placenta Previa
- Placental Abruption
- Polycystic Ovary Syndrome
- Polyhydramnios and Oligohydramnios
- Postpartum Cardiomyopathy
- Postpartum Depression
- Postpartum Endometritis
- Postpartum Hemorrhage
- Postpartum Thyroiditis
- Preeclampsia
- Premature Ovarian Insufficiency
- Premature Rupture of Membranes
- Premenstrual Syndrome / Premenstrual Dysphoric Disorder
- Preterm Labor and Preterm Birth
- Recurrent Pregnancy Loss
- Rh Incompatibility / Hemolytic Disease of the Newborn
- Septic Abortion
- Spontaneous Abortion (Miscarriage)
- Stillbirth (Intrauterine Fetal Demise)
- Trichomoniasis
- Uterine Rupture
- Vaginal Cancer
- Vasa Previa
- Vulvar Cancer
- Vulvovaginal Candidiasis

Pain

- Adjuvant Analgesics
- Atypical Facial Pain

- Central Post-Stroke Pain
- Cervicogenic Headache
- Chronic Low Back Pain
- Chronic Neck Pain
- Chronic Pain vs Acute Pain
- Chronic Pelvic Pain
- Cluster Headache
- Complex Regional Pain Syndrome
- Diabetic Peripheral Neuropathic Pain
- Fibromyalgia
- Glossopharyngeal Neuralgia
- Headache Attributed to Giant Cell Arteritis
- Headache Attributed to Intracranial Hypertension
- Headache Attributed to Intracranial Hypotension
- Headache Attributed to Sinusitis
- Headache in Brain Tumors
- Hemicrania Continua
- Interventional Pain Management
- Medication Overuse Headache
- Migraine
- Myofascial Pain Syndrome
- New Daily Persistent Headache
- Nociception
- Non-Opioid Analgesics
- Occipital Neuralgia
- Opioid Analgesics
- Opioid Therapy Risks
- Other Primary Headache Disorders
- Paroxysmal Hemicrania
- Phantom Limb Pain
- Physical and Rehabilitation Approaches
- Post-Traumatic Headache
- Postherpetic Neuralgia
- Psychological Approaches
- SUNCT and SUNA
- Temporomandibular Joint Disorders
- Tension-Type Headache
- Trigeminal Autonomic Cephalalgias
- Trigeminal Neuralgia
- Types of Pain
- WHO Analgesic Ladder

Pancreas

- Acute Pancreatitis

- Autoimmune Pancreatitis
- Autoimmune Pancreatitis (AIP)
- Chronic Pancreatitis
- Diabetic Ketoacidosis
- Gestational Diabetes Mellitus
- Hyperosmolar Hyperglycemic State
- Hypoglycemia
- Intraductal Papillary Mucinous Neoplasm (IPMN)
- Nesidioblastosis and Congenital Hyperinsulinism
- Pancreatic Adenocarcinoma
- Pancreatic Exocrine Insufficiency
- Pancreatic Neuroendocrine Tumors
- Pancreatic Pseudocyst
- Type 1 Diabetes Mellitus
- Type 2 Diabetes Mellitus

Pediatric

- Acute Lymphoblastic Leukemia (ALL) in Children
- Acute Otitis Media
- Ambiguous Genitalia / Disorders of Sex Development
- Angelman Syndrome
- Brain Tumors in Children
- Bronchiolitis (RSV)
- Bronchopulmonary Dysplasia (BPD)
- Cleft Lip and Palate
- Congenital Diaphragmatic Hernia
- Congenital Heart Disease (Overview)
- Congenital Hip Dysplasia
- Congenital Talipes Equinovarus (Clubfoot)
- Cri-du-Chat Syndrome
- Croup (Laryngotracheobronchitis)
- Cryptorchidism (Undescended Testis)
- Down Syndrome (Trisomy 21)
- Edwards Syndrome (Trisomy 18)
- Epiglottitis
- Esophageal Atresia / Tracheoesophageal Fistula
- Ewing Sarcoma in Children
- Fragile X Syndrome
- Henoch-Schönlein Purpura (IgA Vasculitis)
- Hepatoblastoma
- Hirschsprung Disease
- Human Metapneumovirus (hMPV)
- Hypertrophic Pyloric Stenosis
- Intraventricular Hemorrhage (IVH)

- Intussusception
- Kawasaki Disease
- Klinefelter Syndrome (47,XXY)
- Meconium Aspiration Syndrome
- Necrotizing Enterocolitis (NEC)
- Neonatal Abstinence Syndrome
- Neonatal Hypoglycemia
- Neonatal Jaundice / Hyperbilirubinemia / Kernicterus
- Neonatal Sepsis (Early-Onset and Late-Onset)
- Neural Tube Defects
- Neuroblastoma
- Osteosarcoma in Children
- Patau Syndrome (Trisomy 13)
- Patent Ductus Arteriosus (PDA) in Prematurity
- Pediatric Appendicitis
- Perinatal Asphyxia / Hypoxic-Ischemic Encephalopathy (HIE)
- Periventricular Leukomalacia (PVL)
- Persistent Pulmonary Hypertension of the Newborn (PPHN)
- Prader-Willi Syndrome
- Preterm Birth
- Respiratory Distress Syndrome (RDS)
- Retinoblastoma
- Retinopathy of Prematurity (ROP)
- Rhabdomyosarcoma
- Rotavirus
- Turner Syndrome (45,X)
- Viral Exanthems
- Williams Syndrome
- Wilms Tumor (Nephroblastoma)

Psychiatric

- Acute Stress Disorder
- Adjustment Disorder
- Agoraphobia
- Alcohol Use Disorder
- Anorexia Nervosa
- Antisocial Personality Disorder (ASPD)
- Attention-Deficit/Hyperactivity Disorder (ADHD)
- Autism Spectrum Disorder (ASD)
- Binge-Eating Disorder
- Bipolar I Disorder
- Bipolar II Disorder
- Body Dysmorphic Disorder
- Borderline Personality Disorder (BPD)

- Bulimia Nervosa
- Cyclothymic Disorder
- Delusional Disorder
- Dissociative Amnesia
- Dissociative Identity Disorder
- Generalized Anxiety Disorder (GAD)
- Hoarding Disorder
- Illness Anxiety Disorder (Hypochondriasis)
- Major Depressive Disorder (MDD)
- Narcissistic Personality Disorder
- Obsessive-Compulsive Disorder (OCD)
- Opioid Use Disorder
- Panic Disorder / Panic Attacks
- Persistent Depressive Disorder (Dysthymia)
- Post-Traumatic Stress Disorder (PTSD)
- Schizoaffective Disorder
- Schizophrenia
- Social Anxiety Disorder (Social Phobia)
- Somatic Symptom Disorder
- Specific Phobias
- Stimulant Use Disorder (Cocaine, Amphetamines)
- Tourette Syndrome

Reproductive

- Atrophic Vaginitis (Genitourinary Syndrome of Menopause)
- Balanitis and Balanoposthitis
- Benign Prostatic Hyperplasia
- Cervical Cancer
- Endometriosis
- Erectile Dysfunction
- Male Infertility
- Ovarian Cancer
- Pelvic Inflammatory Disease
- Penile Cancer
- Penile Fracture
- Peyronie's Disease
- Phimosis and Paraphimosis
- Polycystic Ovary Syndrome
- Priapism
- Prostate Cancer
- Testicular Cancer
- Uterine Fibroids
- Vaginal Cancer
- Vulvar Cancer

Skin

- Abscess
- Acne Vulgaris
- Actinic Keratosis
- Alopecia Areata
- Atopic Dermatitis
- Basal Cell Carcinoma
- Bullous Pemphigoid
- Cellulitis
- Contact Dermatitis
- Dermatitis Herpetiformis
- Dermatomyositis
- Epidermolysis Bullosa
- Head Lice (Pediculosis Capitis)
- Herpes Simplex
- Herpes Zoster
- Impetigo
- Melanoma
- Melasma
- Molluscum Contagiosum
- Nummular Dermatitis
- Oculocutaneous Albinism
- Pemphigus Vulgaris
- Pilonidal Sinus
- Psoriasis
- Psoriatic Arthritis
- Rosacea
- Scabies
- Seborrheic Dermatitis
- Squamous Cell Carcinoma
- Systemic Lupus Erythematosus (Skin Manifestations)
- Tinea Infections
- Urticaria
- Vitiligo
- Warts
- Wound Infections and Surgical Site Infections

Sleep

- Acute/Short-Term Insomnia
- Advanced Sleep-Wake Phase Disorder
- Central Sleep Apnea
- Chronic Insomnia Disorder
- Circadian Rhythm Regulation

- Complex Sleep Apnea
- Delayed Sleep-Wake Phase Disorder
- Idiopathic Hypersomnia
- Idiopathic Insomnia
- Insomnia in the Context of Medical/Psychiatric Disorders
- Irregular Sleep-Wake Rhythm Disorder
- Jet Lag Disorder
- Kleine-Levin Syndrome
- Narcolepsy Type 1
- Narcolepsy Type 2
- Non-24-Hour Sleep-Wake Rhythm Disorder
- NREM Parasomnias: Confusional Arousals
- NREM Parasomnias: Sleep Terrors
- NREM Parasomnias: Sleep-Related Eating Disorder
- NREM Parasomnias: Sleepwalking
- Obesity Hypoventilation Syndrome
- Obstructive Sleep Apnea
- Other Parasomnias: Exploding Head Syndrome
- Other Parasomnias: Sleep Enuresis
- Other Parasomnias: Sleep-Related Hallucinations
- Periodic Limb Movement Disorder
- Propriospinal Myoclonus at Sleep Onset
- Psychophysiologic Insomnia
- REM Parasomnias: Nightmare Disorder
- REM Parasomnias: Recurrent Isolated Sleep Paralysis
- REM Parasomnias: REM Sleep Behavior Disorder
- Restless Legs Syndrome
- Shift Work Sleep Disorder
- Sleep Architecture Across the Lifespan
- Sleep Stages (NREM N1–N3, REM)
- Sleep-Related Bruxism
- Sleep-Related Rhythmic Movement Disorder

Spleen

- Asplenia
- Asplenia and Hyposplenism
- Hypersplenism
- Massive Splenomegaly
- Splenic Abscess
- Splenic Artery Aneurysm
- Splenic Cyst
- Splenic Cysts
- Splenic Infarction
- Splenic Lymphoma

- Splenic Rupture
- Splenomegaly

Stomach

- Achalasia
- Acute Gastritis
- Barrett Esophagus
- Cholecystitis
- Cholelithiasis
- Chronic Gastritis
- Duodenal Ulcer
- Esophageal Varices
- Gastric Adenocarcinoma
- Gastric Lymphoma
- Gastric Ulcer
- Gastroesophageal Reflux Disease
- Gastrointestinal Stromal Tumor
- Gastroparesis
- Helicobacter Pylori Infection
- Peptic Ulcer Bleeding
- Peptic Ulcer Disease
- Zollinger-Ellison Syndrome

Throat

- Aphthous Ulcers (Canker Sores)
- Diphtheria
- Epiglottitis
- Gingivitis and Periodontitis
- Laryngitis
- Ludwig Angina
- Obstructive Sleep Apnea
- Oral Candidiasis
- Oral Leukoplakia
- Oral Lichen Planus
- Oral Squamous Cell Carcinoma
- Pharyngitis
- Tonsillar Hypertrophy
- Tonsillitis
- Vocal Cord Nodules
- Vocal Cord Paralysis
- Zenker Diverticulum

Toxicology

- Acetaminophen (Paracetamol) Overdose
- Alcohol Intoxication and Withdrawal
- Altitude Sickness
- Antidotes Overview
- Arsenic Poisoning
- Benzodiazepine Overdose
- Beta-Blocker Overdose
- Cadmium Toxicity
- Calcium Channel Blocker Overdose
- Cannabis Toxicity and Cannabinoid Hyperemesis Syndrome
- Carbon Monoxide Poisoning
- Cyanide Toxicity
- Digoxin Toxicity
- Drowning (Submersion Injury)
- Electrical Injury and Lightning Strike
- Haff Disease
- Hallucinogen Toxicity
- Harmful Algal Blooms
- Heat Stroke and Hyperthermia
- Hydrogen Sulfide Toxicity
- Hymenoptera Stings
- Hypothermia
- Initial Stabilization and Decontamination
- Iron Overdose
- Lead Poisoning (Plumbism)
- Lithium Toxicity
- Mammalian Bites
- Marine Envenomation
- MDMA (Ecstasy) Toxicity
- Mercury Toxicity
- Metformin Overdose
- Methyl Parathion and Other Organophosphate Insecticides
- Novel Synthetic Drugs
- Opioid Overdose
- Organophosphate and Carbamate Insecticide Poisoning
- Radiation Sickness (Acute Radiation Syndrome)
- Salicylate (Aspirin) Overdose
- Scorpion Stings
- Snake Envenomation
- Spider Bites
- SSRI Overdose and Serotonin Syndrome
- Stimulant Overdose (Cocaine, Methamphetamine)
- Theophylline Overdose
- Toxicokinetics and Toxicodynamics

- Tricyclic Antidepressant (TCA) Overdose

Abbreviations Used

Abbreviation	Full Form
DNA	Deoxyribonucleic acid
RNA	Ribonucleic acid
CT	Computed tomography
MRI	Magnetic resonance imaging
ECG/EKG	Electrocardiogram
EEG	Electroencephalogram
EMG	Electromyography
ANA	Antinuclear antibody
HIV	Human immunodeficiency virus
HPV	Human papillomavirus
TNF	Tumor necrosis factor
IL	Interleukin
Ig	Immunoglobulin
CBC	Complete blood count
CMP	Comprehensive metabolic panel
TSH	Thyroid-stimulating hormone
ACTH	Adrenocorticotropic hormone
BMR	Basal metabolic rate
GFR	Glomerular filtration rate
FEV1	Forced expiratory volume in 1 second

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